SE 3002 - Software Quality Engineering

FAST School of Computing - Department of Software Engineering - FAST NUCES Islamabad

ASSIGNMENT 3 - Student Submission

Black-Box Testing: Equivalence Partitioning & Boundary Value Analysis on Testfolio (testfol.io)

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Test Account Used	i243166@isb.nu.edu.pk
Browser & Version	Google Chrome 142.0.0.0 on Ubuntu Linux

Summary of this submission: 11 forms, 51 input fields, 159 equivalence classes, 48 boundaries, 165 EP + Strong EP test cases and 106 BVA test cases (271 executed with screenshots, 0 marked Not Executed); 14 defect observations.

D1 - Test Plan [20 marks]

TP-1 Test Plan Metadata

Test Plan ID	TP-A3-24I-3166
Version	1.0
Author (Roll No.)	24i-3166 (Muhammad Bilal Tahir)
Status	Final

TP-2 Introduction

Testfolio (testfol.io) is a web-based portfolio back-testing and analysis platform. Users build portfolios from tickers and allocation percentages, run historical back-tests over chosen dates with a starting amount, configure rebalancing, cashflows and benchmarks, and view charts and statistics. It also offers Monte Carlo simulation, safe-withdrawal-rate analysis, an optimizer, an efficient-frontier tool, a Lump Sum vs DCA tool, a time-value-of-money calculator and a rebalancing-sensitivity tool. This plan describes black-box testing of the input validation of eleven forms using Equivalence Partitioning (EP) and Boundary Value Analysis (BVA), with every test case executed against the live system on my own account and backed by screenshots.

TP-3 Scope

In-scope features (all 11 forms and their input fields):

- 1 Sign Up (testfol.io/start): Email, Password
- 2 Sign In (testfol.io/login): Email, Password
- 3 Portfolio Backtester (testfol.io/): Start Date, End Date, Starting Value, Rolling Window, Annual cashflow growth, Cashflow, Drag, Ticker, Allocation %, Rebalance frequency (critical form)
- 4 Monte Carlo Simulator: Start/End Date, Number of years, Starting Value, Min block, Max block, Random seed, Allocation %
- 5 Safe Withdrawal Rate Calculator: Retirement horizon, Phase 1 end year, Phase 1 and Phase 2 withdrawal %, Start/End Date, Allocation %
- 6 Backtest Optimizer: Tickers, Start/End Date, Starting Value, Rolling Window, Max holdings, Min weight to include, Max vol
- 7 Efficient Frontier: Tickers, Start/End Date, Max holdings, Min weight to include
- 8 Contact Form: Name, Email, Subject, Message (and topic selector)
- 9 Lump Sum vs DCA: Starting Value, DCA horizon, Ticker, Allocation % (premium tool on the Free plan)
- 10 Time Value of Money Calculator: Starting Value, Monthly contribution
- 11 Rebalancing Sensitivity: Start/End Date, Drag, Allocation % (premium tool on the Free plan)

TP-4 Test Objectives

- Identify the valid and invalid equivalence classes of every input field and verify that each class behaves consistently (valid input accepted, invalid input rejected with a clear message).
- Locate and probe every numeric, length and date boundary with 2-point BVA on all forms and 3-point BVA (below/at/above) on the critical form (Portfolio Backtester).
- Reveal defects such as silent acceptance of invalid values, silent value rewriting, inconsistent messages and validation that is bypassed by plan gating.
- Produce an evidence-backed traceability matrix: every executed test case has an ID, expected result, actual result, pass/fail verdict and screenshot(s).

TP-5 Test Approach

Probe strategy: before designing any test case I explored each form with a probe log (Appendix A): empty values, zero, negative, decimal, very large, very long strings, special characters (apostrophes, spaces, unicode, SQL fragment), reversed/equal/future/ancient dates and allocations that do not sum to 100. I recorded the complete input set of every probe, the exact message shown and the inference. Different messages for different violations were treated as different equivalence classes.

Equivalence classes: for every input field I split the input domain into valid classes (VEC) and invalid classes (IEC) using the rules inferred from the probes, gave each a sequential EC ID (EC-001…), a representative value and the type. Weak EP with robustness (one test case per EC, including every IEC) was applied to all eleven forms; each TC-EP cites the EC it exercises.

Critical form: I designated the Portfolio Backtester as the critical form because it has the most inputs (dates, amounts, windows, percentages, cashflows, drag, tickers and allocations), the richest validation rules and the highest risk (money values and date logic). It receives Strong EP (TC-EP-S1..S6 combine several classes in one test) and 3-point BVA (below/at/above for every boundary).

Boundaries: boundaries were found by binary search on the live system (for example Starting Value accepted at 999,999,999,999 and rejected at 1,000,000,000,000; Rolling Window accepted at 11999 and rejected at 12000). Fields excluded from BVA: Ticker, Email and Password format, selects and the Rebalance option (symbolic data with no numeric boundary; a very long ticker is rejected as an unknown symbol, not by a length rule), Start/End Date on forms without a minimum span (only the ordering boundary is meaningful and is tested on the critical form), and Cashflow amount (no limit found by probing).

Execution: each test case was run on a fresh page load, the inputs were applied, the action button was clicked and the observed response (toast, browser validation message, run result, plan-limit dialog or pricing redirect) was recorded and screenshotted with the roll-number note visible.

Designated critical form: Portfolio Backtester (testfol.io/). Justification: it has the largest number of inputs, the most validation rules, cross-field rules (date order, allocation total) and the highest risk (money values and date logic). It receives Strong EP and 3-point BVA.

TP-6 Test Environment

Browser & Version	Google Chrome 142.0.0.0

Operating System	Ubuntu Linux (GNOME, Wayland)
Screen Resolution	1920 x 1080 screen; browser viewport 1854 x 875
Test Account (email)	i243166@isb.nu.edu.pk
Network / VPN	No VPN; standard home network connection
Plan	Testfolio Free plan (all testing on my own account)

TP-7 Entry & Exit Criteria

Entry criteria:

- Testfolio account created and signed in on the Free plan; all 11 in-scope forms reachable.
- Chrome set up with a consistent viewport/zoom and the roll-number note visible on screen.
- Handout and template read; probe log of at least 15 probes completed before test design.

Exit criteria:

- All EP partition tables and all BVA boundary tables completed with unique EC and BC IDs.
- Every test case executed (or marked Blocked with the reason) with screenshot evidence; at least 35 EP and 25 BVA test cases executed.
- Traceability matrix complete and consistent with the execution log; defect observations recorded.

TP-8 Test Schedule

Activity	Notes / Dates	Est. Hours	Done?
Probe / explore all 11 Testfolio forms	Exploration with probe log; binary search of boundaries	5.0	Yes
Complete EP partition tables (Layer 1)	Classes inferred from probes	2.0	Yes
Design EP test cases (Layer 2)	Weak EP + robustness; Strong EP on Backtester	2.5	Yes
Complete BVA boundary tables (Layer 1)	Boundaries bracketed by probes	1.5	Yes
Design BVA test cases (Layer 2)	2-point all forms, 3-point on Backtester	2.0	Yes
Execute test cases & capture screenshots	One fresh page load per test case	8.0	Yes
Complete traceability matrix	Cross-checked against execution log	1.0	Yes
Compile PDF and submit	Template sections, roll number in header	1.0	Yes

TP-9 Risks & Mitigations

Risk	Likelihood / Impact	Mitigation
Testfolio goes offline or returns intermittent server errors during testing (seen: 'Backtest failed', 'An error occurred while running Monte Carlo')	Medium / High	Run each case on a fresh page, repeat a case once if a transient error appears and record both attempts.

Risk	Likelihood / Impact	Mitigation
Free-plan limits and premium gating hide validation rules (Lump Sum vs DCA, Rebalancing Sensitivity, Monte Carlo years > 15, Efficient Frontier > 2 tickers)	High / Medium	Record the gate behaviour itself as the observed result, mark such cases Blocked and test the field-level browser validation separately.
Validation rules are undocumented, so inferred boundaries may be wrong	Medium / High	Bracket every boundary by binary search with values below, at and above and re-run the boundary cases on fresh pages.
Live data and results change over time (market data refresh) so actual numbers may differ on re-run	Medium / Low	Verdicts depend on acceptance/rejection behaviour and messages, not on exact result numbers; numbers are recorded as supporting evidence only.
Forms with CAPTCHA/bot protection or real side effects (Contact, Sign Up) cannot be fully automated	High / Medium	Run those cases manually where required and avoid sending real messages or creating extra accounts.

TP-10 Probe Log - Appendix A

Every probe lists all inputs entered (context fields included). Probes were run on my own account before any test case was designed.

Probe ID	Form	Test Inputs (field: value, ...)	System Response	Inference / Note
PR-001	Backtester	Starting Value: 0; tickers SPY 60 / AGG 40; dates empty; other fields default	Browser bubble: "Value must be greater than or equal to 1."	Minimum Starting Value is 1; zero is an invalid class.
PR-002	Backtester	Starting Value: empty; SPY 60 / AGG 40	Toast: "Invalid starting value"	Empty is a separate invalid class (app message, not browser).
PR-003	Backtester	Starting Value: 1000000000; SPY 60 / AGG 40	Runs; Ending Value $6,222,004,217.23	Large values are accepted; an upper limit may exist.
PR-004	Backtester	Starting Value: 1000000000000 (1e12); SPY 60 / AGG 40	Toast: "Starting value is too high"	Upper limit lies below 1e12.
PR-005	Backtester	Starting Value: 999999999999; SPY 60 / AGG 40	Runs; Ending Value $6,222,004,217,219.58	Maximum accepted Starting Value is 999,999,999,999.
PR-006	Backtester	SPY 60 / AGG 60 (total 120); defaults	Toast: "Portfolio 1: Allocation percentages must sum to 100"; total box red	Total is validated; upper bound 100.
PR-007	Backtester	SPY 60 / AGG 39.99 (total 99.99)	Same sum-to-100 toast	Exact equality to 100 is required, no tolerance.
PR-008	Backtester	SPY -10 / AGG 110 (total 100)	Runs; Ending Value $15,590.61	Negative weight silently accepted - possible defect.
PR-009	Backtester	Start Date 2020-01-01, End Date 2019-01-01	Toast: "Start date must be before end date"	Date order is validated.
PR-010	Backtester	Start Date 2020-01-01, End Date 2020-01-01	Same toast	Start must be strictly before End.
PR-011	Backtester	Start Date 1990-01-01, End Date 2020-01-01	Runs; header shows 2003-09-26 - 2019-12-31	Start earlier than data is silently clamped; also dates shown one day earlier than entered.
PR-012	Backtester	Start Date 2099-01-01, End Date 2100-01-01	Toast: "No overlapping time period found for the specified tickers and start/end dates"	Future ranges are a distinct invalid class.
PR-013	Backtester	Ticker ZZZZZZ (60%), AGG 40	Toast: "Invalid ticker ZZZZZZ"	Unknown symbols rejected server-side.
PR-014	Backtester	Ticker " OR 1=1 --" style SQL fragment (60%), AGG 40	Toast: "Invalid ticker ' OR 1=1 --"	Special characters are rejected safely; no error leakage.
PR-015	Backtester	Rolling Window: 12000; SPY 60 / AGG 40	Toast: "Rolling window is too large"	Upper limit exists; 11999 is accepted.
PR-016	Backtester	Annual cashflow growth: -100 / 100.01	Bubble ">= -99.99" / "<= 100"	Range is -99.99 to 100.
PR-017	Backtester	Drag: 101 and 1000	Runs (up to 16 s), Ending Value $0.00, no message	No upper validation on drag - defect candidate.
PR-018	Monte Carlo	Number of years: 0 / 101 (others default)	Bubble + toast "Number of years must be positive" / "must be <= 100"	Range 1 to 100 in the form.
PR-019	Monte Carlo	Number of years: 16	Dialog "Upgrade for higher Monte Carlo limits - Free supports up to 15 years and 500 simulations"	Free-plan limit is a separate class.
PR-020	Monte Carlo	Min block: 0.04	Run completes ("Ran 500 simulations") although the field minimum is 0.05	Min block minimum is not enforced - defect candidate.
PR-021	Monte Carlo	Random seed: -1 and 1.5	Field silently rewritten to 0 and 2; run completes	Seed is silently corrected, not rejected.

Probe ID	Form	Test Inputs (field: value, ...)	System Response	Inference / Note
PR-022	Safe Withdrawal Rate	Retirement horizon 0 and 51 (defaults otherwise)	Toast "Retirement horizon must be greater than 0 and at most 50 years"	Range 1 to 50.
PR-023	Safe Withdrawal Rate	Phase 1 end year 30 with horizon 30	Toast "Phase 2 must end after the previous phase"	Phase 1 end must be below the horizon.
PR-024	Safe Withdrawal Rate	Phase 1 withdrawal 101 and -5	Toast "Phase 1 withdrawal rate must be between 0% and 100%"	Range 0 to 100 percent.
PR-025	Backtest Optimizer	Tickers "SPY,AGG,GLD" (comma separated)	Toast "You must specify at least two tickers"	Tickers must be space-separated; minimum two tickers.
PR-026	Backtest Optimizer	Min weight to include 101	Bubble "Value must be less than or equal to 100."	Range 0 to 100.
PR-027	Efficient Frontier	Tickers "SPY AGG GLD" (three)	Redirect to /pricing "You've hit the max tickers limit ... current limit is 2 tickers"	Free plan allows 2 tickers.
PR-028	Lump Sum vs DCA	Valid inputs, click Analyze	Redirect to /pricing "Premium tool - Requires Pro or higher"	Premium gate; validation order must be checked.
PR-029	Lump Sum vs DCA	Starting Value 50 (bubble) then Analyze	Bubble "Value must be greater than or equal to 100." but click redirects to pricing	Premium gate fires before validation.
PR-030	Rebalancing Sensitivity	Valid inputs / reversed dates / allocation 90, click Analyze	Always redirected to /pricing (premium tool)	No validation feedback is observable on the Free plan.
PR-031	Time Value of Money	Starting Value -1 and 5000001	Toast "Starting Value must be between $0 and $5,000,000."	Range 0 to 5,000,000.
PR-032	Time Value of Money	Monthly contribution -1 and 100001	Toast "Contribution must be between $0 and $100,000."	Range 0 to 100,000.

D2 - Equivalence Partitioning [25 marks]

Layer 1 - EP Partition Tables (one per input field)

Sign Up - Email

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-001	Valid, unregistered email address format	newuser@example.com	VEC
EC-002	Empty email	(empty)	IEC
EC-003	Email without @ symbol	notanemail	IEC
EC-004	Email without domain suffix	a@b	IEC
EC-005	Email already registered	(your registered email)	IEC

Sign Up - Password

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-006	Password of valid length	Test@1234	VEC
EC-007	Empty password	(empty)	IEC
EC-008	Password shorter than minimum length	abcde	IEC
EC-009	Password longer than maximum length	73 x a	IEC

Sign In - Email

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-010	Registered email	(your registered email)	VEC
EC-011	Empty email	(empty)	IEC
EC-012	Malformed email	notanemail	IEC
EC-013	Unregistered email	nobody@x.com	IEC

Sign In - Password

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-014	Correct password for the account	(correct password)	VEC
EC-015	Empty password	(empty)	IEC
EC-016	Wrong password	wrongpass1234	IEC

Portfolio Backtester - Start Date

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-017	Valid start date inside data range, before end date	2020-01-01 (end 2021-01-01)	VEC
EC-018	Start date left empty (full history)	(empty)	VEC
EC-019	Start date after end date	2020-01-01 (end 2019-01-01)	IEC

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-020	Start date equal to end date	2020-01-01 (end 2020-01-01)	IEC
EC-021	Start date earlier than first available data	1990-01-01 (end 2020-01-01)	IEC
EC-022	Start date in the far future (no data)	2099-01-01 (end 2100-01-01)	IEC

Portfolio Backtester - End Date

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-023	Valid end date with start empty	2020-01-01 (start empty)	VEC
EC-024	Start only, end date empty (runs to latest data)	2020-01-01 (end empty)	VEC
EC-025	End date later than last available data	2020-01-01 to 2030-01-01	IEC

Portfolio Backtester - Starting Value

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-026	Valid starting value (>=1, below upper limit)	10000	VEC
EC-027	Zero starting value	0	IEC
EC-028	Negative starting value	-5	IEC
EC-029	Fractional value below minimum	0.5	IEC
EC-030	Empty starting value	(empty)	IEC
EC-031	Starting value above the upper limit	5000000000000	IEC

Portfolio Backtester - Rolling Window

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-032	Valid rolling window in months	60	VEC
EC-033	Zero rolling window	0	IEC
EC-034	Negative rolling window	-1	IEC
EC-035	Non-integer rolling window	1.5	IEC
EC-036	Rolling window above upper limit	20000	IEC

Portfolio Backtester - Annual cashflow growth

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-037	Valid growth percentage	2	VEC
EC-038	Growth below minimum	-100	IEC
EC-039	Growth above maximum	101	IEC

Portfolio Backtester - Cashflow amount

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-040	Positive cashflow (contribution)	1000	VEC
EC-041	Negative cashflow (withdrawal)	-500	VEC
EC-042	Zero cashflow (no cashflow)	0	VEC

Portfolio Backtester - Drag (annual fee %)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-043	Valid drag between 0 and 100	0.5	VEC
EC-044	Negative drag	-1	IEC
EC-045	Drag above 100%	150	IEC

Portfolio Backtester - Ticker

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-046	Valid ticker (upper case)	SPY	VEC
EC-047	Valid ticker in lower case	spy	VEC
EC-048	Unknown ticker	ZZZZZZ	IEC
EC-049	Empty ticker with non-zero allocation	(empty)	IEC
EC-050	SQL fragment as ticker	' OR 1=1 --	IEC
EC-051	Non-ASCII (unicode) ticker	ÄÖÜ	IEC
EC-052	Ticker containing a space	S PY	IEC
EC-053	Very long ticker (50 characters)	ABCDEFGHIJ x5	IEC

Portfolio Backtester - Allocation % (portfolio total)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-054	Allocations sum to exactly 100 across 2 assets	SPY 60 / AGG 40	VEC
EC-055	Single asset at 100%	SPY 100 / AGG 0	VEC
EC-056	Allocations sum below 100	SPY 60 / AGG 30 (90)	IEC
EC-057	Allocations sum above 100	SPY 60 / AGG 60 (120)	IEC
EC-058	Negative allocation with offsetting excess	SPY -10 / AGG 110	IEC

Portfolio Backtester - Rebalance frequency

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-059	Valid rebalance option (Yearly, default)	Yearly	VEC

Monte Carlo Simulator - Start / End Date

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-060	Dates empty (full history)	(empty)	VEC
EC-061	Start date after end date	2020-01-01 / 2019-01-01	IEC

Monte Carlo Simulator - Number of years

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-062	Valid horizon within free limit	10	VEC
EC-063	Horizon below minimum	0	IEC

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-064	Horizon above free-plan limit but within 100	50	IEC
EC-065	Horizon above absolute maximum	150	IEC
EC-066	Decimal horizon	2.5	VEC

Monte Carlo Simulator - Starting Value

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-067	Valid starting value	10000	VEC
EC-068	Zero starting value	0	IEC

Monte Carlo Simulator - Min block (years)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-069	Valid minimum block length	1	VEC
EC-070	Minimum block below allowed minimum	0.01	IEC

Monte Carlo Simulator - Max block (years)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-071	Valid maximum block length	5	VEC
EC-072	Maximum block above allowed maximum	25	IEC
EC-073	Maximum block below allowed minimum	0.01	IEC

Monte Carlo Simulator - Random seed

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-074	Seed left empty (random)	(empty)	VEC
EC-075	Valid whole-number seed	42	VEC
EC-076	Negative seed	-1	IEC
EC-077	Non-integer seed	1.5	IEC

Monte Carlo Simulator - Allocation % (portfolio total)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-078	Allocations sum to 100 (50/25/25)	50/25/25	VEC
EC-079	Allocations sum below 100	50/25/20 (95)	IEC

Safe Withdrawal Rate Calculator - Retirement horizon (years)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-080	Valid horizon	30	VEC
EC-081	Zero horizon	0	IEC
EC-082	Horizon above maximum	60	IEC

Safe Withdrawal Rate Calculator - Phase 1 end year

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-083	Valid end year before horizon	10	VEC
EC-084	End year zero	0	IEC
EC-085	End year after the horizon	40 (horizon 30)	IEC

Safe Withdrawal Rate Calculator - Phase 1 annual withdrawal %

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-086	Valid withdrawal rate	4	VEC
EC-087	Withdrawal rate negative	-5	IEC
EC-088	Withdrawal rate above 100	150	IEC

Safe Withdrawal Rate Calculator - Phase 2 annual withdrawal %

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-089	Valid withdrawal rate	6	VEC
EC-090	Withdrawal rate negative	-5	IEC
EC-091	Withdrawal rate above 100	150	IEC

Safe Withdrawal Rate Calculator - Start / End Date

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-092	Dates empty (full history)	(empty)	VEC
EC-093	Start date after end date	2020-01-01 / 2019-01-01	IEC

Safe Withdrawal Rate Calculator - Allocation % (portfolio total)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-094	Allocations sum to 100 (50/30/20)	50/30/20	VEC
EC-095	Allocations sum below 100	50/30/10 (90)	IEC

Backtest Optimizer - Tickers

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-096	Three valid tickers separated by spaces	SPY AGG GLD	VEC
EC-097	Exactly two valid tickers	SPY AGG	VEC
EC-098	Only one ticker	SPY	IEC
EC-099	Empty tickers	(empty)	IEC
EC-100	Contains an unknown ticker	SPY ZZZZZZ	IEC
EC-101	Comma-separated tickers	SPY,AGG,GLD	IEC

Backtest Optimizer - Start / End Date

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-102	Dates empty (full history)	(empty)	VEC
EC-103	Start date after end date	2020-01-01 / 2019-01-01	IEC

Backtest Optimizer - Starting Value

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-104	Valid starting value	10000	VEC
EC-105	Zero starting value	0	IEC

Backtest Optimizer - Rolling Window

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-106	Valid rolling window	60	VEC
EC-107	Zero rolling window	0	IEC

Backtest Optimizer - Max holdings

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-108	Valid max holdings	2	VEC
EC-109	Zero max holdings	0	IEC

Backtest Optimizer - Min weight to include (%)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-110	Valid minimum weight	10	VEC
EC-111	Negative minimum weight	-5	IEC
EC-112	Minimum weight above 100	150	IEC

Backtest Optimizer - Max vol (%)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-113	Valid maximum volatility	20	VEC
EC-114	Negative maximum volatility	-5	IEC

Efficient Frontier - Tickers

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-115	Two valid tickers	SPY AGG	VEC
EC-116	Only one ticker	SPY	IEC
EC-117	Empty tickers	(empty)	IEC
EC-118	Three tickers (above free-plan limit)	SPY AGG GLD	IEC

Efficient Frontier - Start / End Date

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-119	Dates empty (full history)	(empty)	VEC
EC-120	Start date after end date	2020-01-01 / 2019-01-01	IEC

Efficient Frontier - Max holdings

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-121	Valid max holdings	2	VEC
EC-122	Zero max holdings	0	IEC

Efficient Frontier - Min weight to include (%)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-123	Valid minimum weight	10	VEC
EC-124	Negative minimum weight	-5	IEC
EC-125	Minimum weight above 100	150	IEC

Contact Form - Name

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-126	Valid name (1-120 chars)	Muhammad Bilal Tahir	VEC
EC-127	Empty name	(empty)	IEC
EC-128	Name longer than 120 characters	121 x a	IEC

Contact Form - Email

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-129	Valid email format	student@example.com	VEC
EC-130	Empty email	(empty)	IEC
EC-131	Malformed email	notanemail	IEC

Contact Form - Subject

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-132	Valid subject (1-120 chars)	Question about backtests	VEC
EC-133	Empty subject	(empty)	IEC
EC-134	Subject longer than 120 characters	121 x a	IEC

Contact Form - Message

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-135	Valid message (1-5000 chars)	Hello, test message	VEC
EC-136	Empty message	(empty)	IEC

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-137	Message longer than 5000 characters	5001 x a	IEC

Lump Sum vs DCA - Starting Value

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-138	Valid starting value (>=100)	10000	VEC
EC-139	Starting value below minimum	50	IEC

Lump Sum vs DCA - DCA horizon (months)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-140	Valid horizon	12	VEC
EC-141	Horizon below minimum	0	IEC
EC-142	Horizon above maximum	200	IEC

Lump Sum vs DCA - Ticker

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-143	Valid ticker	SPY	VEC
EC-144	Unknown ticker	ZZZZZZ	IEC

Lump Sum vs DCA - Allocation % (portfolio total)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-145	Allocations sum to 100	60/40	VEC
EC-146	Allocations sum below 100	60/30 (90)	IEC

Time Value of Money Calculator - Starting Value

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-147	Valid starting value	10000	VEC
EC-148	Negative starting value	-500	IEC
EC-149	Starting value above maximum	9000000	IEC
EC-150	Empty starting value	(empty)	IEC

Time Value of Money Calculator - Monthly contribution

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-151	Valid monthly contribution	1000	VEC
EC-152	Negative contribution	-50	IEC
EC-153	Contribution above maximum	500000	IEC

Rebalancing Sensitivity - Start / End Date

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-154	Dates empty (full history)	(empty)	VEC
EC-155	Start date after end date	2020-01-01 / 2019-01-01	IEC

Rebalancing Sensitivity - Drag (annual fee %)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-156	Valid drag	0.5	VEC
EC-157	Negative drag	-2	IEC

Rebalancing Sensitivity - Allocation % (portfolio total)

EC ID	Equivalence Class Description	Representative Value	Type (VEC / IEC)
EC-158	Allocations sum to 100	60/40	VEC
EC-159	Allocations sum below 100	60/30 (90)	IEC

Layer 2 - EP Test Case Table (Weak EP with robustness: one TC per EC)

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-001	EC-001	Sign Up	Email	newuser@example.com	Account created / verification step shown	ACCEPTED | Valid unregistered email newuser@example.com accepted: toast 'Code sent' and the verification step 'Enter the verification code sent to newuser@example.com' appears (account is only created after the code is verified, which was not done)	Pass
TC-EP-002	EC-002	Sign Up	Email	(empty)	Rejected; email required	REJECTED (no submission) | Empty email: Confirm my email click does nothing; no message, no account request	Pass
TC-EP-003	EC-003	Sign Up	Email	notanemail	Rejected; invalid email format	Rejected (browser validation message) | Please include an '@' in the email address. 'notanemail' is missing an '@'.	Pass
TC-EP-004	EC-004	Sign Up	Email	a@b	Rejected; invalid email format	ACCEPTED | 'a@b' passes the browser email check (no domain suffix needed); submit then stops silently because the password is empty	Fail
TC-EP-005	EC-005	Sign Up	Email	(your registered email)	Rejected; account exists	RAN | Already-registered email was not rejected: toast 'Code sent' and the dialog moved to 'Enter the verification code sent to ... to create your account' (no 'account exists' message; consistent with anti-enumeration design)	Fail
TC-EP-006	EC-006	Sign Up	Password	Test@1234	Accepted	ACCEPTED | Password of valid length (Test@1234, 9 chars) accepted: 'Code sent' toast and verification step shown	Pass
TC-EP-007	EC-007	Sign Up	Password	(empty)	Rejected; password required	REJECTED (no submission) | Valid email with empty password: Confirm my email click does nothing; no message shown	Pass
TC-EP-008	EC-008	Sign Up	Password	abcde	Rejected; too short	REJECTED (no submission) | 5-character password with matching confirmation: Confirm my email click does nothing and no message explains why	Pass
TC-EP-009	EC-009	Sign Up	Password	73 x a	Rejected or truncated; too long	RAN | 73-character password was NOT rejected or truncated: 'Code sent' toast and the verification-code step appeared (no maximum length enforced at this step)	Fail
TC-EP-010	EC-010	Sign In	Email	(your registered email)	Accepted	ACCEPTED | Registered email accepted in the Sign in dialog	Pass
TC-EP-011	EC-011	Sign In	Email	(empty)	Rejected; email required	REJECTED (no submission) | Empty email and password: Sign in click does nothing, no request is made	Pass
TC-EP-012	EC-012	Sign In	Email	notanemail	Rejected; invalid email format	Rejected (browser validation message) | Please include an '@' in the email address. 'notanemail' is missing an '@'.	Pass
TC-EP-013	EC-013	Sign In	Email	nobody@x.com	Rejected; invalid credentials	Rejected (toast) | toast: Email or password wasn't recognized. Try again, use a one-time code, or create/reset your password. (generic message, does not reveal whether the email exists)	Pass

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-014	EC-014	Sign In	Password	(correct password)	Signed in	RAN | Correct password: signed in, 'Finish setup' dialog appears and the account shows Free plan	Pass
TC-EP-015	EC-015	Sign In	Password	(empty)	Rejected; password required	REJECTED (no submission) | Registered email with empty password: Sign in click does nothing, no message shown	Pass
TC-EP-016	EC-016	Sign In	Password	wrongpass1234	Rejected; invalid credentials	Rejected (toast) | toast: Email or password wasn't recognized. Try again, use a one-time code, or create/reset your password.	Pass
TC-EP-017	EC-017	Portfolio Backtester	Start Date	2020-01-01 (end 2021-01-01)	Backtest runs for the entered period	RAN | EV $11,401.48, CAGR 13.98% | 1.00 years: 2019-12-31 - 2020-12-31	Pass
TC-EP-018	EC-018	Portfolio Backtester	Start Date	(empty)	Backtest runs from earliest common date	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-019	EC-019	Portfolio Backtester	Start Date	2020-01-01 (end 2019-01-01)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Start date must be before end date	Pass
TC-EP-020	EC-020	Portfolio Backtester	Start Date	2020-01-01 (end 2020-01-01)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Start date must be before end date	Pass
TC-EP-021	EC-021	Portfolio Backtester	Start Date	1990-01-01 (end 2020-01-01)	Rejected, or user told the start is adjusted	RAN | EV $33,672.37, CAGR 7.75% | 16.26 years: 2003-09-26 - 2019-12-31	Fail
TC-EP-022	EC-022	Portfolio Backtester	Start Date	2099-01-01 (end 2100-01-01)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: No overlapping time period found for the specified tickers and start/end dates	Pass
TC-EP-023	EC-023	Portfolio Backtester	End Date	2020-01-01 (start empty)	Backtest runs up to the entered date	RAN | EV $33,672.37, CAGR 7.75% | 16.26 years: 2003-09-26 - 2019-12-31	Pass
TC-EP-024	EC-024	Portfolio Backtester	End Date	2020-01-01 (end empty)	Backtest runs to latest data	RAN | EV $18,478.07, CAGR 9.45% | 6.80 years: 2019-12-31 - 2026-10-19	Pass
TC-EP-025	EC-025	Portfolio Backtester	End Date	2020-01-01 to 2030-01-01	Rejected, or user told the end is adjusted	RAN | EV $18,478.07, CAGR 9.45% | 6.80 years: 2019-12-31 - 2026-10-19	Fail
TC-EP-026	EC-026	Portfolio Backtester	Starting Value	10000	Accepted; calculation/analysis completes and results are shown	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-027	EC-027	Portfolio Backtester	Starting Value	0	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Starting Value: Value must be greater than or equal to 1.	Pass
TC-EP-028	EC-028	Portfolio Backtester	Starting Value	-5	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Starting Value: Value must be greater than or equal to 1.	Pass
TC-EP-029	EC-029	Portfolio Backtester	Starting Value	0.5	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Starting Value: Value must be greater than or equal to 1.	Pass
TC-EP-030	EC-030	Portfolio Backtester	Starting Value	(empty)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Invalid starting value	Pass
TC-EP-031	EC-031	Portfolio Backtester	Starting Value	5000000000000	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Starting value is too high	Pass
TC-EP-032	EC-032	Portfolio Backtester	Rolling Window	60	Accepted; calculation/analysis completes and results are shown	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-033	EC-033	Portfolio Backtester	Rolling Window	0	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Rolling Window: Value must be greater than or equal to 1.	Pass
TC-EP-034	EC-034	Portfolio Backtester	Rolling Window	-1	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Rolling Window: Value must be greater than or equal to 1.	Pass

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-035	EC-035	Portfolio Backtester	Rolling Window	1.5	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Rolling Window: Please enter a valid value. The two nearest valid values are 1 and 2.	Pass
TC-EP-036	EC-036	Portfolio Backtester	Rolling Window	20000	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Rolling window is too large	Pass
TC-EP-037	EC-037	Portfolio Backtester	Annual cashflow growth	2	Accepted; calculation/analysis completes and results are shown	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-038	EC-038	Portfolio Backtester	Annual cashflow growth	-100	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Annual cashflow growth: Value must be greater than or equal to -99.99.	Pass
TC-EP-039	EC-039	Portfolio Backtester	Annual cashflow growth	101	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Annual cashflow growth: Value must be less than or equal to 100.	Pass
TC-EP-040	EC-040	Portfolio Backtester	Cashflow amount	1000	Accepted; calculation/analysis completes and results are shown	RAN | EV $135,115.00, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-041	EC-041	Portfolio Backtester	Cashflow amount	-500	Accepted; calculation/analysis completes and results are shown	RAN | EV $25,772.56 (total contributions -$1,500.00), CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-042	EC-042	Portfolio Backtester	Cashflow amount	0	Accepted; calculation/analysis completes and results are shown	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-043	EC-043	Portfolio Backtester	Drag (annual fee %)	0.5	Accepted; calculation/analysis completes and results are shown	RAN | EV $55,457.11, CAGR 7.71% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-044	EC-044	Portfolio Backtester	Drag (annual fee %)	-1	Rejected with a validation message; no calculation is run	RAN | EV $78,320.13, CAGR 9.33% | 23.06 years: 2003-09-26 - 2026-10-19	Fail
TC-EP-045	EC-045	Portfolio Backtester	Drag (annual fee %)	150	Rejected with a validation message; no calculation is run	RAN | EV $0.00, CAGR -75.88% | 23.06 years: 2003-09-26 - 2026-10-19	Fail
TC-EP-046	EC-046	Portfolio Backtester	Ticker	SPY	Accepted; calculation/analysis completes and results are shown	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-047	EC-047	Portfolio Backtester	Ticker	spy	Accepted; calculation/analysis completes and results are shown	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-048	EC-048	Portfolio Backtester	Ticker	ZZZZZZ	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Invalid ticker ZZZZZZ	Pass
TC-EP-049	EC-049	Portfolio Backtester	Ticker	(empty)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Portfolio 1: Non-zero allocation has empty ticker	Pass
TC-EP-050	EC-050	Portfolio Backtester	Ticker	' OR 1=1 --	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Invalid ticker ' OR 1=1 --	Pass
TC-EP-051	EC-051	Portfolio Backtester	Ticker	ÄÖÜ	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Invalid ticker ÄÖÜ	Pass
TC-EP-052	EC-052	Portfolio Backtester	Ticker	S PY	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Invalid ticker S PY	Pass

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-053	EC-053	Portfolio Backtester	Ticker	ABCDEFGHIJ x5	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Invalid ticker ABCDEFGHIJABCDEFGHIJABCDEFGHIJABCDEFGHIJABCDEFGHIJ	Pass
TC-EP-054	EC-054	Portfolio Backtester	Allocation % (portfolio total)	SPY 60 / AGG 40	Accepted; calculation/analysis completes and results are shown	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-055	EC-055	Portfolio Backtester	Allocation % (portfolio total)	SPY 100 / AGG 0	Accepted; calculation/analysis completes and results are shown	RAN | single asset SPY EV $318,820.13, CAGR 10.81% | 33.72 years: 1993-01-29 - 2026-10-19	Pass
TC-EP-056	EC-056	Portfolio Backtester	Allocation % (portfolio total)	SPY 60 / AGG 30 (90)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Portfolio 1: Allocation percentages must sum to 100	Pass
TC-EP-057	EC-057	Portfolio Backtester	Allocation % (portfolio total)	SPY 60 / AGG 60 (120)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Portfolio 1: Allocation percentages must sum to 100	Pass
TC-EP-058	EC-058	Portfolio Backtester	Allocation % (portfolio total)	SPY -10 / AGG 110	Rejected with a validation message; no calculation is run	RAN | EV $15,590.61, CAGR 1.94% | 23.06 years: 2003-09-26 - 2026-10-19	Fail
TC-EP-059	EC-059	Portfolio Backtester	Rebalance frequency	Yearly	Accepted; calculation/analysis completes and results are shown	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-060	EC-060	Monte Carlo Simulator	Start / End Date	(empty)	Simulation runs	RAN | | 500 simulations of 15 years	Pass
TC-EP-061	EC-061	Monte Carlo Simulator	Start / End Date	2020-01-01 / 2019-01-01	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Start date must be before end date	Pass
TC-EP-062	EC-062	Monte Carlo Simulator	Number of years	10	Accepted; calculation/analysis completes and results are shown	RAN | | 500 simulations of 10 years	Pass
TC-EP-063	EC-063	Monte Carlo Simulator	Number of years	0	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Number of years: Value must be greater than or equal to 1. | toast: Number of years must be positive	Pass
TC-EP-064	EC-064	Monte Carlo Simulator	Number of years	50	Rejected or plan-limit message	PLAN-LIMIT DIALOG | "Upgrade for higher Monte Carlo limits - Free supports up to 15 years and 500 simulations" (years 50)	Pass
TC-EP-065	EC-065	Monte Carlo Simulator	Number of years	150	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Number of years: Value must be less than or equal to 100. | toast: Number of years must be <= 100	Pass
TC-EP-066	EC-066	Monte Carlo Simulator	Number of years	2.5	Accepted; calculation/analysis completes and results are shown	RAN | | 500 simulations of 2.5 years	Pass
TC-EP-067	EC-067	Monte Carlo Simulator	Starting Value	10000	Accepted; calculation/analysis completes and results are shown	RAN | | 500 simulations of 15 years	Pass
TC-EP-068	EC-068	Monte Carlo Simulator	Starting Value	0	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Starting Value: Value must be greater than or equal to 1. | toast: Starting value must be positive	Pass
TC-EP-069	EC-069	Monte Carlo Simulator	Min block (years)	1	Accepted; calculation/analysis completes and results are shown	RAN | | 500 simulations of 15 years	Pass
TC-EP-070	EC-070	Monte Carlo Simulator	Min block (years)	0.01	Rejected with a validation message; no calculation is run	RAN (DEFECT: browser reports min 0.05 but app accepts 0.01) | Min block 0.01 -> "Ran 500 simulations in 2.7 seconds", random seed auto-filled 1275341795	Fail

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-071	EC-071	Monte Carlo Simulator	Max block (years)	5	Accepted; calculation/analysis completes and results are shown	RAN | | 500 simulations of 15 years	**Pass**
TC-EP-072	EC-072	Monte Carlo Simulator	Max block (years)	25	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Max block (in years): Value must be less than or equal to 20. | toast: Max block length must be <= 20 years	**Pass**
TC-EP-073	EC-073	Monte Carlo Simulator	Max block (years)	0.01	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Max block (in years): Value must be greater than or equal to 0.05. | toast: Min block length must be <= max block length	**Pass**
TC-EP-074	EC-074	Monte Carlo Simulator	Random seed	(empty)	Accepted; calculation/analysis completes and results are shown	RAN | | 500 simulations of 15 years	**Pass**
TC-EP-075	EC-075	Monte Carlo Simulator	Random seed	42	Accepted; calculation/analysis completes and results are shown	RAN | | 500 simulations of 15 years (seed 42)	**Pass**
TC-EP-076	EC-076	Monte Carlo Simulator	Random seed	-1	Rejected with a validation message; no calculation is run	RAN (silent correction: seed -1 reset to 0) | Random seed field shows 0 after click; "Ran 500 simulations in 0.9 seconds"; no message	**Fail**
TC-EP-077	EC-077	Monte Carlo Simulator	Random seed	1.5	Rejected with a validation message; no calculation is run	RAN (silent correction: seed 1.5 rounded to 2) | Random seed field shows 2; "Ran 500 simulations in 1.0 seconds"; no message	**Fail**
TC-EP-078	EC-078	Monte Carlo Simulator	Allocation % (portfolio total)	50/25/25	Accepted; calculation/analysis completes and results are shown	RAN | | 500 simulations of 15 years	**Pass**
TC-EP-079	EC-079	Monte Carlo Simulator	Allocation % (portfolio total)	50/25/20 (95)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Portfolio allocations must sum to 100 (total box 95 in red)	**Pass**
TC-EP-080	EC-080	Safe Withdrawal Rate Calculator	Retirement horizon (years)	30	Accepted; calculation/analysis completes and results are shown	RAN | 58.55 years: 1968-04-01 - 2026-10-19	**Pass**
TC-EP-081	EC-081	Safe Withdrawal Rate Calculator	Retirement horizon (years)	0	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Retirement horizon must be greater than 0 and at most 50 years	**Pass**
TC-EP-082	EC-082	Safe Withdrawal Rate Calculator	Retirement horizon (years)	60	Rejected with a validation message; no calculation is run	REJECTED (native browser validation + toast; field auto-clamped to 50) | Retirement horizon: Value must be less than or equal to 50. | toast: Retirement horizon must be greater than 0 and at most 50 years	**Pass**
TC-EP-083	EC-083	Safe Withdrawal Rate Calculator	Phase 1 end year	10	Accepted; calculation/analysis completes and results are shown	RAN | 58.55 years: 1968-04-01 - 2026-10-19	**Pass**
TC-EP-084	EC-084	Safe Withdrawal Rate Calculator	Phase 1 end year	0	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Phase 1 end year must be greater than 0 and at most 50	**Pass**
TC-EP-085	EC-085	Safe Withdrawal Rate Calculator	Phase 1 end year	40 (horizon 30)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Phase 1 cannot end after the retirement horizon	**Pass**
TC-EP-086	EC-086	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	4	Accepted; calculation/analysis completes and results are shown	RAN | 58.55 years: 1968-04-01 - 2026-10-19 (Phase 1 4%)	**Pass**
TC-EP-087	EC-087	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	-5	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Phase 1 annual withdrawal: Value must be greater than or equal to 0. | toast: Phase 1 withdrawal rate must be between 0% and 100%	**Pass**

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-088	EC-088	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	150	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Phase 1 annual withdrawal: Value must be less than or equal to 100.	Pass
TC-EP-089	EC-089	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	6	Accepted; calculation/analysis completes and results are shown	RAN | 58.55 years: 1968-04-01 - 2026-10-19 (Phase 2 6%)	Pass
TC-EP-090	EC-090	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	-5	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Phase 2 annual withdrawal: Value must be greater than or equal to 0. | toast: Phase 2 withdrawal rate must be between 0% and 100%	Pass
TC-EP-091	EC-091	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	150	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Phase 2 annual withdrawal: Value must be less than or equal to 100.	Pass
TC-EP-092	EC-092	Safe Withdrawal Rate Calculator	Start / End Date	(empty)	Accepted; calculation/analysis completes and results are shown	RAN | 58.55 years: 1968-04-01 - 2026-10-19	Pass
TC-EP-093	EC-093	Safe Withdrawal Rate Calculator	Start / End Date	2020-01-01 / 2019-01-01	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Invalid request: Value error, Start date must be before end date	Pass
TC-EP-094	EC-094	Safe Withdrawal Rate Calculator	Allocation % (portfolio total)	50/30/20	Accepted; calculation/analysis completes and results are shown	RAN | 58.55 years: 1968-04-01 - 2026-10-19	Pass
TC-EP-095	EC-095	Safe Withdrawal Rate Calculator	Allocation % (portfolio total)	50/30/10 (90)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Portfolio 1 weights must add up to 100%	Pass
TC-EP-096	EC-096	Backtest Optimizer	Tickers	SPY AGG GLD	Accepted; calculation/analysis completes and results are shown	RAN | 21.92 years: 2004-11-18 - 2026-10-19 (SPY 58.56%, AGG 0%, GLD 41.44%)	Pass
TC-EP-097	EC-097	Backtest Optimizer	Tickers	SPY AGG	Accepted; calculation/analysis completes and results are shown	RAN | 23.06 years: 2003-09-26 - 2026-10-19 (SPY 100%, AGG 0%)	Pass
TC-EP-098	EC-098	Backtest Optimizer	Tickers	SPY	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: You must specify at least two tickers	Pass
TC-EP-099	EC-099	Backtest Optimizer	Tickers	(empty)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: You must specify at least two tickers	Pass
TC-EP-100	EC-100	Backtest Optimizer	Tickers	SPY ZZZZZZ	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Invalid ticker ZZZZZZ	Pass
TC-EP-101	EC-101	Backtest Optimizer	Tickers	SPY,AGG,GLD	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: You must specify at least two tickers (comma delimiter not recognised)	Pass
TC-EP-102	EC-102	Backtest Optimizer	Start / End Date	(empty)	Accepted; calculation/analysis completes and results are shown	RAN | 21.92 years: 2004-11-18 - 2026-10-19	Pass
TC-EP-103	EC-103	Backtest Optimizer	Start / End Date	2020-01-01 / 2019-01-01	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: End date must be after start date	Pass
TC-EP-104	EC-104	Backtest Optimizer	Starting Value	10000	Accepted; calculation/analysis completes and results are shown	RAN | 21.92 years: 2004-11-18 - 2026-10-19	Pass
TC-EP-105	EC-105	Backtest Optimizer	Starting Value	0	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Starting Value: Value must be greater than or equal to 1.	Pass

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-106	EC-106	Backtest Optimizer	Rolling Window	60	Accepted; calculation/analysis completes and results are shown	RAN | 21.92 years: 2004-11-18 - 2026-10-19	Pass
TC-EP-107	EC-107	Backtest Optimizer	Rolling Window	0	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Rolling Window: Value must be greater than or equal to 1.	Pass
TC-EP-108	EC-108	Backtest Optimizer	Max holdings	2	Accepted; calculation/analysis completes and results are shown	RAN | 21.92 years: 2004-11-18 - 2026-10-19	Pass
TC-EP-109	EC-109	Backtest Optimizer	Max holdings	0	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Max holdings: Value must be greater than or equal to 1.	Pass
TC-EP-110	EC-110	Backtest Optimizer	Min weight to include (%)	10	Accepted; calculation/analysis completes and results are shown	RAN | 21.92 years: 2004-11-18 - 2026-10-19	Pass
TC-EP-111	EC-111	Backtest Optimizer	Min weight to include (%)	-5	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Min weight to include: Value must be greater than or equal to 0.	Pass
TC-EP-112	EC-112	Backtest Optimizer	Min weight to include (%)	150	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Min weight to include: Value must be less than or equal to 100.	Pass
TC-EP-113	EC-113	Backtest Optimizer	Max vol (%)	20	Accepted; calculation/analysis completes and results are shown	RAN | 21.92 years: 2004-11-18 - 2026-10-19	Pass
TC-EP-114	EC-114	Backtest Optimizer	Max vol (%)	-5	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Max vol: Value must be greater than or equal to 0.	Pass
TC-EP-115	EC-115	Efficient Frontier	Tickers	SPY AGG	Accepted; calculation/analysis completes and results are shown	RAN | | Efficient frontier computed 23.06 years: 2003-09-26 - 2026-10-19 (SPY 39.49% / AGG 60.51%)	Pass
TC-EP-116	EC-116	Efficient Frontier	Tickers	SPY	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: You must specify at least two tickers	Pass
TC-EP-117	EC-117	Efficient Frontier	Tickers	(empty)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: You must specify at least two tickers	Pass
TC-EP-118	EC-118	Efficient Frontier	Tickers	SPY AGG GLD	Rejected or plan-limit message	PLAN-LIMIT PAGE | Redirected to /pricing (reason=limit, tool=efficient_frontier, limit=max_efficient_frontier_tickers): "You've hit the max tickers limit ... Your current limit is 2 tickers"	Pass
TC-EP-119	EC-119	Efficient Frontier	Start / End Date	(empty)	Accepted; calculation/analysis completes and results are shown	RAN | | Efficient frontier computed 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-EP-120	EC-120	Efficient Frontier	Start / End Date	2020-01-01 / 2019-01-01	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: End date must be after start date	Pass
TC-EP-121	EC-121	Efficient Frontier	Max holdings	2	Accepted; calculation/analysis completes and results are shown	RAN | | Efficient frontier computed 23.06 years: 2003-09-26 - 2026-10-19 (max holdings 2)	Pass
TC-EP-122	EC-122	Efficient Frontier	Max holdings	0	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Max holdings: Value must be greater than or equal to 1.	Pass
TC-EP-123	EC-123	Efficient Frontier	Min weight to include (%)	10	Accepted; calculation/analysis completes and results are shown	RAN | | Efficient frontier computed 23.06 years: 2003-09-26 - 2026-10-19 (min weight 10)	Pass
TC-EP-124	EC-124	Efficient Frontier	Min weight to include (%)	-5	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Min weight to include: Value must be greater than or equal to 0.	Pass

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-125	EC-125	Efficient Frontier	Min weight to include (%)	150	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Min weight to include: Value must be less than or equal to 100.	Pass
TC-EP-126	EC-126	Contact Form	Name	Muhammad Bilal Tahir	Accepted	ACCEPTED | Valid name accepted (field shows 'Muhammad Bilal Tahir')	Pass
TC-EP-127	EC-127	Contact Form	Name	(empty)	Rejected; name required	RAN | Empty Name was NOT rejected: submit succeeded and the form cleared (Name is optional; no error shown)	Fail
TC-EP-128	EC-128	Contact Form	Name	121 x a	Input limited/rejected at 120	REJECTED (input limited) | Typing a 121st character is ignored; Name stays at 120 chars	Pass
TC-EP-129	EC-129	Contact Form	Email	student@example.com	Accepted	ACCEPTED | Valid email accepted	Pass
TC-EP-130	EC-130	Contact Form	Email	(empty)	Rejected; email required	RAN | Empty Email was NOT rejected: submit succeeded and the form cleared (no error shown)	Fail
TC-EP-131	EC-131	Contact Form	Email	notanemail	Rejected; invalid format	RAN | Malformed email 'notanemail' was NOT rejected: toast 'Message sent. We will get back to you soon.' and the form cleared	Fail
TC-EP-132	EC-132	Contact Form	Subject	Question about backtests	Accepted	ACCEPTED | Valid subject accepted (27/120 counter)	Pass
TC-EP-133	EC-133	Contact Form	Subject	(empty)	Rejected; subject required	Rejected (toast) | toast: Subject is required.	Pass
TC-EP-134	EC-134	Contact Form	Subject	121 x a	Input limited/rejected at 120	REJECTED (input limited) | Typing beyond 120 chars is ignored; counter stays 120 / 120	Pass
TC-EP-135	EC-135	Contact Form	Message	Hello, test message	Accepted	ACCEPTED | Valid message accepted (52/5000 counter)	Pass
TC-EP-136	EC-136	Contact Form	Message	(empty)	Rejected; message required	Rejected (toast) | toast: Message must be at least 10 characters.	Pass
TC-EP-137	EC-137	Contact Form	Message	5001 x a	Input limited/rejected at 5000	REJECTED (input limited) | Typing beyond 5000 chars is ignored; message stays 5000 chars	Pass
TC-EP-138	EC-138	Lump Sum vs DCA	Starting Value	10000	Input accepted by validation (no validation error); analysis would run	PLAN-LIMIT PAGE (premium tool) | After clicking Analyze with valid inputs: redirected to /pricing?reason=premium_tool&tool=lumpsum_dca_sensitivity: "Premium tool - Lump Sum vs DCA: Requires Pro or higher."	Blocked
TC-EP-139	EC-139	Lump Sum vs DCA	Starting Value	50	Rejected with a validation message; no calculation is run	BLOCKED (browser validation message shown for the field, but clicking Analyze redirects to /pricing?reason=premium_tool) | Starting Value: Value must be greater than or equal to 100. (bubble via field validation); click -> premium-tool pricing page	Blocked
TC-EP-140	EC-140	Lump Sum vs DCA	DCA horizon (months)	12	Input accepted by validation (no validation error); analysis would run	PLAN-LIMIT PAGE (premium tool) | Valid horizon 12 -> Analyze redirects to /pricing?reason=premium_tool&tool=lumpsum_dca_sensitivity ("Requires Pro or higher")	Blocked
TC-EP-141	EC-141	Lump Sum vs DCA	DCA horizon (months)	0	Rejected with a validation message; no calculation is run	BLOCKED (browser validation message shown for the field, but clicking Analyze redirects to pricing) | DCA horizon (months): Value must be greater than or equal to 1.	Blocked
TC-EP-142	EC-142	Lump Sum vs DCA	DCA horizon (months)	200	Rejected with a validation message; no calculation is run	BLOCKED (browser validation message shown for the field, but clicking Analyze redirects to pricing) | DCA horizon (months): Value must be less than or equal to 120.	Blocked

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-143	EC-143	Lump Sum vs DCA	Ticker	SPY	Input accepted by validation (no validation error); analysis would run	PLAN-LIMIT PAGE (premium tool) | Valid ticker SPY -> Analyze redirects to /pricing?reason=premium_tool&tool=lumpsum_dca_sensitivity	Blocked
TC-EP-144	EC-144	Lump Sum vs DCA	Ticker	ZZZZZZ	Validation message shown	BLOCKED (premium gate before validation) | Unknown ticker ZZZZZZ -> Analyze redirects to /pricing (no "Invalid ticker" message shown)	Blocked
TC-EP-145	EC-145	Lump Sum vs DCA	Allocation % (portfolio total)	60/40	Input accepted by validation (no validation error); analysis would run	PLAN-LIMIT PAGE (premium tool) | Allocation total 100 (60/40) -> Analyze redirects to /pricing?reason=premium_tool&tool=lumpsum_dca_sensitivity	Blocked
TC-EP-146	EC-146	Lump Sum vs DCA	Allocation % (portfolio total)	60/30 (90)	Validation message shown	BLOCKED (premium gate before validation) | Allocation total 90 (box turns red) -> Analyze redirects to /pricing (no allocation message shown)	Blocked
TC-EP-147	EC-147	Time Value of Money Calculator	Starting Value	10000	Accepted; calculation/analysis completes and results are shown	RAN | | sampled 1805 feasible solutions (charts shown)	Pass
TC-EP-148	EC-148	Time Value of Money Calculator	Starting Value	-500	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Starting Value must be between $0 and $5,000,000.	Pass
TC-EP-149	EC-149	Time Value of Money Calculator	Starting Value	9000000	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Starting Value must be between $0 and $5,000,000.	Pass
TC-EP-150	EC-150	Time Value of Money Calculator	Starting Value	(empty)	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Starting Value must be between $0 and $5,000,000.	Pass
TC-EP-151	EC-151	Time Value of Money Calculator	Monthly contribution	1000	Accepted; calculation/analysis completes and results are shown	RAN | | sampled 1805 feasible solutions (charts shown)	Pass
TC-EP-152	EC-152	Time Value of Money Calculator	Monthly contribution	-50	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Contribution must be between $0 and $100,000.	Pass
TC-EP-153	EC-153	Time Value of Money Calculator	Monthly contribution	500000	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Contribution must be between $0 and $100,000.	Pass
TC-EP-154	EC-154	Rebalancing Sensitivity	Start / End Date	(empty)	Input accepted by validation (no validation error); analysis would run	PLAN-LIMIT PAGE (premium tool) | Valid inputs, dates empty -> Analyze redirects to /pricing?reason=premium_tool&tool=rebalancing_sensitivity ("Rebalancing Sensitivity: Requires Pro or higher")	Blocked
TC-EP-155	EC-155	Rebalancing Sensitivity	Start / End Date	2020-01-01 / 2019-01-01	Validation message shown	BLOCKED (premium gate before validation) | Start 01/01/2020 after End 01/01/2019 -> Analyze redirects to /pricing (no date message shown)	Blocked
TC-EP-156	EC-156	Rebalancing Sensitivity	Drag (annual fee %)	0.5	Input accepted by validation (no validation error); analysis would run	PLAN-LIMIT PAGE (premium tool) | Valid drag 0.5 -> Analyze redirects to /pricing?reason=premium_tool&tool=rebalancing_sensitivity	Blocked
TC-EP-157	EC-157	Rebalancing Sensitivity	Drag (annual fee %)	-2	Validation message shown	BLOCKED (premium gate before validation) | Negative drag -2 -> Analyze redirects to /pricing (no drag message shown)	Blocked
TC-EP-158	EC-158	Rebalancing Sensitivity	Allocation % (portfolio total)	60/40	Input accepted by validation (no validation error); analysis would run	PLAN-LIMIT PAGE (premium tool) | Allocation total 100 -> Analyze redirects to /pricing (premium_tool)	Blocked
TC-EP-159	EC-159	Rebalancing Sensitivity	Allocation % (portfolio total)	60/30 (90)	Validation message shown	BLOCKED (premium gate before validation) | Allocation total 90 (box turns red) -> Analyze redirects to /pricing (no allocation message shown)	Blocked

Strong EP - Critical form: Portfolio Backtester (several classes covered in one test case)

TC-EP ID	EC IDs Covered	Feature / Form	Input Field(s)	Test Value(s)	Expected Result	Actual Result	P/F
TC-EP-S1	All valid classes: start/end dates, Starting Value, Rolling Window, growth, cashflow, drag, ticker and allocation	Portfolio Backtester	multiple fields	Start 2015-01-01, End 2020-01-01, SV 50000, RW 36, growth 2, cashflow 100, drag 0.5, SPY60/AGG40	Accepted; calculation/analysis completes and results are shown	RAN | EV $72,951.87, CAGR 7.69% | 5.00 years: 2014-12-31 - 2019-12-31	**Pass**
TC-EP-S2	IEC of Starting Value + IEC of Rolling Window + IEC of cashflow growth	Portfolio Backtester	multiple fields	SV 0, RW 0, growth -100	All three invalid fields are flagged; no run	Rejected (browser validation message) | Starting Value: Value must be greater than or equal to 1. ; Rolling Window: Value must be greater than or equal to 1. ; Annual cashflow growth: Value must be greater than or equal to -99.99. (all three fields invalid; browser shows the first)	**Pass**
TC-EP-S3	IEC unknown ticker + IEC allocation sum below 100	Portfolio Backtester	multiple fields	ticker ZZZZZZ, alloc 60/30	Both problems are reported; no run	Rejected (toast) | toast: Portfolio 1: Allocation percentages must sum to 100 (unknown-ticker error not reported)	**Fail**
TC-EP-S4	IEC start after end + IEC empty Starting Value + IEC unknown ticker + IEC allocation sum above 100	Portfolio Backtester	multiple fields	2020-01-01/2019-01-01, SV empty, ZZZZZZ, 60/60	All problems are reported; no run	Rejected (toast) | toast: Portfolio 1: Allocation percentages must sum to 100 / Invalid starting value / Start date must be before end date (unknown-ticker error not reported)	**Fail**
TC-EP-S5	VEC dates and numerics + IEC negative allocation	Portfolio Backtester	multiple fields	2015-01-01/2020-01-01, SV 50000, alloc -10/110	Rejected with a validation message; no calculation is run	RAN | EV $55,329.87, CAGR 2.05% | 5.00 years: 2014-12-31 - 2019-12-31	**Fail**
TC-EP-S6	IEC empty ticker + IEC drag above 100	Portfolio Backtester	multiple fields	ticker empty, drag 150	Both problems are reported; no run	Rejected (toast) | toast: Portfolio 1: Non-zero allocation has empty ticker (drag 150 not reported)	**Fail**

D3 - Boundary Value Analysis [25 marks]

Judgment on field selection: BVA was applied to fields with numeric, length or ordering boundaries discovered by probing. Fields excluded: Ticker, Email and Password format, Rebalance select (symbolic input, no boundary); Cashflow amount (no limit found); date fields on forms other than the ordering rule on the critical form.

Layer 1 - BVA Boundary Tables

Sign Up - Password

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-001	Sign Up	Password	EC-008	Minimum password length	8 characters

Portfolio Backtester - Starting Value

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-002	Portfolio Backtester	Starting Value	EC-026	Minimum starting value	1
BC-003	Portfolio Backtester	Starting Value	EC-031	Maximum starting value (largest accepted)	999,999,999,999

Portfolio Backtester - Rolling Window

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-004	Portfolio Backtester	Rolling Window	EC-033	Minimum rolling window (months)	1
BC-005	Portfolio Backtester	Rolling Window	EC-032	Maximum rolling window (largest accepted)	11999

Portfolio Backtester - Annual cashflow growth

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-006	Portfolio Backtester	Annual cashflow growth	EC-038	Minimum annual cashflow growth (%)	-99.99
BC-007	Portfolio Backtester	Annual cashflow growth	EC-039	Maximum annual cashflow growth (%)	100

Portfolio Backtester - Allocation % (portfolio total)

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-008	Portfolio Backtester	Allocation % (portfolio total)	EC-054	Required allocation total (%)	100

Portfolio Backtester - Start Date / End Date

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-009	Portfolio Backtester	Start Date / End Date	EC-017	Start date must be strictly before end date (end relative to start 2020-01-01)	end = start + 1 day

Portfolio Backtester - Drag (annual fee %)

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-010	Portfolio Backtester	Drag (annual fee %)	EC-044	Lower limit of drag (expected 0%)	0
BC-011	Portfolio Backtester	Drag (annual fee %)	EC-045	Upper limit of drag (expected 100%)	100

Monte Carlo Simulator - Number of years

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-012	Monte Carlo Simulator	Number of years	EC-063	Minimum horizon (years)	1
BC-013	Monte Carlo Simulator	Number of years	EC-064	Free-plan maximum horizon (years)	15
BC-014	Monte Carlo Simulator	Number of years	EC-065	Absolute maximum horizon (years)	100

Monte Carlo Simulator - Starting Value

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-015	Monte Carlo Simulator	Starting Value	EC-068	Minimum starting value	1

Monte Carlo Simulator - Min block (years)

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-016	Monte Carlo Simulator	Min block (years)	EC-069	Minimum value of Min block (years)	0.05

Monte Carlo Simulator - Max block (years)

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-017	Monte Carlo Simulator	Max block (years)	EC-071	Maximum value of Max block (years)	20

Monte Carlo Simulator - Random seed

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-018	Monte Carlo Simulator	Random seed	EC-076	Minimum random seed	0

Safe Withdrawal Rate Calculator - Retirement horizon (years)

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-019	Safe Withdrawal Rate Calculator	Retirement horizon (years)	EC-081	Minimum retirement horizon (years)	1
BC-020	Safe Withdrawal Rate Calculator	Retirement horizon (years)	EC-082	Maximum retirement horizon (years)	50

Safe Withdrawal Rate Calculator - Phase 1 end year

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-021	Safe Withdrawal Rate Calculator	Phase 1 end year	EC-084	Minimum Phase 1 end year	1

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-022	Safe Withdrawal Rate Calculator	Phase 1 end year	EC-083	Phase 1 end year must be below the horizon (horizon 30)	29

Safe Withdrawal Rate Calculator - Phase 1 annual withdrawal %

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-023	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	EC-086	Minimum Phase 1 withdrawal rate (%)	0
BC-024	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	EC-088	Maximum Phase 1 withdrawal rate (%)	100

Safe Withdrawal Rate Calculator - Phase 2 annual withdrawal %

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-025	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	EC-089	Minimum Phase 2 withdrawal rate (%)	0
BC-026	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	EC-091	Maximum Phase 2 withdrawal rate (%)	100

Backtest Optimizer - Starting Value

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-027	Backtest Optimizer	Starting Value	EC-105	Minimum starting value	1

Backtest Optimizer - Rolling Window

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-028	Backtest Optimizer	Rolling Window	EC-107	Minimum rolling window (months)	1

Backtest Optimizer - Max holdings

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-029	Backtest Optimizer	Max holdings	EC-109	Minimum max holdings	1

Backtest Optimizer - Min weight to include (%)

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-030	Backtest Optimizer	Min weight to include (%)	EC-111	Minimum value of min weight (%)	0
BC-031	Backtest Optimizer	Min weight to include (%)	EC-112	Maximum value of min weight (%)	100

Backtest Optimizer - Max vol (%)

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-032	Backtest Optimizer	Max vol (%)	EC-113	Minimum value of max vol (%)	0

Backtest Optimizer - Tickers

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-033	Backtest Optimizer	Tickers	EC-098	Minimum number of tickers	2

Efficient Frontier - Tickers

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-034	Efficient Frontier	Tickers	EC-116	Minimum number of tickers	2
BC-035	Efficient Frontier	Tickers	EC-115	Free-plan maximum number of tickers	2

Efficient Frontier - Max holdings

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-036	Efficient Frontier	Max holdings	EC-122	Minimum max holdings	1

Efficient Frontier - Min weight to include (%)

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-037	Efficient Frontier	Min weight to include (%)	EC-124	Minimum value of min weight (%)	0
BC-038	Efficient Frontier	Min weight to include (%)	EC-125	Maximum value of min weight (%)	100

Contact Form - Name

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-039	Contact Form	Name	EC-128	Maximum name length	120

Contact Form - Subject

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-040	Contact Form	Subject	EC-134	Maximum subject length	120

Contact Form - Message

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-041	Contact Form	Message	EC-137	Maximum message length	5000

Lump Sum vs DCA - Starting Value

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-042	Lump Sum vs DCA	Starting Value	EC-139	Minimum starting value	100

Lump Sum vs DCA - DCA horizon (months)

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-043	Lump Sum vs DCA	DCA horizon (months)	EC-141	Minimum DCA horizon (months)	1

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-044	Lump Sum vs DCA	DCA horizon (months)	EC-142	Maximum DCA horizon (months)	120

Time Value of Money Calculator - Starting Value

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-045	Time Value of Money Calculator	Starting Value	EC-148	Minimum starting value	0
BC-046	Time Value of Money Calculator	Starting Value	EC-149	Maximum starting value	5,000,000

Time Value of Money Calculator - Monthly contribution

BC ID	Feature / Form	Input Field	EC ID	Boundary Description	Boundary Value
BC-047	Time Value of Money Calculator	Monthly contribution	EC-152	Minimum monthly contribution	0
BC-048	Time Value of Money Calculator	Monthly contribution	EC-153	Maximum monthly contribution	100,000

Layer 2 - BVA Test Case Table (2-point on all forms; 3-point on the critical form)

TC-BVA ID	BC IDs	Feature / Form	Input Field	Test Value	Position	Expected Result	Actual Result	P/F
TC-BVA-001	BC-001	Sign Up	Password	7 chars	Below	Rejected; too short	REJECTED (button disabled) | 7-character password (abcdefg) in both fields: the 'Confirm my email' button stays disabled; no message. (6 chars, shot s2_220, is also disabled)	**Pass**
TC-BVA-002	BC-001	Sign Up	Password	8 chars	At	Accepted	ACCEPTED | 8-character password (abcdefgh): 'Confirm my email' button becomes enabled (the form was not submitted)	**Pass**
TC-BVA-003	BC-002	Portfolio Backtester	Starting Value	0.99	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Starting Value: Value must be greater than or equal to 1.	**Pass**
TC-BVA-004	BC-002	Portfolio Backtester	Starting Value	1	At	Accepted; calculation completes	RAN | EV $6.22, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-005	BC-002	Portfolio Backtester	Starting Value	2	Above	Accepted; calculation completes	RAN | EV $12.44, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-006	BC-003	Portfolio Backtester	Starting Value	999,999,999,998	Below	Accepted; calculation completes	RAN | EV $6,222,004,217,213.32, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-007	BC-003	Portfolio Backtester	Starting Value	999,999,999,999	At	Accepted; calculation completes	RAN | EV $6,222,004,217,219.58, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-008	BC-003	Portfolio Backtester	Starting Value	1,000,000,000,000	Above	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Starting value is too high	**Pass**
TC-BVA-009	BC-004	Portfolio Backtester	Rolling Window	0	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Rolling Window: Value must be greater than or equal to 1.	**Pass**
TC-BVA-010	BC-004	Portfolio Backtester	Rolling Window	1	At	Accepted; calculation completes	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-011	BC-004	Portfolio Backtester	Rolling Window	2	Above	Accepted; calculation completes	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-012	BC-005	Portfolio Backtester	Rolling Window	11998	Below	Accepted; calculation completes	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-013	BC-005	Portfolio Backtester	Rolling Window	11999	At	Accepted; calculation completes	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-014	BC-005	Portfolio Backtester	Rolling Window	12000	Above	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Rolling window is too large	**Pass**
TC-BVA-015	BC-006	Portfolio Backtester	Annual cashflow growth	-100	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Annual cashflow growth: Value must be greater than or equal to -99.99.	**Pass**
TC-BVA-016	BC-006	Portfolio Backtester	Annual cashflow growth	-99.99	At	Accepted; calculation completes	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-017	BC-006	Portfolio Backtester	Annual cashflow growth	-99.98	Above	Accepted; calculation completes	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-018	BC-007	Portfolio Backtester	Annual cashflow growth	99.99	Below	Accepted; calculation completes	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**
TC-BVA-019	BC-007	Portfolio Backtester	Annual cashflow growth	100	At	Accepted; calculation completes	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	**Pass**

TC-BVA ID	BC IDs	Feature / Form	Input Field	Test Value	Position	Expected Result	Actual Result	P/F
TC-BVA-020	BC-007	Portfolio Backtester	Annual cashflow growth	100.01	Above	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Annual cashflow growth: Value must be less than or equal to 100.	Pass
TC-BVA-021	BC-008	Portfolio Backtester	Allocation % (portfolio total)	99.99 (60/39.99)	Below	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Portfolio 1: Allocation percentages must sum to 100	Pass
TC-BVA-022	BC-008	Portfolio Backtester	Allocation % (portfolio total)	100 (60/40)	At	Accepted; calculation completes	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-BVA-023	BC-008	Portfolio Backtester	Allocation % (portfolio total)	100.01 (60/40.01)	Above	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Portfolio 1: Allocation percentages must sum to 100	Pass
TC-BVA-024	BC-009	Portfolio Backtester	Start Date / End Date	end 2019-12-31	Below	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Start date must be before end date	Pass
TC-BVA-025	BC-009	Portfolio Backtester	Start Date / End Date	end 2020-01-01	At	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Start date must be before end date	Pass
TC-BVA-026	BC-009	Portfolio Backtester	Start Date / End Date	end 2020-01-02	Above	Accepted; calculation completes	RAN | EV $10,067.15, CAGR 239.45% | 0.01 years: 2019-12-31 - 2020-01-02	Pass
TC-BVA-027	BC-010	Portfolio Backtester	Drag (annual fee %)	-1	Below	Rejected with a validation message; no calculation is run	RAN | EV $78,320.13, CAGR 9.33% | 23.06 years: 2003-09-26 - 2026-10-19	Fail
TC-BVA-028	BC-010	Portfolio Backtester	Drag (annual fee %)	0	At	Accepted; calculation completes	RAN | EV $62,220.04, CAGR 8.25% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-BVA-029	BC-010	Portfolio Backtester	Drag (annual fee %)	1	Above	Accepted; calculation completes	RAN | EV $49,429.16, CAGR 7.17% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-BVA-030	BC-011	Portfolio Backtester	Drag (annual fee %)	99	Below	Accepted; calculation completes	RAN | EV $0.00, CAGR -59.77% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-BVA-031	BC-011	Portfolio Backtester	Drag (annual fee %)	100	At	Accepted; calculation completes	RAN | EV $0.00, CAGR -60.17% | 23.06 years: 2003-09-26 - 2026-10-19	Pass
TC-BVA-032	BC-011	Portfolio Backtester	Drag (annual fee %)	101	Above	Rejected with a validation message; no calculation is run	RAN | EV $0.00, CAGR -60.57% | 23.06 years: 2003-09-26 - 2026-10-19	Fail
TC-BVA-033	BC-012	Monte Carlo Simulator	Number of years	0	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Number of years: Value must be greater than or equal to 1. | toast: Number of years must be positive	Pass
TC-BVA-034	BC-012	Monte Carlo Simulator	Number of years	1	At	Accepted; calculation completes	RAN | | 500 simulations of 1 years	Pass
TC-BVA-035	BC-013	Monte Carlo Simulator	Number of years	15	At	Accepted; calculation completes	RAN | | 500 simulations of 15 years	Pass
TC-BVA-036	BC-013	Monte Carlo Simulator	Number of years	16	Above	Rejected or plan-limit message	PLAN-LIMIT DIALOG | "Upgrade for higher Monte Carlo limits - Free supports up to 15 years and 500 simulations" (years 16)	Pass
TC-BVA-037	BC-014	Monte Carlo Simulator	Number of years	100	At	Accepted (paid plan) or plan-limit message on Free	PLAN-LIMIT DIALOG | "Upgrade for higher Monte Carlo limits - Free supports up to 15 years and 500 simulations" (years 100)	Pass
TC-BVA-038	BC-014	Monte Carlo Simulator	Number of years	101	Above	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Number of years: Value must be less than or equal to 100. | toast: Number of years must be <= 100	Pass

TC-BVA ID	BC IDs	Feature / Form	Input Field	Test Value	Position	Expected Result	Actual Result	P/F
TC-BVA-039	BC-015	Monte Carlo Simulator	Starting Value	0	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Starting Value: Value must be greater than or equal to 1. | toast: Starting value must be positive	Pass
TC-BVA-040	BC-015	Monte Carlo Simulator	Starting Value	1	At	Accepted; calculation completes	RAN | | 500 simulations of 15 years (starting value 1)	Pass
TC-BVA-041	BC-016	Monte Carlo Simulator	Min block (years)	0.04	Below	Rejected with a validation message; no calculation is run	RAN (DEFECT: browser reports min 0.05 but app accepts 0.04) | Min block 0.04 -> "Ran 500 simulations in 0.8 seconds", random seed auto-filled 85952096	Fail
TC-BVA-042	BC-016	Monte Carlo Simulator	Min block (years)	0.05	At	Accepted; calculation completes	RAN | | 500 simulations of 15 years	Pass
TC-BVA-043	BC-017	Monte Carlo Simulator	Max block (years)	20	At	Accepted; calculation completes	RAN | | 500 simulations of 15 years (max block 20)	Pass
TC-BVA-044	BC-017	Monte Carlo Simulator	Max block (years)	21	Above	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Max block (in years): Value must be less than or equal to 20. | toast: Max block length must be <= 20 years	Pass
TC-BVA-045	BC-018	Monte Carlo Simulator	Random seed	-1	Below	Rejected with a validation message; no calculation is run	RAN (silent correction: seed -1 reset to 0) | Random seed field shows 0; "Ran 500 simulations in 0.7 seconds"; no message	Fail
TC-BVA-046	BC-018	Monte Carlo Simulator	Random seed	0	At	Accepted; calculation completes	RAN | | 500 simulations of 15 years (seed 0). NOTE first attempt (shots 182,183) returned transient toast "An error occurred while running Monte Carlo"; repeat OK	Pass
TC-BVA-047	BC-019	Safe Withdrawal Rate Calculator	Retirement horizon (years)	0	Below	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Retirement horizon must be greater than 0 and at most 50 years	Pass
TC-BVA-048	BC-019	Safe Withdrawal Rate Calculator	Retirement horizon (years)	2 (Phase 1 end 1)	At	Accepted; calculation completes	RAN | 58.55 years: 1968-04-01 - 2026-10-19 (horizon 2, phase 1 end 1)	Pass
TC-BVA-049	BC-020	Safe Withdrawal Rate Calculator	Retirement horizon (years)	50	At	Accepted; calculation completes	RAN | 58.55 years: 1968-04-01 - 2026-10-19 (horizon 50)	Pass
TC-BVA-050	BC-020	Safe Withdrawal Rate Calculator	Retirement horizon (years)	51	Above	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Retirement horizon: Value must be less than or equal to 50. | toast: Retirement horizon must be greater than 0 and at most 50 years	Pass
TC-BVA-051	BC-021	Safe Withdrawal Rate Calculator	Phase 1 end year	0	Below	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Phase 1 end year must be greater than 0 and at most 50	Pass
TC-BVA-052	BC-021	Safe Withdrawal Rate Calculator	Phase 1 end year	1	At	Accepted; calculation completes	RAN | 58.55 years: 1968-04-01 - 2026-10-19 (phase 1 end 1)	Pass
TC-BVA-053	BC-022	Safe Withdrawal Rate Calculator	Phase 1 end year	29	At	Accepted; calculation completes	RAN | 58.55 years: 1968-04-01 - 2026-10-19 (phase 1 end 29)	Pass
TC-BVA-054	BC-022	Safe Withdrawal Rate Calculator	Phase 1 end year	30	Above	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Phase 2 must end after the previous phase (phase 1 end = 30 = horizon)	Pass
TC-BVA-055	BC-023	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	-1	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Phase 1 annual withdrawal: Value must be greater than or equal to 0. | toast: Phase 1 withdrawal rate must be between 0% and 100%	Pass
TC-BVA-056	BC-023	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	0	At	Accepted; calculation completes	RAN | 58.55 years: 1968-04-01 - 2026-10-19 (phase 1 withdrawal 0%)	Pass
TC-BVA-057	BC-024	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	100	At	Accepted; calculation completes	RAN | 58.55 years: 1968-04-01 - 2026-10-19 (phase 1 withdrawal 100%; survival 0.00%)	Pass

TC-BVA ID	BC IDs	Feature / Form	Input Field	Test Value	Position	Expected Result	Actual Result	P/F
TC-BVA-058	BC-024	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	101	Above	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Phase 1 annual withdrawal: Value must be less than or equal to 100. | toast: Phase 1 withdrawal rate must be between 0% and 100%	Pass
TC-BVA-059	BC-025	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	-1	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation + toast) | Phase 2 annual withdrawal: Value must be greater than or equal to 0. | toast: Phase 2 withdrawal rate must be between 0% and 100%	Pass
TC-BVA-060	BC-025	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	0	At	Accepted; calculation completes	RAN | 58.55 years: 1968-04-01 - 2026-10-19 (phase 2 withdrawal 0%)	Pass
TC-BVA-061	BC-026	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	100	At	Accepted; calculation completes	RAN | 58.55 years: 1968-04-01 - 2026-10-19 (phase 2 withdrawal 100%; survival 0.00%)	Pass
TC-BVA-062	BC-026	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	101	Above	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Phase 2 annual withdrawal: Value must be less than or equal to 100. (no run; results area unchanged)	Pass
TC-BVA-063	BC-027	Backtest Optimizer	Starting Value	0	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Starting Value: Value must be greater than or equal to 1.	Pass
TC-BVA-064	BC-027	Backtest Optimizer	Starting Value	1	At	Accepted; calculation completes	RAN | 21.92 years: 2004-11-18 - 2026-10-19 (starting value 1)	Pass
TC-BVA-065	BC-028	Backtest Optimizer	Rolling Window	0	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Rolling Window: Value must be greater than or equal to 1.	Pass
TC-BVA-066	BC-028	Backtest Optimizer	Rolling Window	1	At	Accepted; calculation completes	RAN | 21.92 years: 2004-11-18 - 2026-10-19 (rolling window 1)	Pass
TC-BVA-067	BC-029	Backtest Optimizer	Max holdings	0	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Max holdings: Value must be greater than or equal to 1.	Pass
TC-BVA-068	BC-029	Backtest Optimizer	Max holdings	1	At	Accepted; calculation completes	RAN | 21.92 years: 2004-11-18 - 2026-10-19 (max holdings 1: SPY 100%)	Pass
TC-BVA-069	BC-030	Backtest Optimizer	Min weight to include (%)	-1	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Min weight to include: Value must be greater than or equal to 0.	Pass
TC-BVA-070	BC-030	Backtest Optimizer	Min weight to include (%)	0	At	Accepted; calculation completes	RAN | 21.92 years: 2004-11-18 - 2026-10-19 (min weight 0)	Pass
TC-BVA-071	BC-031	Backtest Optimizer	Min weight to include (%)	100	At	Accepted; calculation completes	RAN | 21.92 years: 2004-11-18 - 2026-10-19 (min weight 100: SPY 100%)	Pass
TC-BVA-072	BC-031	Backtest Optimizer	Min weight to include (%)	101	Above	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Min weight to include: Value must be less than or equal to 100.	Pass
TC-BVA-073	BC-032	Backtest Optimizer	Max vol (%)	-1	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Max vol: Value must be greater than or equal to 0.	Pass
TC-BVA-074	BC-032	Backtest Optimizer	Max vol (%)	0	At	Accepted (may report no feasible solution)	ACCEPTED (no validation error); optimizer reports no feasible solution | toast: Optimization failed. Try relaxing constraints. (max vol 0)	Pass
TC-BVA-075	BC-033	Backtest Optimizer	Tickers	1 ticker (SPY)	Below	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: You must specify at least two tickers	Pass

TC-BVA ID	BC IDs	Feature / Form	Input Field	Test Value	Position	Expected Result	Actual Result	P/F
TC-BVA-076	BC-033	Backtest Optimizer	Tickers	2 tickers (SPY AGG)	At	Accepted; calculation completes	RAN | 23.06 years: 2003-09-26 - 2026-10-19 (two tickers SPY AGG)	Pass
TC-BVA-077	BC-034	Efficient Frontier	Tickers	1 ticker (SPY)	Below	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: You must specify at least two tickers	Pass
TC-BVA-078	BC-034	Efficient Frontier	Tickers	2 tickers (SPY AGG)	At	Accepted; calculation completes	RAN | | Efficient frontier computed 23.06 years: 2003-09-26 - 2026-10-19 (2 tickers)	Pass
TC-BVA-079	BC-035	Efficient Frontier	Tickers	2 tickers (SPY AGG)	At	Accepted; calculation completes	RAN | | Efficient frontier computed 23.06 years: 2003-09-26 - 2026-10-19 (2 tickers = free-plan maximum)	Pass
TC-BVA-080	BC-035	Efficient Frontier	Tickers	3 tickers (SPY AGG GLD)	Above	Rejected or plan-limit message	PLAN-LIMIT PAGE | 3 tickers -> redirected to /pricing?reason=limit&tool=efficient_frontier&limit=max_efficient_frontier_tickers ("You've hit the max tickers limit ... current limit is 2 tickers")	Pass
TC-BVA-081	BC-036	Efficient Frontier	Max holdings	0	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Max holdings: Value must be greater than or equal to 1.	Pass
TC-BVA-082	BC-036	Efficient Frontier	Max holdings	1	At	Accepted; calculation completes	RAN | | Efficient frontier computed 23.06 years: 2003-09-26 - 2026-10-19 (max holdings 1: SPY 100%)	Pass
TC-BVA-083	BC-037	Efficient Frontier	Min weight to include (%)	-1	Below	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Min weight to include: Value must be greater than or equal to 0.	Pass
TC-BVA-084	BC-037	Efficient Frontier	Min weight to include (%)	0	At	Accepted; calculation completes	RAN | | Efficient frontier computed 23.06 years: 2003-09-26 - 2026-10-19 (min weight 0)	Pass
TC-BVA-085	BC-038	Efficient Frontier	Min weight to include (%)	100	At	Accepted; calculation completes	RAN | | Efficient frontier computed 23.06 years: 2003-09-26 - 2026-10-19 (min weight 100: SPY 100%)	Pass
TC-BVA-086	BC-038	Efficient Frontier	Min weight to include (%)	101	Above	Rejected with a validation message; no calculation is run	Rejected (browser validation message) | Min weight to include: Value must be less than or equal to 100.	Pass
TC-BVA-087	BC-039	Contact Form	Name	120 chars	At	Accepted	ACCEPTED | 120-char name accepted	Pass
TC-BVA-088	BC-039	Contact Form	Name	121 chars	Above	Rejected or truncated at 120	REJECTED (input limited) | 121st character ignored; Name stays 120	Pass
TC-BVA-089	BC-040	Contact Form	Subject	120 chars	At	Accepted	ACCEPTED | 120-char subject accepted (counter 120 / 120)	Pass
TC-BVA-090	BC-040	Contact Form	Subject	121 chars	Above	Rejected or truncated at 120	REJECTED (input limited) | 121st character ignored; Subject stays 120	Pass
TC-BVA-091	BC-041	Contact Form	Message	5000 chars	At	Accepted	ACCEPTED | 5000-char message accepted	Pass
TC-BVA-092	BC-041	Contact Form	Message	5001 chars	Above	Rejected or truncated at 5000	REJECTED (input limited) | 5001st character ignored; Message stays 5000	Pass
TC-BVA-093	BC-042	Lump Sum vs DCA	Starting Value	99	Below	Rejected with a validation message; no calculation is run	BLOCKED (browser validation message shown for the field, but clicking Analyze redirects to pricing) | Starting Value: Value must be greater than or equal to 100.	Blocked
TC-BVA-094	BC-042	Lump Sum vs DCA	Starting Value	100	At	Input accepted by validation	PLAN-LIMIT PAGE (premium tool) | Starting Value 100 accepted by validation; Analyze redirects to /pricing (premium_tool)	Blocked
TC-BVA-095	BC-043	Lump Sum vs DCA	DCA horizon (months)	0	Below	Rejected with a validation message; no calculation is run	BLOCKED (browser validation message shown for the field, but clicking Analyze redirects to pricing) | DCA horizon (months): Value must be greater than or equal to 1.	Blocked
TC-BVA-096	BC-043	Lump Sum vs DCA	DCA horizon (months)	1	At	Input accepted by validation	PLAN-LIMIT PAGE (premium tool) | DCA horizon 1 accepted by validation; Analyze redirects to /pricing (premium_tool)	Blocked

TC-BVA ID	BC IDs	Feature / Form	Input Field	Test Value	Position	Expected Result	Actual Result	P/F
TC-BVA-097	BC-044	Lump Sum vs DCA	DCA horizon (months)	120	At	Input accepted by validation	PLAN-LIMIT PAGE (premium tool) | DCA horizon 120 accepted by validation; Analyze redirects to /pricing (premium_tool)	**Blocked**
TC-BVA-098	BC-044	Lump Sum vs DCA	DCA horizon (months)	121	Above	Rejected with a validation message; no calculation is run	BLOCKED (browser validation message shown for the field, but clicking Analyze redirects to pricing) | DCA horizon (months): Value must be less than or equal to 120.	**Blocked**
TC-BVA-099	BC-045	Time Value of Money Calculator	Starting Value	-1	Below	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Starting Value must be between $0 and $5,000,000.	**Pass**
TC-BVA-100	BC-045	Time Value of Money Calculator	Starting Value	0	At	Accepted; calculation completes	RAN | | sampled 1805 feasible solutions (starting value 0)	**Pass**
TC-BVA-101	BC-046	Time Value of Money Calculator	Starting Value	5,000,000	At	Accepted; calculation completes	RAN | | sampled 1805 feasible solutions (starting value 5,000,000)	**Pass**
TC-BVA-102	BC-046	Time Value of Money Calculator	Starting Value	5,000,001	Above	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Starting Value must be between $0 and $5,000,000.	**Pass**
TC-BVA-103	BC-047	Time Value of Money Calculator	Monthly contribution	-1	Below	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Contribution must be between $0 and $100,000.	**Pass**
TC-BVA-104	BC-047	Time Value of Money Calculator	Monthly contribution	0	At	Accepted; calculation completes	RAN | | sampled 1805 feasible solutions (contribution 0)	**Pass**
TC-BVA-105	BC-048	Time Value of Money Calculator	Monthly contribution	100,000	At	Accepted; calculation completes	RAN | | sampled 1805 feasible solutions (contribution 100,000)	**Pass**
TC-BVA-106	BC-048	Time Value of Money Calculator	Monthly contribution	100,001	Above	Rejected with a validation message; no calculation is run	Rejected (toast) | toast: Contribution must be between $0 and $100,000.	**Pass**

D4 - Traceability Matrix [10 marks]

Pass = behaviour matches the expected result; Fail = behaviour deviates (defect observation); Blocked = the Free plan redirects to the pricing page before the validation can be observed. Every test case from D2 and D3 is listed.

TC ID	Feature / Form	Input Field	Technique	Class / Boundary Description	Status	Result
TC-EP-001	Sign Up	Email	Weak EP + robustness	EC-001: Valid, unregistered email address format	Executed	Pass
TC-EP-002	Sign Up	Email	Weak EP + robustness	EC-002: Empty email	Executed	Pass
TC-EP-003	Sign Up	Email	Weak EP + robustness	EC-003: Email without @ symbol	Executed	Pass
TC-EP-004	Sign Up	Email	Weak EP + robustness	EC-004: Email without domain suffix	Executed	Fail
TC-EP-005	Sign Up	Email	Weak EP + robustness	EC-005: Email already registered	Executed	Fail
TC-EP-006	Sign Up	Password	Weak EP + robustness	EC-006: Password of valid length	Executed	Pass
TC-EP-007	Sign Up	Password	Weak EP + robustness	EC-007: Empty password	Executed	Pass
TC-EP-008	Sign Up	Password	Weak EP + robustness	EC-008: Password shorter than minimum length	Executed	Pass
TC-EP-009	Sign Up	Password	Weak EP + robustness	EC-009: Password longer than maximum length	Executed	Fail
TC-EP-010	Sign In	Email	Weak EP + robustness	EC-010: Registered email	Executed	Pass
TC-EP-011	Sign In	Email	Weak EP + robustness	EC-011: Empty email	Executed	Pass
TC-EP-012	Sign In	Email	Weak EP + robustness	EC-012: Malformed email	Executed	Pass
TC-EP-013	Sign In	Email	Weak EP + robustness	EC-013: Unregistered email	Executed	Pass
TC-EP-014	Sign In	Password	Weak EP + robustness	EC-014: Correct password for the account	Executed	Pass
TC-EP-015	Sign In	Password	Weak EP + robustness	EC-015: Empty password	Executed	Pass
TC-EP-016	Sign In	Password	Weak EP + robustness	EC-016: Wrong password	Executed	Pass
TC-EP-017	Portfolio Backtester	Start Date	Weak EP + robustness	EC-017: Valid start date inside data range, before end date	Executed	Pass
TC-EP-018	Portfolio Backtester	Start Date	Weak EP + robustness	EC-018: Start date left empty (full history)	Executed	Pass
TC-EP-019	Portfolio Backtester	Start Date	Weak EP + robustness	EC-019: Start date after end date	Executed	Pass
TC-EP-020	Portfolio Backtester	Start Date	Weak EP + robustness	EC-020: Start date equal to end date	Executed	Pass
TC-EP-021	Portfolio Backtester	Start Date	Weak EP + robustness	EC-021: Start date earlier than first available data	Executed	Fail
TC-EP-022	Portfolio Backtester	Start Date	Weak EP + robustness	EC-022: Start date in the far future (no data)	Executed	Pass
TC-EP-023	Portfolio Backtester	End Date	Weak EP + robustness	EC-023: Valid end date with start empty	Executed	Pass
TC-EP-024	Portfolio Backtester	End Date	Weak EP + robustness	EC-024: Start only, end date empty (runs to latest data)	Executed	Pass
TC-EP-025	Portfolio Backtester	End Date	Weak EP + robustness	EC-025: End date later than last available data	Executed	Fail
TC-EP-026	Portfolio Backtester	Starting Value	Weak EP + robustness	EC-026: Valid starting value (>=1, below upper limit)	Executed	Pass
TC-EP-027	Portfolio Backtester	Starting Value	Weak EP + robustness	EC-027: Zero starting value	Executed	Pass
TC-EP-028	Portfolio Backtester	Starting Value	Weak EP + robustness	EC-028: Negative starting value	Executed	Pass
TC-EP-029	Portfolio Backtester	Starting Value	Weak EP + robustness	EC-029: Fractional value below minimum	Executed	Pass
TC-EP-030	Portfolio Backtester	Starting Value	Weak EP + robustness	EC-030: Empty starting value	Executed	Pass
TC-EP-031	Portfolio Backtester	Starting Value	Weak EP + robustness	EC-031: Starting value above the upper limit	Executed	Pass
TC-EP-032	Portfolio Backtester	Rolling Window	Weak EP + robustness	EC-032: Valid rolling window in months	Executed	Pass
TC-EP-033	Portfolio Backtester	Rolling Window	Weak EP + robustness	EC-033: Zero rolling window	Executed	Pass
TC-EP-034	Portfolio Backtester	Rolling Window	Weak EP + robustness	EC-034: Negative rolling window	Executed	Pass
TC-EP-035	Portfolio Backtester	Rolling Window	Weak EP + robustness	EC-035: Non-integer rolling window	Executed	Pass
TC-EP-036	Portfolio Backtester	Rolling Window	Weak EP + robustness	EC-036: Rolling window above upper limit	Executed	Pass
TC-EP-037	Portfolio Backtester	Annual cashflow growth	Weak EP + robustness	EC-037: Valid growth percentage	Executed	Pass
TC-EP-038	Portfolio Backtester	Annual cashflow growth	Weak EP + robustness	EC-038: Growth below minimum	Executed	Pass
TC-EP-039	Portfolio Backtester	Annual cashflow growth	Weak EP + robustness	EC-039: Growth above maximum	Executed	Pass
TC-EP-040	Portfolio Backtester	Cashflow amount	Weak EP + robustness	EC-040: Positive cashflow (contribution)	Executed	Pass
TC-EP-041	Portfolio Backtester	Cashflow amount	Weak EP + robustness	EC-041: Negative cashflow (withdrawal)	Executed	Pass
TC-EP-042	Portfolio Backtester	Cashflow amount	Weak EP + robustness	EC-042: Zero cashflow (no cashflow)	Executed	Pass

TC ID	Feature / Form	Input Field	Technique	Class / Boundary Description	Status	Result
TC-EP-043	Portfolio Backtester	Drag (annual fee %)	Weak EP + robustness	EC-043: Valid drag between 0 and 100	Executed	Pass
TC-EP-044	Portfolio Backtester	Drag (annual fee %)	Weak EP + robustness	EC-044: Negative drag	Executed	Fail
TC-EP-045	Portfolio Backtester	Drag (annual fee %)	Weak EP + robustness	EC-045: Drag above 100%	Executed	Fail
TC-EP-046	Portfolio Backtester	Ticker	Weak EP + robustness	EC-046: Valid ticker (upper case)	Executed	Pass
TC-EP-047	Portfolio Backtester	Ticker	Weak EP + robustness	EC-047: Valid ticker in lower case	Executed	Pass
TC-EP-048	Portfolio Backtester	Ticker	Weak EP + robustness	EC-048: Unknown ticker	Executed	Pass
TC-EP-049	Portfolio Backtester	Ticker	Weak EP + robustness	EC-049: Empty ticker with non-zero allocation	Executed	Pass
TC-EP-050	Portfolio Backtester	Ticker	Weak EP + robustness	EC-050: SQL fragment as ticker	Executed	Pass
TC-EP-051	Portfolio Backtester	Ticker	Weak EP + robustness	EC-051: Non-ASCII (unicode) ticker	Executed	Pass
TC-EP-052	Portfolio Backtester	Ticker	Weak EP + robustness	EC-052: Ticker containing a space	Executed	Pass
TC-EP-053	Portfolio Backtester	Ticker	Weak EP + robustness	EC-053: Very long ticker (50 characters)	Executed	Pass
TC-EP-054	Portfolio Backtester	Allocation % (portfolio total)	Weak EP + robustness	EC-054: Allocations sum to exactly 100 across 2 assets	Executed	Pass
TC-EP-055	Portfolio Backtester	Allocation % (portfolio total)	Weak EP + robustness	EC-055: Single asset at 100%	Executed	Pass
TC-EP-056	Portfolio Backtester	Allocation % (portfolio total)	Weak EP + robustness	EC-056: Allocations sum below 100	Executed	Pass
TC-EP-057	Portfolio Backtester	Allocation % (portfolio total)	Weak EP + robustness	EC-057: Allocations sum above 100	Executed	Pass
TC-EP-058	Portfolio Backtester	Allocation % (portfolio total)	Weak EP + robustness	EC-058: Negative allocation with offsetting excess	Executed	Fail
TC-EP-059	Portfolio Backtester	Rebalance frequency	Weak EP + robustness	EC-059: Valid rebalance option (Yearly, default)	Executed	Pass
TC-EP-060	Monte Carlo Simulator	Start / End Date	Weak EP + robustness	EC-060: Dates empty (full history)	Executed	Pass
TC-EP-061	Monte Carlo Simulator	Start / End Date	Weak EP + robustness	EC-061: Start date after end date	Executed	Pass
TC-EP-062	Monte Carlo Simulator	Number of years	Weak EP + robustness	EC-062: Valid horizon within free limit	Executed	Pass
TC-EP-063	Monte Carlo Simulator	Number of years	Weak EP + robustness	EC-063: Horizon below minimum	Executed	Pass
TC-EP-064	Monte Carlo Simulator	Number of years	Weak EP + robustness	EC-064: Horizon above free-plan limit but within 100	Executed	Pass
TC-EP-065	Monte Carlo Simulator	Number of years	Weak EP + robustness	EC-065: Horizon above absolute maximum	Executed	Pass
TC-EP-066	Monte Carlo Simulator	Number of years	Weak EP + robustness	EC-066: Decimal horizon	Executed	Pass
TC-EP-067	Monte Carlo Simulator	Starting Value	Weak EP + robustness	EC-067: Valid starting value	Executed	Pass
TC-EP-068	Monte Carlo Simulator	Starting Value	Weak EP + robustness	EC-068: Zero starting value	Executed	Pass
TC-EP-069	Monte Carlo Simulator	Min block (years)	Weak EP + robustness	EC-069: Valid minimum block length	Executed	Pass
TC-EP-070	Monte Carlo Simulator	Min block (years)	Weak EP + robustness	EC-070: Minimum block below allowed minimum	Executed	Fail
TC-EP-071	Monte Carlo Simulator	Max block (years)	Weak EP + robustness	EC-071: Valid maximum block length	Executed	Pass
TC-EP-072	Monte Carlo Simulator	Max block (years)	Weak EP + robustness	EC-072: Maximum block above allowed maximum	Executed	Pass
TC-EP-073	Monte Carlo Simulator	Max block (years)	Weak EP + robustness	EC-073: Maximum block below allowed minimum	Executed	Pass
TC-EP-074	Monte Carlo Simulator	Random seed	Weak EP + robustness	EC-074: Seed left empty (random)	Executed	Pass
TC-EP-075	Monte Carlo Simulator	Random seed	Weak EP + robustness	EC-075: Valid whole-number seed	Executed	Pass
TC-EP-076	Monte Carlo Simulator	Random seed	Weak EP + robustness	EC-076: Negative seed	Executed	Fail
TC-EP-077	Monte Carlo Simulator	Random seed	Weak EP + robustness	EC-077: Non-integer seed	Executed	Fail
TC-EP-078	Monte Carlo Simulator	Allocation % (portfolio total)	Weak EP + robustness	EC-078: Allocations sum to 100 (50/25/25)	Executed	Pass
TC-EP-079	Monte Carlo Simulator	Allocation % (portfolio total)	Weak EP + robustness	EC-079: Allocations sum below 100	Executed	Pass
TC-EP-080	Safe Withdrawal Rate Calculator	Retirement horizon (years)	Weak EP + robustness	EC-080: Valid horizon	Executed	Pass
TC-EP-081	Safe Withdrawal Rate Calculator	Retirement horizon (years)	Weak EP + robustness	EC-081: Zero horizon	Executed	Pass
TC-EP-082	Safe Withdrawal Rate Calculator	Retirement horizon (years)	Weak EP + robustness	EC-082: Horizon above maximum	Executed	Pass
TC-EP-083	Safe Withdrawal Rate Calculator	Phase 1 end year	Weak EP + robustness	EC-083: Valid end year before horizon	Executed	Pass
TC-EP-084	Safe Withdrawal Rate Calculator	Phase 1 end year	Weak EP + robustness	EC-084: End year zero	Executed	Pass

TC ID	Feature / Form	Input Field	Technique	Class / Boundary Description	Status	Result
TC-EP-085	Safe Withdrawal Rate Calculator	Phase 1 end year	Weak EP + robustness	EC-085: End year after the horizon	Executed	**Pass**
TC-EP-086	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	Weak EP + robustness	EC-086: Valid withdrawal rate	Executed	**Pass**
TC-EP-087	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	Weak EP + robustness	EC-087: Withdrawal rate negative	Executed	**Pass**
TC-EP-088	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	Weak EP + robustness	EC-088: Withdrawal rate above 100	Executed	**Pass**
TC-EP-089	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	Weak EP + robustness	EC-089: Valid withdrawal rate	Executed	**Pass**
TC-EP-090	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	Weak EP + robustness	EC-090: Withdrawal rate negative	Executed	**Pass**
TC-EP-091	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	Weak EP + robustness	EC-091: Withdrawal rate above 100	Executed	**Pass**
TC-EP-092	Safe Withdrawal Rate Calculator	Start / End Date	Weak EP + robustness	EC-092: Dates empty (full history)	Executed	**Pass**
TC-EP-093	Safe Withdrawal Rate Calculator	Start / End Date	Weak EP + robustness	EC-093: Start date after end date	Executed	**Pass**
TC-EP-094	Safe Withdrawal Rate Calculator	Allocation % (portfolio total)	Weak EP + robustness	EC-094: Allocations sum to 100 (50/30/20)	Executed	**Pass**
TC-EP-095	Safe Withdrawal Rate Calculator	Allocation % (portfolio total)	Weak EP + robustness	EC-095: Allocations sum below 100	Executed	**Pass**
TC-EP-096	Backtest Optimizer	Tickers	Weak EP + robustness	EC-096: Three valid tickers separated by spaces	Executed	**Pass**
TC-EP-097	Backtest Optimizer	Tickers	Weak EP + robustness	EC-097: Exactly two valid tickers	Executed	**Pass**
TC-EP-098	Backtest Optimizer	Tickers	Weak EP + robustness	EC-098: Only one ticker	Executed	**Pass**
TC-EP-099	Backtest Optimizer	Tickers	Weak EP + robustness	EC-099: Empty tickers	Executed	**Pass**
TC-EP-100	Backtest Optimizer	Tickers	Weak EP + robustness	EC-100: Contains an unknown ticker	Executed	**Pass**
TC-EP-101	Backtest Optimizer	Tickers	Weak EP + robustness	EC-101: Comma-separated tickers	Executed	**Pass**
TC-EP-102	Backtest Optimizer	Start / End Date	Weak EP + robustness	EC-102: Dates empty (full history)	Executed	**Pass**
TC-EP-103	Backtest Optimizer	Start / End Date	Weak EP + robustness	EC-103: Start date after end date	Executed	**Pass**
TC-EP-104	Backtest Optimizer	Starting Value	Weak EP + robustness	EC-104: Valid starting value	Executed	**Pass**
TC-EP-105	Backtest Optimizer	Starting Value	Weak EP + robustness	EC-105: Zero starting value	Executed	**Pass**
TC-EP-106	Backtest Optimizer	Rolling Window	Weak EP + robustness	EC-106: Valid rolling window	Executed	**Pass**
TC-EP-107	Backtest Optimizer	Rolling Window	Weak EP + robustness	EC-107: Zero rolling window	Executed	**Pass**
TC-EP-108	Backtest Optimizer	Max holdings	Weak EP + robustness	EC-108: Valid max holdings	Executed	**Pass**
TC-EP-109	Backtest Optimizer	Max holdings	Weak EP + robustness	EC-109: Zero max holdings	Executed	**Pass**
TC-EP-110	Backtest Optimizer	Min weight to include (%)	Weak EP + robustness	EC-110: Valid minimum weight	Executed	**Pass**
TC-EP-111	Backtest Optimizer	Min weight to include (%)	Weak EP + robustness	EC-111: Negative minimum weight	Executed	**Pass**
TC-EP-112	Backtest Optimizer	Min weight to include (%)	Weak EP + robustness	EC-112: Minimum weight above 100	Executed	**Pass**
TC-EP-113	Backtest Optimizer	Max vol (%)	Weak EP + robustness	EC-113: Valid maximum volatility	Executed	**Pass**
TC-EP-114	Backtest Optimizer	Max vol (%)	Weak EP + robustness	EC-114: Negative maximum volatility	Executed	**Pass**
TC-EP-115	Efficient Frontier	Tickers	Weak EP + robustness	EC-115: Two valid tickers	Executed	**Pass**
TC-EP-116	Efficient Frontier	Tickers	Weak EP + robustness	EC-116: Only one ticker	Executed	**Pass**
TC-EP-117	Efficient Frontier	Tickers	Weak EP + robustness	EC-117: Empty tickers	Executed	**Pass**
TC-EP-118	Efficient Frontier	Tickers	Weak EP + robustness	EC-118: Three tickers (above free-plan limit)	Executed	**Pass**
TC-EP-119	Efficient Frontier	Start / End Date	Weak EP + robustness	EC-119: Dates empty (full history)	Executed	**Pass**
TC-EP-120	Efficient Frontier	Start / End Date	Weak EP + robustness	EC-120: Start date after end date	Executed	**Pass**
TC-EP-121	Efficient Frontier	Max holdings	Weak EP + robustness	EC-121: Valid max holdings	Executed	**Pass**

TC ID	Feature / Form	Input Field	Technique	Class / Boundary Description	Status	Result
TC-EP-122	Efficient Frontier	Max holdings	Weak EP + robustness	EC-122: Zero max holdings	Executed	Pass
TC-EP-123	Efficient Frontier	Min weight to include (%)	Weak EP + robustness	EC-123: Valid minimum weight	Executed	Pass
TC-EP-124	Efficient Frontier	Min weight to include (%)	Weak EP + robustness	EC-124: Negative minimum weight	Executed	Pass
TC-EP-125	Efficient Frontier	Min weight to include (%)	Weak EP + robustness	EC-125: Minimum weight above 100	Executed	Pass
TC-EP-126	Contact Form	Name	Weak EP + robustness	EC-126: Valid name (1-120 chars)	Executed	Pass
TC-EP-127	Contact Form	Name	Weak EP + robustness	EC-127: Empty name	Executed	Fail
TC-EP-128	Contact Form	Name	Weak EP + robustness	EC-128: Name longer than 120 characters	Executed	Pass
TC-EP-129	Contact Form	Email	Weak EP + robustness	EC-129: Valid email format	Executed	Pass
TC-EP-130	Contact Form	Email	Weak EP + robustness	EC-130: Empty email	Executed	Fail
TC-EP-131	Contact Form	Email	Weak EP + robustness	EC-131: Malformed email	Executed	Fail
TC-EP-132	Contact Form	Subject	Weak EP + robustness	EC-132: Valid subject (1-120 chars)	Executed	Pass
TC-EP-133	Contact Form	Subject	Weak EP + robustness	EC-133: Empty subject	Executed	Pass
TC-EP-134	Contact Form	Subject	Weak EP + robustness	EC-134: Subject longer than 120 characters	Executed	Pass
TC-EP-135	Contact Form	Message	Weak EP + robustness	EC-135: Valid message (1-5000 chars)	Executed	Pass
TC-EP-136	Contact Form	Message	Weak EP + robustness	EC-136: Empty message	Executed	Pass
TC-EP-137	Contact Form	Message	Weak EP + robustness	EC-137: Message longer than 5000 characters	Executed	Pass
TC-EP-138	Lump Sum vs DCA	Starting Value	Weak EP + robustness	EC-138: Valid starting value (>=100)	Executed	Blocked
TC-EP-139	Lump Sum vs DCA	Starting Value	Weak EP + robustness	EC-139: Starting value below minimum	Executed	Blocked
TC-EP-140	Lump Sum vs DCA	DCA horizon (months)	Weak EP + robustness	EC-140: Valid horizon	Executed	Blocked
TC-EP-141	Lump Sum vs DCA	DCA horizon (months)	Weak EP + robustness	EC-141: Horizon below minimum	Executed	Blocked
TC-EP-142	Lump Sum vs DCA	DCA horizon (months)	Weak EP + robustness	EC-142: Horizon above maximum	Executed	Blocked
TC-EP-143	Lump Sum vs DCA	Ticker	Weak EP + robustness	EC-143: Valid ticker	Executed	Blocked
TC-EP-144	Lump Sum vs DCA	Ticker	Weak EP + robustness	EC-144: Unknown ticker	Executed	Blocked
TC-EP-145	Lump Sum vs DCA	Allocation % (portfolio total)	Weak EP + robustness	EC-145: Allocations sum to 100	Executed	Blocked
TC-EP-146	Lump Sum vs DCA	Allocation % (portfolio total)	Weak EP + robustness	EC-146: Allocations sum below 100	Executed	Blocked
TC-EP-147	Time Value of Money Calculator	Starting Value	Weak EP + robustness	EC-147: Valid starting value	Executed	Pass
TC-EP-148	Time Value of Money Calculator	Starting Value	Weak EP + robustness	EC-148: Negative starting value	Executed	Pass
TC-EP-149	Time Value of Money Calculator	Starting Value	Weak EP + robustness	EC-149: Starting value above maximum	Executed	Pass
TC-EP-150	Time Value of Money Calculator	Starting Value	Weak EP + robustness	EC-150: Empty starting value	Executed	Pass
TC-EP-151	Time Value of Money Calculator	Monthly contribution	Weak EP + robustness	EC-151: Valid monthly contribution	Executed	Pass
TC-EP-152	Time Value of Money Calculator	Monthly contribution	Weak EP + robustness	EC-152: Negative contribution	Executed	Pass
TC-EP-153	Time Value of Money Calculator	Monthly contribution	Weak EP + robustness	EC-153: Contribution above maximum	Executed	Pass
TC-EP-154	Rebalancing Sensitivity	Start / End Date	Weak EP + robustness	EC-154: Dates empty (full history)	Executed	Blocked
TC-EP-155	Rebalancing Sensitivity	Start / End Date	Weak EP + robustness	EC-155: Start date after end date	Executed	Blocked
TC-EP-156	Rebalancing Sensitivity	Drag (annual fee %)	Weak EP + robustness	EC-156: Valid drag	Executed	Blocked
TC-EP-157	Rebalancing Sensitivity	Drag (annual fee %)	Weak EP + robustness	EC-157: Negative drag	Executed	Blocked
TC-EP-158	Rebalancing Sensitivity	Allocation % (portfolio total)	Weak EP + robustness	EC-158: Allocations sum to 100	Executed	Blocked
TC-EP-159	Rebalancing Sensitivity	Allocation % (portfolio total)	Weak EP + robustness	EC-159: Allocations sum below 100	Executed	Blocked
TC-EP-S1	Portfolio Backtester	multiple fields	Strong EP	Strong EP 1: every field in a valid class simultaneously	Executed	Pass
TC-EP-S2	Portfolio Backtester	multiple fields	Strong EP	Strong EP 2: three numeric fields invalid simultaneously	Executed	Pass
TC-EP-S3	Portfolio Backtester	multiple fields	Strong EP	Strong EP 3: valid numerics + unknown ticker + allocation sum 90	Executed	Fail

TC ID	Feature / Form	Input Field	Technique	Class / Boundary Description	Status	Result
TC-EP-S4	Portfolio Backtester	multiple fields	Strong EP	Strong EP 4: invalid date order + empty starting value + unknown ticker + alloc sum 120	Executed	Fail
TC-EP-S5	Portfolio Backtester	multiple fields	Strong EP	Strong EP 5: valid dates + valid numerics + negative allocation (invalid)	Executed	Fail
TC-EP-S6	Portfolio Backtester	multiple fields	Strong EP	Strong EP 6: empty ticker + drag above 100	Executed	Fail
TC-BVA-001	Sign Up	Password	2-point BVA	BC-001 (Below): Minimum password length	Executed	Pass
TC-BVA-002	Sign Up	Password	2-point BVA	BC-001 (At): Minimum password length	Executed	Pass
TC-BVA-003	Portfolio Backtester	Starting Value	3-point BVA	BC-002 (Below): Minimum starting value	Executed	Pass
TC-BVA-004	Portfolio Backtester	Starting Value	3-point BVA	BC-002 (At): Minimum starting value	Executed	Pass
TC-BVA-005	Portfolio Backtester	Starting Value	3-point BVA	BC-002 (Above): Minimum starting value	Executed	Pass
TC-BVA-006	Portfolio Backtester	Starting Value	3-point BVA	BC-003 (Below): Maximum starting value (largest accepted)	Executed	Pass
TC-BVA-007	Portfolio Backtester	Starting Value	3-point BVA	BC-003 (At): Maximum starting value (largest accepted)	Executed	Pass
TC-BVA-008	Portfolio Backtester	Starting Value	3-point BVA	BC-003 (Above): Maximum starting value (largest accepted)	Executed	Pass
TC-BVA-009	Portfolio Backtester	Rolling Window	3-point BVA	BC-004 (Below): Minimum rolling window (months)	Executed	Pass
TC-BVA-010	Portfolio Backtester	Rolling Window	3-point BVA	BC-004 (At): Minimum rolling window (months)	Executed	Pass
TC-BVA-011	Portfolio Backtester	Rolling Window	3-point BVA	BC-004 (Above): Minimum rolling window (months)	Executed	Pass
TC-BVA-012	Portfolio Backtester	Rolling Window	3-point BVA	BC-005 (Below): Maximum rolling window (largest accepted)	Executed	Pass
TC-BVA-013	Portfolio Backtester	Rolling Window	3-point BVA	BC-005 (At): Maximum rolling window (largest accepted)	Executed	Pass
TC-BVA-014	Portfolio Backtester	Rolling Window	3-point BVA	BC-005 (Above): Maximum rolling window (largest accepted)	Executed	Pass
TC-BVA-015	Portfolio Backtester	Annual cashflow growth	3-point BVA	BC-006 (Below): Minimum annual cashflow growth (%)	Executed	Pass
TC-BVA-016	Portfolio Backtester	Annual cashflow growth	3-point BVA	BC-006 (At): Minimum annual cashflow growth (%)	Executed	Pass
TC-BVA-017	Portfolio Backtester	Annual cashflow growth	3-point BVA	BC-006 (Above): Minimum annual cashflow growth (%)	Executed	Pass
TC-BVA-018	Portfolio Backtester	Annual cashflow growth	3-point BVA	BC-007 (Below): Maximum annual cashflow growth (%)	Executed	Pass
TC-BVA-019	Portfolio Backtester	Annual cashflow growth	3-point BVA	BC-007 (At): Maximum annual cashflow growth (%)	Executed	Pass
TC-BVA-020	Portfolio Backtester	Annual cashflow growth	3-point BVA	BC-007 (Above): Maximum annual cashflow growth (%)	Executed	Pass
TC-BVA-021	Portfolio Backtester	Allocation % (portfolio total)	3-point BVA	BC-008 (Below): Required allocation total (%)	Executed	Pass
TC-BVA-022	Portfolio Backtester	Allocation % (portfolio total)	3-point BVA	BC-008 (At): Required allocation total (%)	Executed	Pass
TC-BVA-023	Portfolio Backtester	Allocation % (portfolio total)	3-point BVA	BC-008 (Above): Required allocation total (%)	Executed	Pass
TC-BVA-024	Portfolio Backtester	Start Date / End Date	3-point BVA	BC-009 (Below): Start date must be strictly before end date (end relative to start 2020-01-01)	Executed	Pass
TC-BVA-025	Portfolio Backtester	Start Date / End Date	3-point BVA	BC-009 (At): Start date must be strictly before end date (end relative to start 2020-01-01)	Executed	Pass
TC-BVA-026	Portfolio Backtester	Start Date / End Date	3-point BVA	BC-009 (Above): Start date must be strictly before end date (end relative to start 2020-01-01)	Executed	Pass
TC-BVA-027	Portfolio Backtester	Drag (annual fee %)	3-point BVA	BC-010 (Below): Lower limit of drag (expected 0%)	Executed	Fail
TC-BVA-028	Portfolio Backtester	Drag (annual fee %)	3-point BVA	BC-010 (At): Lower limit of drag (expected 0%)	Executed	Pass
TC-BVA-029	Portfolio Backtester	Drag (annual fee %)	3-point BVA	BC-010 (Above): Lower limit of drag (expected 0%)	Executed	Pass
TC-BVA-030	Portfolio Backtester	Drag (annual fee %)	3-point BVA	BC-011 (Below): Upper limit of drag (expected 100%)	Executed	Pass
TC-BVA-031	Portfolio Backtester	Drag (annual fee %)	3-point BVA	BC-011 (At): Upper limit of drag (expected 100%)	Executed	Pass
TC-BVA-032	Portfolio Backtester	Drag (annual fee %)	3-point BVA	BC-011 (Above): Upper limit of drag (expected 100%)	Executed	Fail
TC-BVA-033	Monte Carlo Simulator	Number of years	2-point BVA	BC-012 (Below): Minimum horizon (years)	Executed	Pass
TC-BVA-034	Monte Carlo Simulator	Number of years	2-point BVA	BC-012 (At): Minimum horizon (years)	Executed	Pass
TC-BVA-035	Monte Carlo Simulator	Number of years	2-point BVA	BC-013 (At): Free-plan maximum horizon (years)	Executed	Pass
TC-BVA-036	Monte Carlo Simulator	Number of years	2-point BVA	BC-013 (Above): Free-plan maximum horizon (years)	Executed	Pass
TC-BVA-037	Monte Carlo Simulator	Number of years	2-point BVA	BC-014 (At): Absolute maximum horizon (years)	Executed	Pass
TC-BVA-038	Monte Carlo Simulator	Number of years	2-point BVA	BC-014 (Above): Absolute maximum horizon (years)	Executed	Pass
TC-BVA-039	Monte Carlo Simulator	Starting Value	2-point BVA	BC-015 (Below): Minimum starting value	Executed	Pass
TC-BVA-040	Monte Carlo Simulator	Starting Value	2-point BVA	BC-015 (At): Minimum starting value	Executed	Pass
TC-BVA-041	Monte Carlo Simulator	Min block (years)	2-point BVA	BC-016 (Below): Minimum value of Min block (years)	Executed	Fail
TC-BVA-042	Monte Carlo Simulator	Min block (years)	2-point BVA	BC-016 (At): Minimum value of Min block (years)	Executed	Pass

TC ID	Feature / Form	Input Field	Technique	Class / Boundary Description	Status	Result
TC-BVA-043	Monte Carlo Simulator	Max block (years)	2-point BVA	BC-017 (At): Maximum value of Max block (years)	Executed	Pass
TC-BVA-044	Monte Carlo Simulator	Max block (years)	2-point BVA	BC-017 (Above): Maximum value of Max block (years)	Executed	Pass
TC-BVA-045	Monte Carlo Simulator	Random seed	2-point BVA	BC-018 (Below): Minimum random seed	Executed	Fail
TC-BVA-046	Monte Carlo Simulator	Random seed	2-point BVA	BC-018 (At): Minimum random seed	Executed	Pass
TC-BVA-047	Safe Withdrawal Rate Calculator	Retirement horizon (years)	2-point BVA	BC-019 (Below): Minimum retirement horizon (years)	Executed	Pass
TC-BVA-048	Safe Withdrawal Rate Calculator	Retirement horizon (years)	2-point BVA	BC-019 (At): Minimum retirement horizon (years)	Executed	Pass
TC-BVA-049	Safe Withdrawal Rate Calculator	Retirement horizon (years)	2-point BVA	BC-020 (At): Maximum retirement horizon (years)	Executed	Pass
TC-BVA-050	Safe Withdrawal Rate Calculator	Retirement horizon (years)	2-point BVA	BC-020 (Above): Maximum retirement horizon (years)	Executed	Pass
TC-BVA-051	Safe Withdrawal Rate Calculator	Phase 1 end year	2-point BVA	BC-021 (Below): Minimum Phase 1 end year	Executed	Pass
TC-BVA-052	Safe Withdrawal Rate Calculator	Phase 1 end year	2-point BVA	BC-021 (At): Minimum Phase 1 end year	Executed	Pass
TC-BVA-053	Safe Withdrawal Rate Calculator	Phase 1 end year	2-point BVA	BC-022 (At): Phase 1 end year must be below the horizon (horizon 30)	Executed	Pass
TC-BVA-054	Safe Withdrawal Rate Calculator	Phase 1 end year	2-point BVA	BC-022 (Above): Phase 1 end year must be below the horizon (horizon 30)	Executed	Pass
TC-BVA-055	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	2-point BVA	BC-023 (Below): Minimum Phase 1 withdrawal rate (%)	Executed	Pass
TC-BVA-056	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	2-point BVA	BC-023 (At): Minimum Phase 1 withdrawal rate (%)	Executed	Pass
TC-BVA-057	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	2-point BVA	BC-024 (At): Maximum Phase 1 withdrawal rate (%)	Executed	Pass
TC-BVA-058	Safe Withdrawal Rate Calculator	Phase 1 annual withdrawal %	2-point BVA	BC-024 (Above): Maximum Phase 1 withdrawal rate (%)	Executed	Pass
TC-BVA-059	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	2-point BVA	BC-025 (Below): Minimum Phase 2 withdrawal rate (%)	Executed	Pass
TC-BVA-060	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	2-point BVA	BC-025 (At): Minimum Phase 2 withdrawal rate (%)	Executed	Pass
TC-BVA-061	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	2-point BVA	BC-026 (At): Maximum Phase 2 withdrawal rate (%)	Executed	Pass
TC-BVA-062	Safe Withdrawal Rate Calculator	Phase 2 annual withdrawal %	2-point BVA	BC-026 (Above): Maximum Phase 2 withdrawal rate (%)	Executed	Pass
TC-BVA-063	Backtest Optimizer	Starting Value	2-point BVA	BC-027 (Below): Minimum starting value	Executed	Pass
TC-BVA-064	Backtest Optimizer	Starting Value	2-point BVA	BC-027 (At): Minimum starting value	Executed	Pass
TC-BVA-065	Backtest Optimizer	Rolling Window	2-point BVA	BC-028 (Below): Minimum rolling window (months)	Executed	Pass
TC-BVA-066	Backtest Optimizer	Rolling Window	2-point BVA	BC-028 (At): Minimum rolling window (months)	Executed	Pass
TC-BVA-067	Backtest Optimizer	Max holdings	2-point BVA	BC-029 (Below): Minimum max holdings	Executed	Pass
TC-BVA-068	Backtest Optimizer	Max holdings	2-point BVA	BC-029 (At): Minimum max holdings	Executed	Pass
TC-BVA-069	Backtest Optimizer	Min weight to include (%)	2-point BVA	BC-030 (Below): Minimum value of min weight (%)	Executed	Pass
TC-BVA-070	Backtest Optimizer	Min weight to include (%)	2-point BVA	BC-030 (At): Minimum value of min weight (%)	Executed	Pass
TC-BVA-071	Backtest Optimizer	Min weight to include (%)	2-point BVA	BC-031 (At): Maximum value of min weight (%)	Executed	Pass
TC-BVA-072	Backtest Optimizer	Min weight to include (%)	2-point BVA	BC-031 (Above): Maximum value of min weight (%)	Executed	Pass
TC-BVA-073	Backtest Optimizer	Max vol (%)	2-point BVA	BC-032 (Below): Minimum value of max vol (%)	Executed	Pass
TC-BVA-074	Backtest Optimizer	Max vol (%)	2-point BVA	BC-032 (At): Minimum value of max vol (%)	Executed	Pass
TC-BVA-075	Backtest Optimizer	Tickers	2-point BVA	BC-033 (Below): Minimum number of tickers	Executed	Pass

TC ID	Feature / Form	Input Field	Technique	Class / Boundary Description	Status	Result
TC-BVA-076	Backtest Optimizer	Tickers	2-point BVA	BC-033 (At): Minimum number of tickers	Executed	**Pass**
TC-BVA-077	Efficient Frontier	Tickers	2-point BVA	BC-034 (Below): Minimum number of tickers	Executed	**Pass**
TC-BVA-078	Efficient Frontier	Tickers	2-point BVA	BC-034 (At): Minimum number of tickers	Executed	**Pass**
TC-BVA-079	Efficient Frontier	Tickers	2-point BVA	BC-035 (At): Free-plan maximum number of tickers	Executed	**Pass**
TC-BVA-080	Efficient Frontier	Tickers	2-point BVA	BC-035 (Above): Free-plan maximum number of tickers	Executed	**Pass**
TC-BVA-081	Efficient Frontier	Max holdings	2-point BVA	BC-036 (Below): Minimum max holdings	Executed	**Pass**
TC-BVA-082	Efficient Frontier	Max holdings	2-point BVA	BC-036 (At): Minimum max holdings	Executed	**Pass**
TC-BVA-083	Efficient Frontier	Min weight to include (%)	2-point BVA	BC-037 (Below): Minimum value of min weight (%)	Executed	**Pass**
TC-BVA-084	Efficient Frontier	Min weight to include (%)	2-point BVA	BC-037 (At): Minimum value of min weight (%)	Executed	**Pass**
TC-BVA-085	Efficient Frontier	Min weight to include (%)	2-point BVA	BC-038 (At): Maximum value of min weight (%)	Executed	**Pass**
TC-BVA-086	Efficient Frontier	Min weight to include (%)	2-point BVA	BC-038 (Above): Maximum value of min weight (%)	Executed	**Pass**
TC-BVA-087	Contact Form	Name	2-point BVA	BC-039 (At): Maximum name length	Executed	**Pass**
TC-BVA-088	Contact Form	Name	2-point BVA	BC-039 (Above): Maximum name length	Executed	**Pass**
TC-BVA-089	Contact Form	Subject	2-point BVA	BC-040 (At): Maximum subject length	Executed	**Pass**
TC-BVA-090	Contact Form	Subject	2-point BVA	BC-040 (Above): Maximum subject length	Executed	**Pass**
TC-BVA-091	Contact Form	Message	2-point BVA	BC-041 (At): Maximum message length	Executed	**Pass**
TC-BVA-092	Contact Form	Message	2-point BVA	BC-041 (Above): Maximum message length	Executed	**Pass**
TC-BVA-093	Lump Sum vs DCA	Starting Value	2-point BVA	BC-042 (Below): Minimum starting value	Executed	**Blocked**
TC-BVA-094	Lump Sum vs DCA	Starting Value	2-point BVA	BC-042 (At): Minimum starting value	Executed	**Blocked**
TC-BVA-095	Lump Sum vs DCA	DCA horizon (months)	2-point BVA	BC-043 (Below): Minimum DCA horizon (months)	Executed	**Blocked**
TC-BVA-096	Lump Sum vs DCA	DCA horizon (months)	2-point BVA	BC-043 (At): Minimum DCA horizon (months)	Executed	**Blocked**
TC-BVA-097	Lump Sum vs DCA	DCA horizon (months)	2-point BVA	BC-044 (At): Maximum DCA horizon (months)	Executed	**Blocked**
TC-BVA-098	Lump Sum vs DCA	DCA horizon (months)	2-point BVA	BC-044 (Above): Maximum DCA horizon (months)	Executed	**Blocked**
TC-BVA-099	Time Value of Money Calculator	Starting Value	2-point BVA	BC-045 (Below): Minimum starting value	Executed	**Pass**
TC-BVA-100	Time Value of Money Calculator	Starting Value	2-point BVA	BC-045 (At): Minimum starting value	Executed	**Pass**
TC-BVA-101	Time Value of Money Calculator	Starting Value	2-point BVA	BC-046 (At): Maximum starting value	Executed	**Pass**
TC-BVA-102	Time Value of Money Calculator	Starting Value	2-point BVA	BC-046 (Above): Maximum starting value	Executed	**Pass**
TC-BVA-103	Time Value of Money Calculator	Monthly contribution	2-point BVA	BC-047 (Below): Minimum monthly contribution	Executed	**Pass**
TC-BVA-104	Time Value of Money Calculator	Monthly contribution	2-point BVA	BC-047 (At): Minimum monthly contribution	Executed	**Pass**
TC-BVA-105	Time Value of Money Calculator	Monthly contribution	2-point BVA	BC-048 (At): Maximum monthly contribution	Executed	**Pass**
TC-BVA-106	Time Value of Money Calculator	Monthly contribution	2-point BVA	BC-048 (Above): Maximum monthly contribution	Executed	**Pass**

D5 - Execution Log [20 marks]

Screenshots (roll-number note visible) for every executed test case. The verdict, actual result and defect observations are in D2/D3; the defect list below groups them.

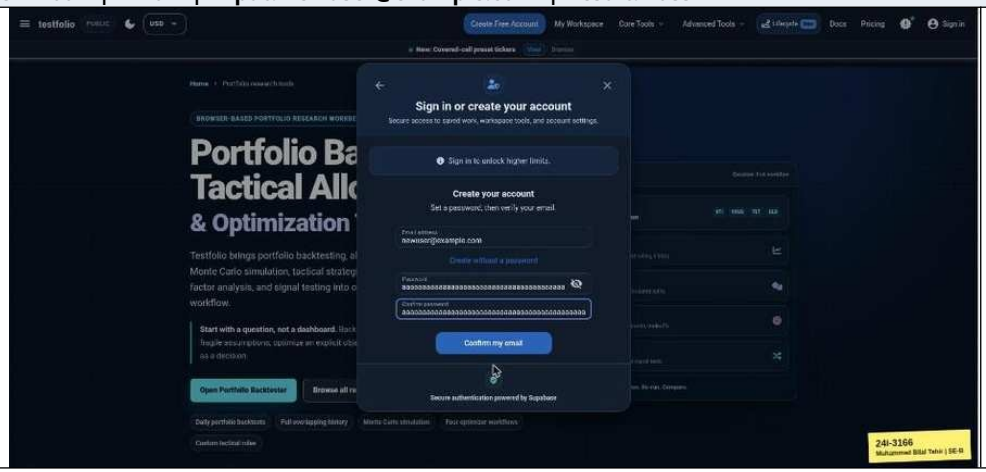
Defect observations

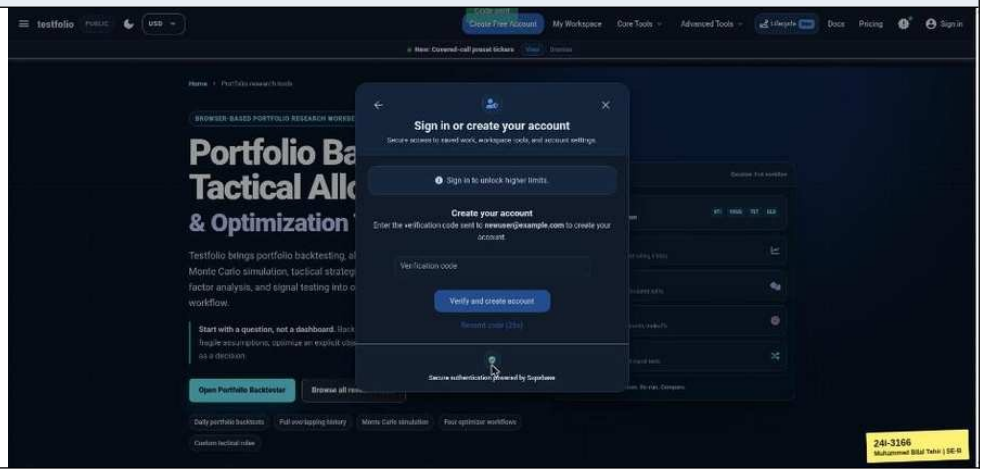
Defect	Form	Observation	Test cases
D-1	Portfolio Backtester	The period reported in the results is one day earlier than the dates entered (entered 2020-01-01 to 2021-01-01, results show 2019-12-31 - 2020-12-31).	TC-EP-017, TC-EP-023, TC-EP-024, TC-EP-S1, TC-BVA-026
D-2	Portfolio Backtester	A start date earlier than the first available data (1990-01-01) or an end date later than the latest data (2030-01-01) is accepted and silently clamped; the user is not told.	TC-EP-021, TC-EP-025
D-3	Portfolio Backtester	A negative allocation (SPY -10 / AGG 110, total 100) is accepted and a back-test runs (short position) with no warning.	TC-EP-058, TC-EP-S5
D-4	Portfolio Backtester	Drag (annual fee %) accepts negative values and values above 100 (e.g. -1, 101, 150) without any validation; 99%-150% produce Ending Value $0.00.	TC-EP-044, TC-EP-045, TC-BVA-027, TC-BVA-032
D-5	Portfolio Backtester	When several fields are invalid at once, the validation reports only some of the problems (first-error-wins): an unknown ticker is not reported when the allocation total is wrong; an empty ticker hides the drag problem.	TC-EP-S3, TC-EP-S4, TC-EP-S6
D-6	Monte Carlo Simulator	Min block below its minimum of 0.05 (0.01 and 0.04) is accepted and the simulation runs, although the field reports a 0.05 minimum.	TC-EP-070, TC-BVA-041
D-7	Monte Carlo Simulator	Invalid random seeds are silently rewritten instead of rejected (-1 becomes 0, 1.5 becomes 2) and the simulation runs without any message.	TC-EP-076, TC-EP-077, TC-BVA-045
D-8	Backtester / Monte Carlo	Intermittent server errors: "Backtest failed" (Starting Value 999,999,998) and "An error occurred while running Monte Carlo" (seed 0) appeared once each and disappeared on repeating the identical input.	TC-BVA-006, TC-BVA-046
D-9	Lump Sum vs DCA / Rebalancing Sensitivity	On the Free plan the premium gate fires before any validation: clicking Analyze with invalid input (unknown ticker, allocation 90, reversed dates, negative drag, DCA horizon 0) redirects to the pricing page, so validation errors are never shown.	TC-EP-139 to TC-EP-146, TC-EP-154 to TC-EP-159, TC-BVA-093 to TC-BVA-098
D-10	Safe Withdrawal Rate Calculator	A horizon above the maximum (60) is silently clamped to 50 while the error toast is shown; the value the user typed is replaced.	TC-EP-082
D-11	Backtest Optimizer	A minimum weight of 100% is accepted with three tickers and returns a single-asset result; the constraint is not flagged as nearly infeasible.	TC-BVA-071
D-12	Contact	Name and Email are not validated: the form submits (toast "Message sent. We will get back to you soon.") and clears with an empty Name, an empty Email or a malformed Email (notanemail). Only Subject (required) and Message (minimum 10 characters) are validated; Message is checked before Name/Email.	TC-EP-127, TC-EP-130, TC-EP-131
D-13	Sign Up	Password rules are not communicated and are inconsistent: passwords below 8 characters silently disable the Confirm button (no message, no hint of the minimum), while a 73-character password is accepted without any maximum-length error.	TC-EP-008, TC-EP-009, TC-BVA-001, TC-BVA-002
D-14	Sign Up / Sign In	Empty or invalid submissions give no feedback: clicking Confirm/Sign in with empty fields does nothing and shows no message; "a@b" passes the email check.	TC-EP-002, TC-EP-004, TC-EP-007, TC-EP-011, TC-EP-015

Screenshots by test case

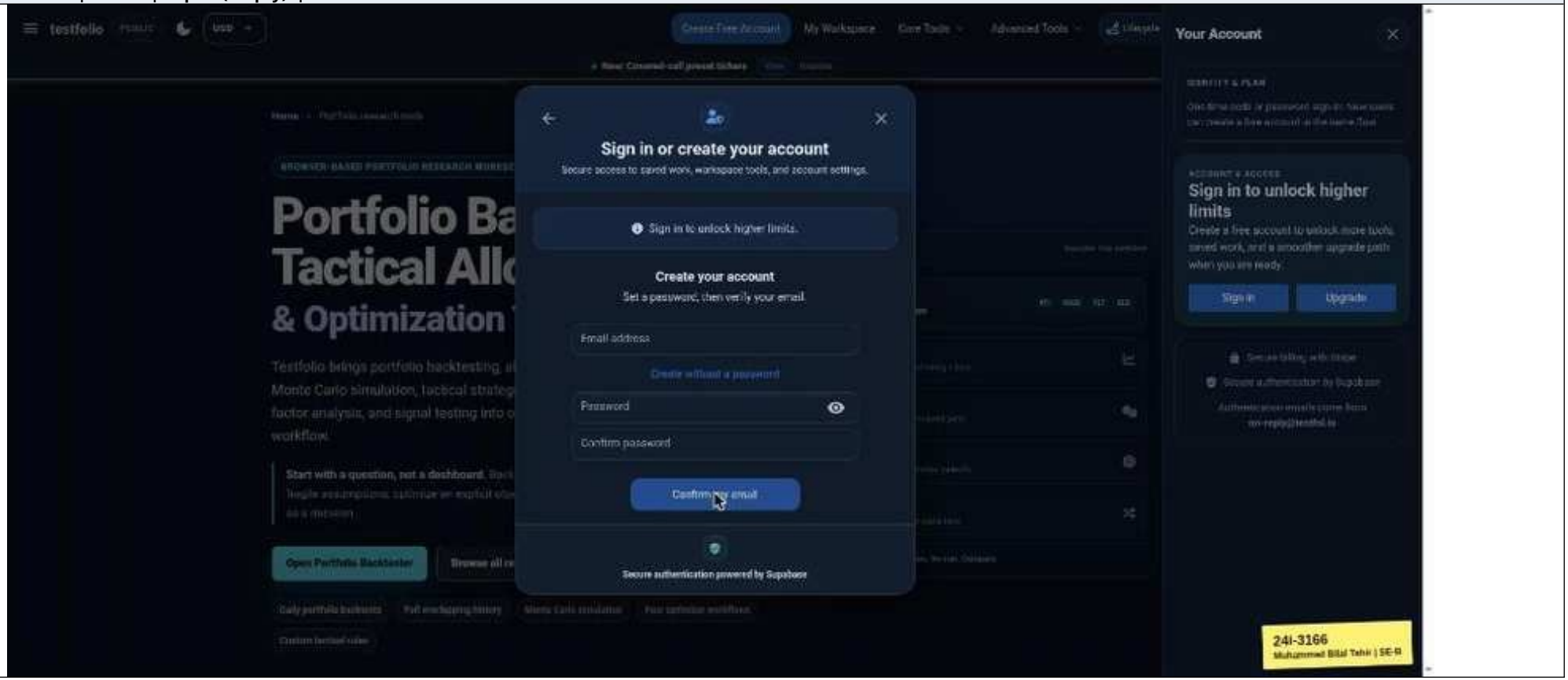
Sign Up (testfol.io/start)

TC-EP-001 | Email | input: newuser@example.com | result: Pass

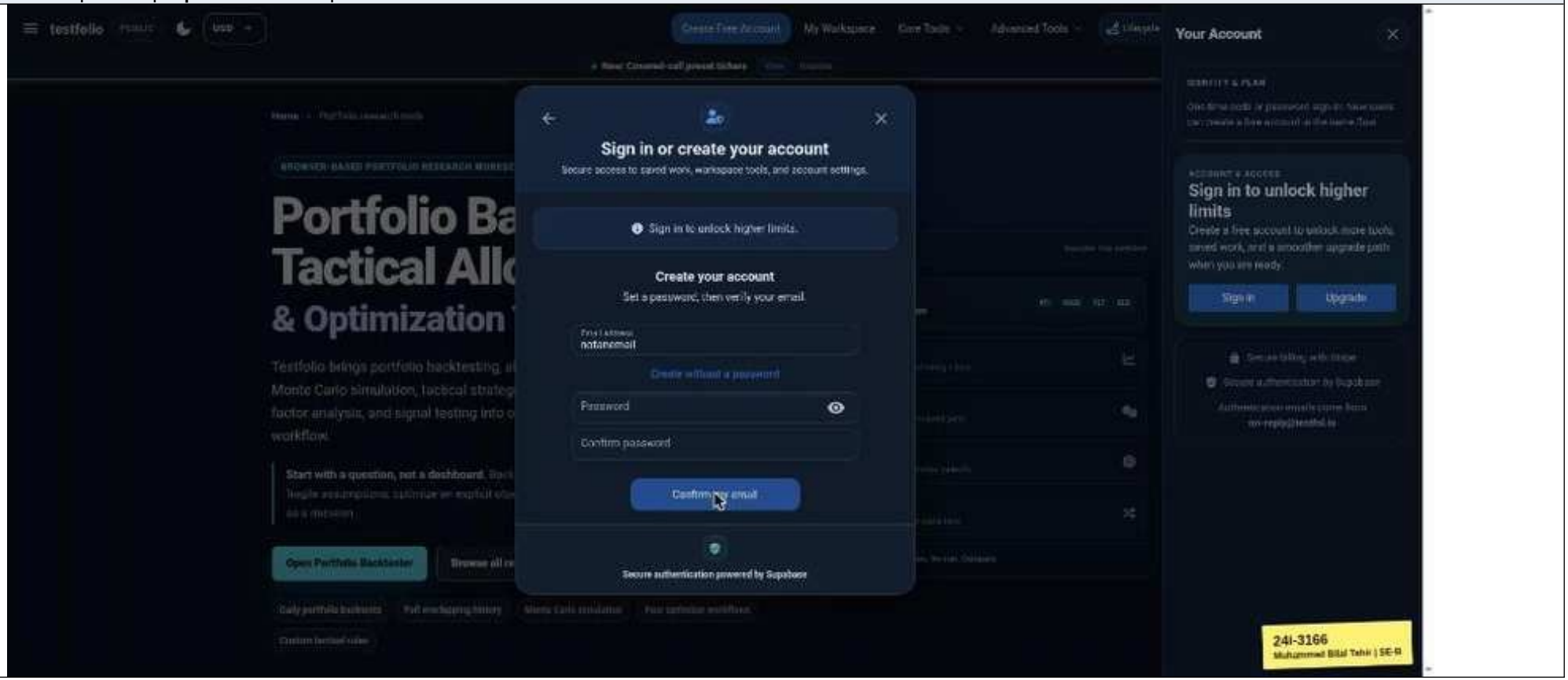




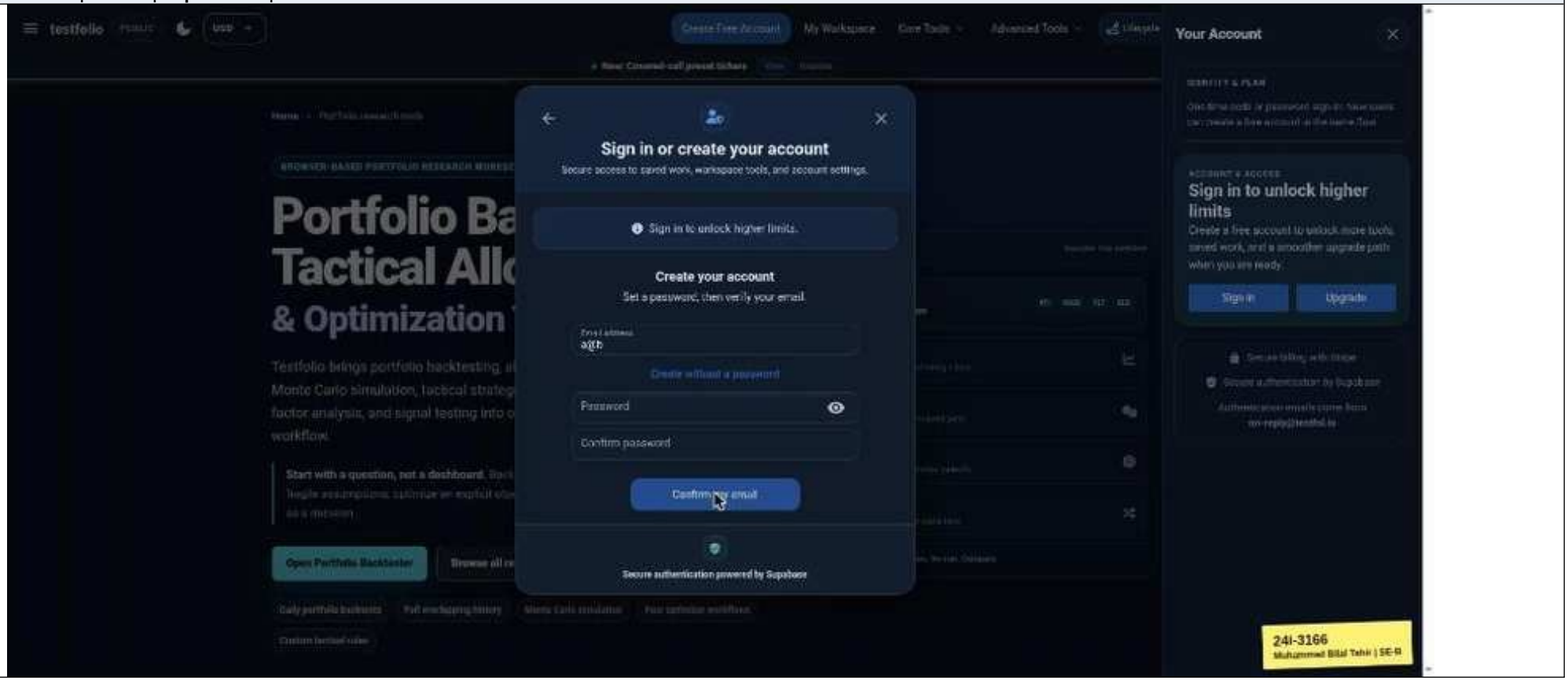
TC-EP-002 | Email | input: (empty) | result: Pass



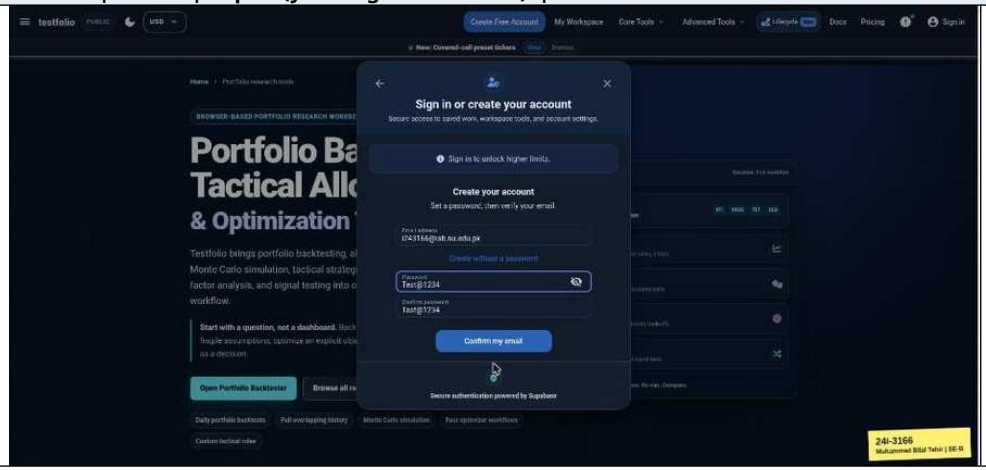
TC-EP-003 | Email | input: notanemail | result: Pass

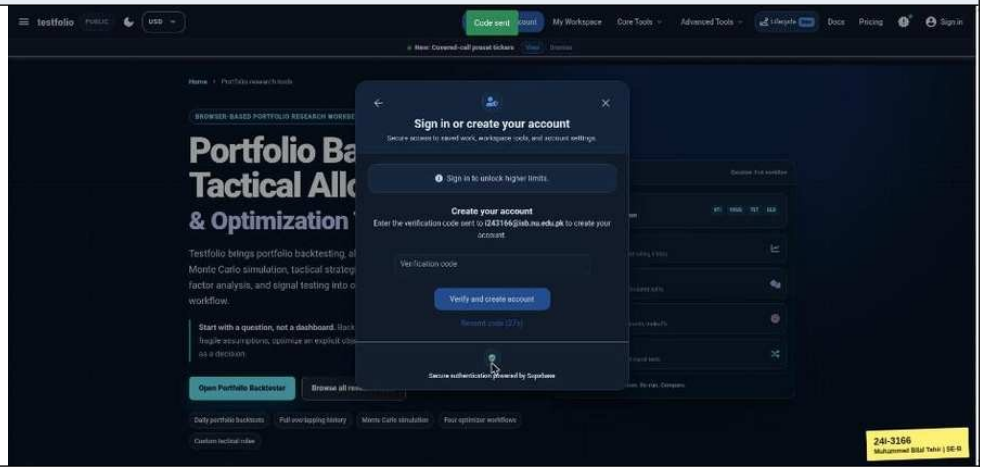


TC-EP-004 | Email | input: a@b | result: Fail

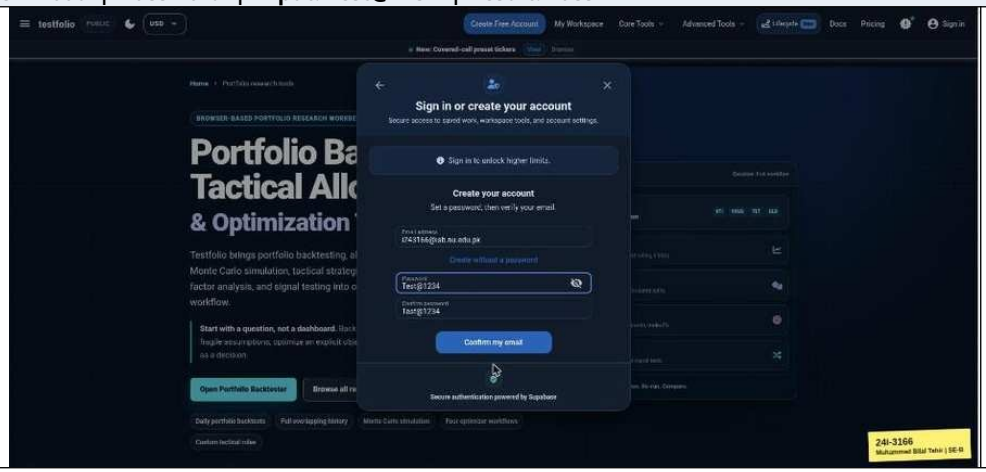


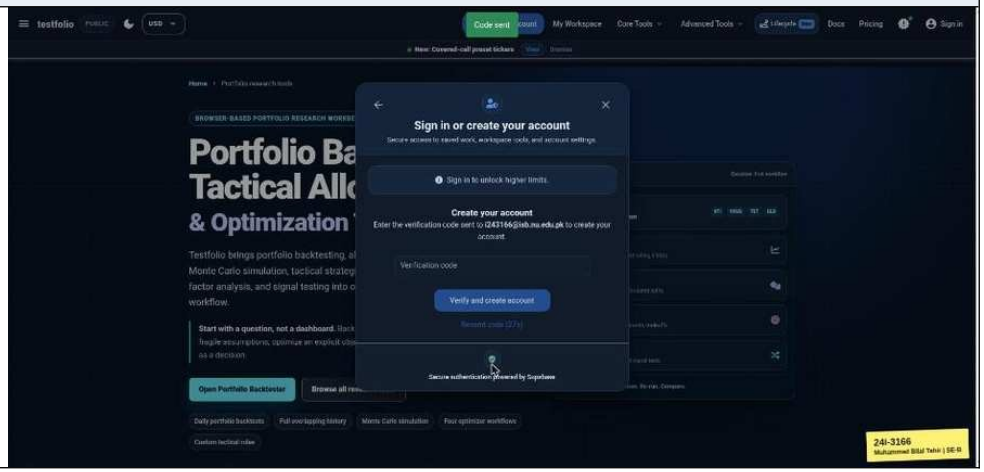
TC-EP-005 | Email | input: (your registered email) | result: Fail



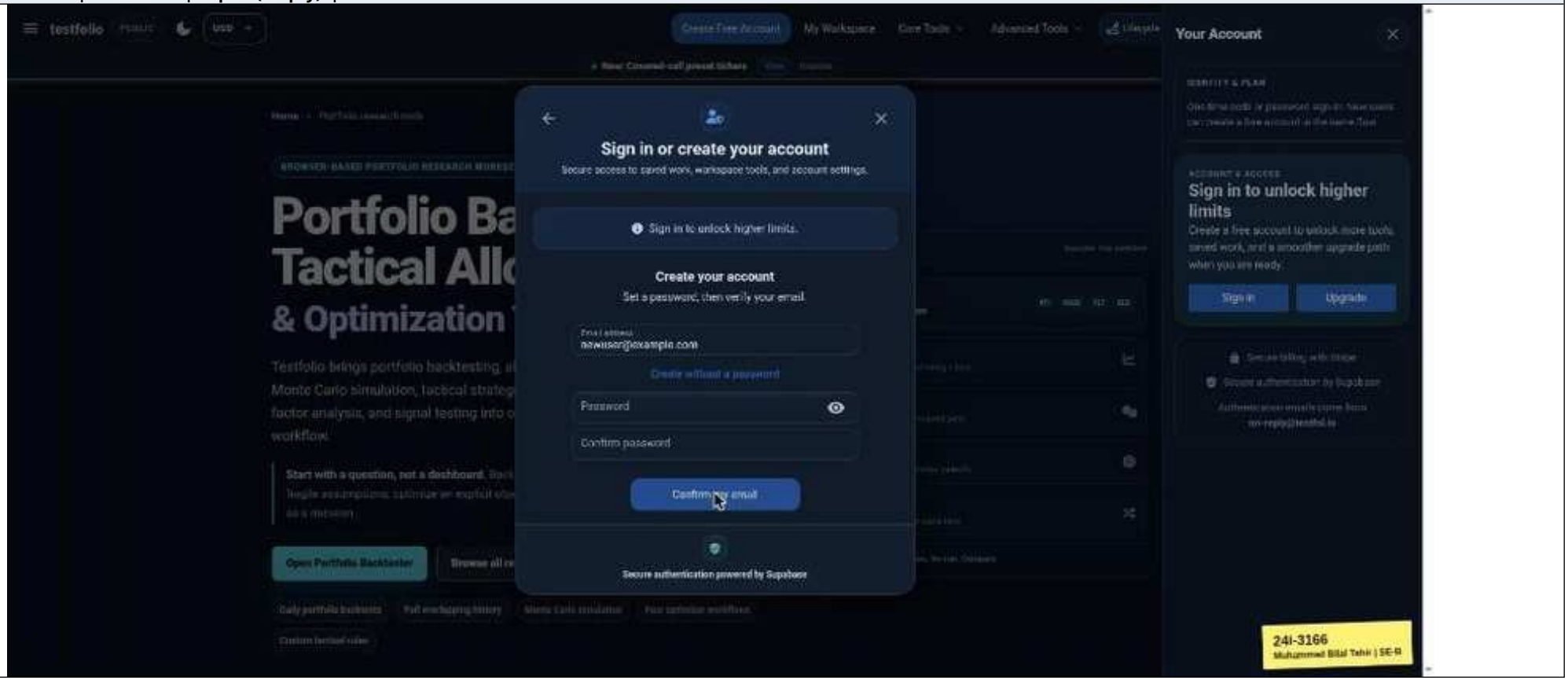


TC-EP-006 | Password | input: Test@1234 | result: Pass

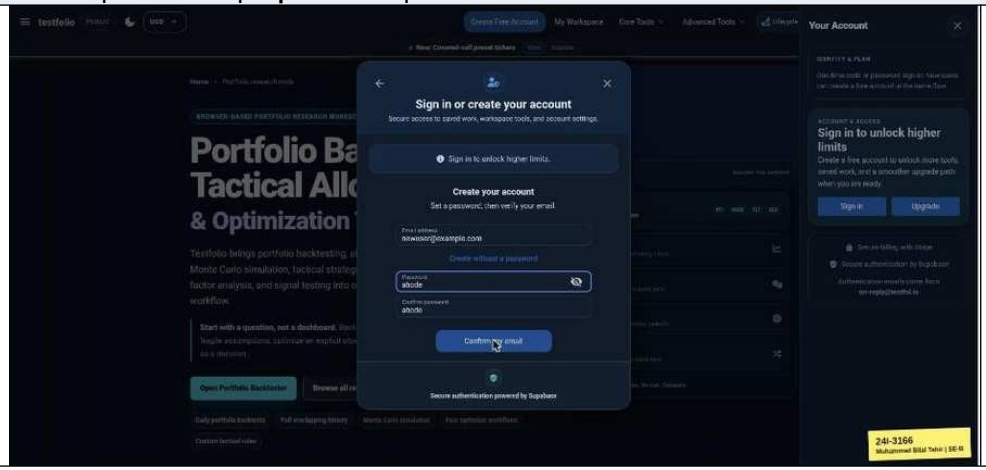


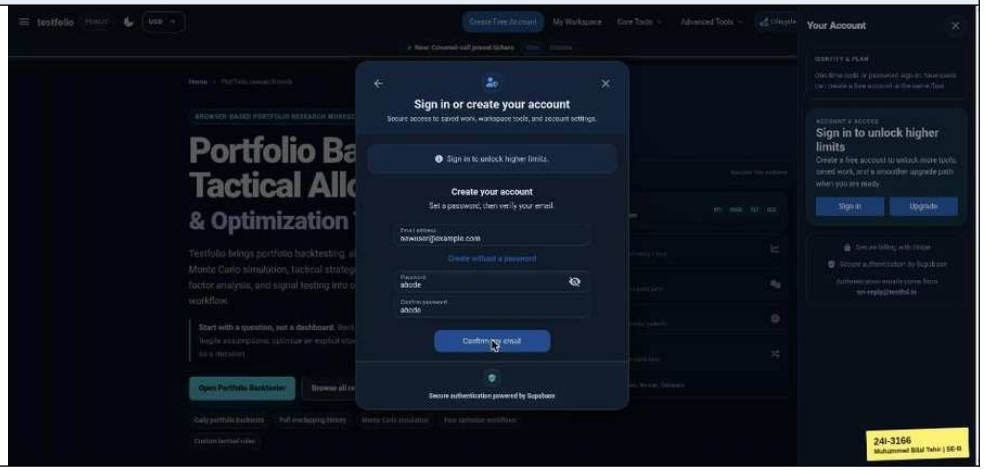


TC-EP-007 | Password | input: (empty) | result: Pass

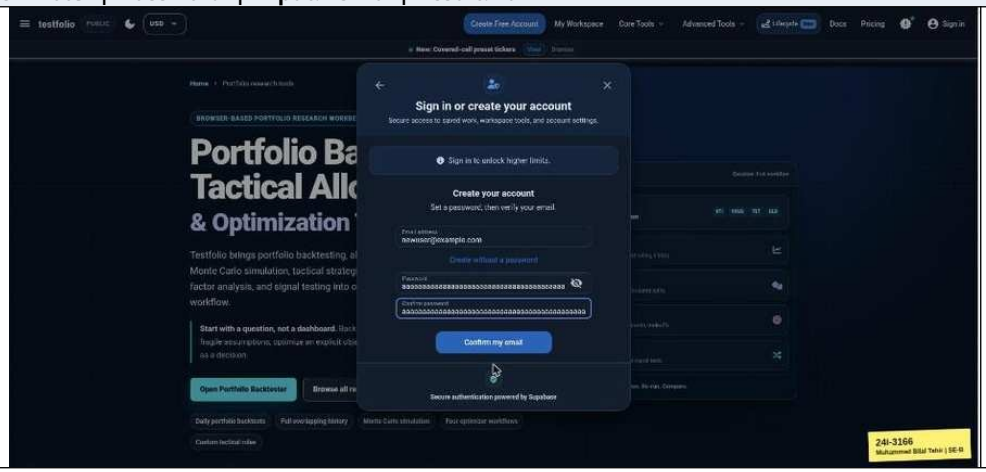


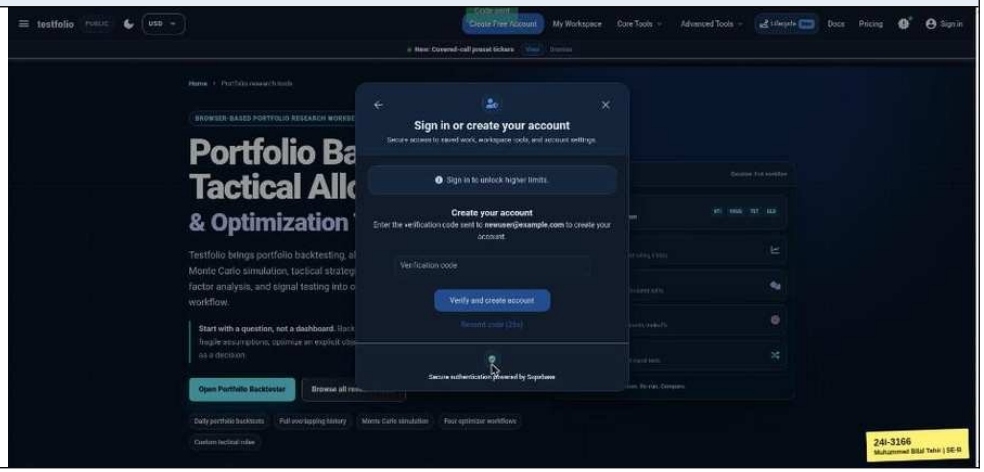
TC-EP-008 | Password | input: abcde | result: Pass





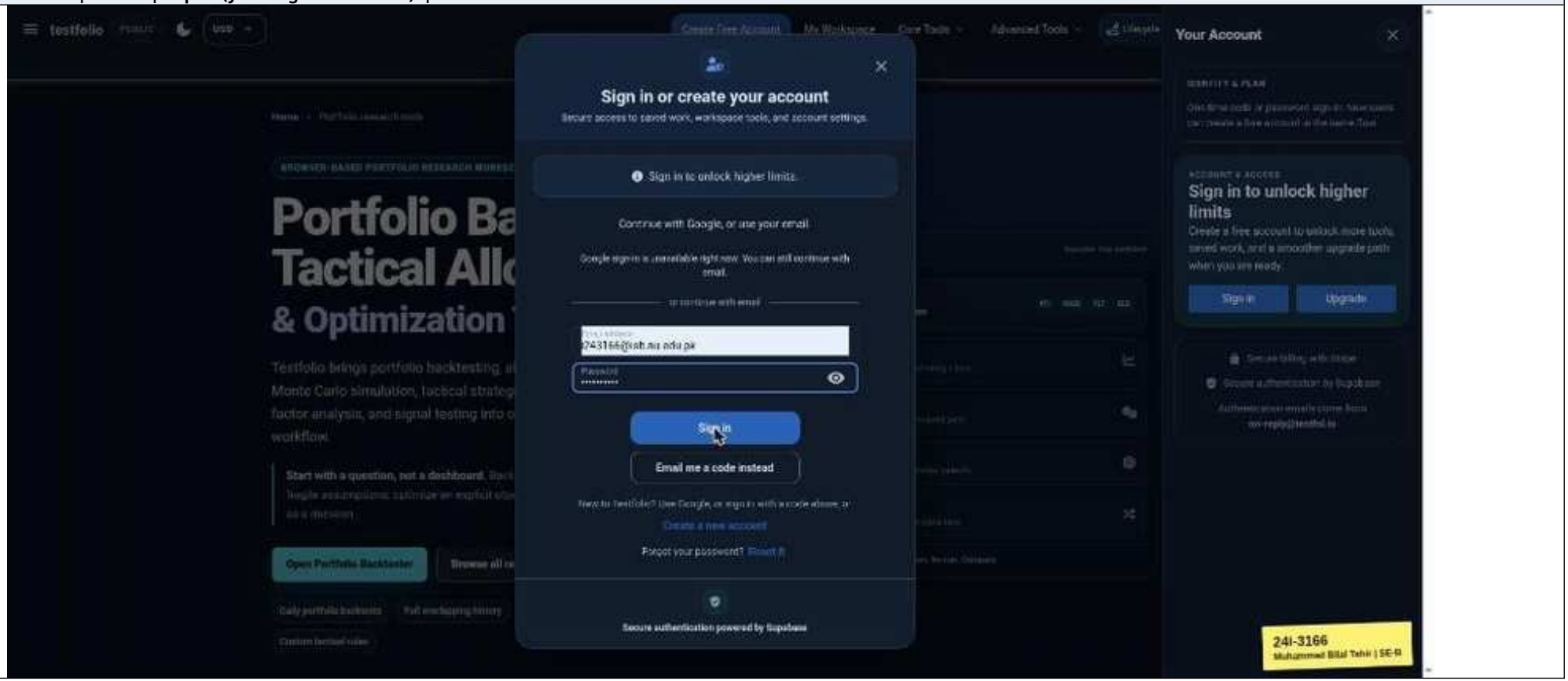
TC-EP-009 | Password | input: 73 x a | result: Fail



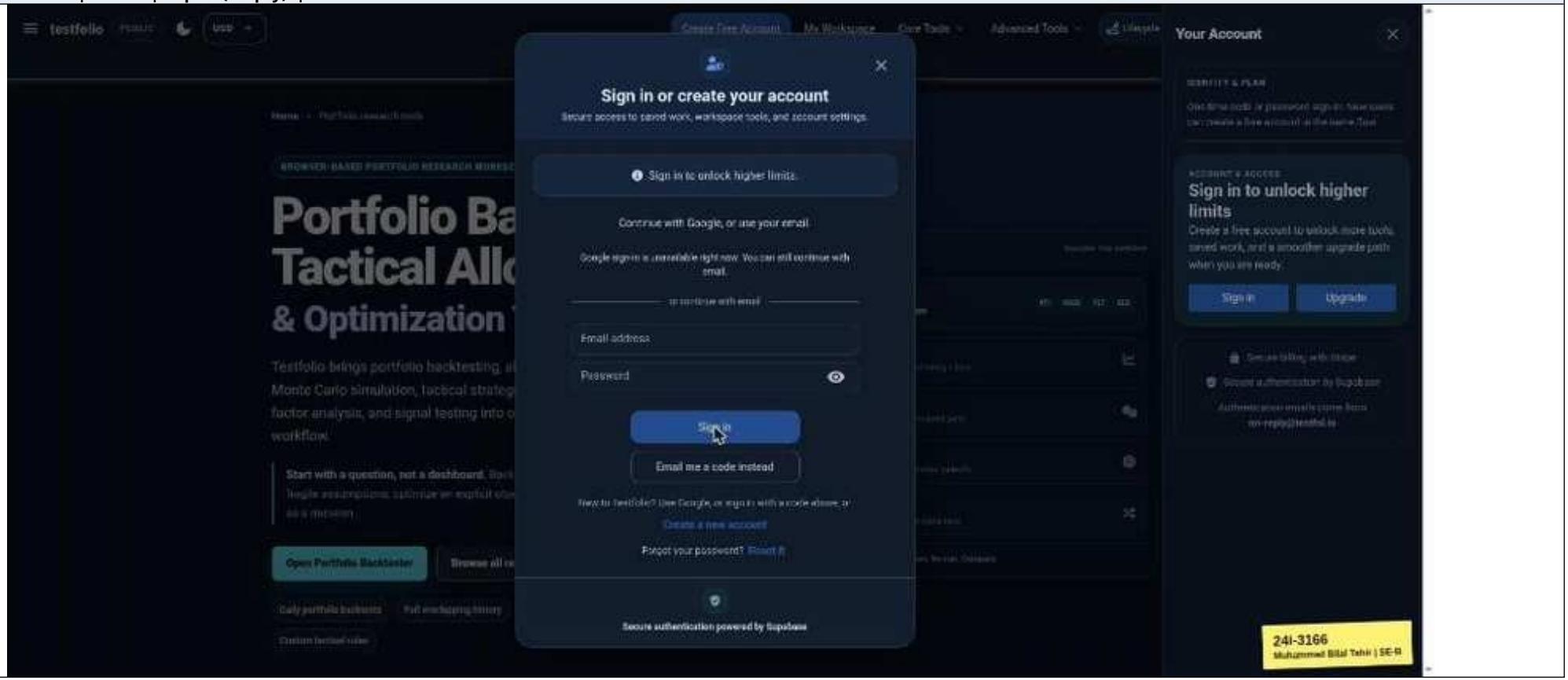


Sign In (testfol.io/login)

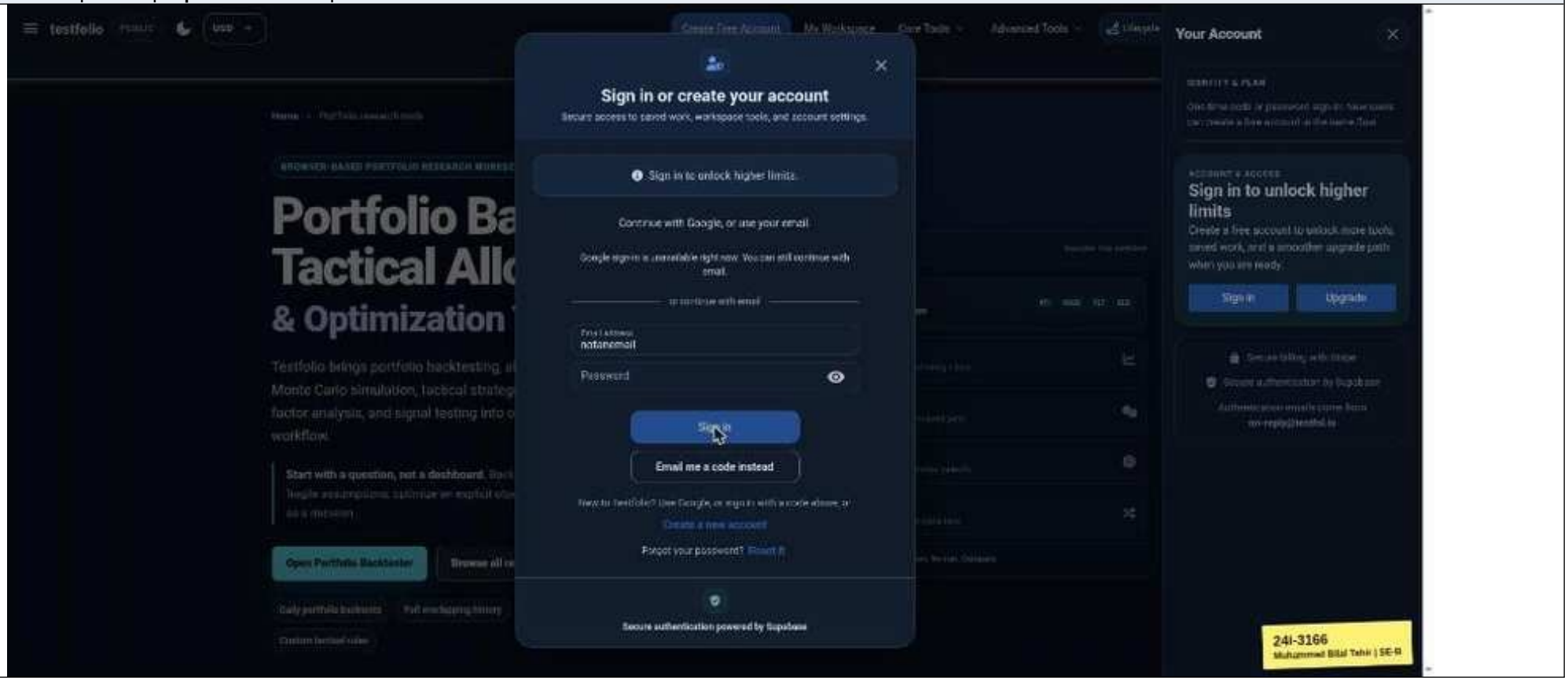
TC-EP-010 | Email | input: (your registered email) | result: Pass



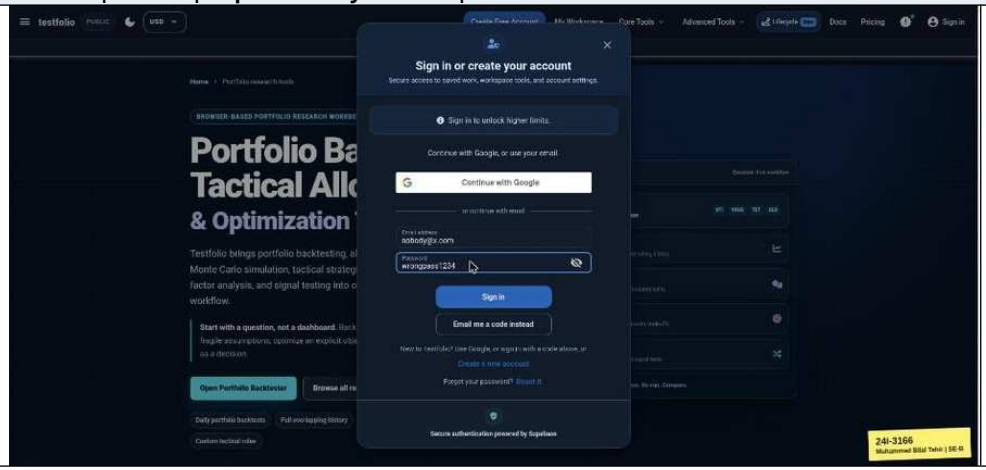
TC-EP-011 | Email | input: (empty) | result: Pass

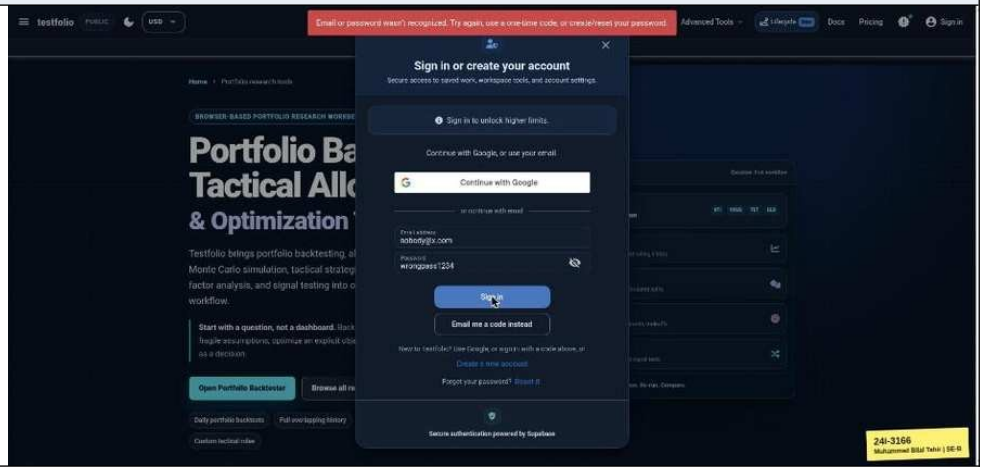


TC-EP-012 | Email | input: notanemail | result: Pass

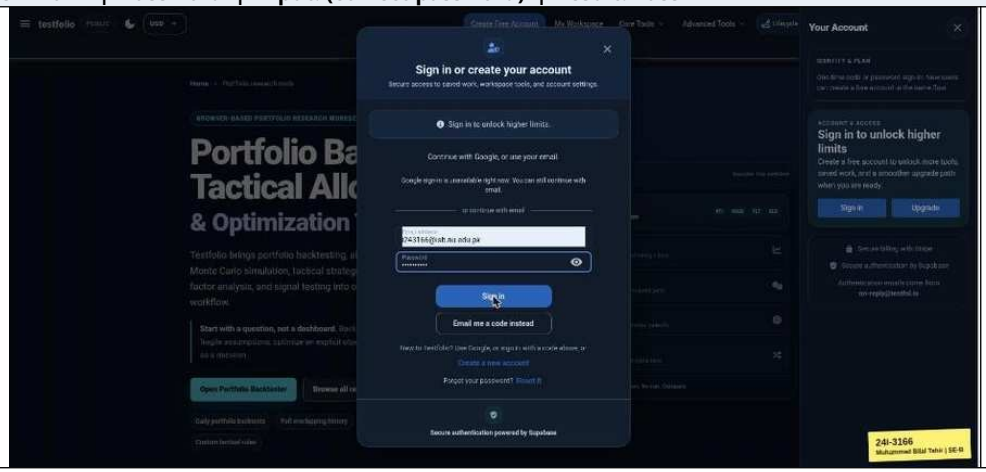


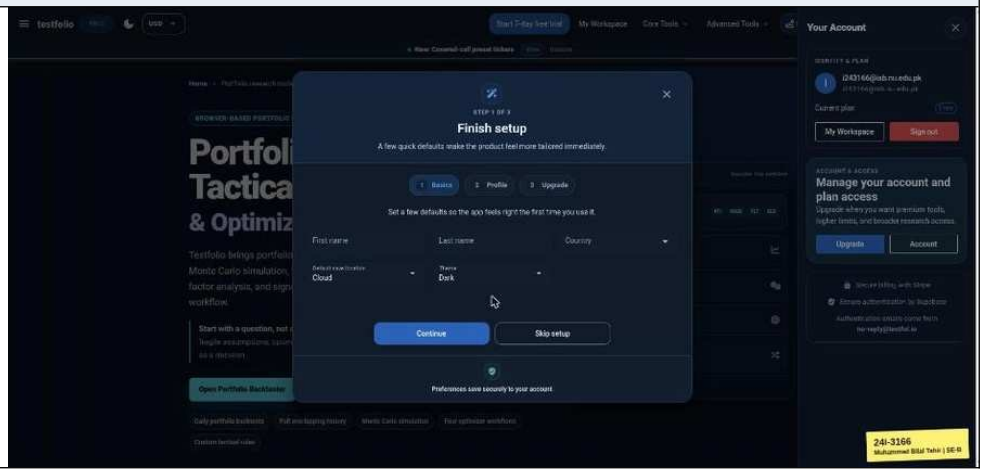
TC-EP-013 | Email | input: nobody@x.com | result: Pass



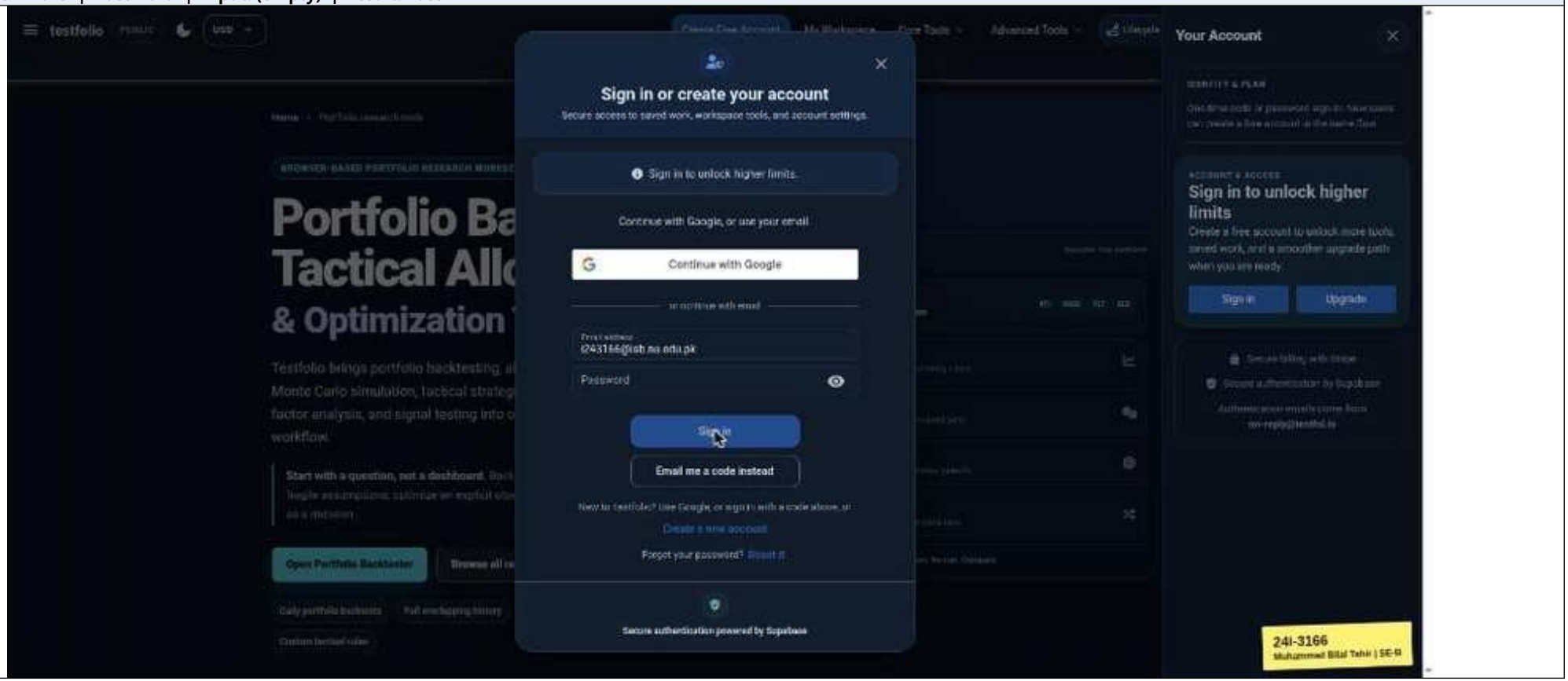


TC-EP-014 | Password | input: (correct password) | result: Pass

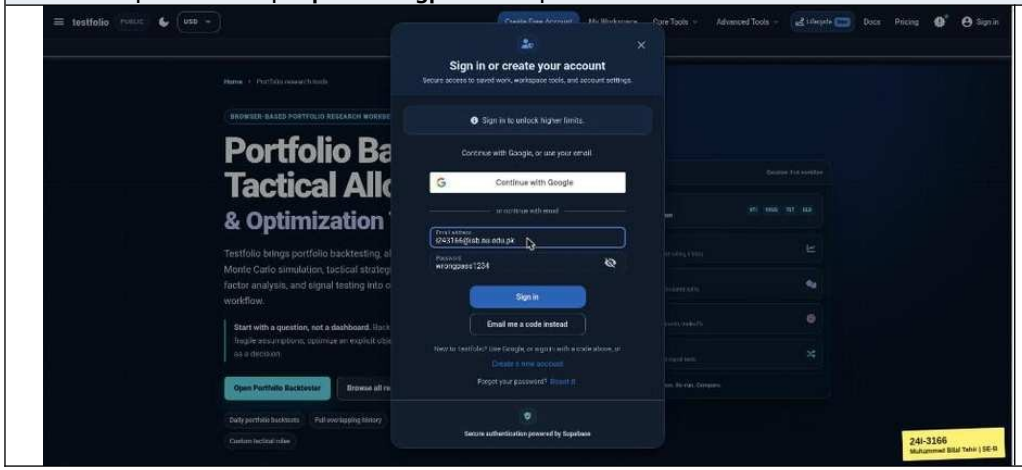


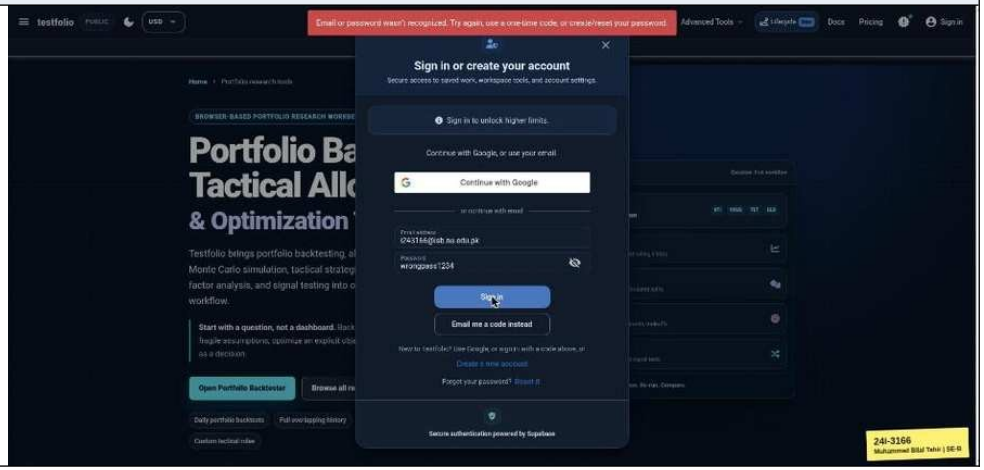


TC-EP-015 | Password | input: (empty) | result: Pass



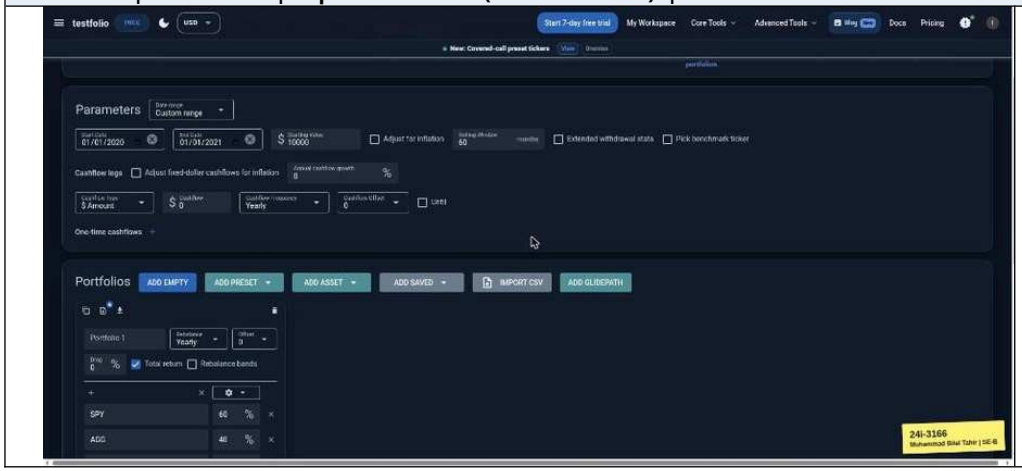
TC-EP-016 | Password | input: wrongpass1234 | result: Pass





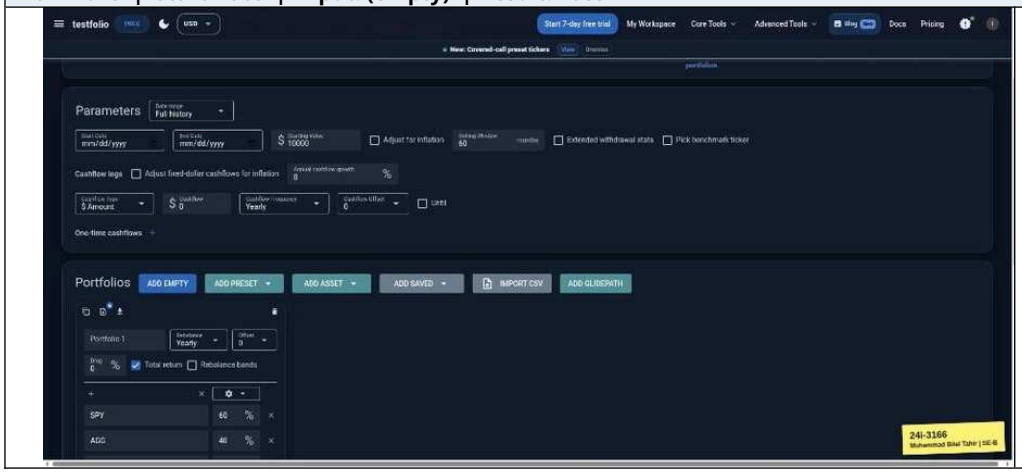
Portfolio Backtester (testfol.io/)

TC-EP-017 | Start Date | input: 2020-01-01 (end 2021-01-01) | result: Pass



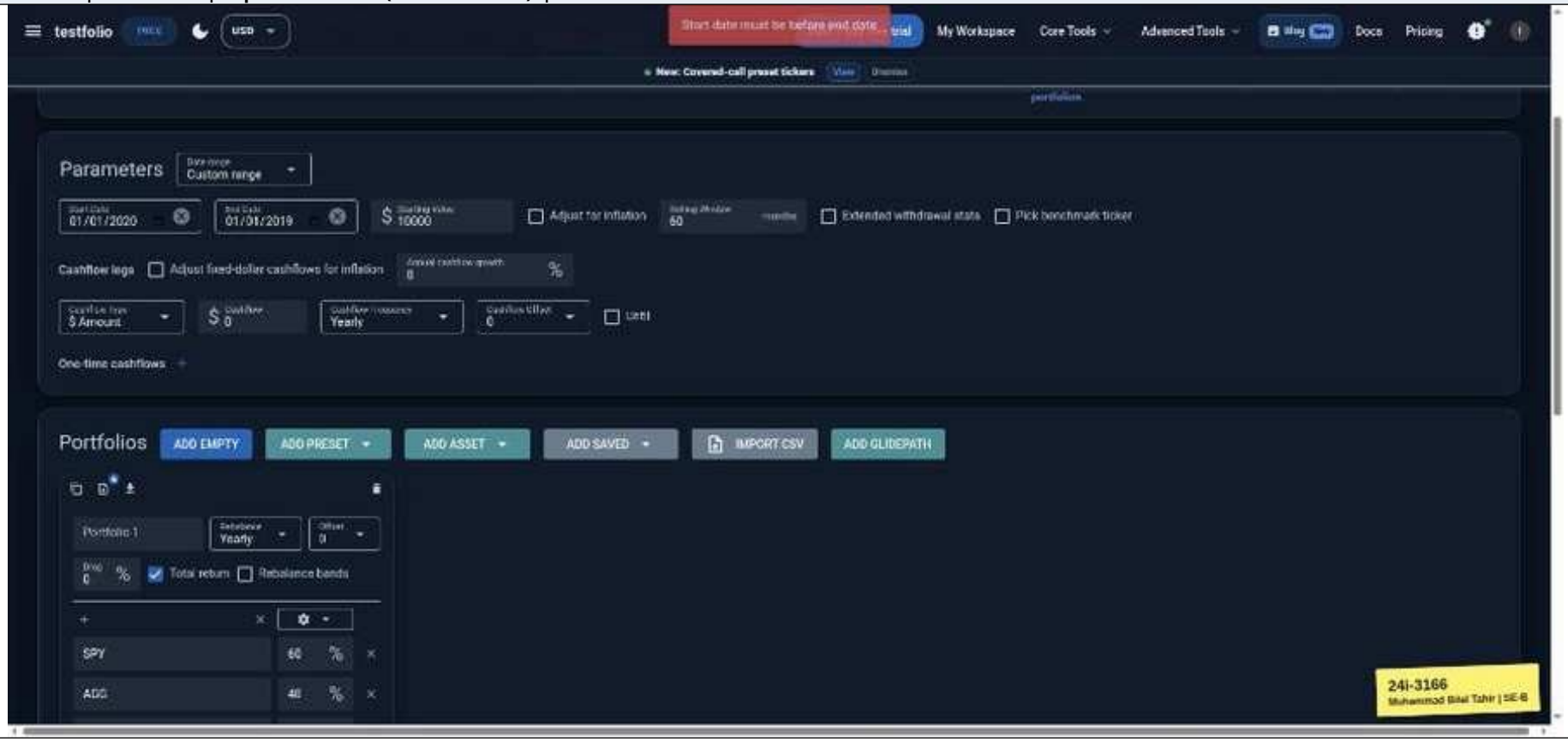


TC-EP-018 | Start Date | input: (empty) | result: Pass

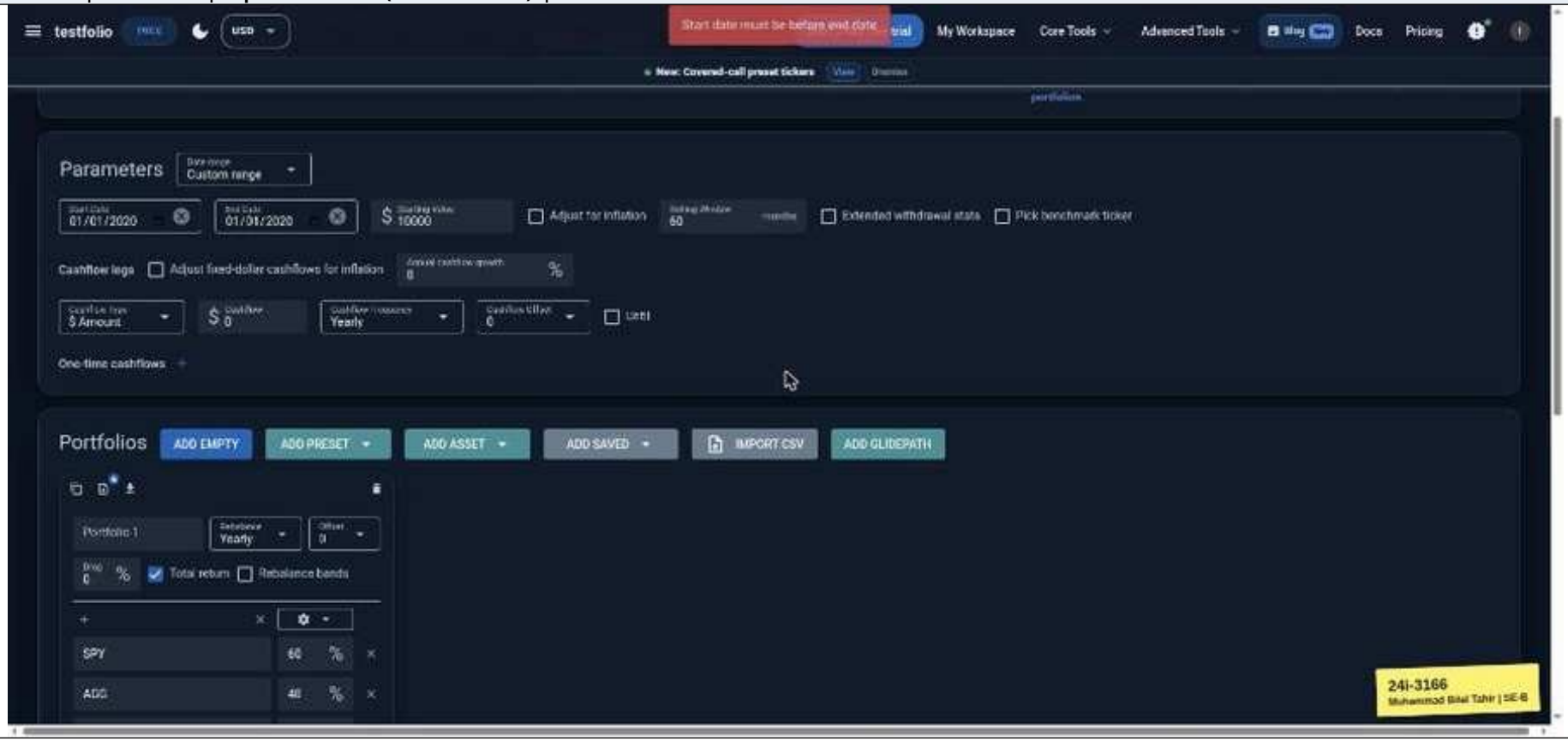




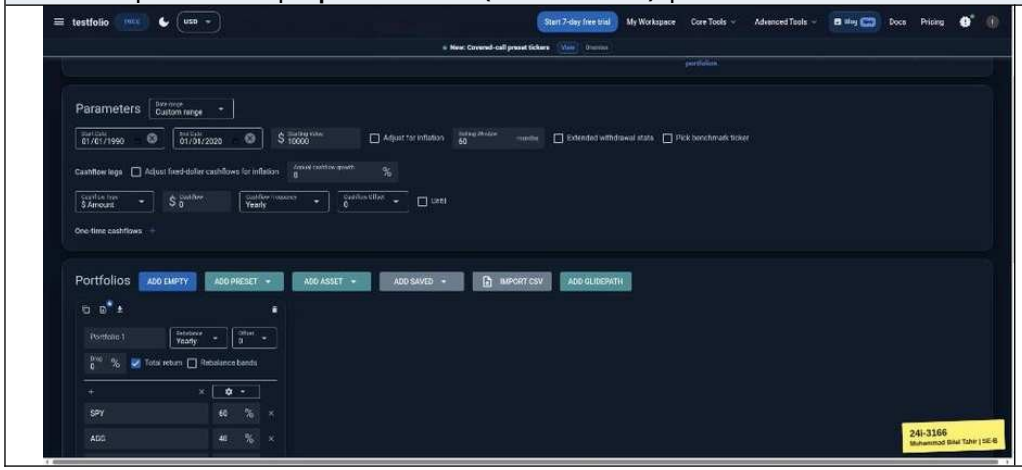
TC-EP-019 | Start Date | input: 2020-01-01 (end 2019-01-01) | result: Pass

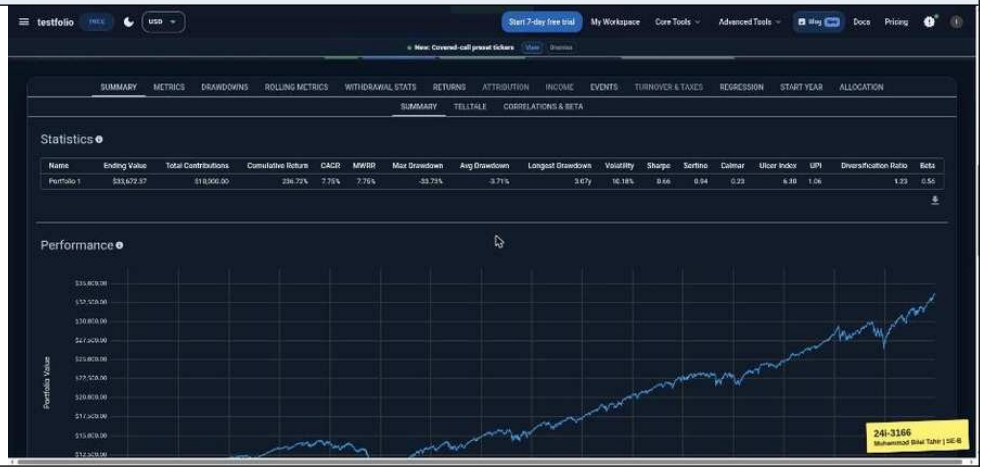


TC-EP-020 | Start Date | input: 2020-01-01 (end 2020-01-01) | result: Pass

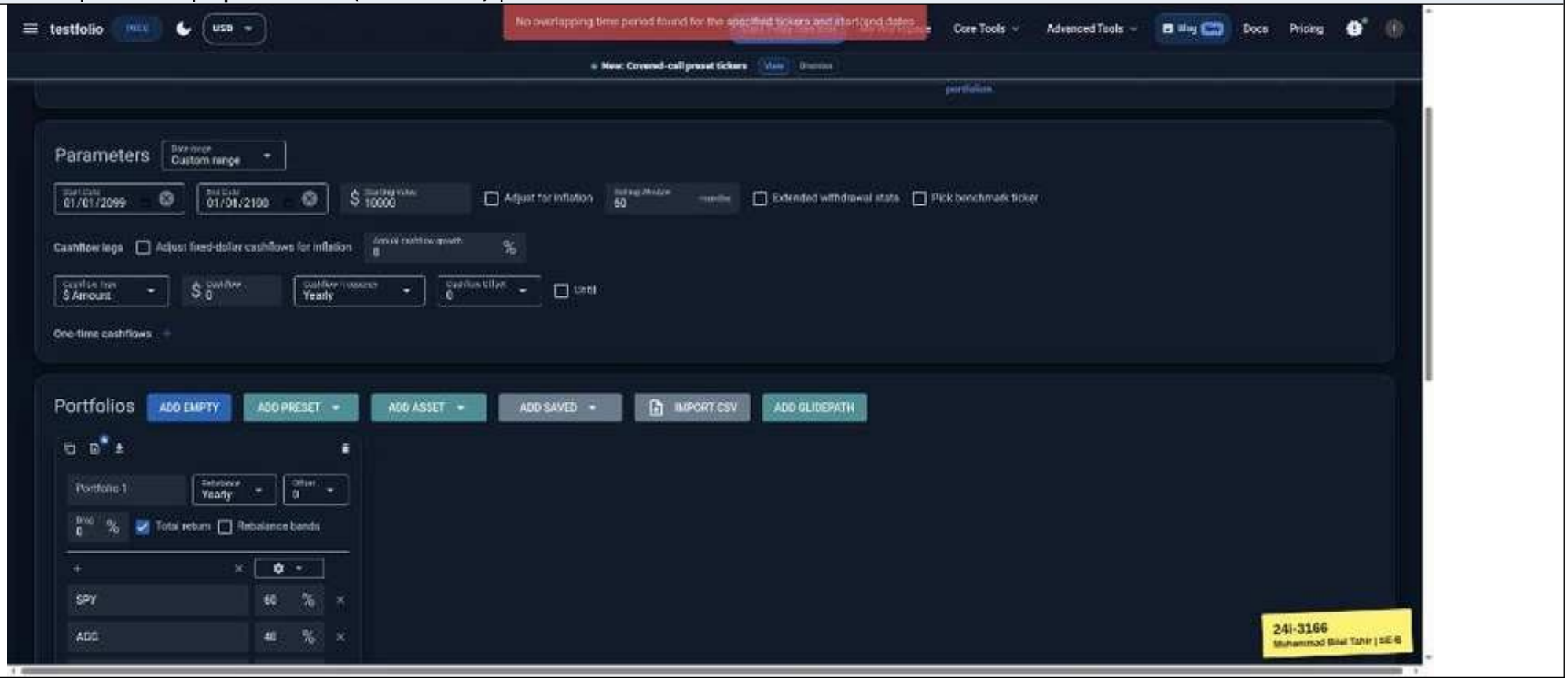


TC-EP-021 | Start Date | input: 1990-01-01 (end 2020-01-01) | result: Fail

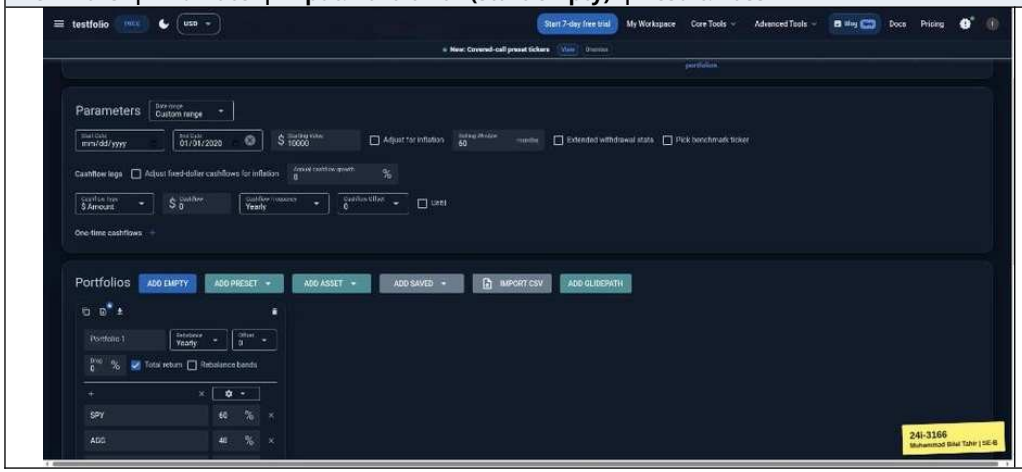


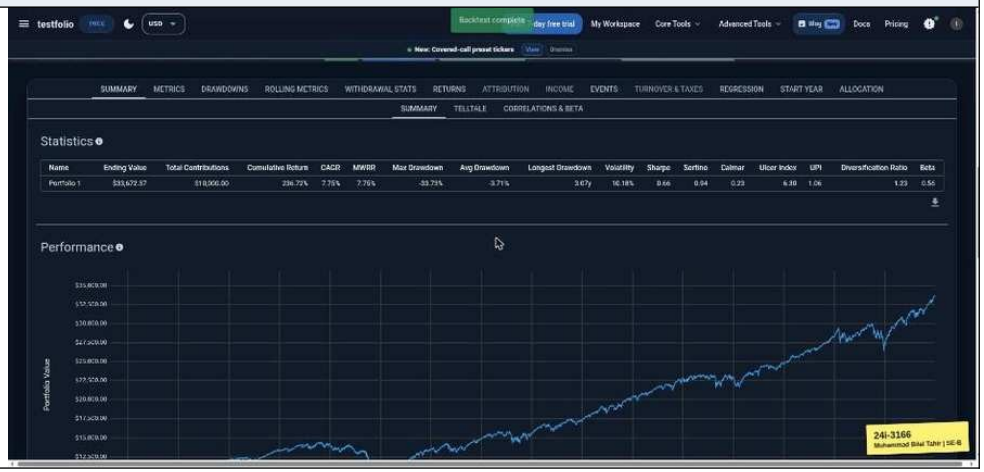


TC-EP-022 | Start Date | input: 2099-01-01 (end 2100-01-01) | result: Pass

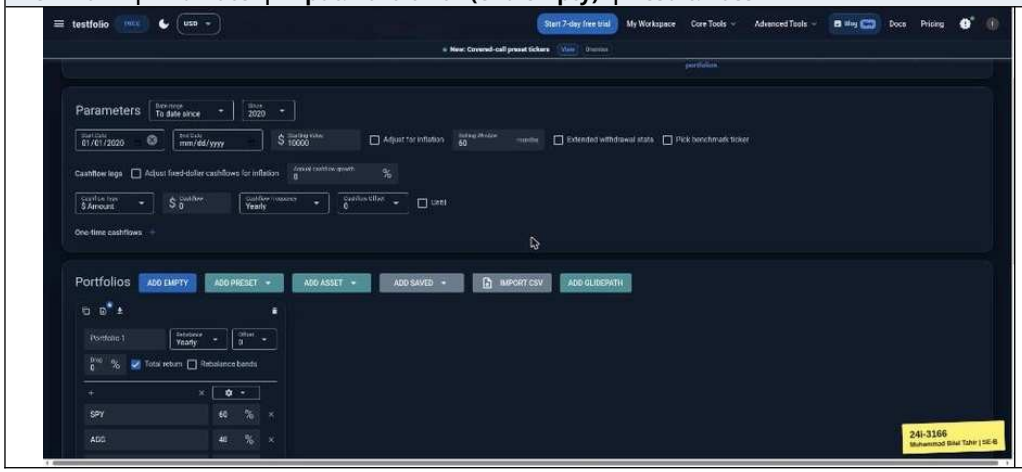


TC-EP-023 | End Date | input: 2020-01-01 (start empty) | result: Pass



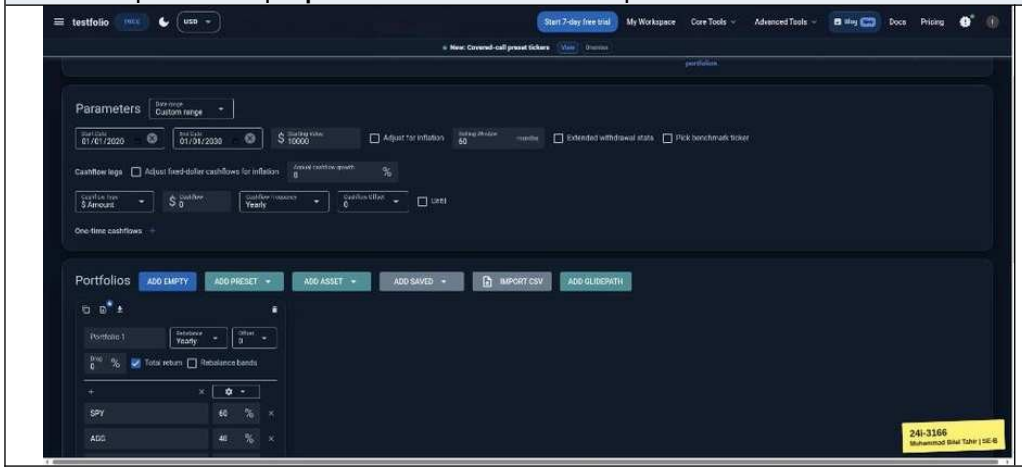


TC-EP-024 | End Date | input: 2020-01-01 (end empty) | result: Pass



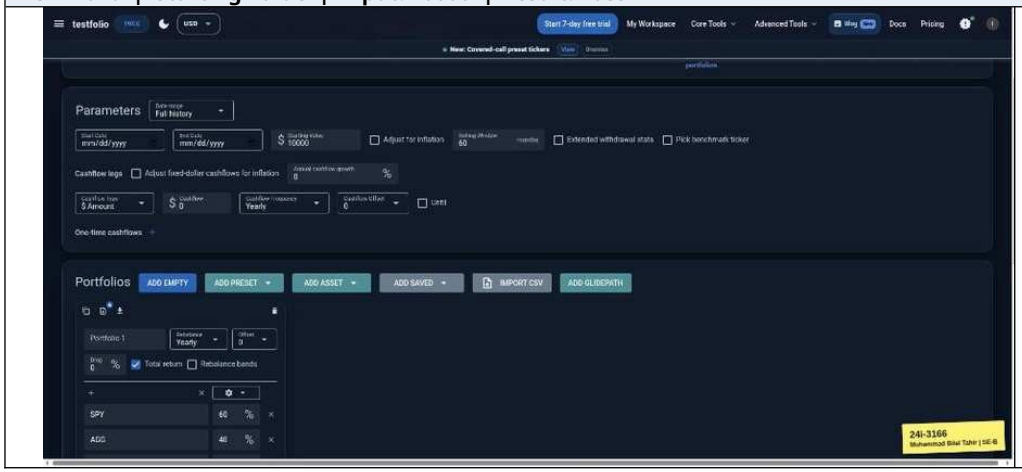


TC-EP-025 | End Date | input: 2020-01-01 to 2030-01-01 | result: Fail



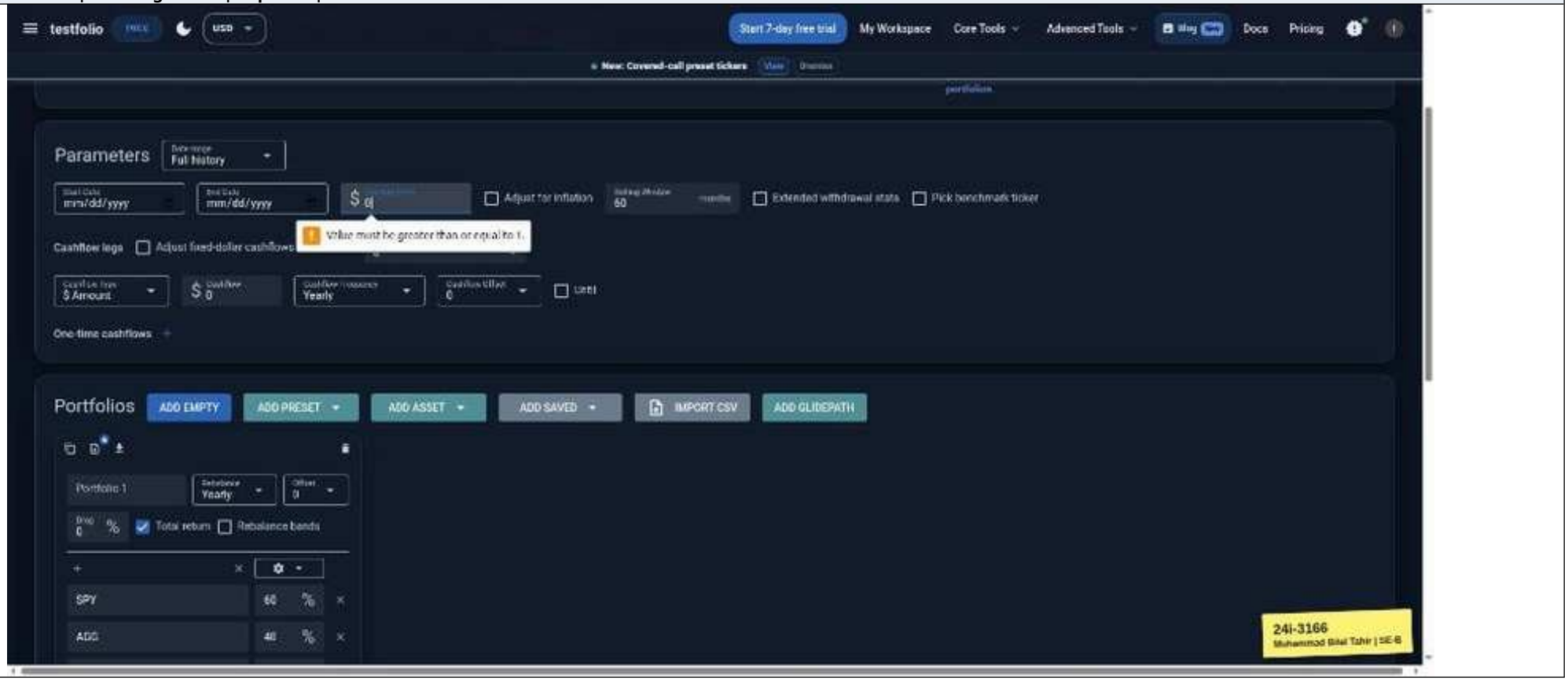


TC-EP-026 | Starting Value | input: 10000 | result: Pass

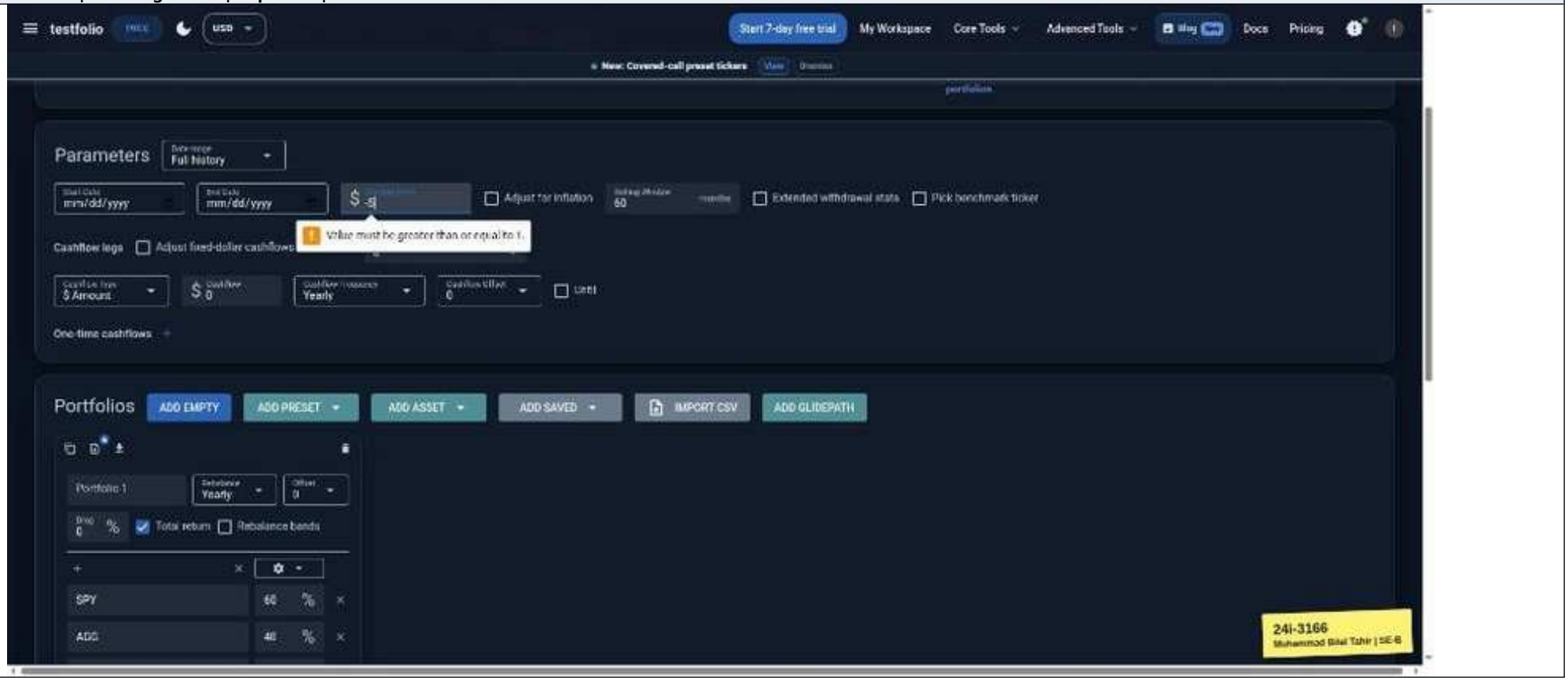




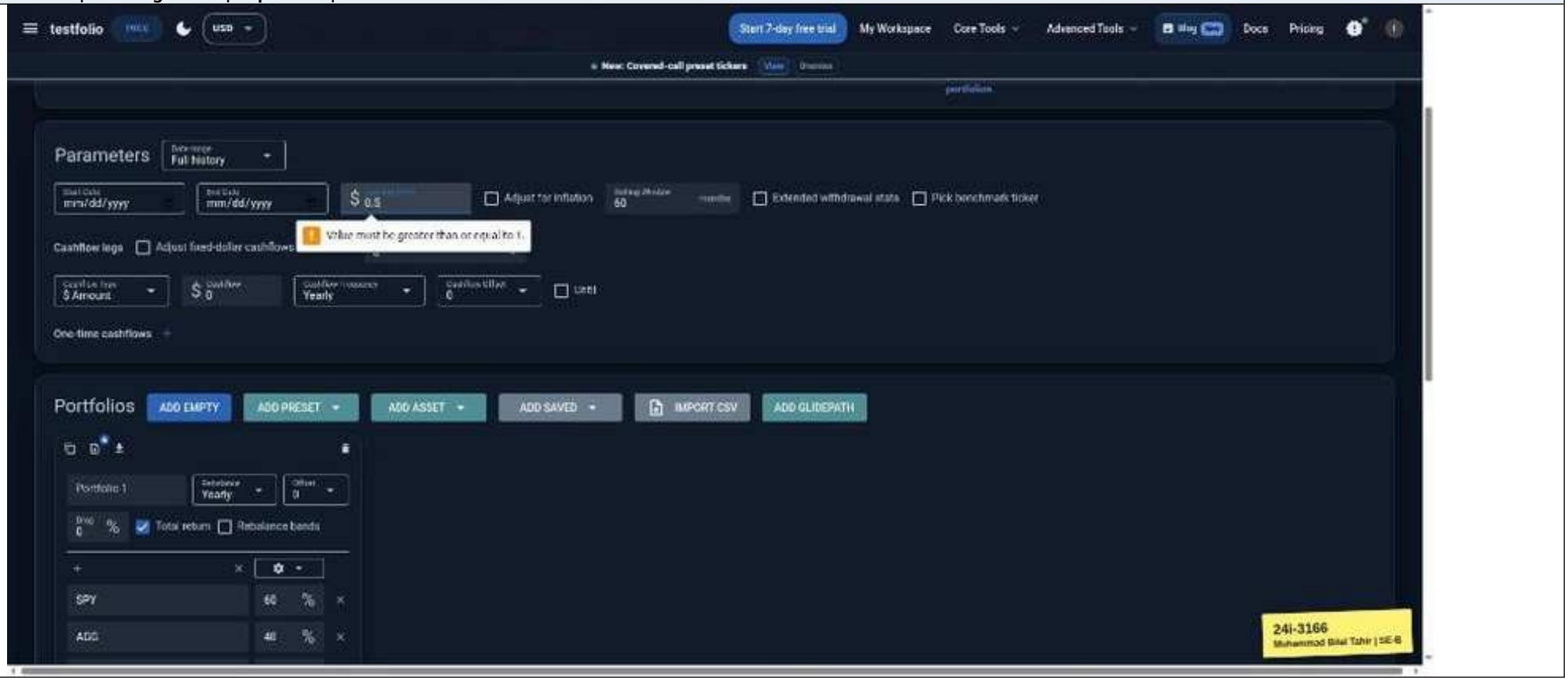
TC-EP-027 | Starting Value | input: 0 | result: Pass



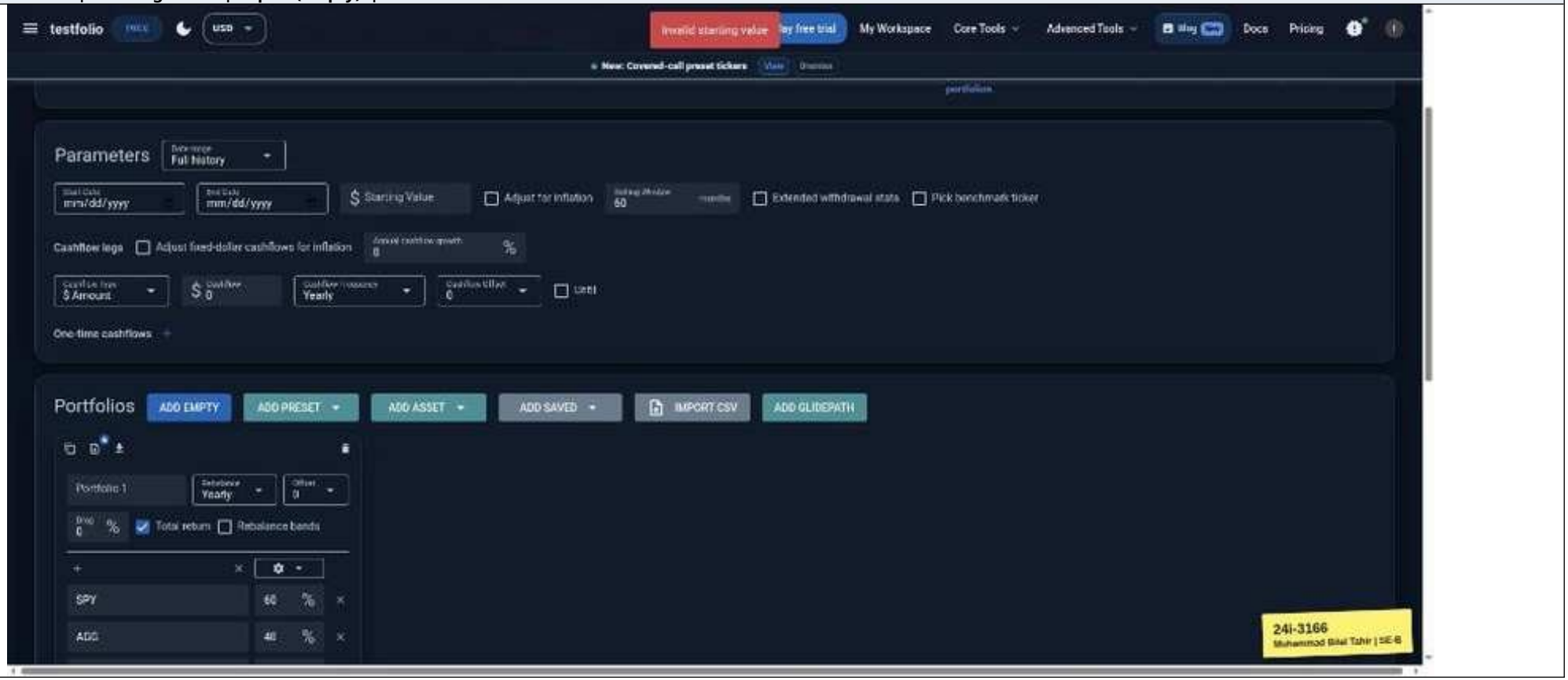
TC-EP-028 | Starting Value | input: -5 | result: Pass



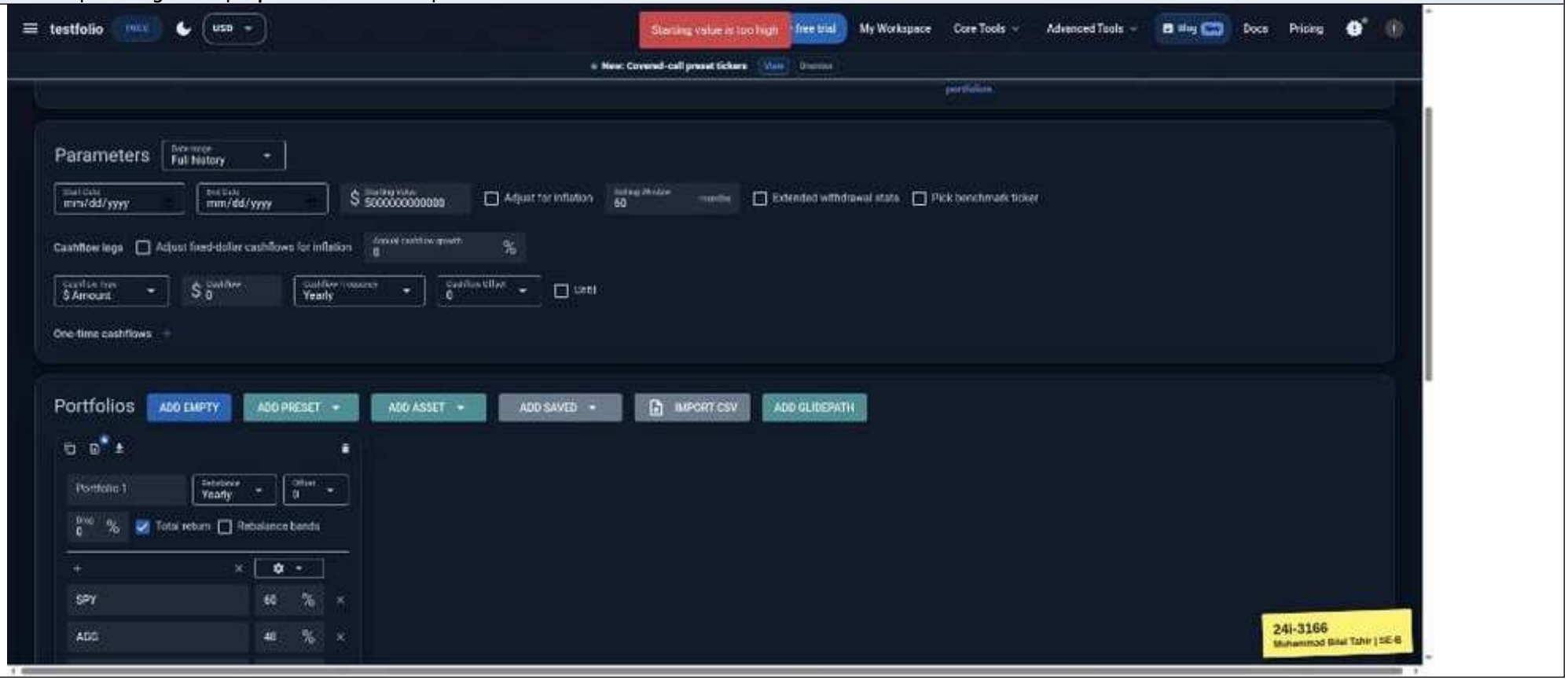
TC-EP-029 | Starting Value | input: 0.5 | result: Pass



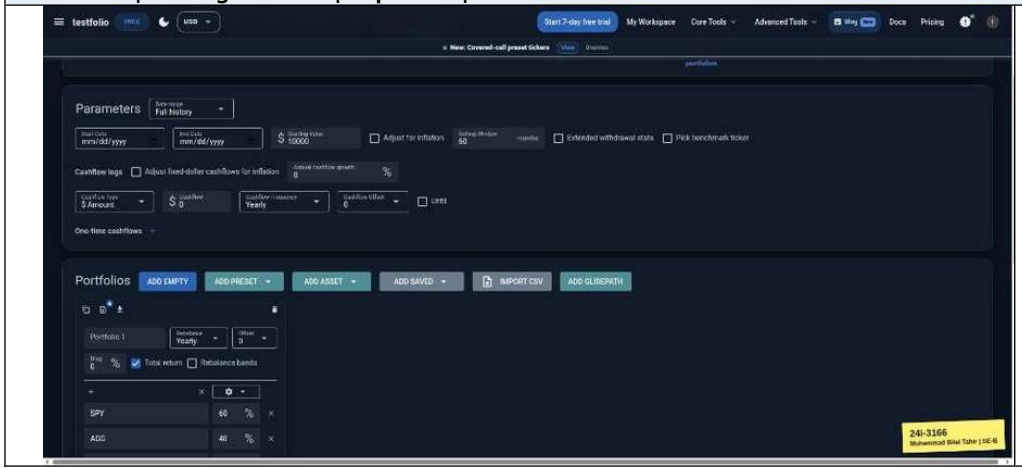
TC-EP-030 | Starting Value | input: (empty) | result: Pass



TC-EP-031 | Starting Value | input: 5000000000000 | result: Pass

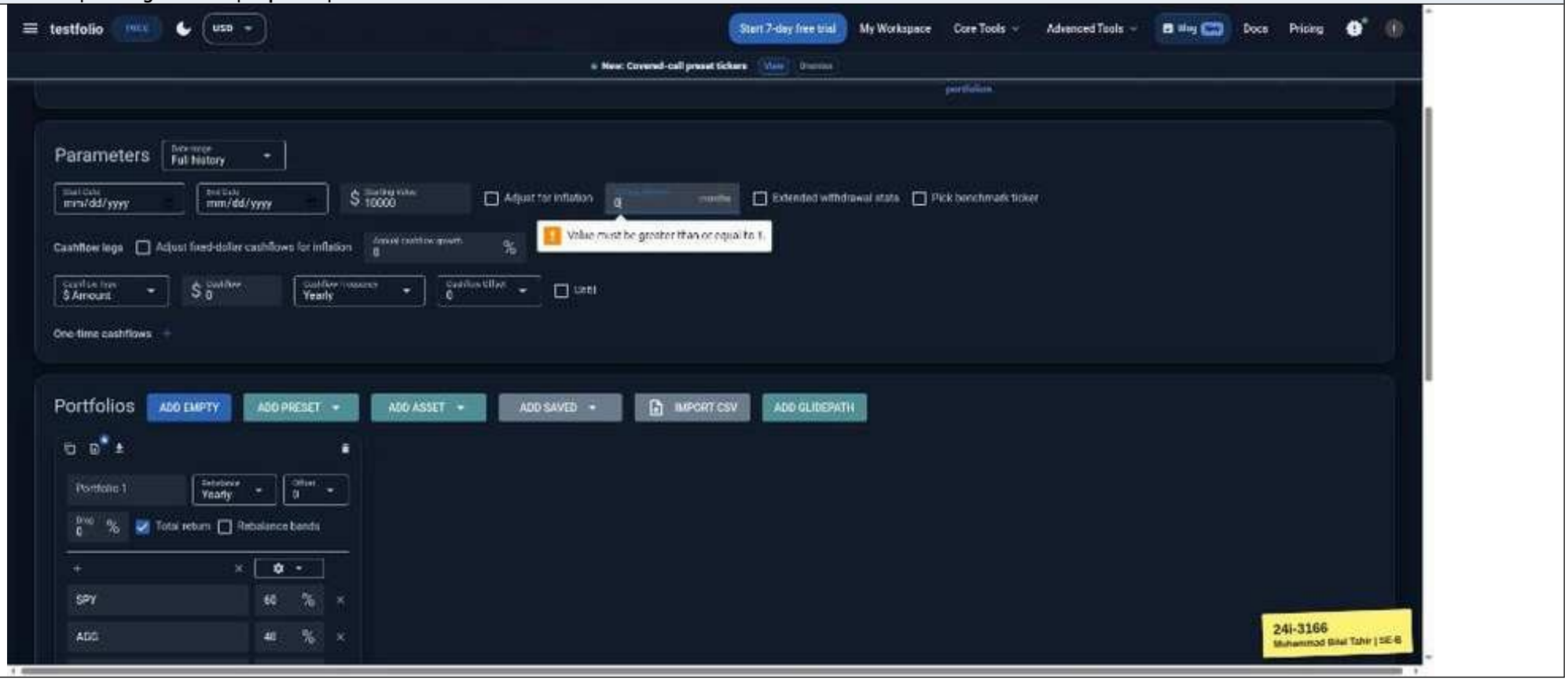


TC-EP-032 | Rolling Window | input: 60 | result: Pass

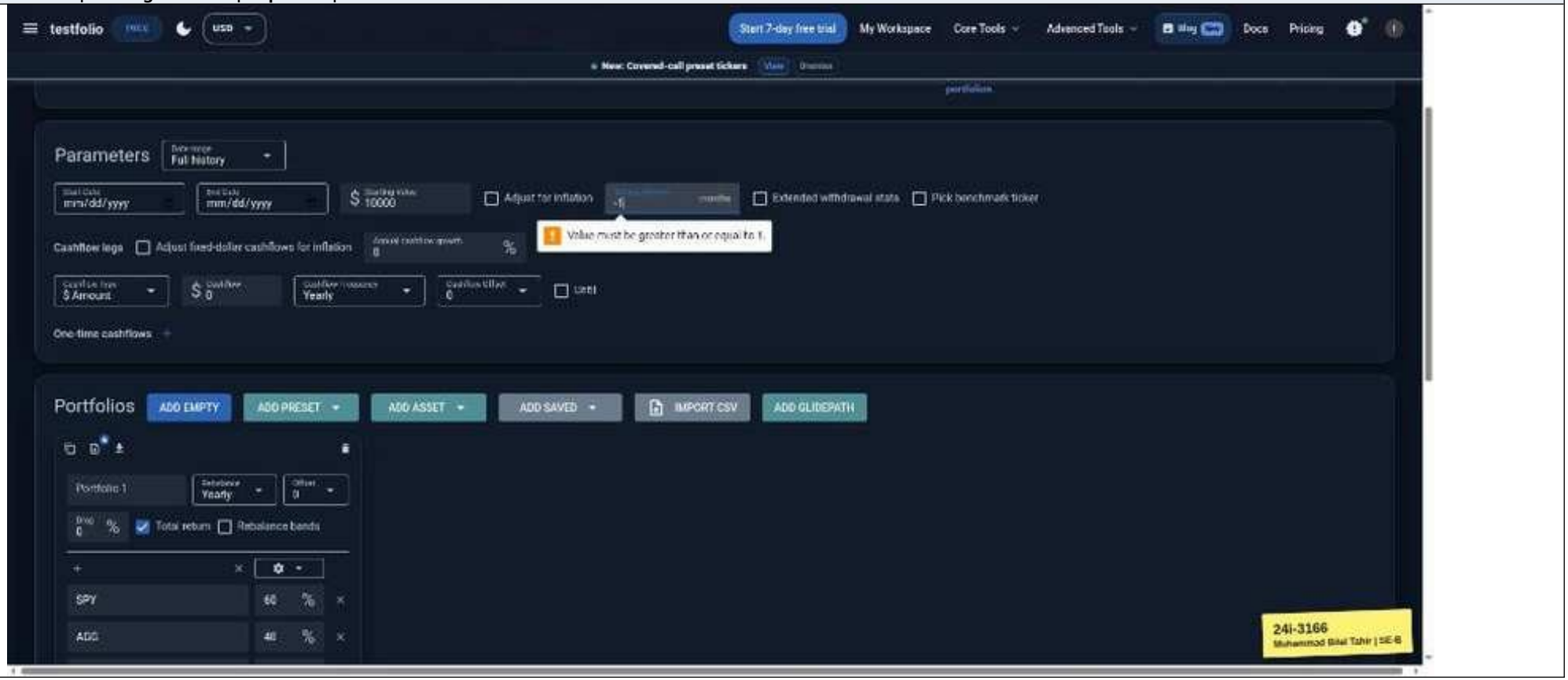




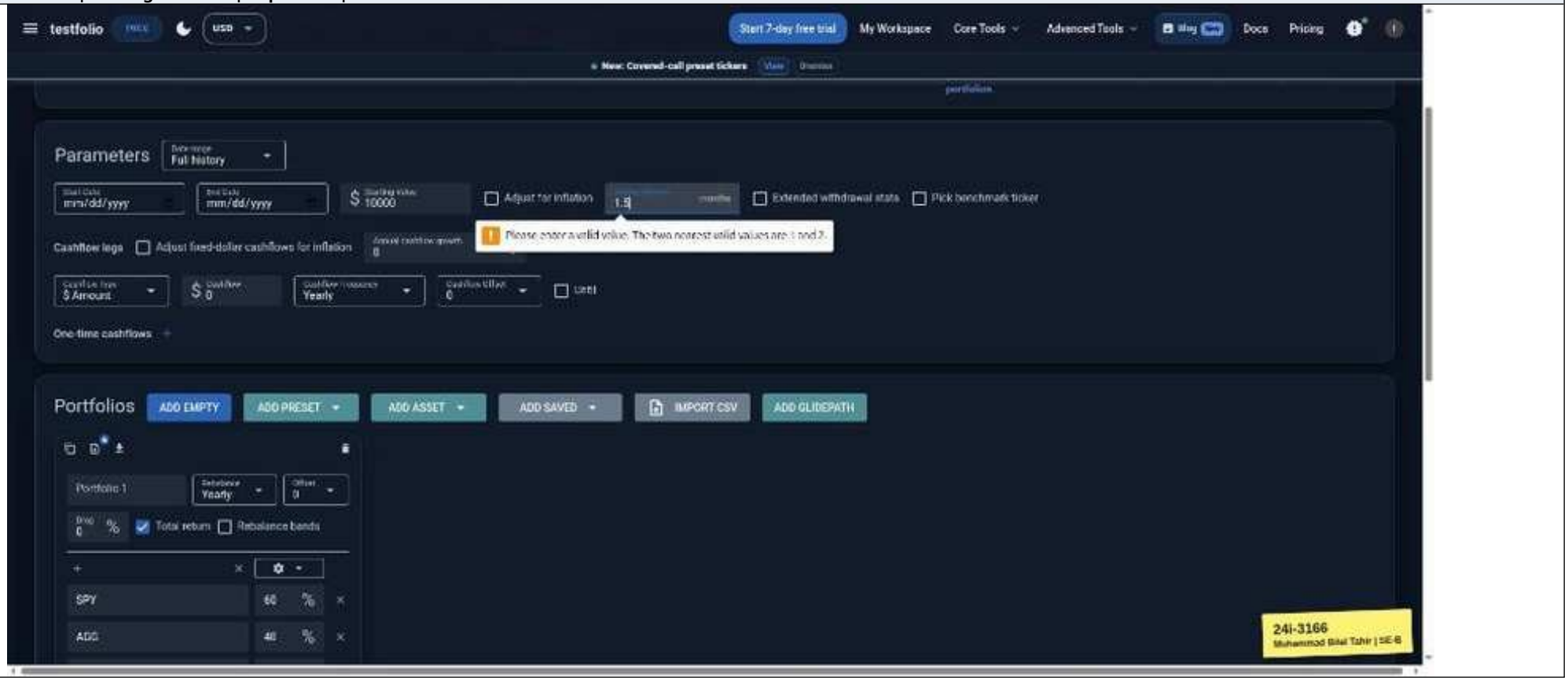
TC-EP-033 | Rolling Window | input: 0 | result: Pass



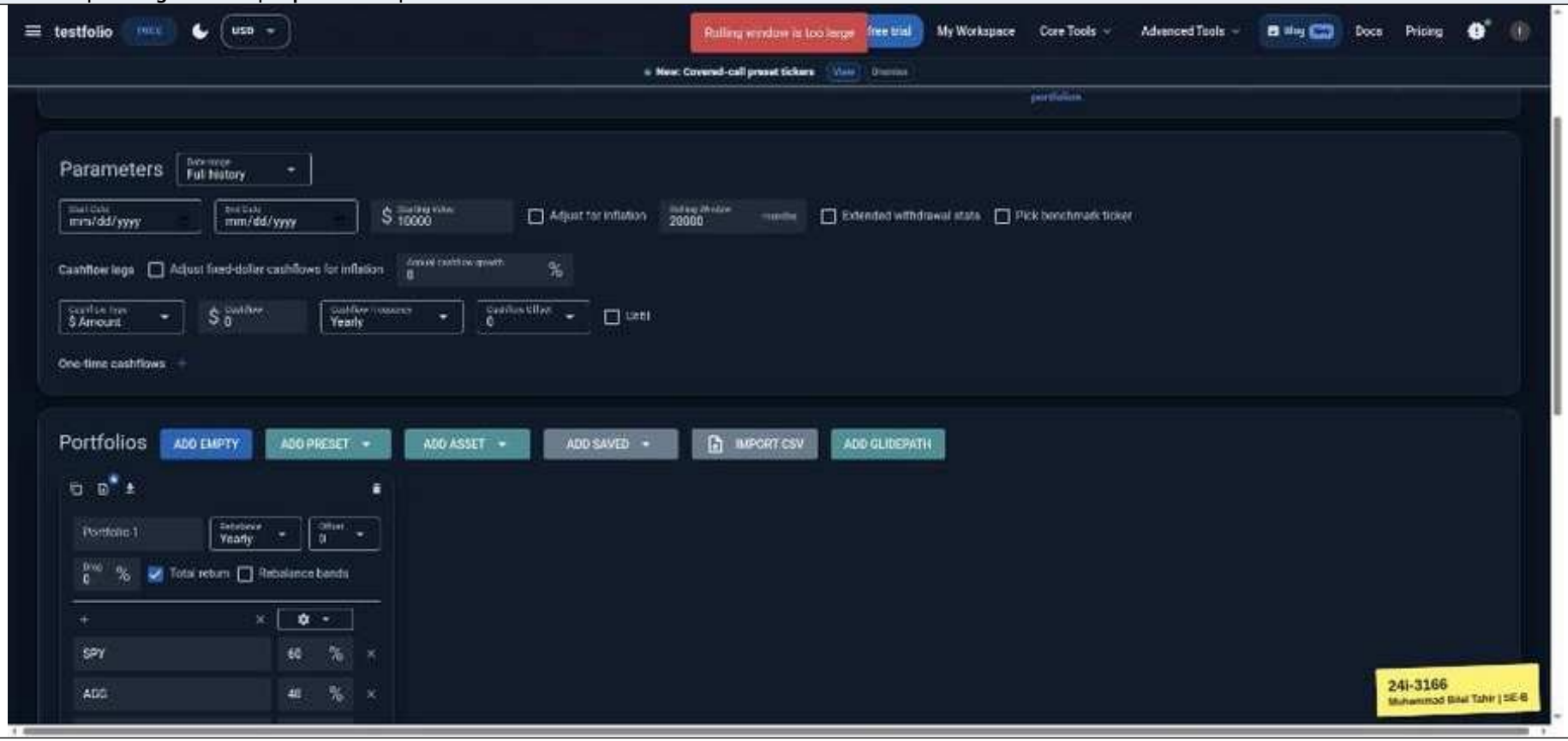
TC-EP-034 | Rolling Window | input: -1 | result: Pass



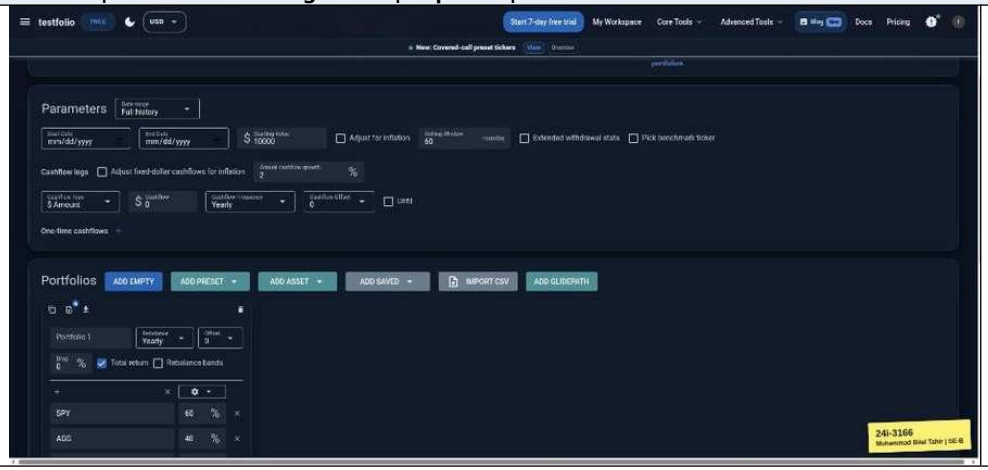
TC-EP-035 | Rolling Window | input: 1.5 | result: Pass



TC-EP-036 | Rolling Window | input: 20000 | result: Pass

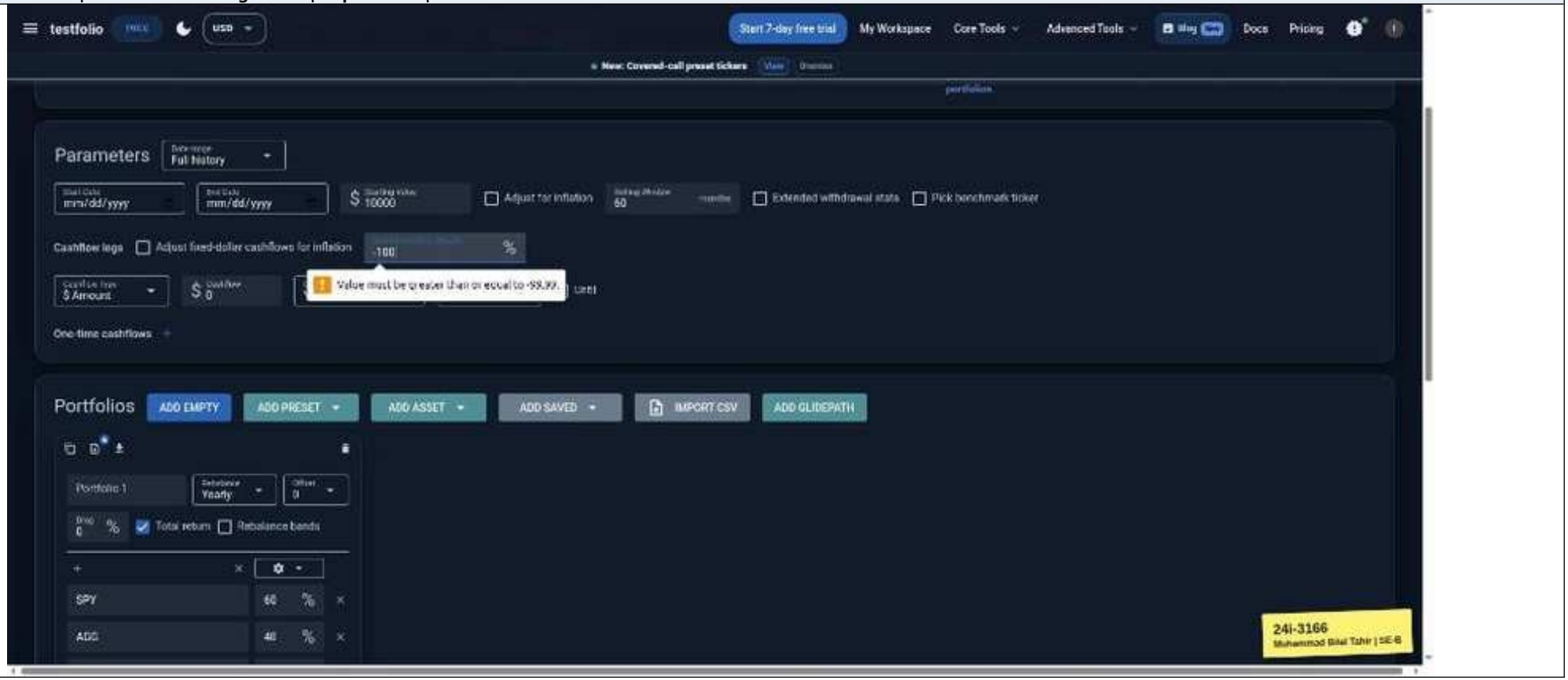


TC-EP-037 | Annual cashflow growth | input: 2 | result: Pass

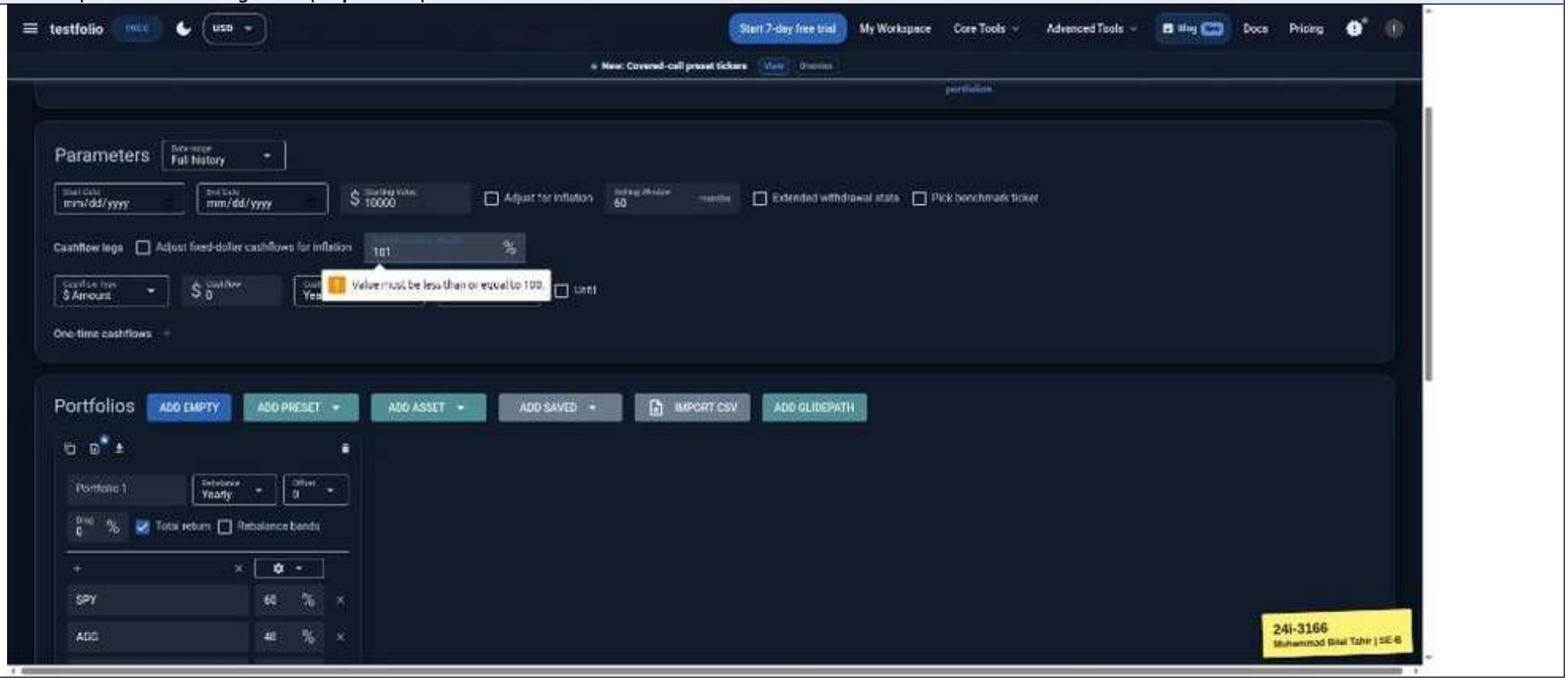




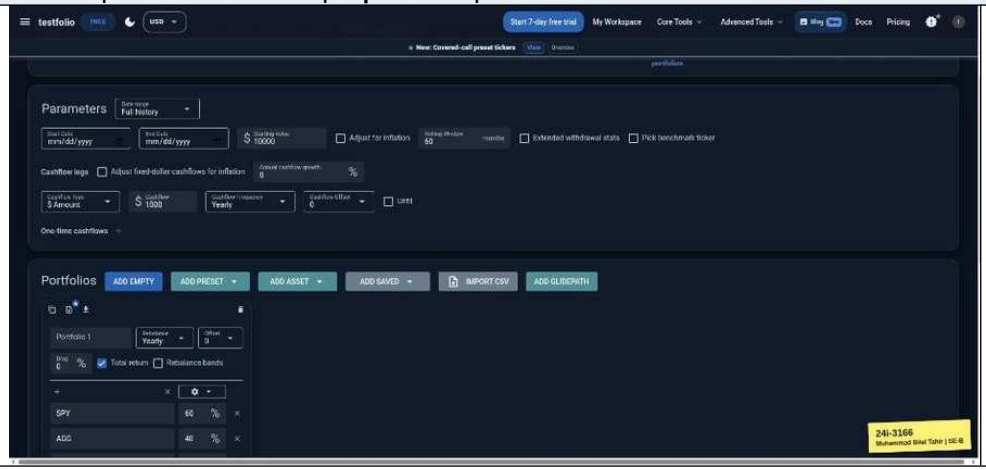
TC-EP-038 | Annual cashflow growth | input: -100 | result: Pass



TC-EP-039 | Annual cashflow growth | input: 101 | result: Pass

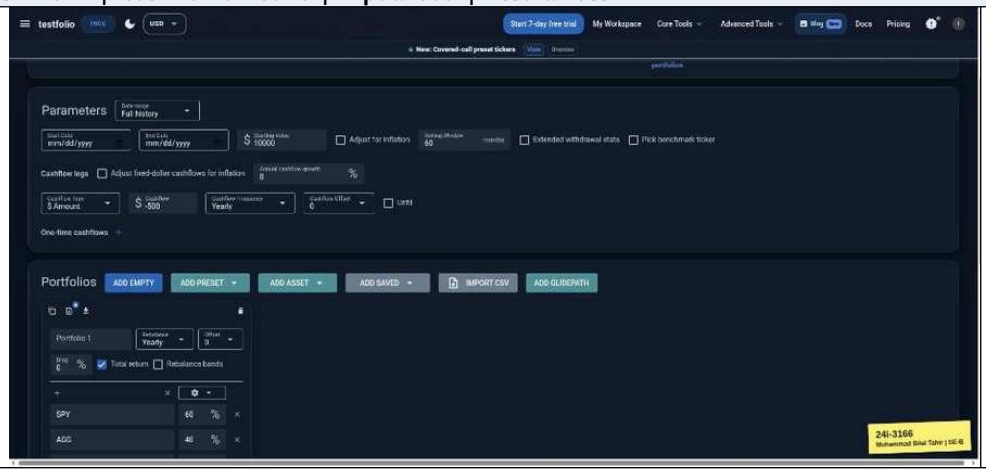


TC-EP-040 | Cashflow amount | input: 1000 | result: Pass



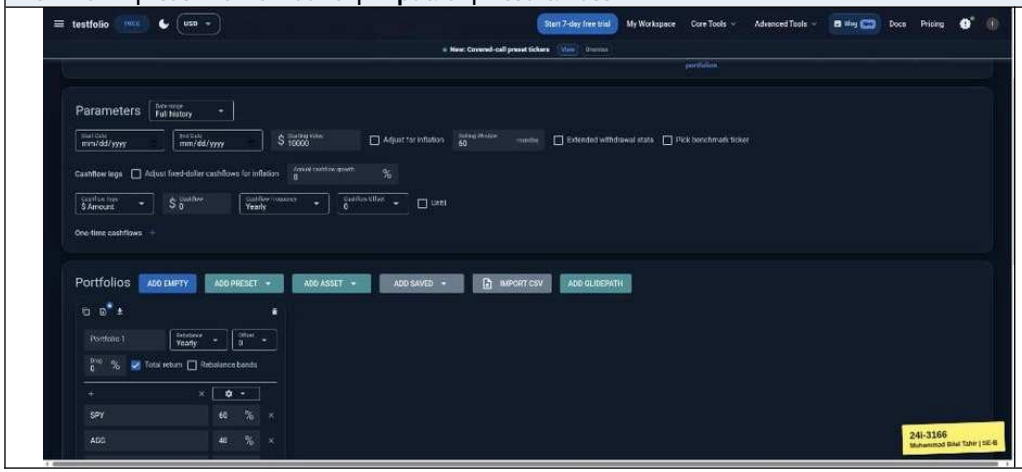


TC-EP-041 | Cashflow amount | input: -500 | result: Pass



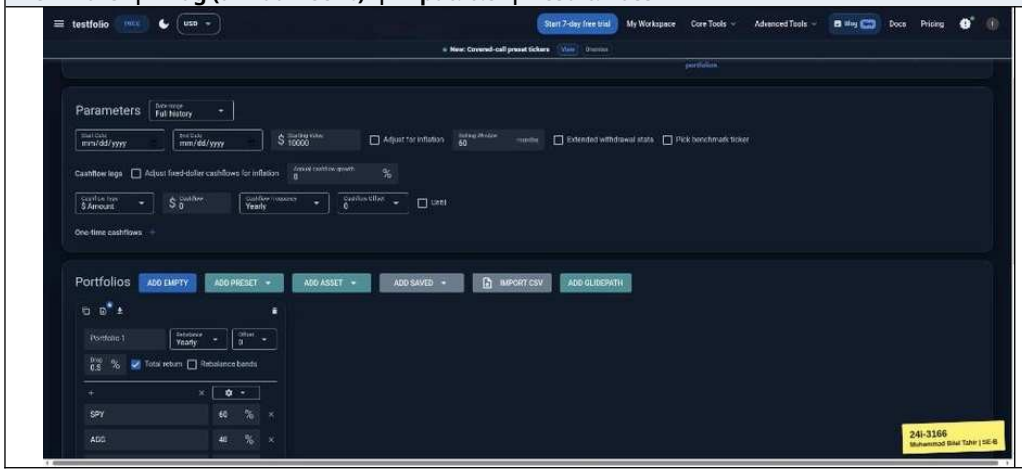


TC-EP-042 | Cashflow amount | input: 0 | result: Pass



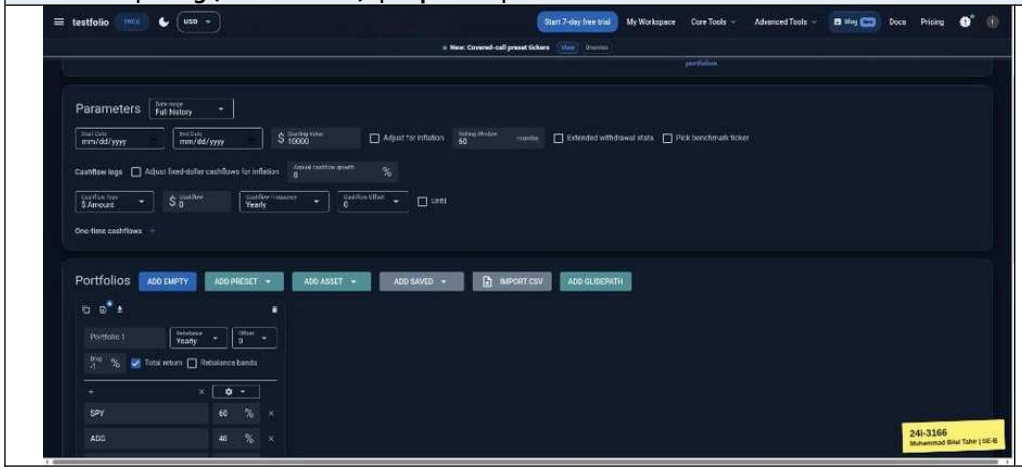


TC-EP-043 | Drag (annual fee %) | input: 0.5 | result: Pass



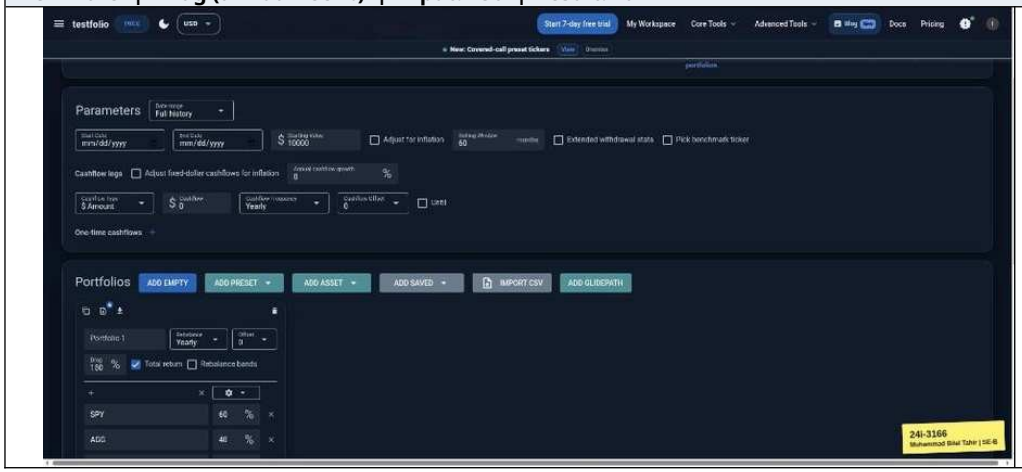


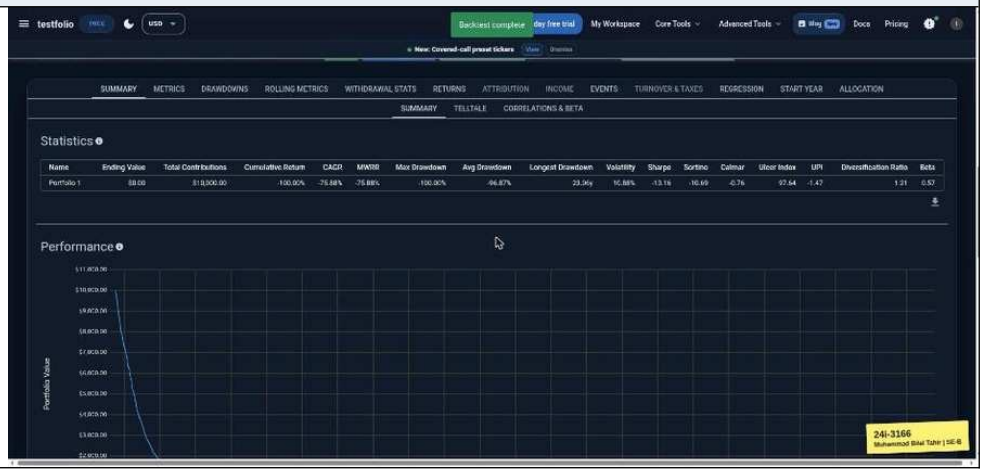
TC-EP-044 | Drag (annual fee %) | input: -1 | result: Fail



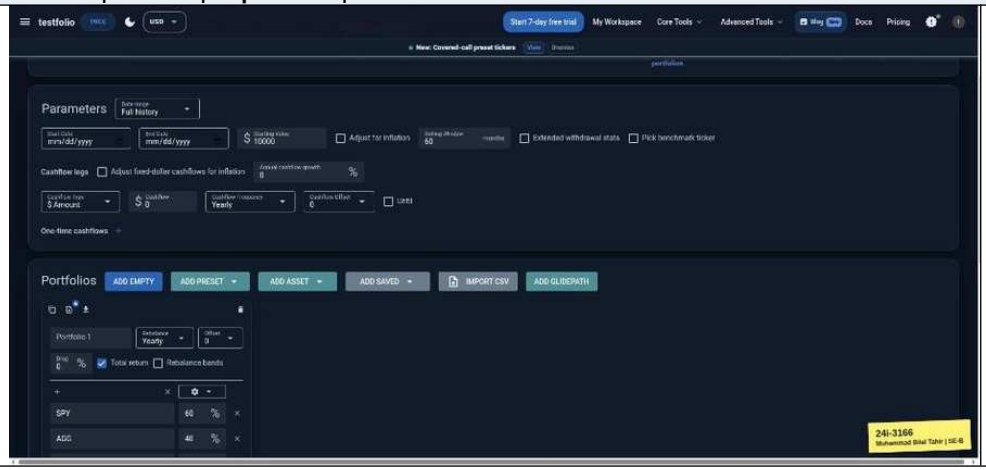


TC-EP-045 | Drag (annual fee %) | input: 150 | result: Fail



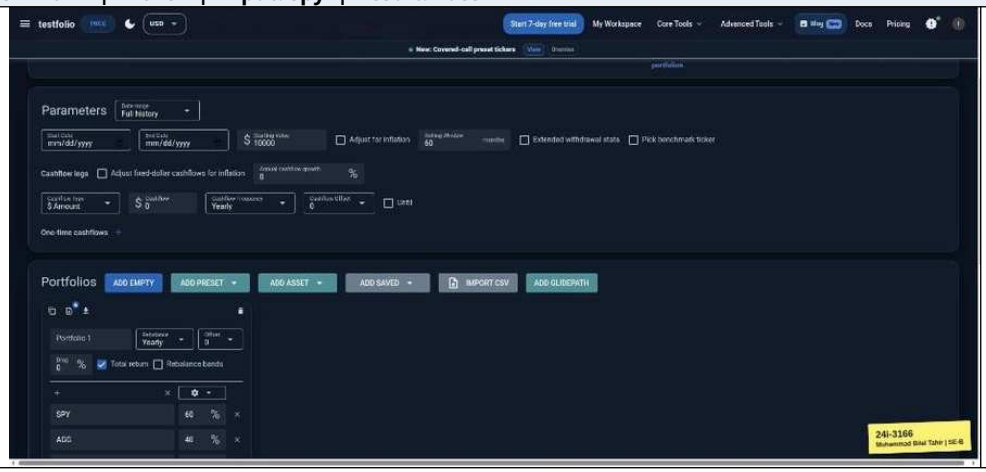


TC-EP-046 | Ticker | input: SPY | result: Pass



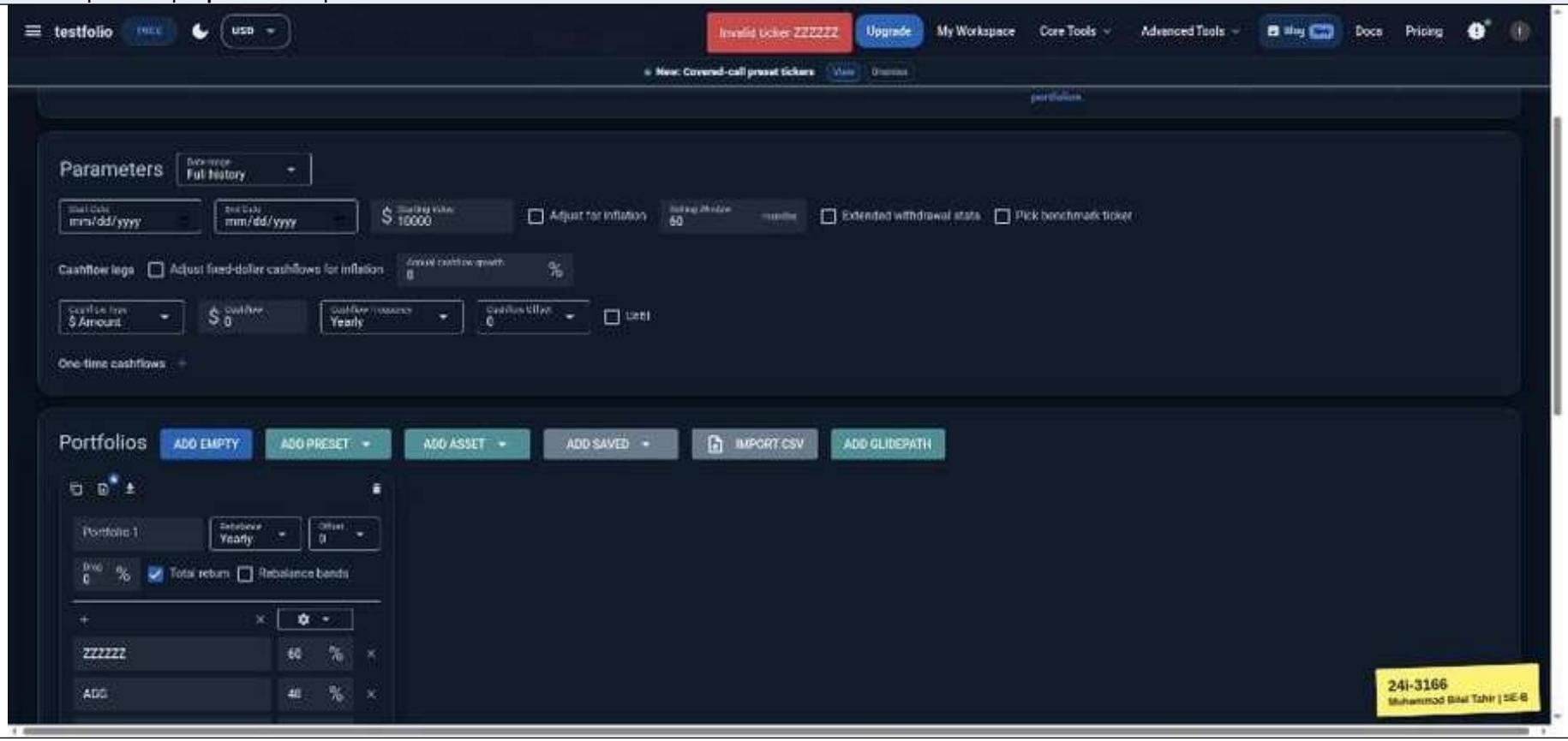


TC-EP-047 | Ticker | input: spy | result: Pass

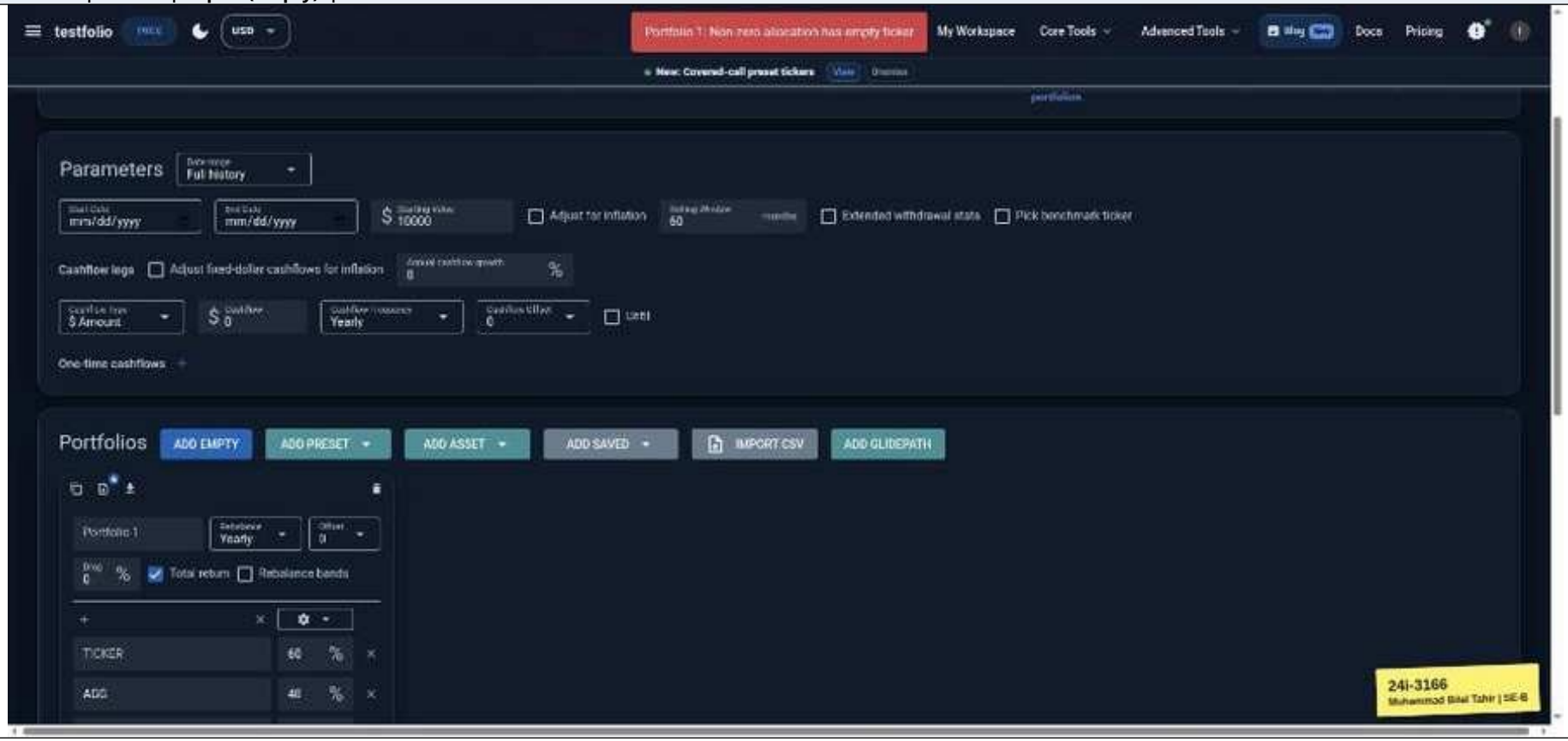




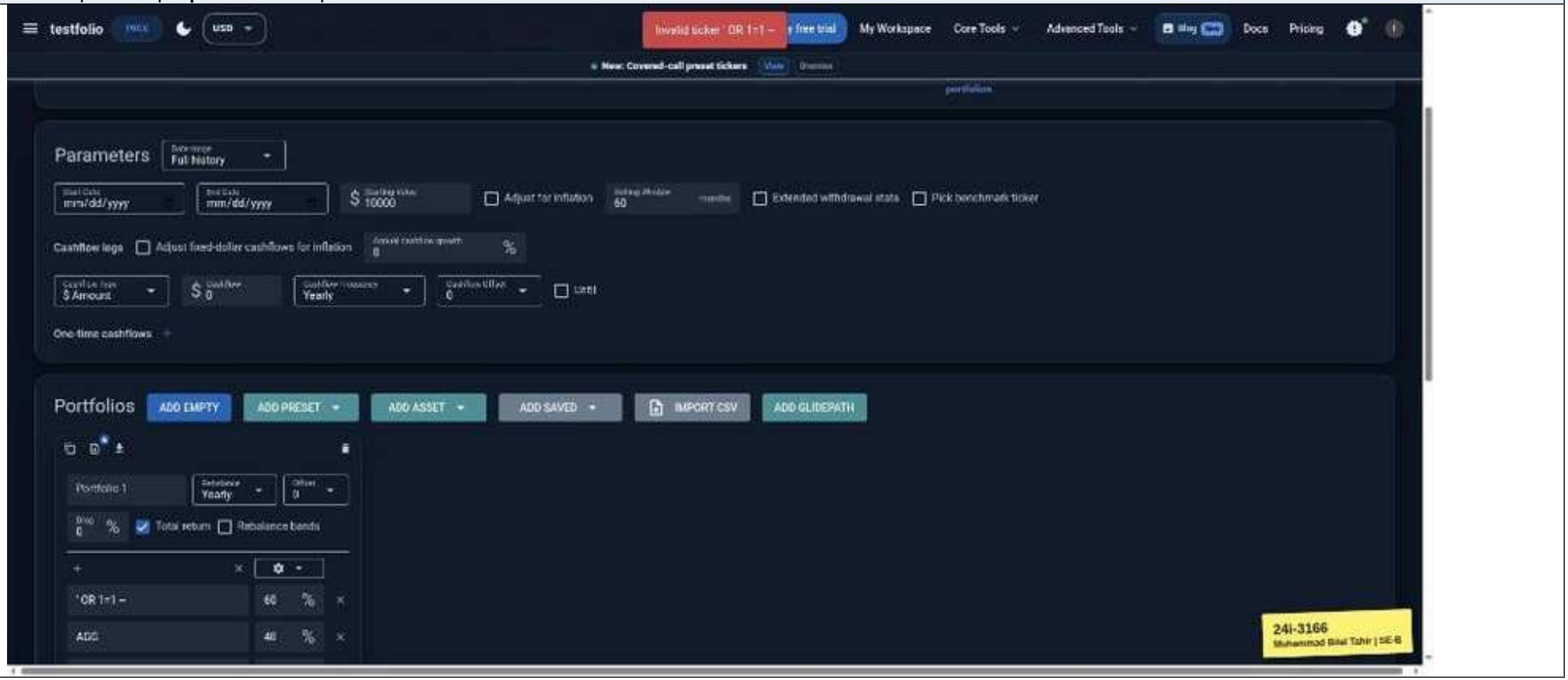
TC-EP-048 | Ticker | input: ZZZZZZ | result: Pass



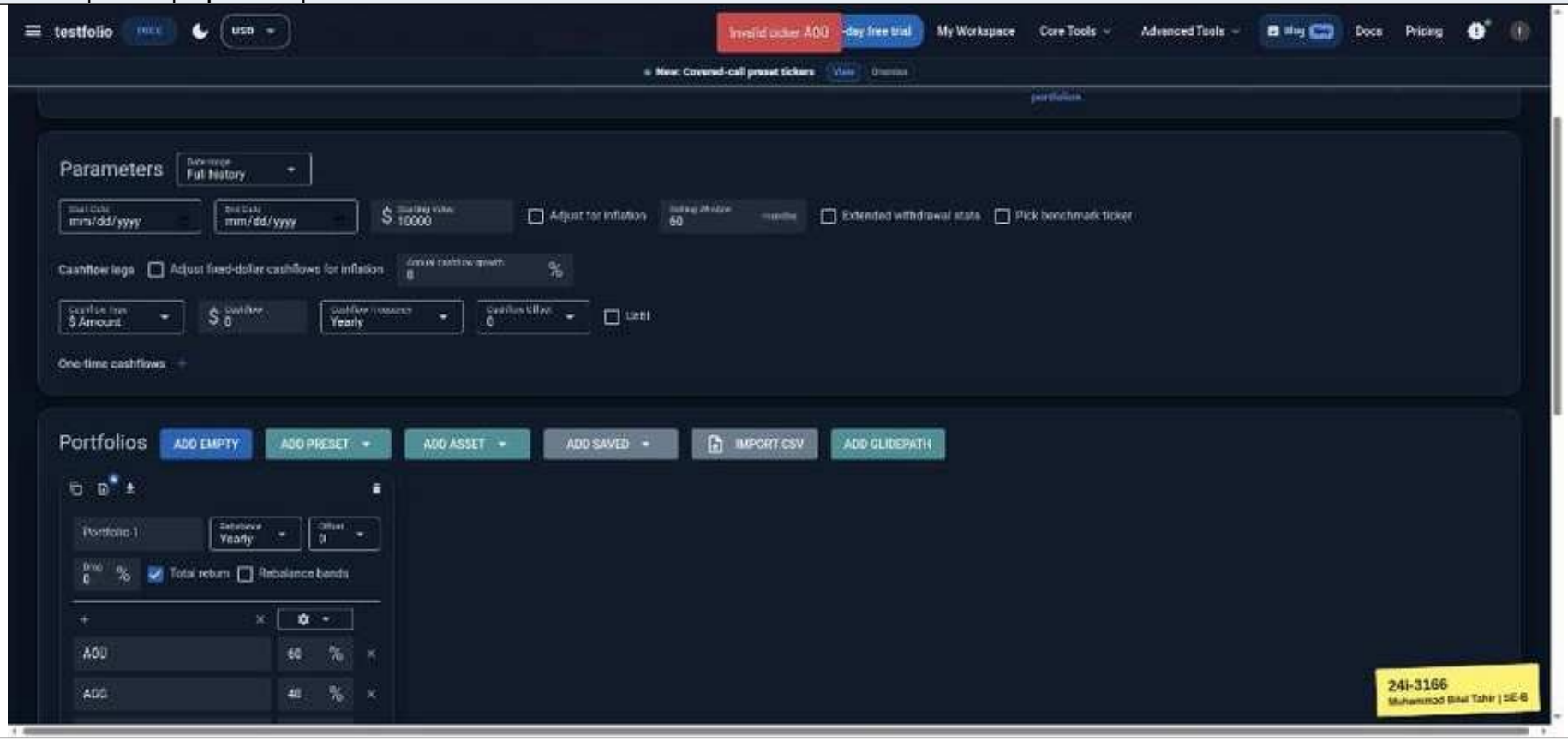
TC-EP-049 | Ticker | input: (empty) | result: Pass



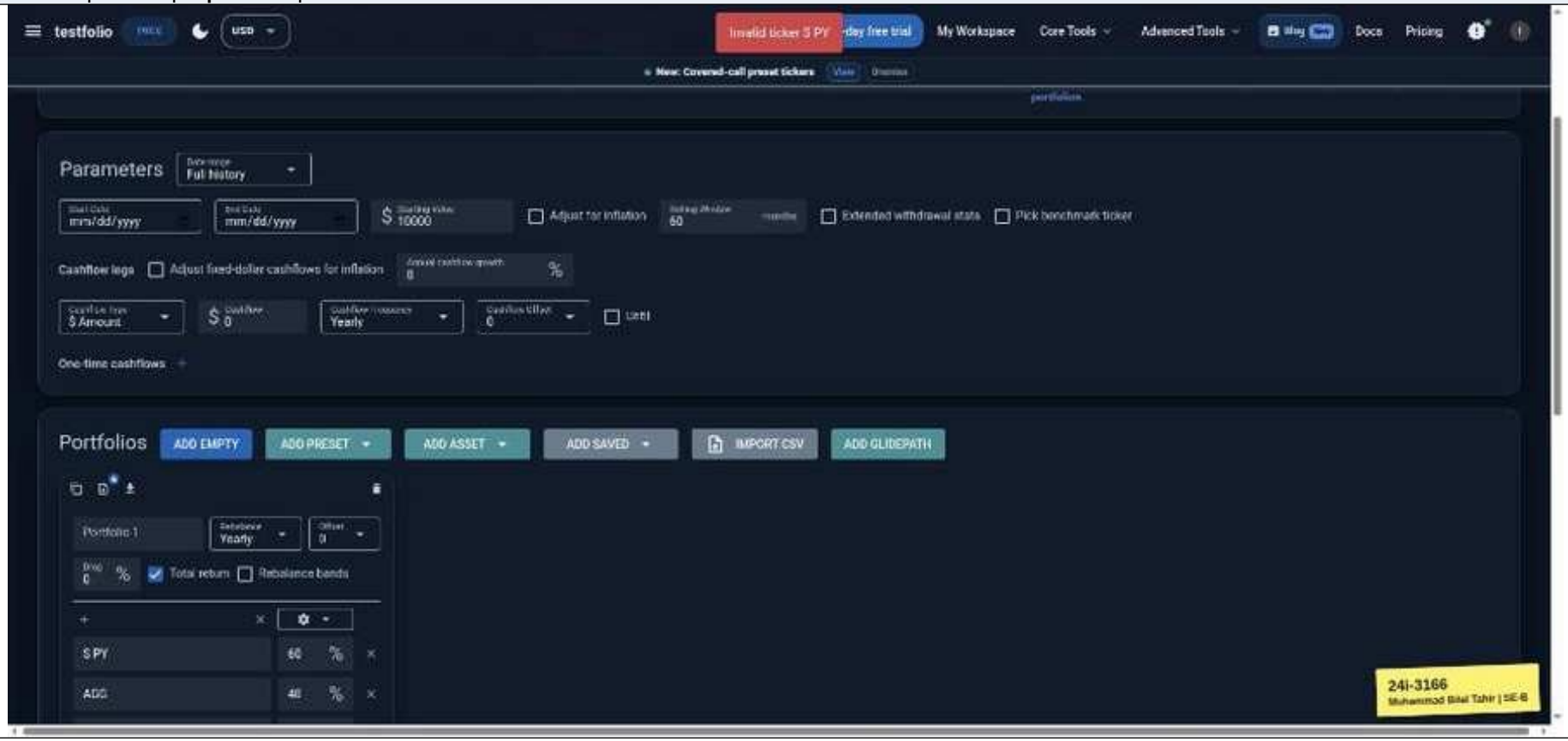
TC-EP-050 | Ticker | input: ' OR 1=1 -- | result: Pass



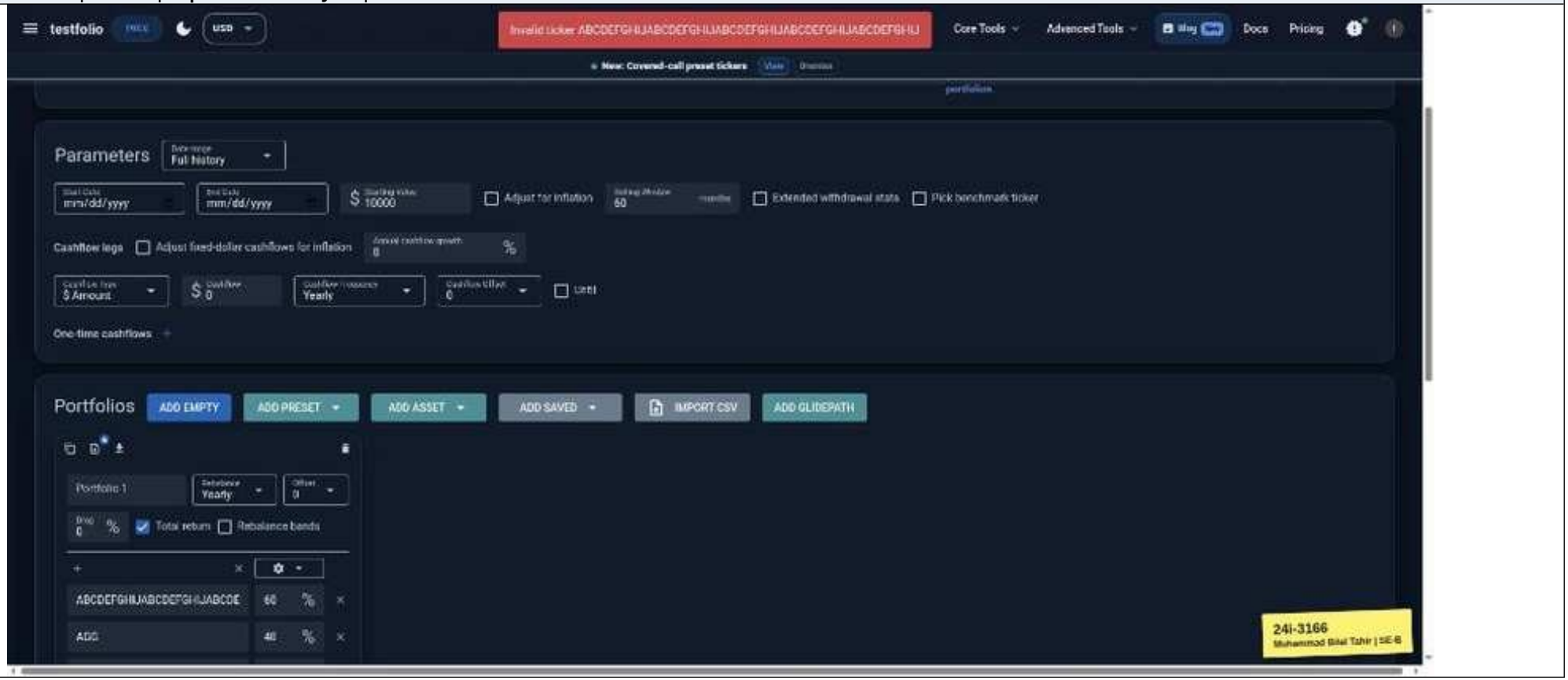
TC-EP-051 | Ticker | input: ÄÖÜ | result: Pass



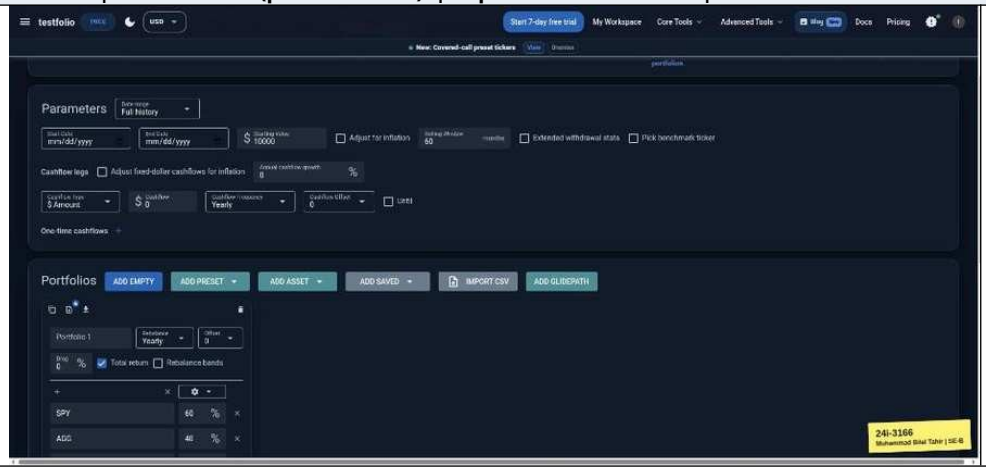
TC-EP-052 | Ticker | input: S PY | result: Pass



TC-EP-053 | Ticker | input: ABCDEFGHIJ x5 | result: Pass

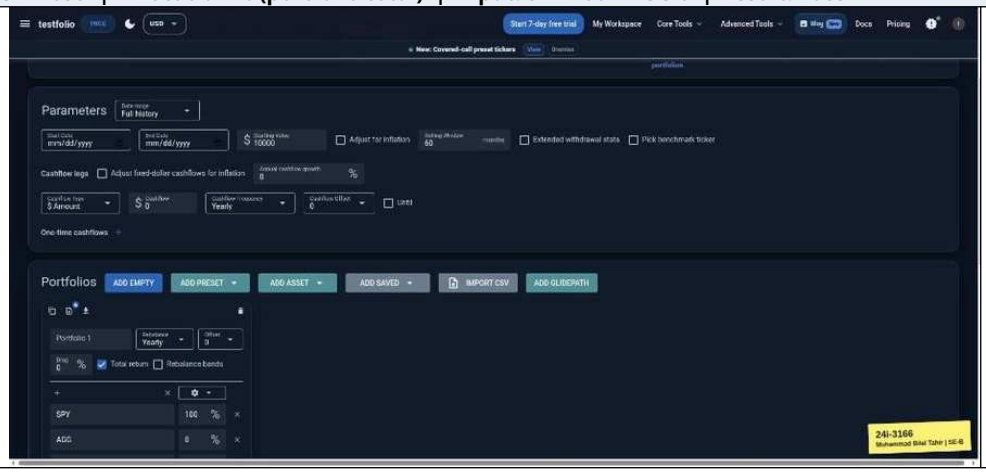


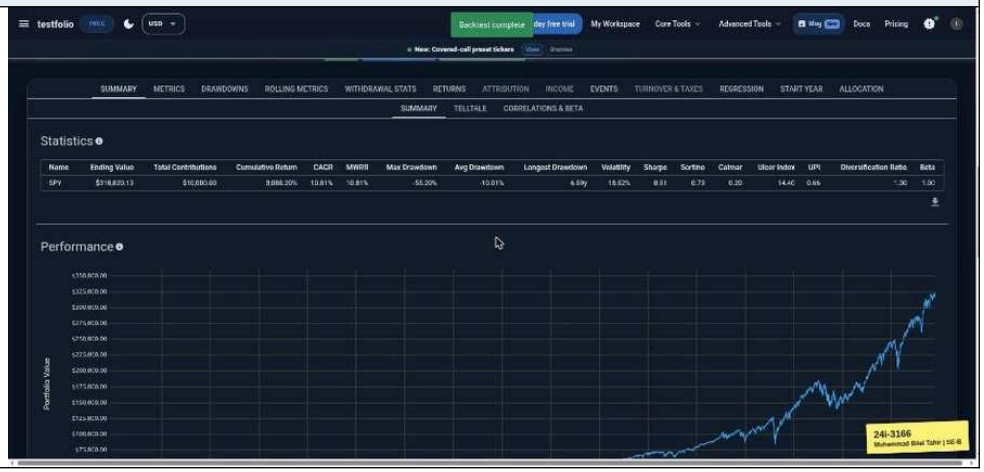
TC-EP-054 | Allocation % (portfolio total) | input: SPY 60 / AGG 40 | result: Pass



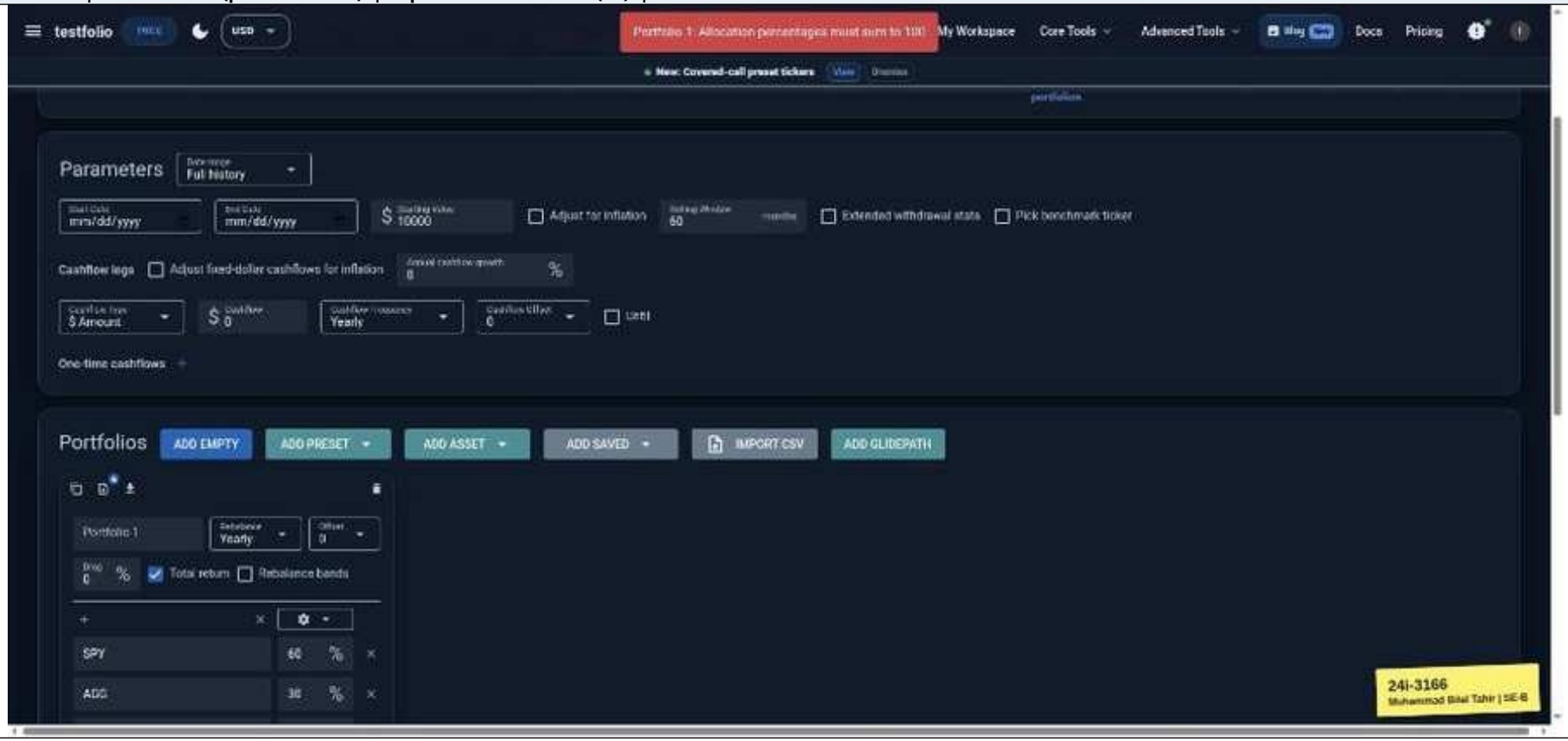


TC-EP-055 | Allocation % (portfolio total) | input: SPY 100 / AGG 0 | result: Pass

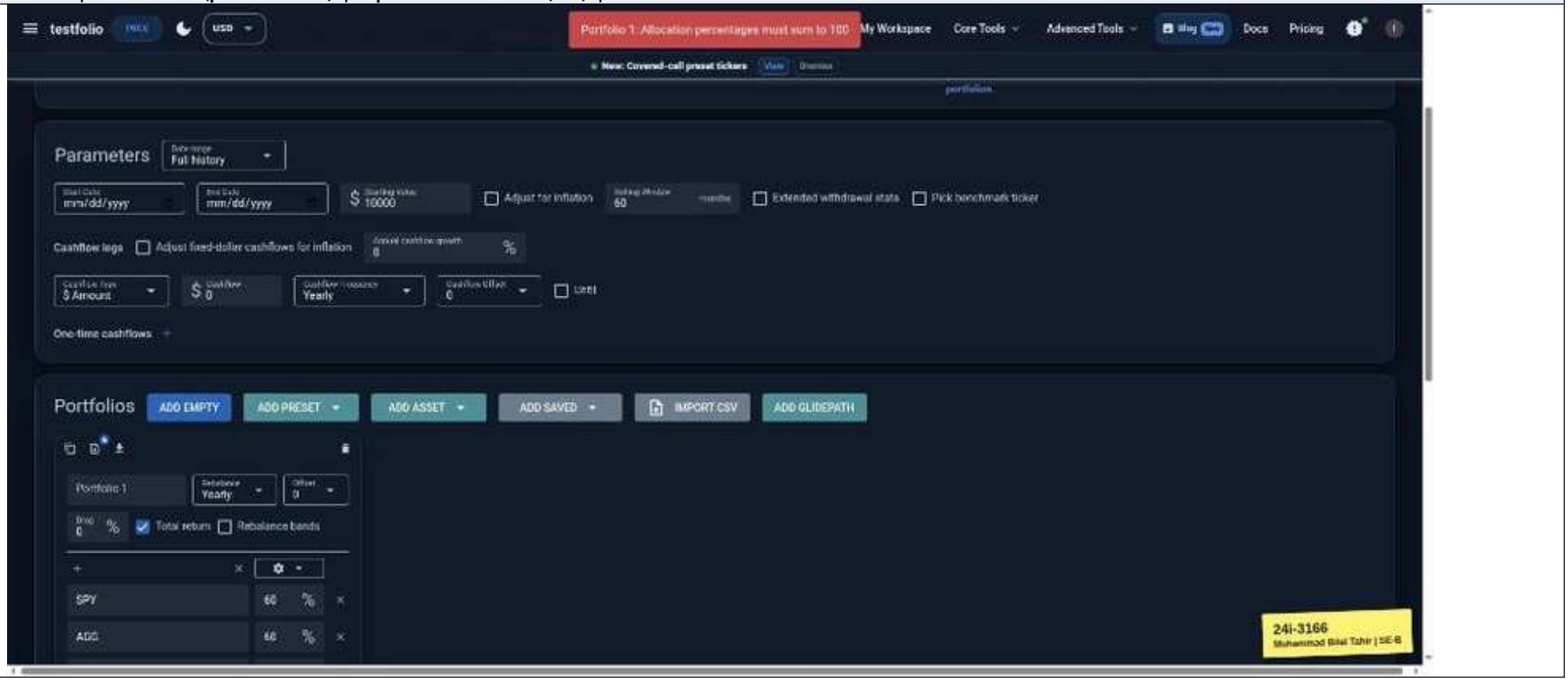




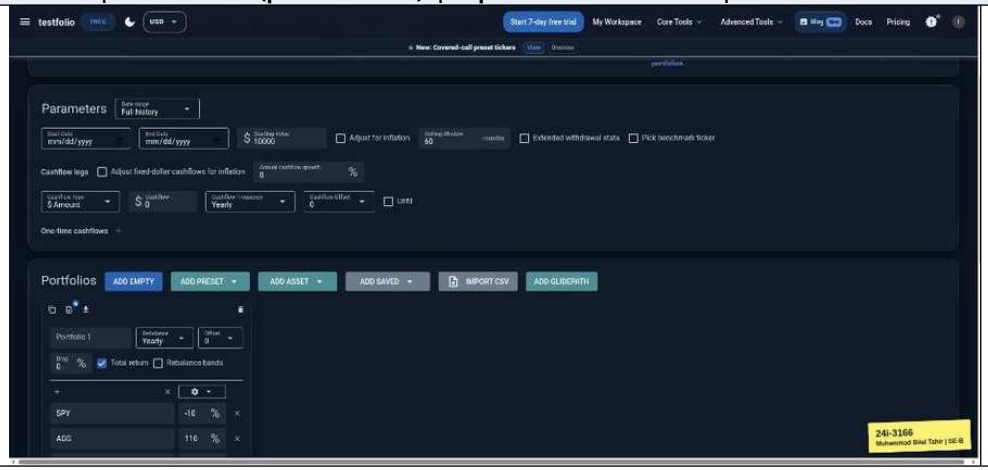
TC-EP-056 | Allocation % (portfolio total) | input: SPY 60 / AGG 30 (90) | result: Pass

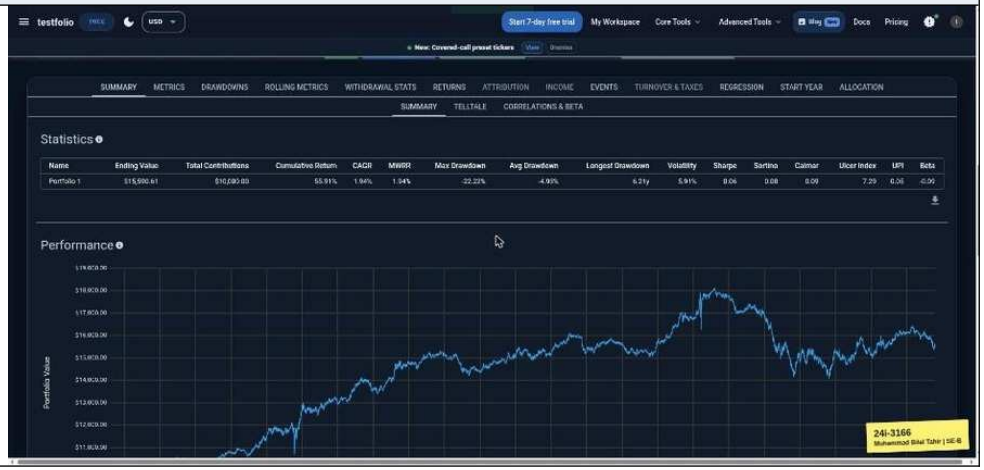


TC-EP-057 | Allocation % (portfolio total) | input: SPY 60 / AGG 60 (120) | result: Pass

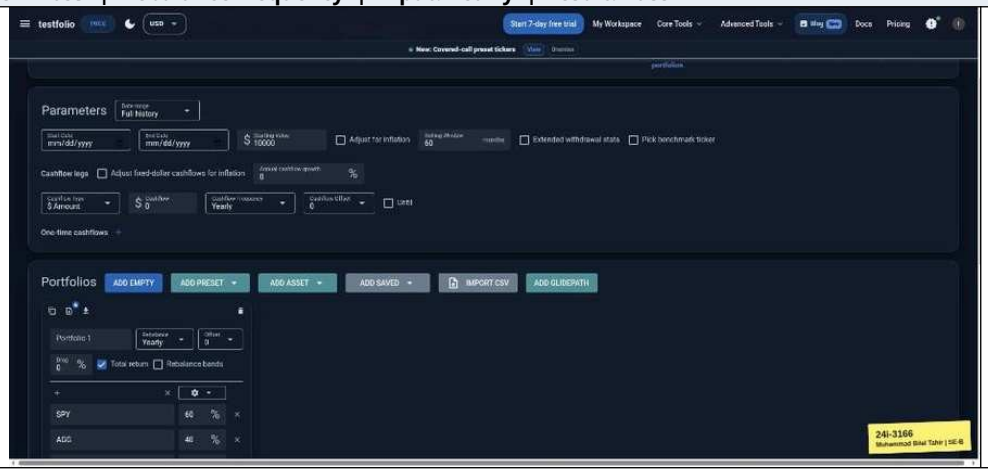


TC-EP-058 | Allocation % (portfolio total) | input: SPY -10 / AGG 110 | result: Fail





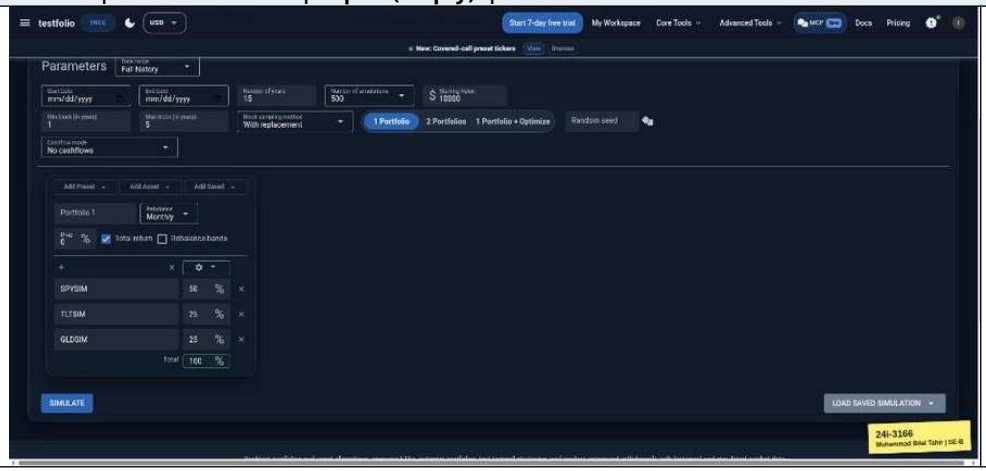
TC-EP-059 | Rebalance frequency | input: Yearly | result: Pass

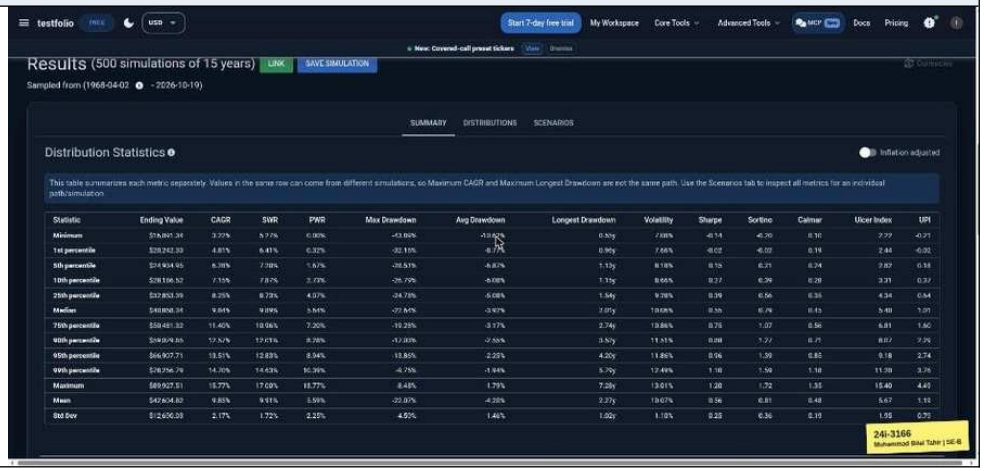




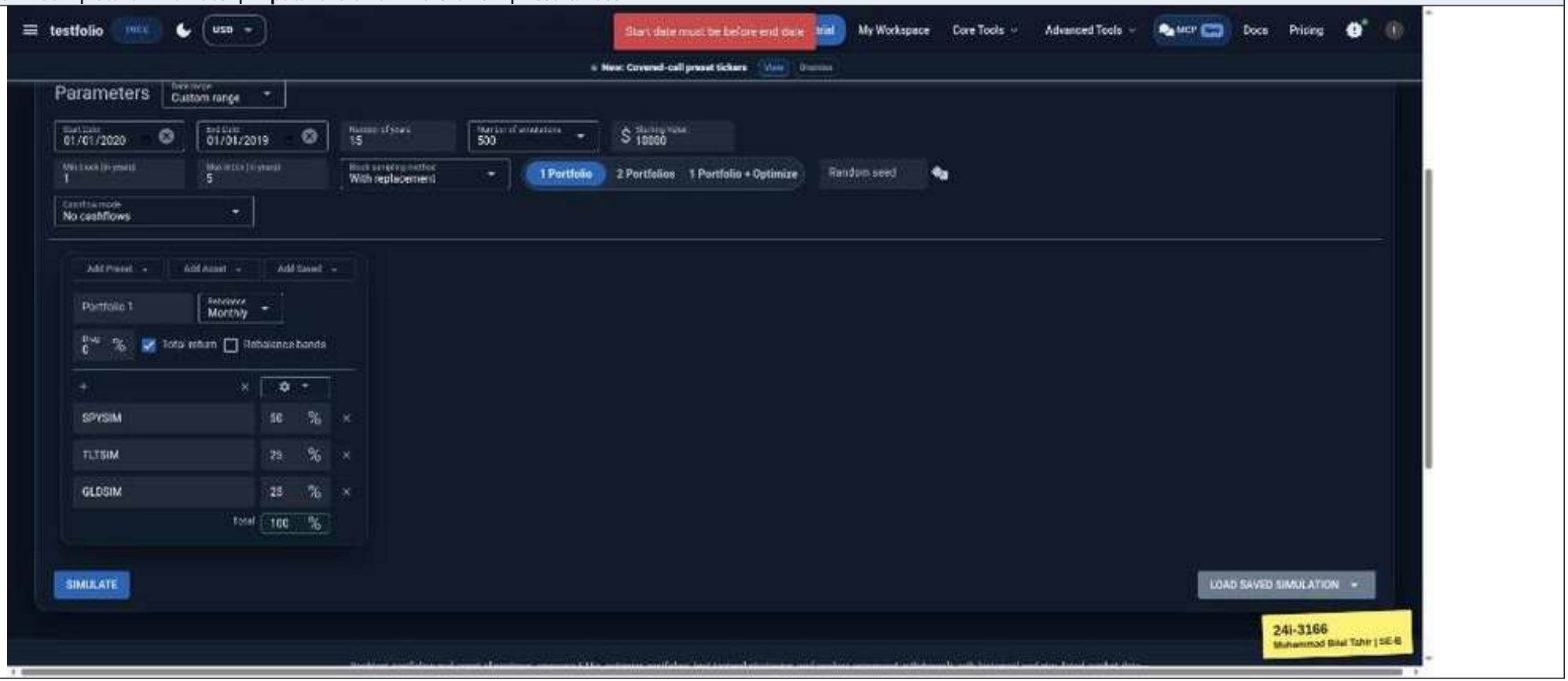
Monte Carlo Simulator (testfol.io/monte-carlo)

TC-EP-060 | Start / End Date | input: (empty) | result: Pass

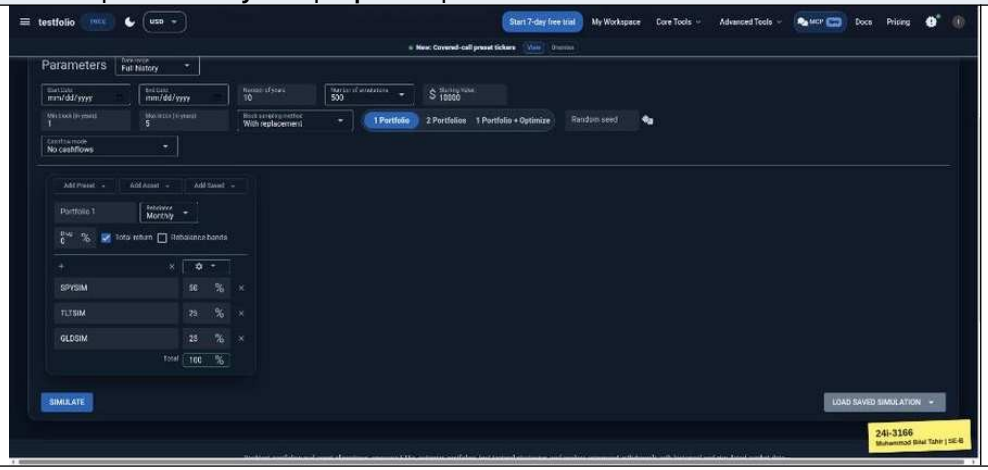


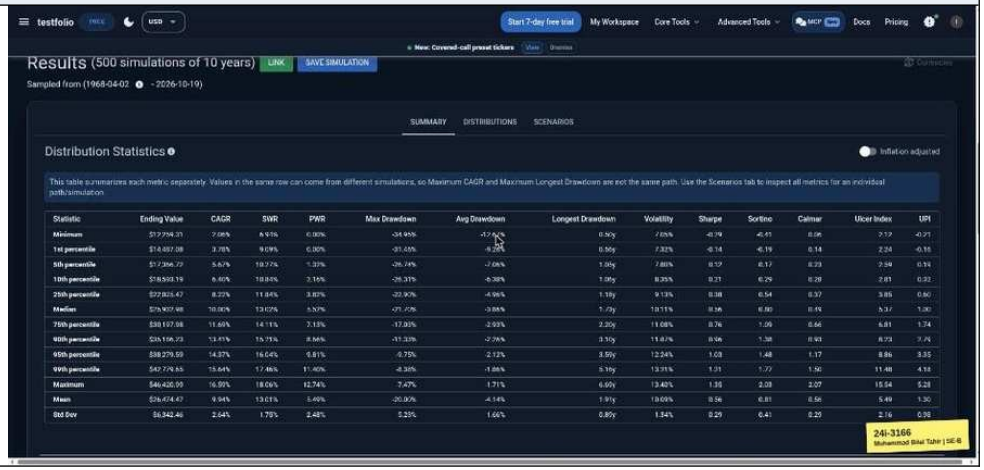


TC-EP-061 | Start / End Date | input: 2020-01-01 / 2019-01-01 | result: Pass

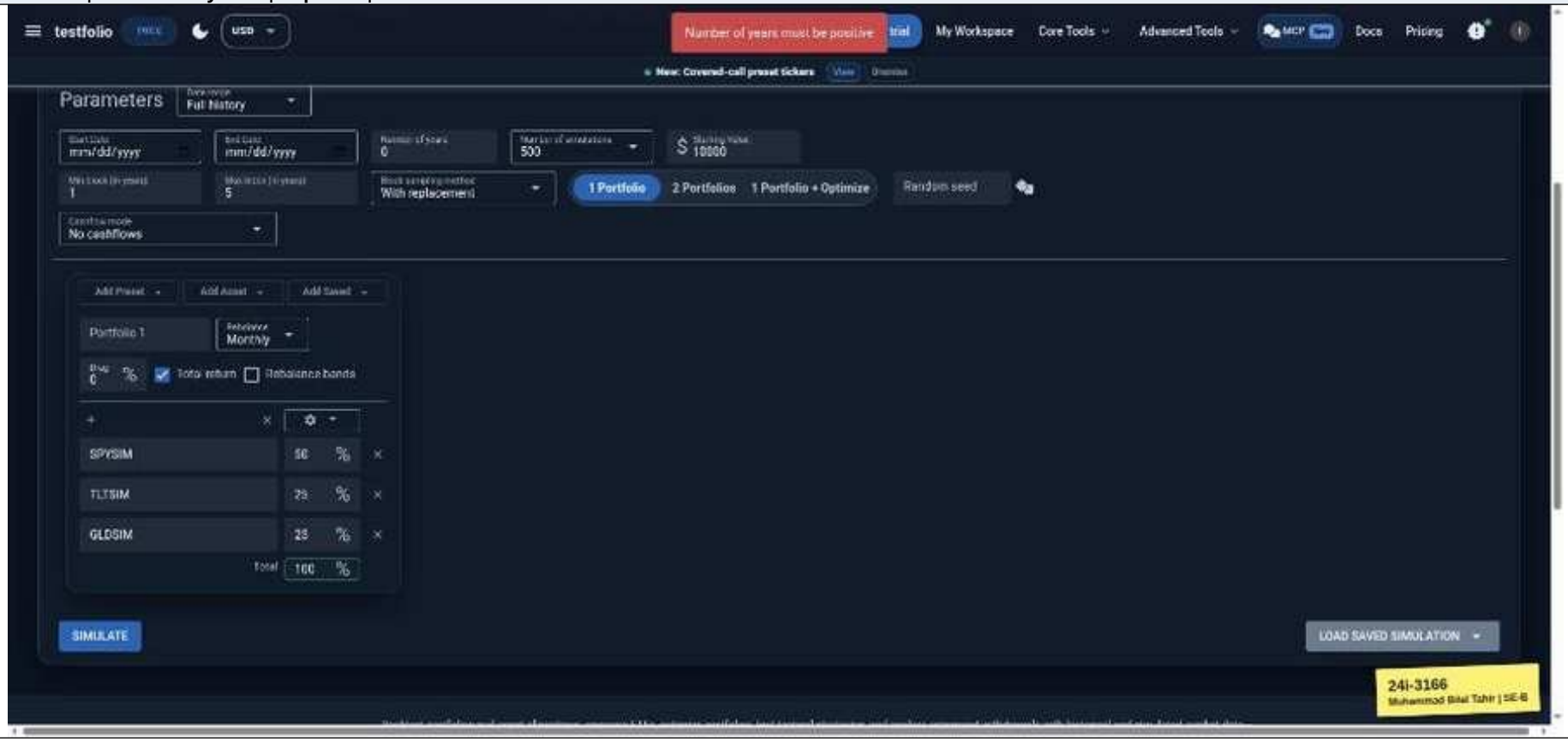


TC-EP-062 | Number of years | input: 10 | result: Pass

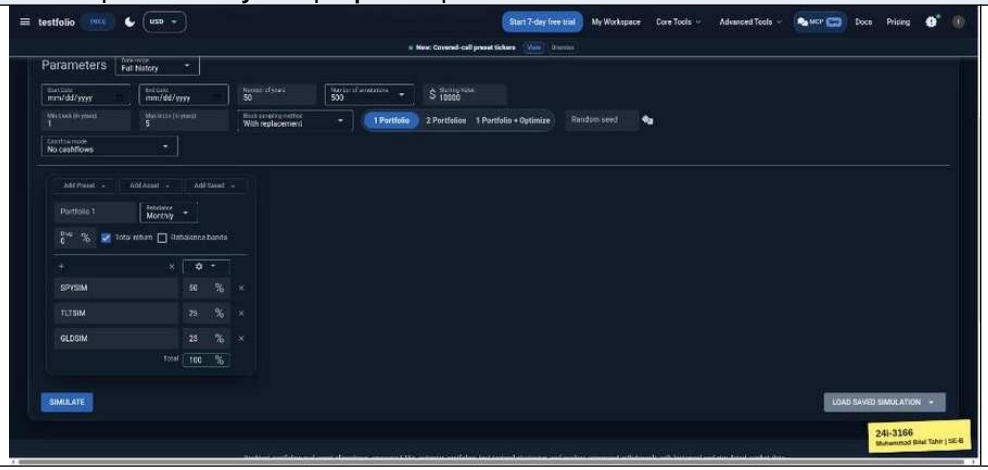


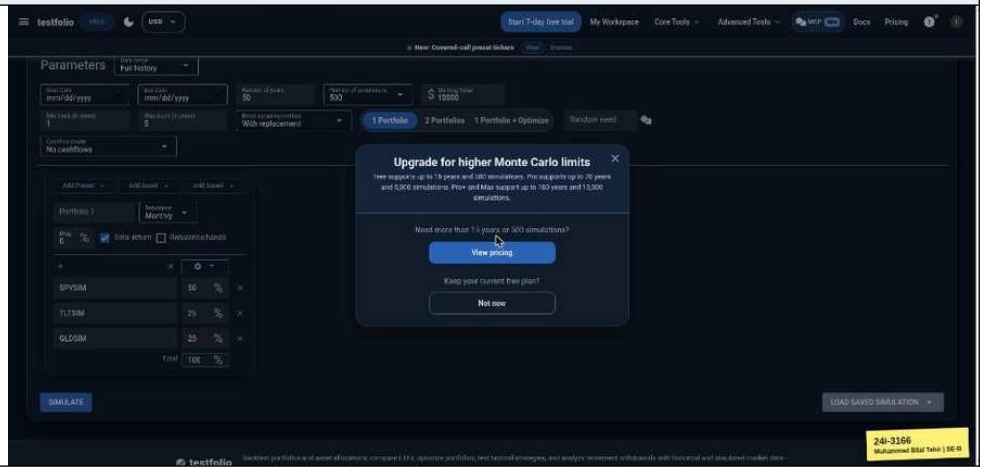


TC-EP-063 | Number of years | input: 0 | result: Pass

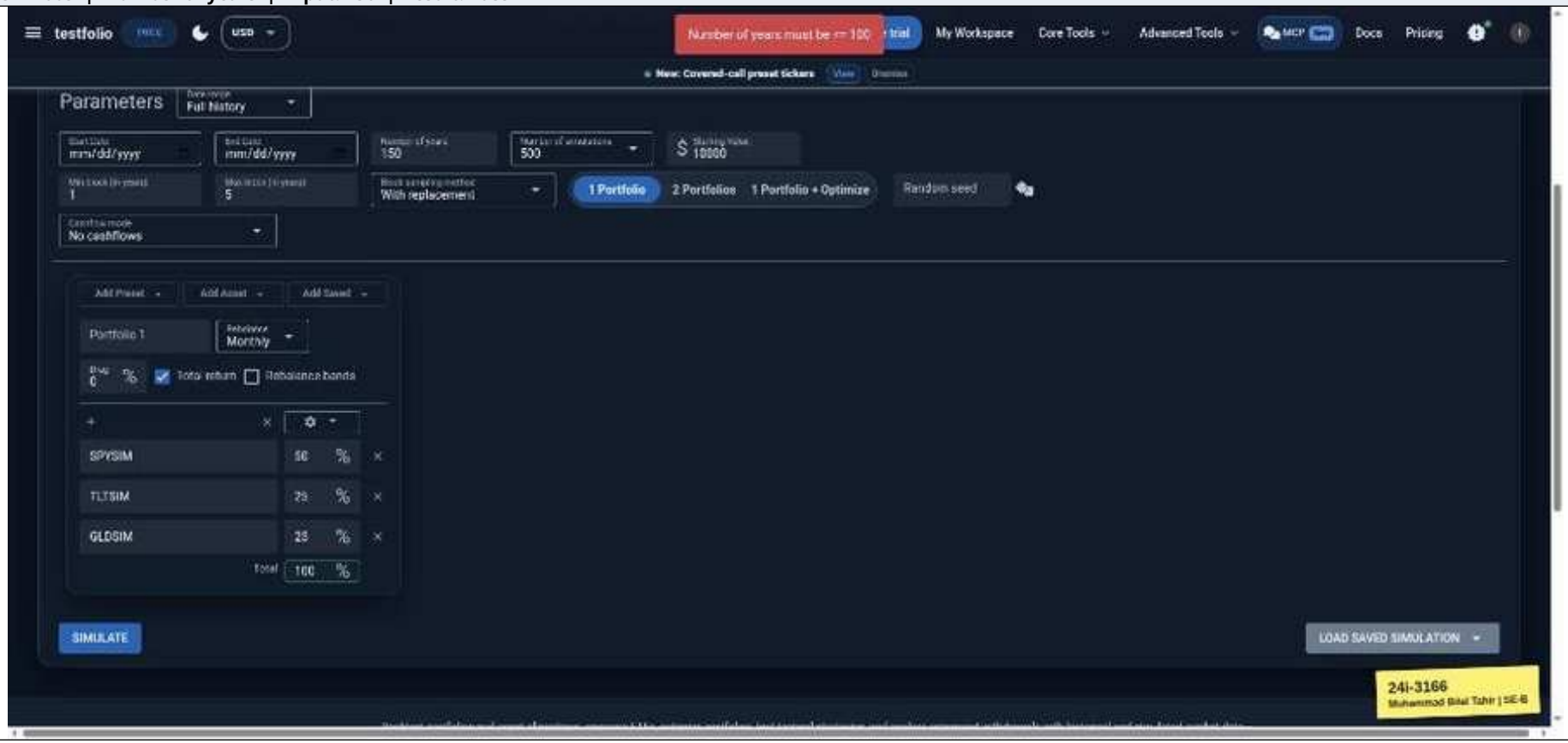


TC-EP-064 | Number of years | input: 50 | result: Pass

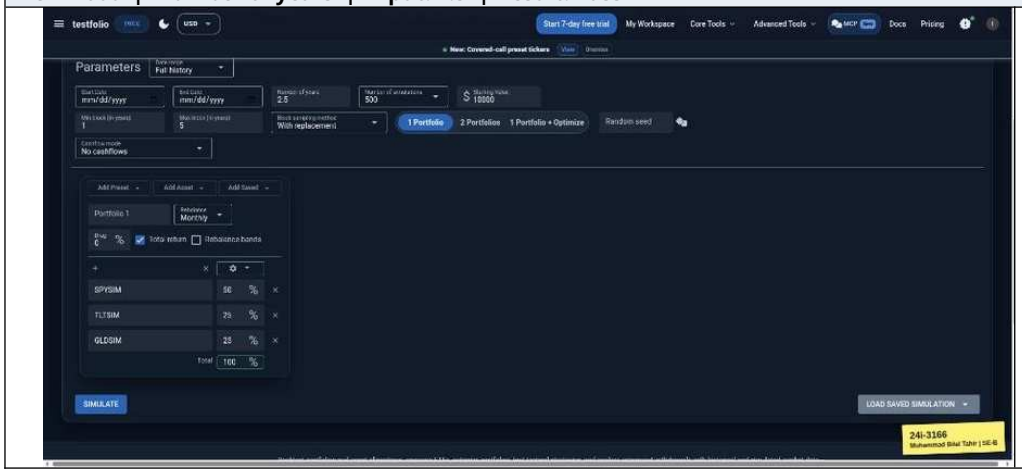


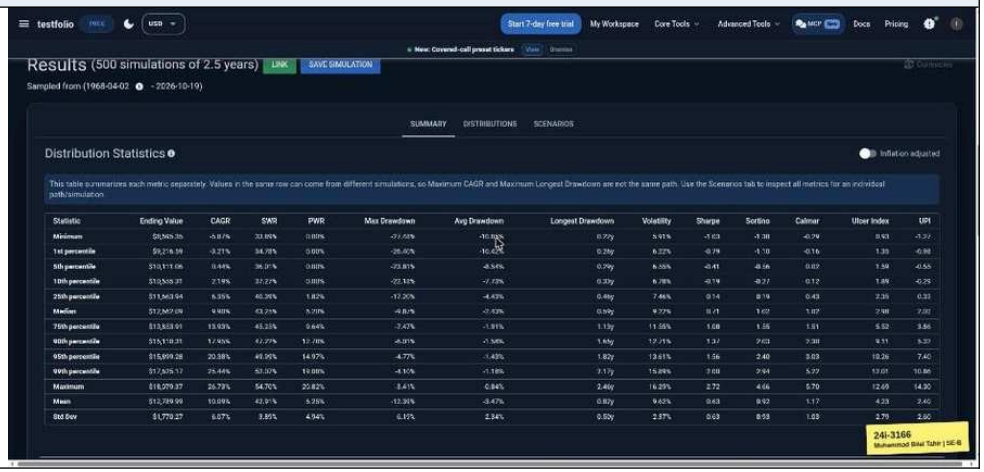


TC-EP-065 | Number of years | input: 150 | result: Pass

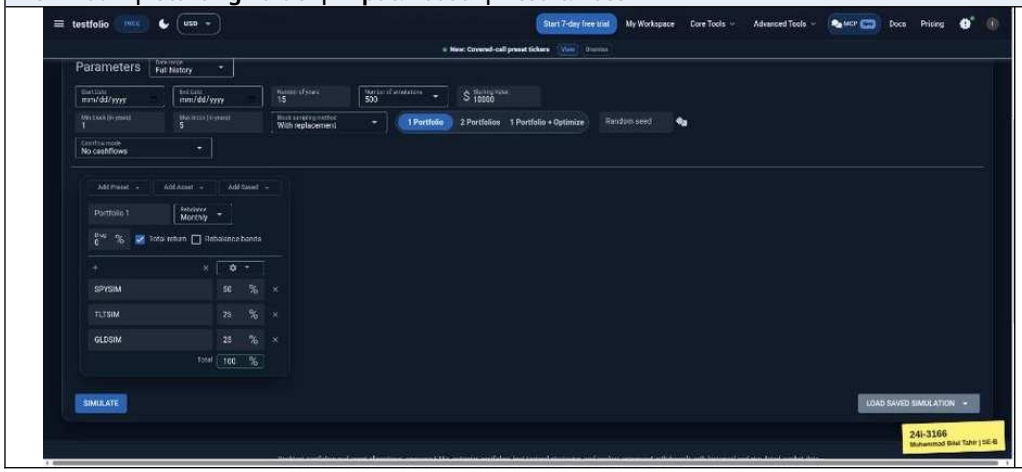


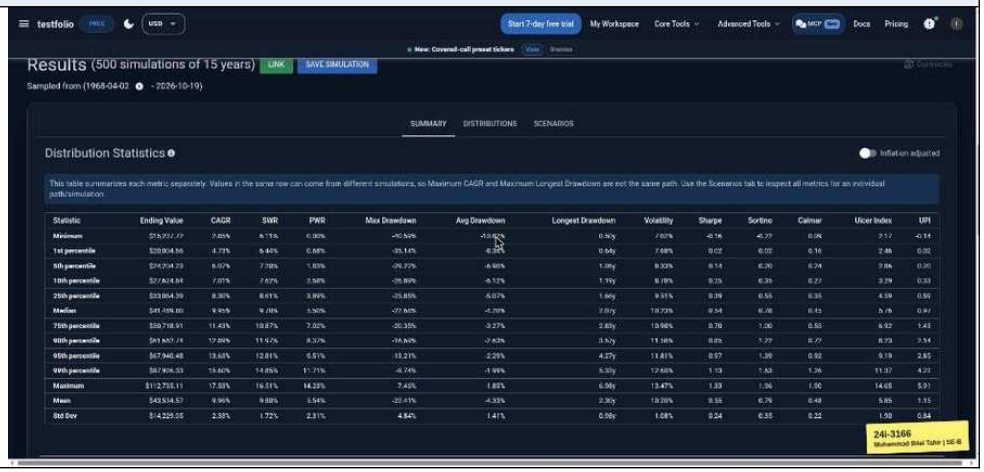
TC-EP-066 | Number of years | input: 2.5 | result: Pass



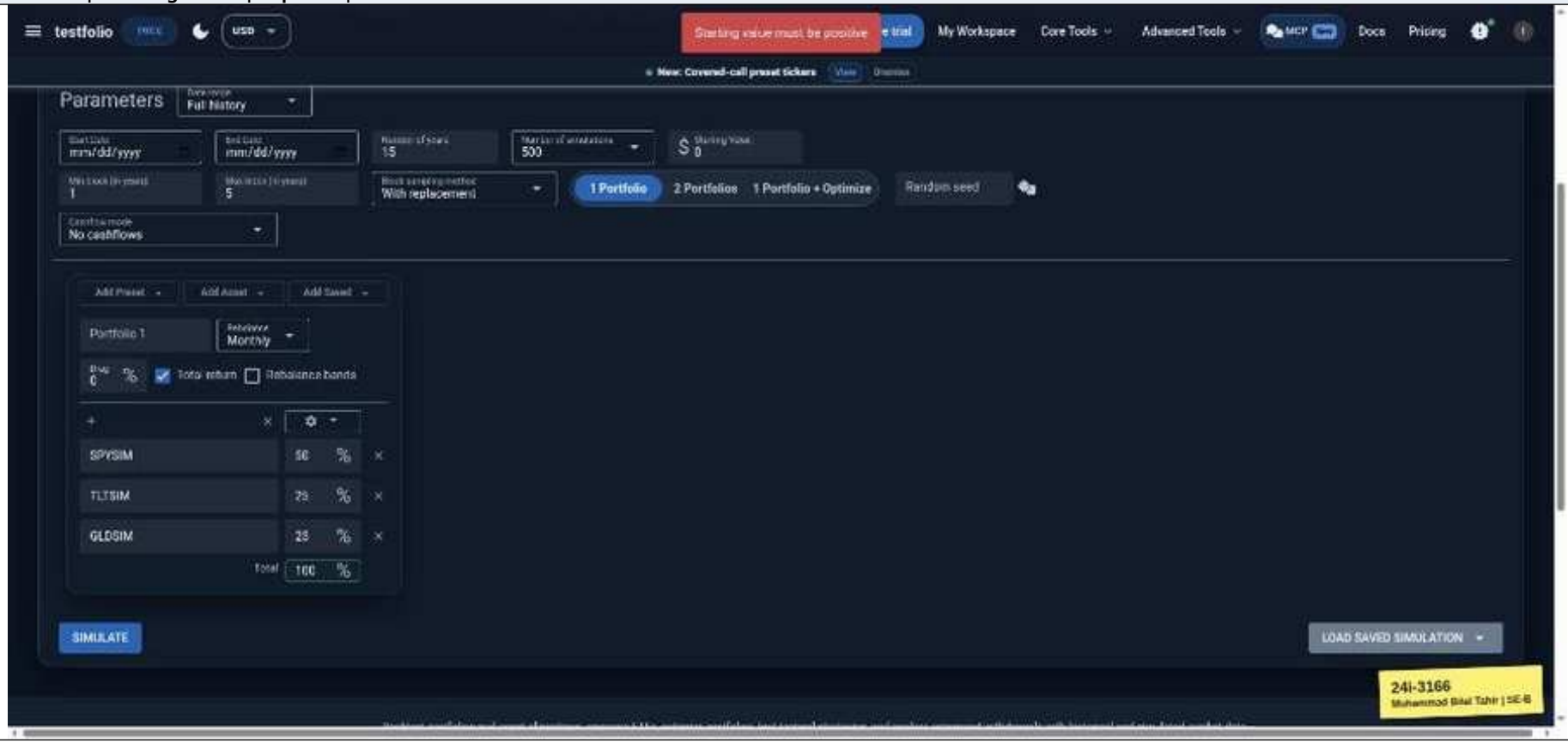


TC-EP-067 | Starting Value | input: 10000 | result: Pass

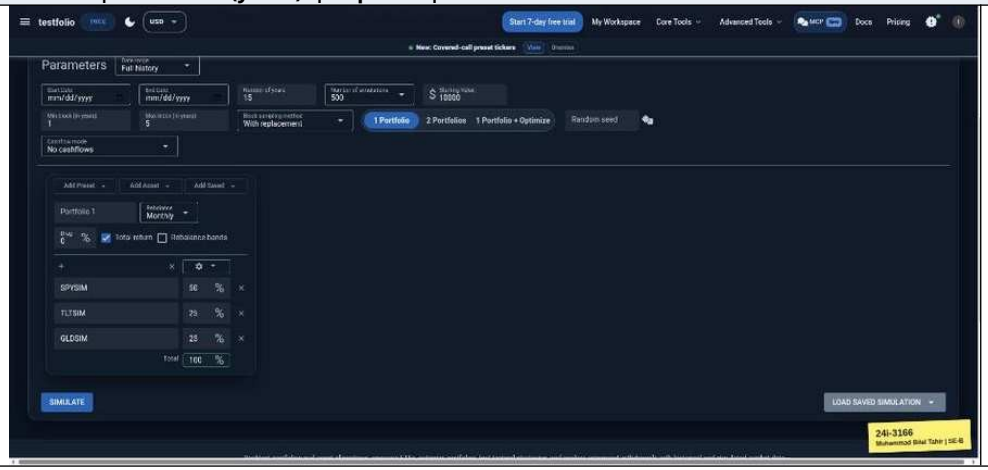


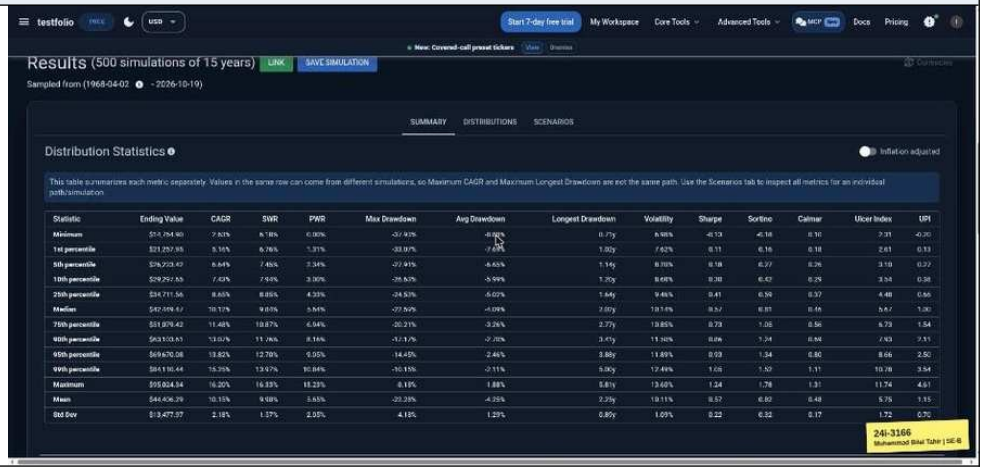


TC-EP-068 | Starting Value | input: 0 | result: Pass

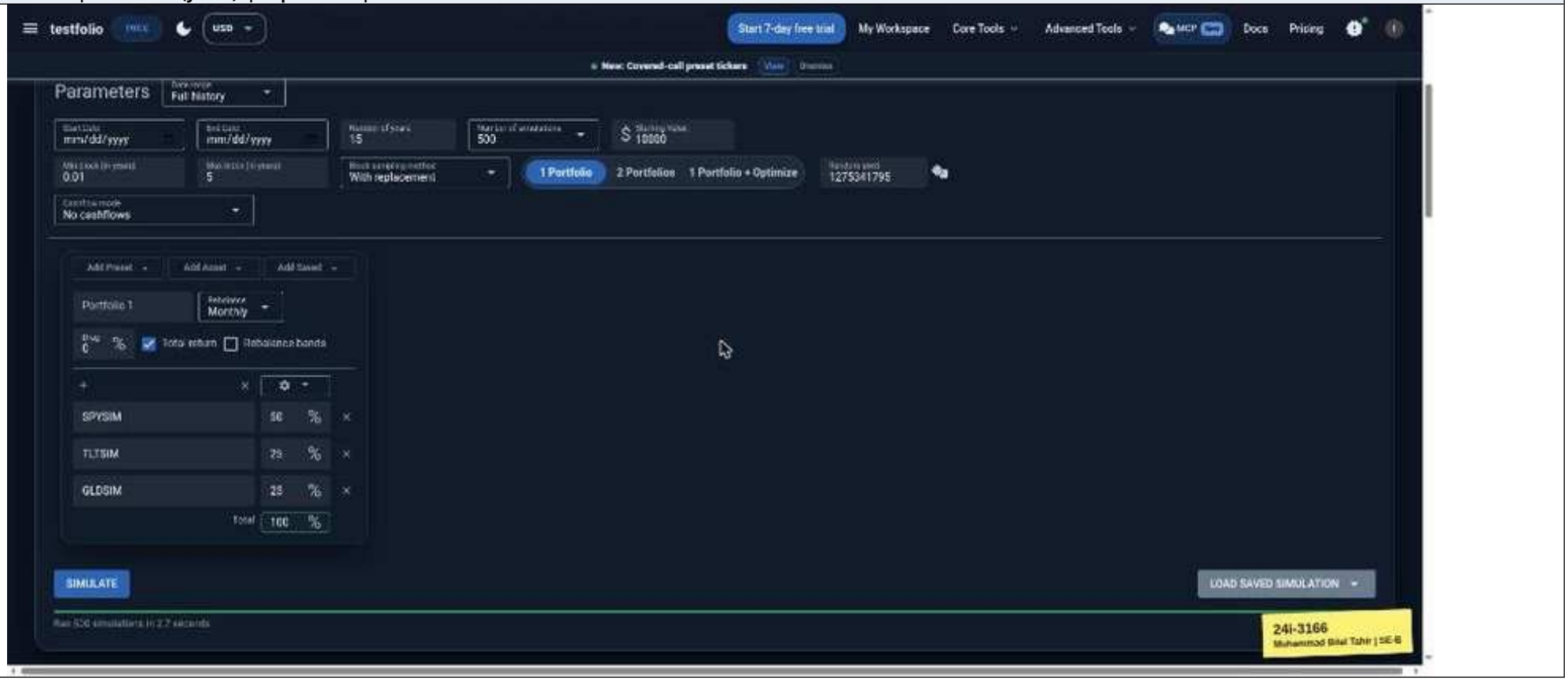


TC-EP-069 | Min block (years) | input: 1 | result: Pass

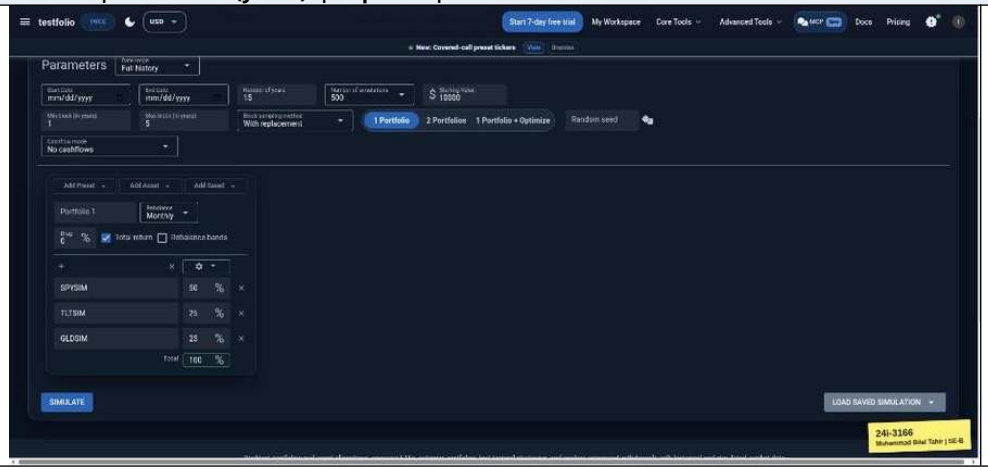


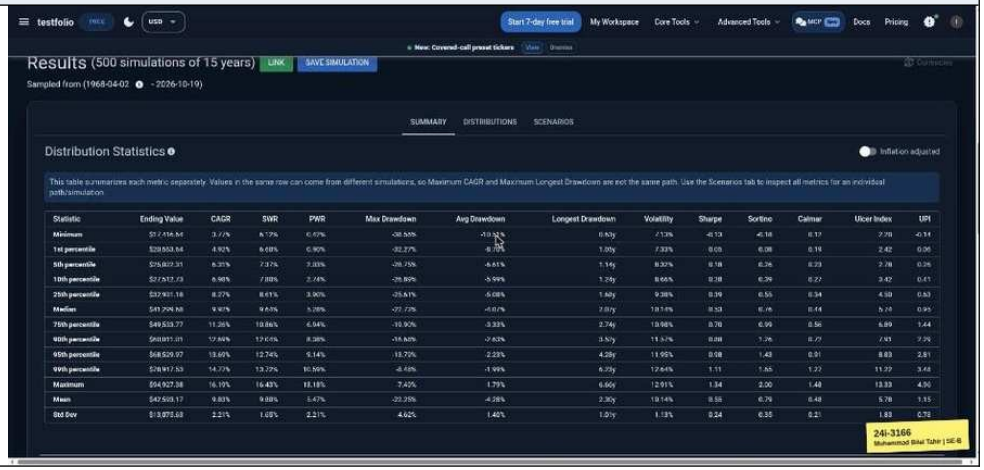


TC-EP-070 | Min block (years) | input: 0.01 | result: Fail

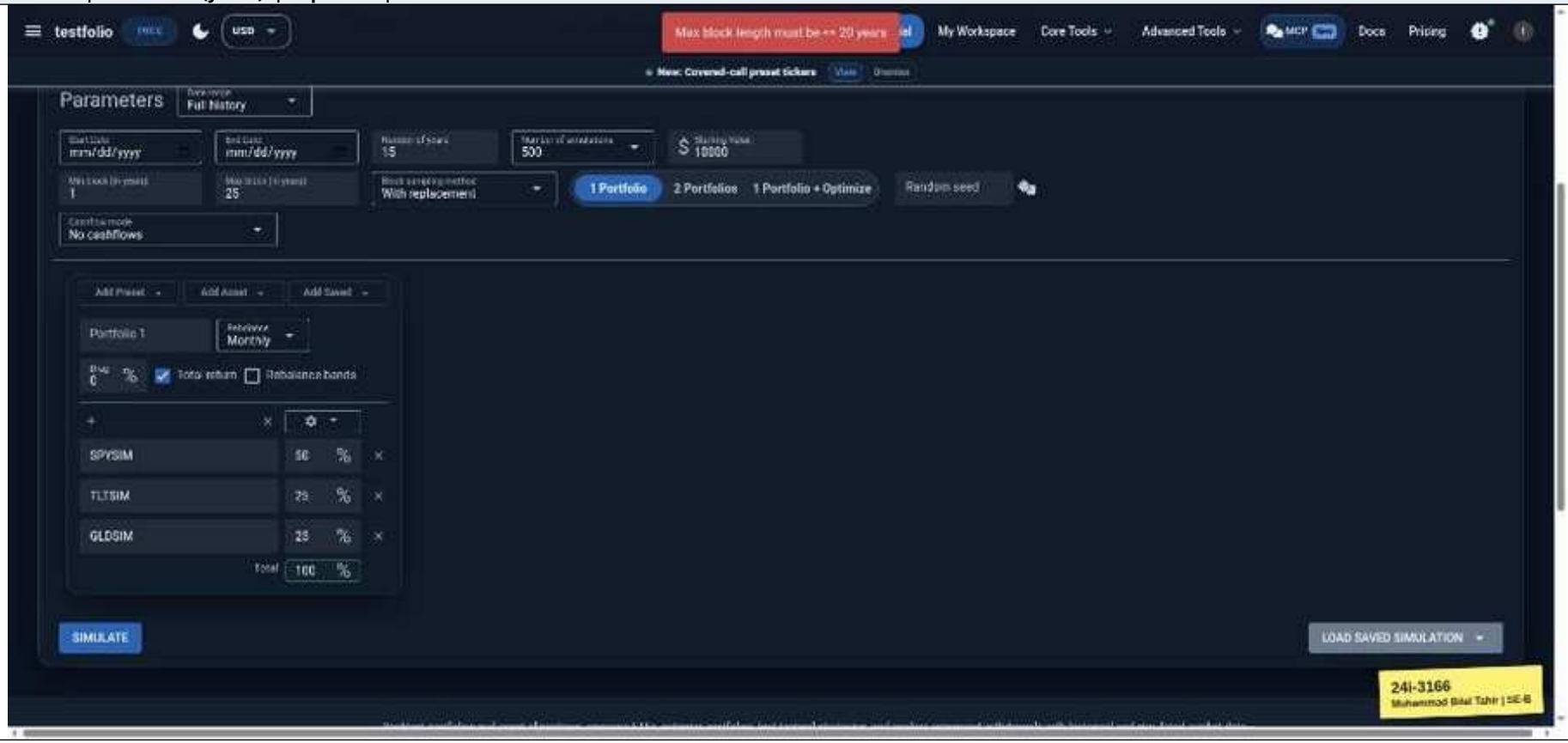


TC-EP-071 | Max block (years) | input: 5 | result: Pass

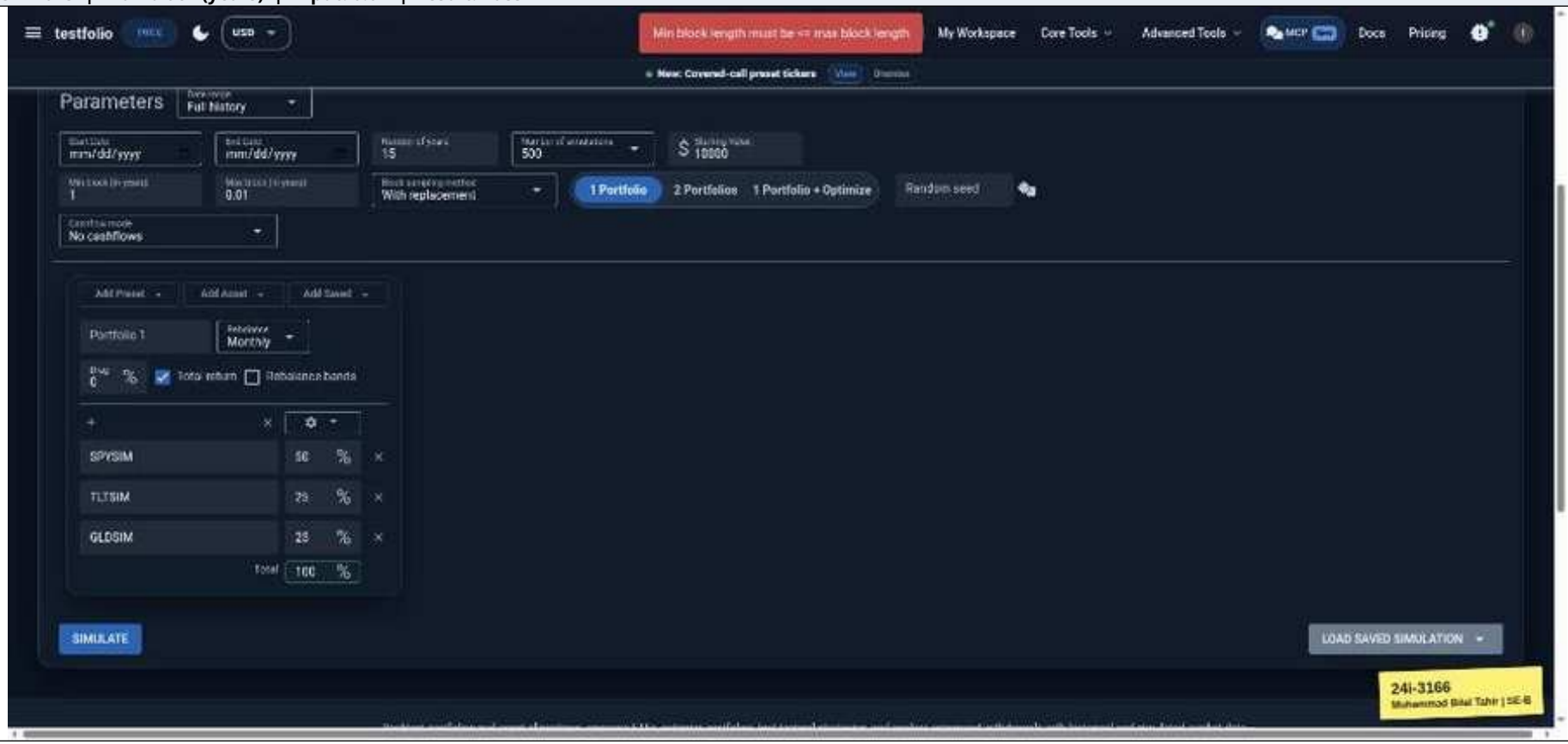




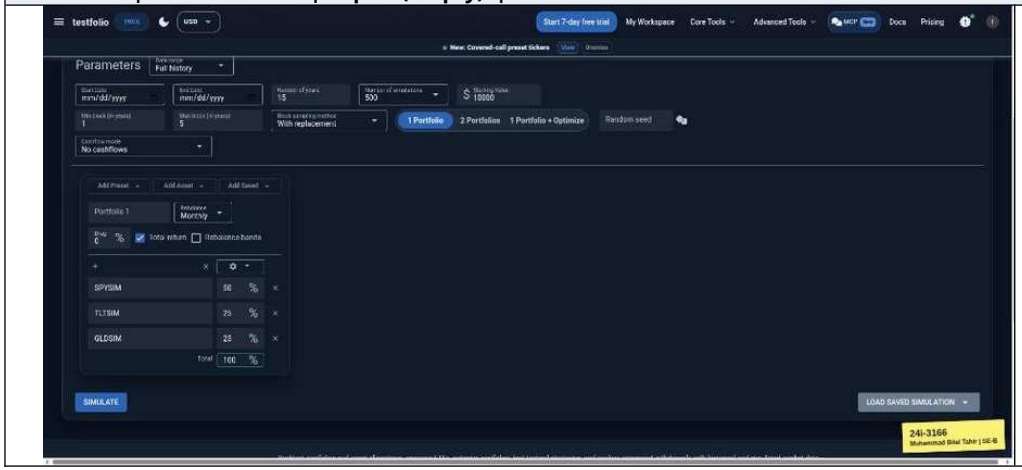
TC-EP-072 | Max block (years) | input: 25 | result: Pass

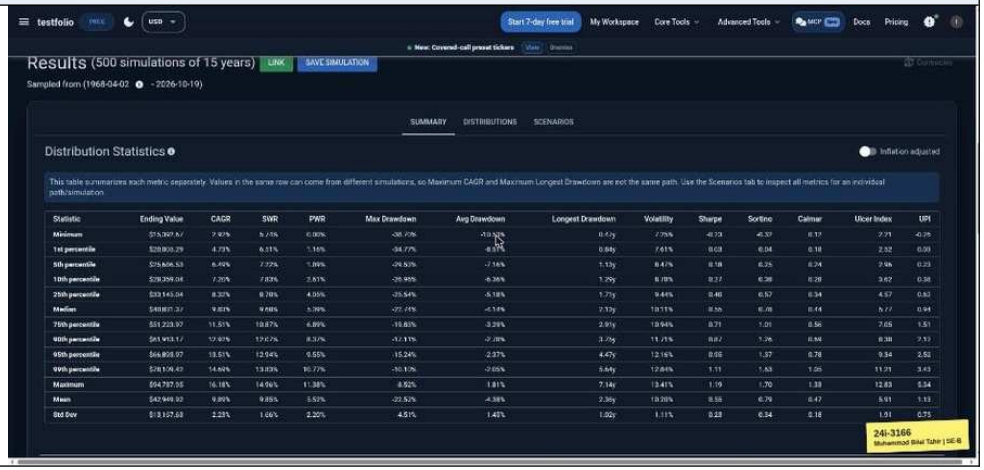


TC-EP-073 | Max block (years) | input: 0.01 | result: Pass

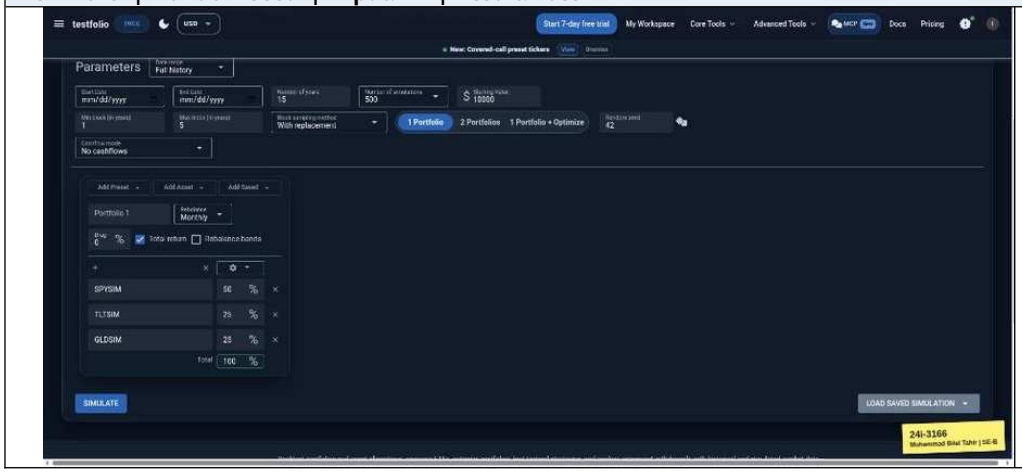


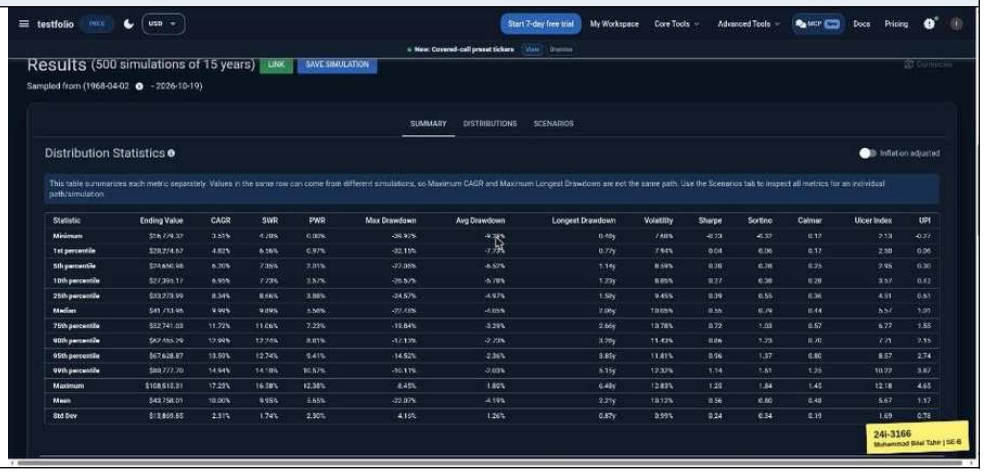
TC-EP-074 | Random seed | input: (empty) | result: Pass



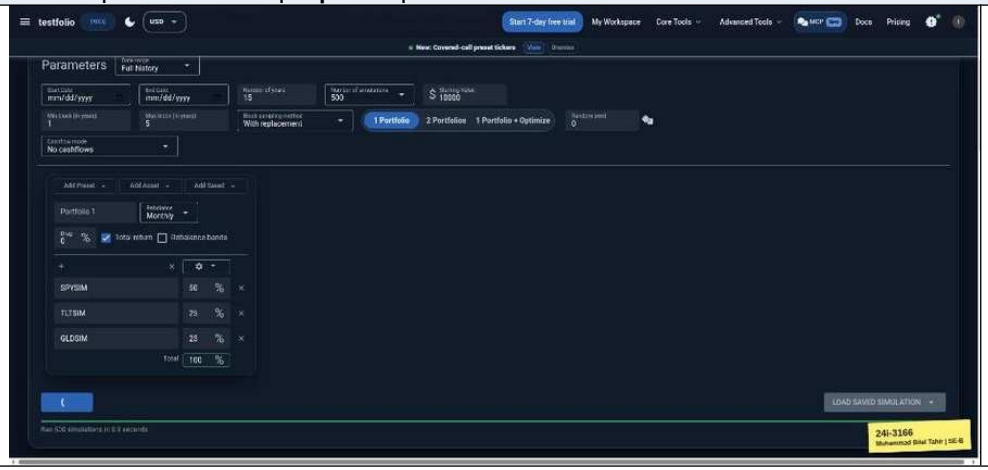


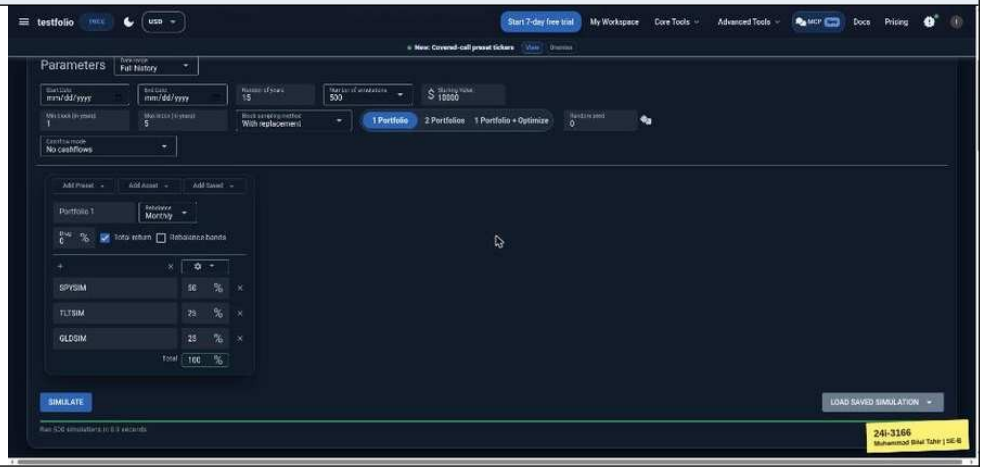
TC-EP-075 | Random seed | input: 42 | result: Pass



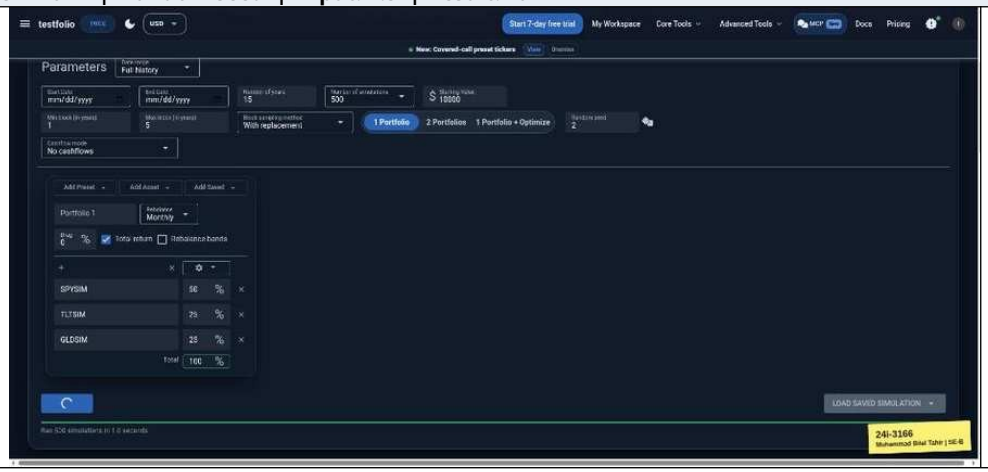


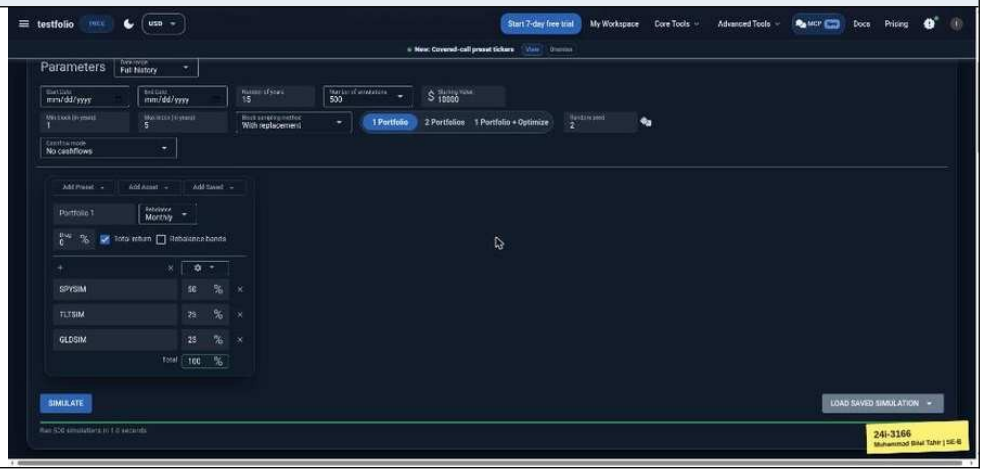
TC-EP-076 | Random seed | input: -1 | result: Fail



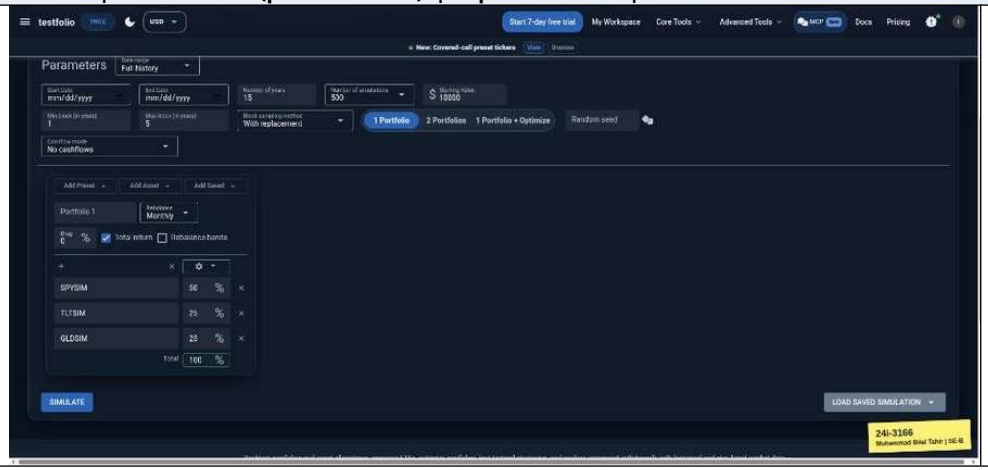


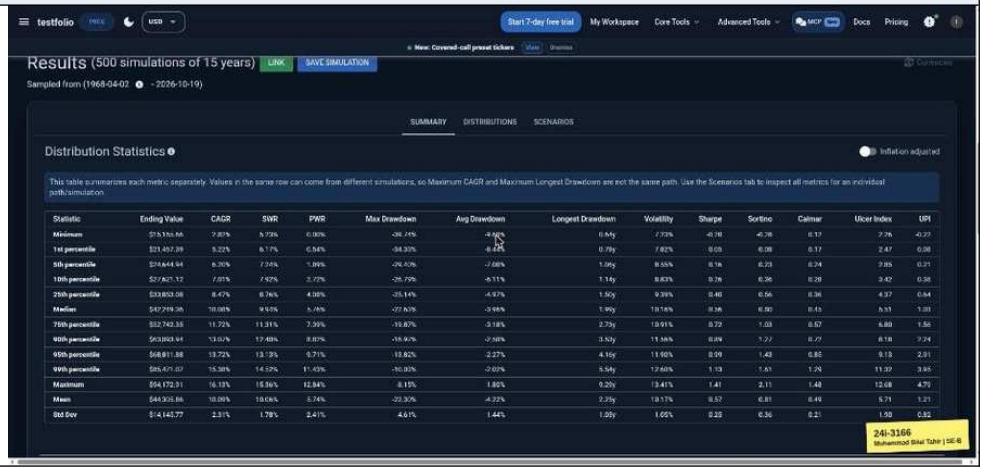
TC-EP-077 | Random seed | input: 1.5 | result: Fail



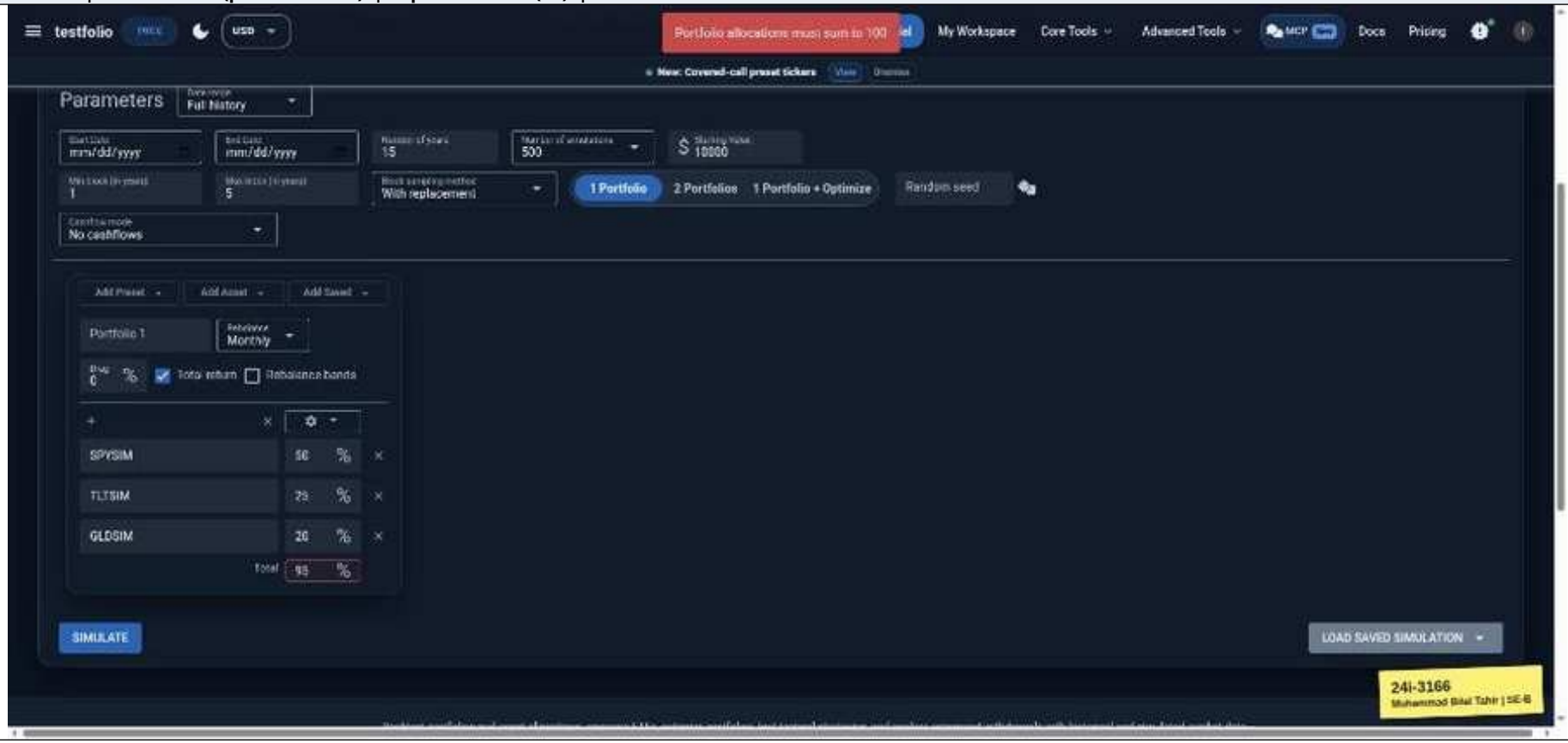


TC-EP-078 | Allocation % (portfolio total) | input: 50/25/25 | result: Pass



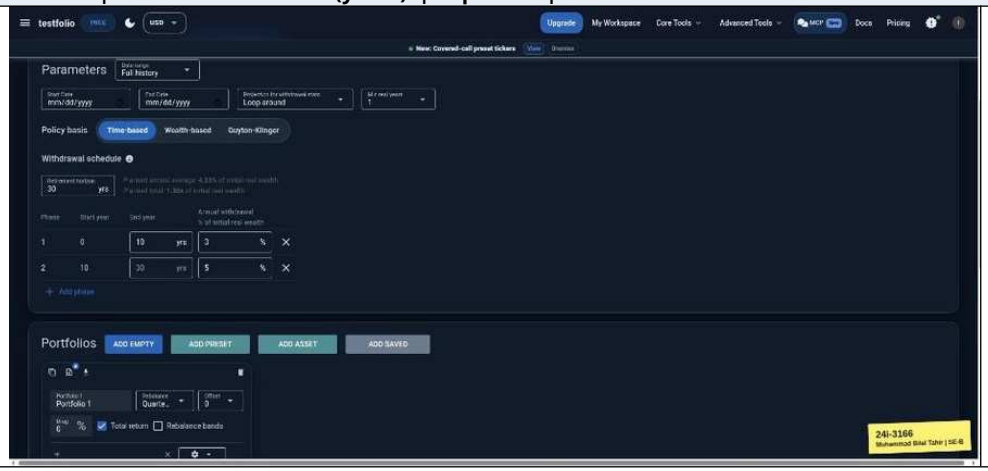


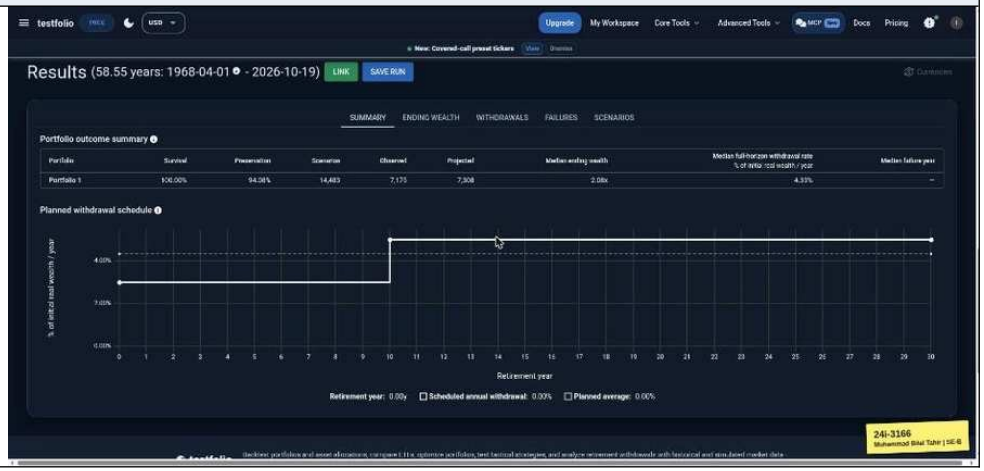
TC-EP-079 | Allocation % (portfolio total) | input: 50/25/20 (95) | result: Pass



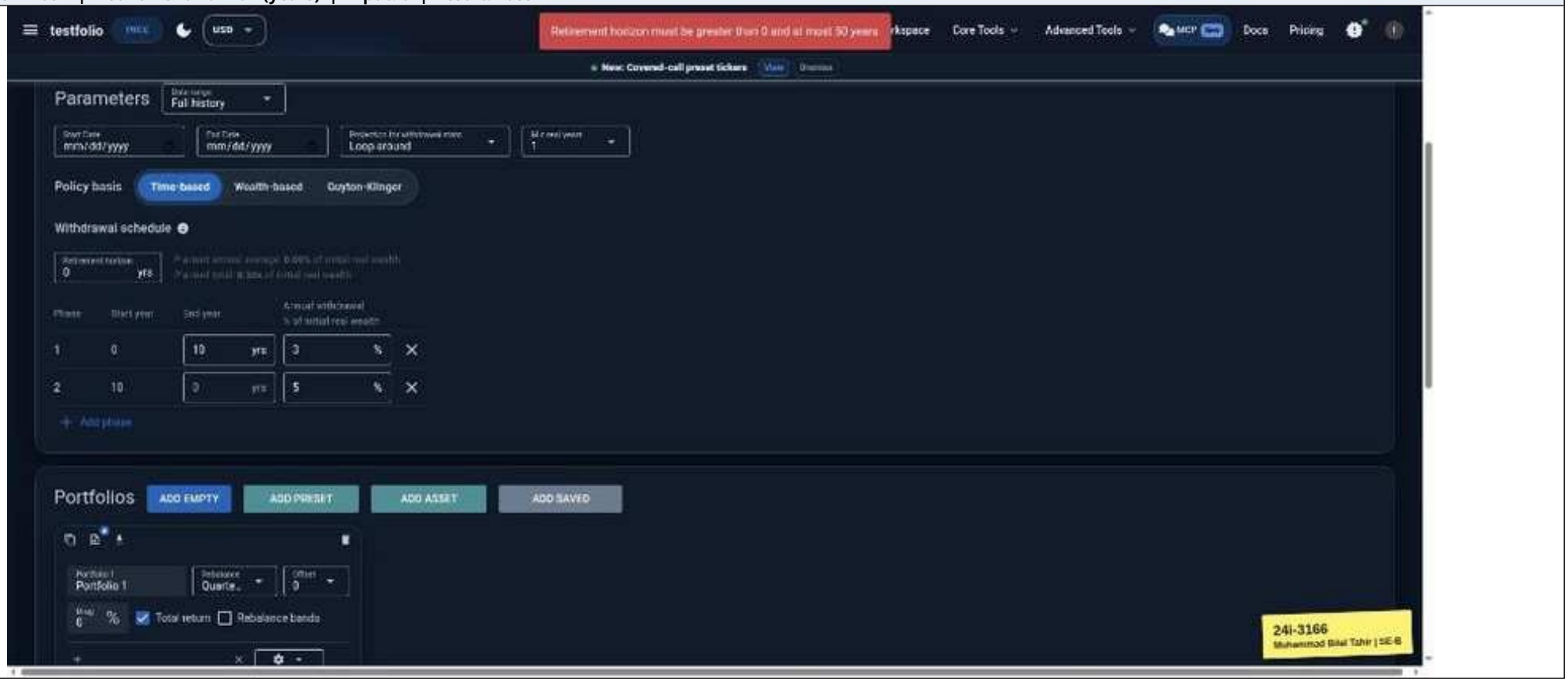
Safe Withdrawal Rate Calculator (testfol.io/safe-withdrawal-rate-calculator)

TC-EP-080 | Retirement horizon (years) | input: 30 | result: Pass

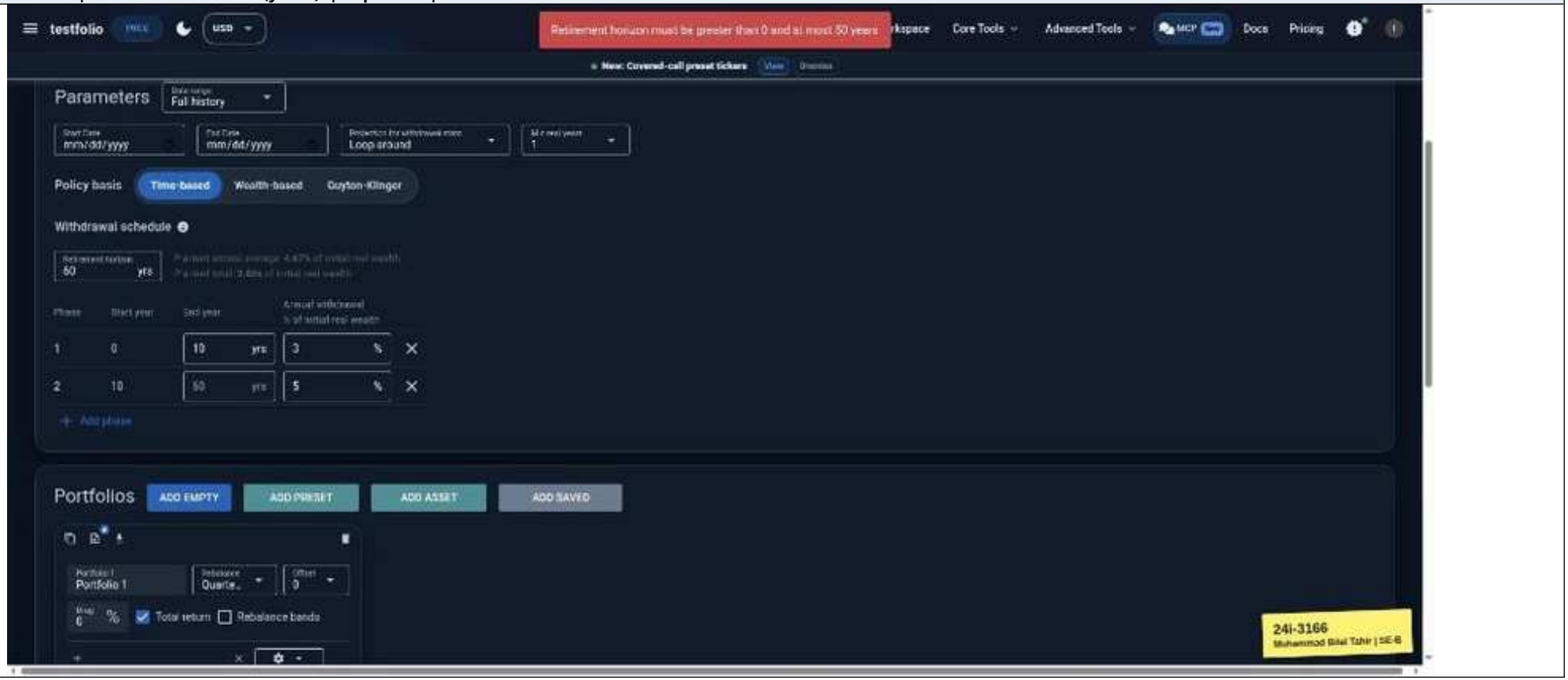




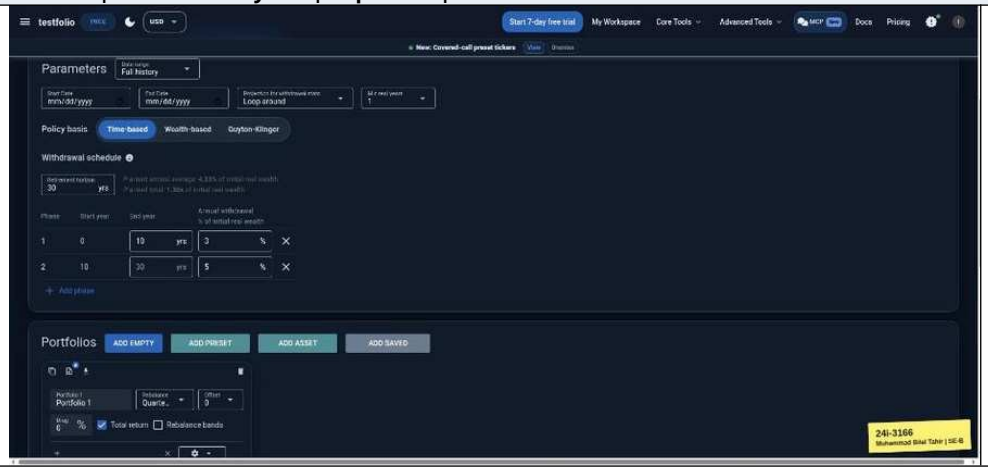
TC-EP-081 | Retirement horizon (years) | input: 0 | result: Pass

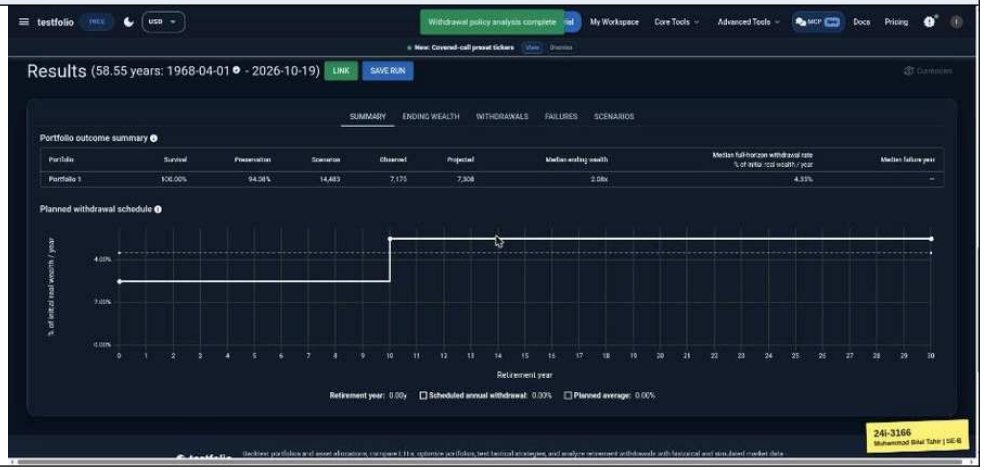


TC-EP-082 | Retirement horizon (years) | input: 60 | result: Pass

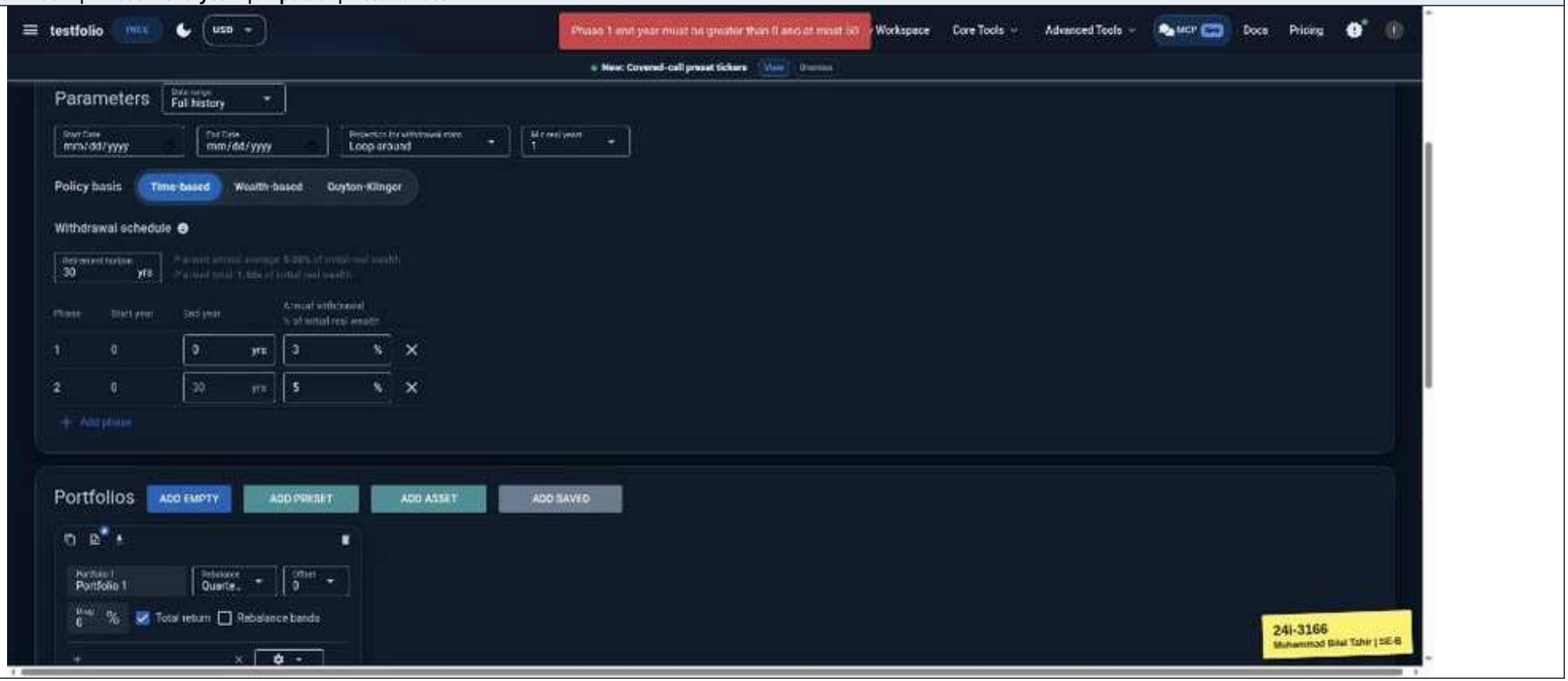


TC-EP-083 | Phase 1 end year | input: 10 | result: Pass

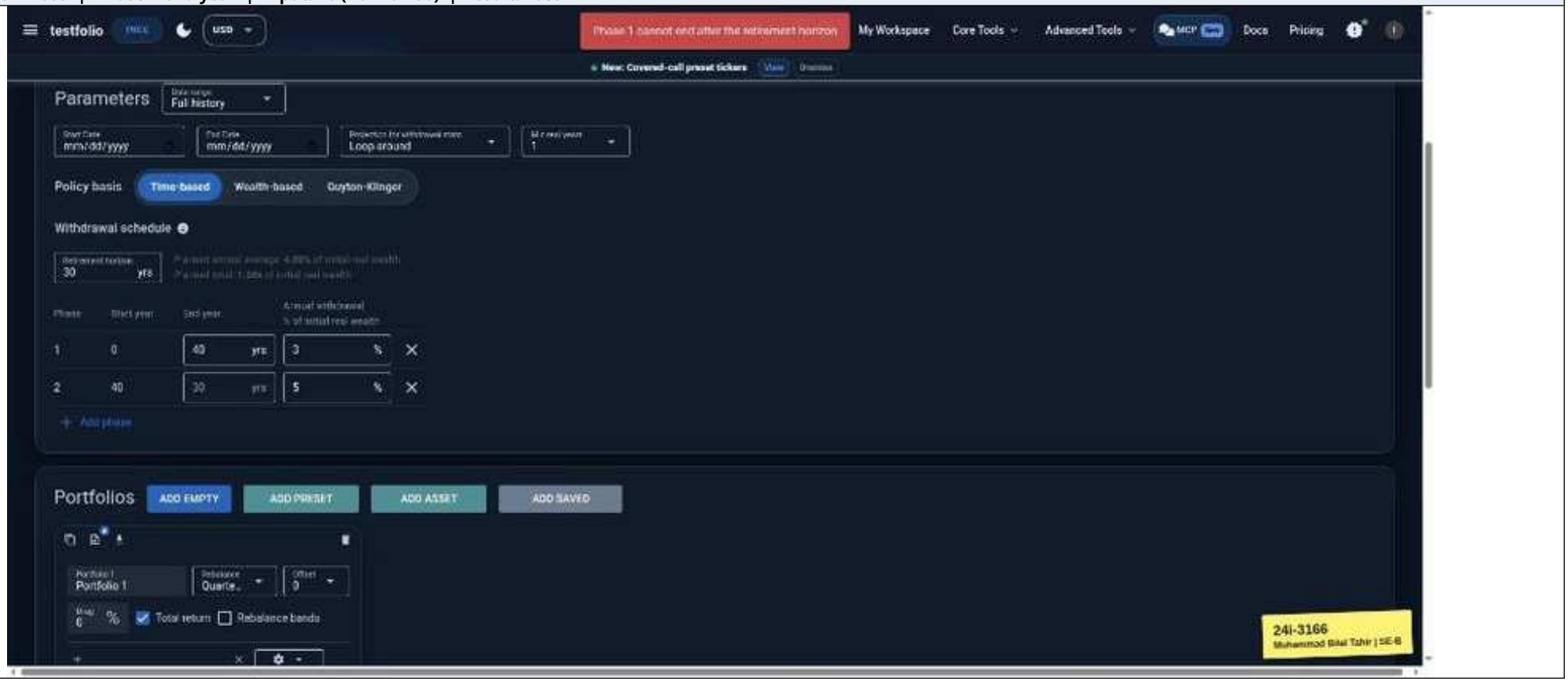




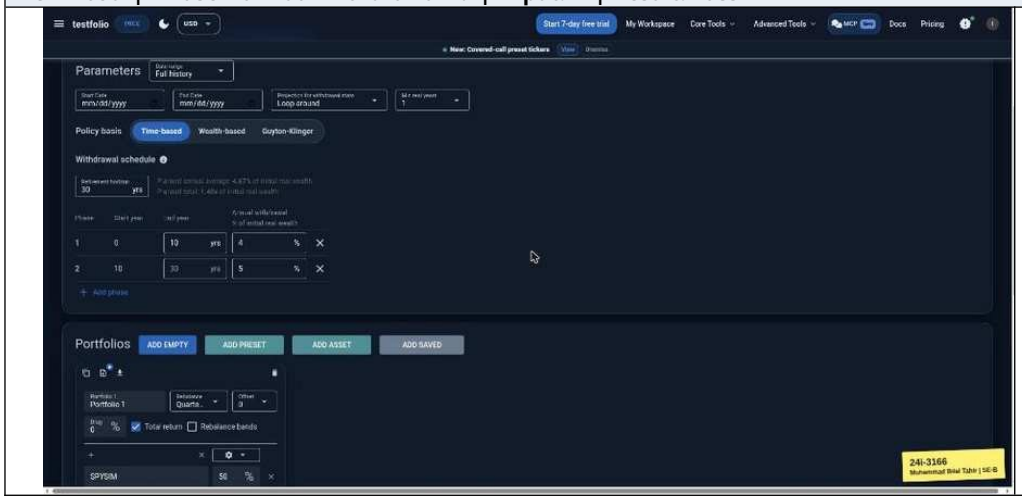
TC-EP-084 | Phase 1 end year | input: 0 | result: Pass

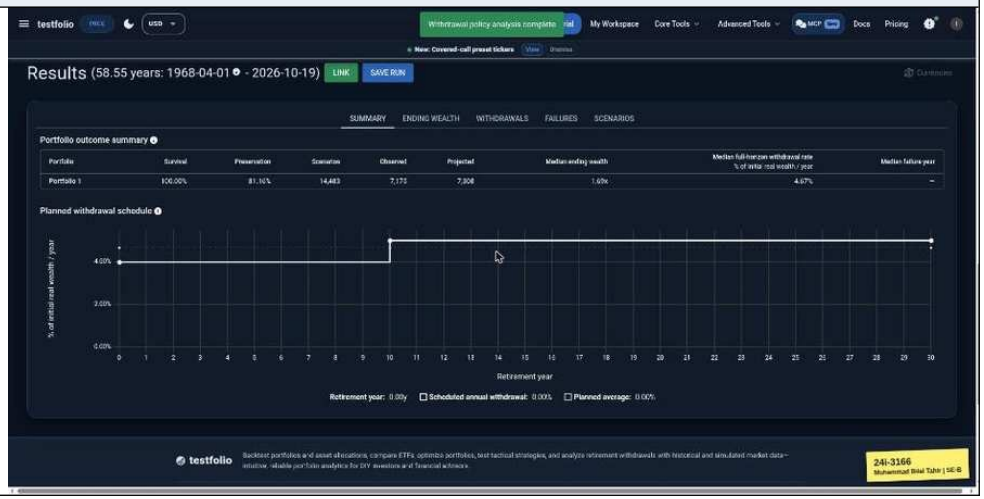


TC-EP-085 | Phase 1 end year | input: 40 (horizon 30) | result: Pass

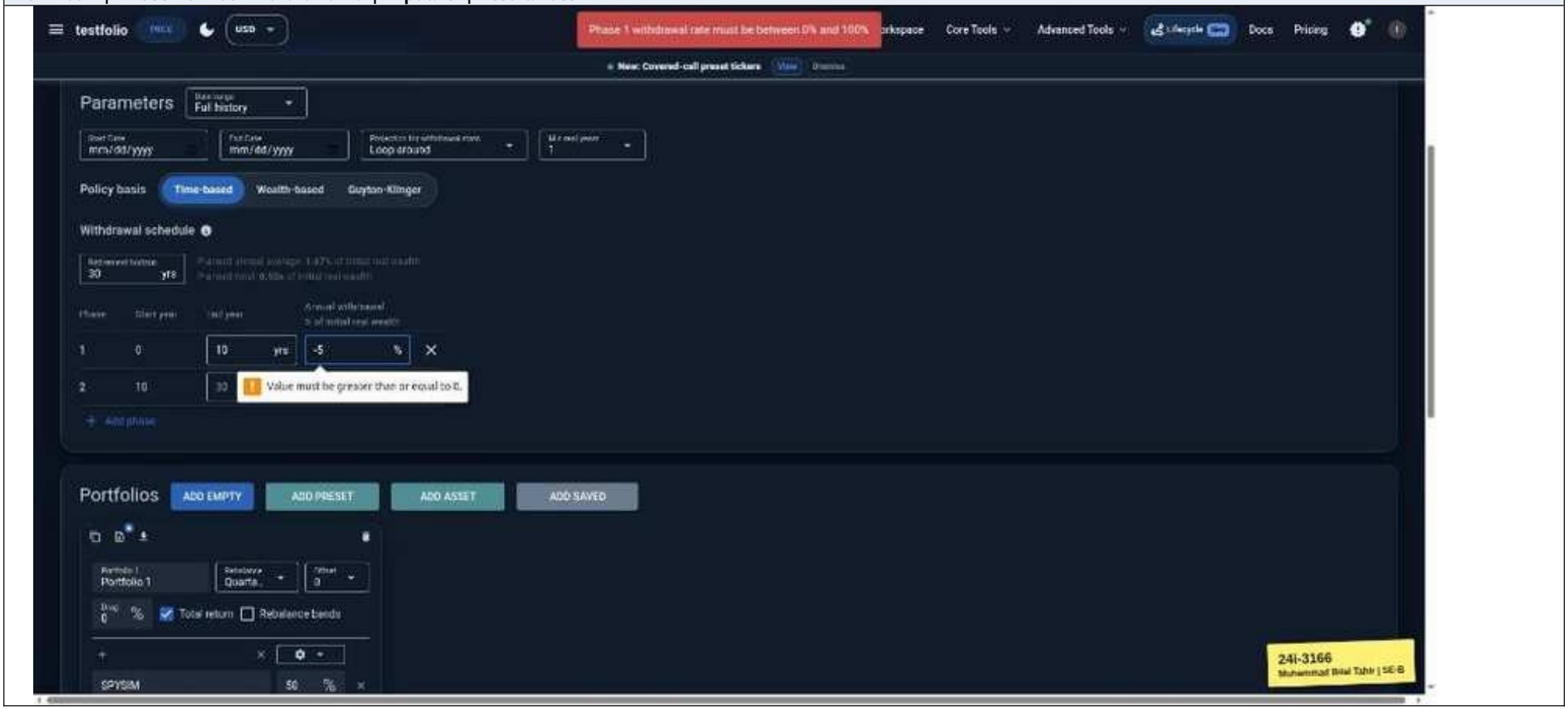


TC-EP-086 | Phase 1 annual withdrawal % | input: 4 | result: Pass

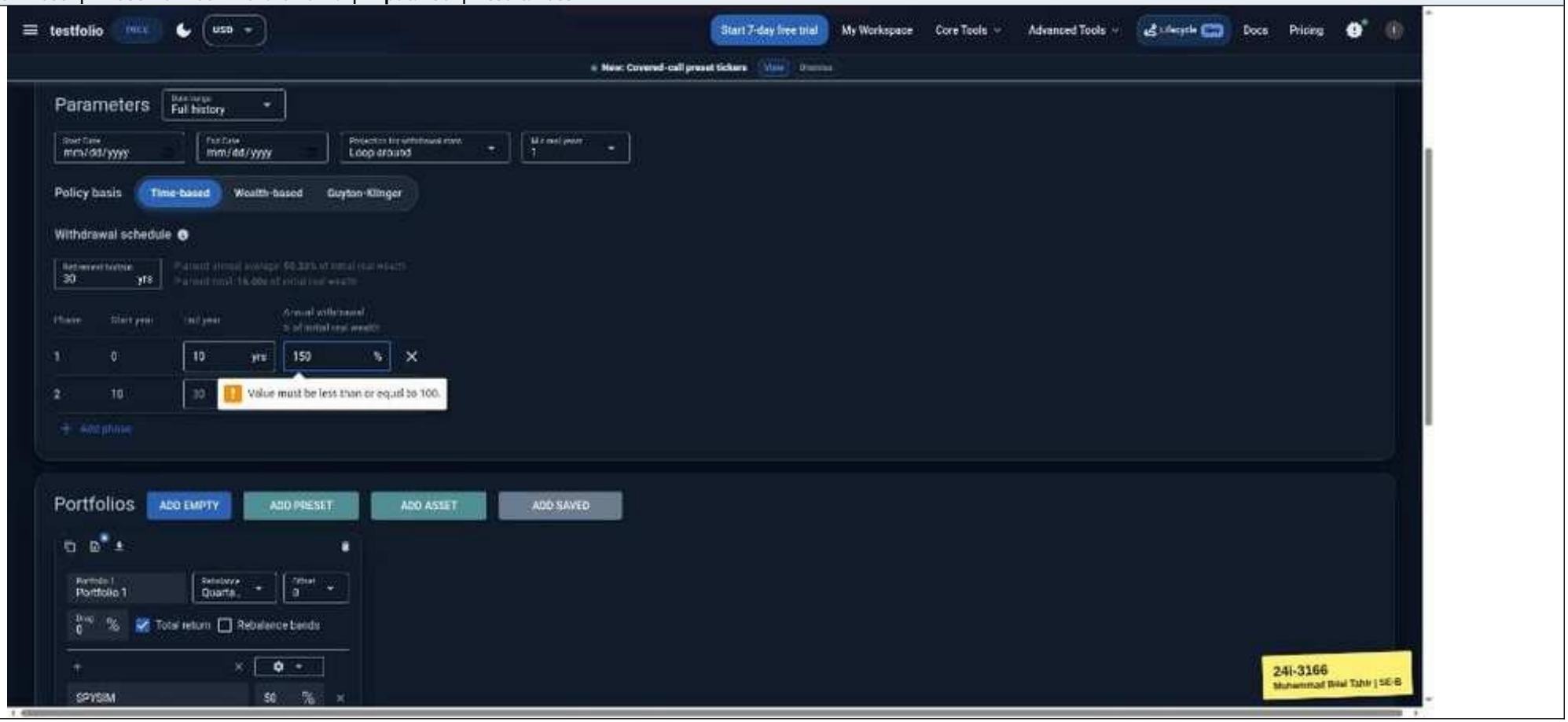




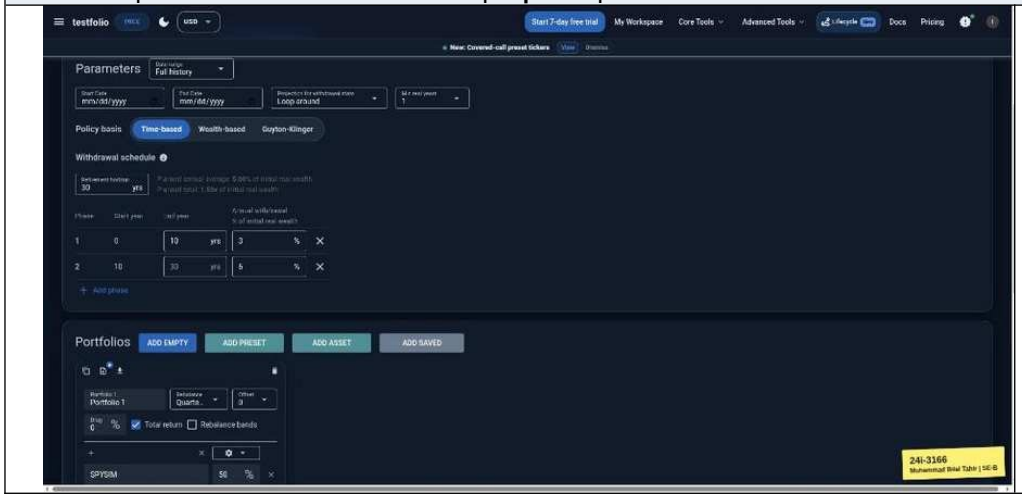
TC-EP-087 | Phase 1 annual withdrawal % | input: -5 | result: Pass

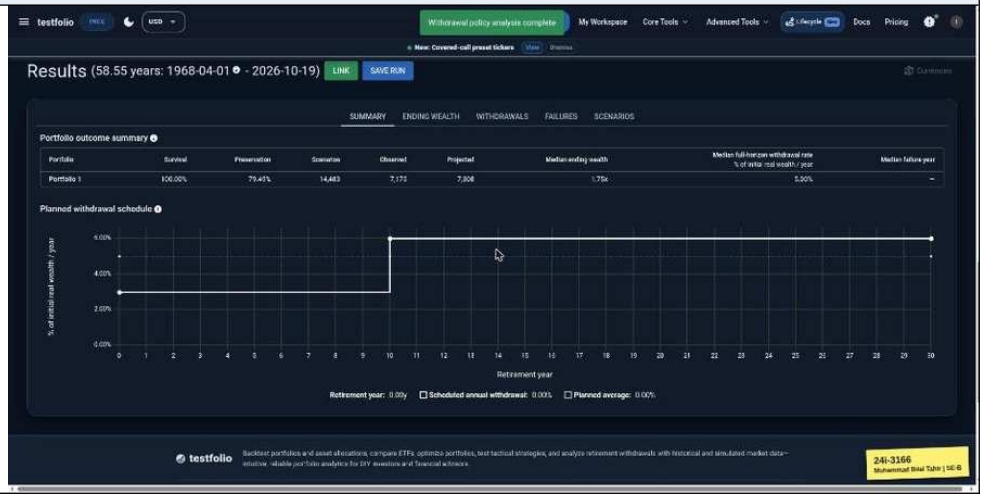


TC-EP-088 | Phase 1 annual withdrawal % | input: 150 | result: Pass

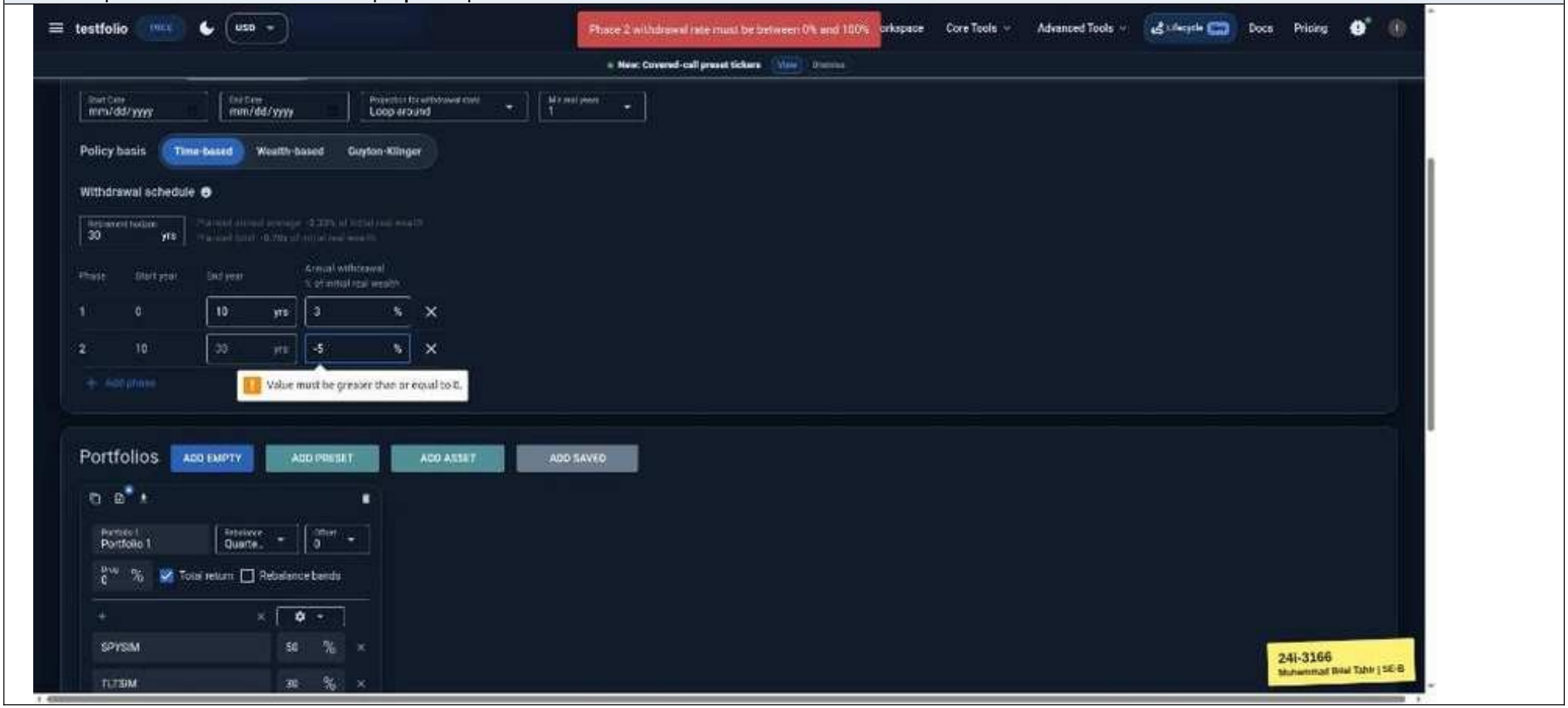


TC-EP-089 | Phase 2 annual withdrawal % | input: 6 | result: Pass

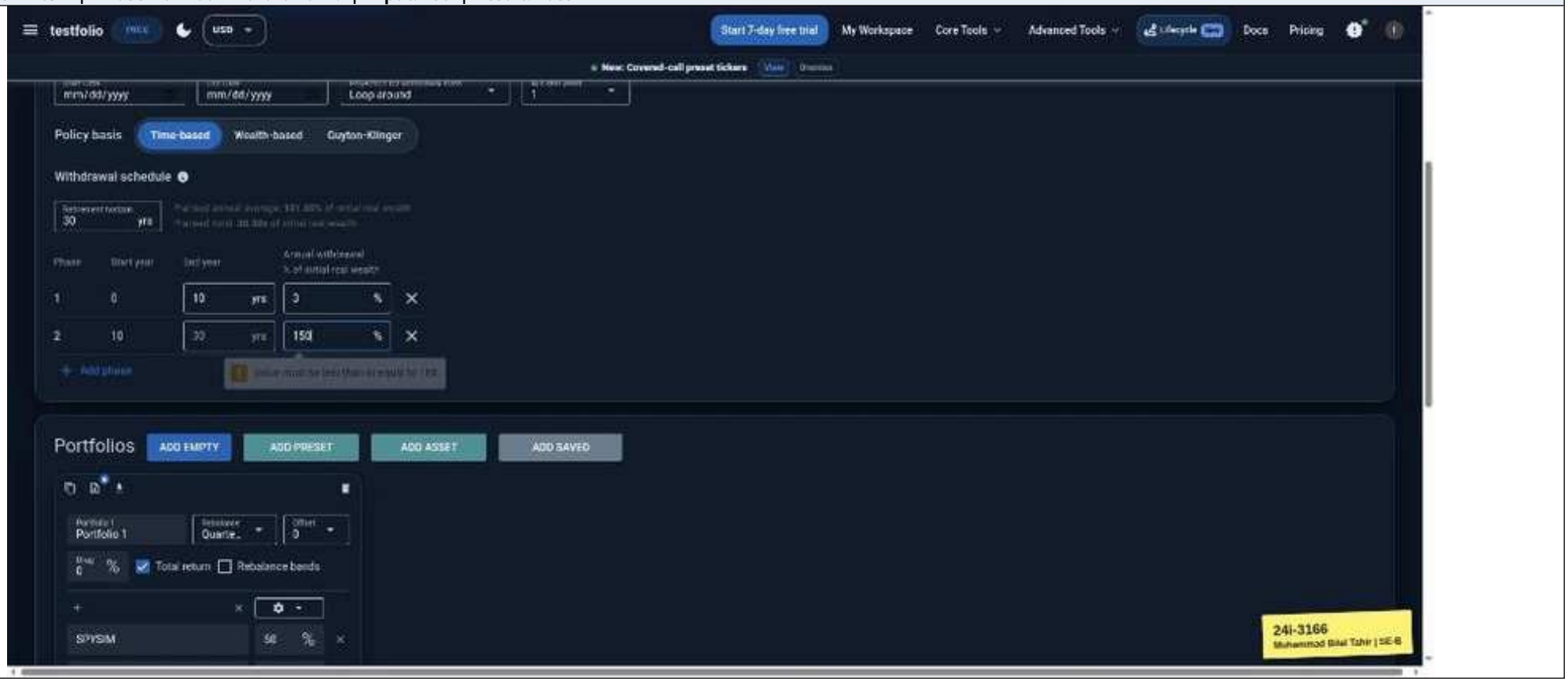




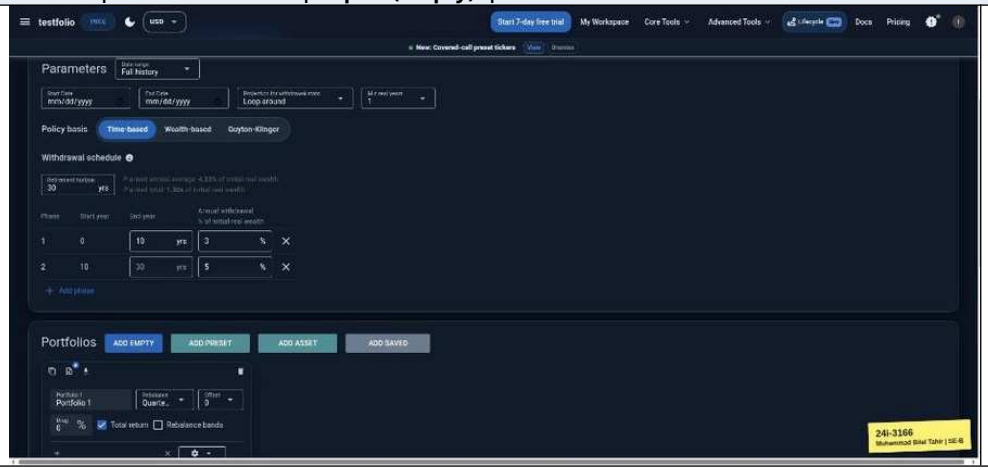
TC-EP-090 | Phase 2 annual withdrawal % | input: -5 | result: Pass

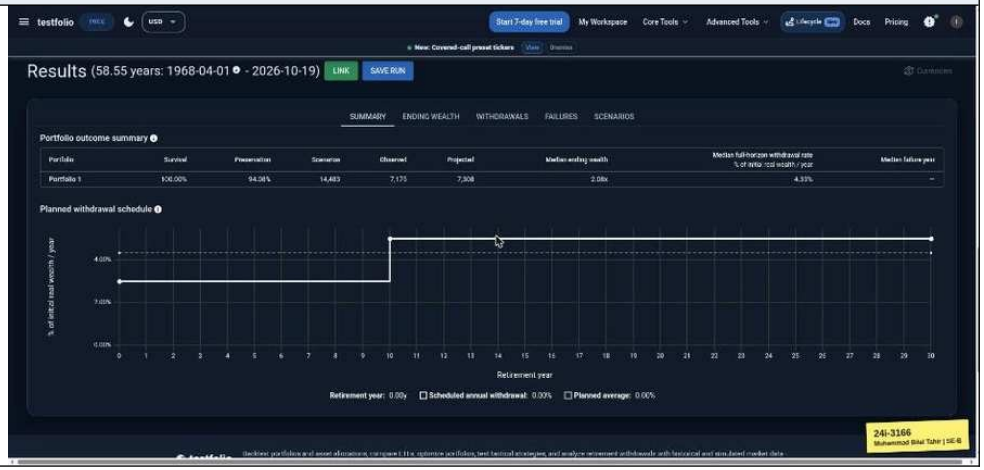


TC-EP-091 | Phase 2 annual withdrawal % | input: 150 | result: Pass

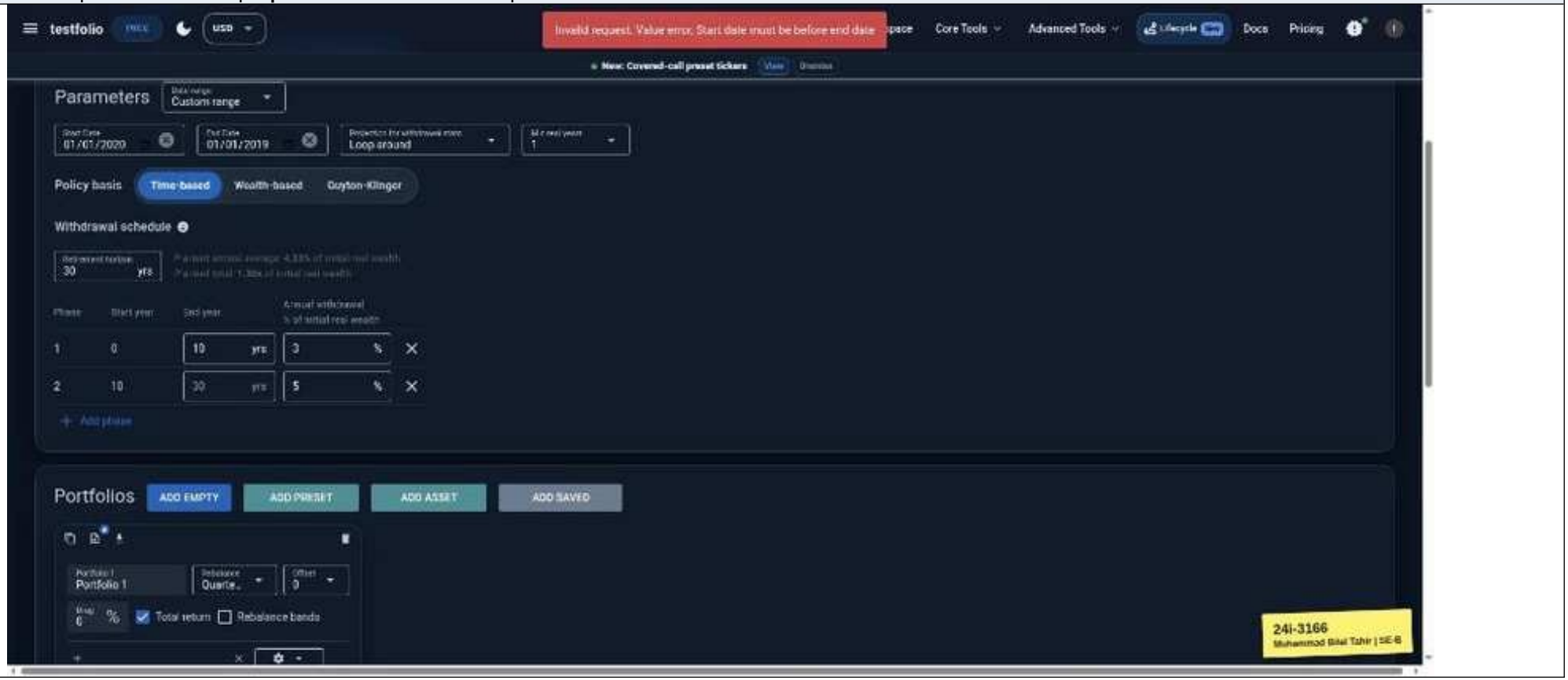


TC-EP-092 | Start / End Date | input: (empty) | result: Pass

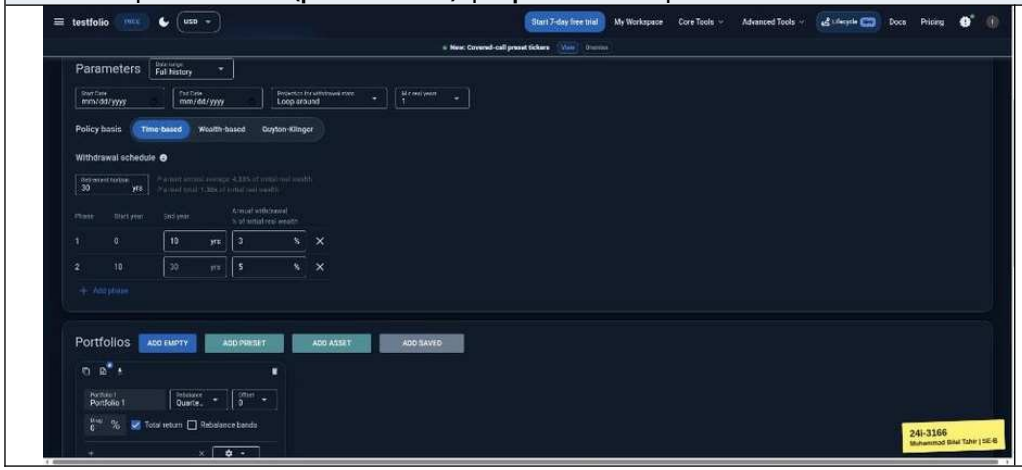


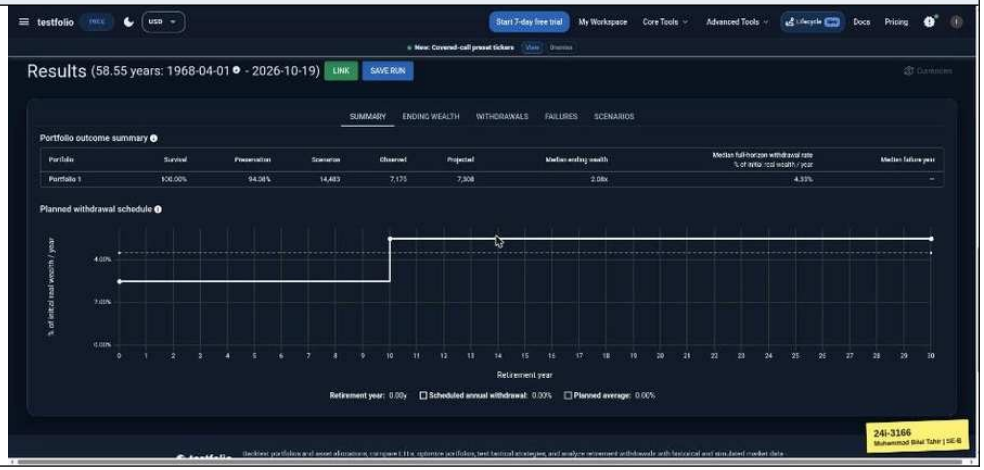


TC-EP-093 | Start / End Date | input: 2020-01-01 / 2019-01-01 | result: Pass

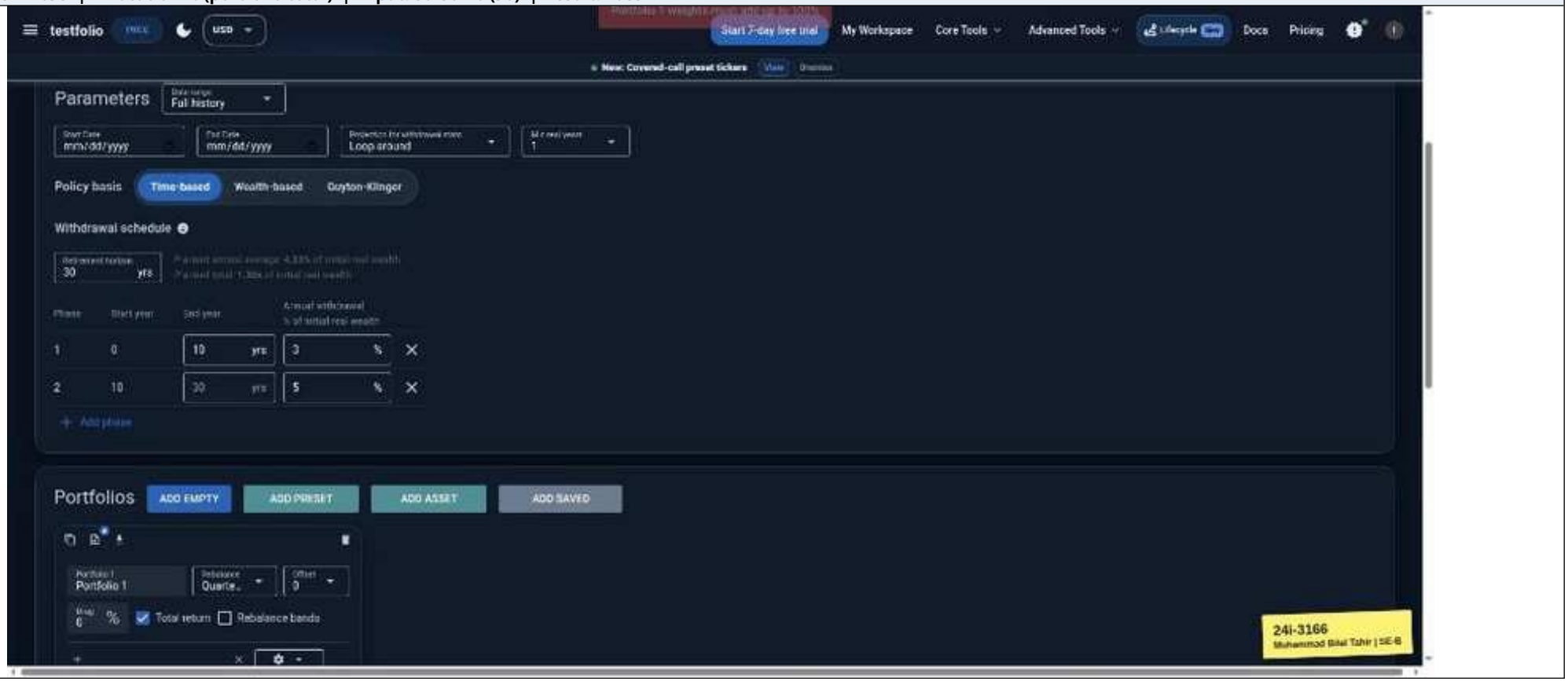


TC-EP-094 | Allocation % (portfolio total) | input: 50/30/20 | result: Pass



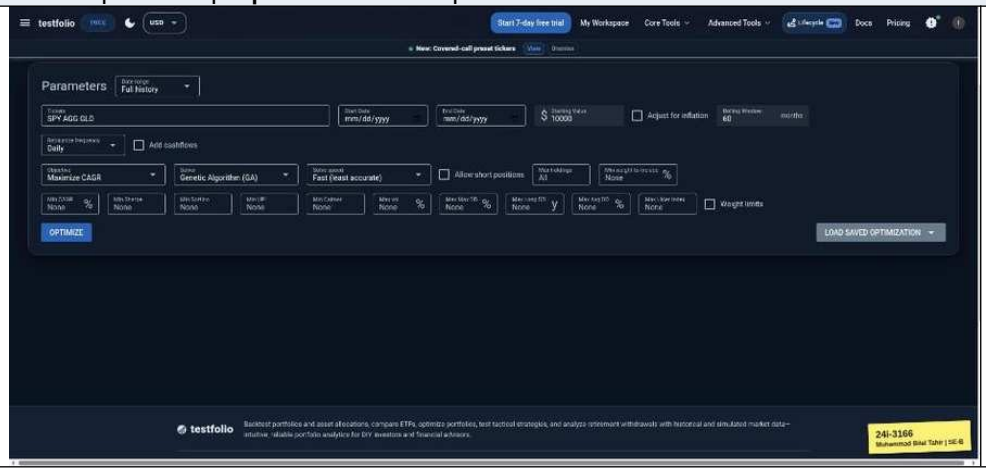


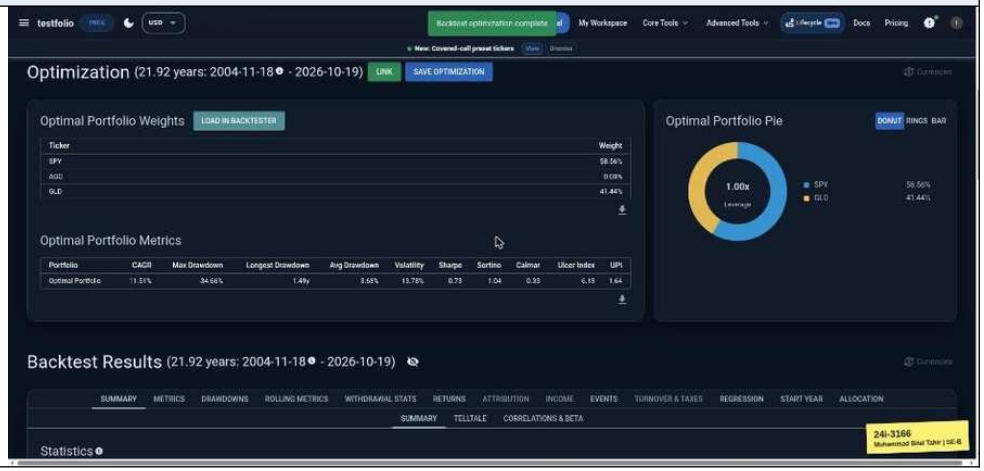
TC-EP-095 | Allocation % (portfolio total) | input: 50/30/10 (90) | result: Pass



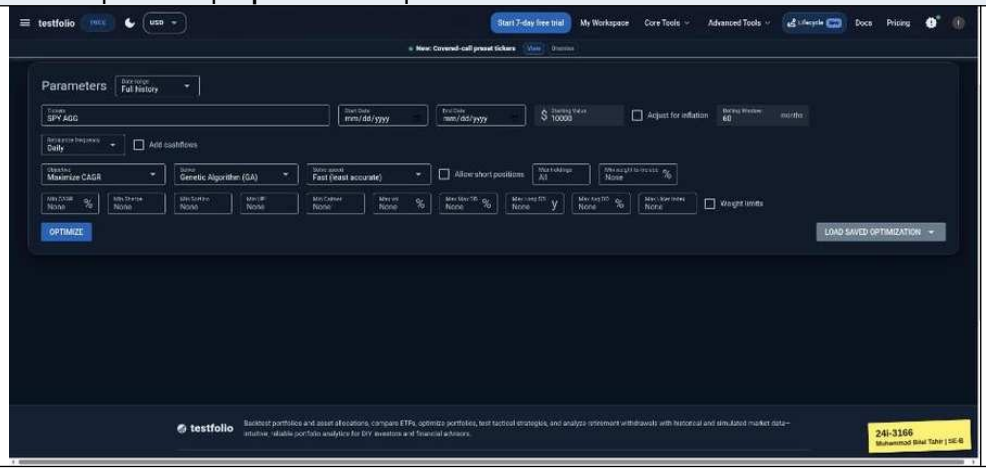
Backtest Optimizer (testfol.io/backtest-optimizer)

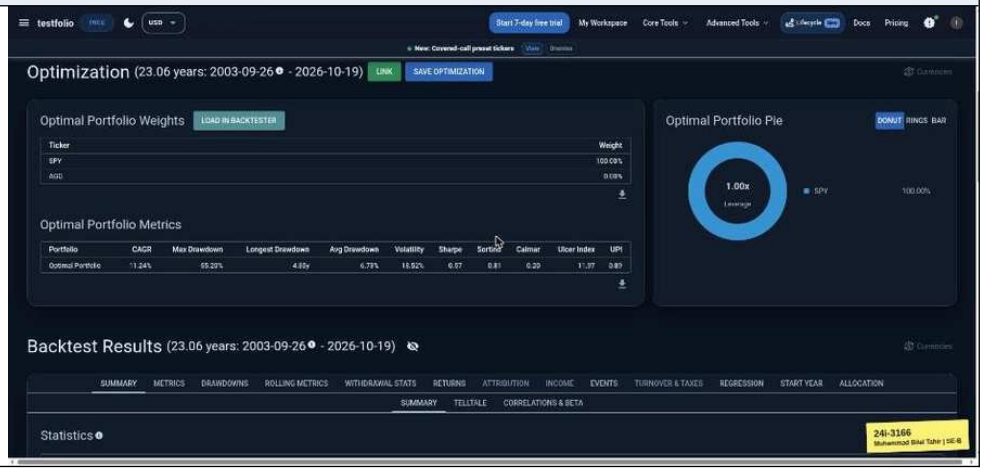
TC-EP-096 | Tickers | input: SPY AGG GLD | result: Pass



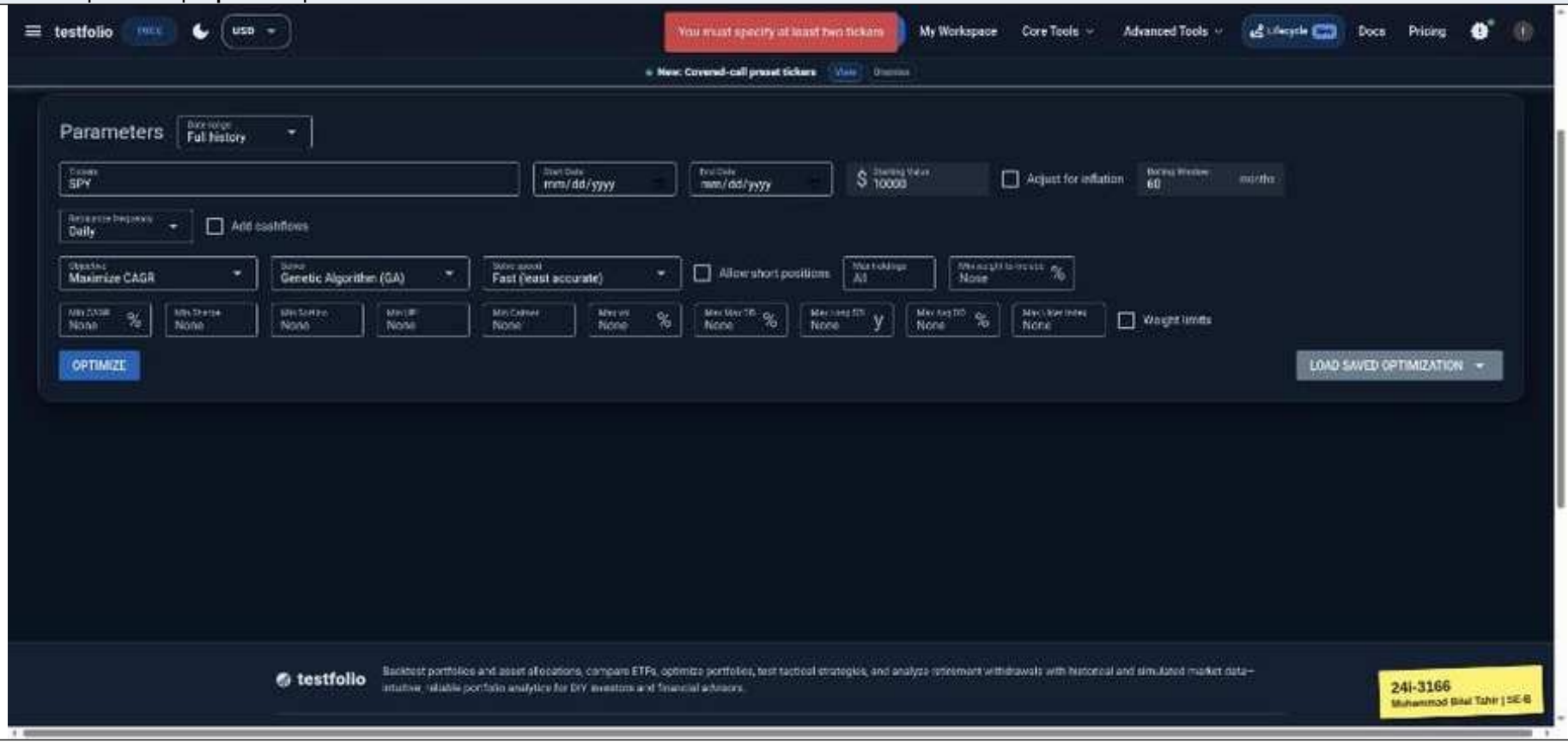


TC-EP-097 | Tickers | input: SPY AGG | result: Pass

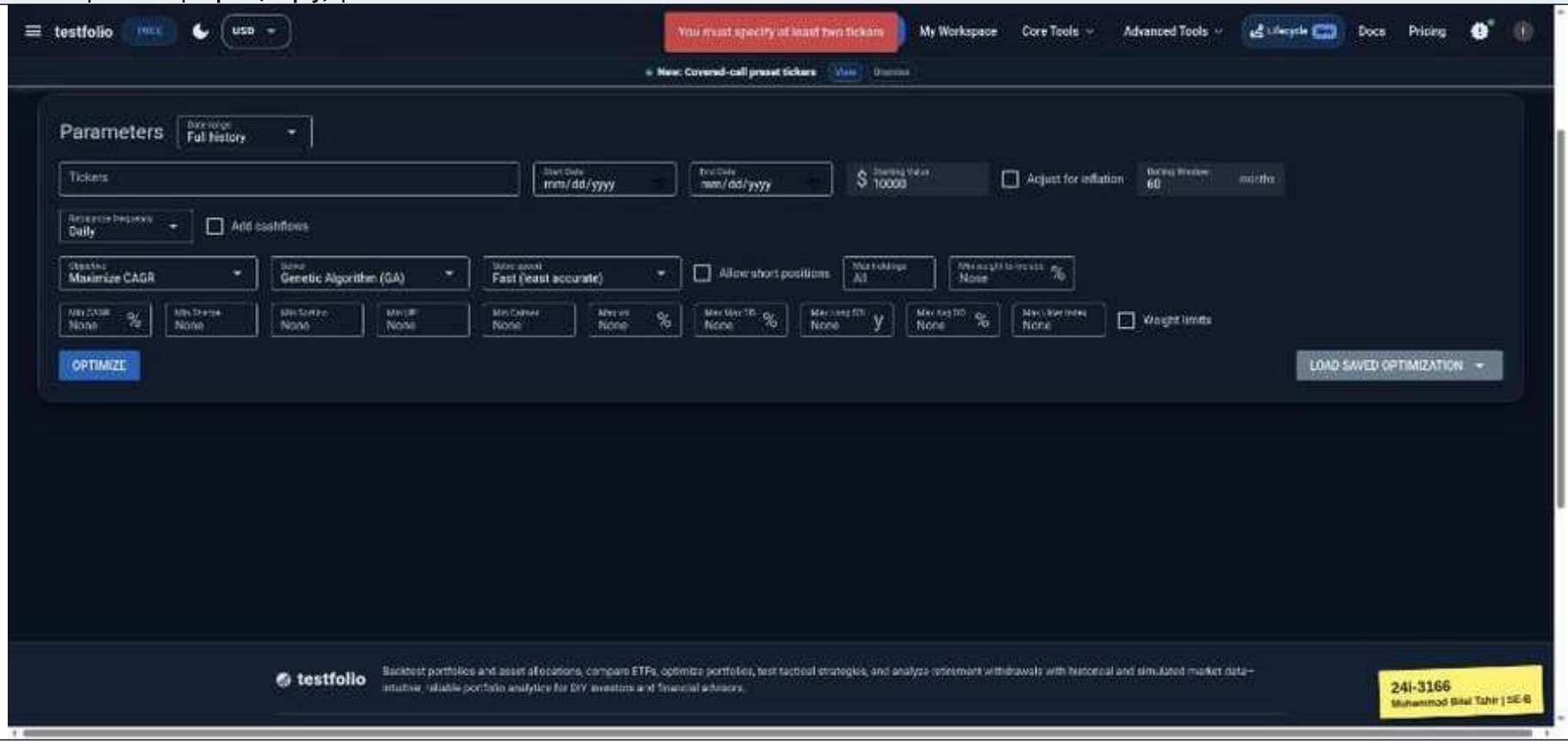




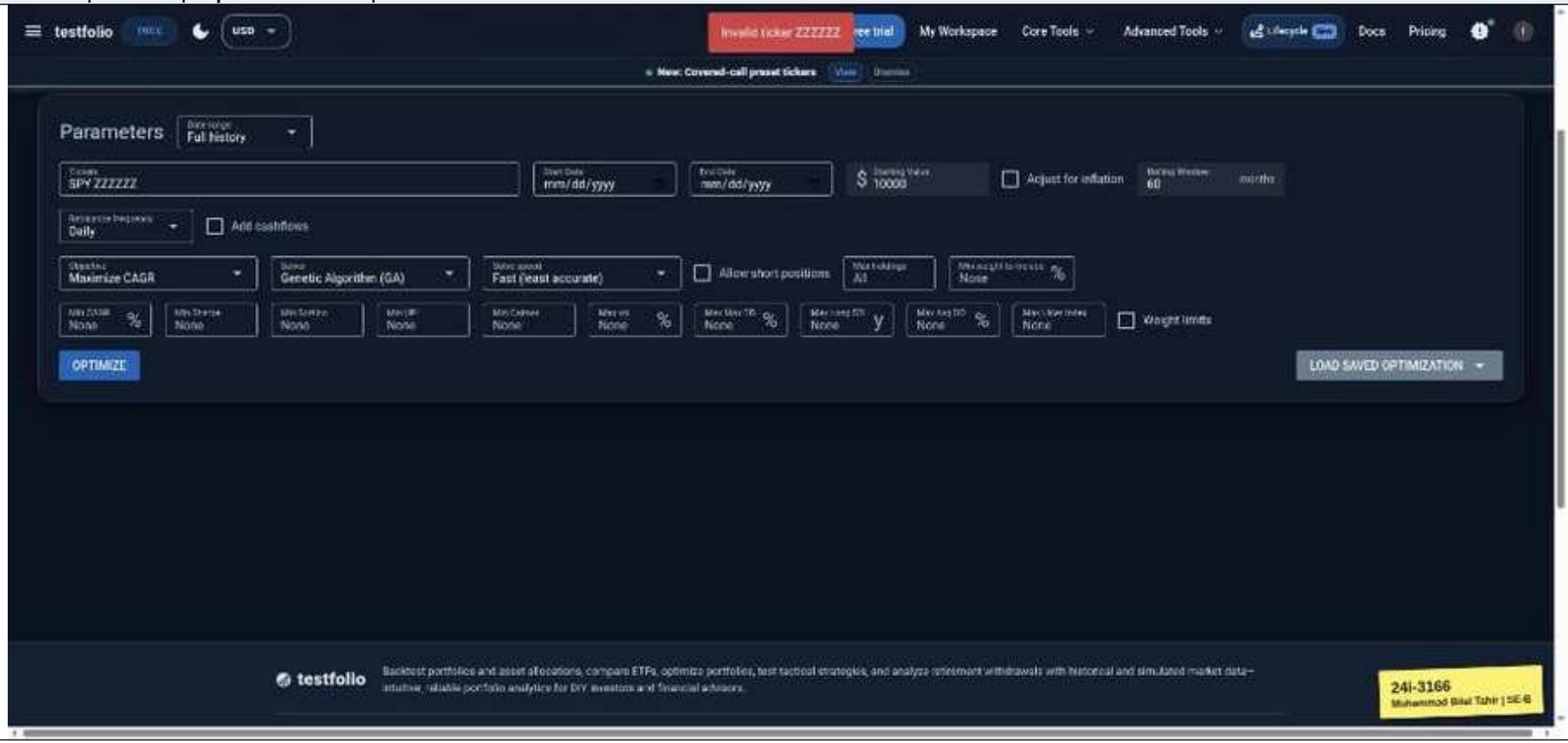
TC-EP-098 | Tickers | input: SPY | result: Pass



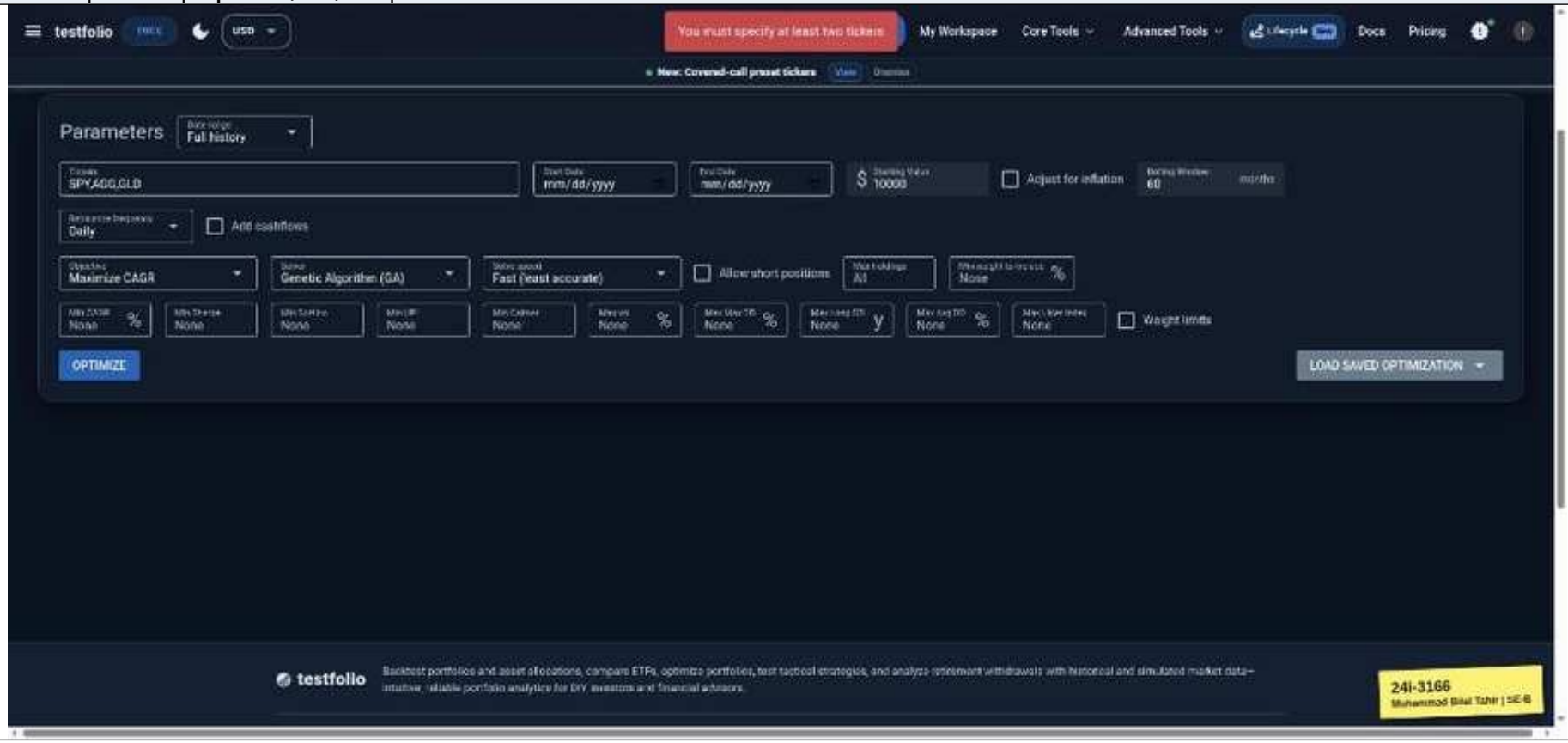
TC-EP-099 | Tickers | input: (empty) | result: Pass



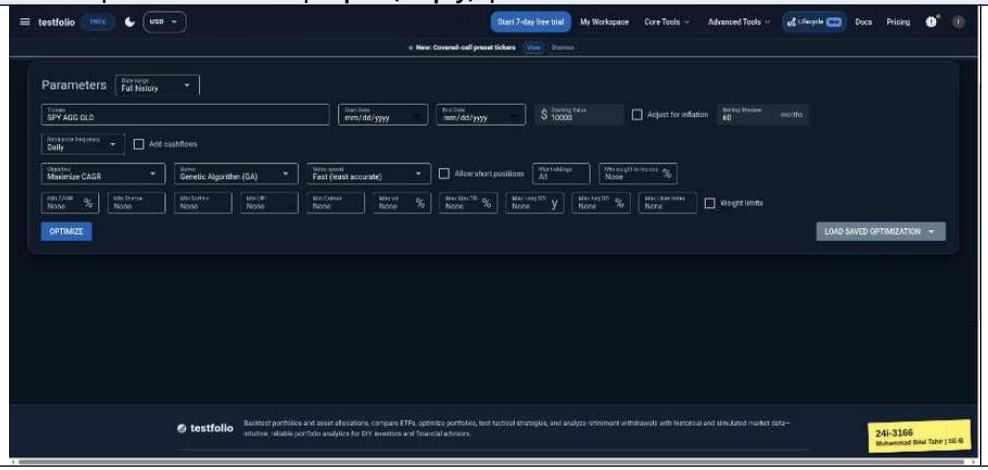
TC-EP-100 | Tickers | input: SPY ZZZZZZ | result: Pass

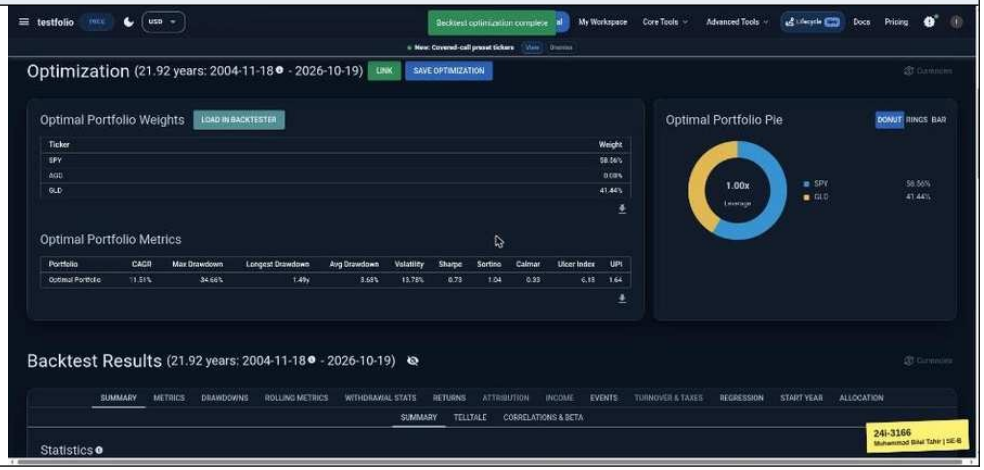


TC-EP-101 | Tickers | input: SPY,AGG,GLD | result: Pass

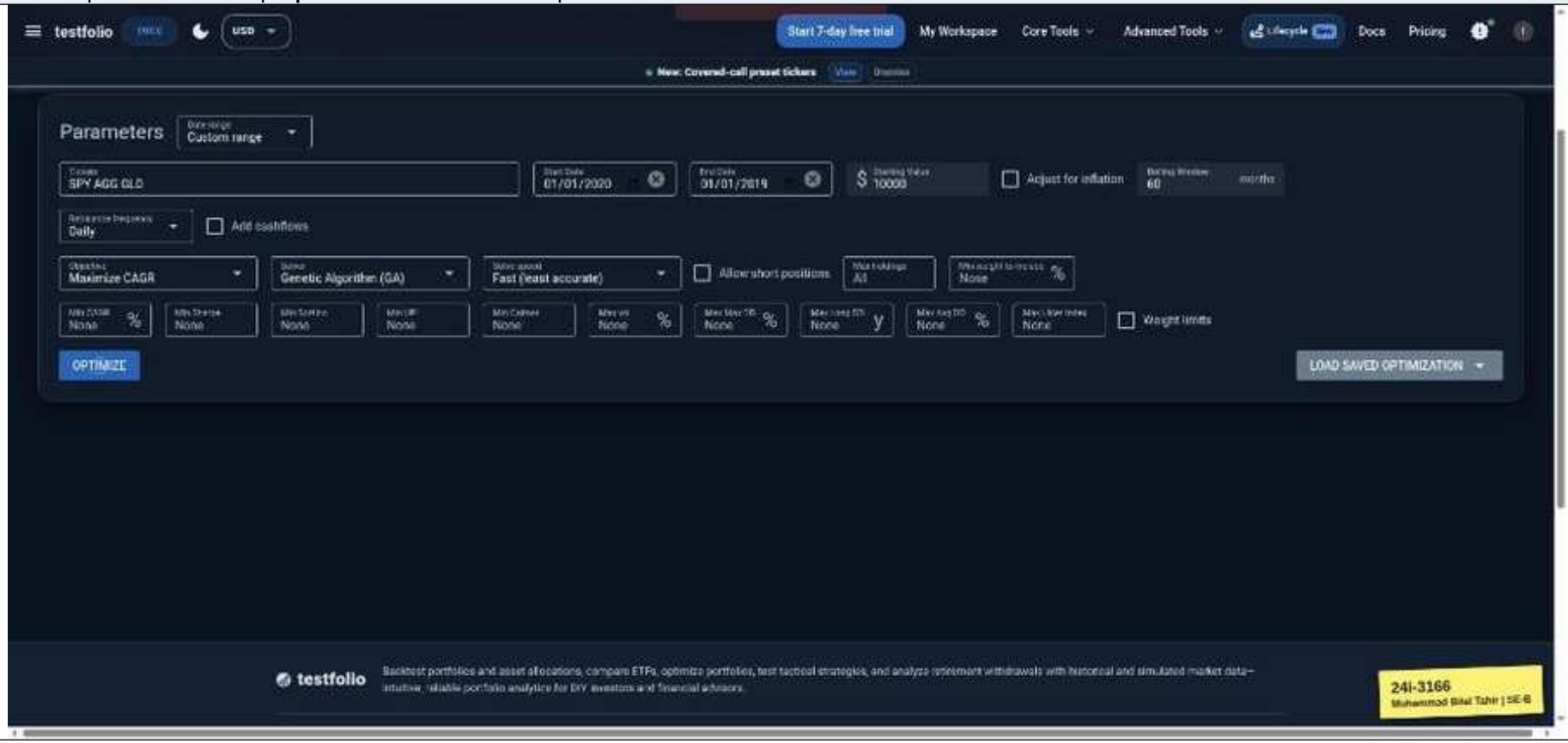


TC-EP-102 | Start / End Date | input: (empty) | result: Pass

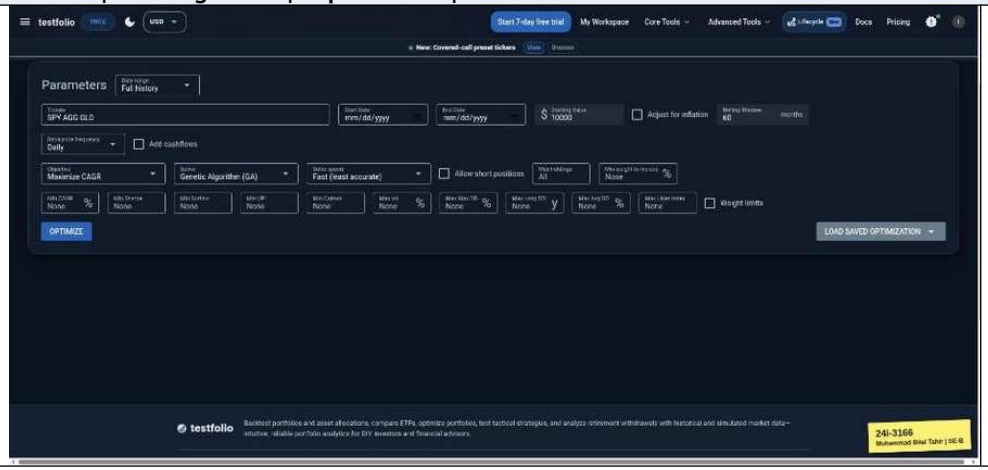


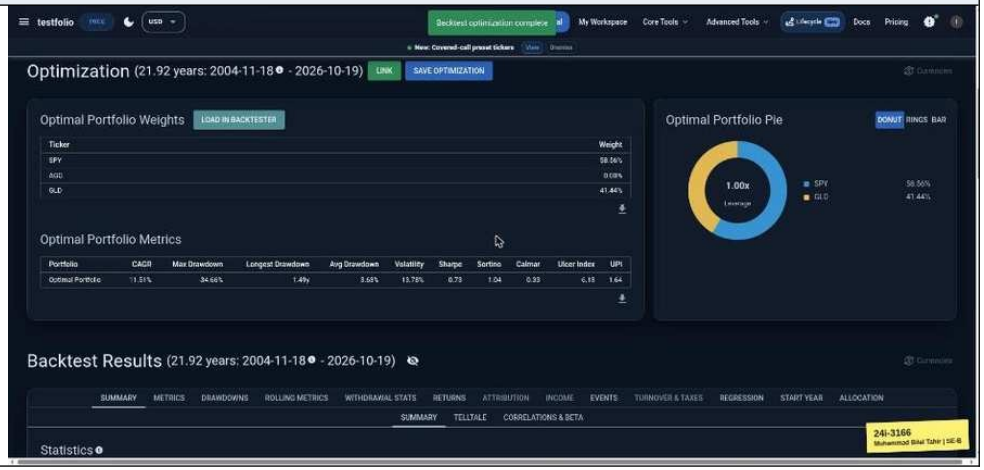


TC-EP-103 | Start / End Date | input: 2020-01-01 / 2019-01-01 | result: Pass

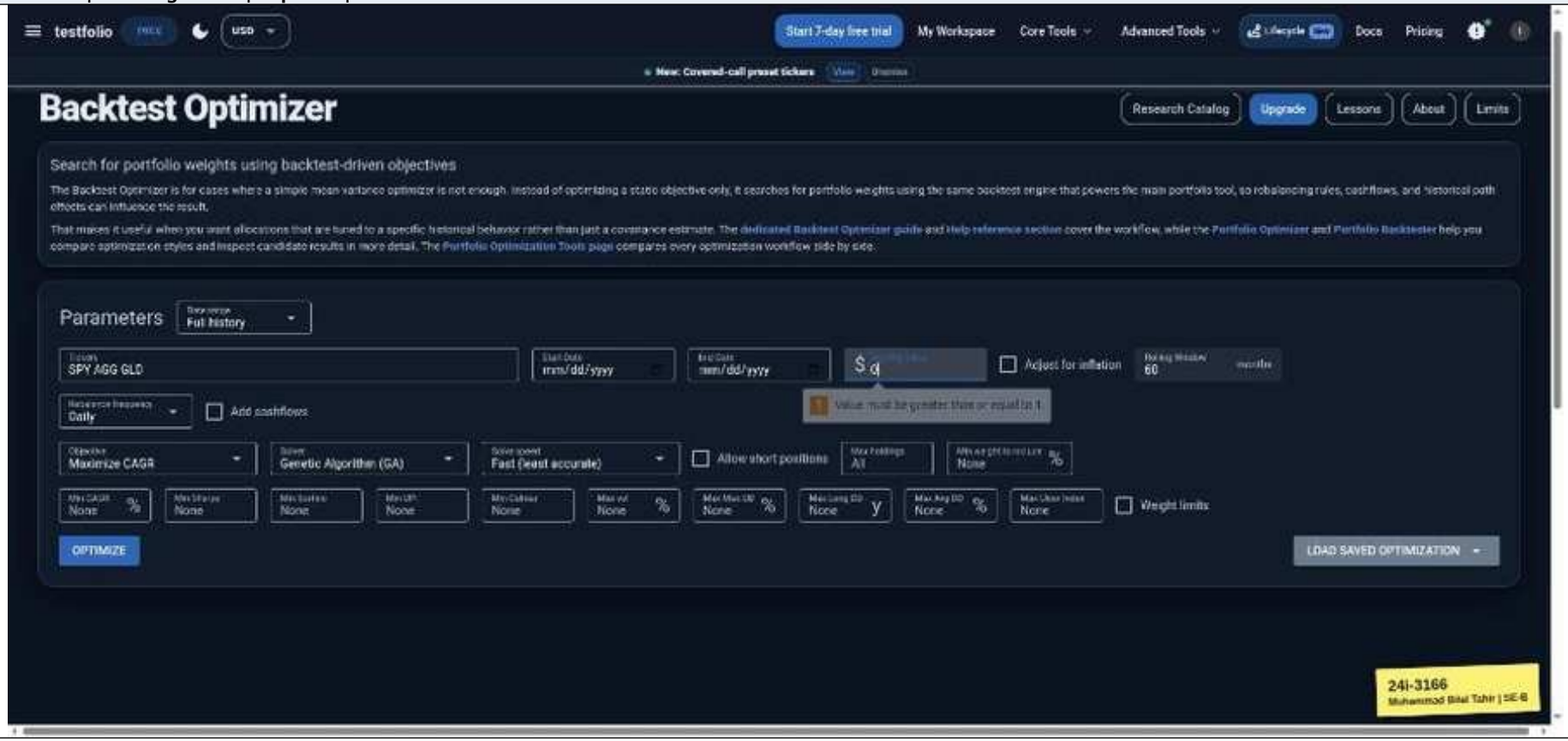


TC-EP-104 | Starting Value | input: 10000 | result: Pass

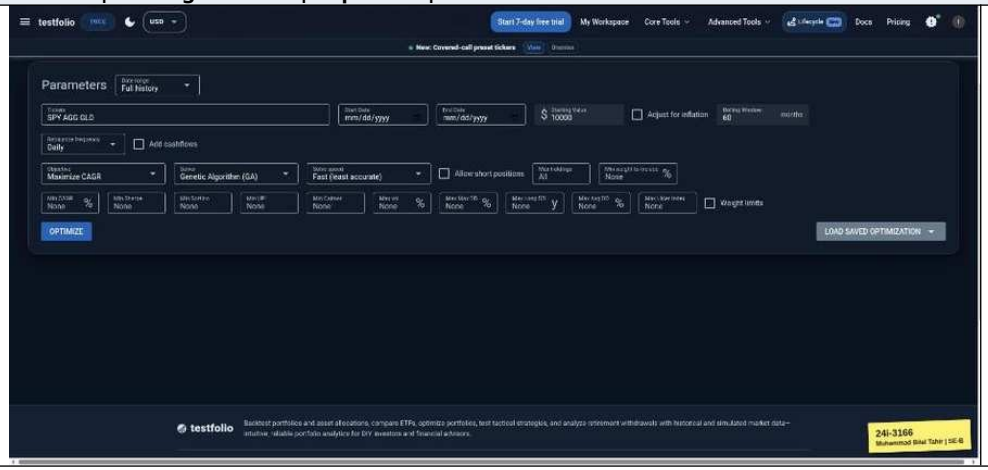


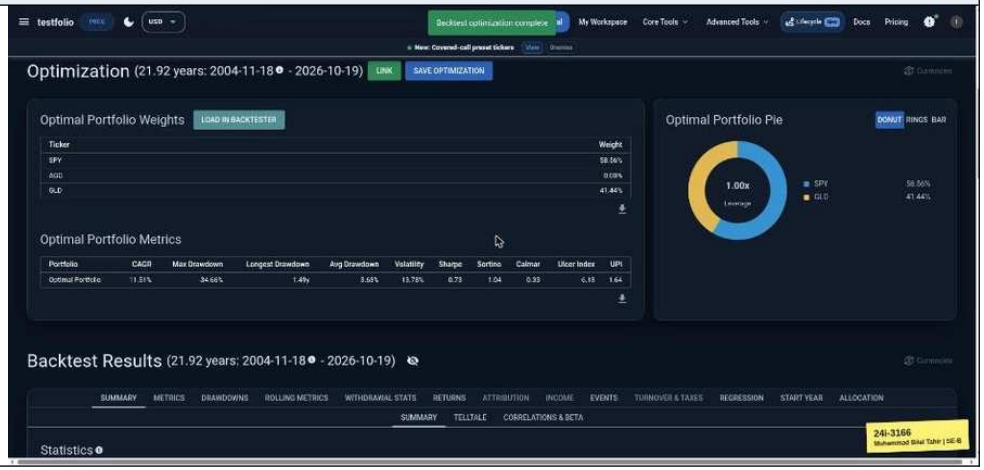


TC-EP-105 | Starting Value | input: 0 | result: Pass

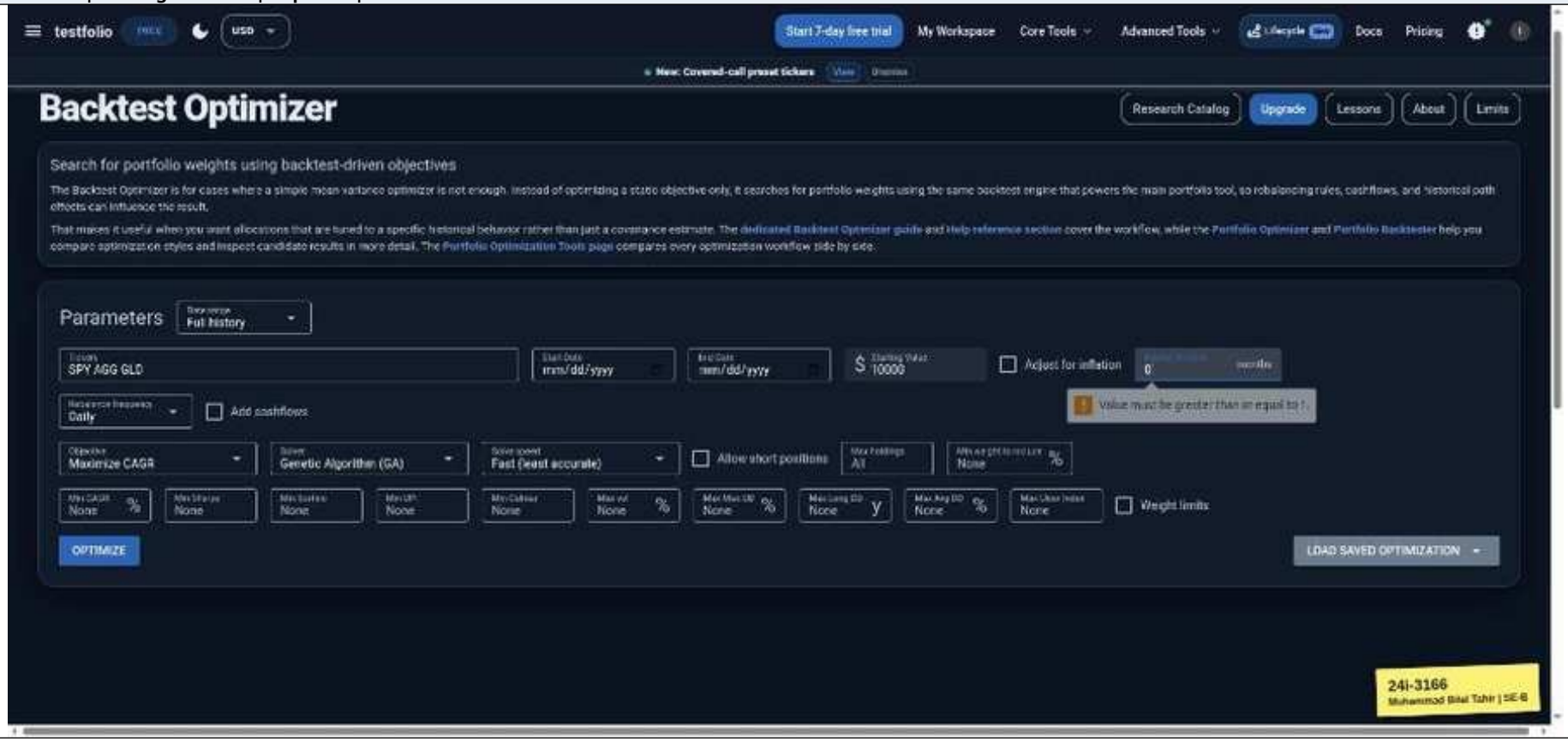


TC-EP-106 | Rolling Window | input: 60 | result: Pass

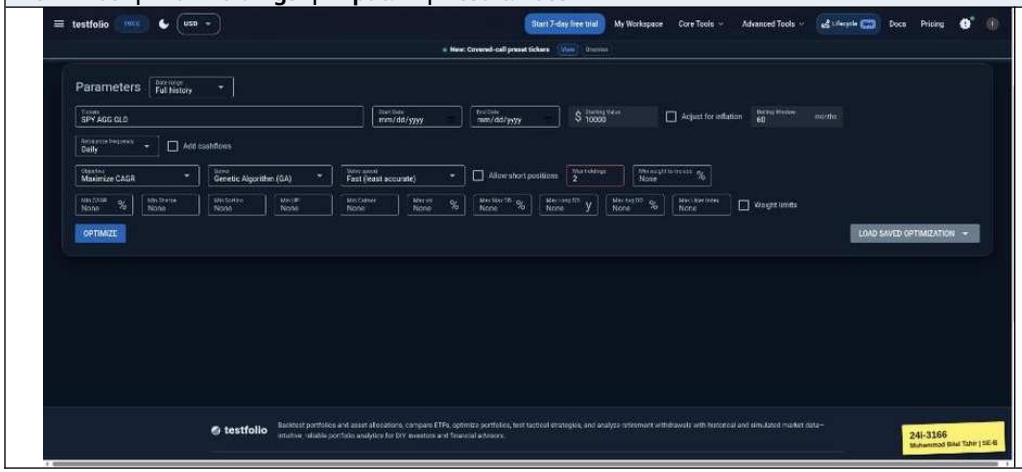


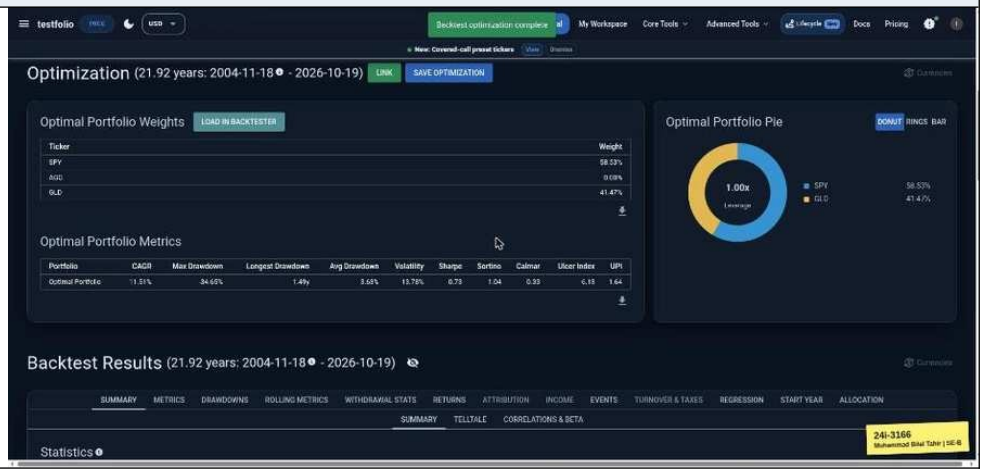


TC-EP-107 | Rolling Window | input: 0 | result: Pass

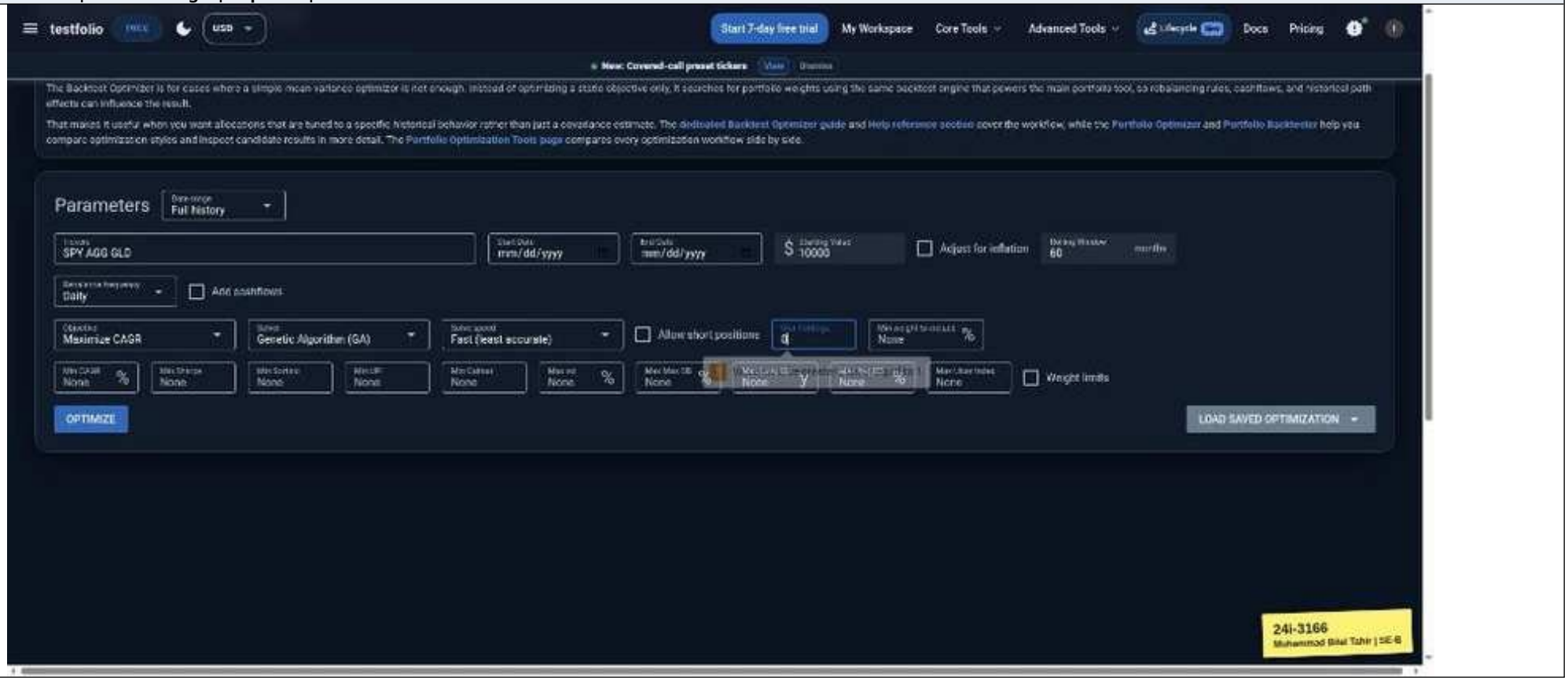


TC-EP-108 | Max holdings | input: 2 | result: Pass

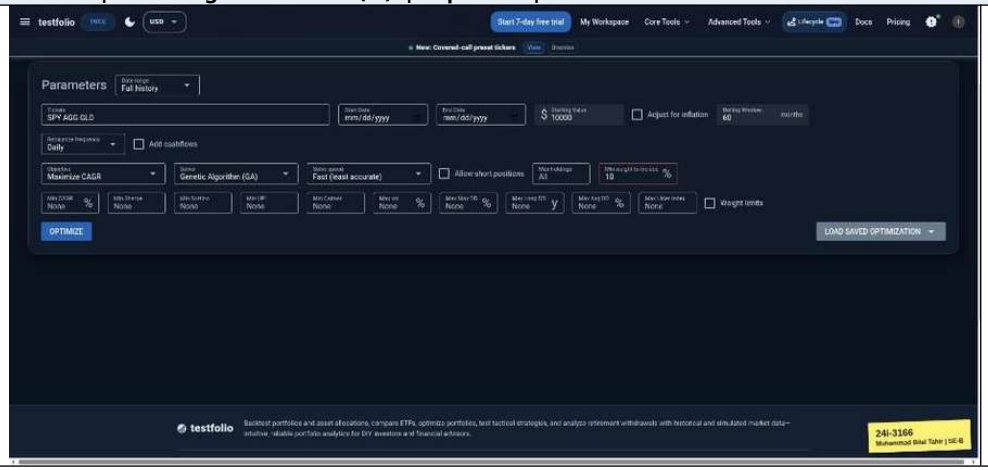


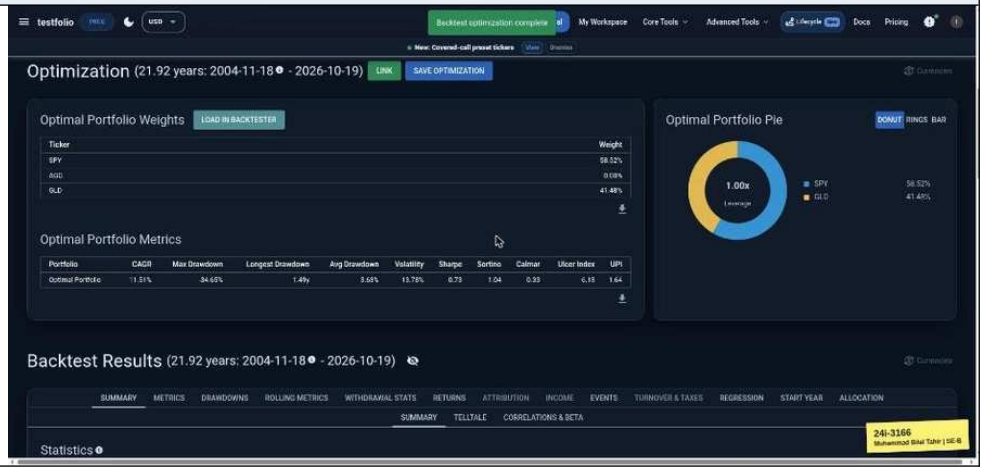


TC-EP-109 | Max holdings | input: 0 | result: Pass

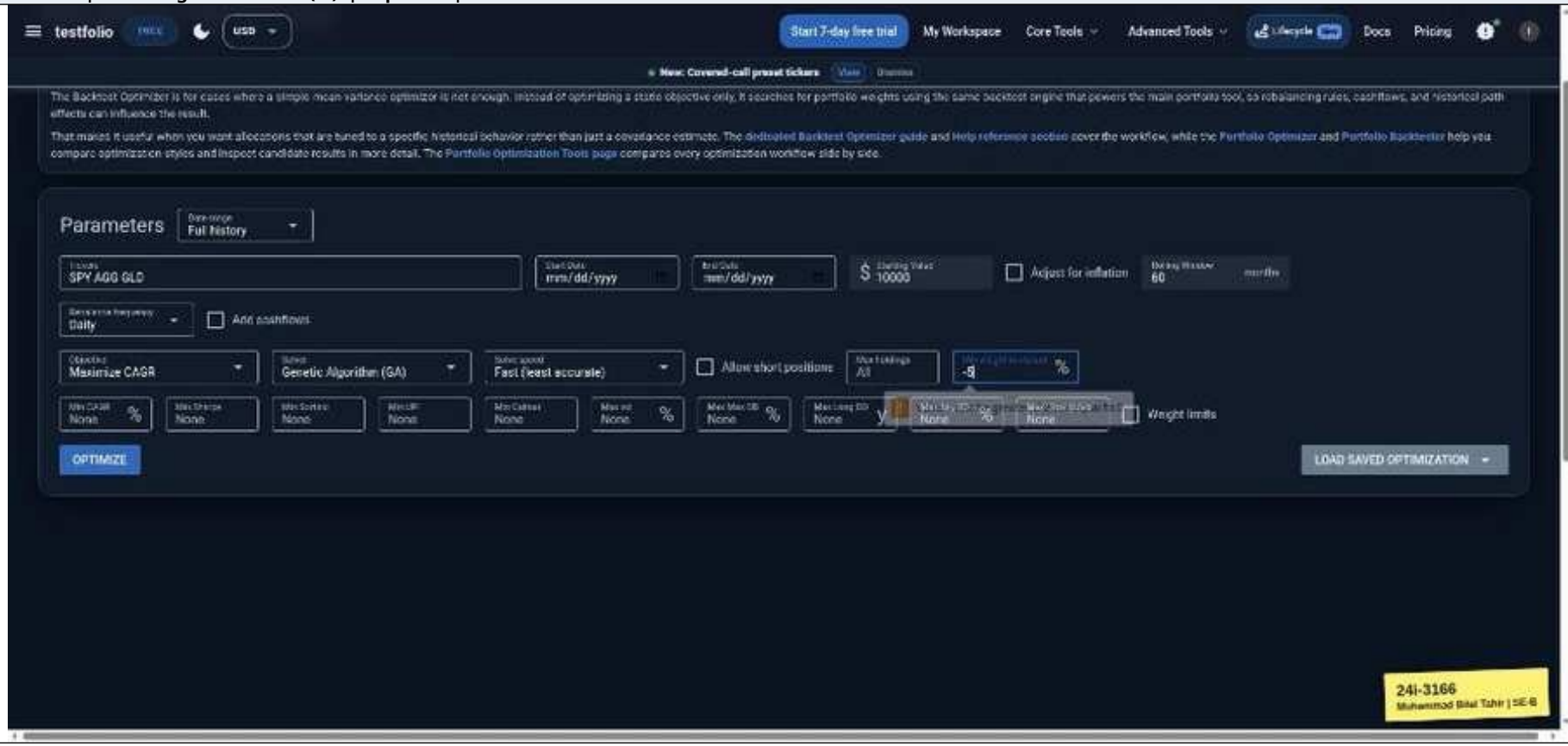


TC-EP-110 | Min weight to include (%) | input: 10 | result: Pass

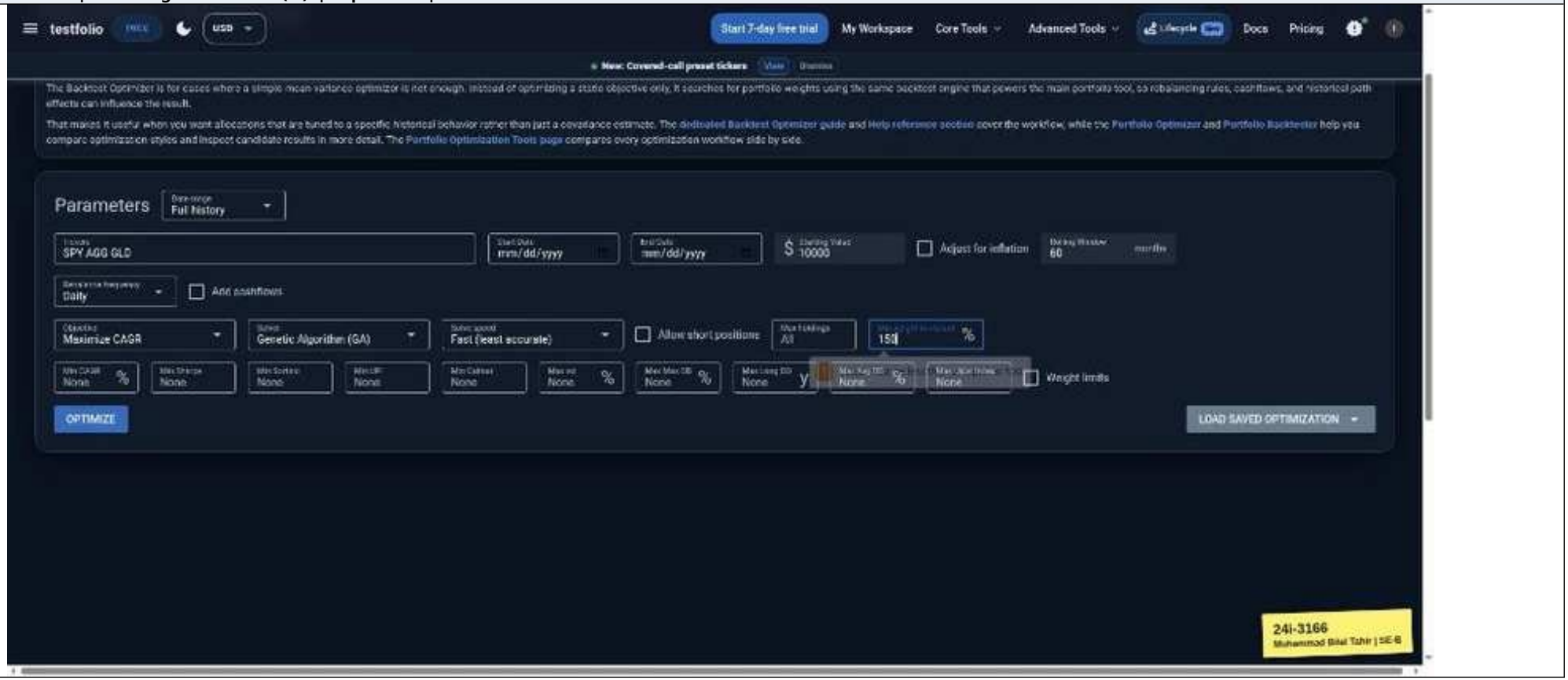




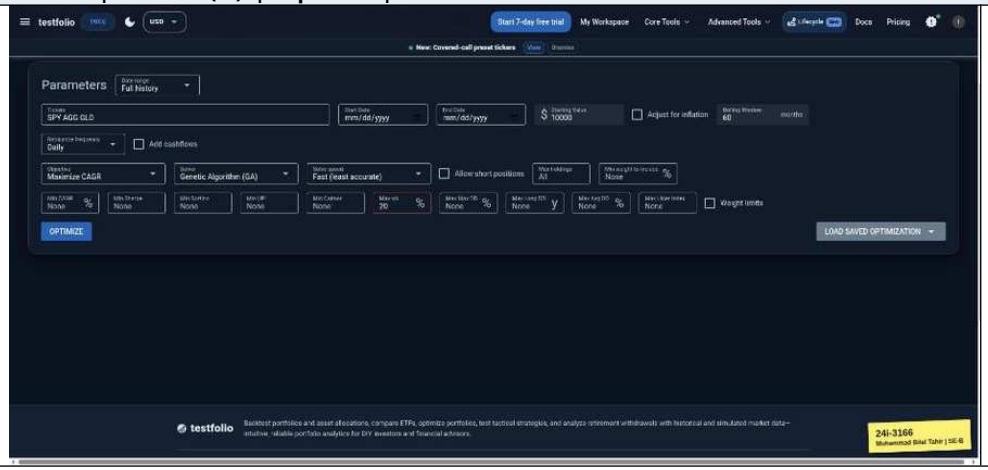
TC-EP-111 | Min weight to include (%) | input: -5 | result: Pass

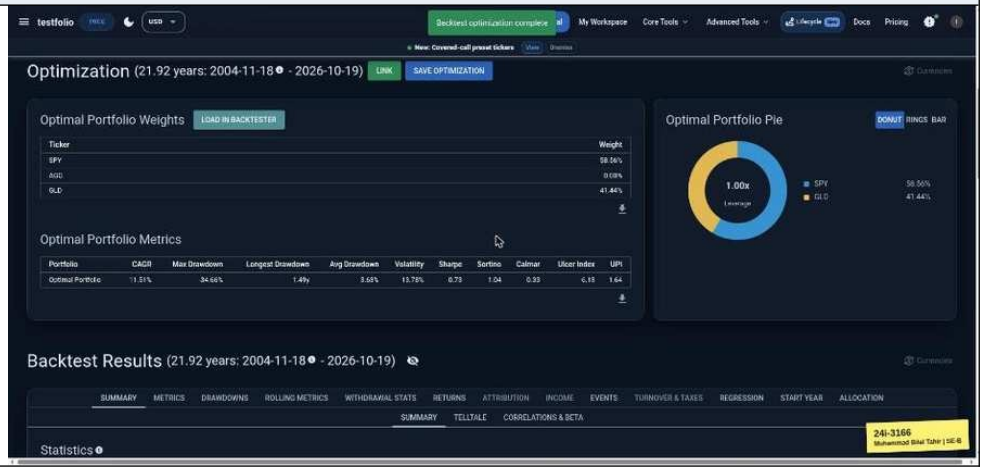


TC-EP-112 | Min weight to include (%) | input: 150 | result: Pass

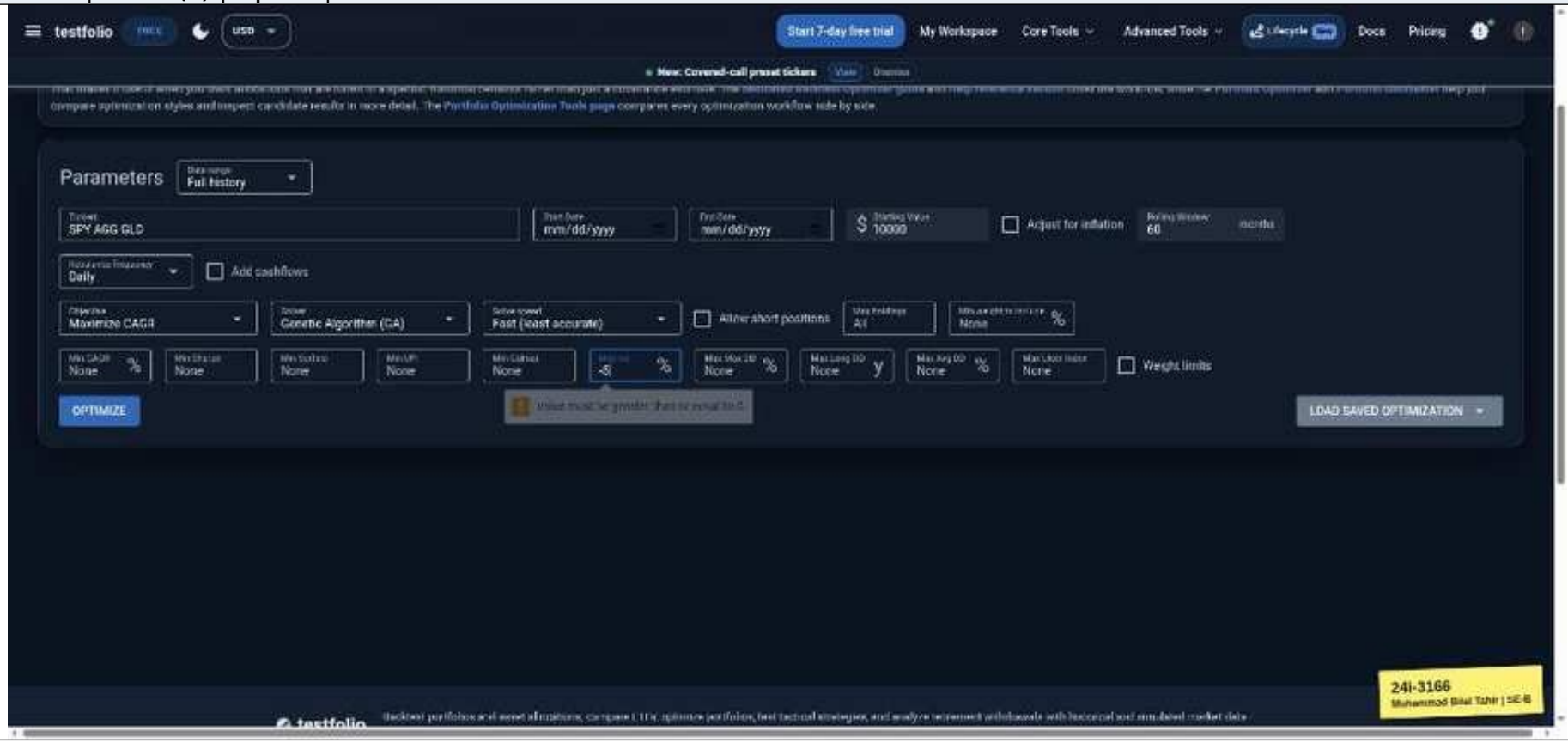


TC-EP-113 | Max vol (%) | input: 20 | result: Pass



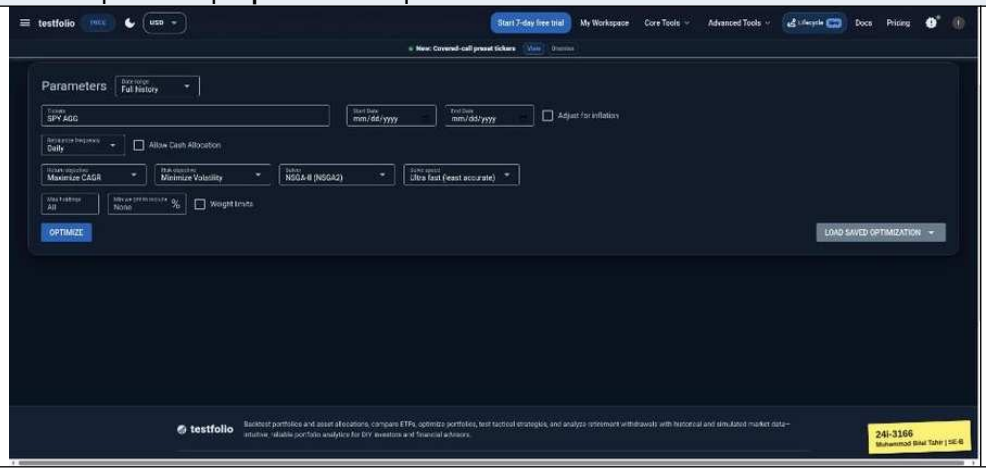


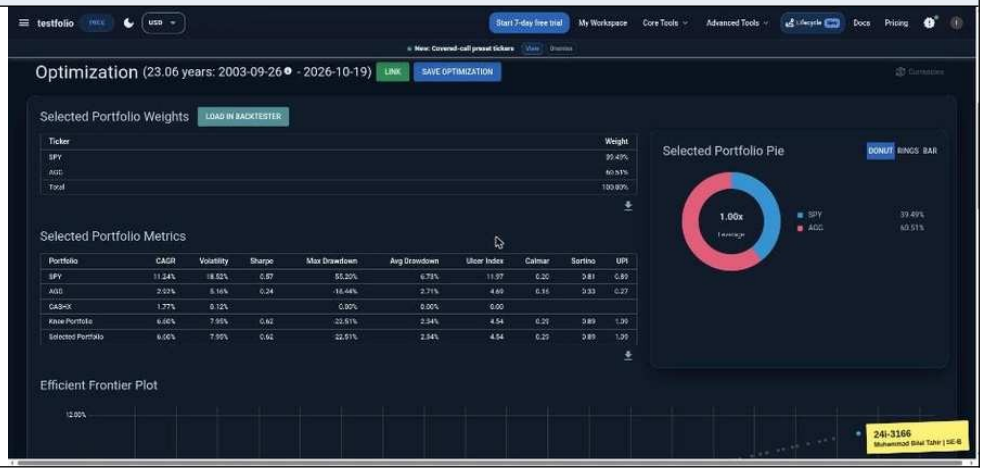
TC-EP-114 | Max vol (%) | input: -5 | result: Pass



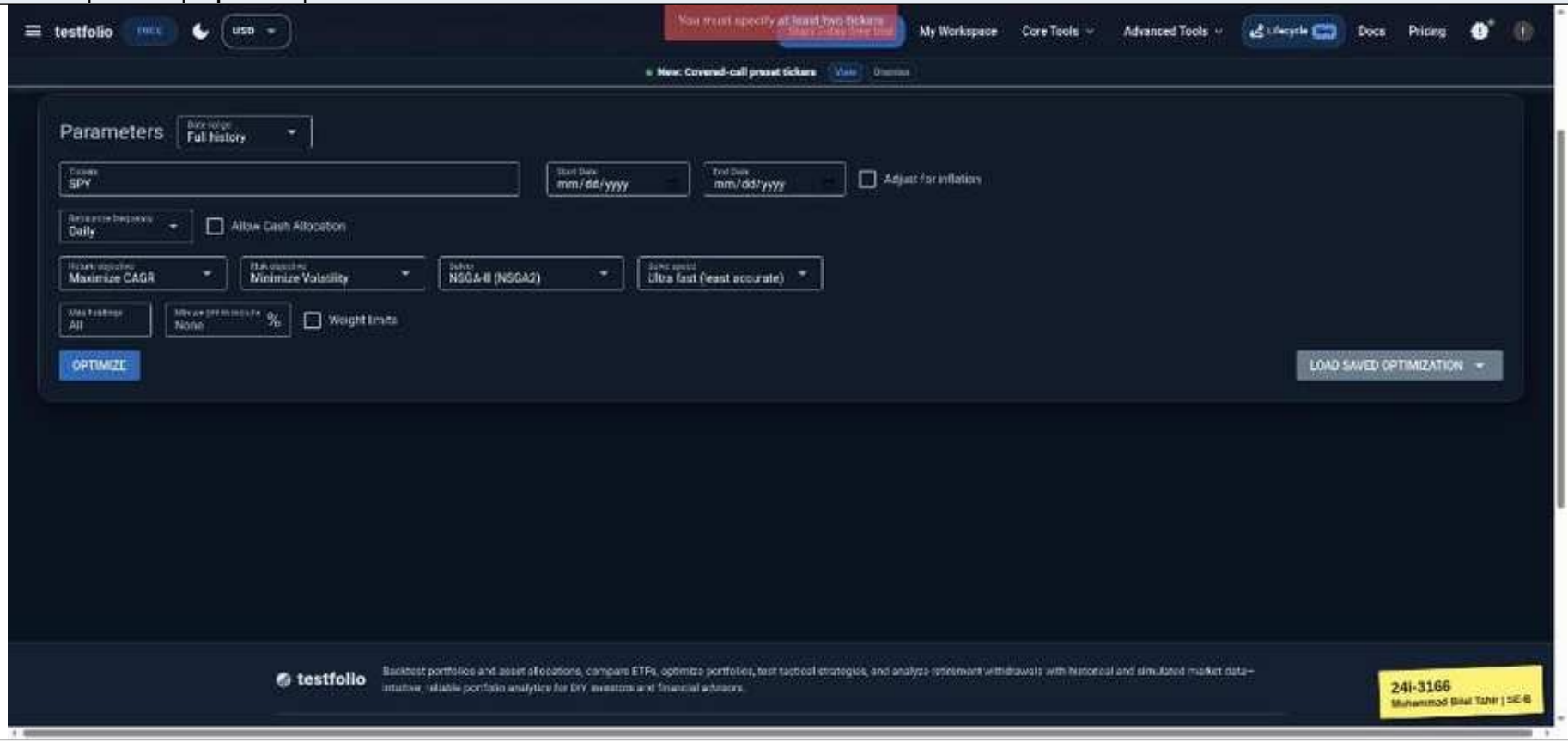
Efficient Frontier (testfol.io/efficient-frontier)

TC-EP-115 | Tickers | input: SPY AGG | result: Pass

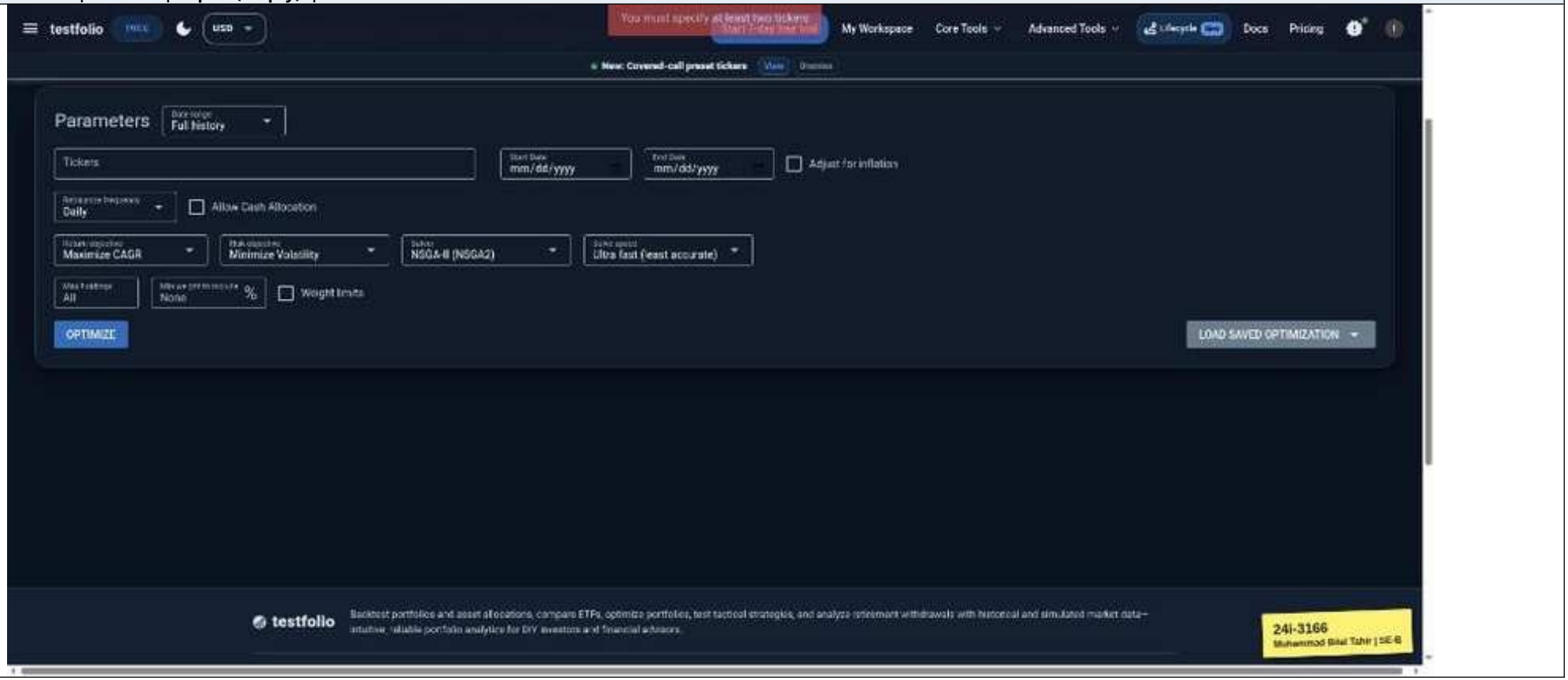




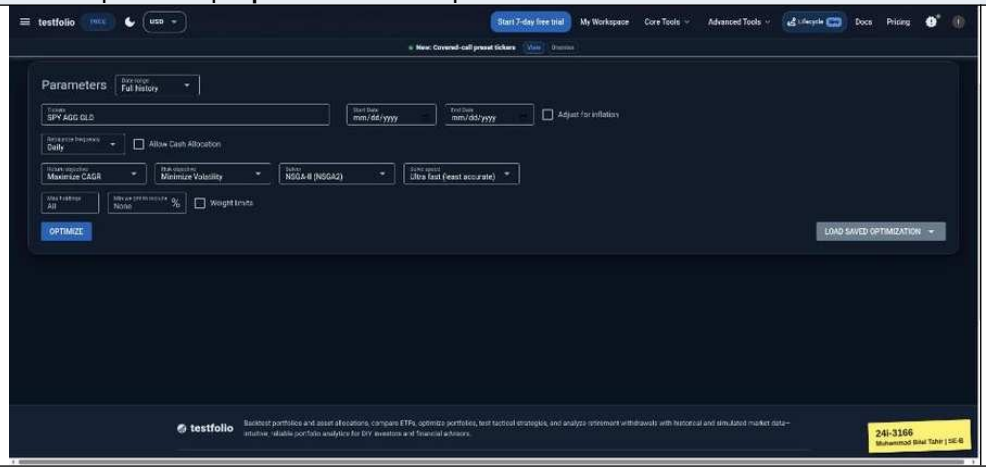
TC-EP-116 | Tickers | input: SPY | result: Pass

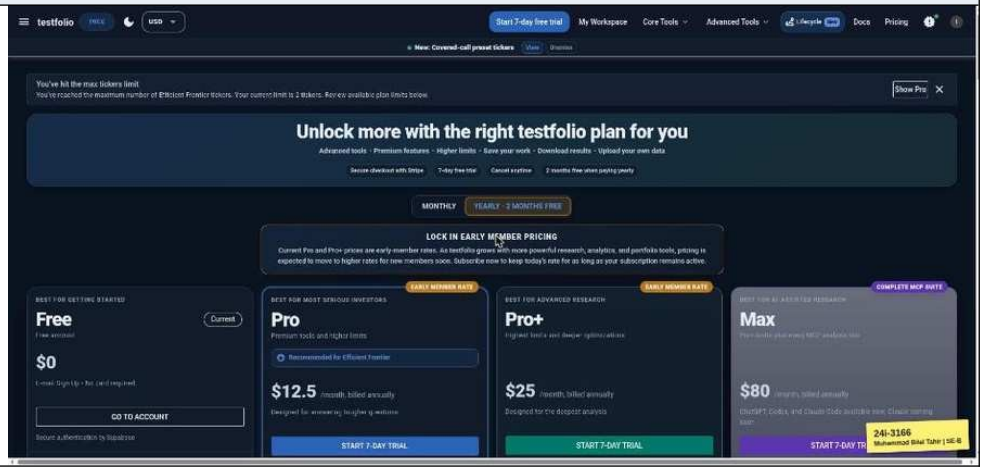


TC-EP-117 | Tickers | input: (empty) | result: Pass

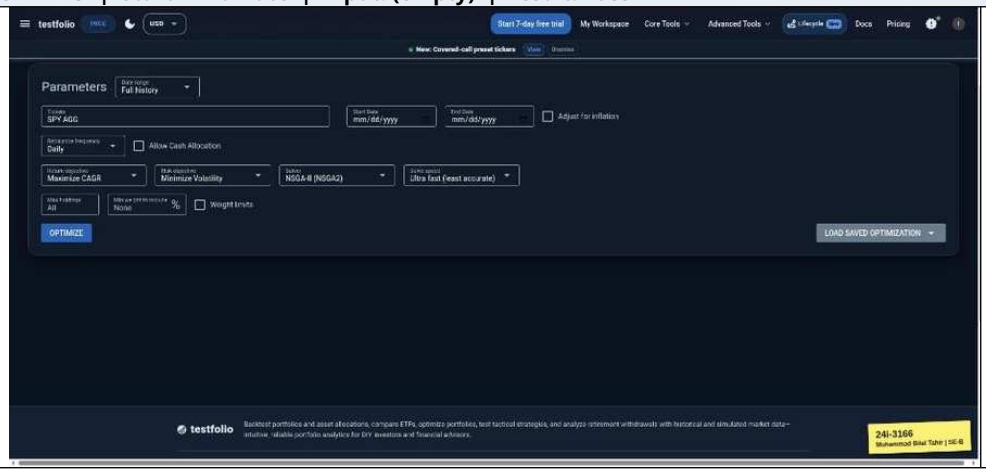


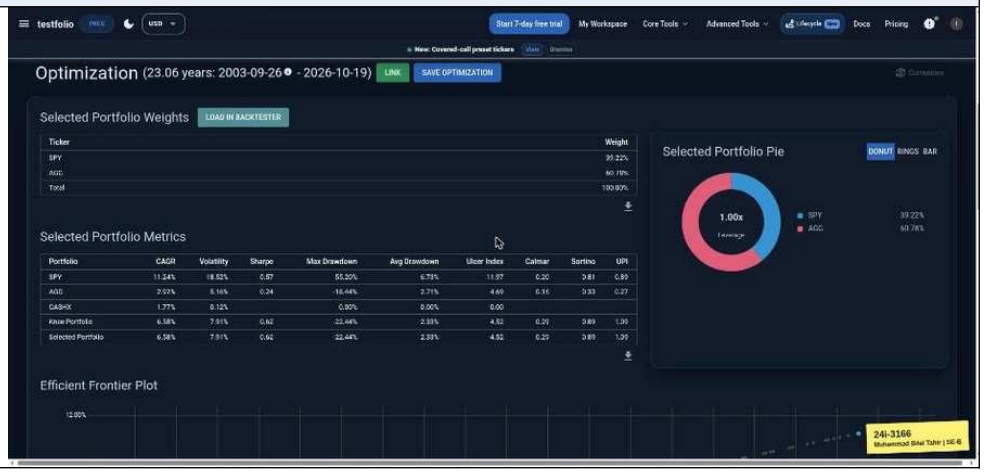
TC-EP-118 | Tickers | input: SPY AGG GLD | result: Pass



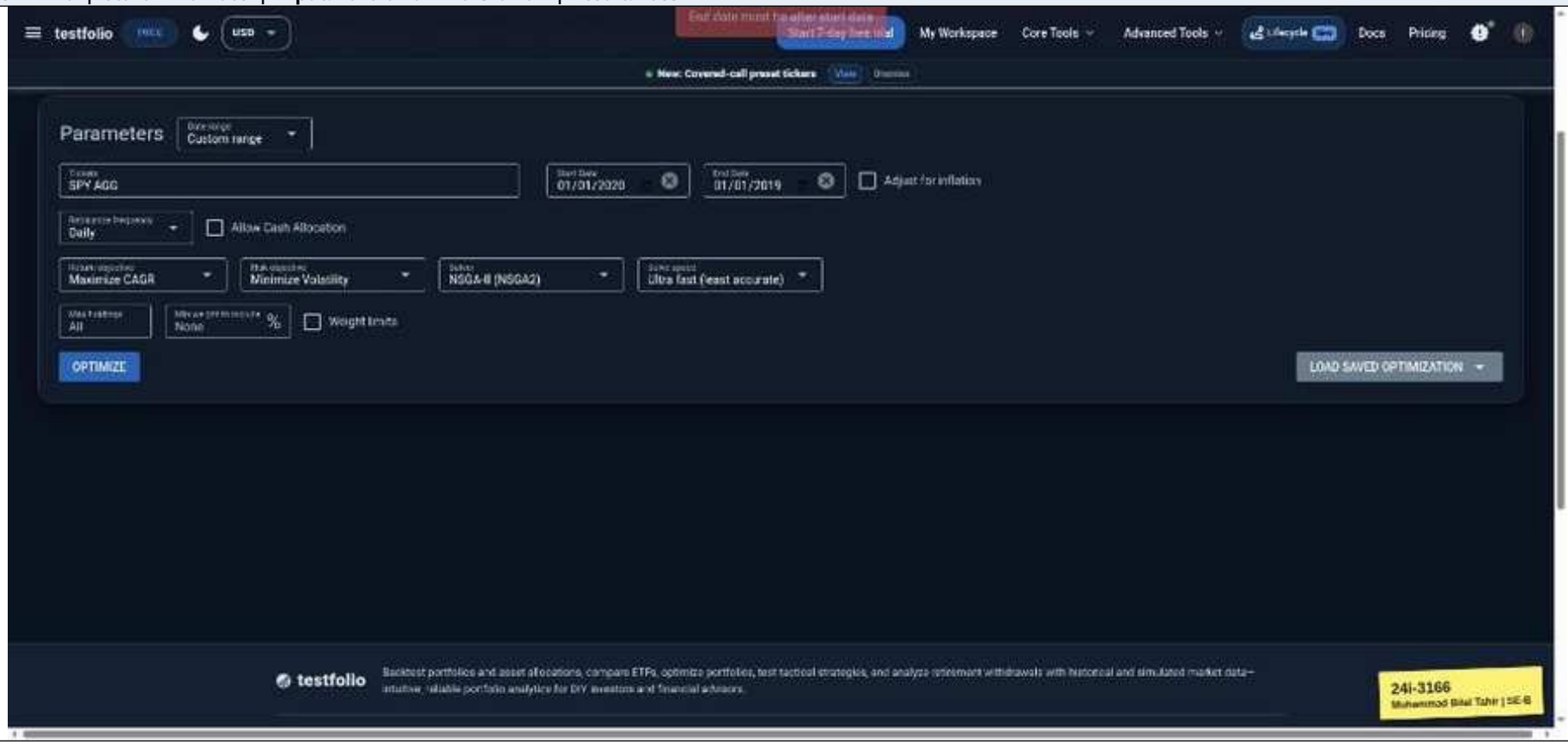


TC-EP-119 | Start / End Date | input: (empty) | result: Pass

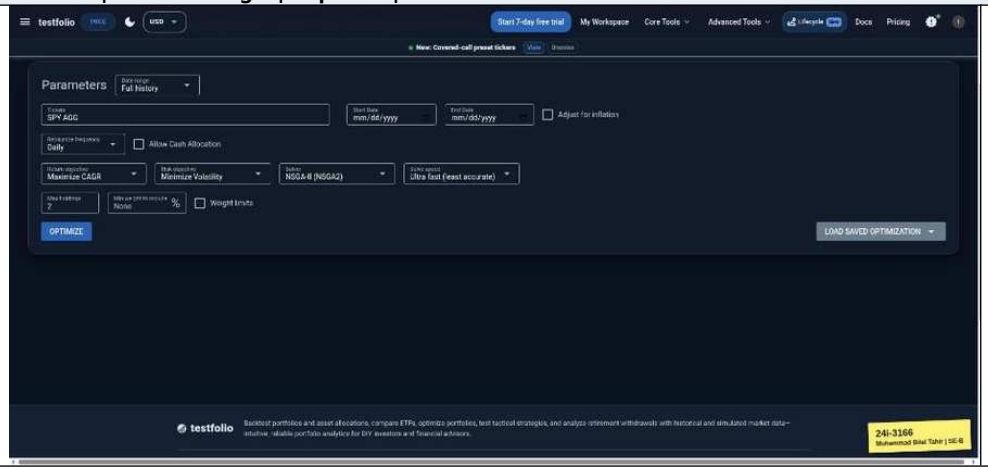


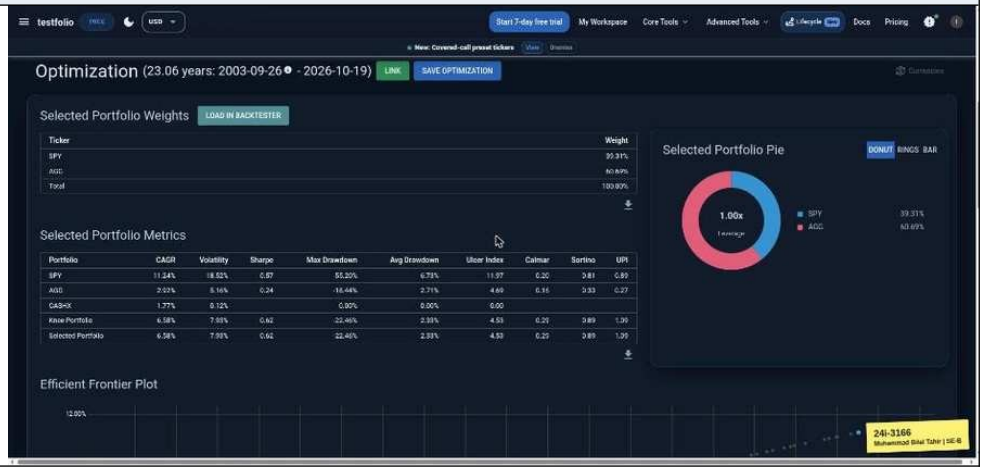


TC-EP-120 | Start / End Date | input: 2020-01-01 / 2019-01-01 | result: Pass

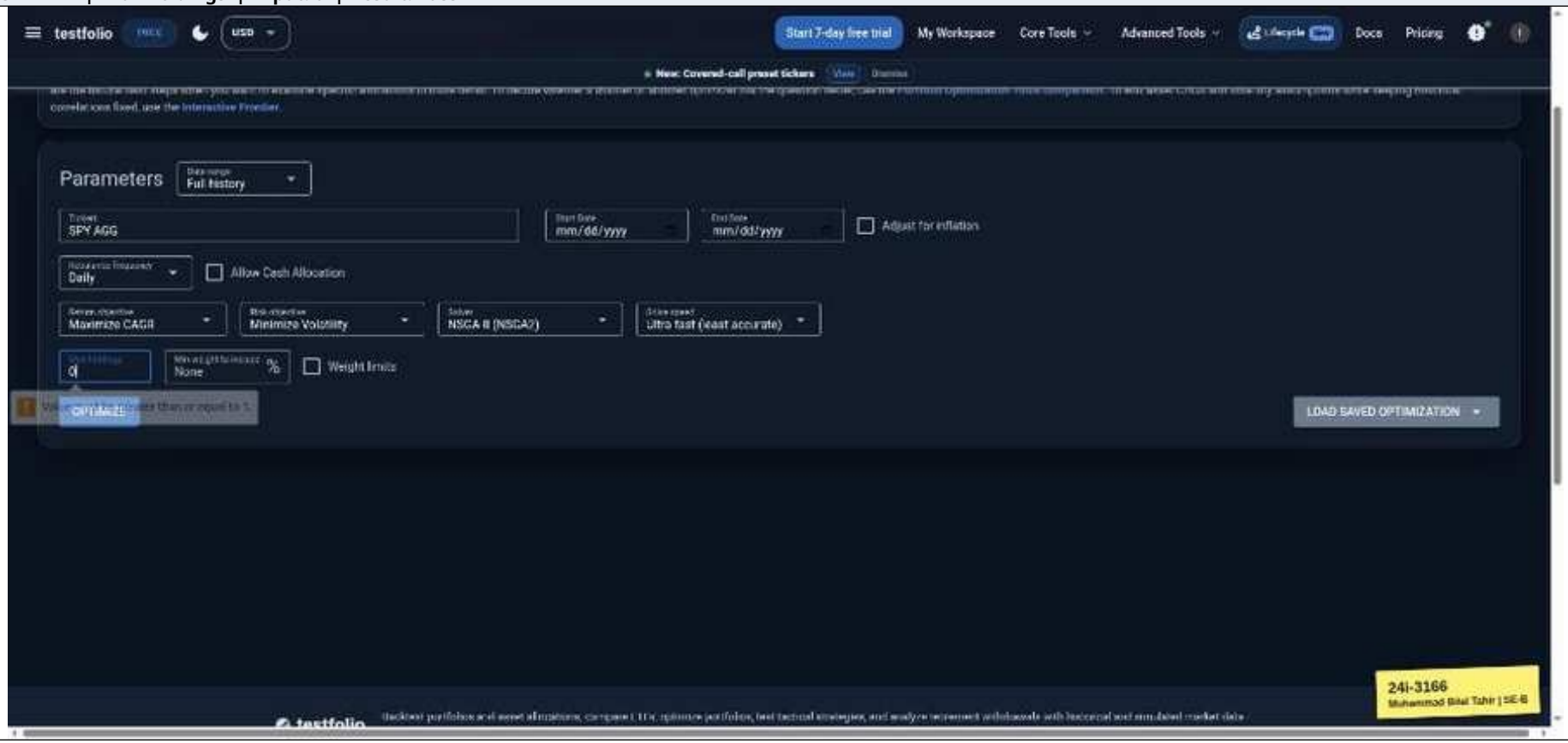


TC-EP-121 | Max holdings | input: 2 | result: Pass

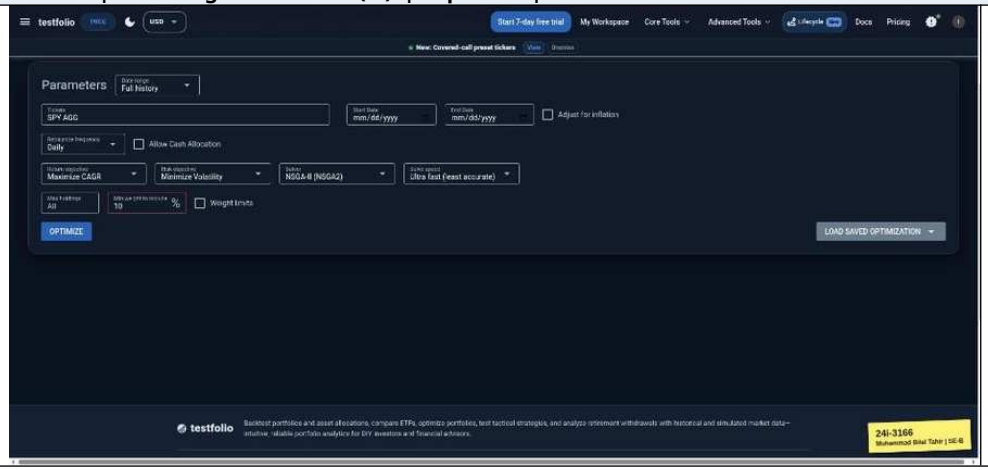


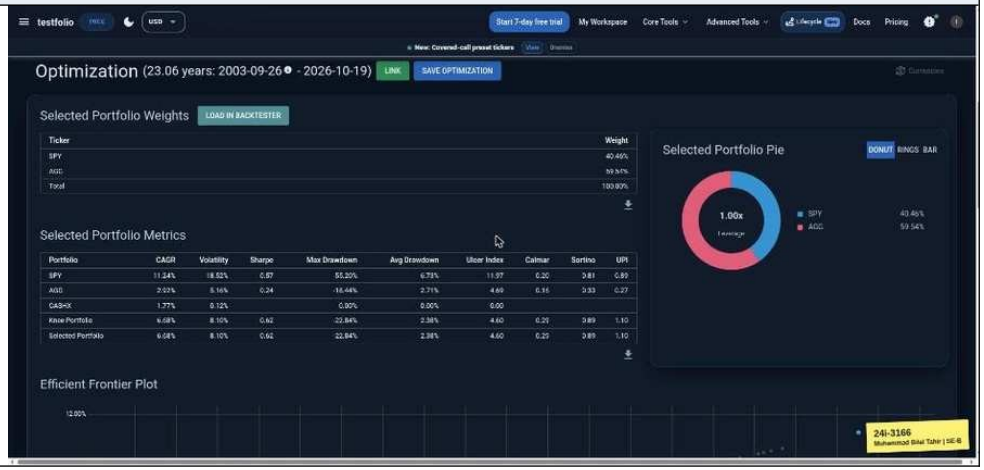


TC-EP-122 | Max holdings | input: 0 | result: Pass

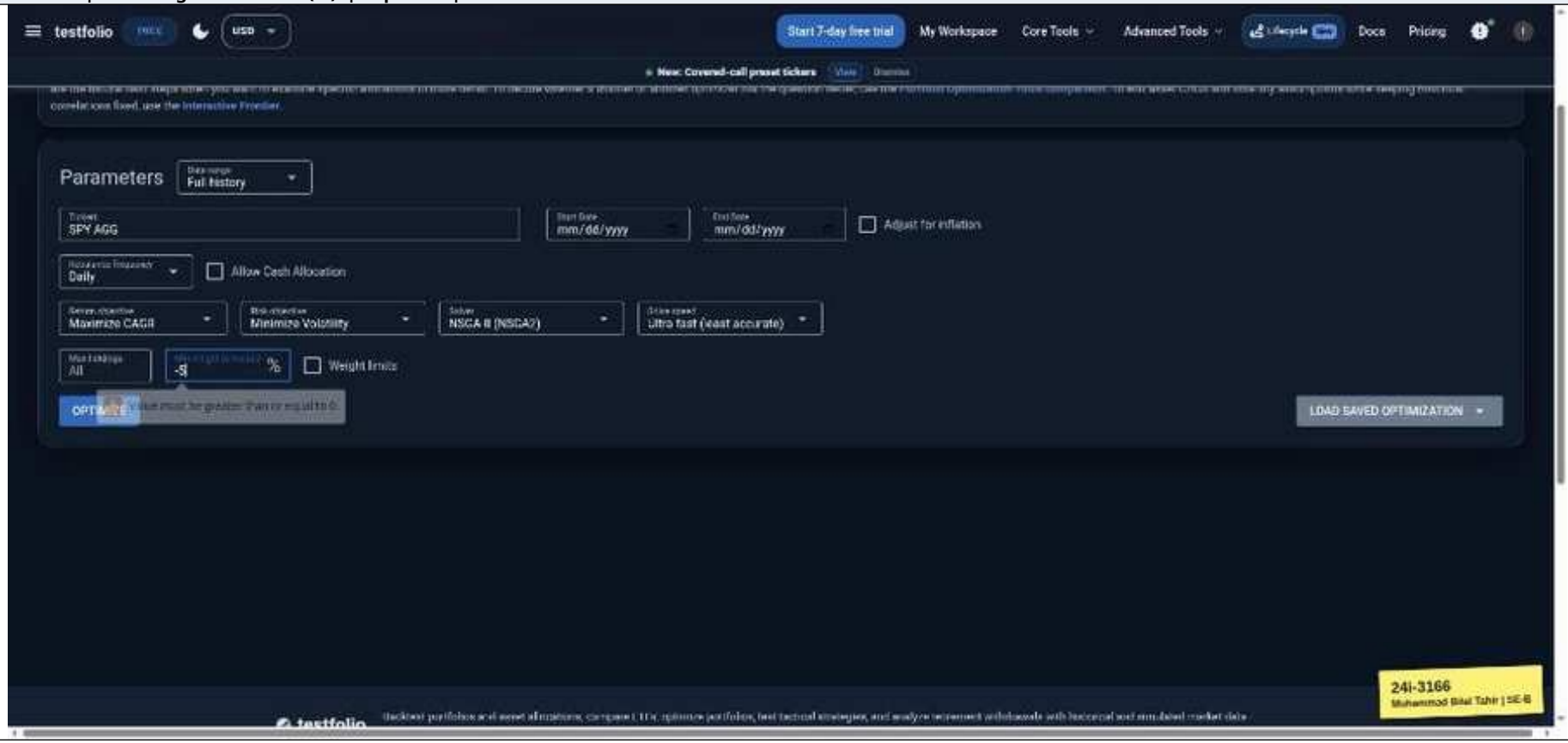


TC-EP-123 | Min weight to include (%) | input: 10 | result: Pass

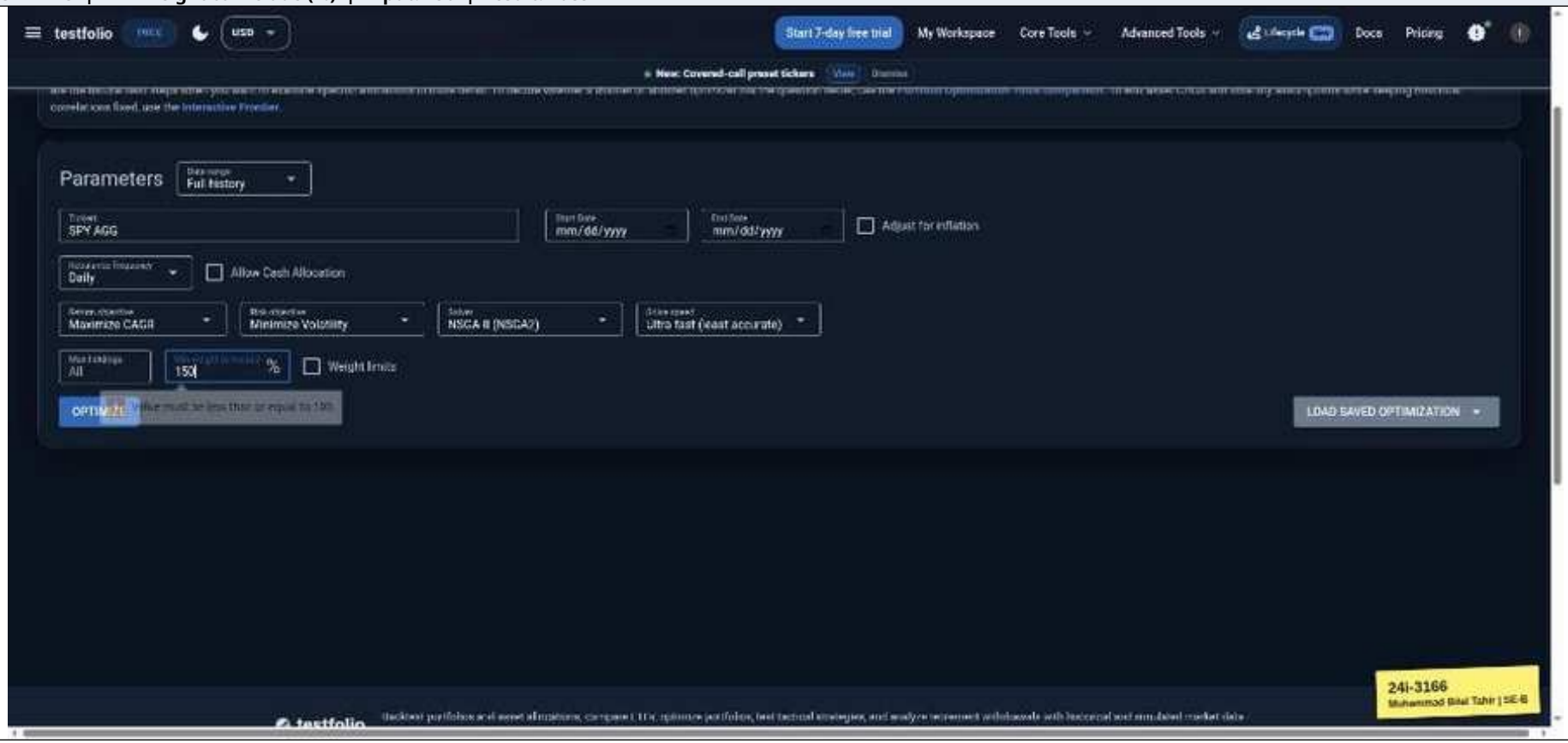




TC-EP-124 | Min weight to include (%) | input: -5 | result: Pass

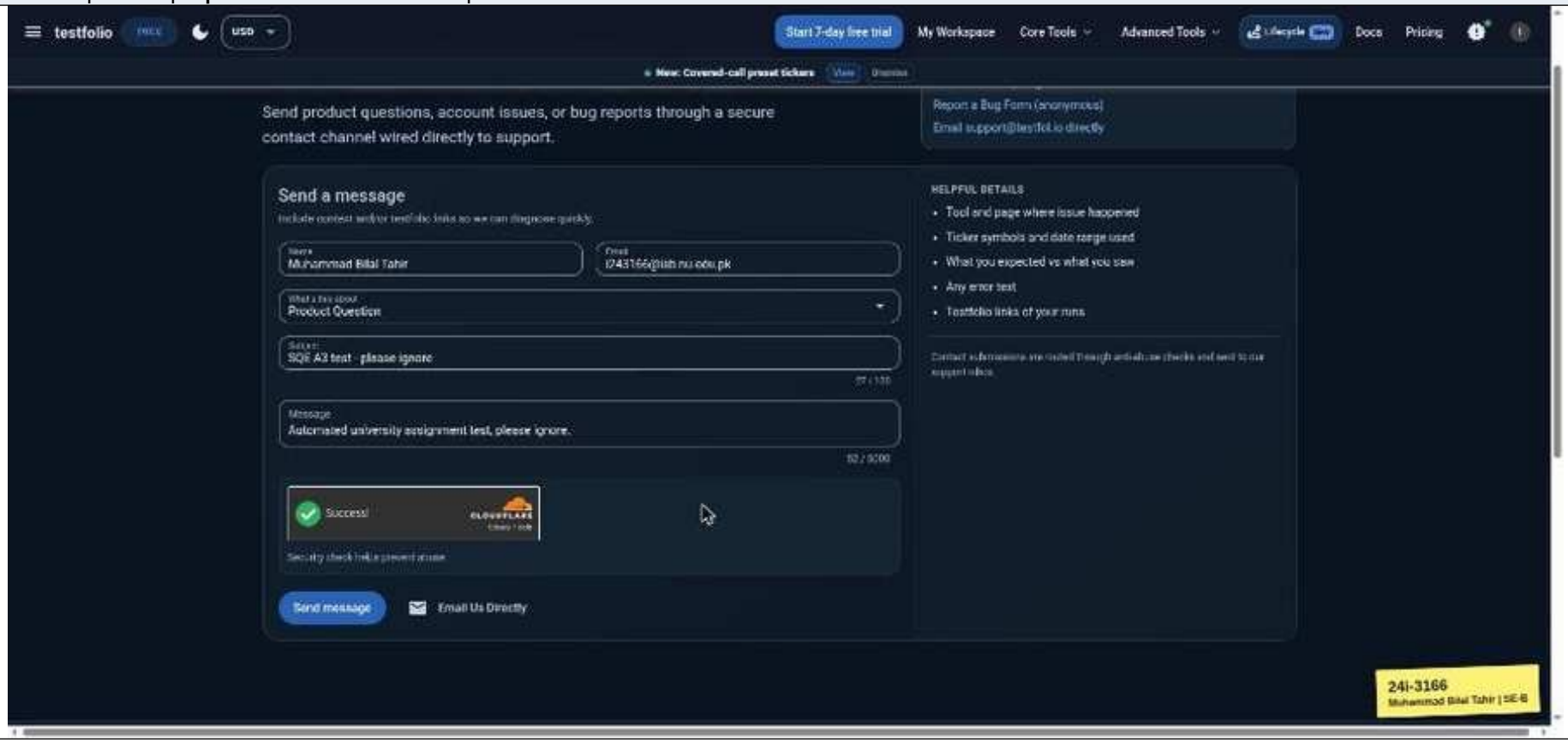


TC-EP-125 | Min weight to include (%) | input: 150 | result: Pass

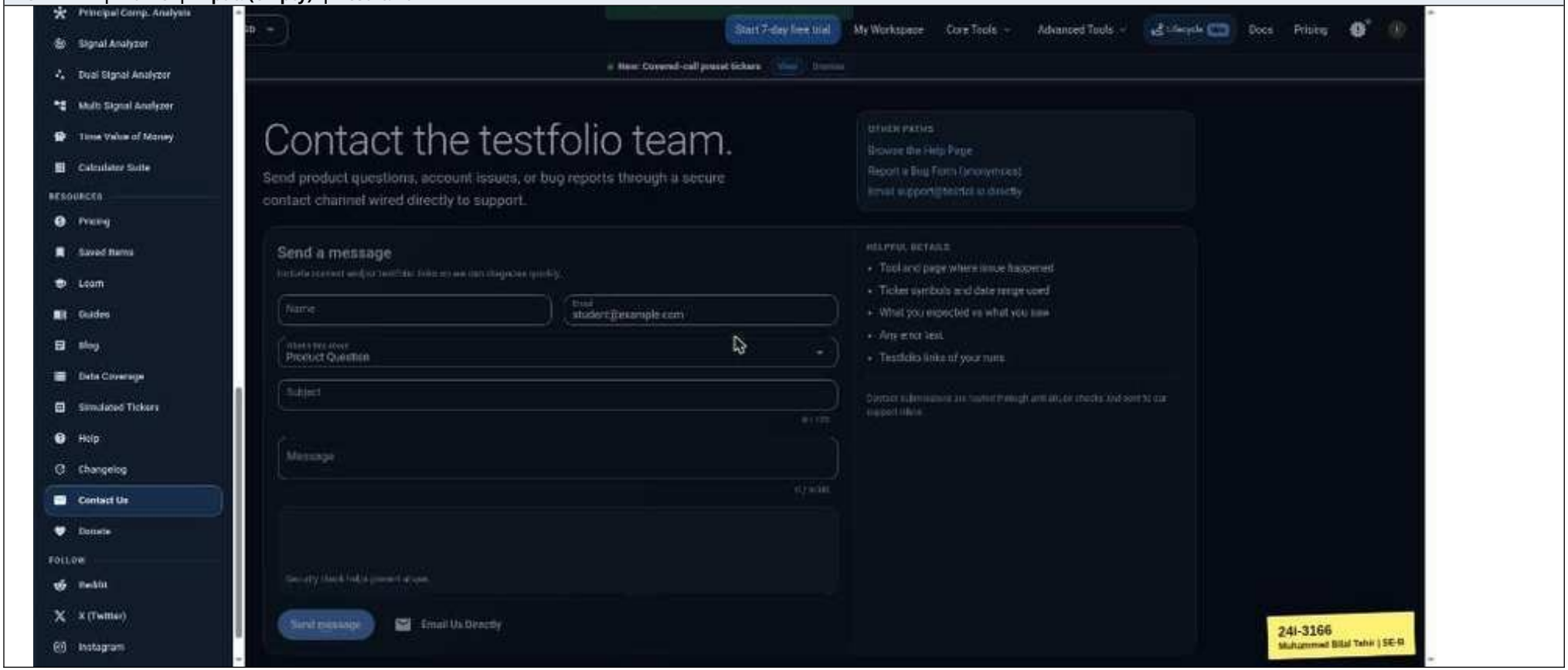


Contact Form (testfol.io/contact)

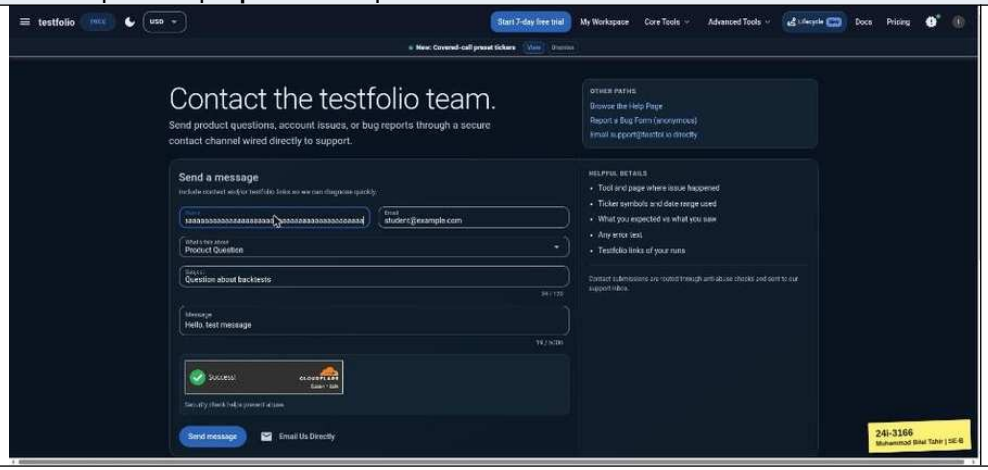
TC-EP-126 | Name | input: Muhammad Bilal Tahir | result: Pass

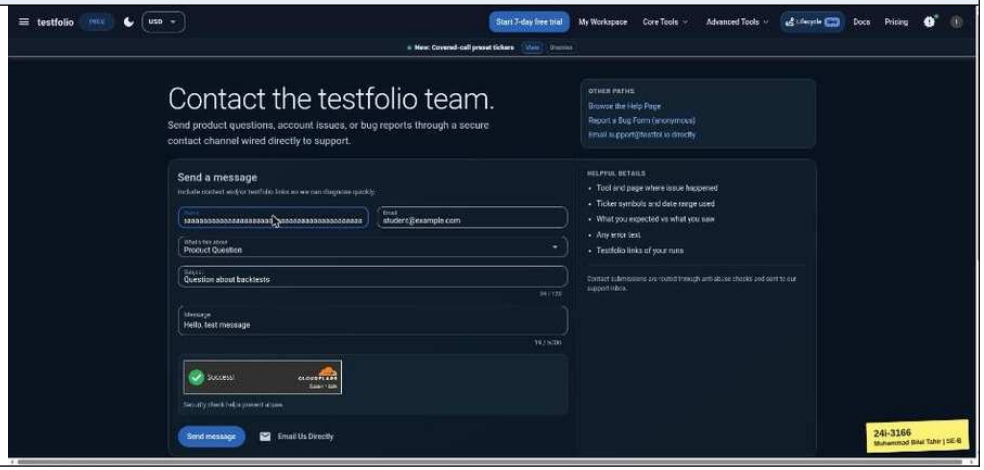


TC-EP-127 | Name | input: (empty) | result: Fail

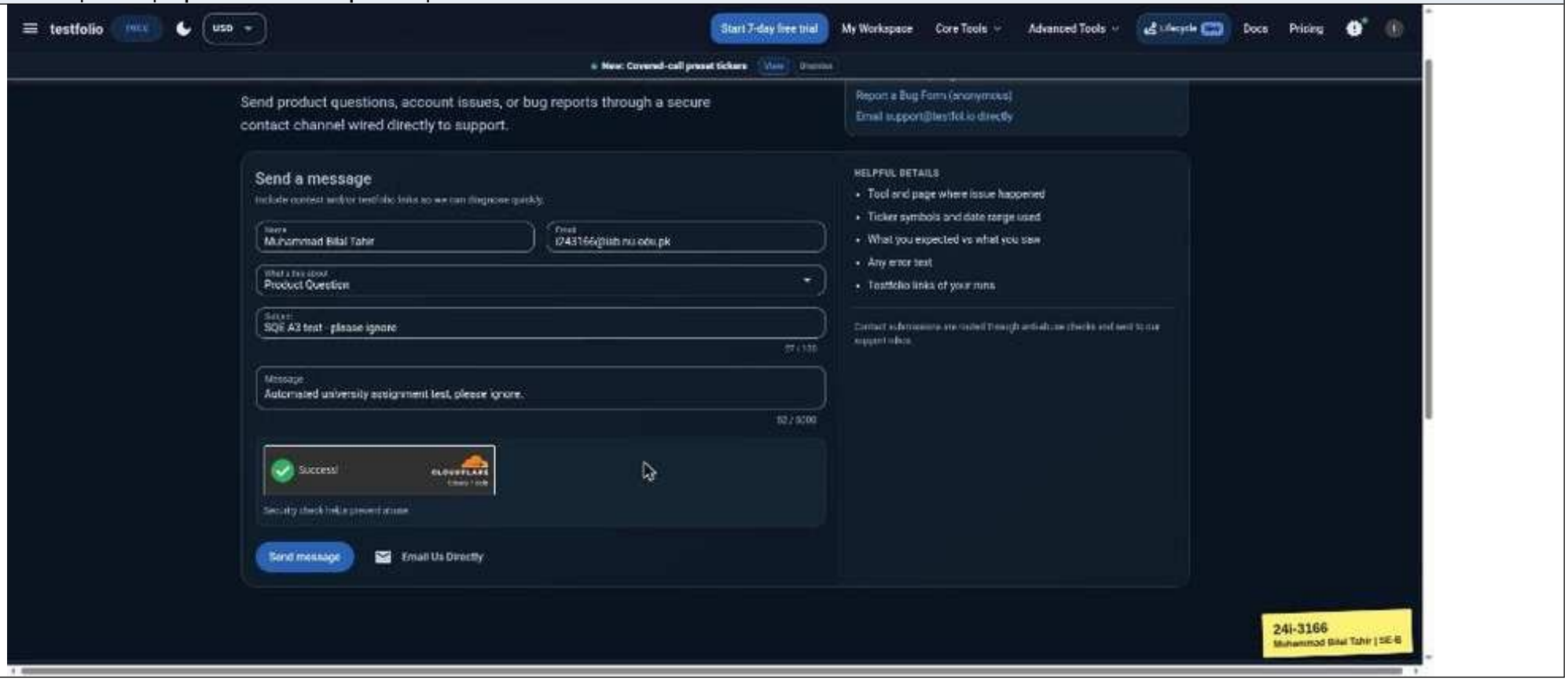


TC-EP-128 | Name | input: 121 x a | result: Pass

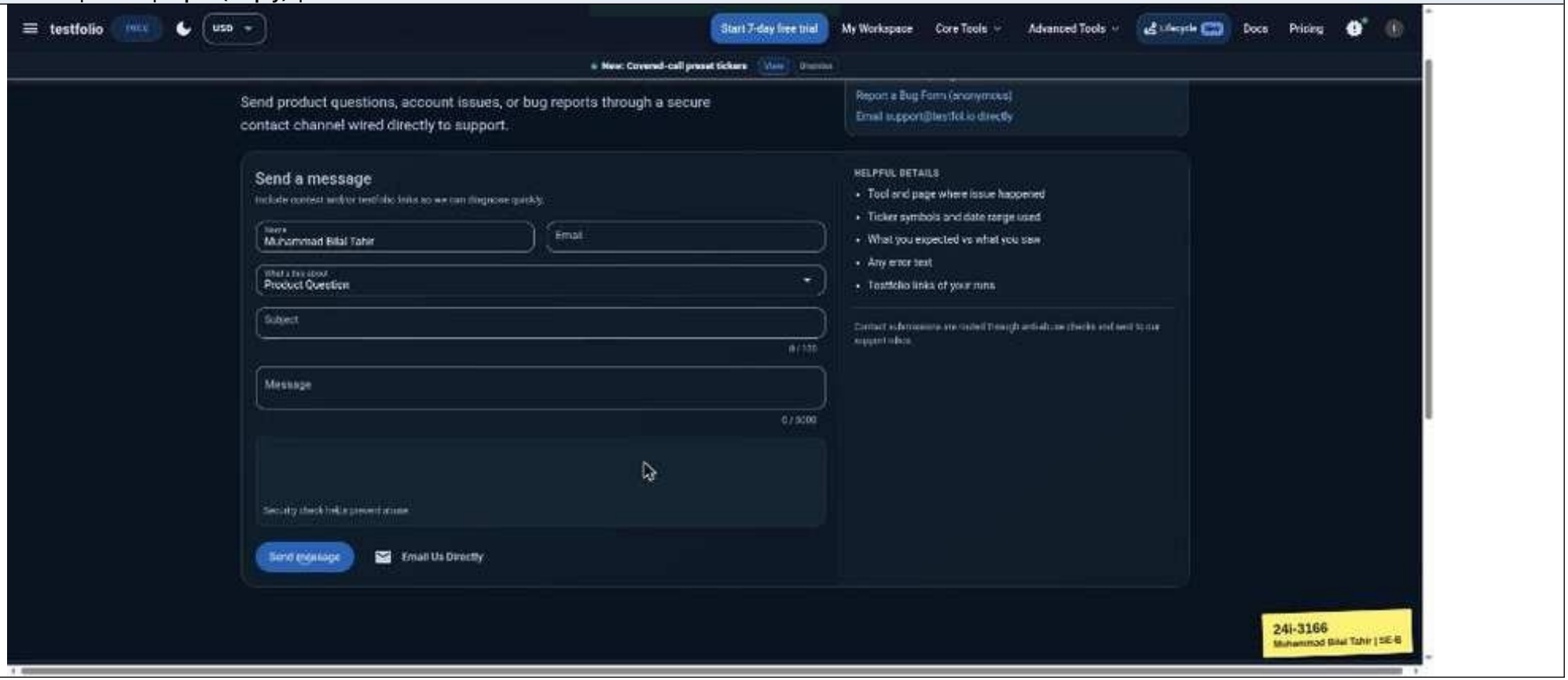




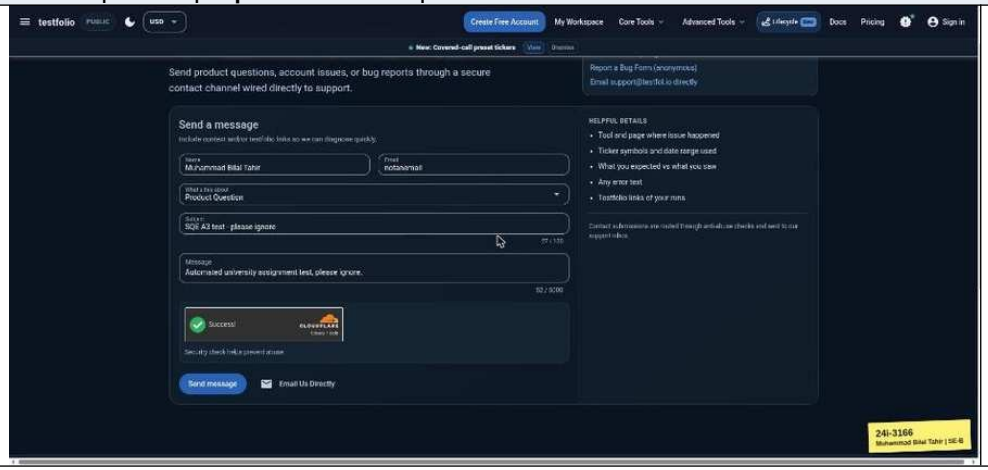
TC-EP-129 | Email | input: student@example.com | result: Pass

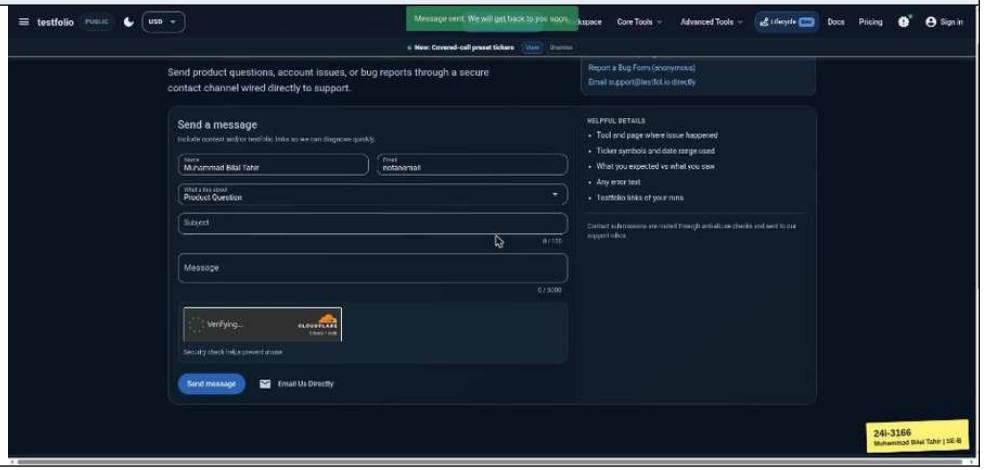


TC-EP-130 | Email | input: (empty) | result: Fail

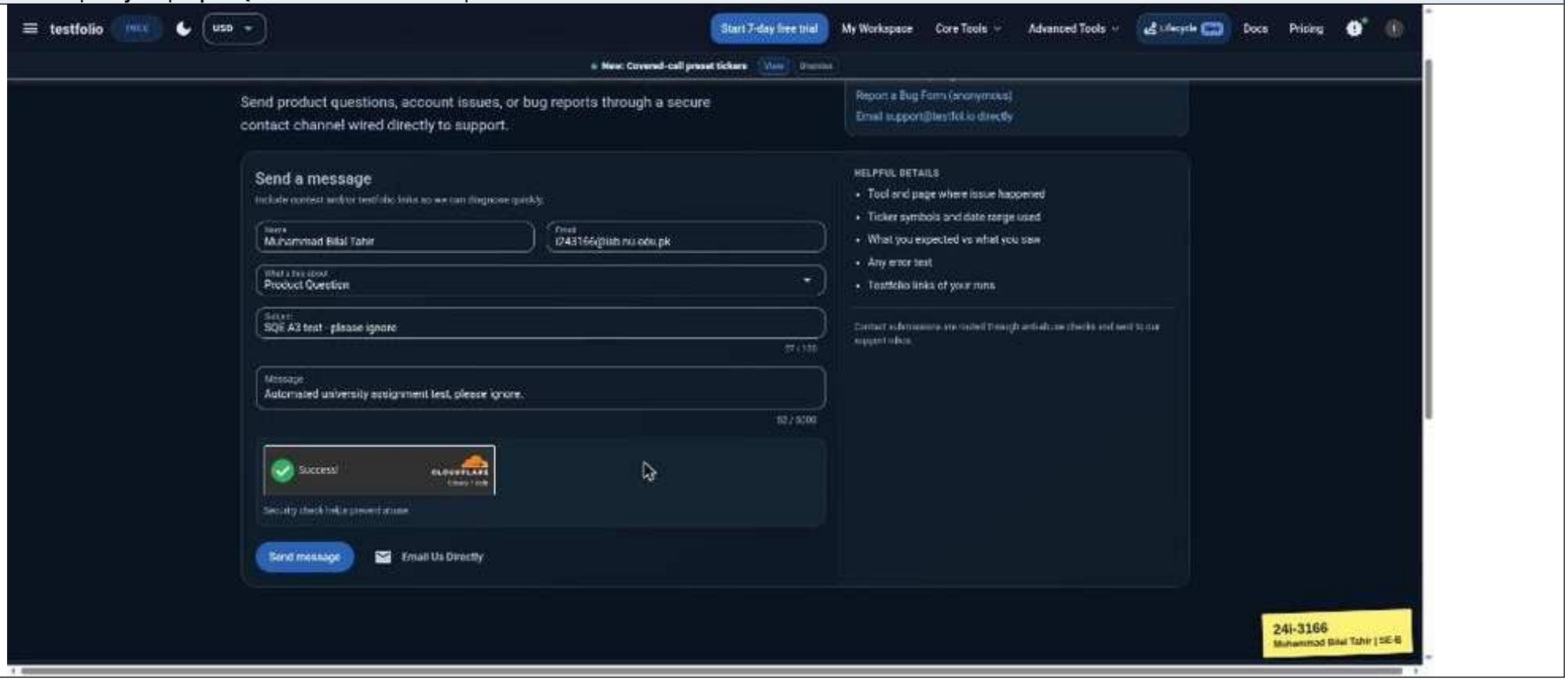


TC-EP-131 | Email | input: notanemail | result: Fail

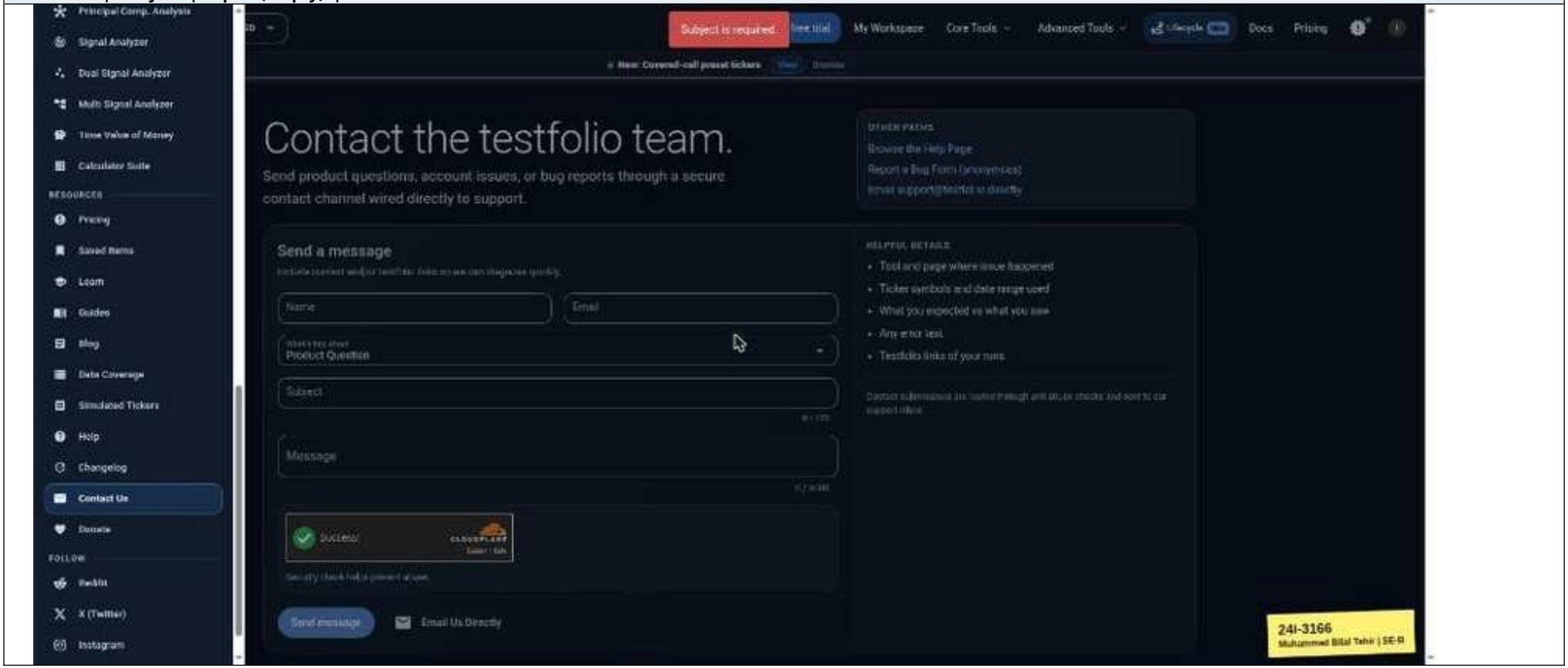




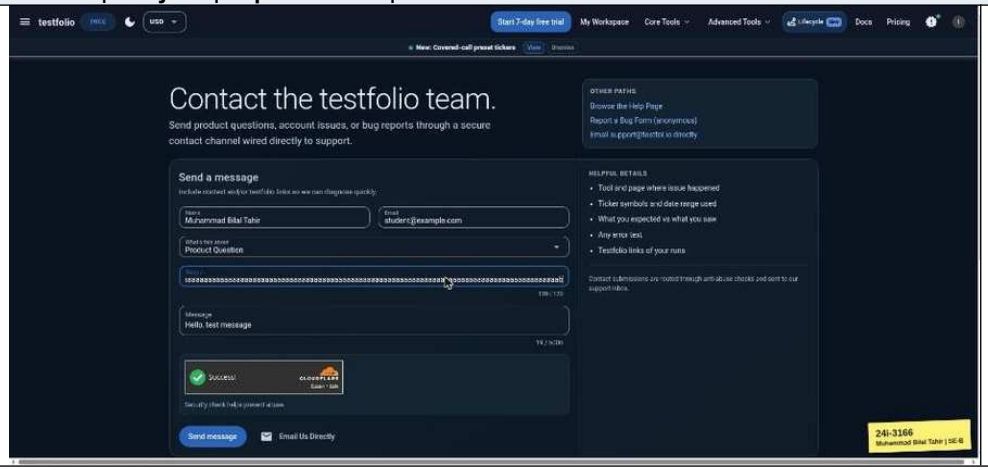
TC-EP-132 | Subject | input: Question about backtests | result: Pass

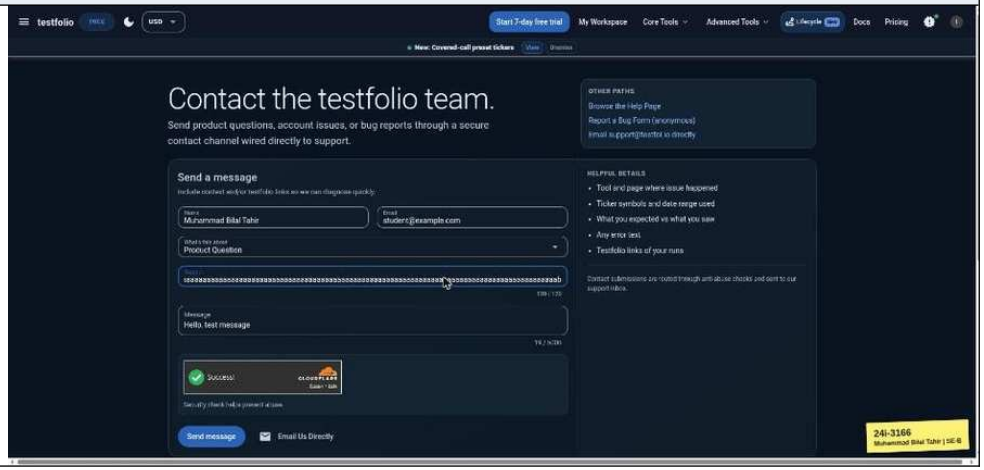


TC-EP-133 | Subject | input: (empty) | result: Pass

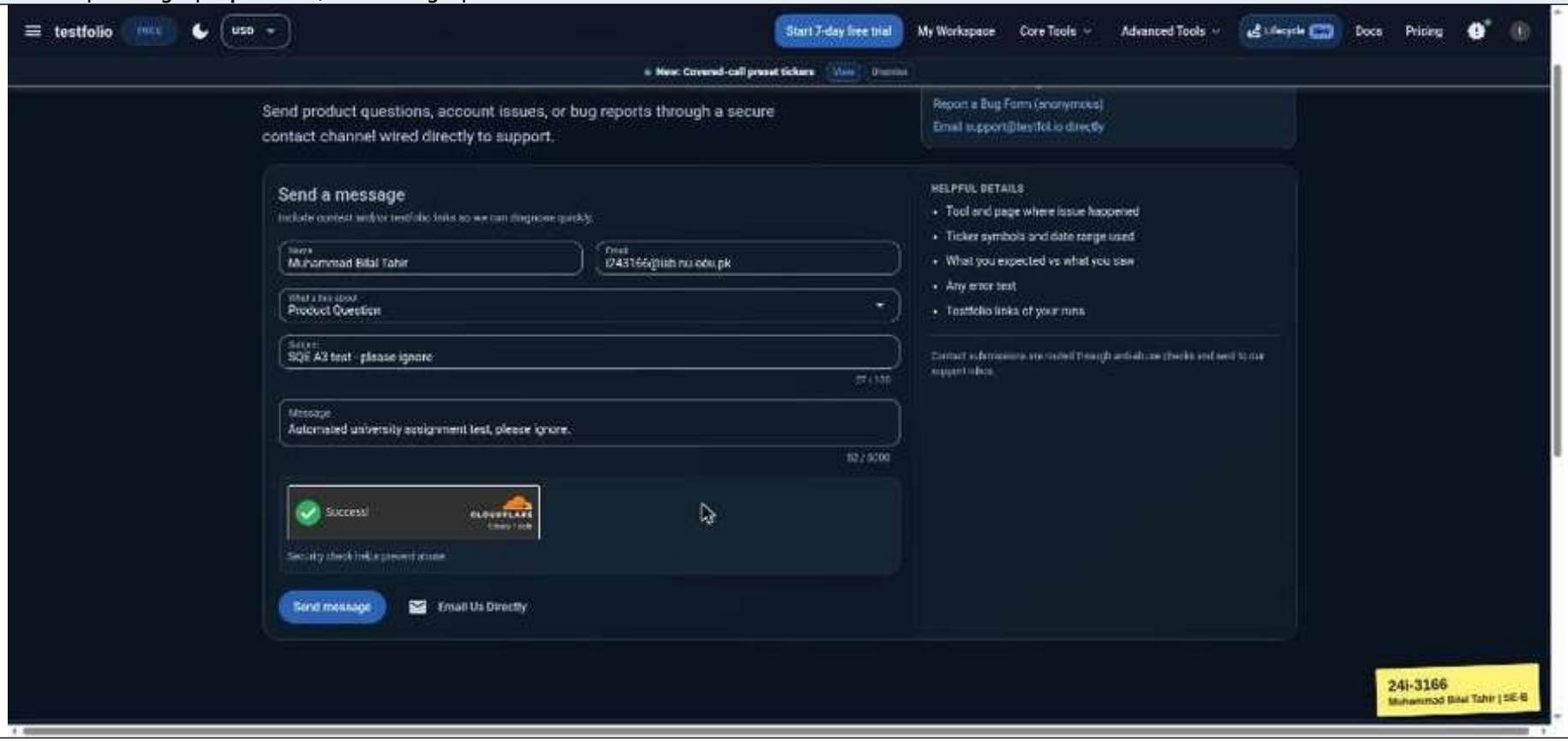


TC-EP-134 | Subject | input: 121 x a | result: Pass

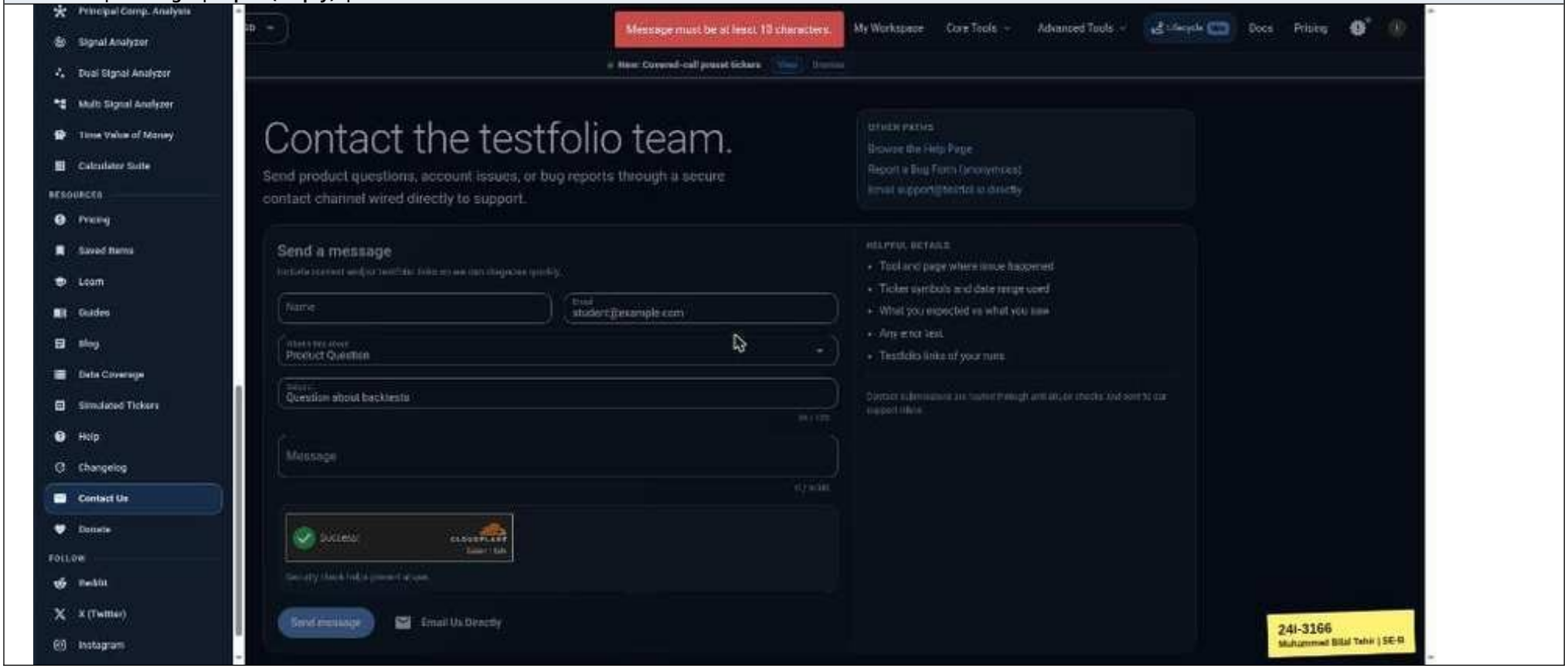




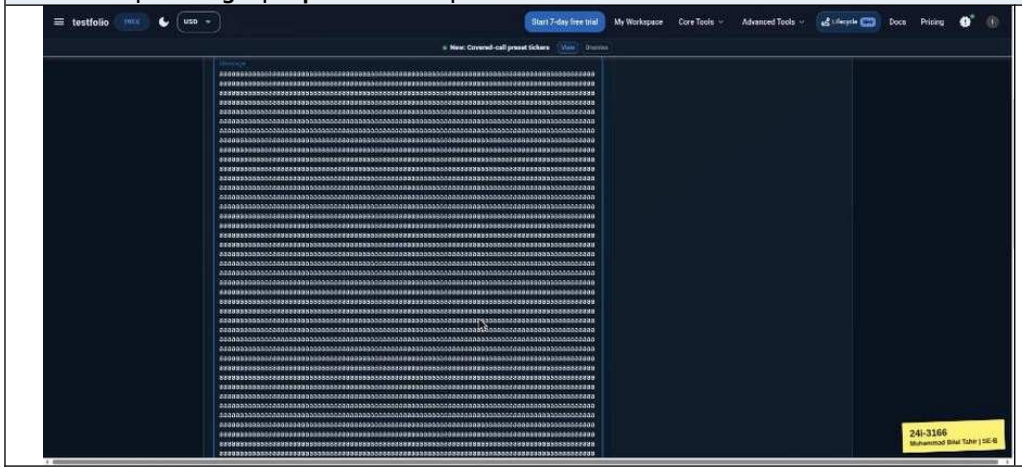
TC-EP-135 | Message | input: Hello, test message | result: Pass

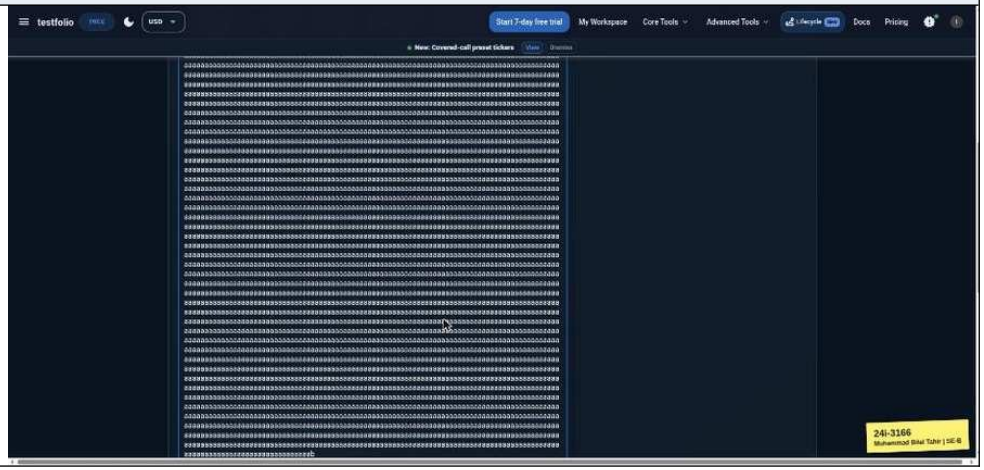


TC-EP-136 | Message | input: (empty) | result: Pass



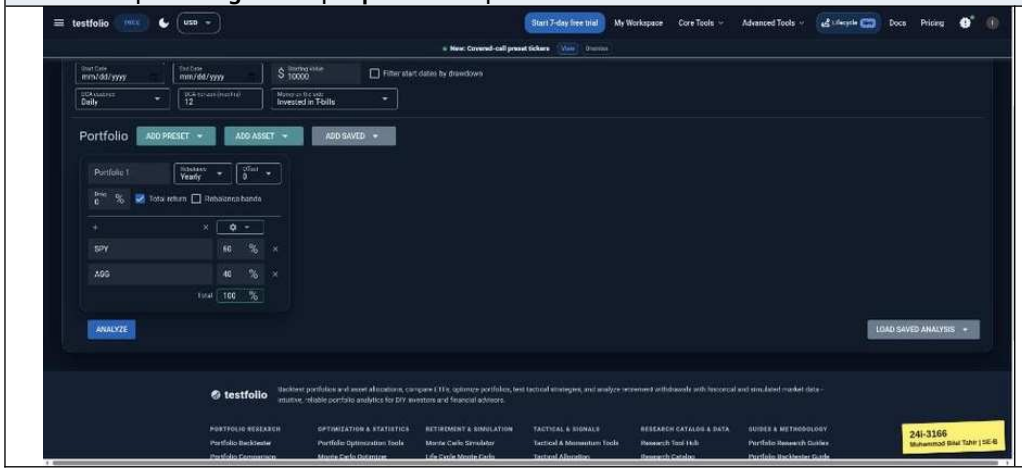
TC-EP-137 | Message | input: 5001 x a | result: Pass

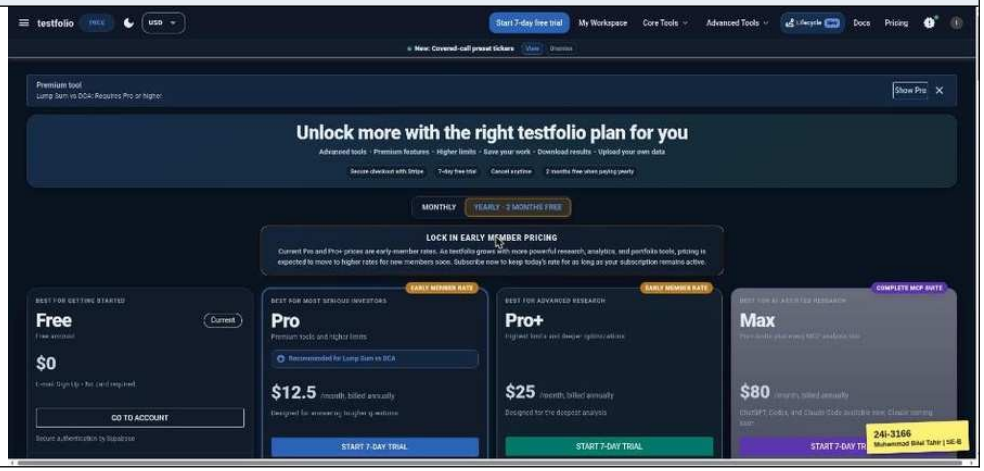




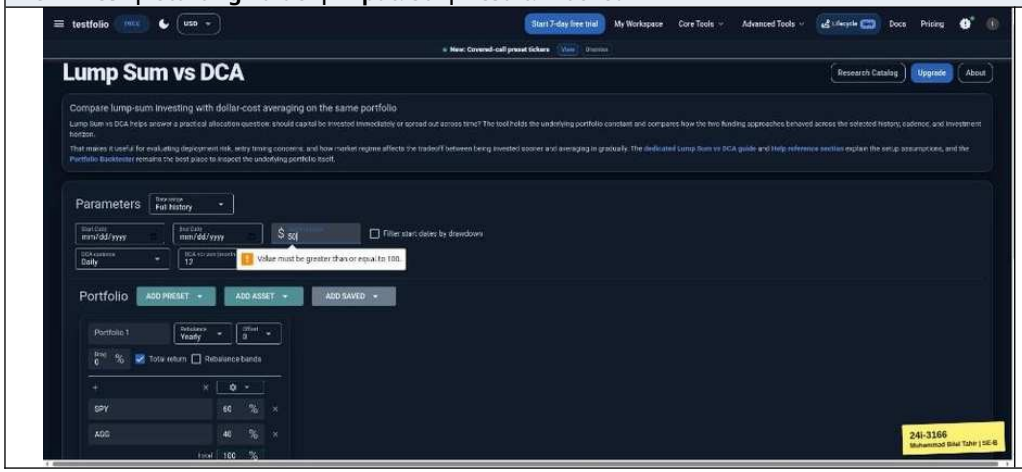
Lump Sum vs DCA (testfol.io/lumpsum-vs-dca)

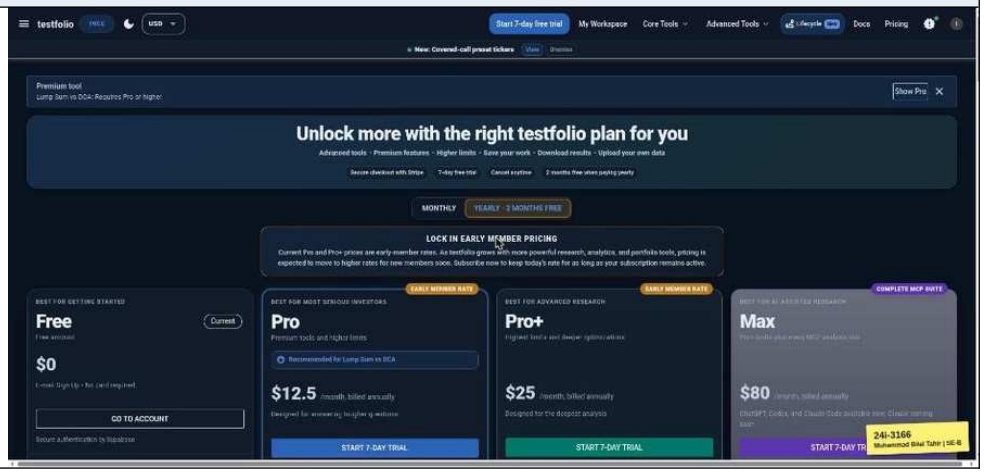
TC-EP-138 | Starting Value | input: 10000 | result: Blocked



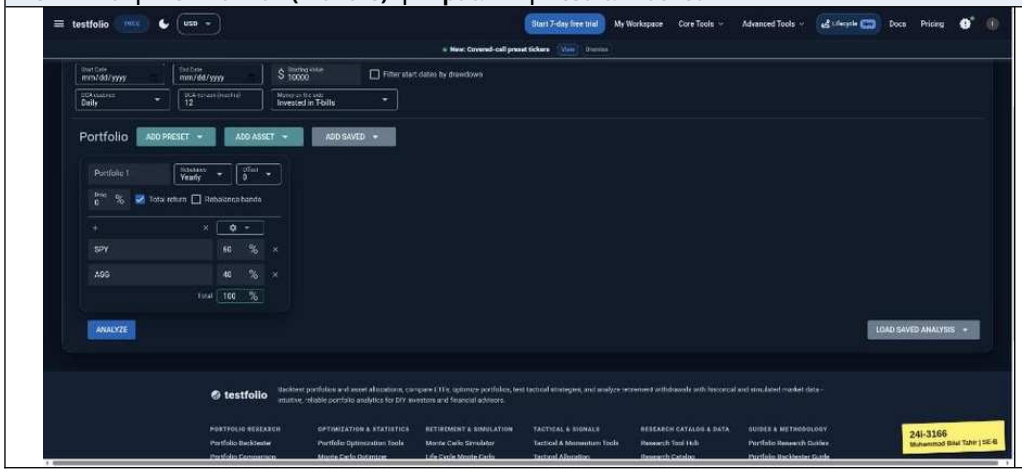


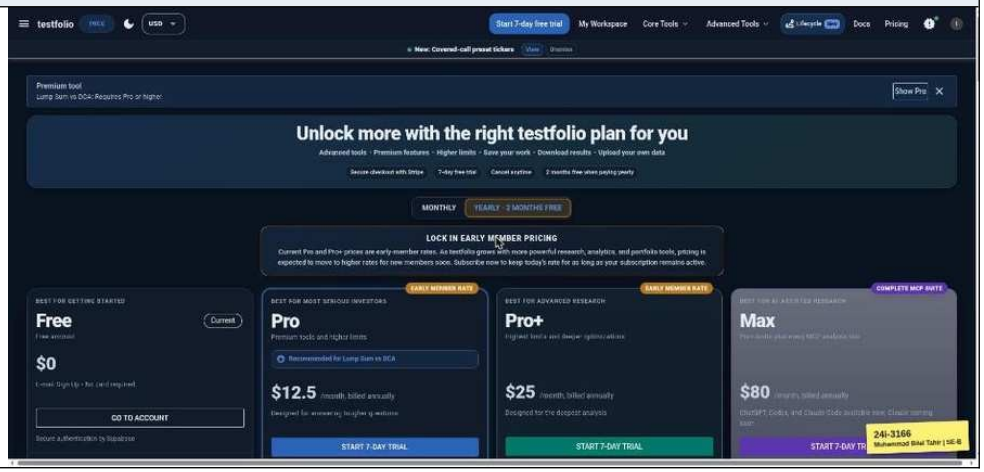
TC-EP-139 | Starting Value | input: 50 | result: Blocked



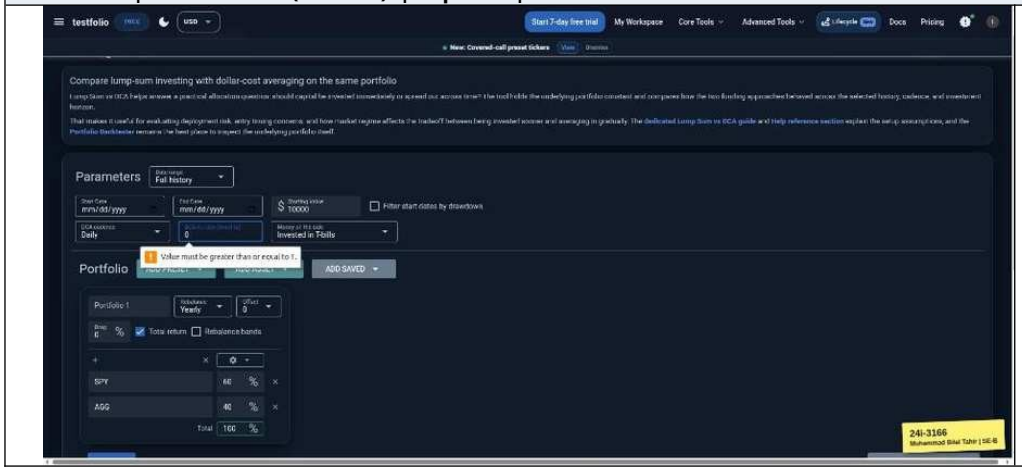


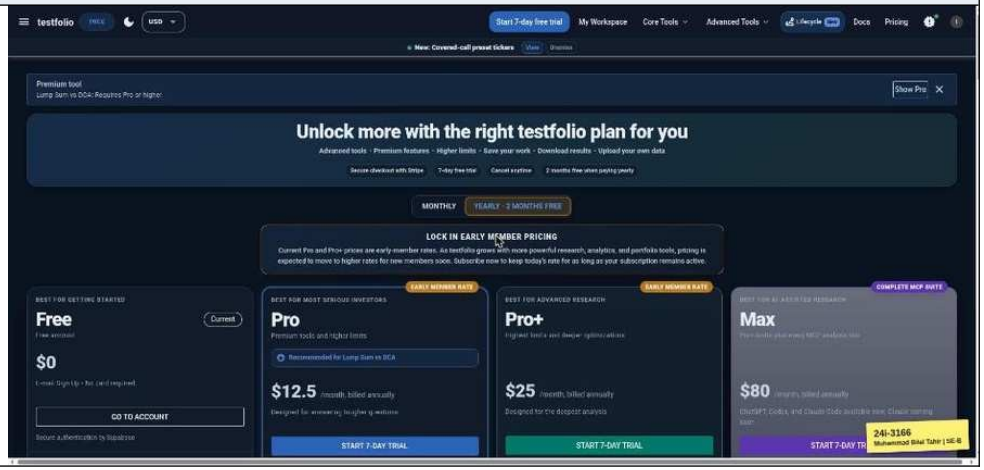
TC-EP-140 | DCA horizon (months) | input: 12 | result: Blocked



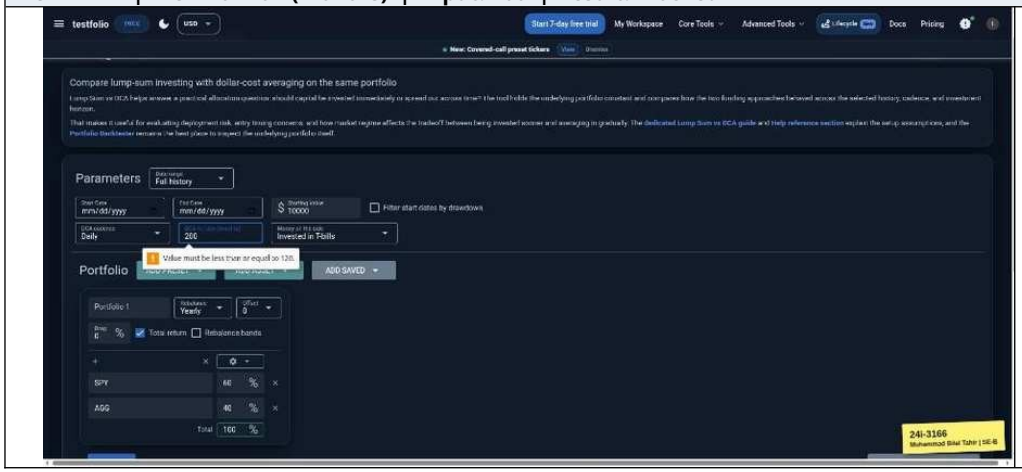


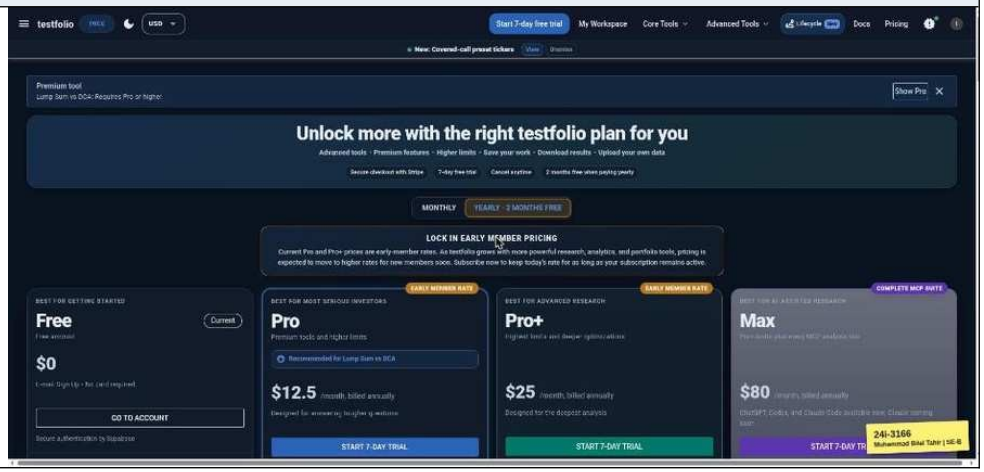
TC-EP-141 | DCA horizon (months) | input: 0 | result: Blocked



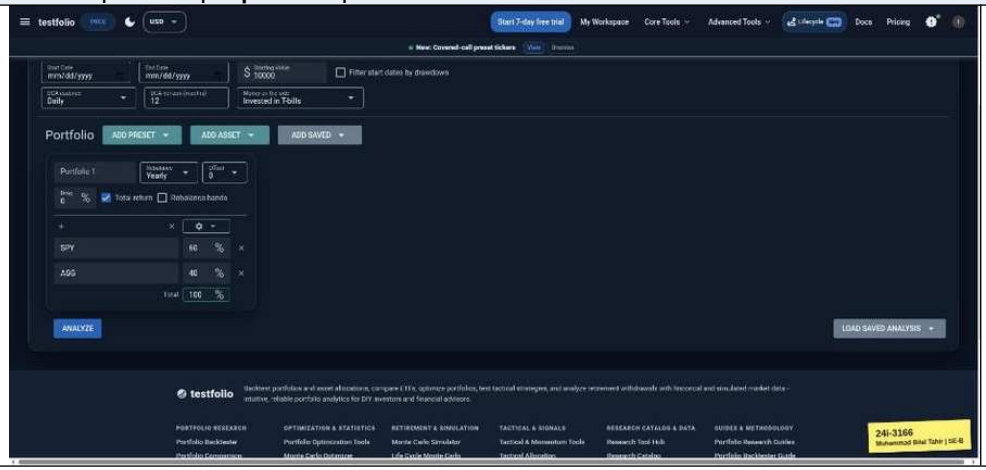


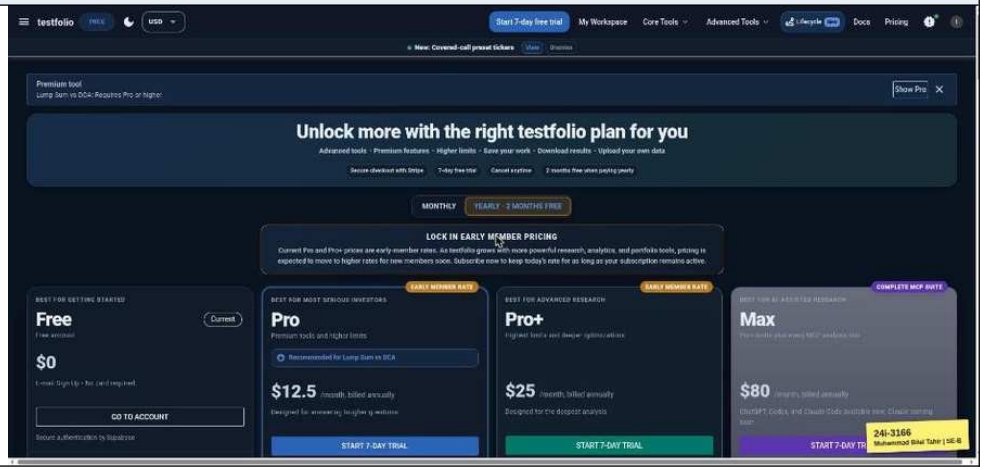
TC-EP-142 | DCA horizon (months) | input: 200 | result: Blocked



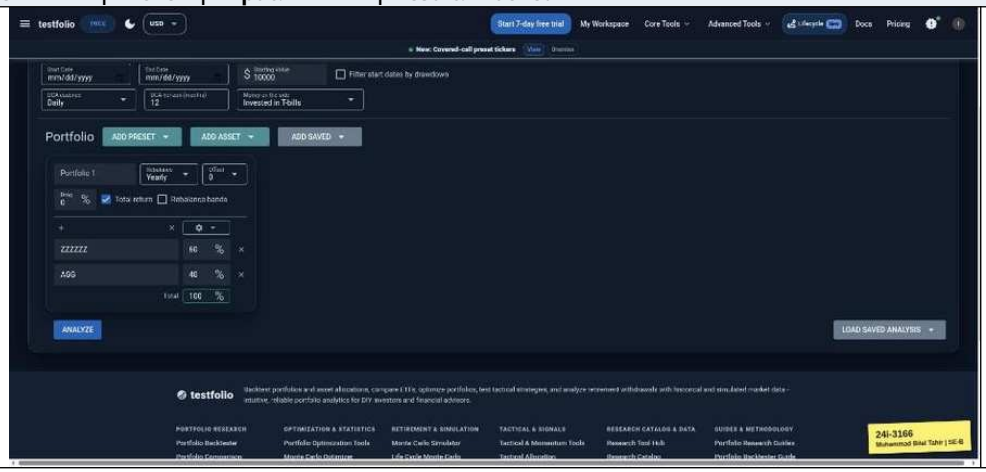


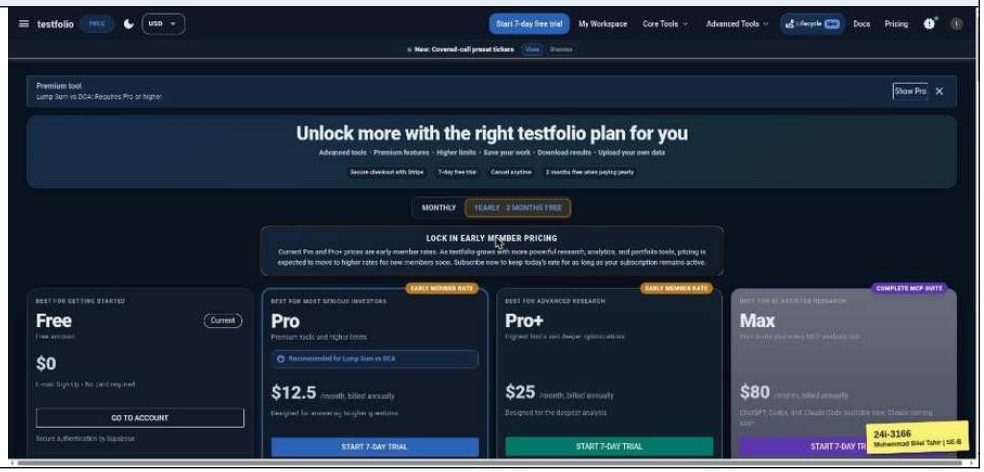
TC-EP-143 | Ticker | input: SPY | result: Blocked



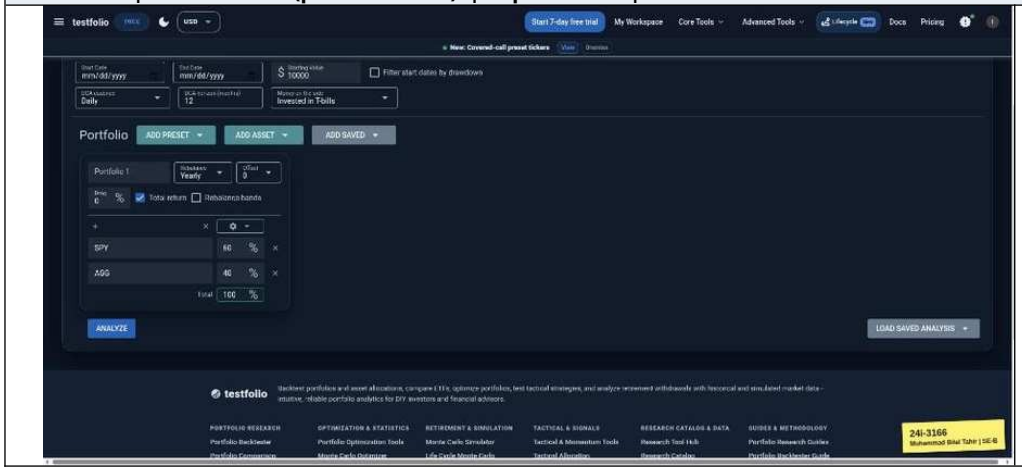


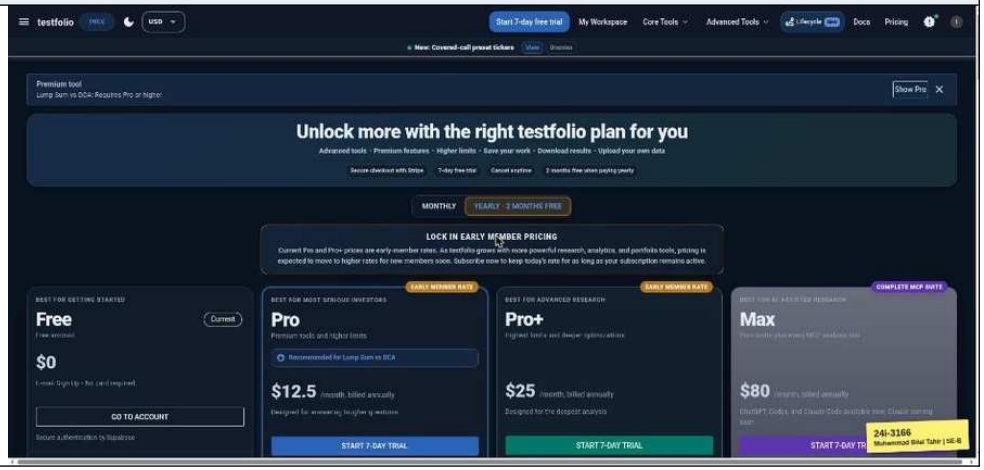
TC-EP-144 | Ticker | input: ZZZZZZ | result: Blocked



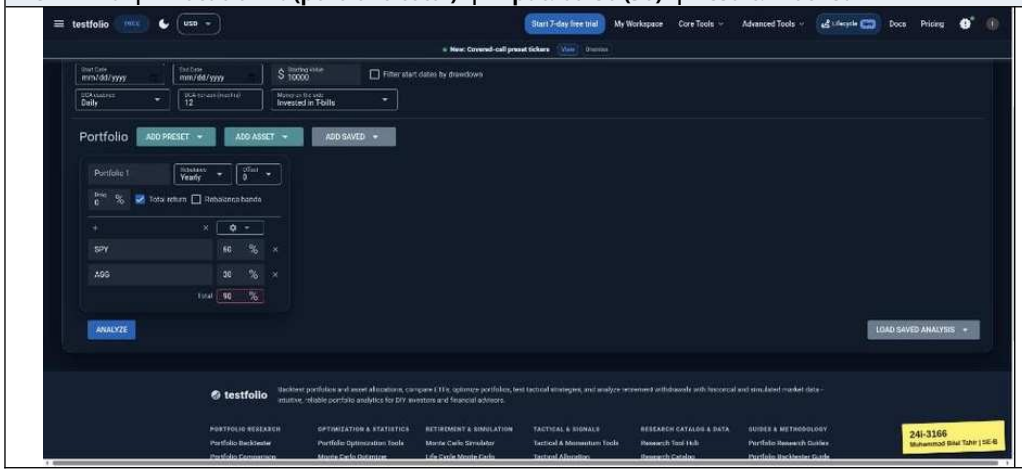


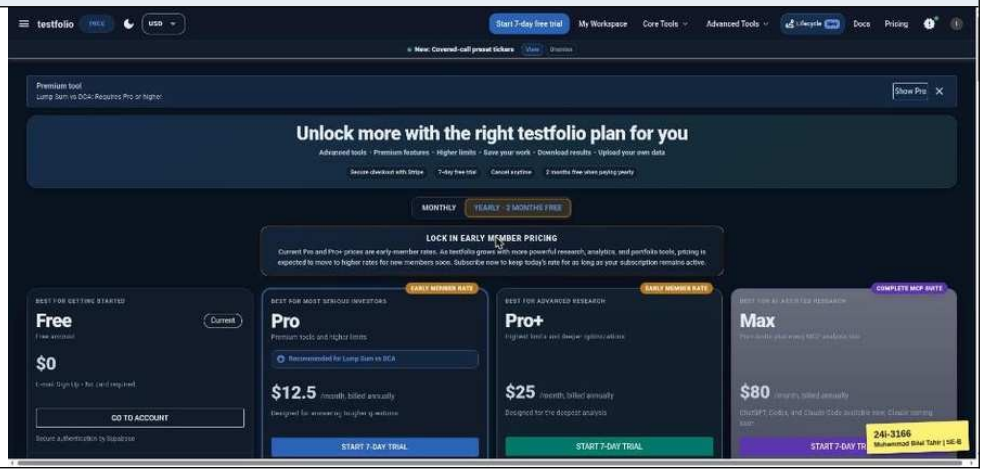
TC-EP-145 | Allocation % (portfolio total) | input: 60/40 | result: Blocked





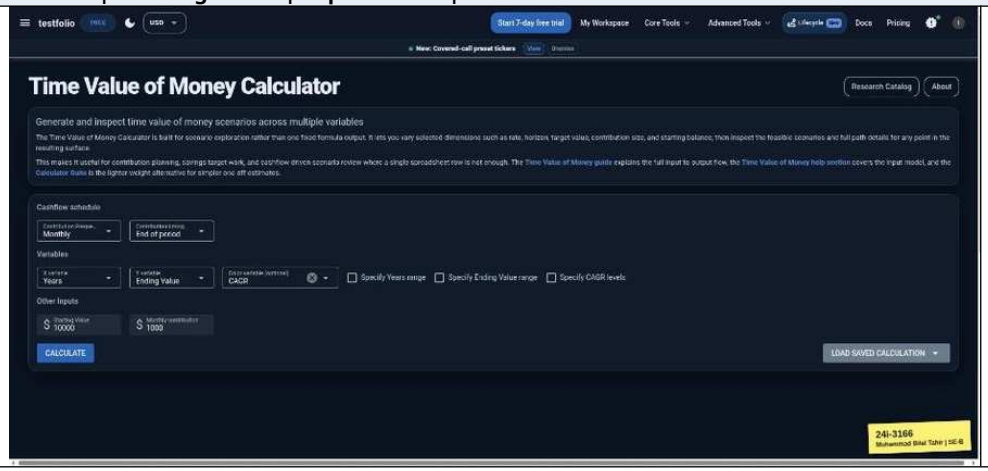
TC-EP-146 | Allocation % (portfolio total) | input: 60/30 (90) | result: Blocked





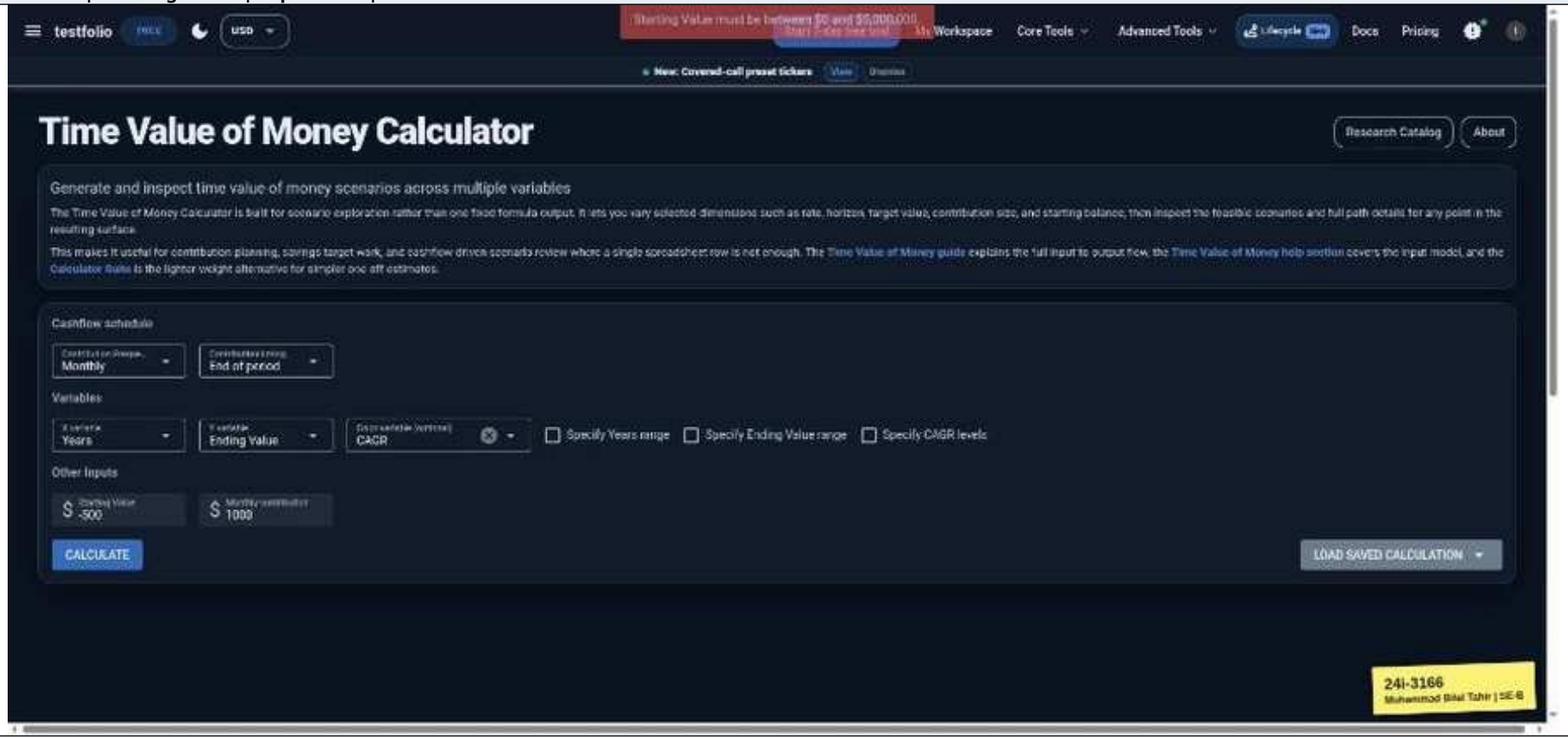
Time Value of Money Calculator (testfol.io/time-value-of-money)

TC-EP-147 | Starting Value | input: 10000 | result: Pass

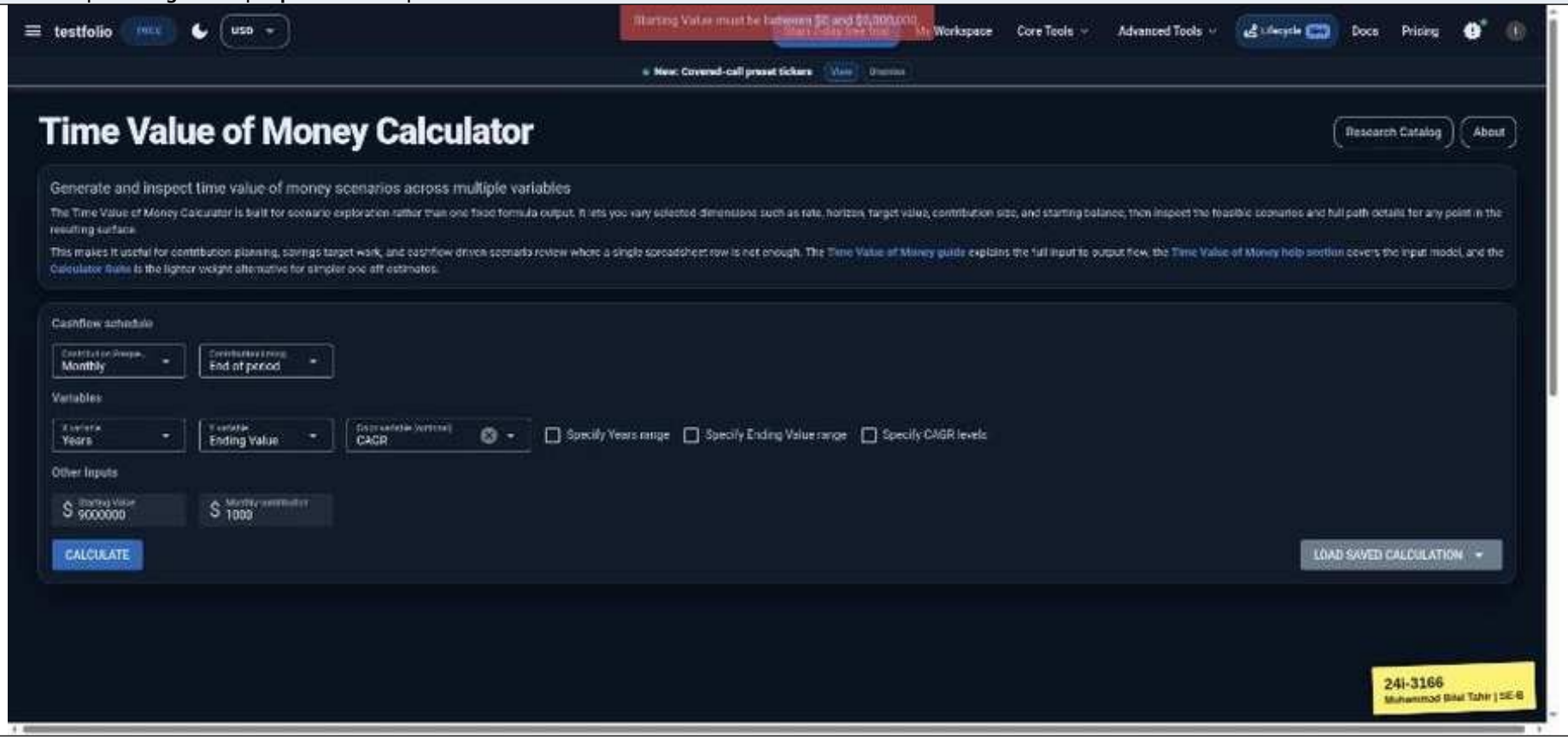




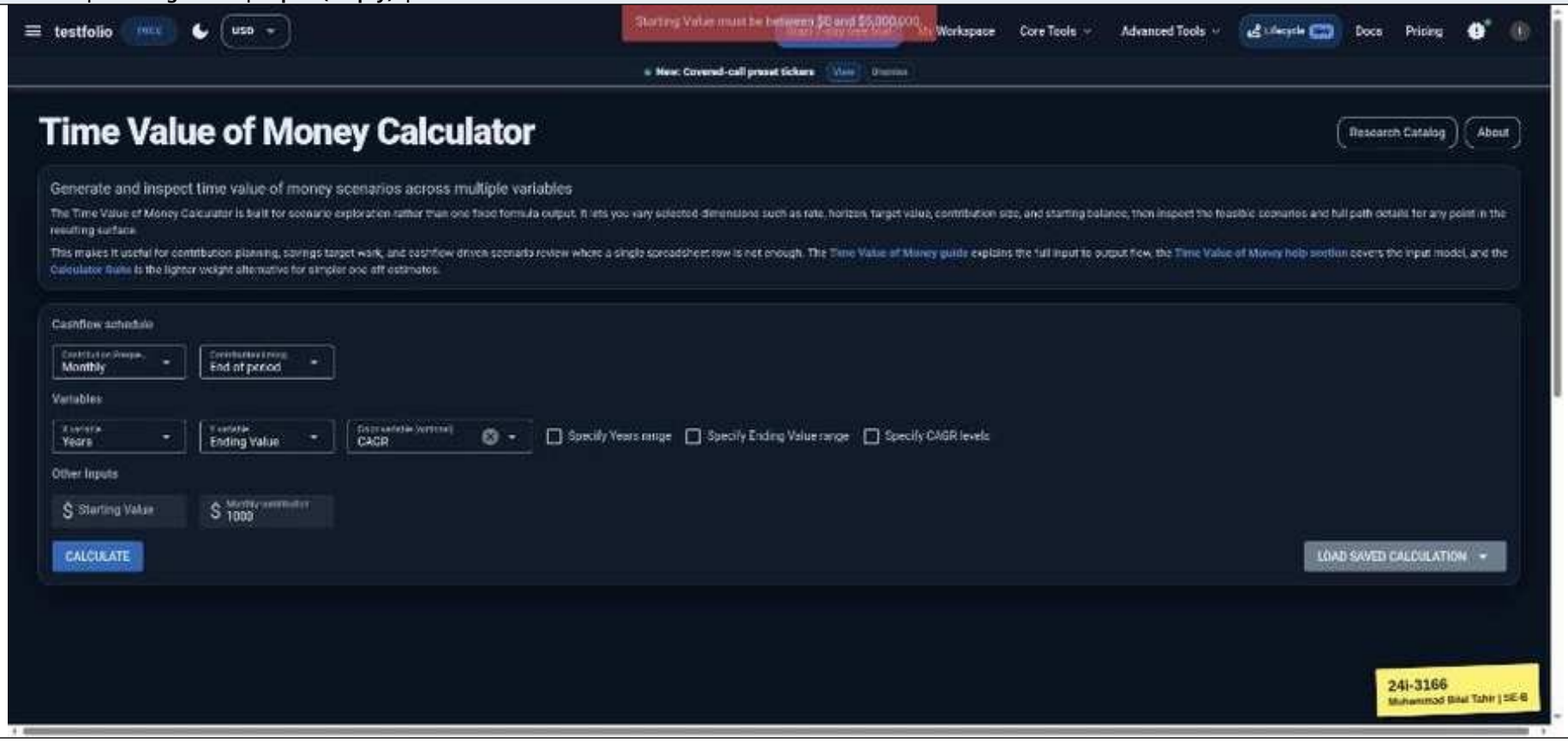
TC-EP-148 | Starting Value | input: -500 | result: Pass



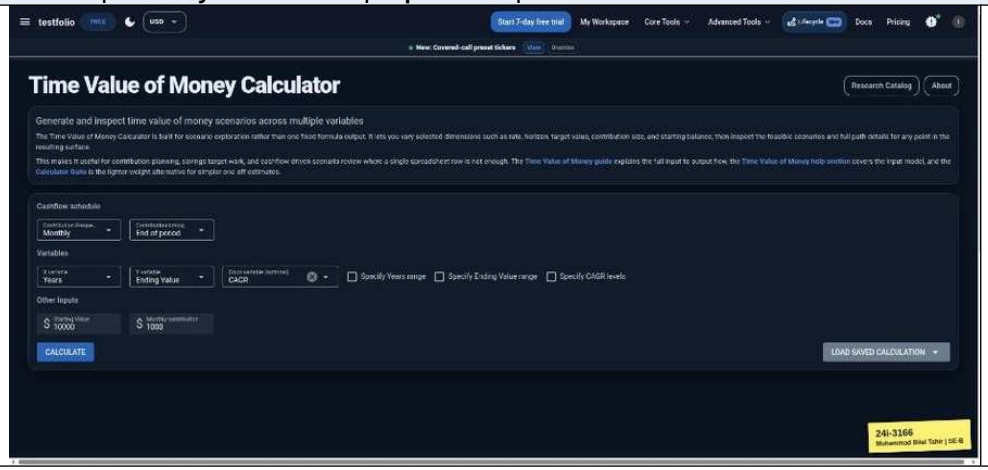
TC-EP-149 | Starting Value | input: 9000000 | result: Pass



TC-EP-150 | Starting Value | input: (empty) | result: Pass

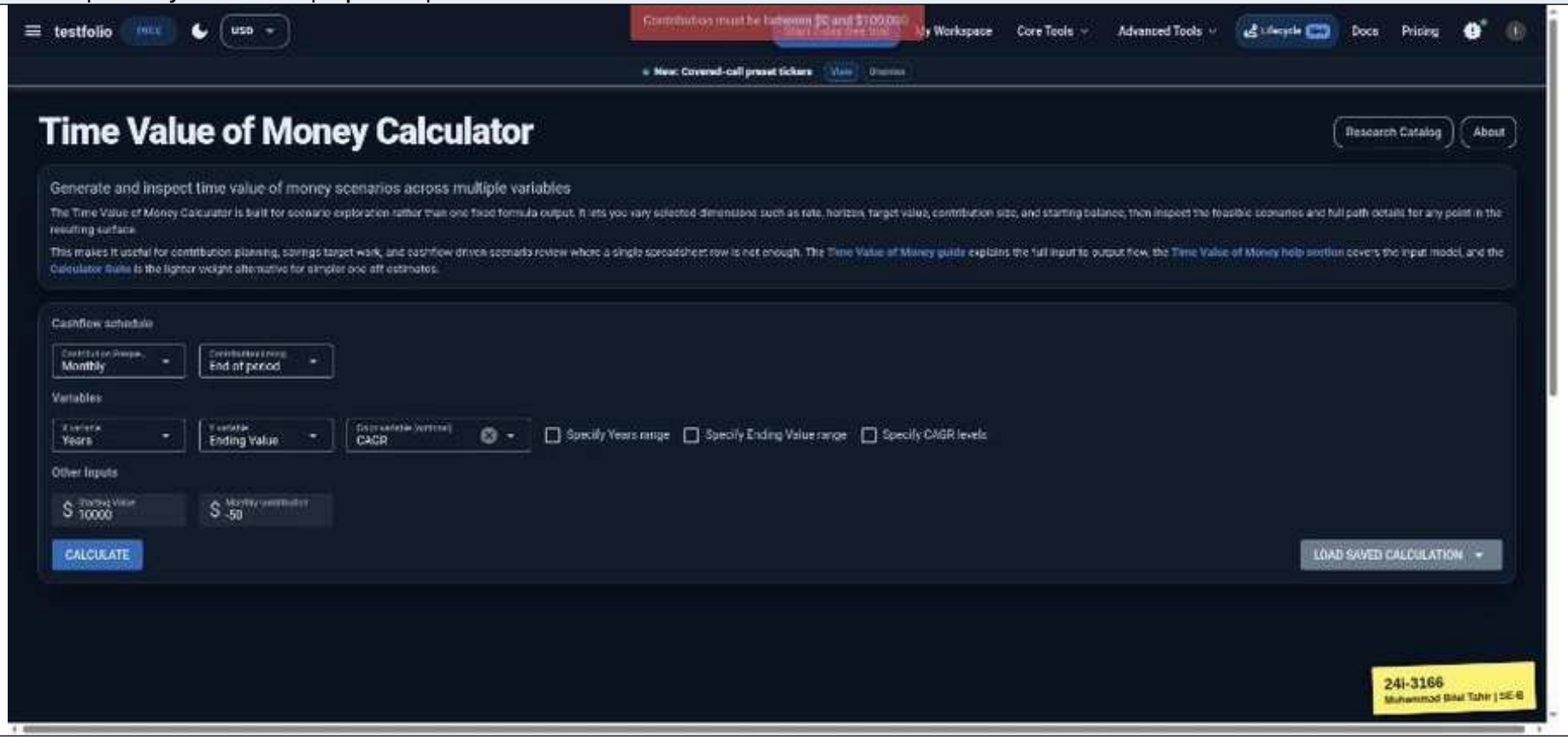


TC-EP-151 | Monthly contribution | input: 1000 | result: Pass

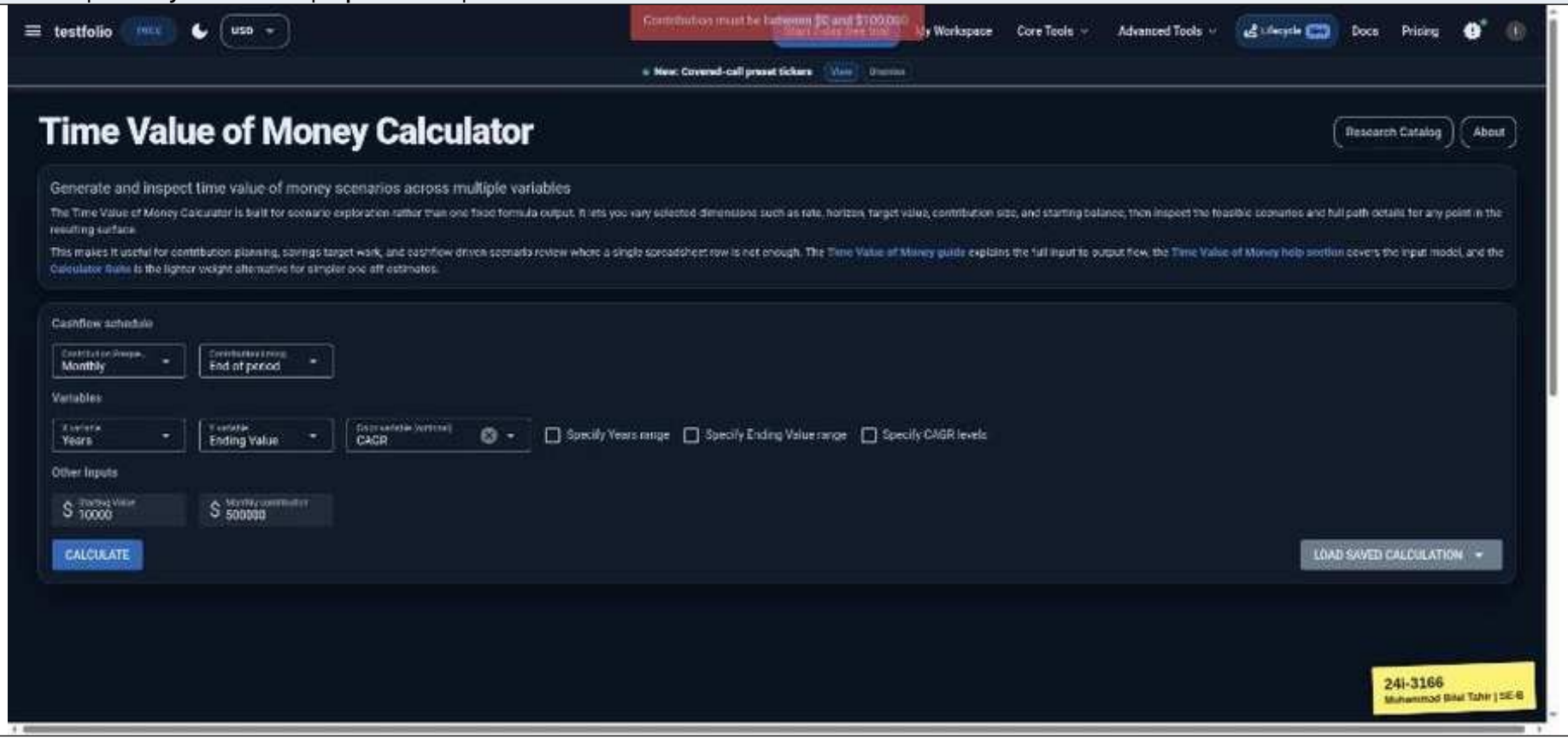




TC-EP-152 | Monthly contribution | input: -50 | result: Pass

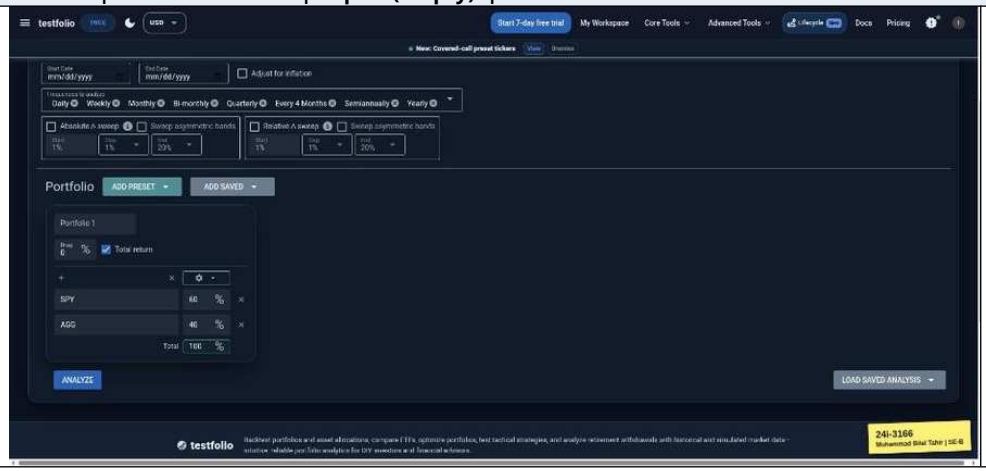


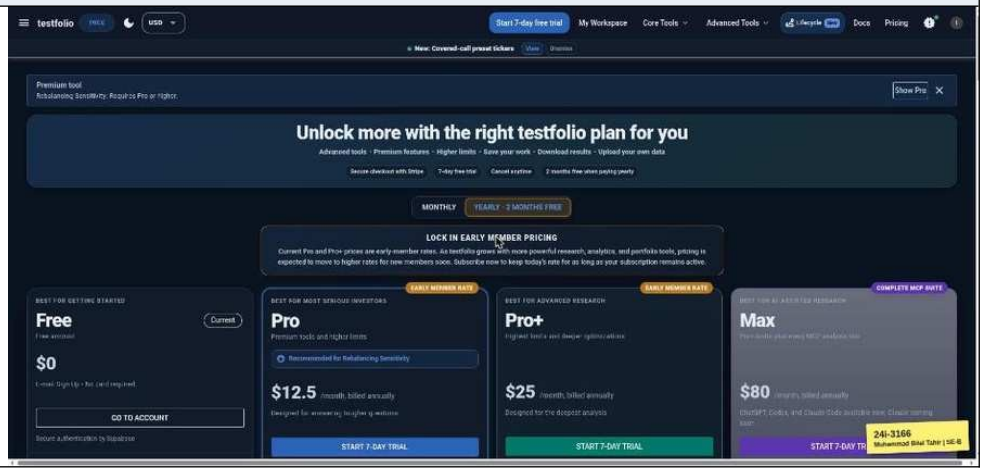
TC-EP-153 | Monthly contribution | input: 500000 | result: Pass



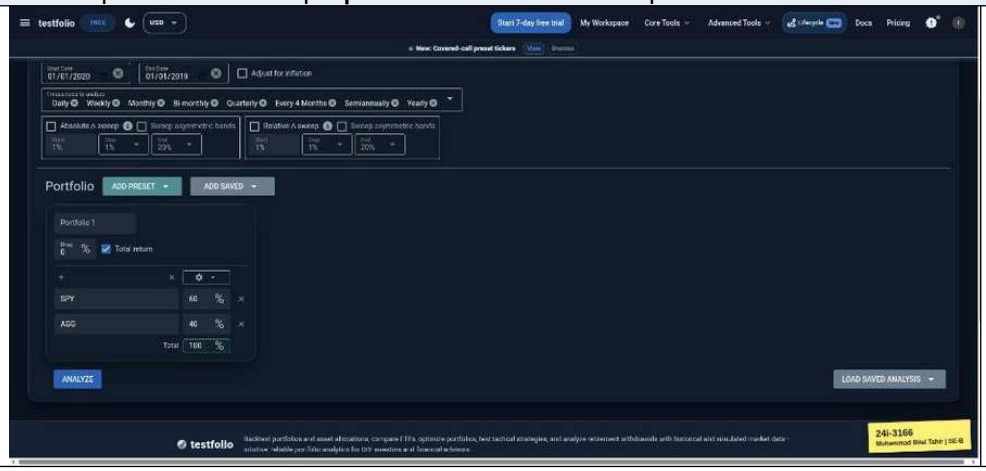
Rebalancing Sensitivity (testfol.io/rebalancing-sensitivity)

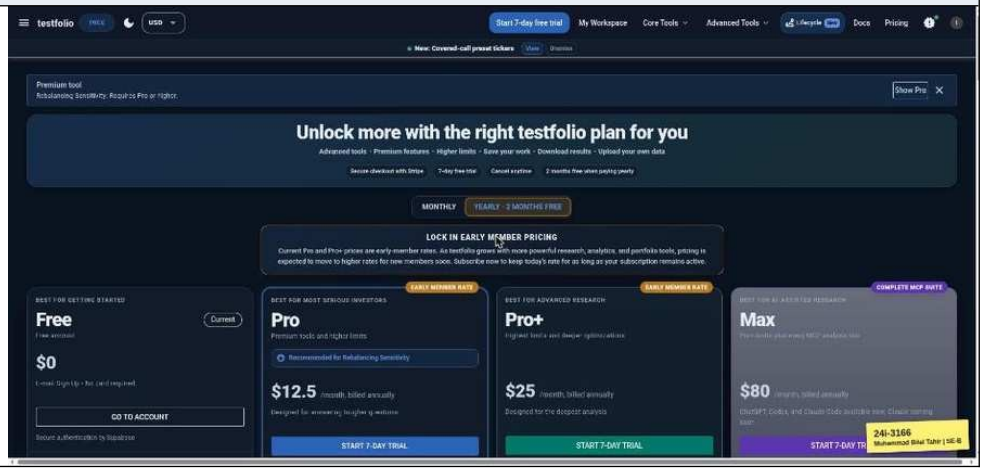
TC-EP-154 | Start / End Date | input: (empty) | result: Blocked



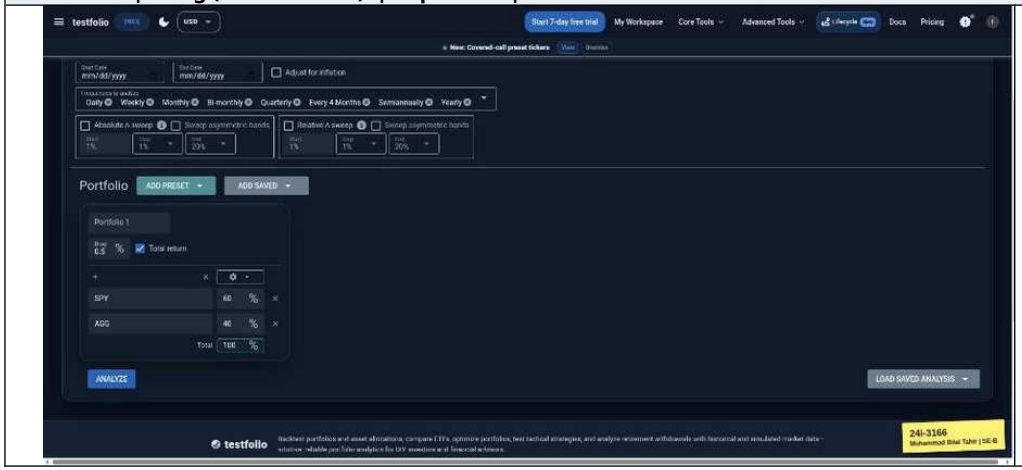


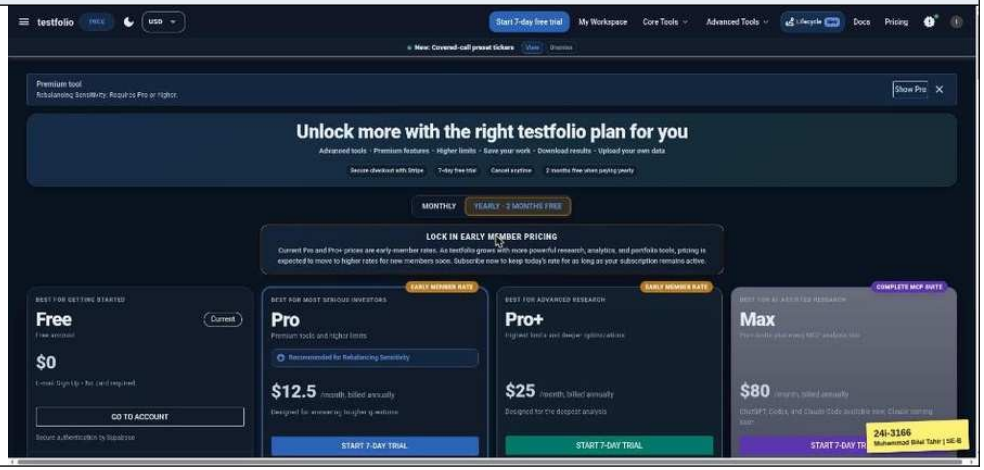
TC-EP-155 | Start / End Date | input: 2020-01-01 / 2019-01-01 | result: Blocked



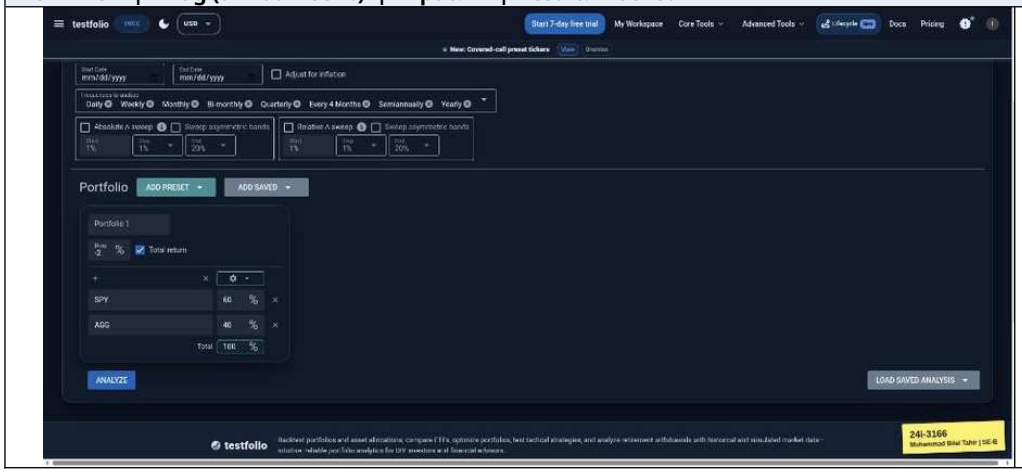


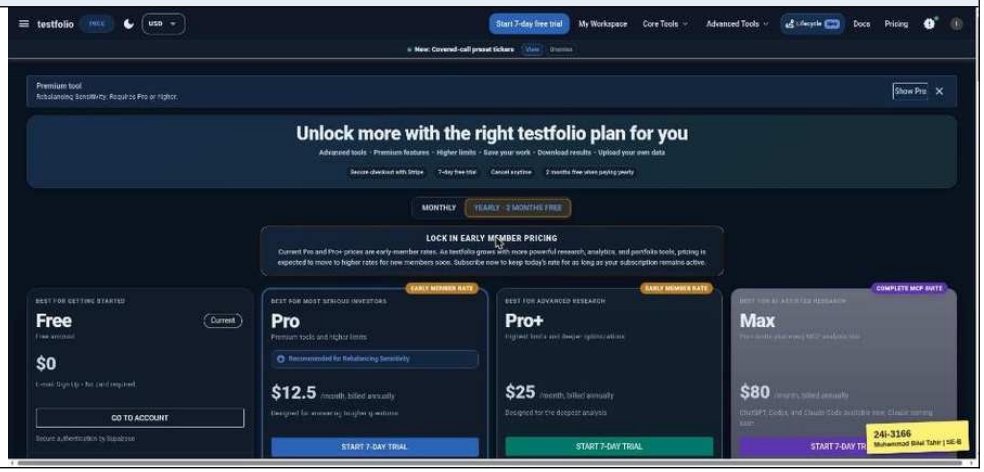
TC-EP-156 | Drag (annual fee %) | input: 0.5 | result: Blocked



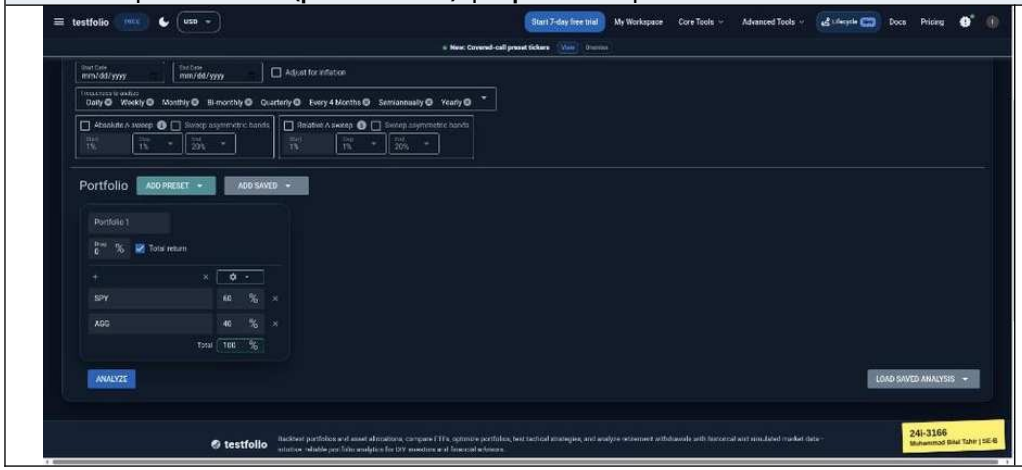


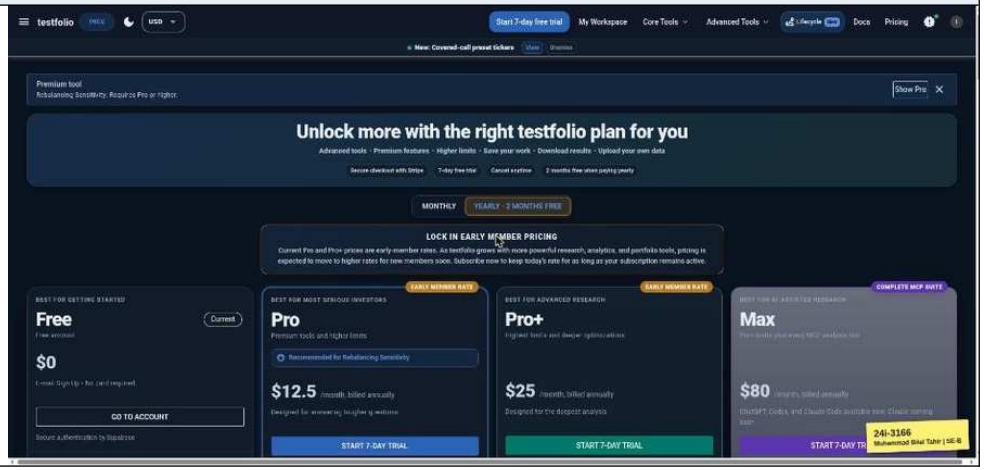
TC-EP-157 | Drag (annual fee %) | input: -2 | result: Blocked



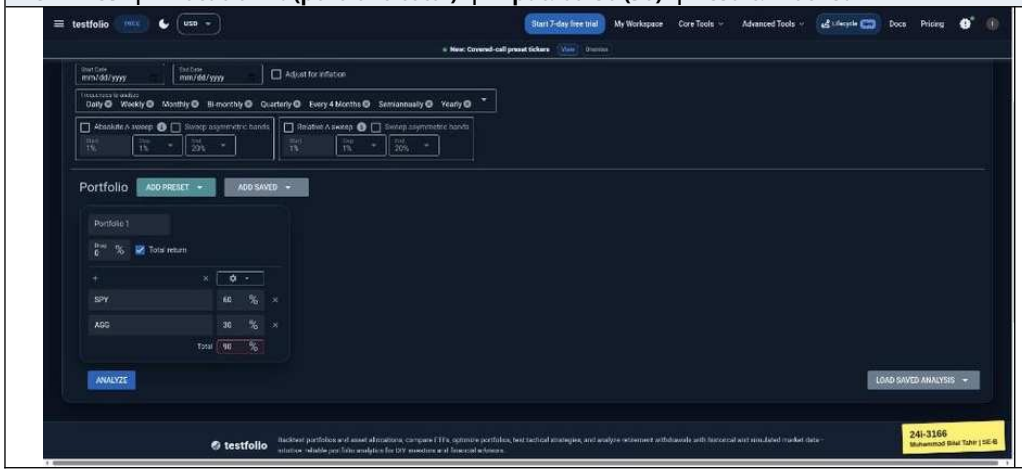


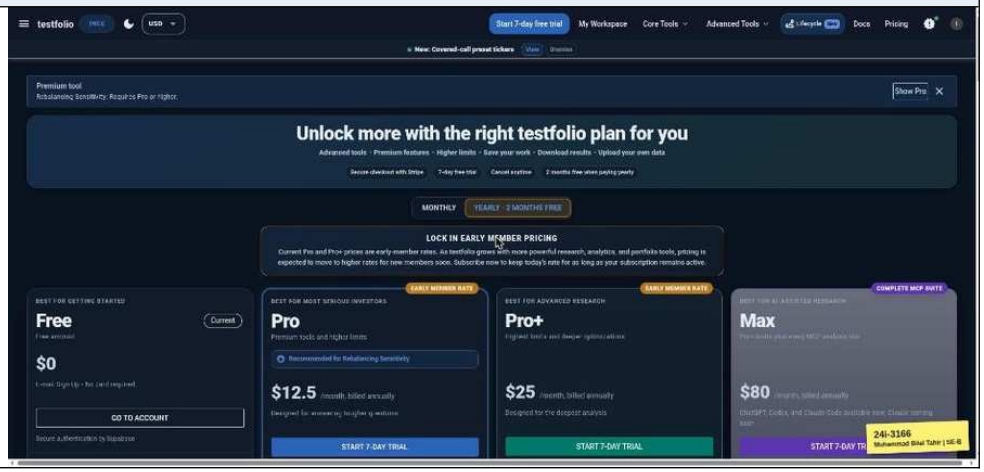
TC-EP-158 | Allocation % (portfolio total) | input: 60/40 | result: Blocked





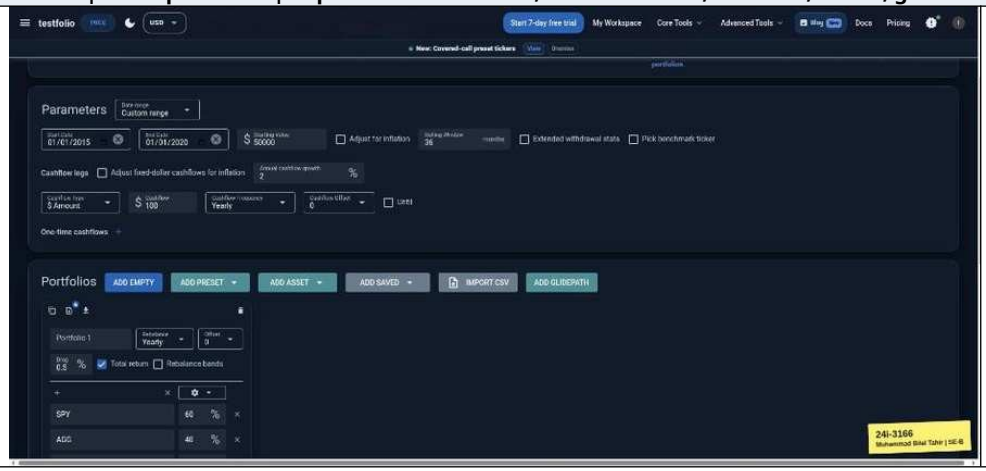
TC-EP-159 | Allocation % (portfolio total) | input: 60/30 (90) | result: Blocked





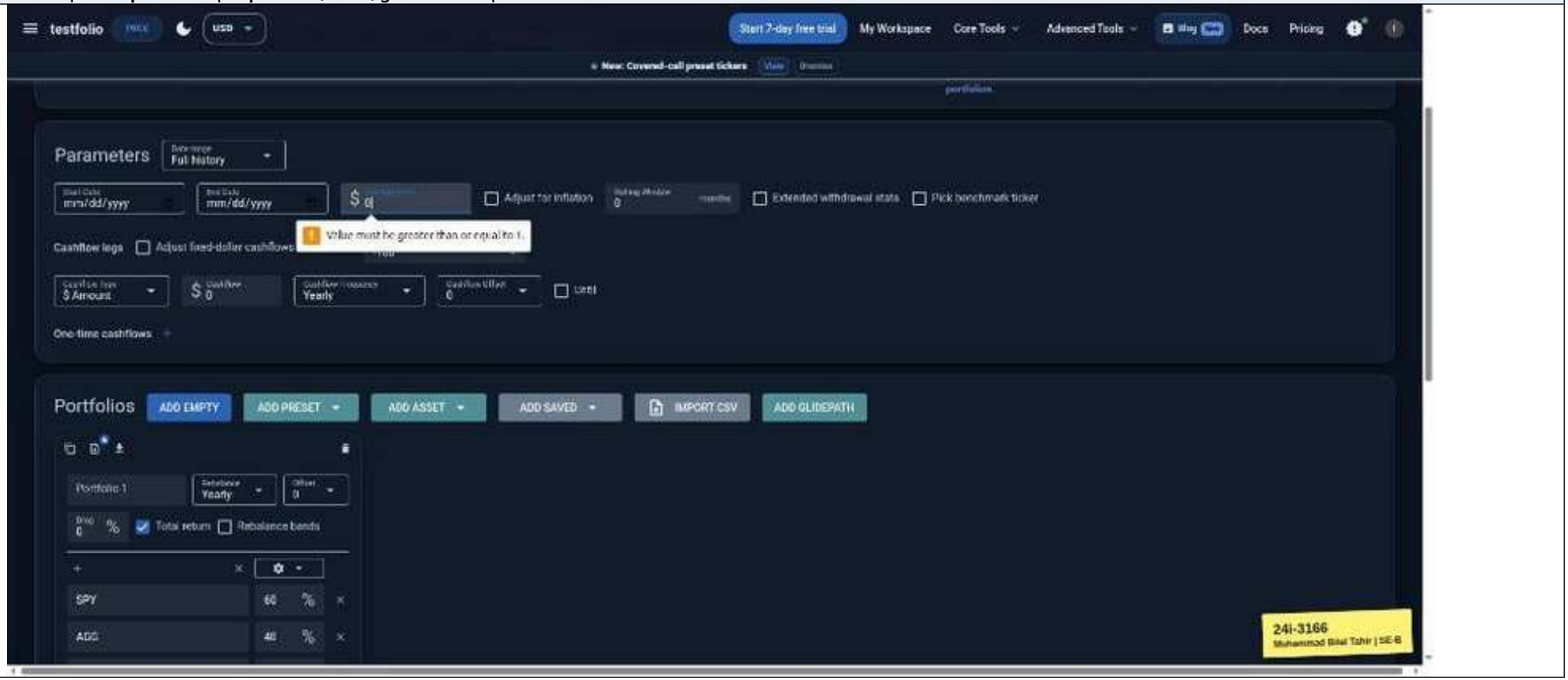
Portfolio Backtester (testfol.io/)

TC-EP-S1 | multiple fields | input: Start 2015-01-01, End 2020-01-01, SV 50000, RW 36, growth 2, cashflow 100, drag 0.5, SPY60/AGG40 | result: Pass

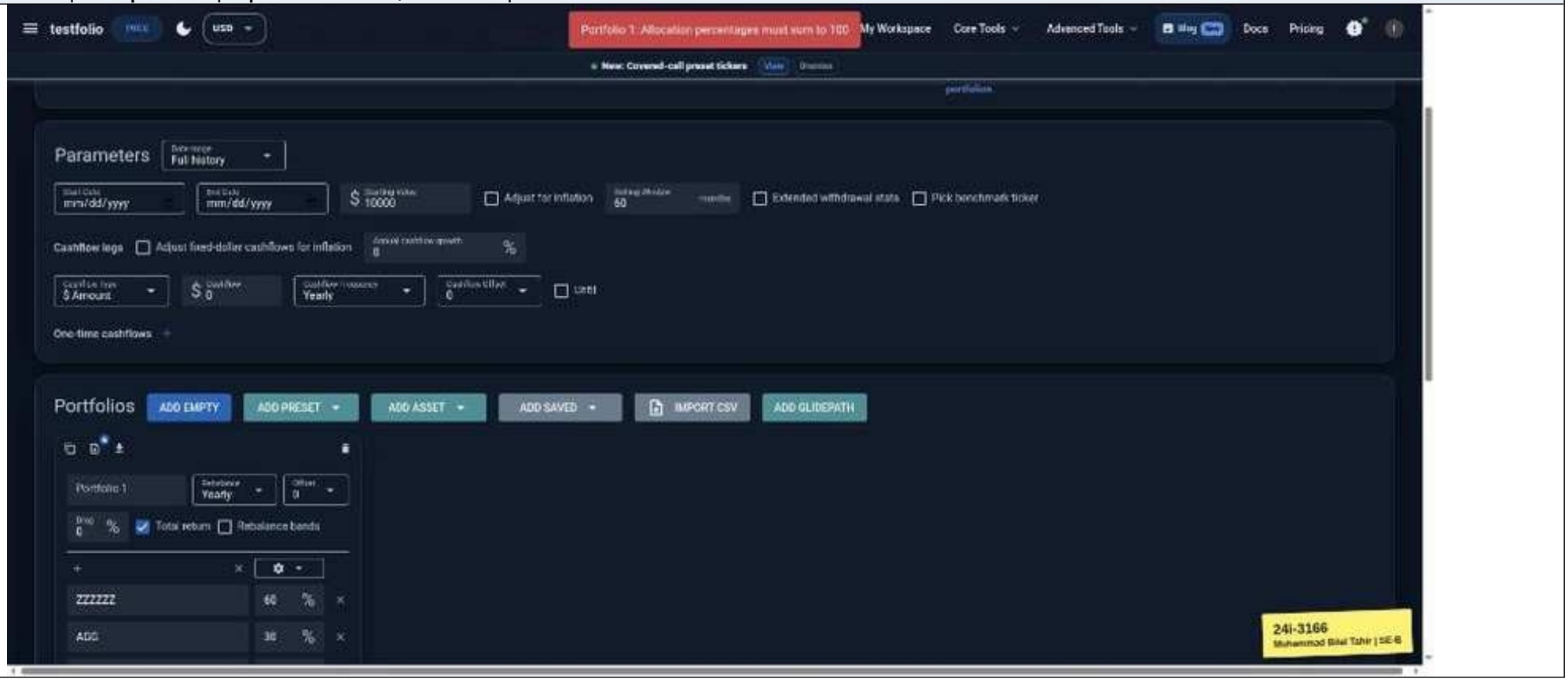




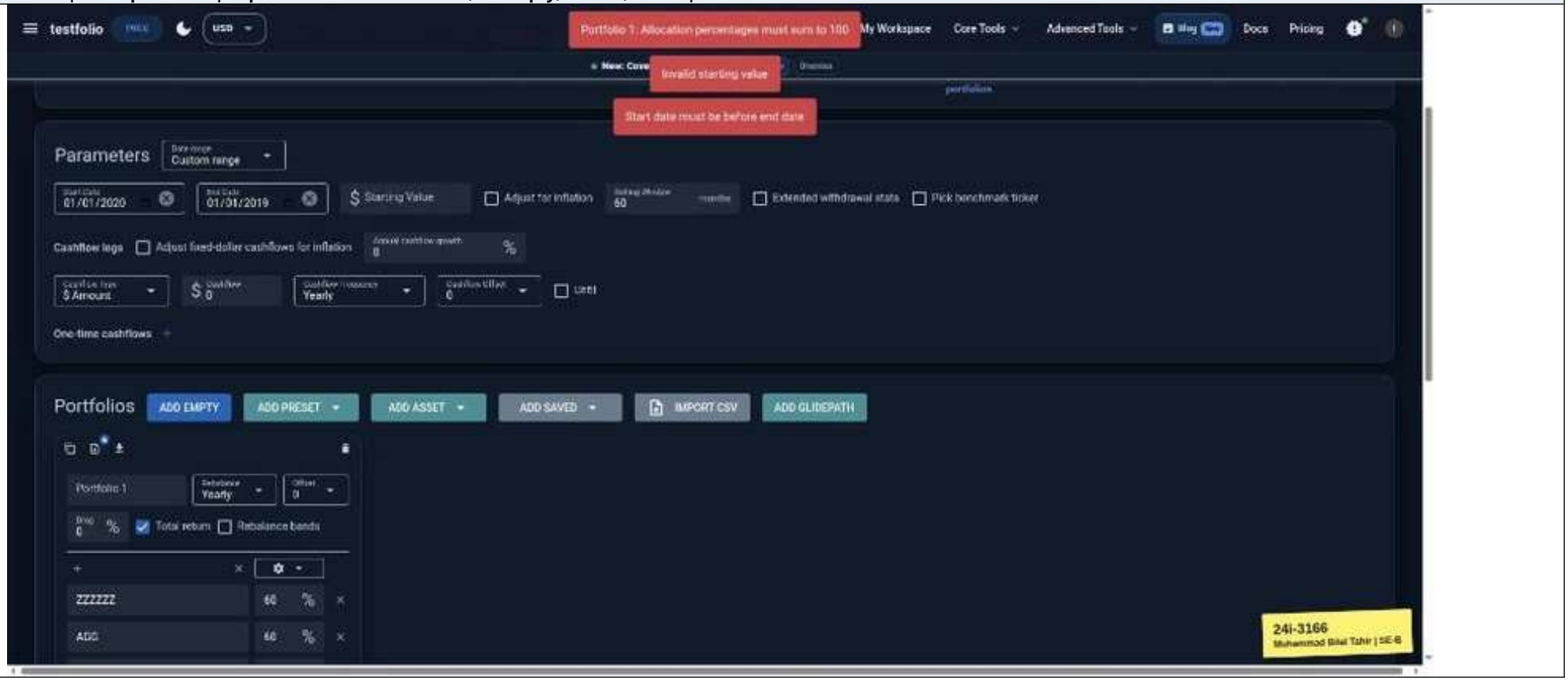
TC-EP-S2 | multiple fields | input: SV 0, RW 0, growth -100 | result: Pass



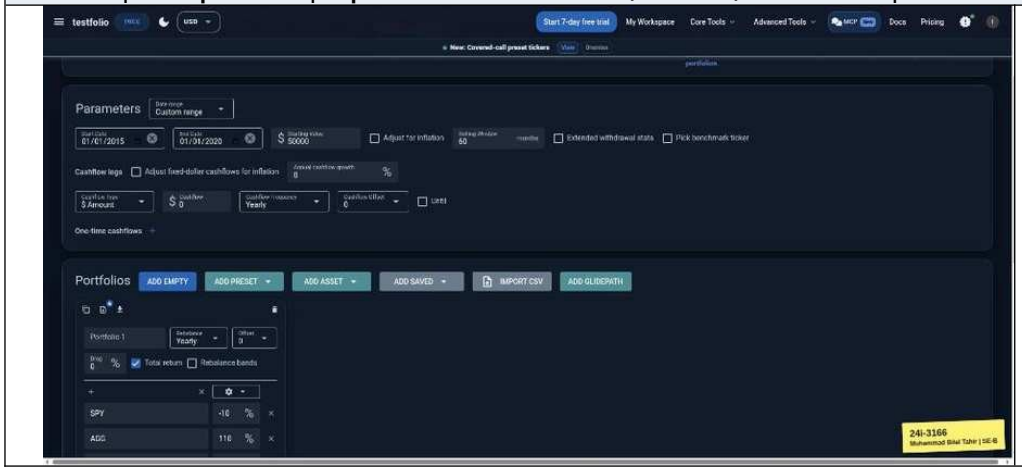
TC-EP-S3 | multiple fields | input: ticker ZZZZZZ, alloc 60/30 | result: Fail

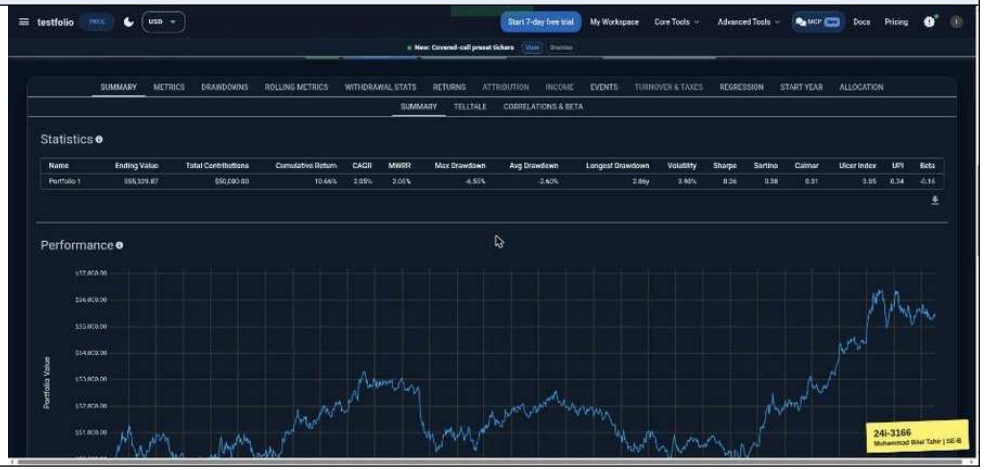


TC-EP-S4 | multiple fields | input: 2020-01-01/2019-01-01, SV empty, ZZZZZZ, 60/60 | result: Fail

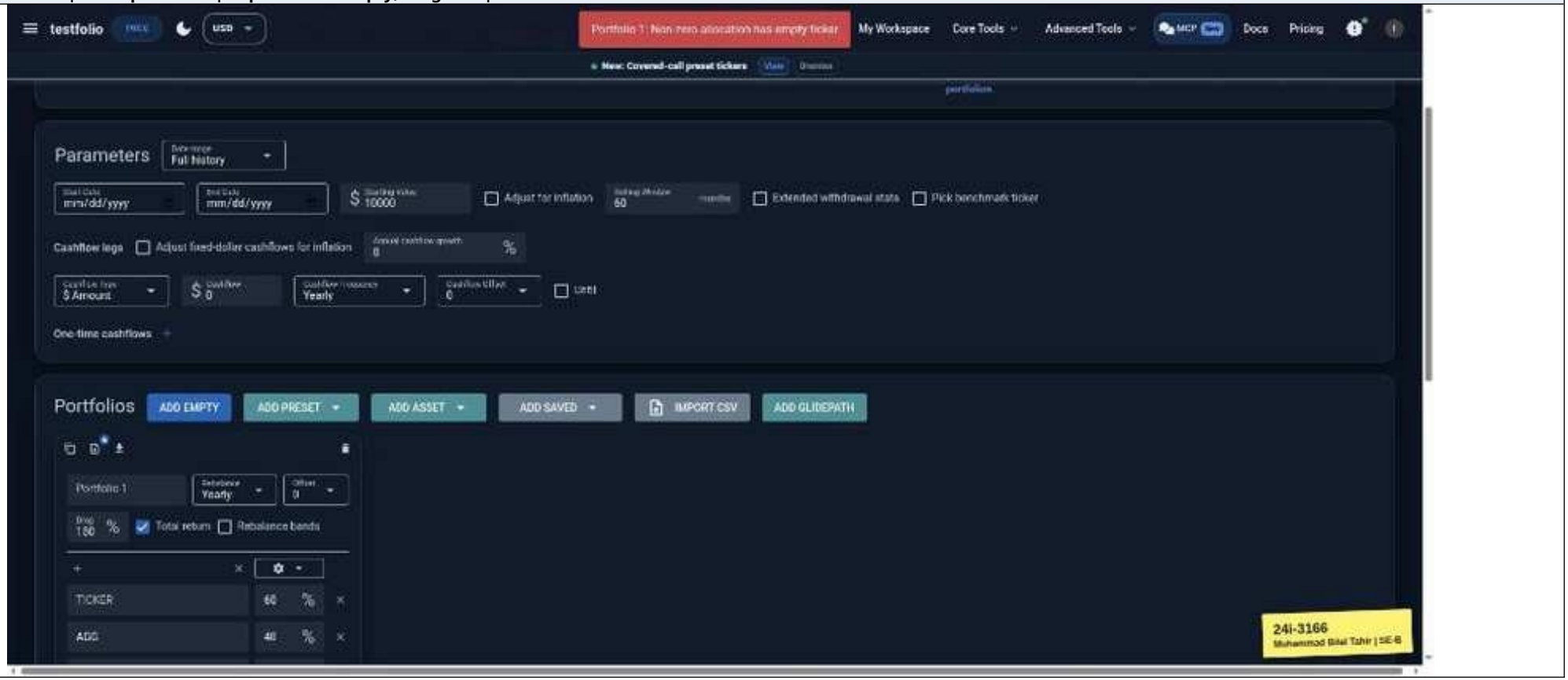


TC-EP-S5 | multiple fields | input: 2015-01-01/2020-01-01, SV 50000, alloc -10/110 | result: Fail

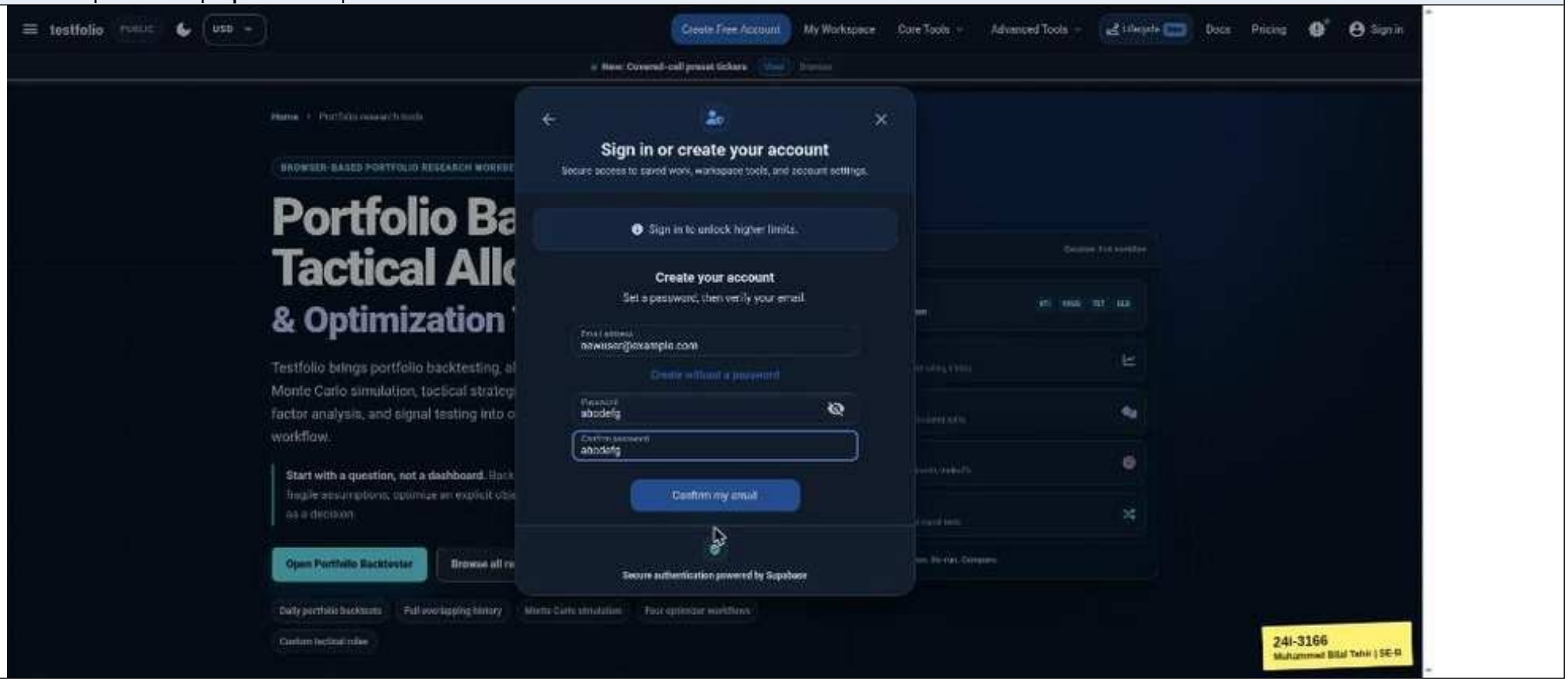




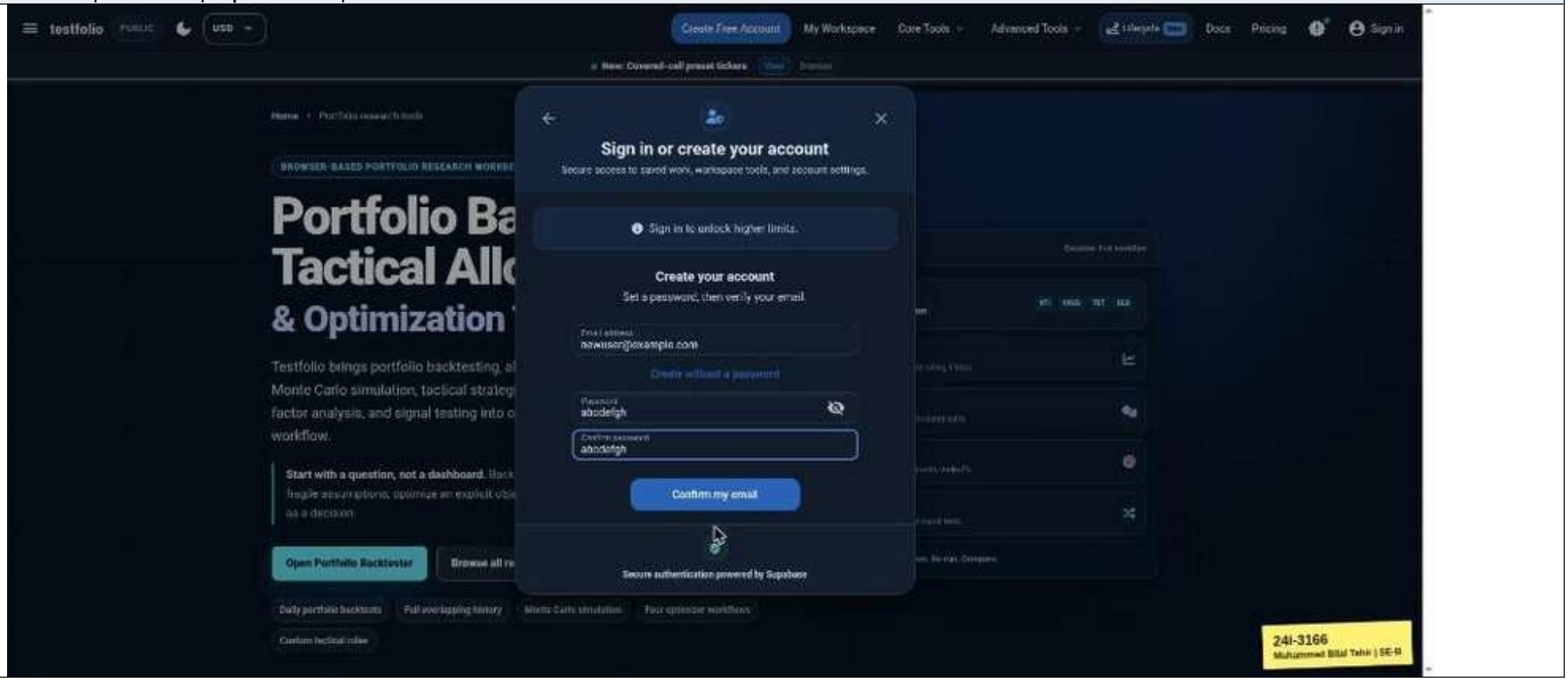
TC-EP-S6 | multiple fields | input: ticker empty, drag 150 | result: Fail



Sign Up (testfol.io/start)

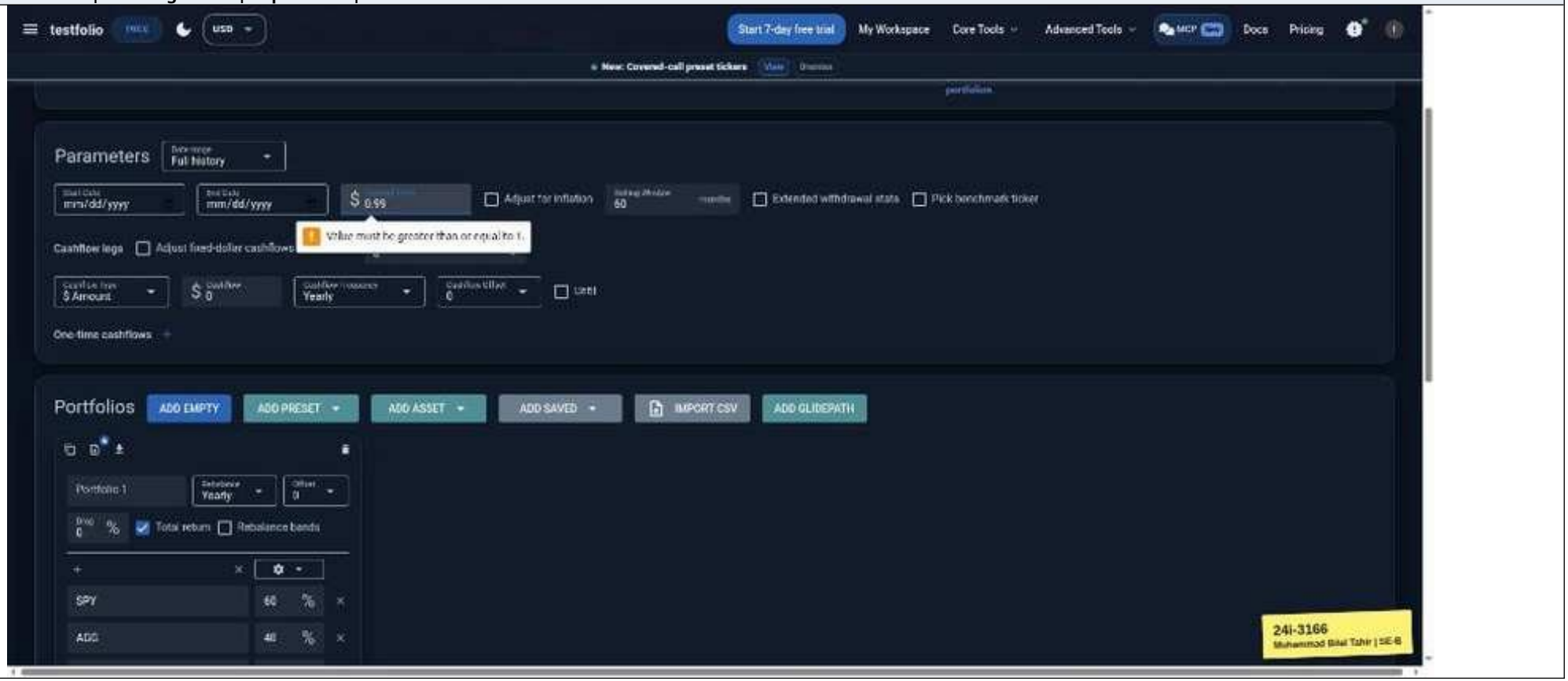
TC-BVA-001 | Password | input: 7 chars | result: Pass


TC-BVA-002 | Password | input: 8 chars | result: Pass

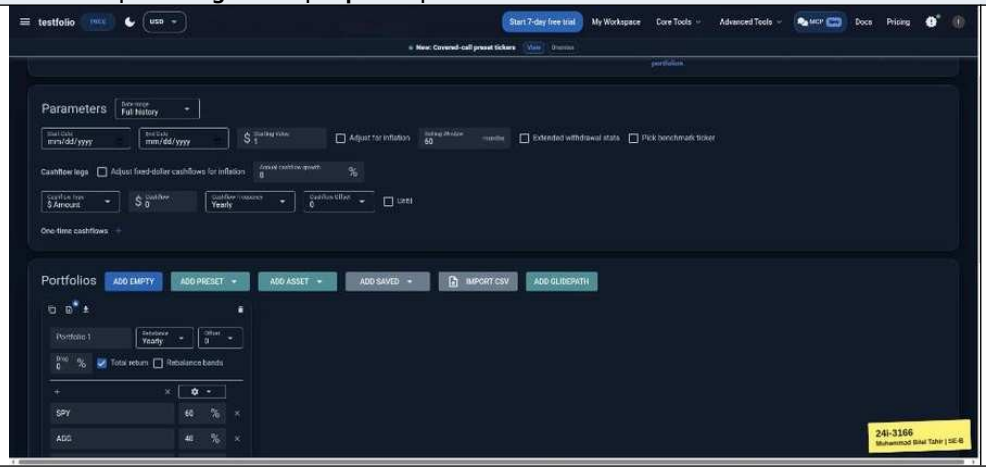


Portfolio Backtester (testfol.io/)

TC-BVA-003 | Starting Value | input: 0.99 | result: Pass

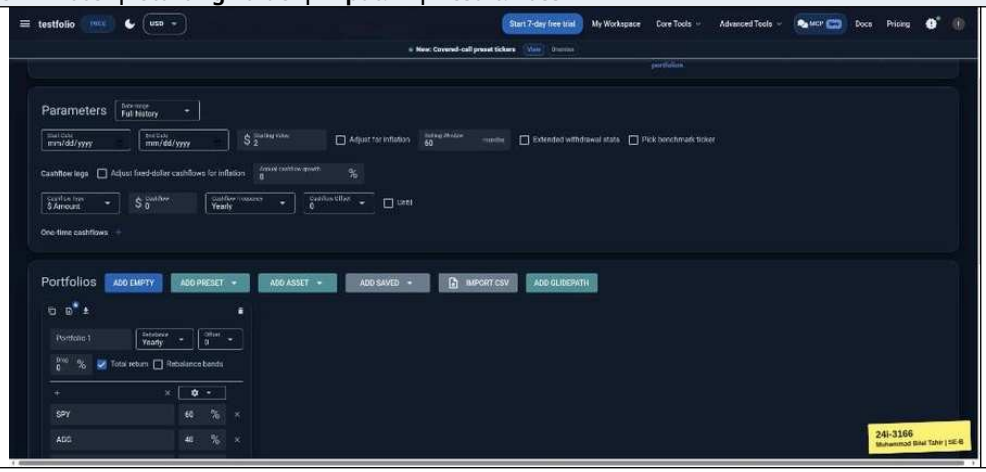


TC-BVA-004 | Starting Value | input: 1 | result: Pass



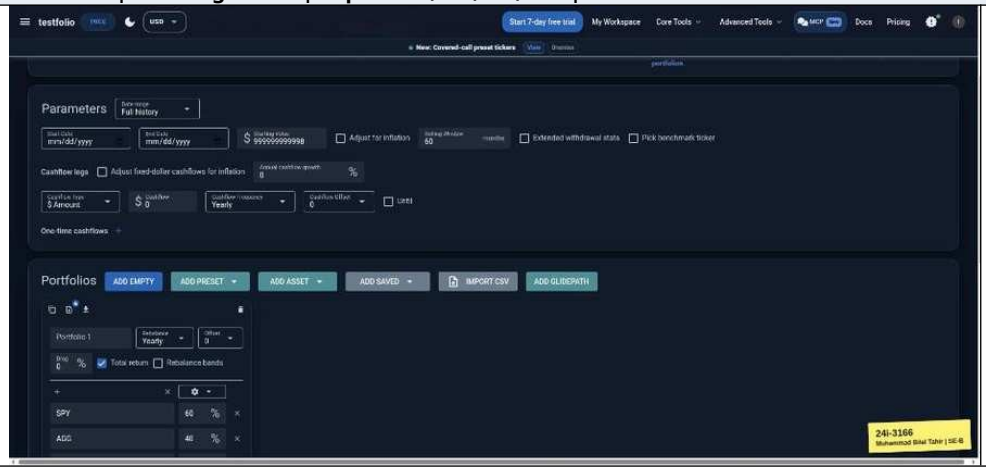


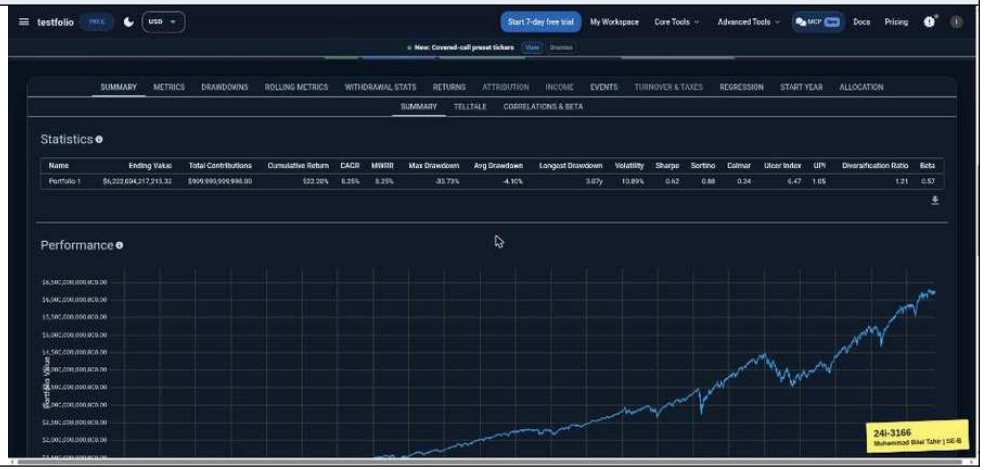
TC-BVA-005 | Starting Value | input: 2 | result: Pass



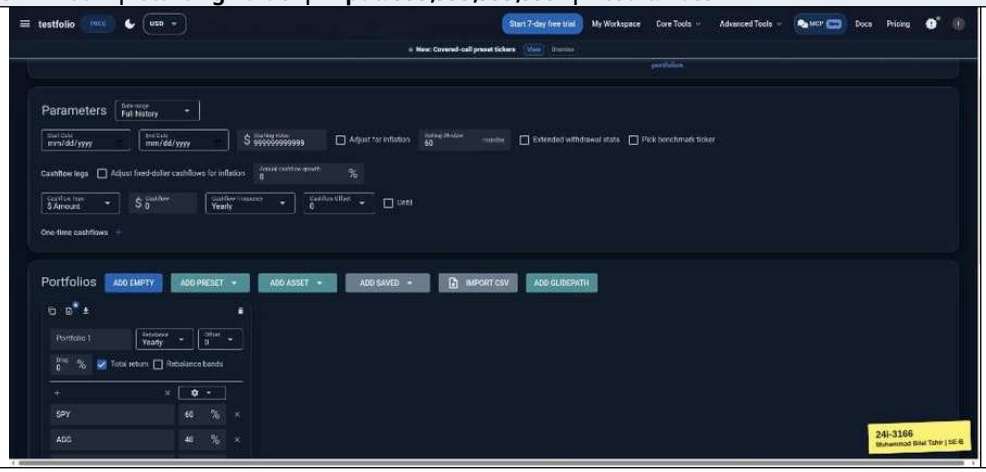


TC-BVA-006 | Starting Value | input: 999,999,999,998 | result: Pass



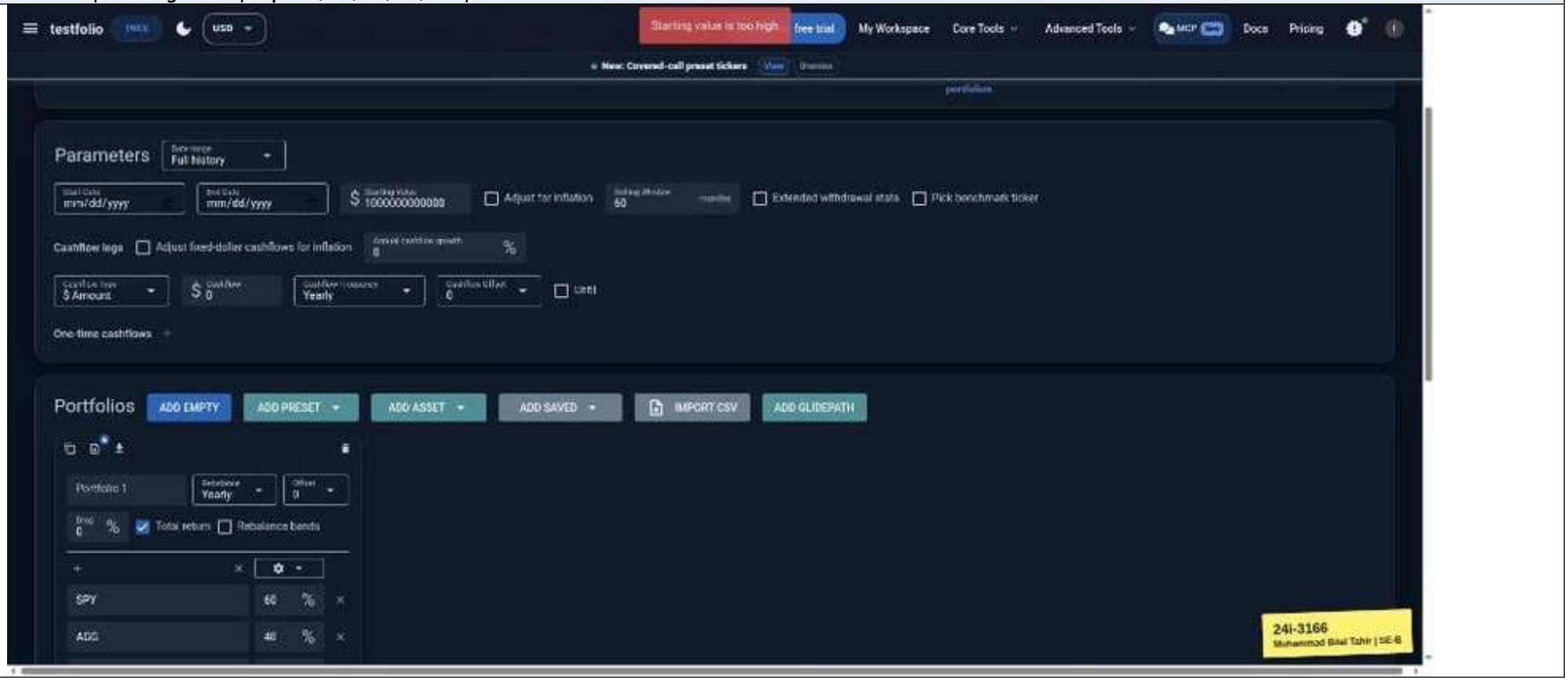


TC-BVA-007 | Starting Value | input: 999,999,999,999 | result: Pass

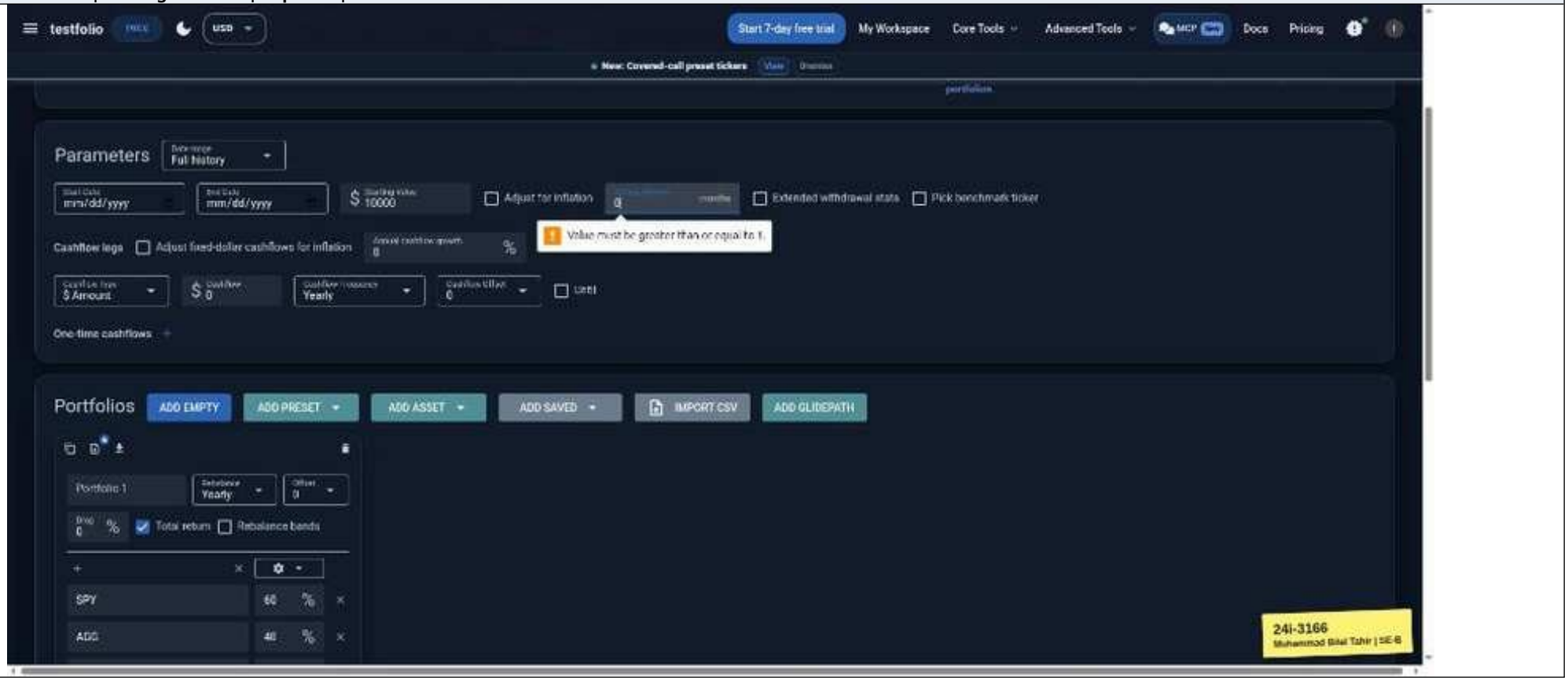




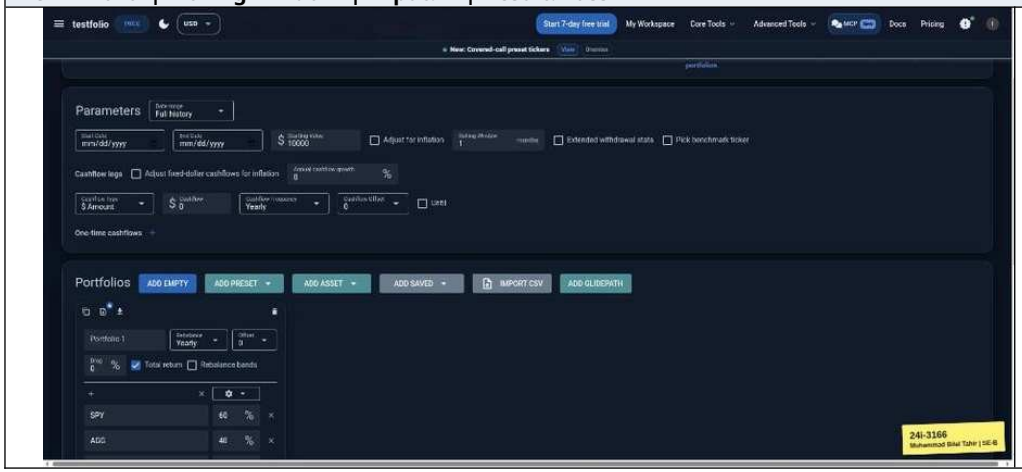
TC-BVA-008 | Starting Value | input: 1,000,000,000,000 | result: Pass



TC-BVA-009 | Rolling Window | input: 0 | result: Pass

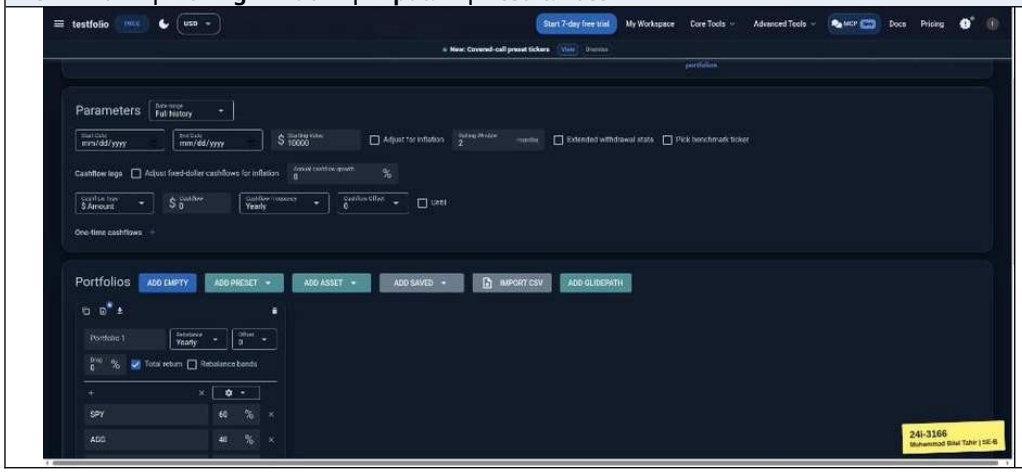


TC-BVA-010 | Rolling Window | input: 1 | result: Pass



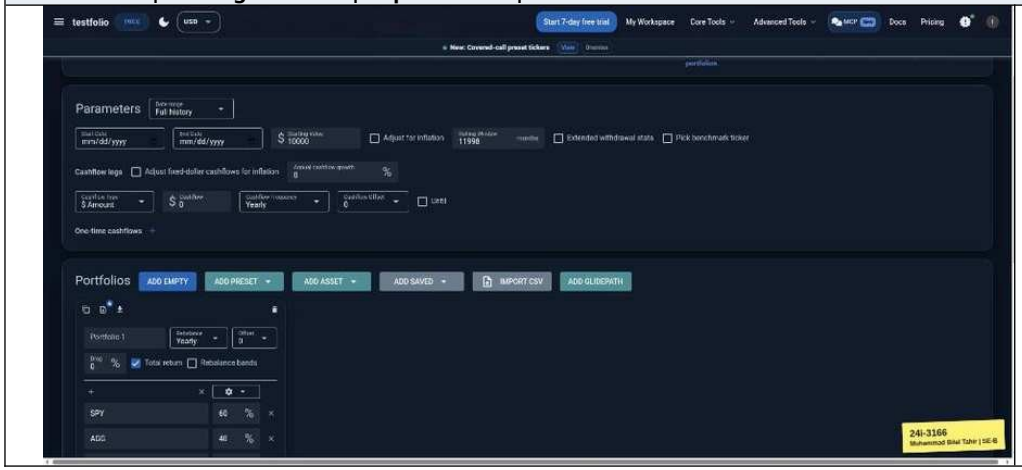


TC-BVA-011 | Rolling Window | input: 2 | result: Pass



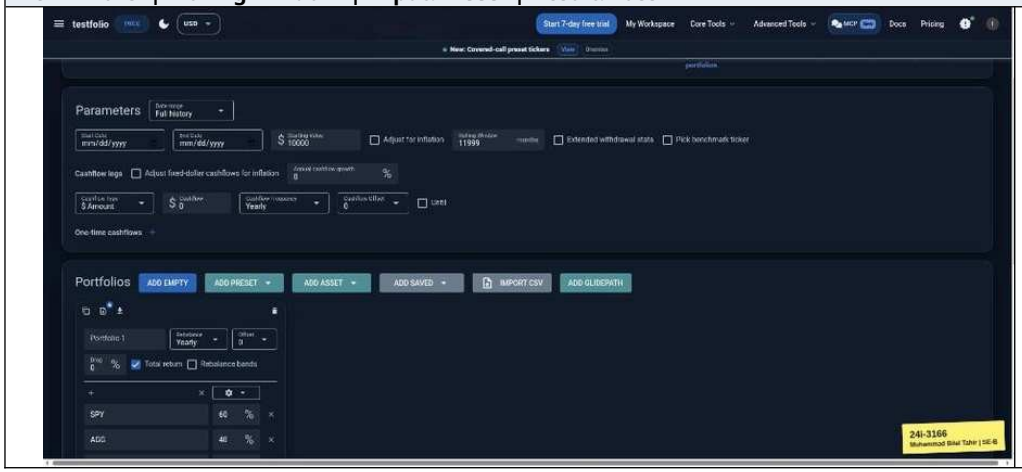


TC-BVA-012 | Rolling Window | input: 11998 | result: Pass



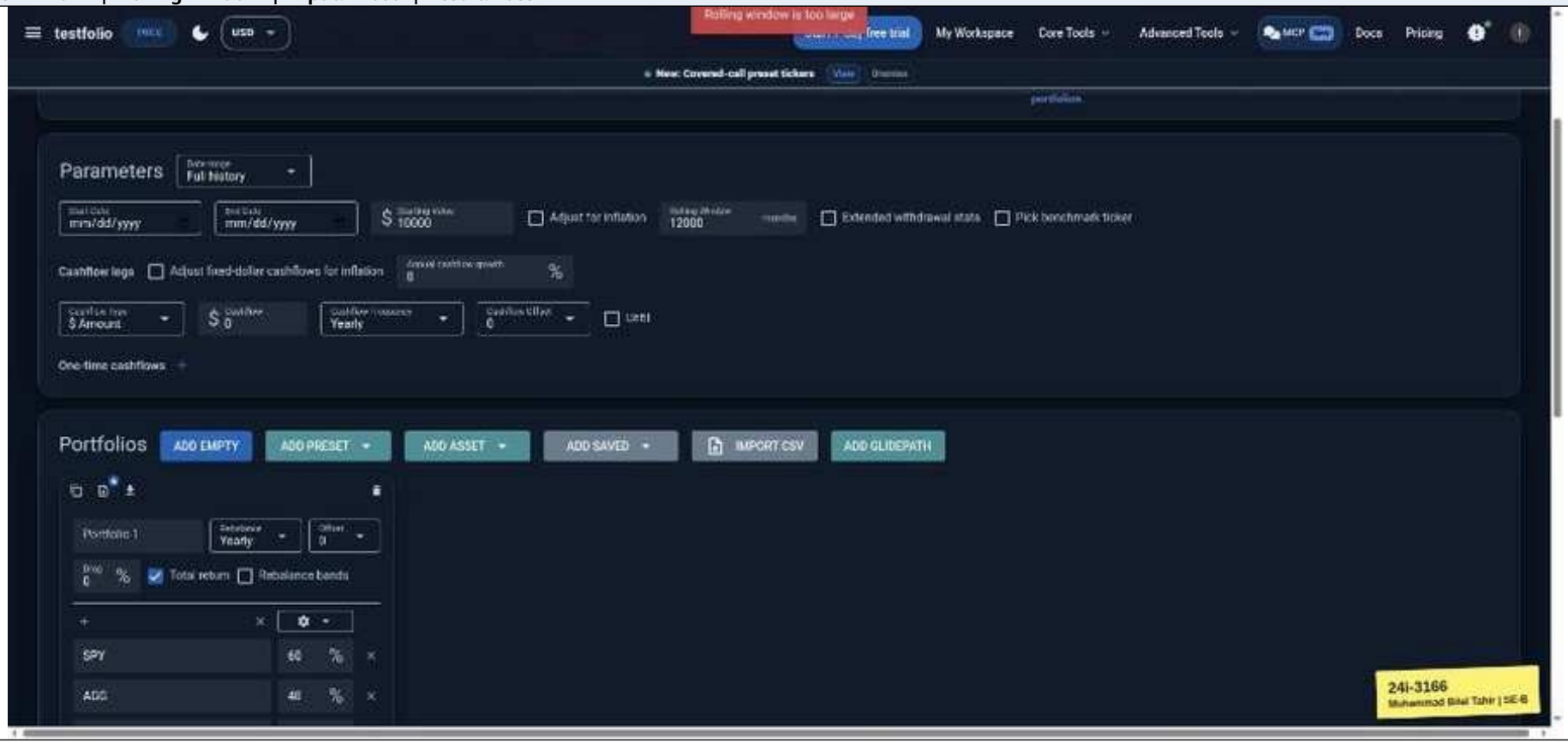


TC-BVA-013 | Rolling Window | input: 11999 | result: Pass

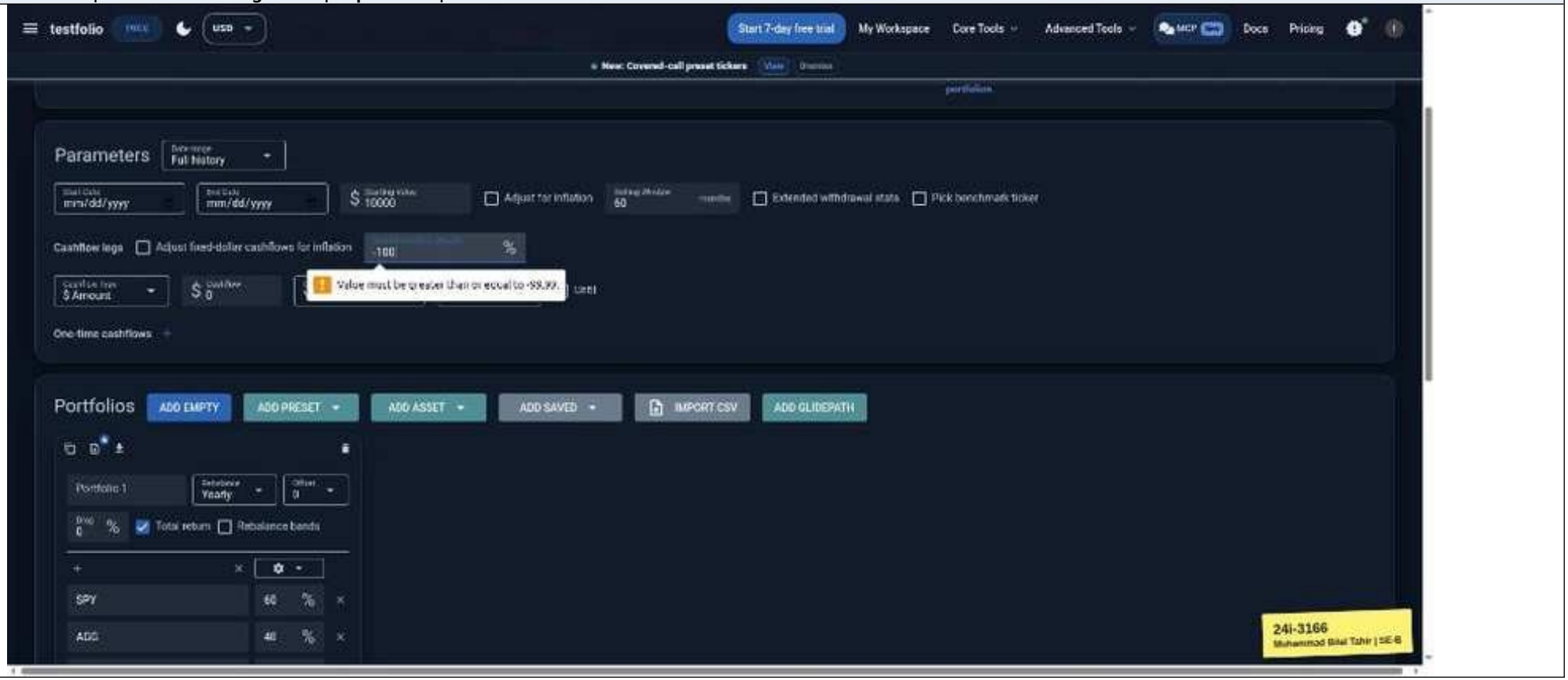




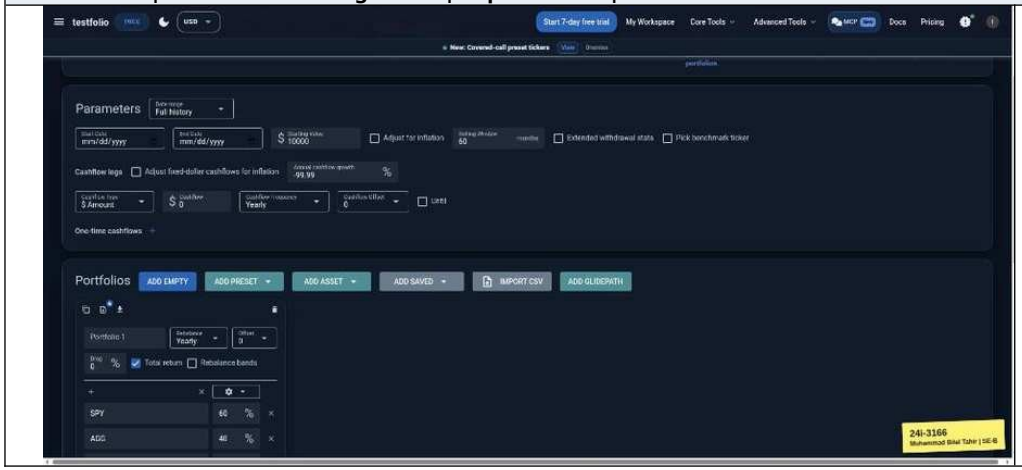
TC-BVA-014 | Rolling Window | input: 12000 | result: Pass



TC-BVA-015 | Annual cashflow growth | input: -100 | result: Pass

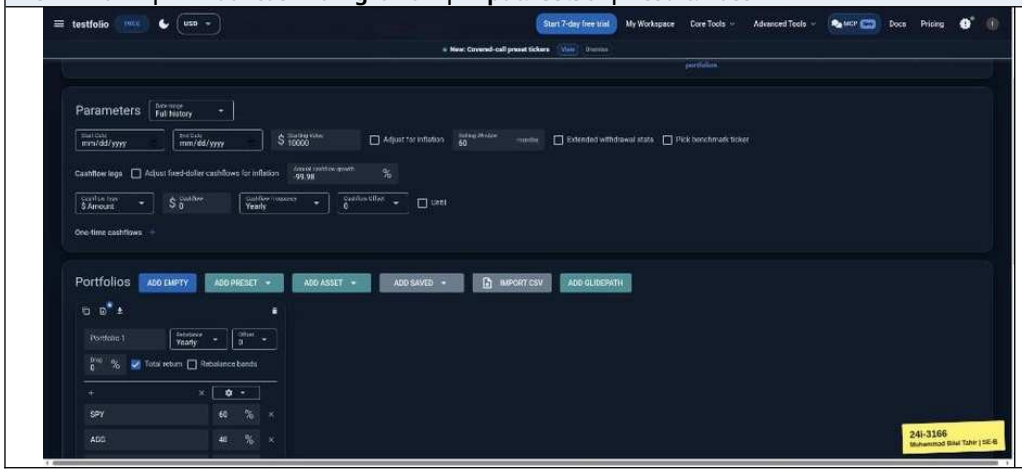


TC-BVA-016 | Annual cashflow growth | input: -99.99 | result: Pass



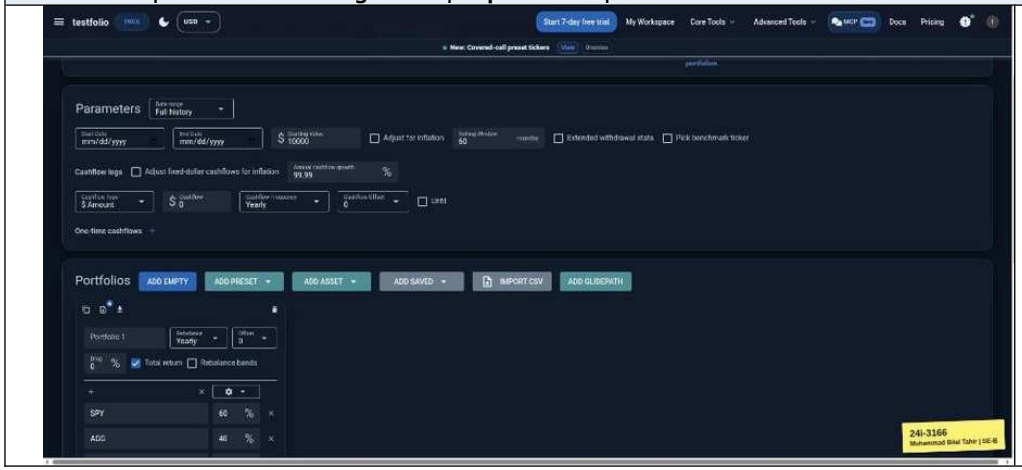


TC-BVA-017 | Annual cashflow growth | input: -99.98 | result: Pass



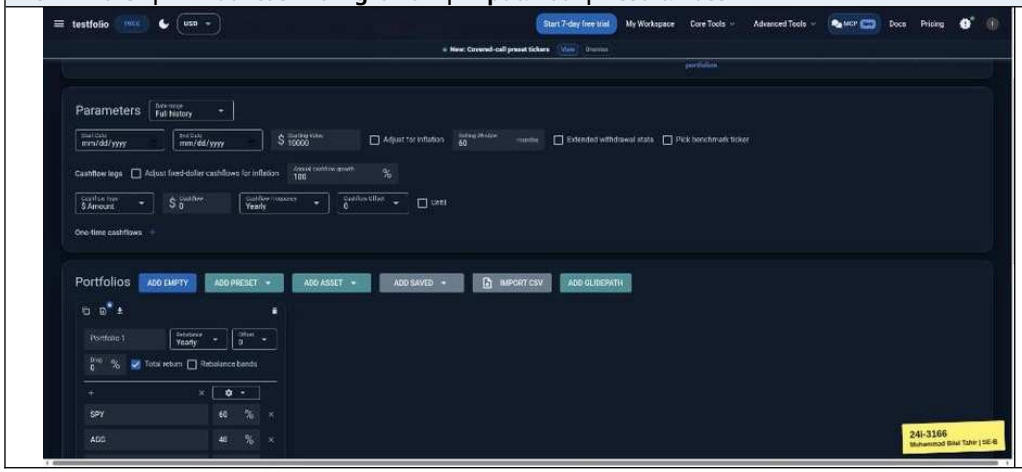


TC-BVA-018 | Annual cashflow growth | input: 99.99 | result: Pass



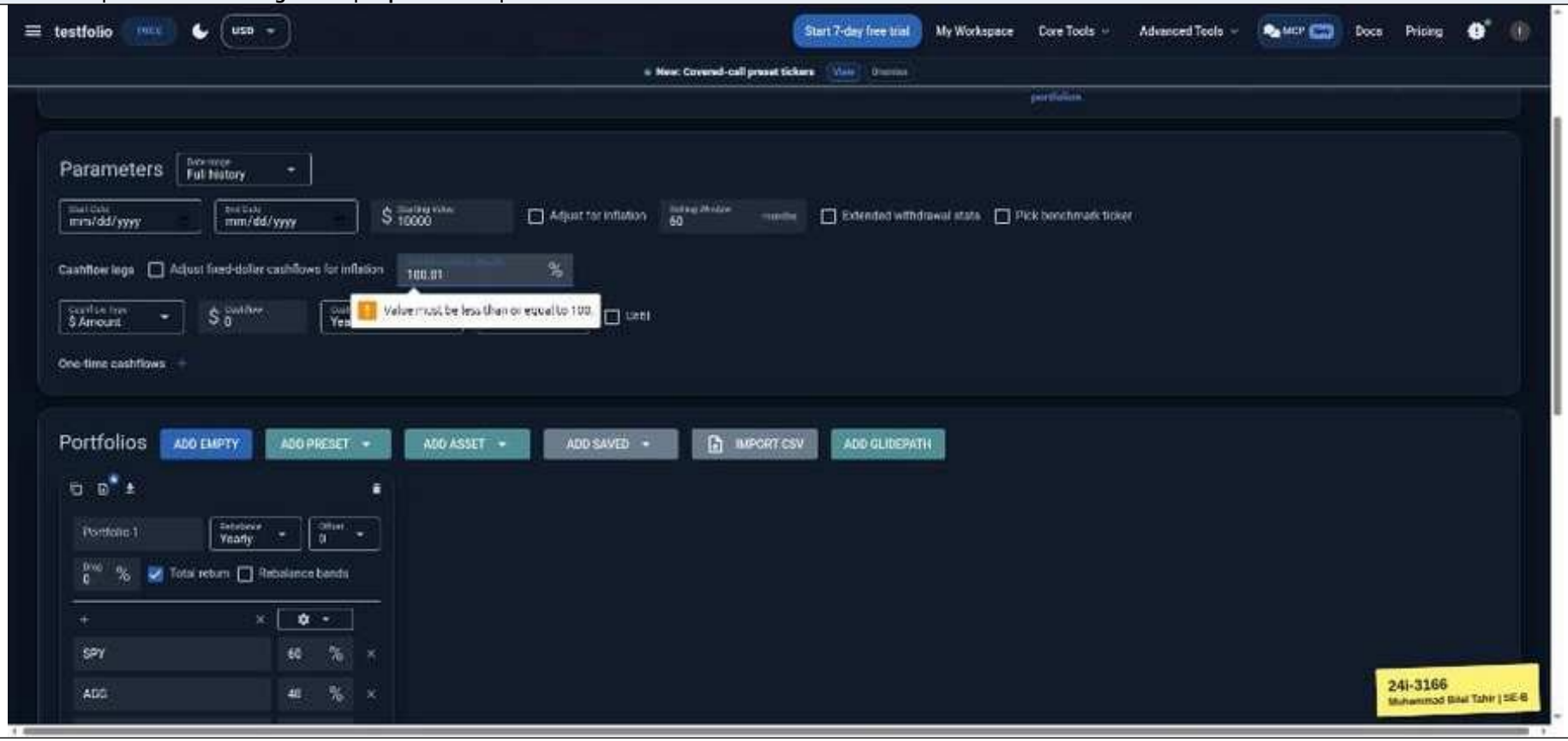


TC-BVA-019 | Annual cashflow growth | input: 100 | result: Pass

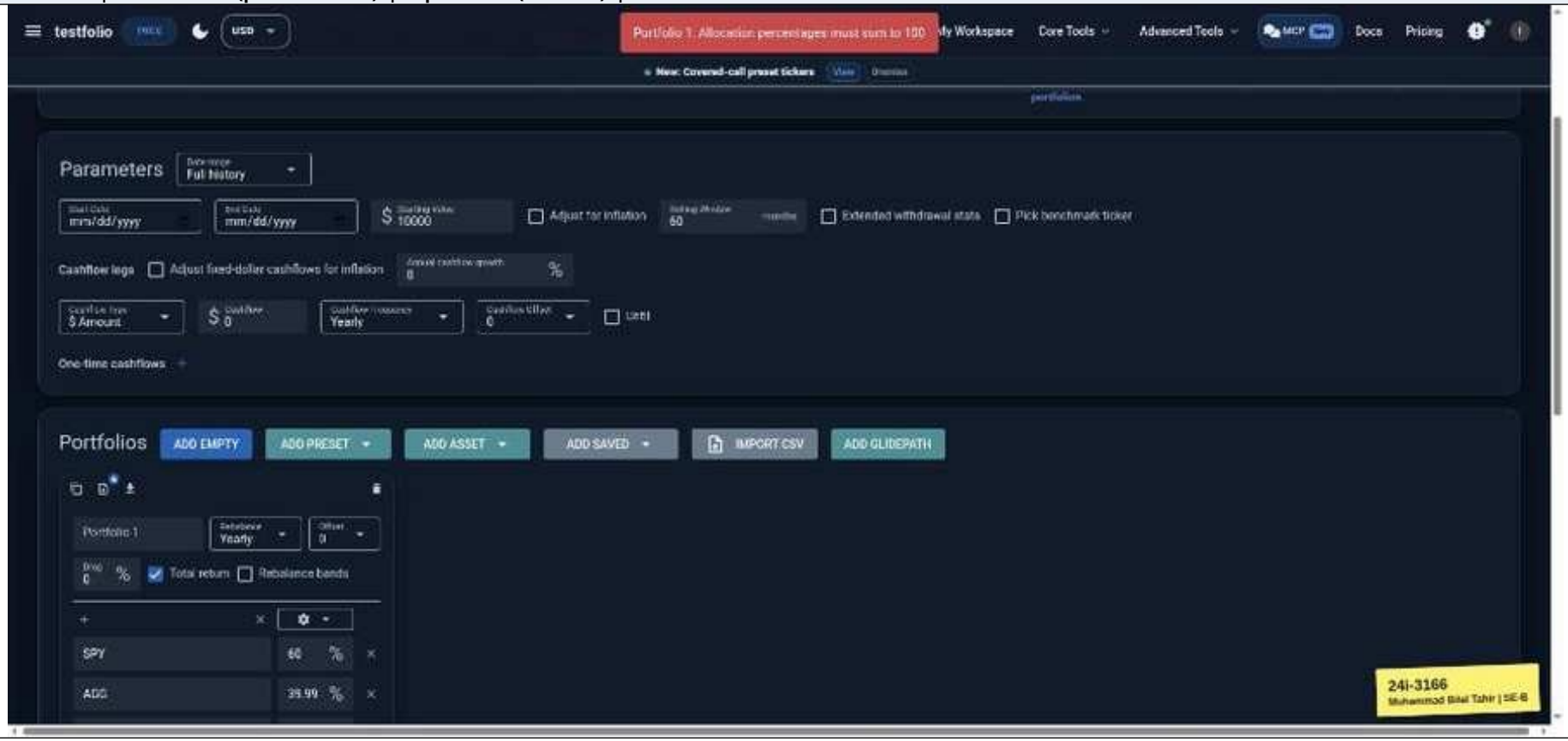




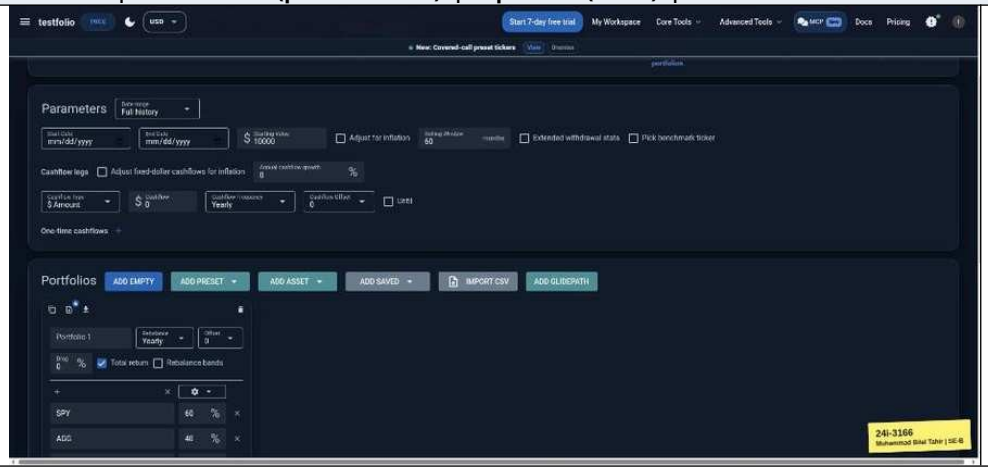
TC-BVA-020 | Annual cashflow growth | input: 100.01 | result: Pass



TC-BVA-021 | Allocation % (portfolio total) | input: 99.99 (60/39.99) | result: Pass

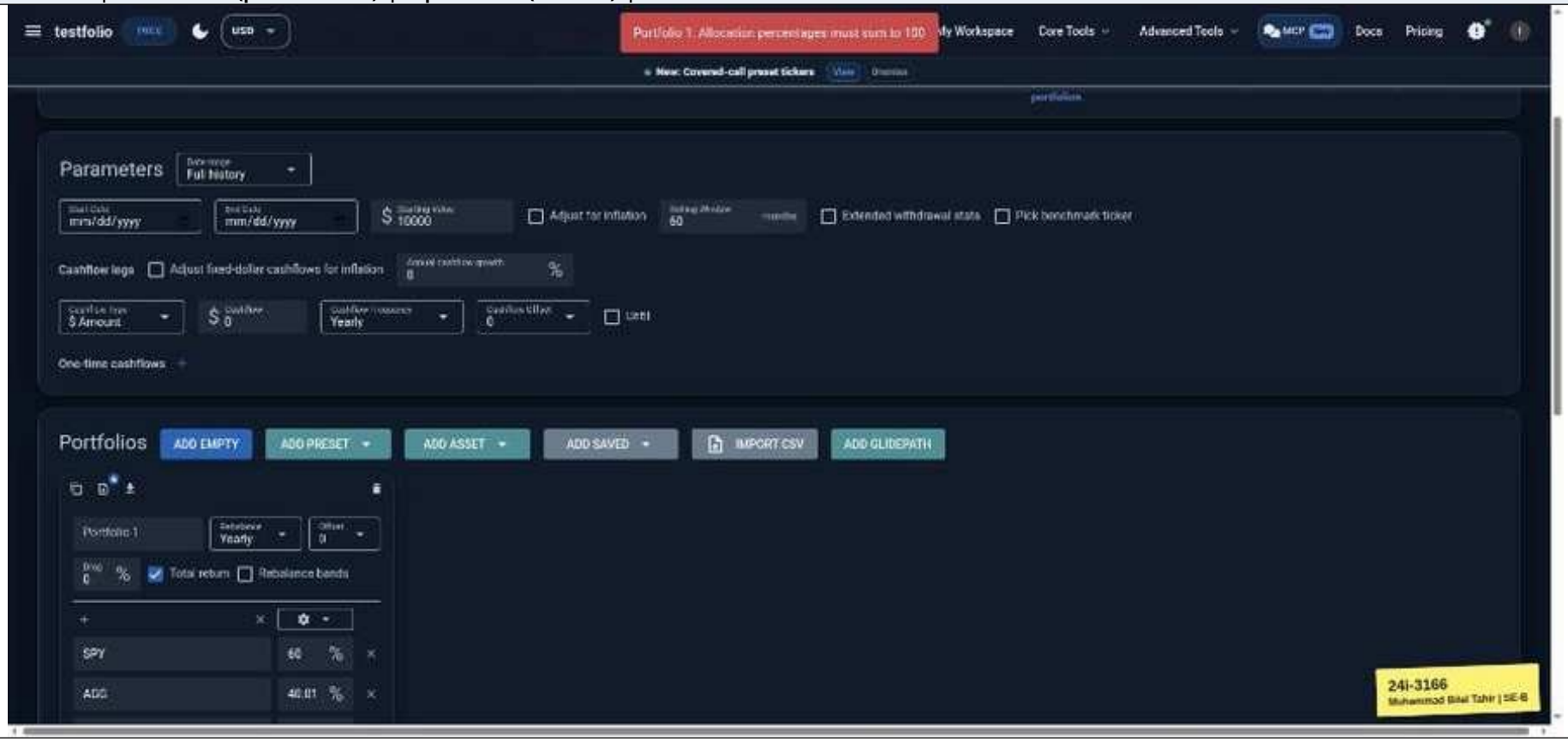


TC-BVA-022 | Allocation % (portfolio total) | input: 100 (60/40) | result: Pass

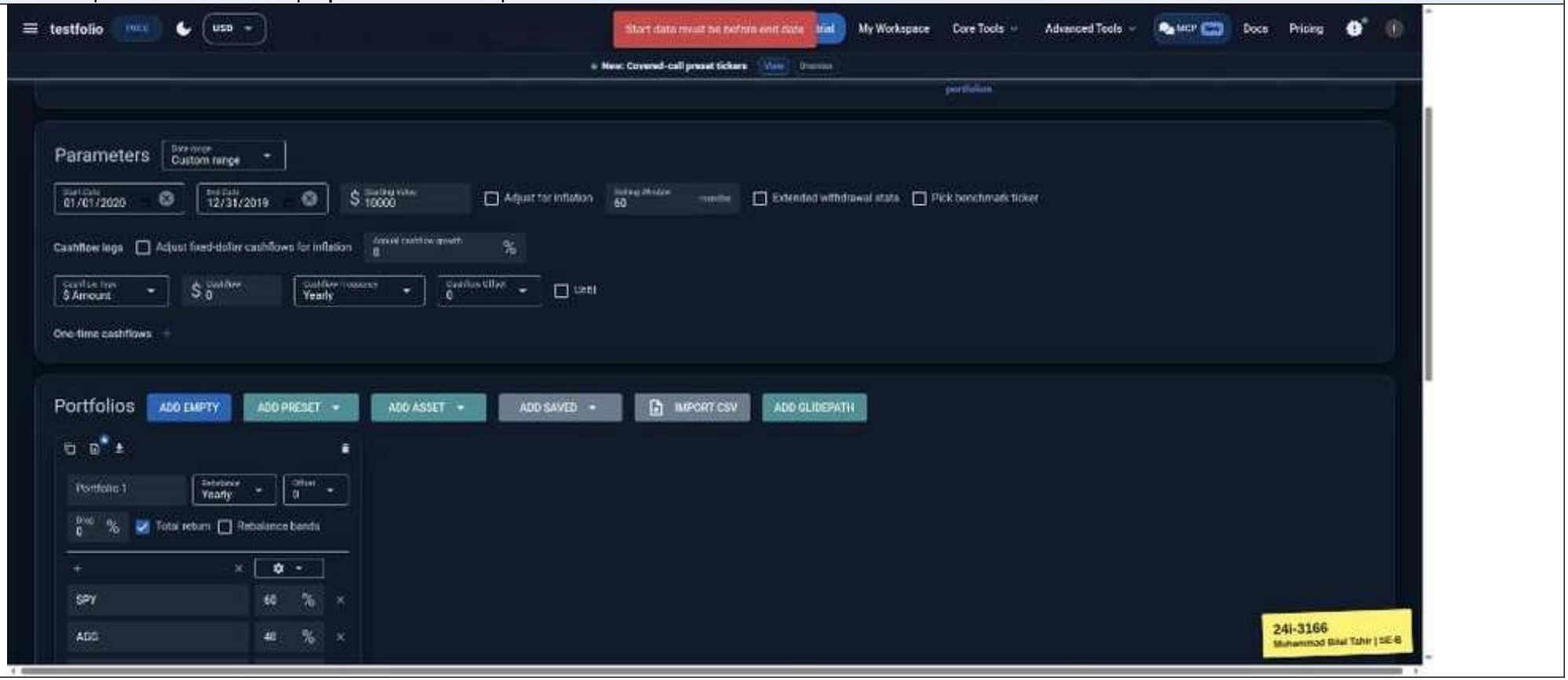




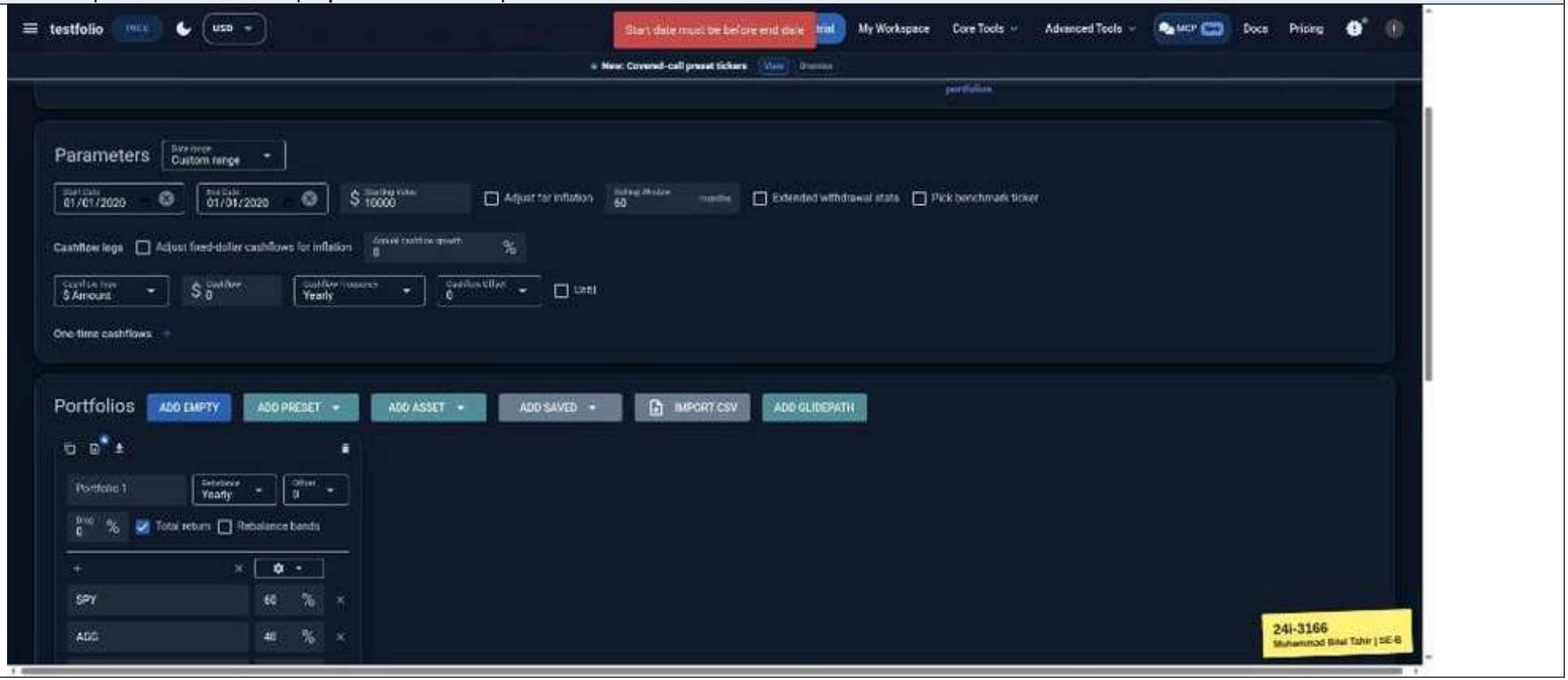
TC-BVA-023 | Allocation % (portfolio total) | input: 100.01 (60/40.01) | result: Pass



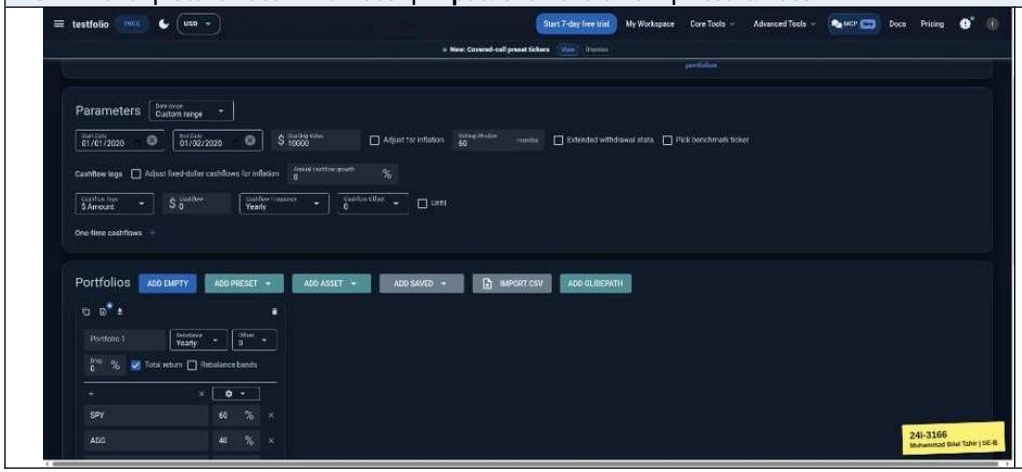
TC-BVA-024 | Start Date / End Date | input: end 2019-12-31 | result: Pass

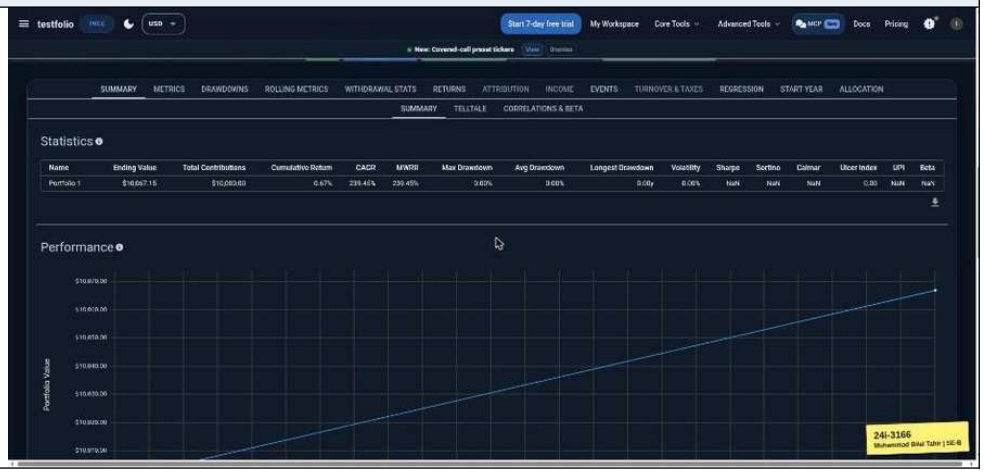


TC-BVA-025 | Start Date / End Date | input: end 2020-01-01 | result: Pass

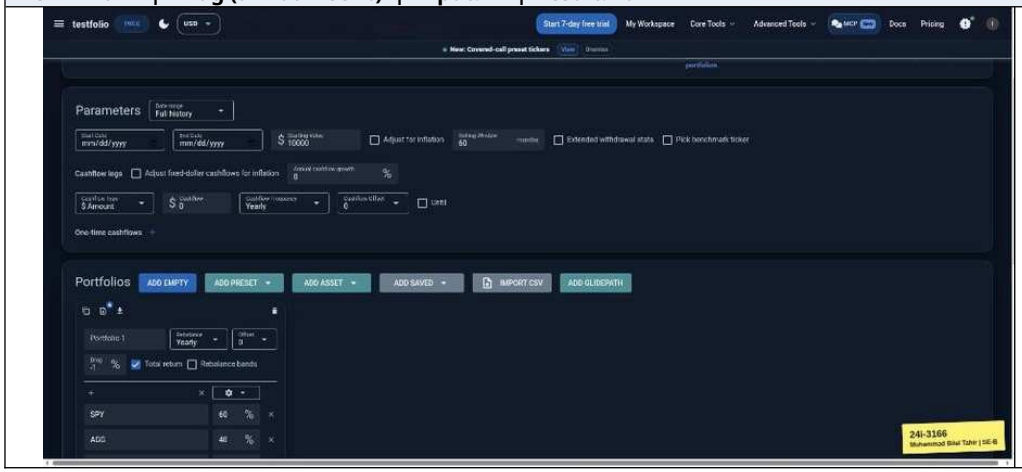


TC-BVA-026 | Start Date / End Date | input: end 2020-01-02 | result: Pass



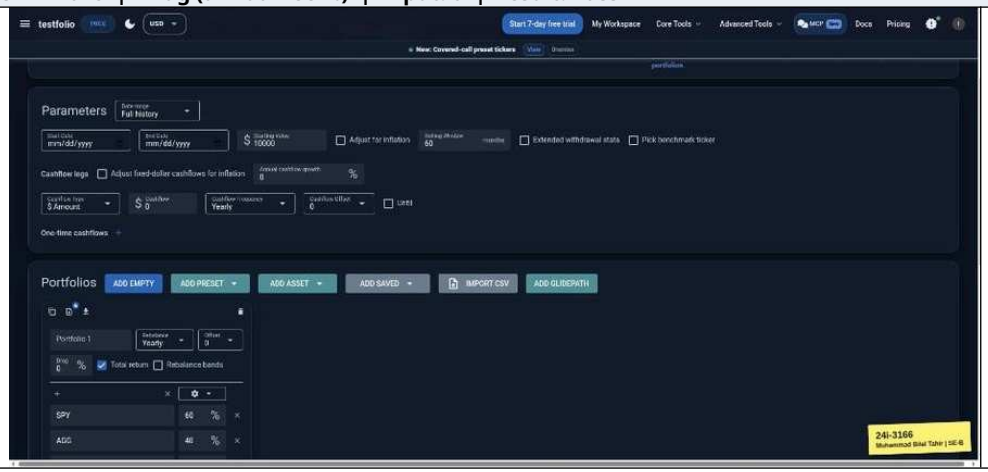


TC-BVA-027 | Drag (annual fee %) | input: -1 | result: Fail



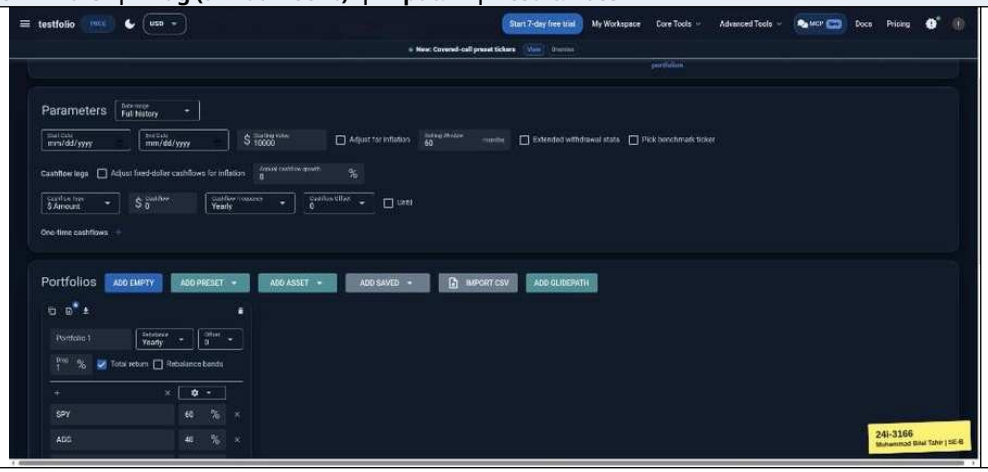


TC-BVA-028 | Drag (annual fee %) | input: 0 | result: Pass



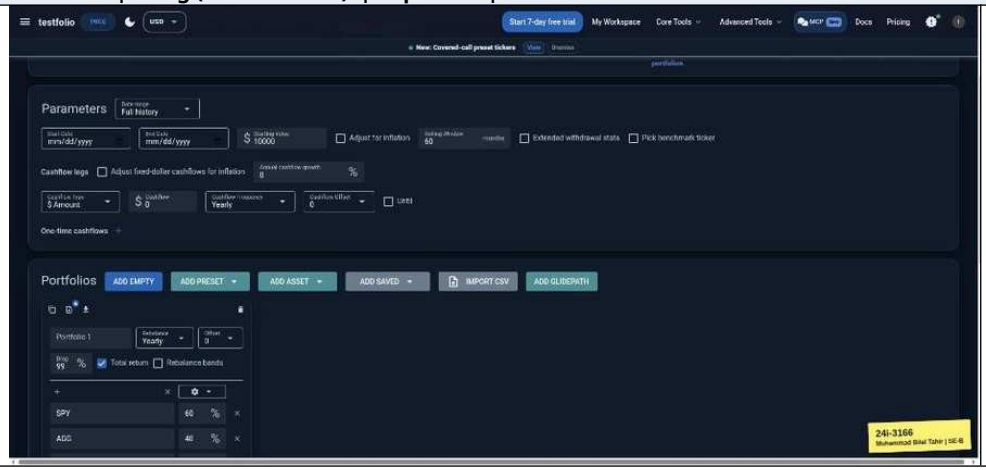


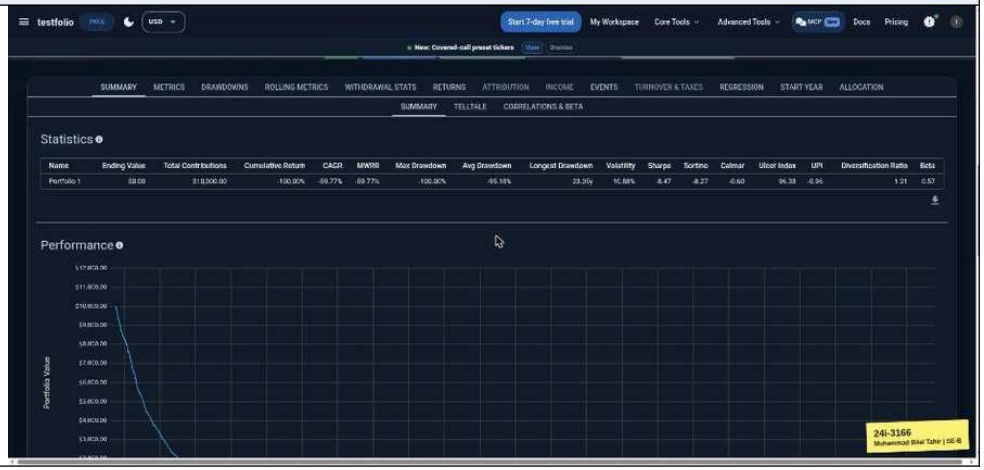
TC-BVA-029 | Drag (annual fee %) | input: 1 | result: Pass



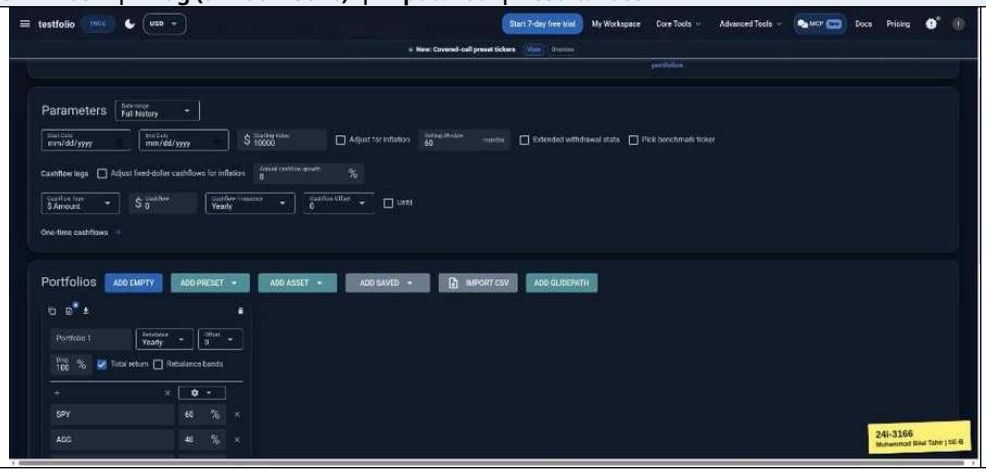


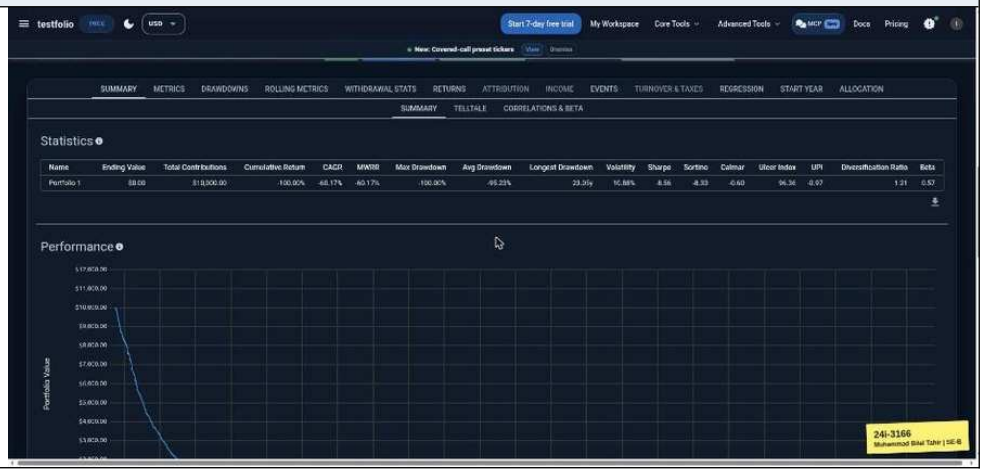
TC-BVA-030 | Drag (annual fee %) | input: 99 | result: Pass



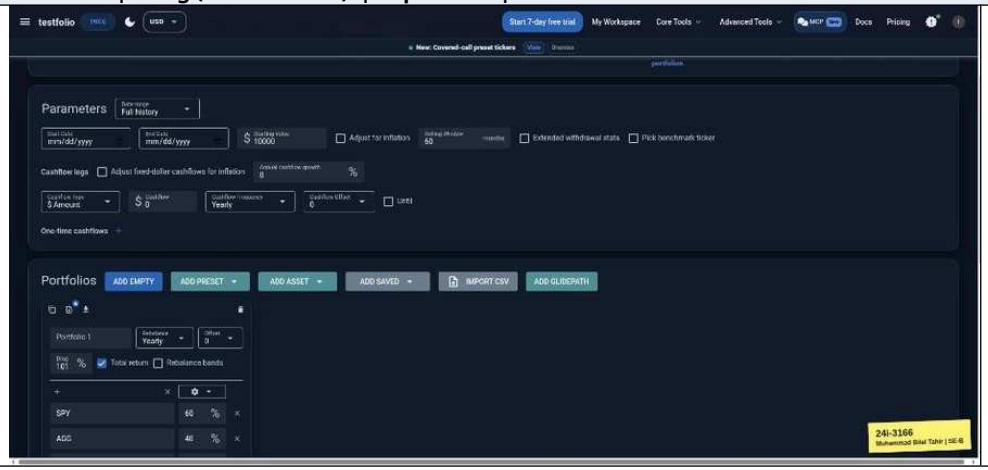


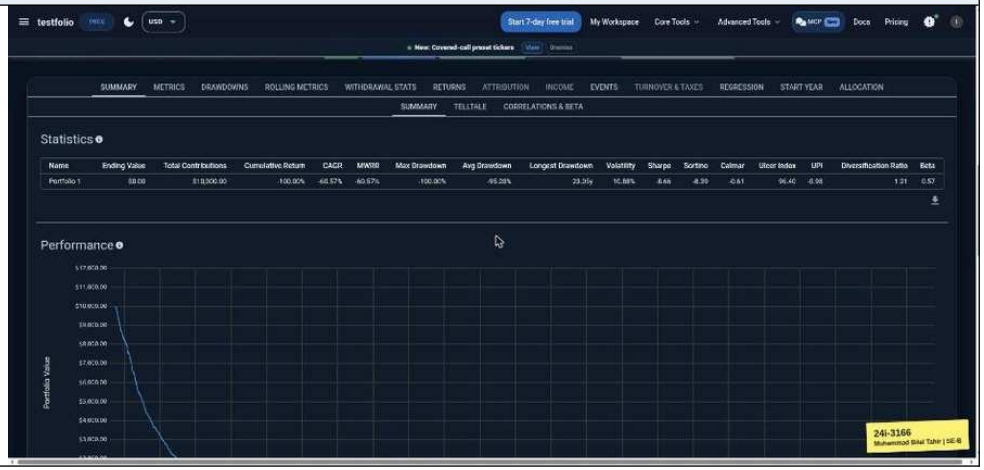
TC-BVA-031 | Drag (annual fee %) | input: 100 | result: Pass





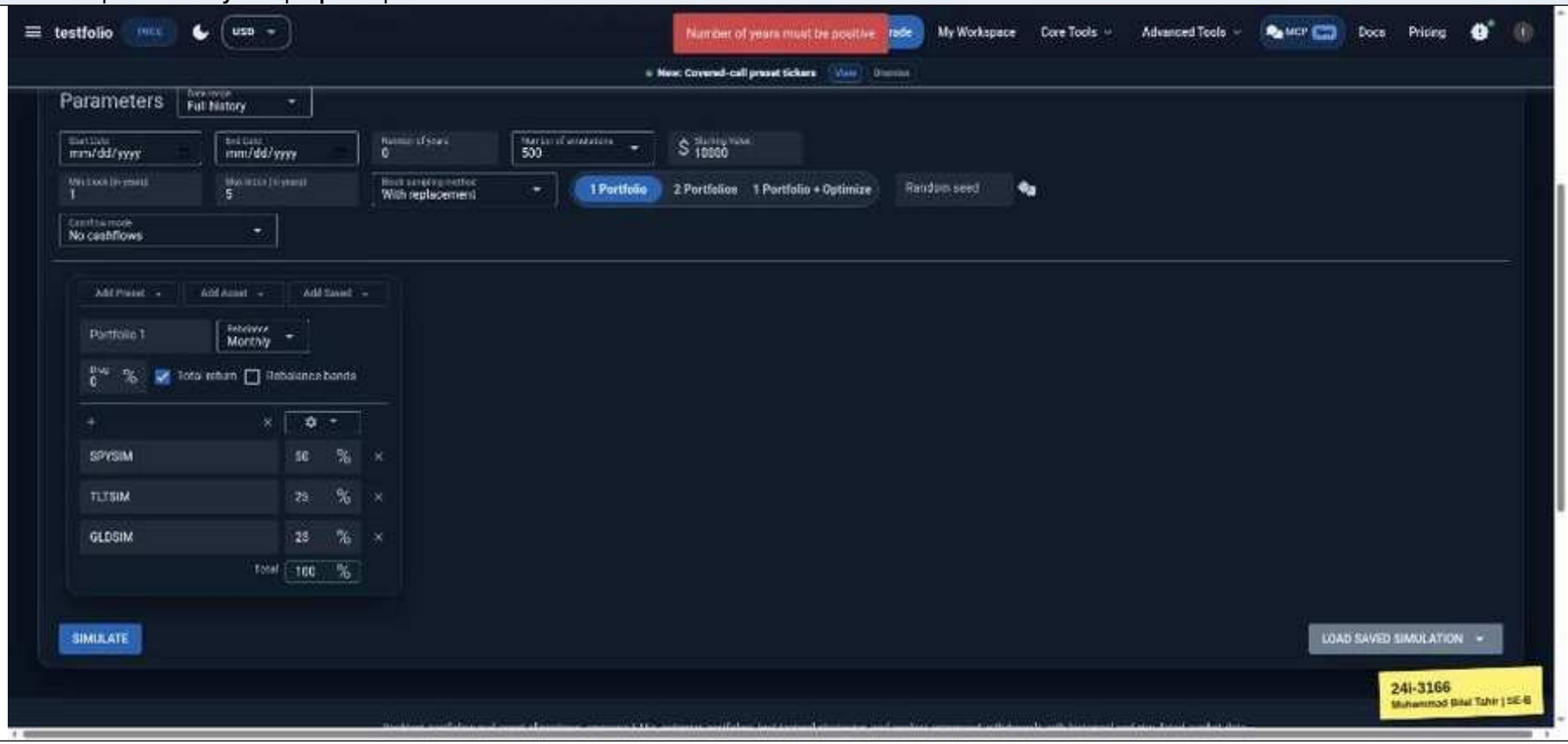
TC-BVA-032 | Drag (annual fee %) | input: 101 | result: Fail



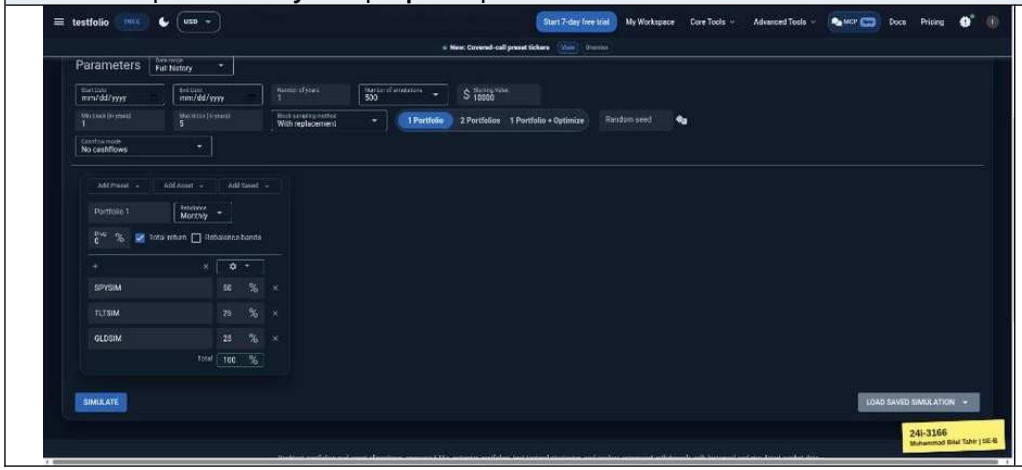


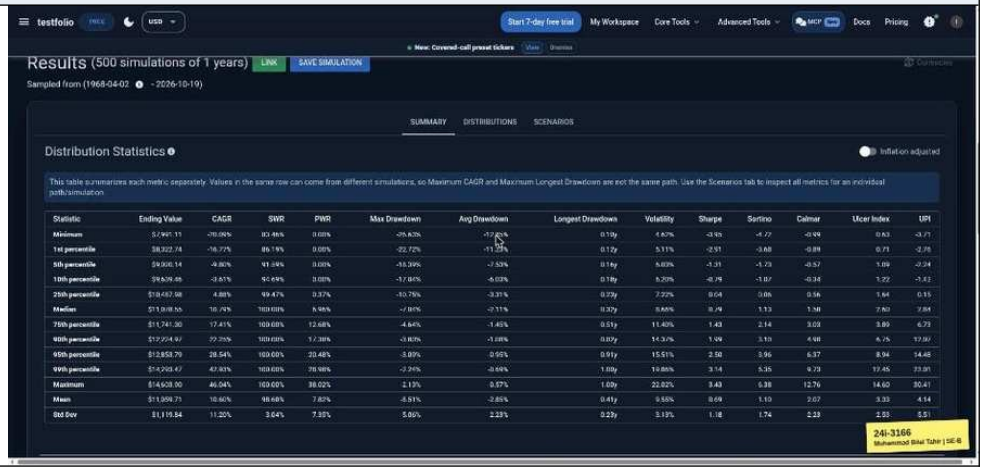
Monte Carlo Simulator (testfol.io/monte-carlo)

TC-BVA-033 | Number of years | input: 0 | result: Pass

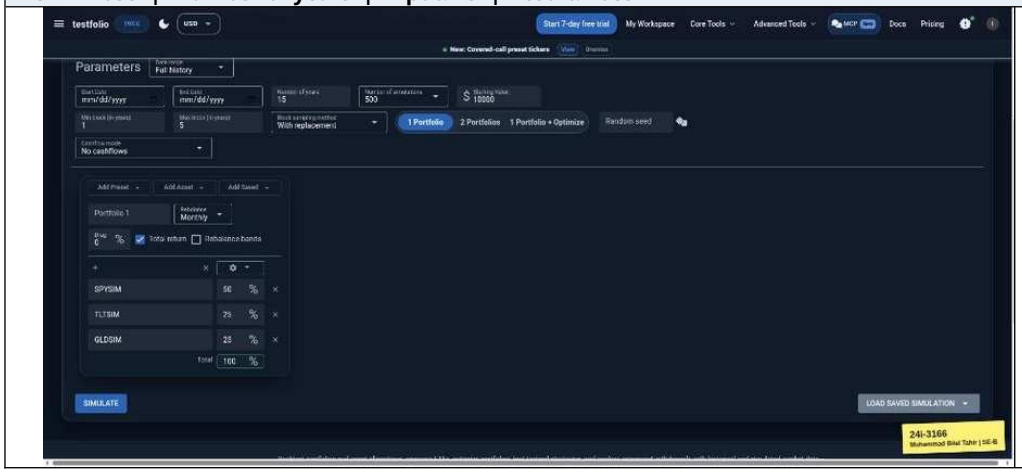


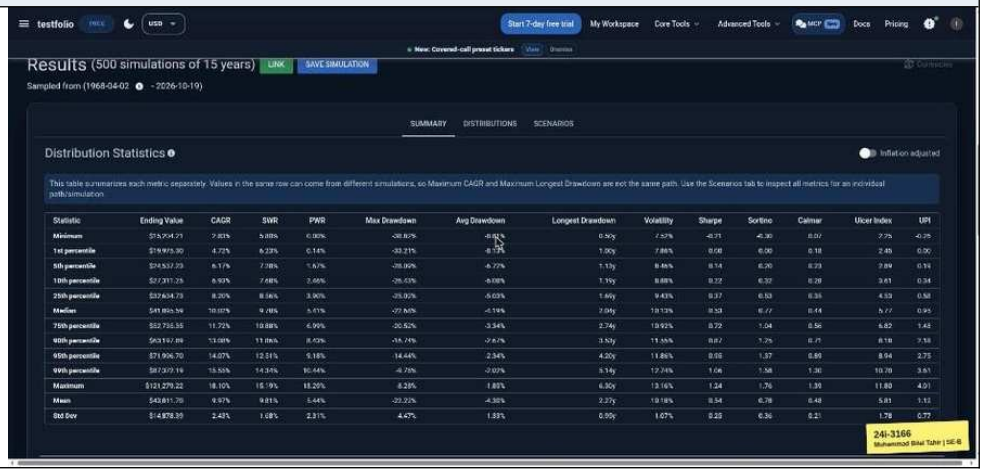
TC-BVA-034 | Number of years | input: 1 | result: Pass



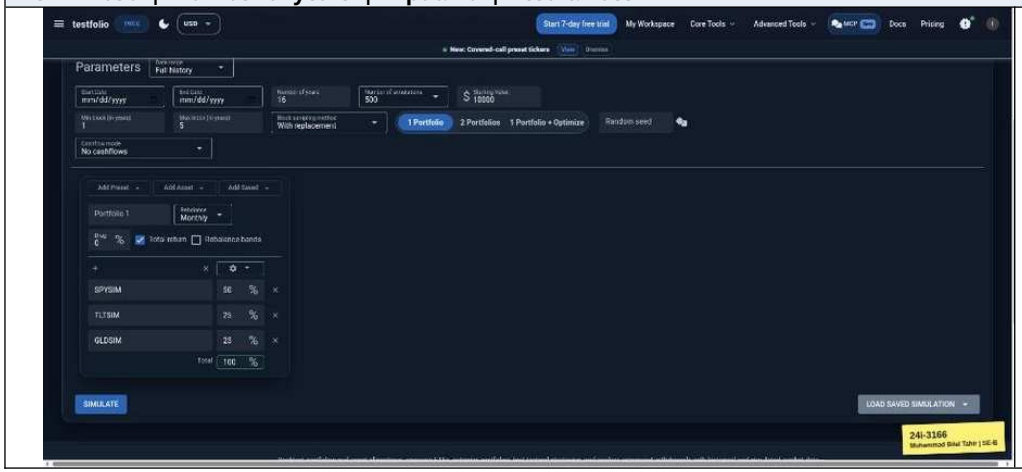


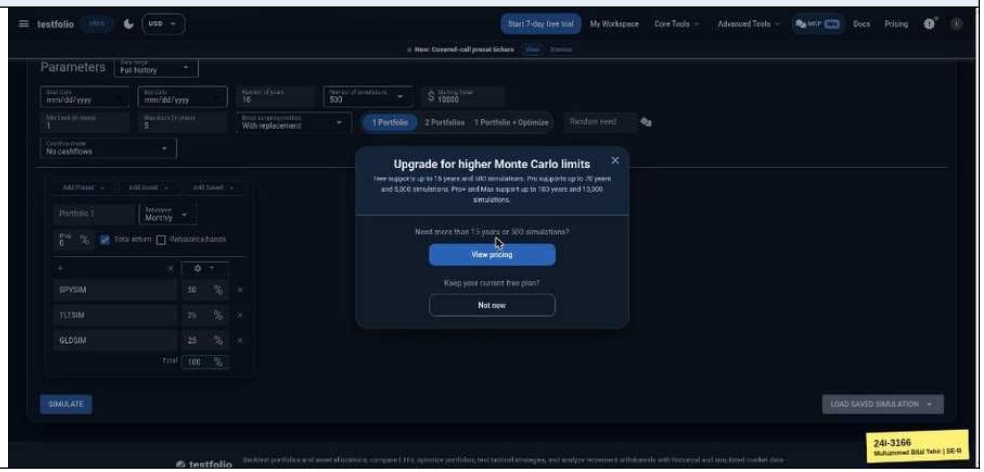
TC-BVA-035 | Number of years | input: 15 | result: Pass



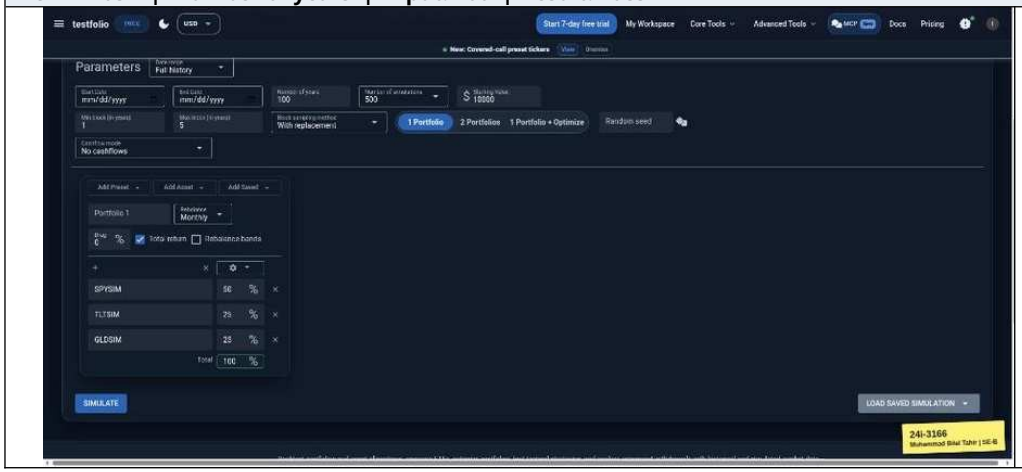


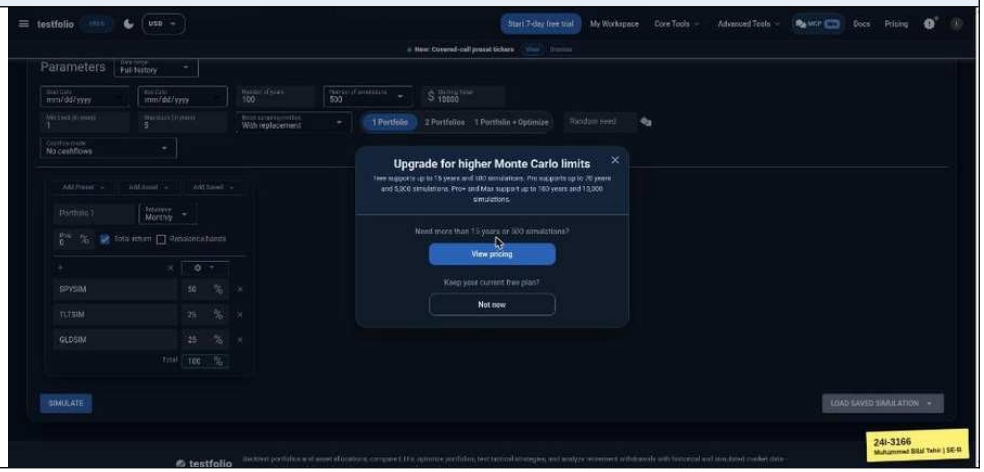
TC-BVA-036 | Number of years | input: 16 | result: Pass



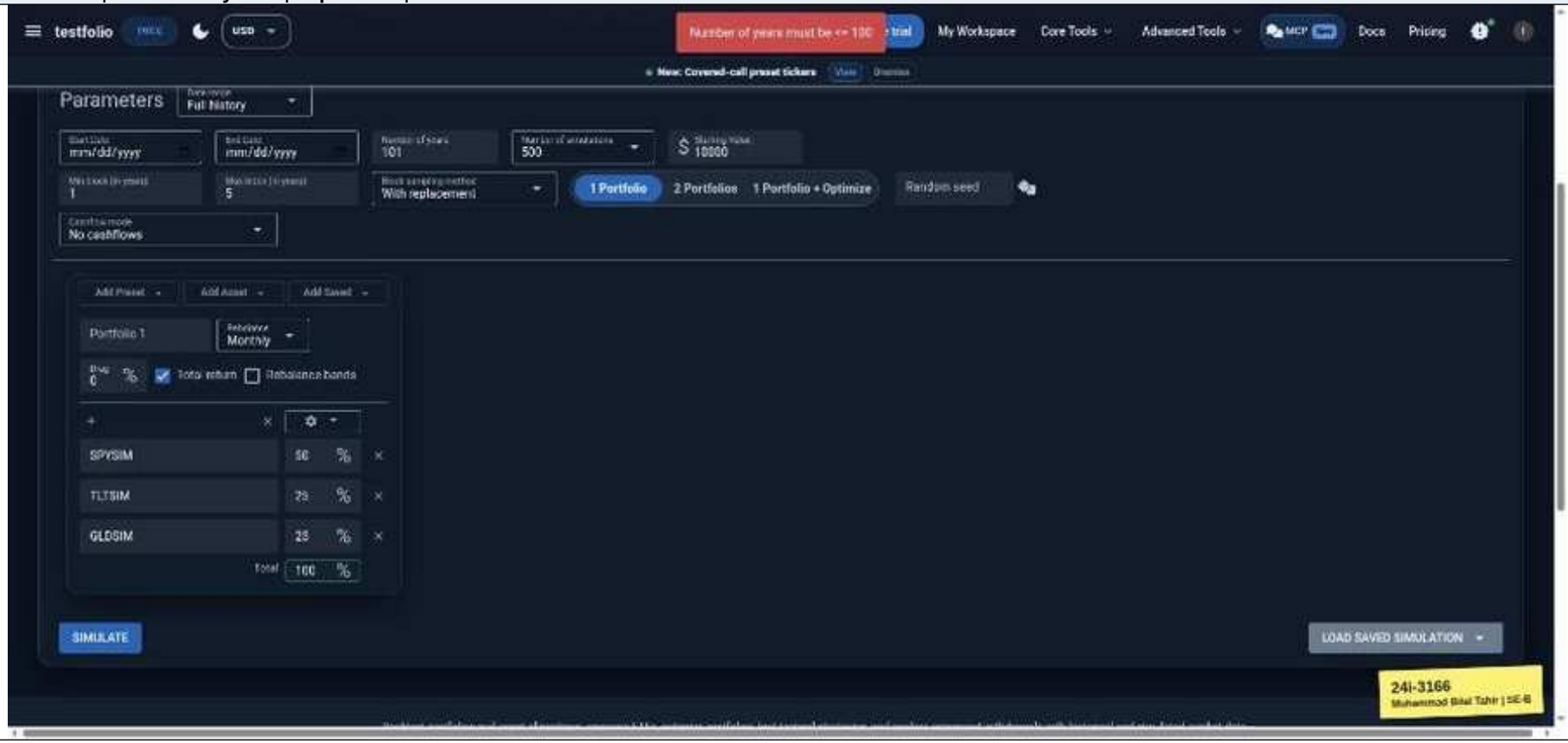


TC-BVA-037 | Number of years | input: 100 | result: Pass

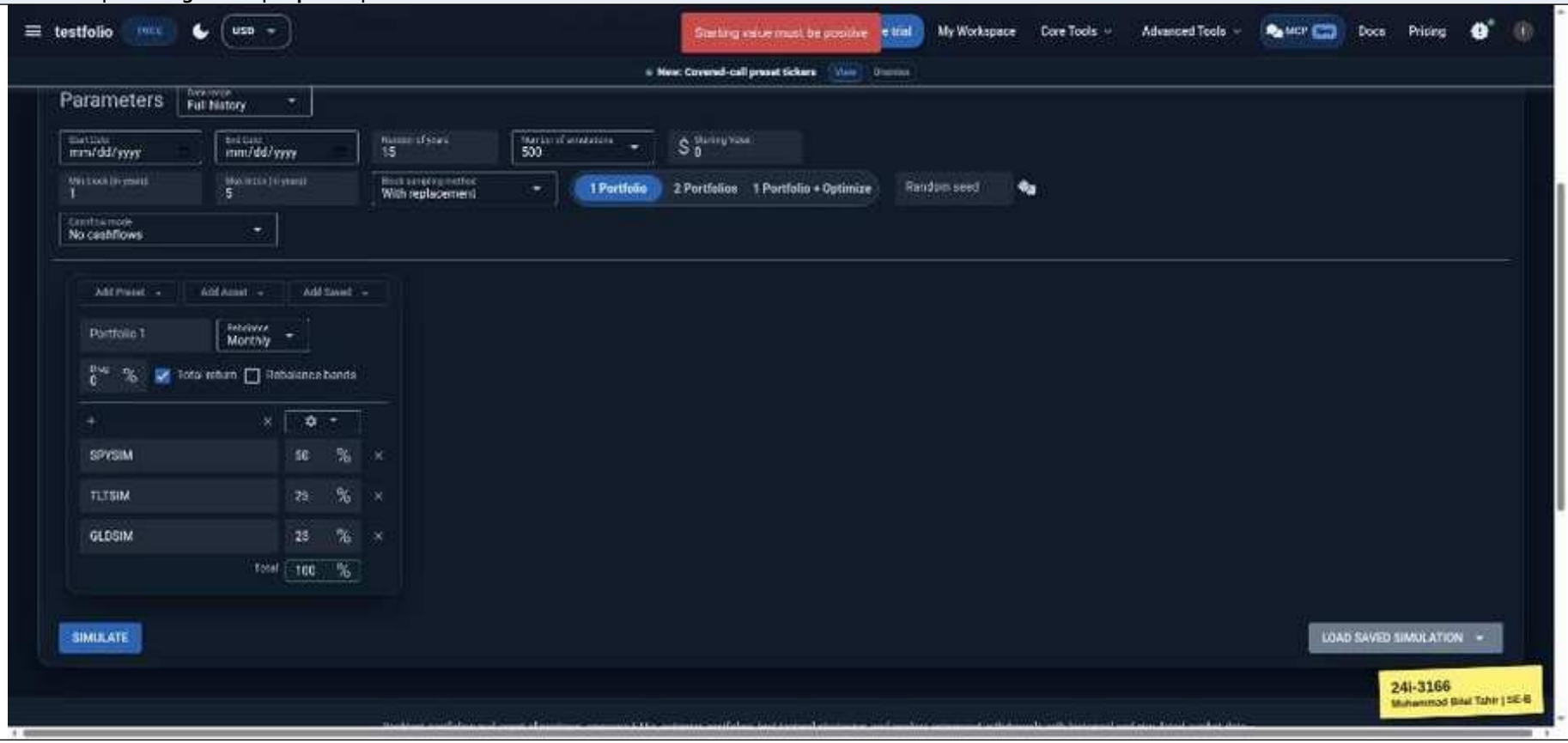




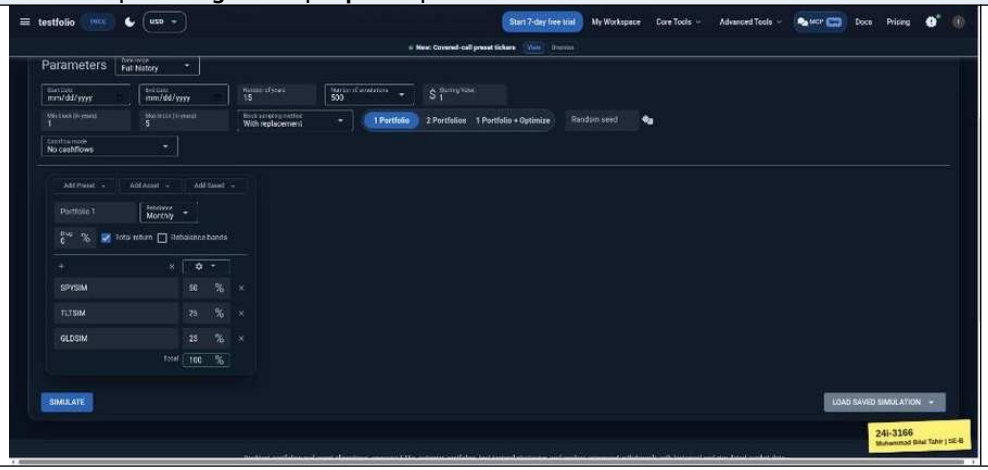
TC-BVA-038 | Number of years | input: 101 | result: Pass

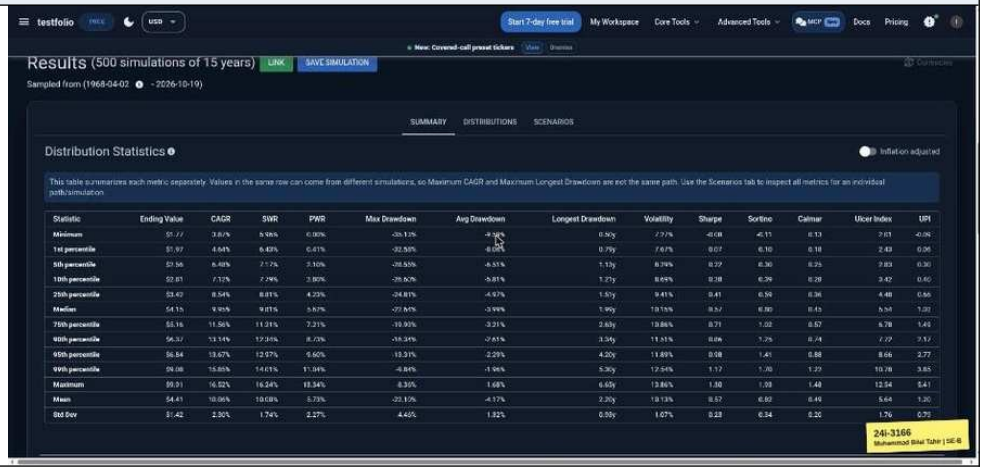


TC-BVA-039 | Starting Value | input: 0 | result: Pass

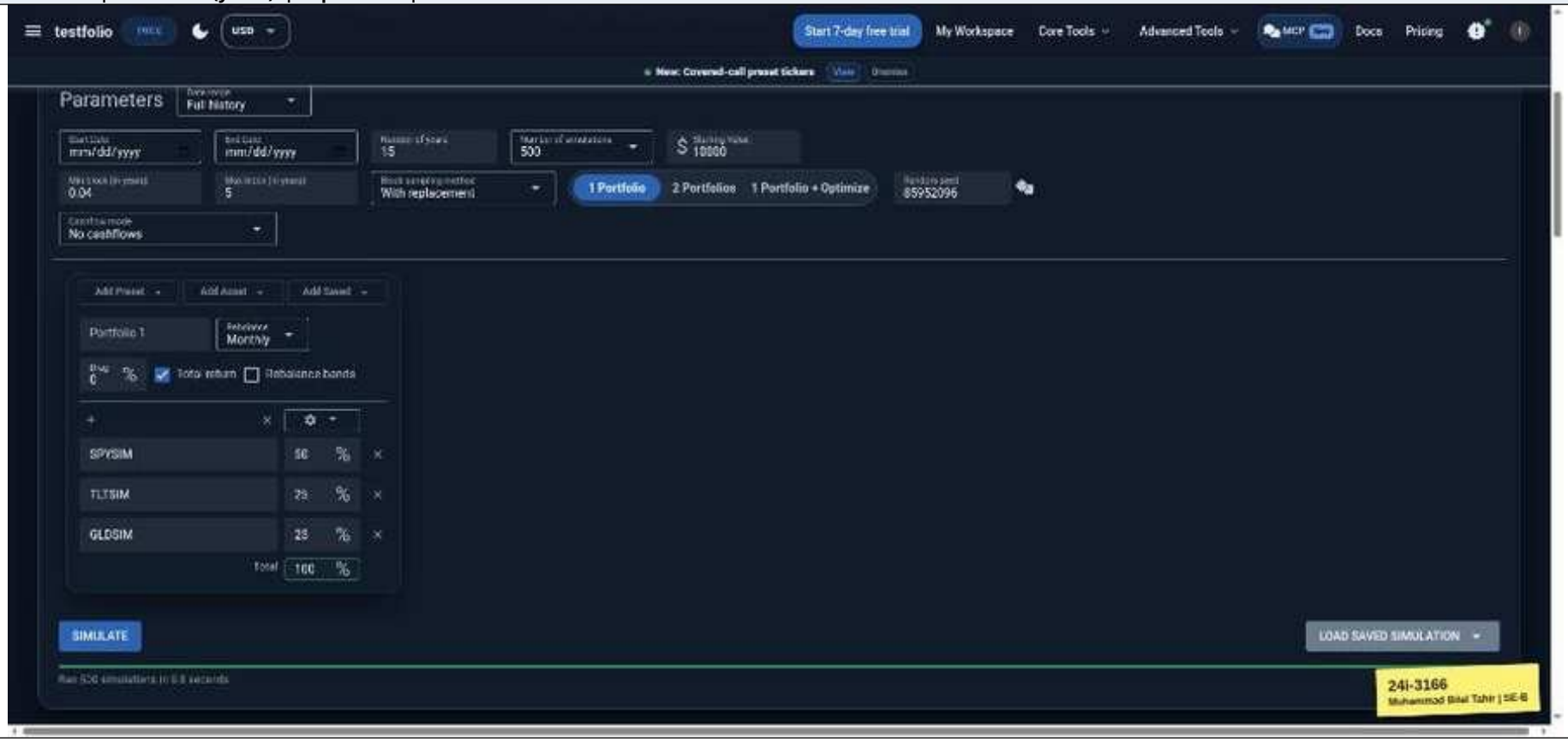


TC-BVA-040 | Starting Value | input: 1 | result: Pass

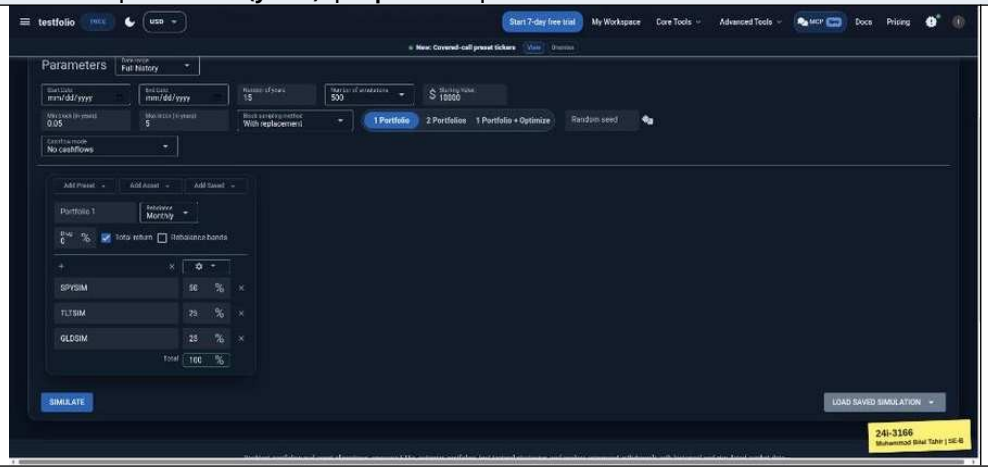


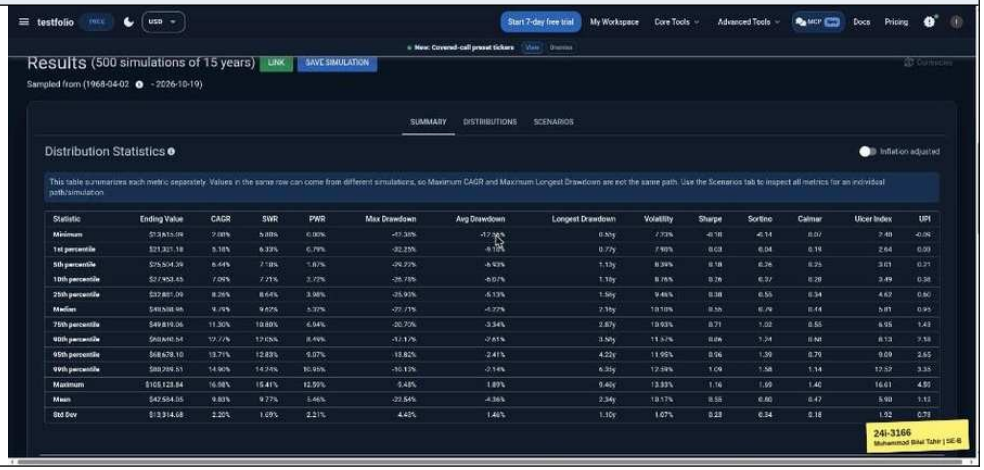


TC-BVA-041 | Min block (years) | input: 0.04 | result: Fail

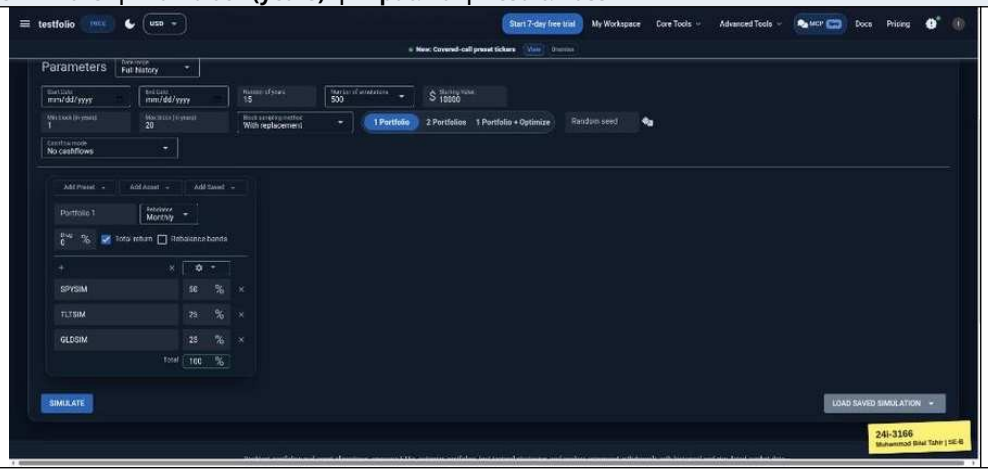


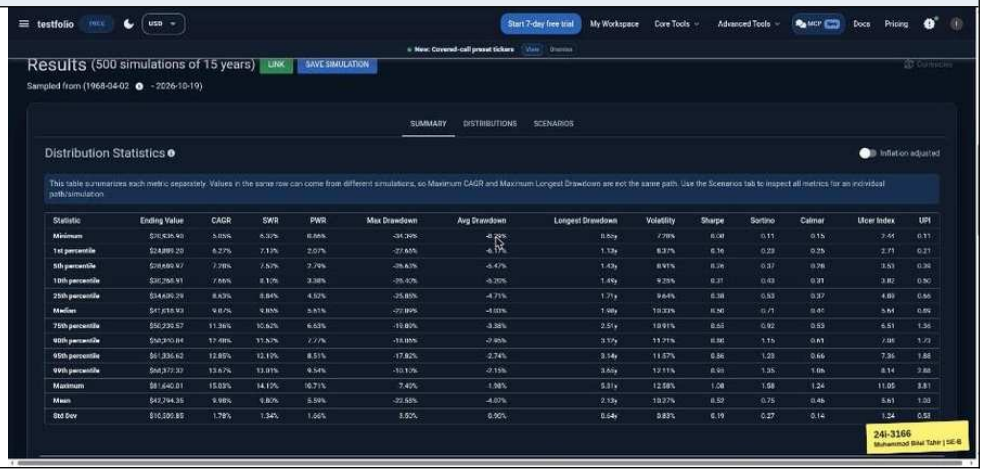
TC-BVA-042 | Min block (years) | input: 0.05 | result: Pass



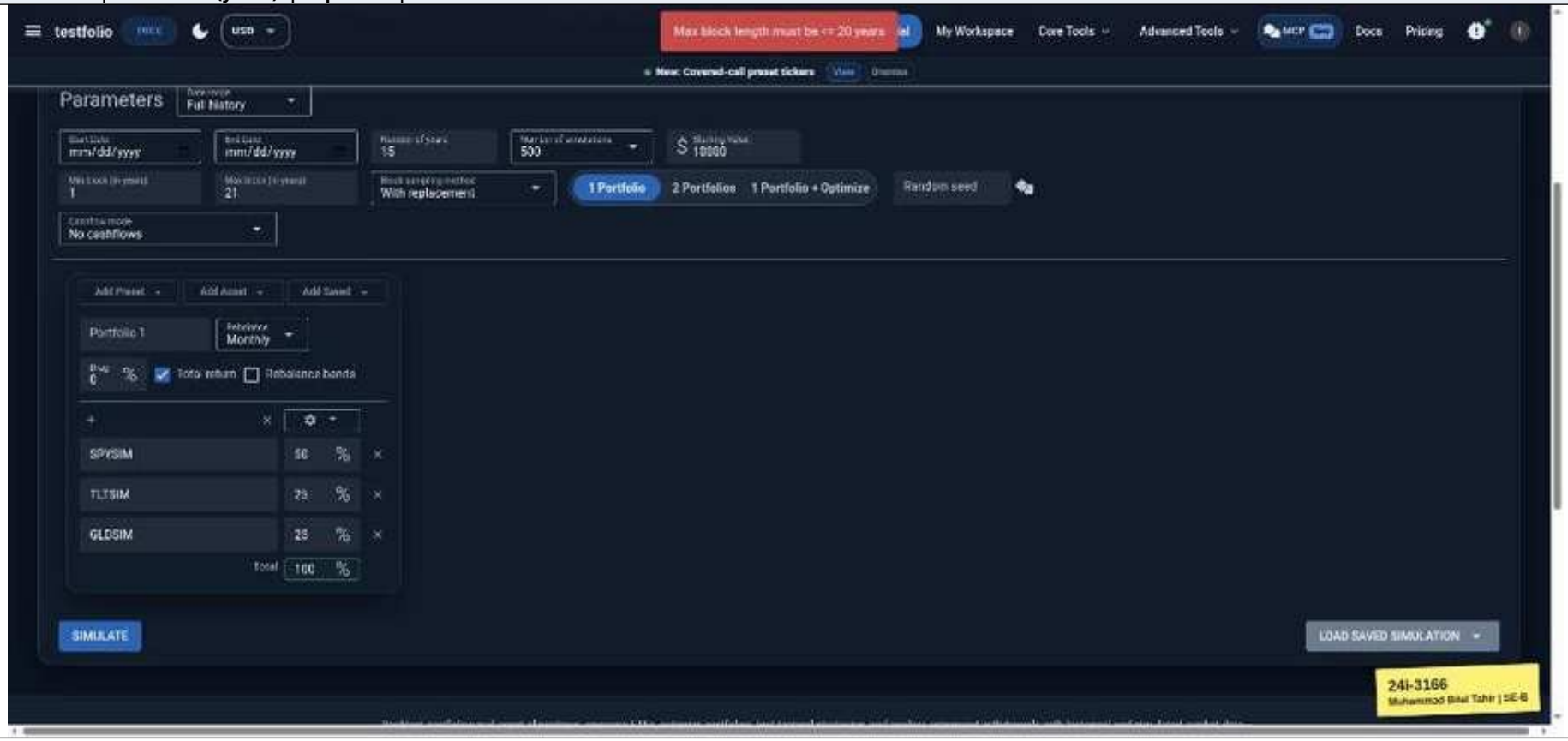


TC-BVA-043 | Max block (years) | input: 20 | result: Pass

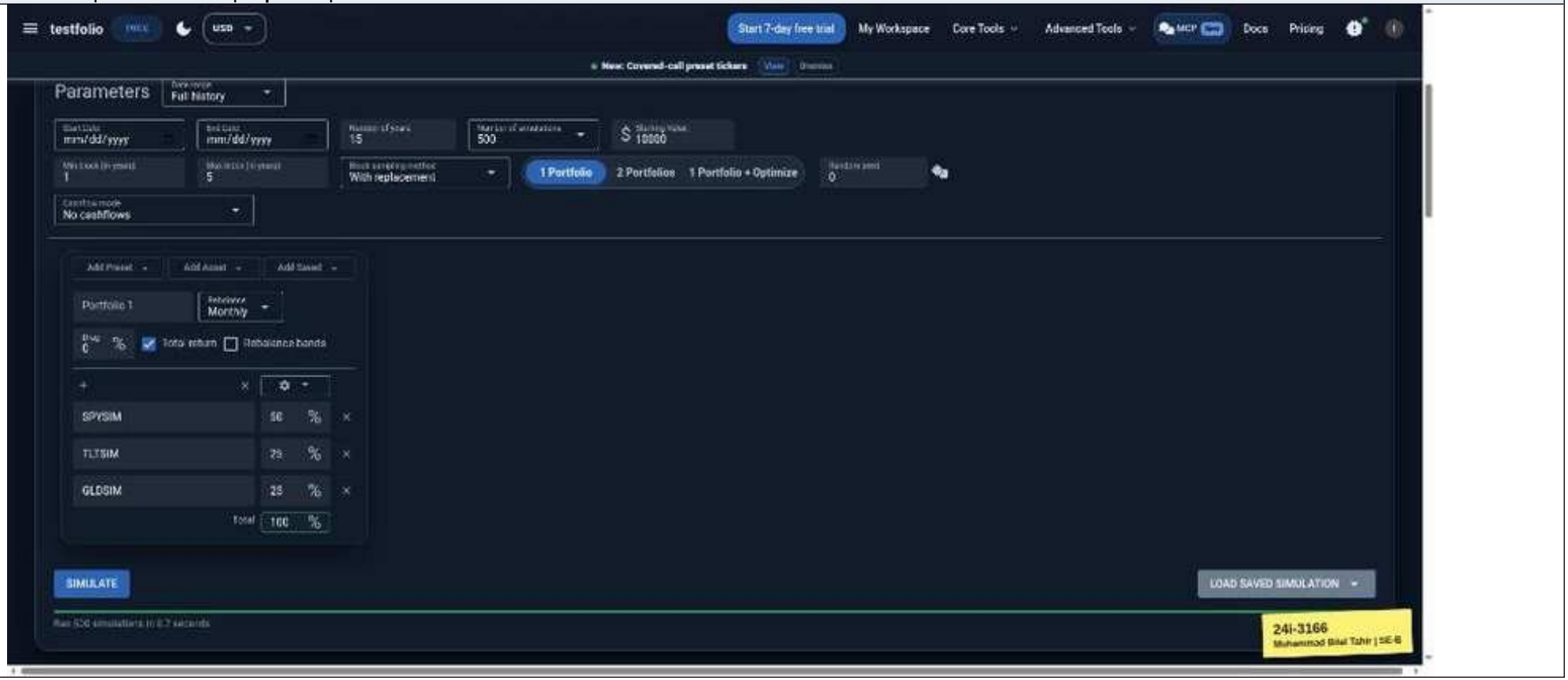




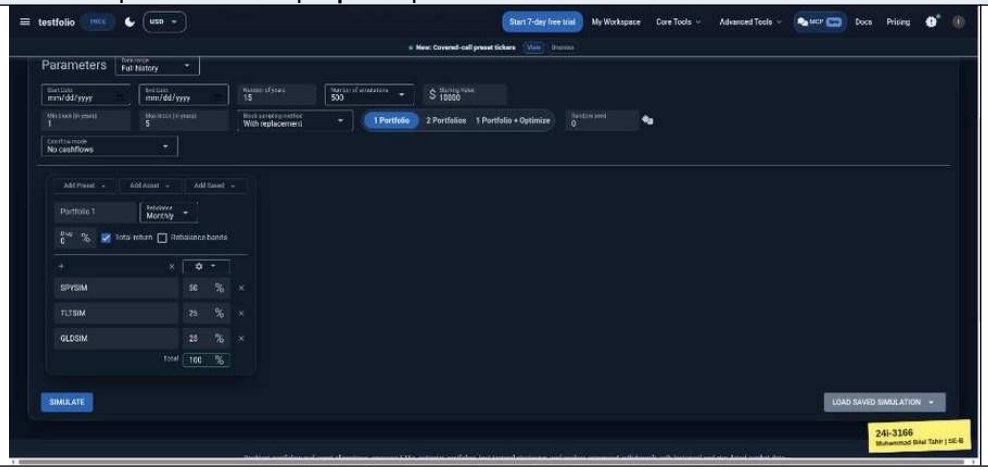
TC-BVA-044 | Max block (years) | input: 21 | result: Pass

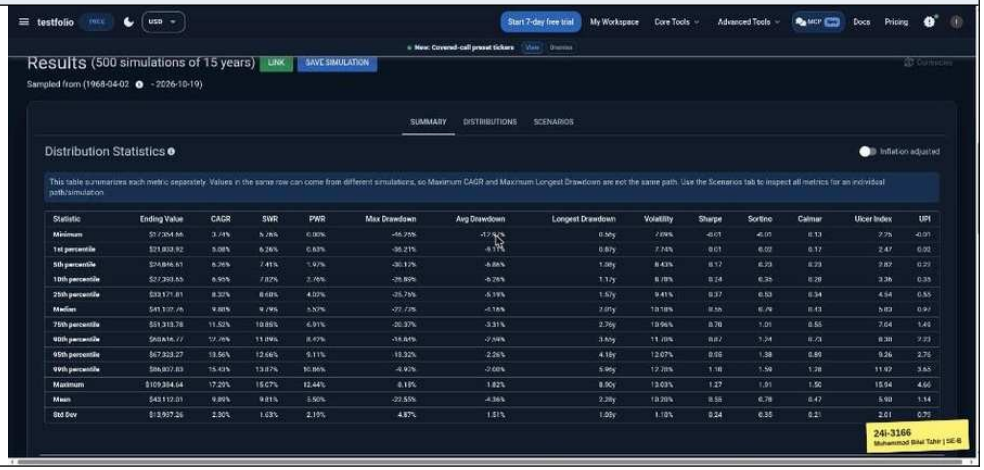


TC-BVA-045 | Random seed | input: -1 | result: Fail



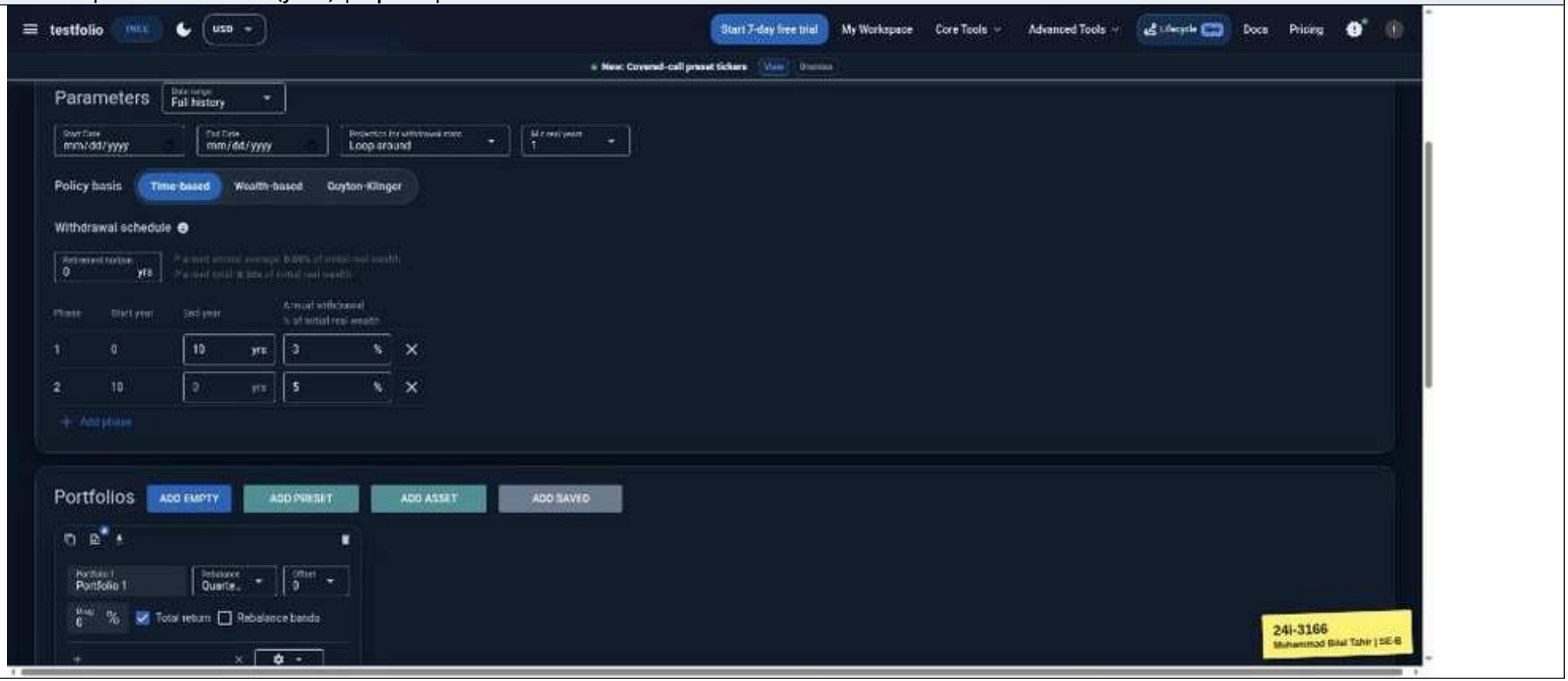
TC-BVA-046 | Random seed | input: 0 | result: Pass



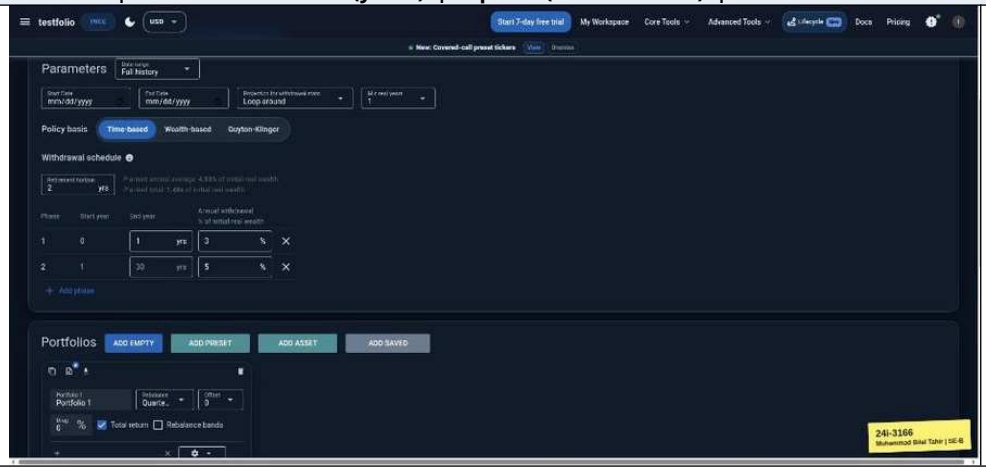


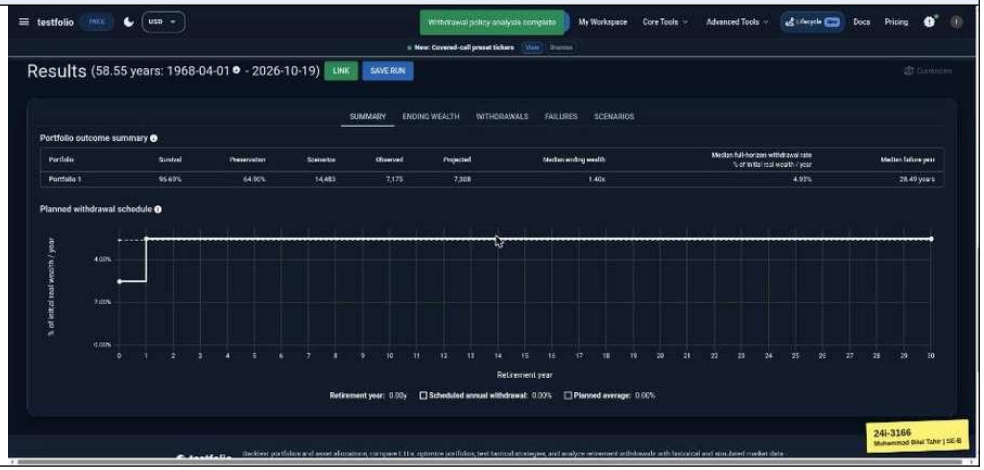
Safe Withdrawal Rate Calculator (testfol.io/safe-withdrawal-rate-calculator)

TC-BVA-047 | Retirement horizon (years) | input: 0 | result: Pass

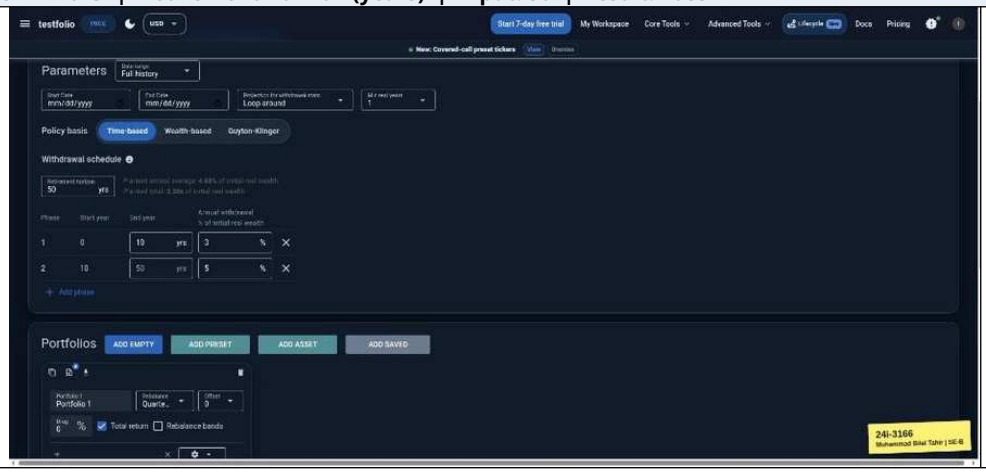


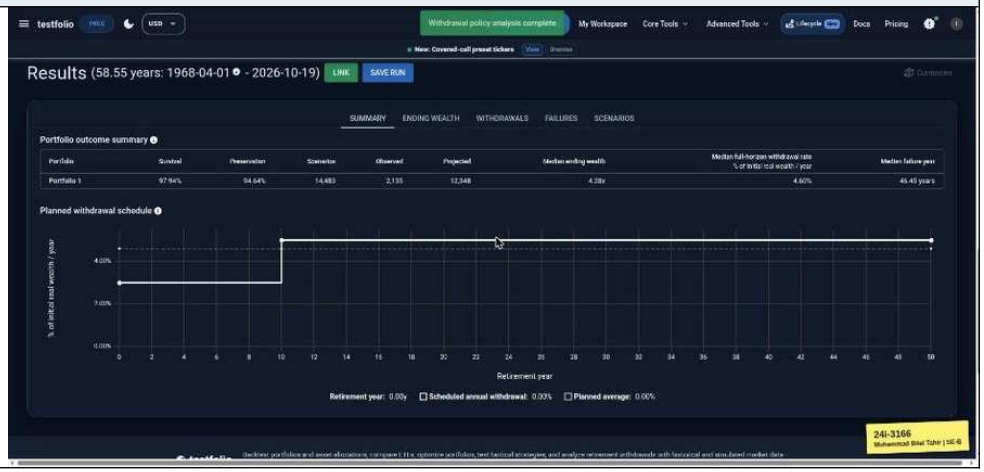
TC-BVA-048 | Retirement horizon (years) | input: 2 (Phase 1 end 1) | result: Pass



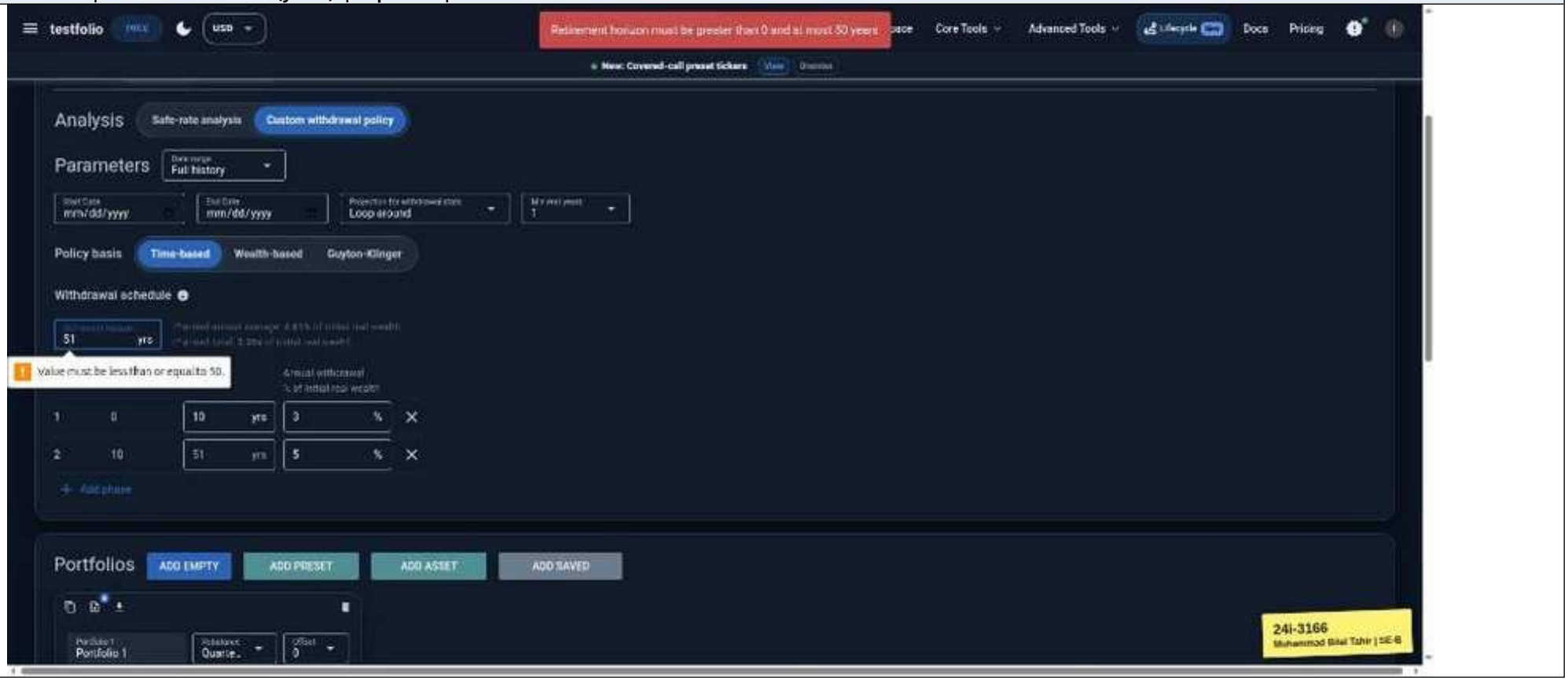


TC-BVA-049 | Retirement horizon (years) | input: 50 | result: Pass

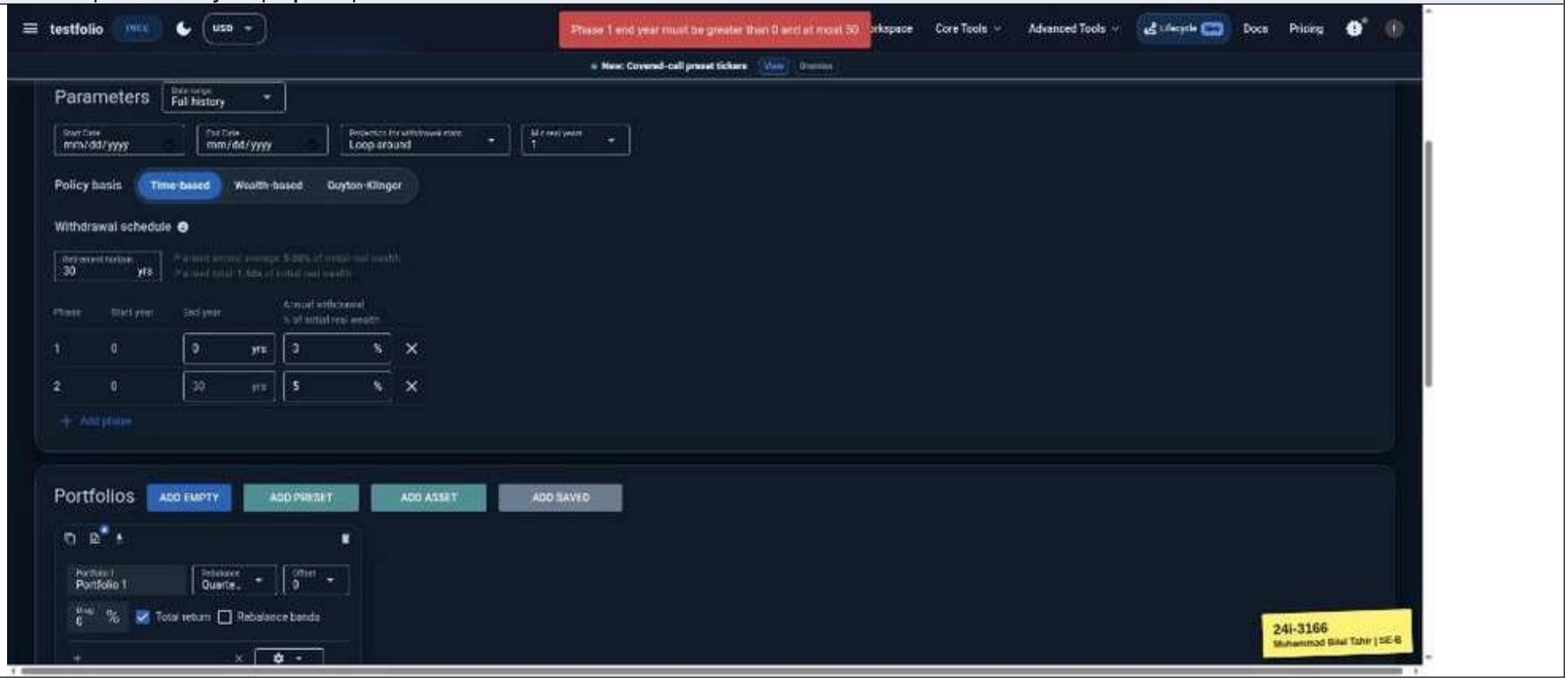




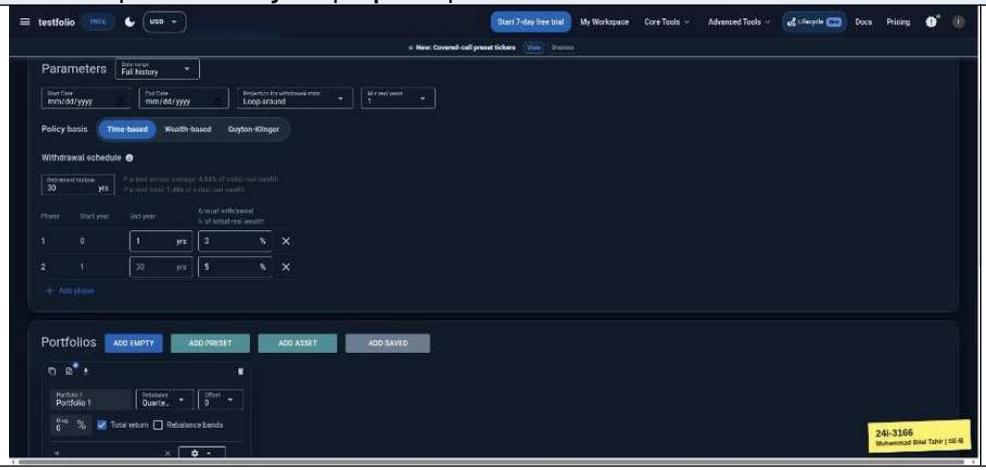
TC-BVA-050 | Retirement horizon (years) | input: 51 | result: Pass

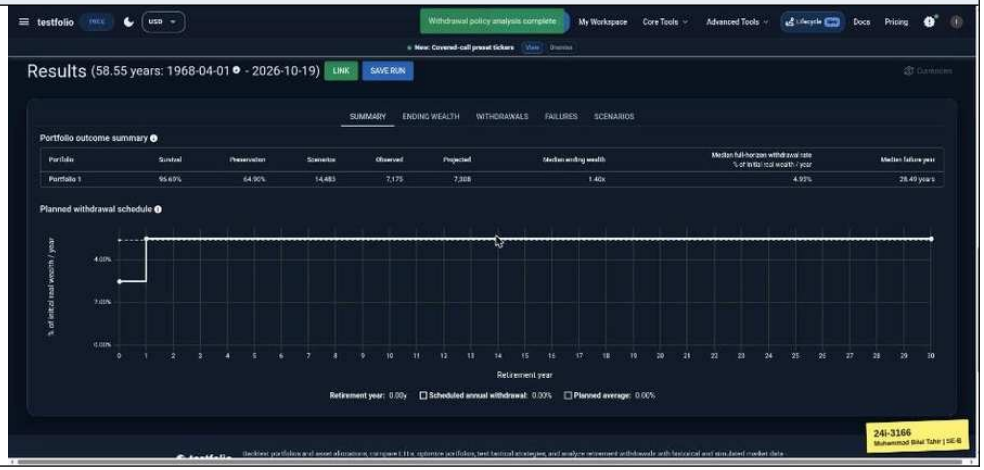


TC-BVA-051 | Phase 1 end year | input: 0 | result: Pass

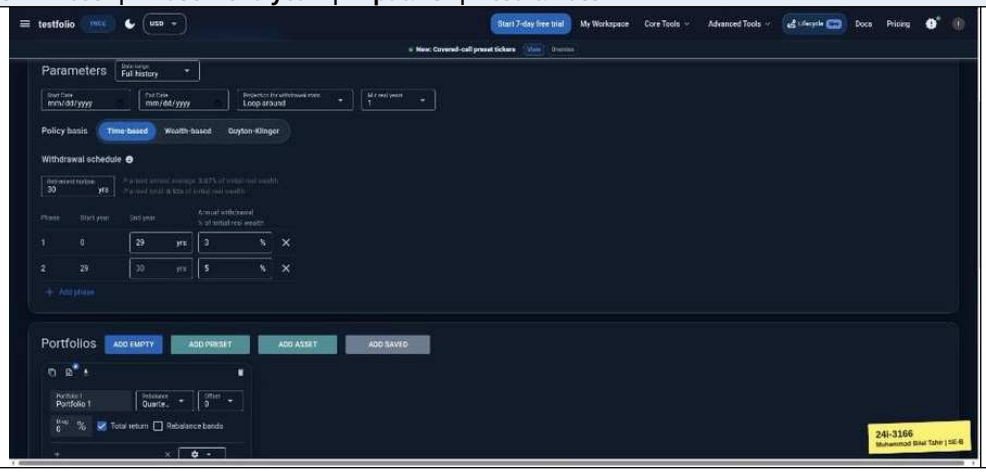


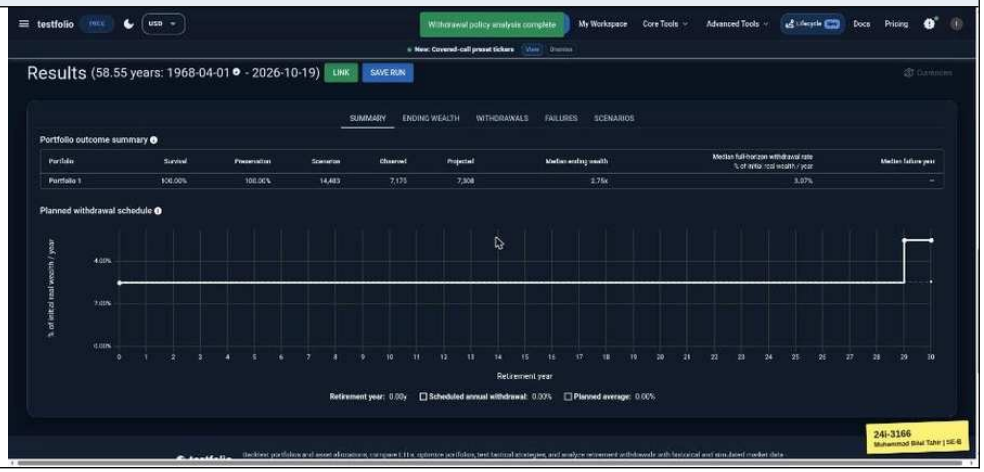
TC-BVA-052 | Phase 1 end year | input: 1 | result: Pass



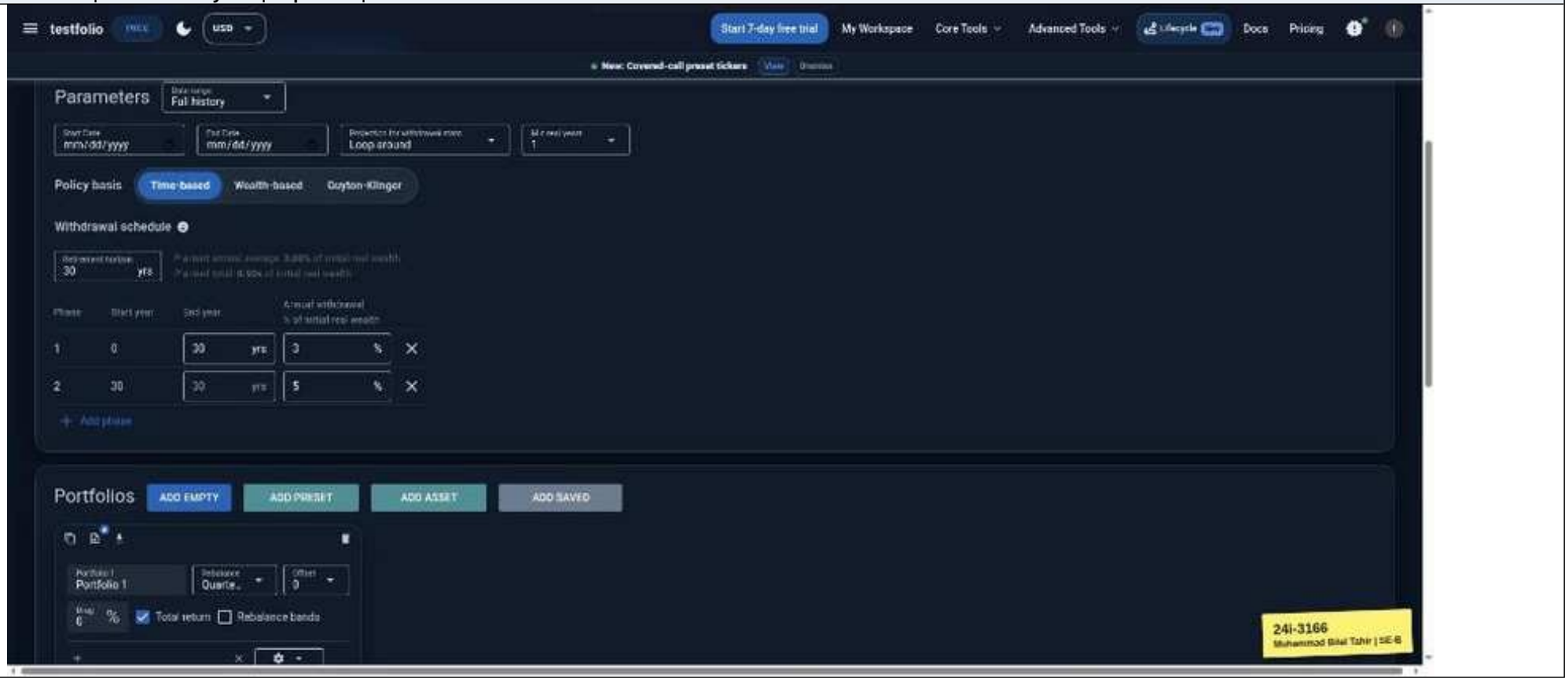


TC-BVA-053 | Phase 1 end year | input: 29 | result: Pass

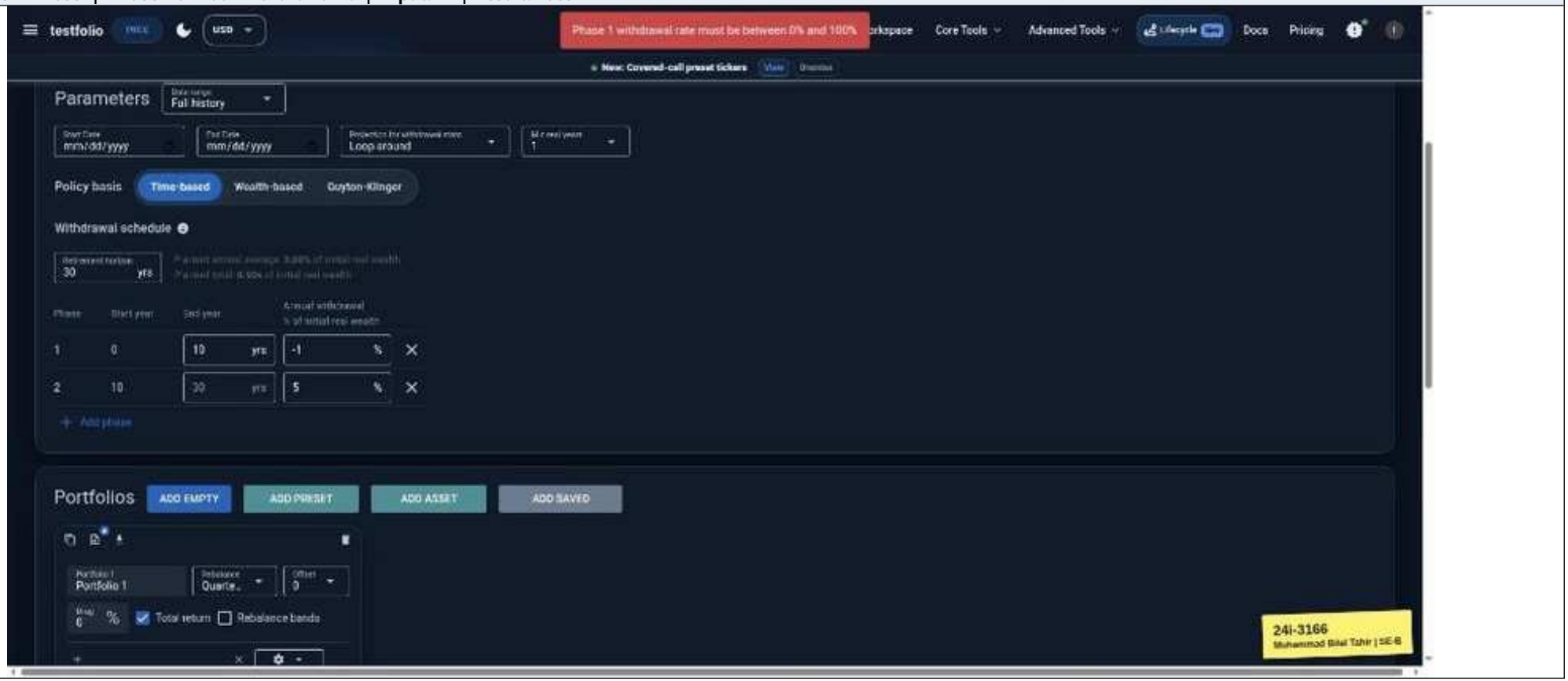




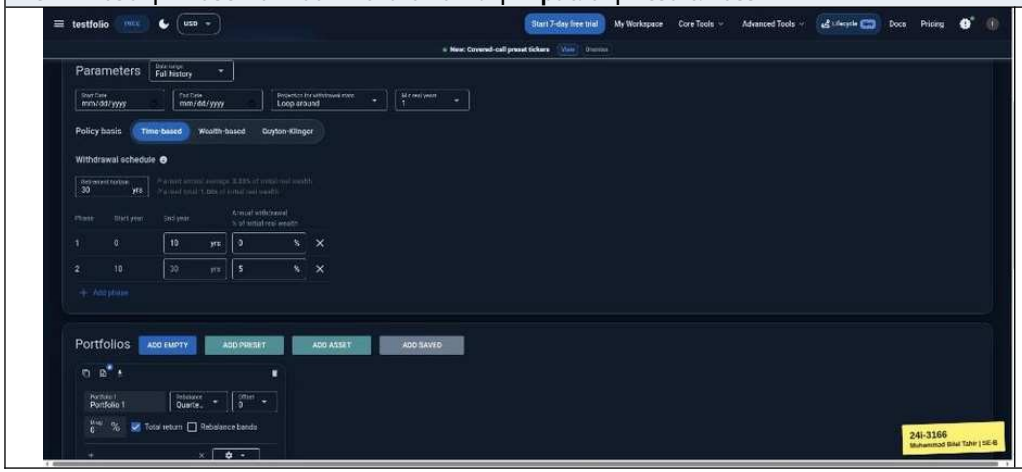
TC-BVA-054 | Phase 1 end year | input: 30 | result: Pass

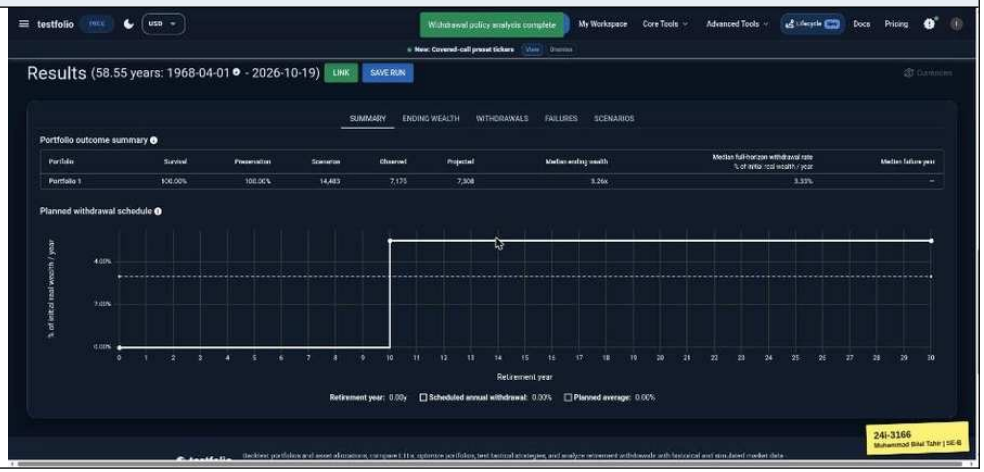


TC-BVA-055 | Phase 1 annual withdrawal % | input: -1 | result: Pass

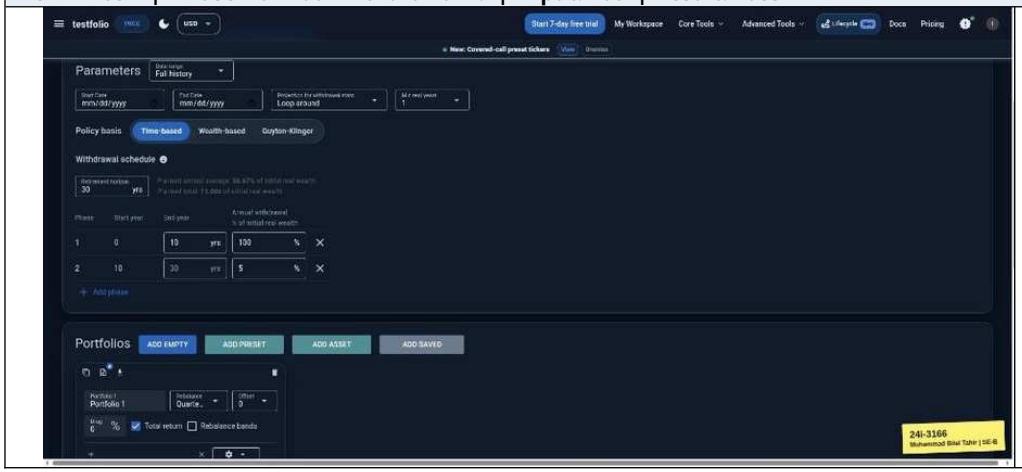


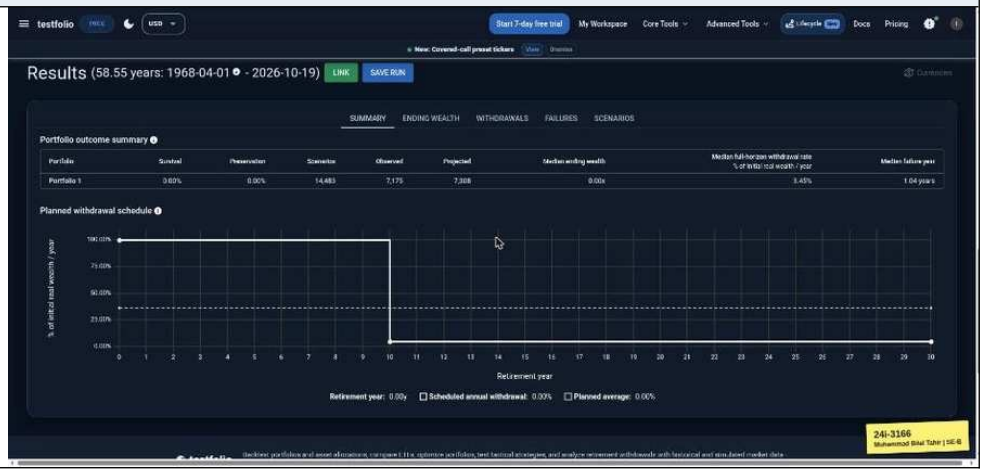
TC-BVA-056 | Phase 1 annual withdrawal % | input: 0 | result: Pass



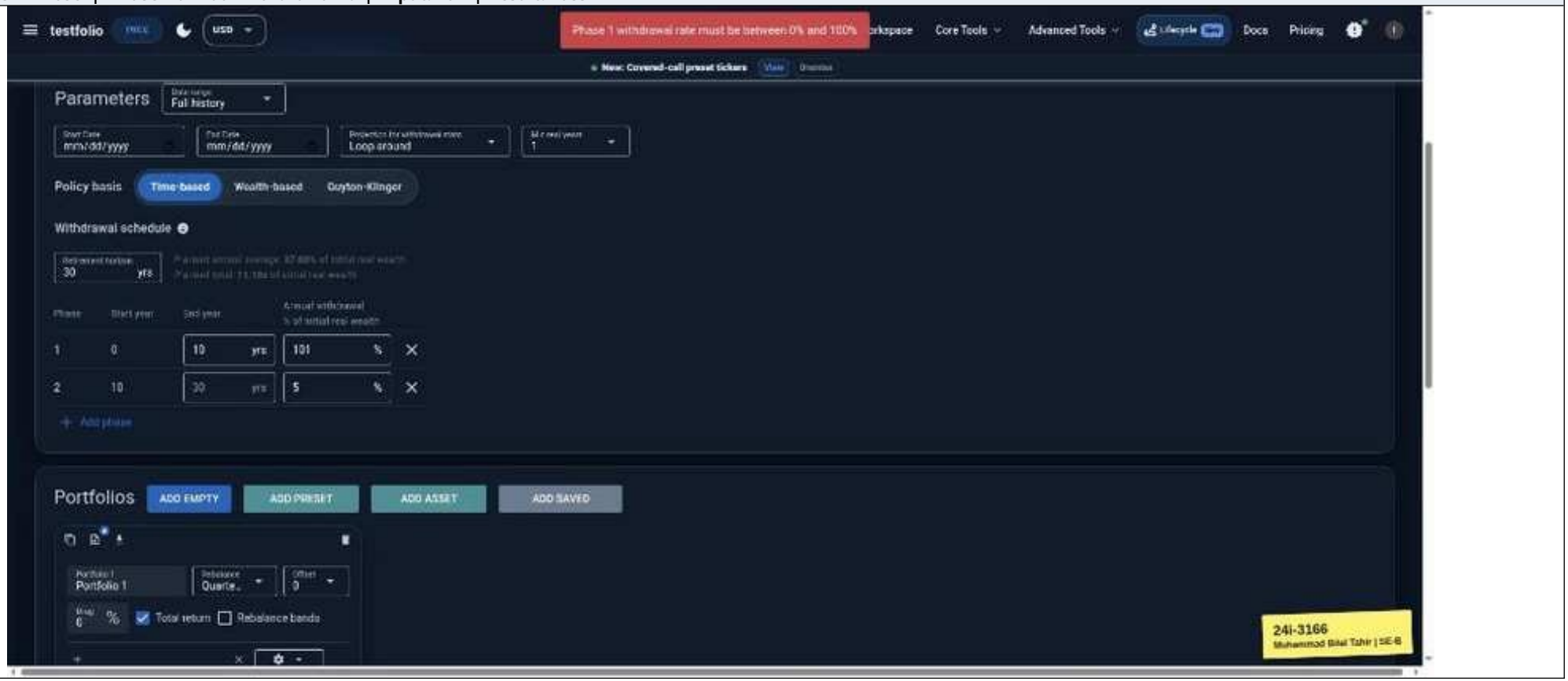


TC-BVA-057 | Phase 1 annual withdrawal % | input: 100 | result: Pass

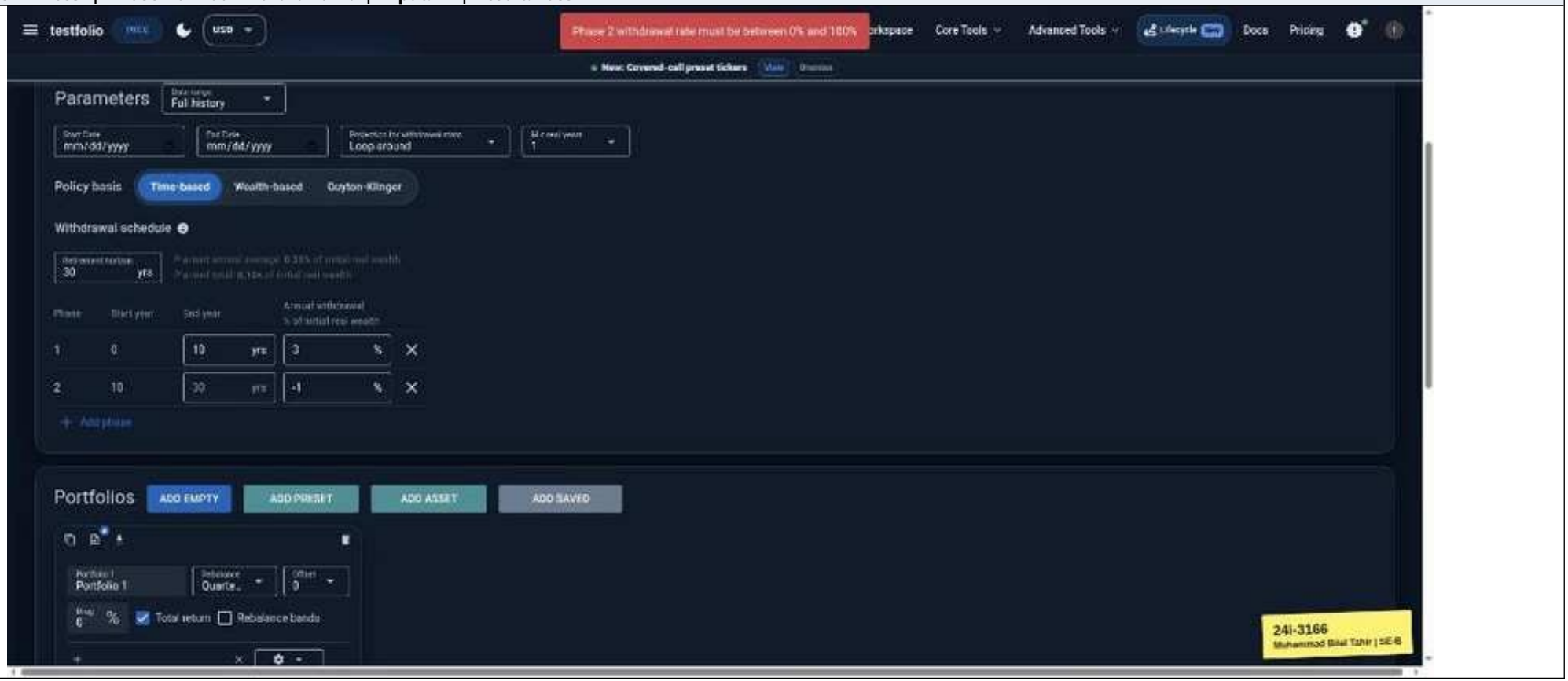




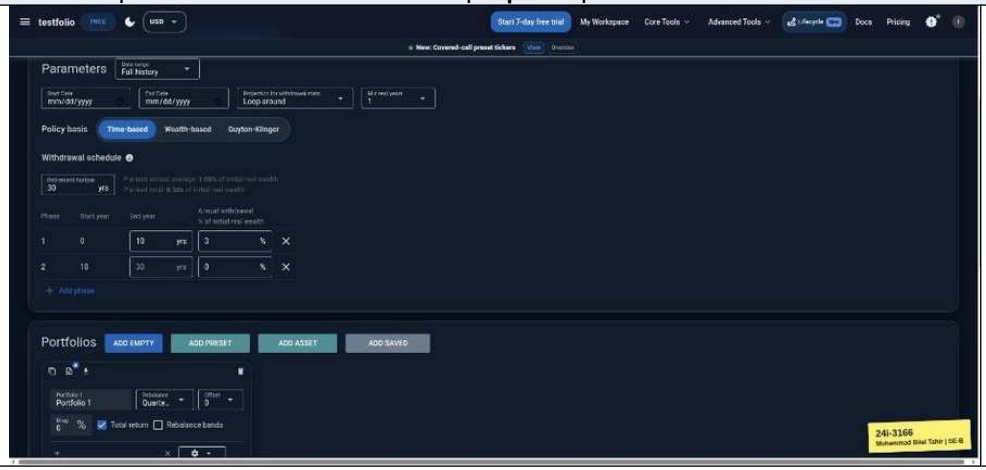
TC-BVA-058 | Phase 1 annual withdrawal % | input: 101 | result: Pass

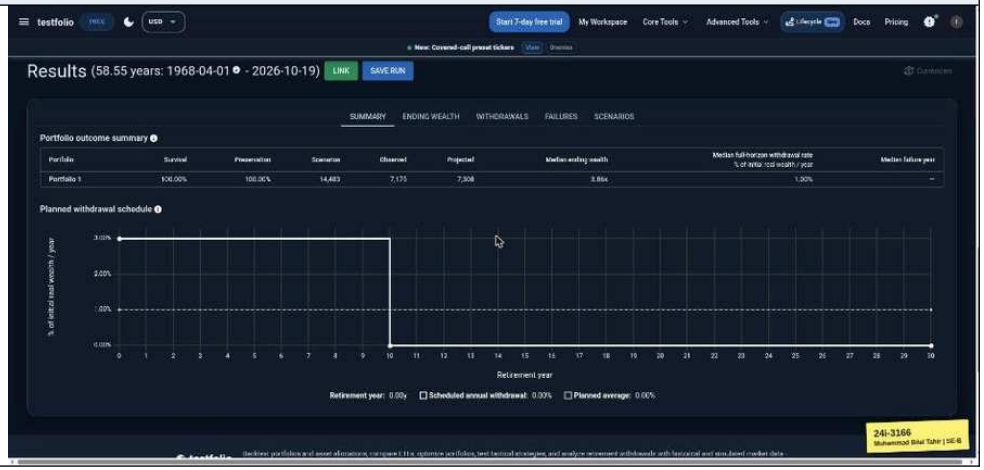


TC-BVA-059 | Phase 2 annual withdrawal % | input: -1 | result: Pass

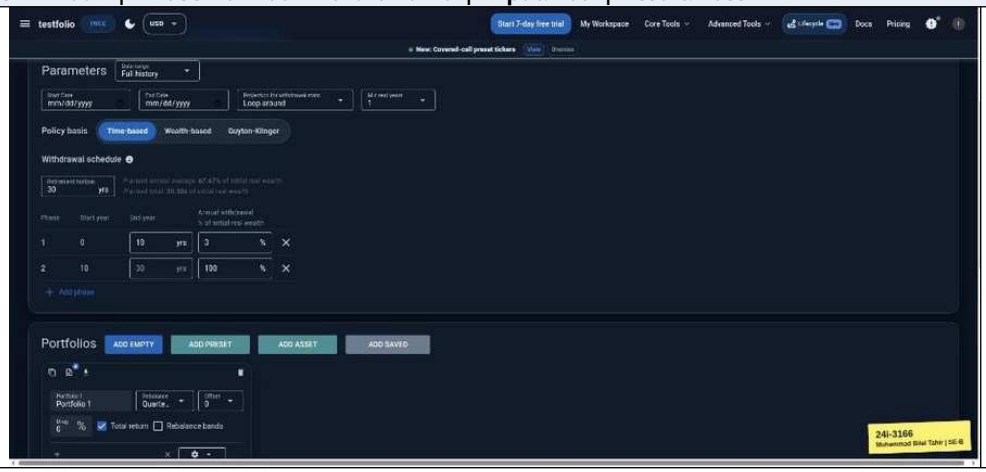


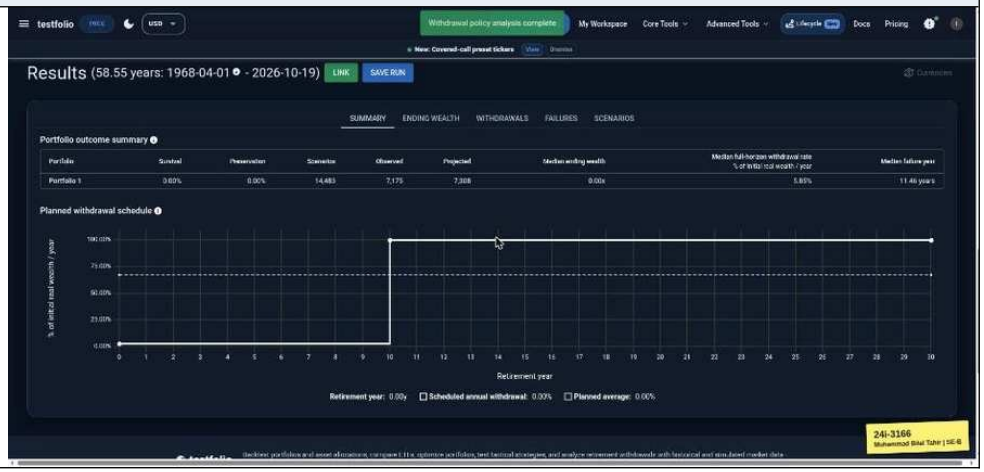
TC-BVA-060 | Phase 2 annual withdrawal % | input: 0 | result: Pass



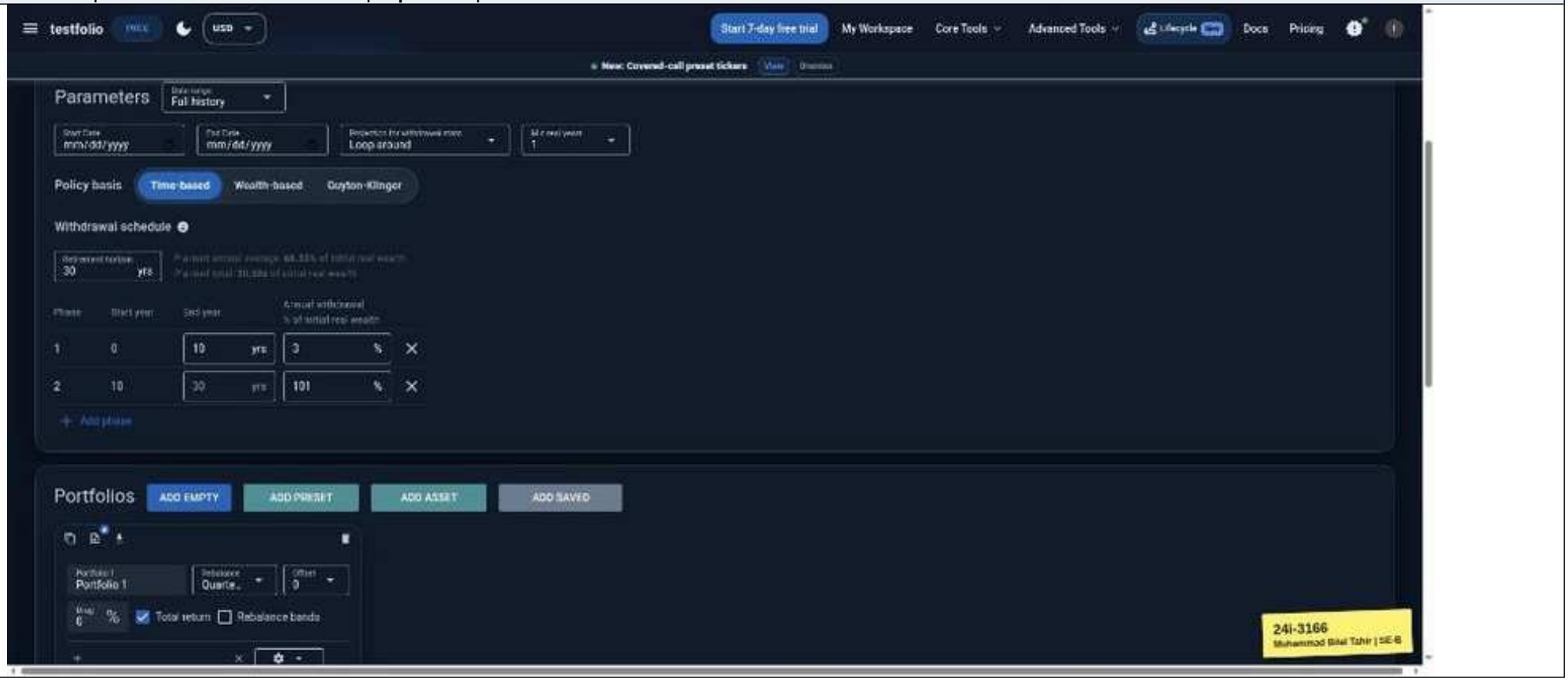


TC-BVA-061 | Phase 2 annual withdrawal % | input: 100 | result: Pass



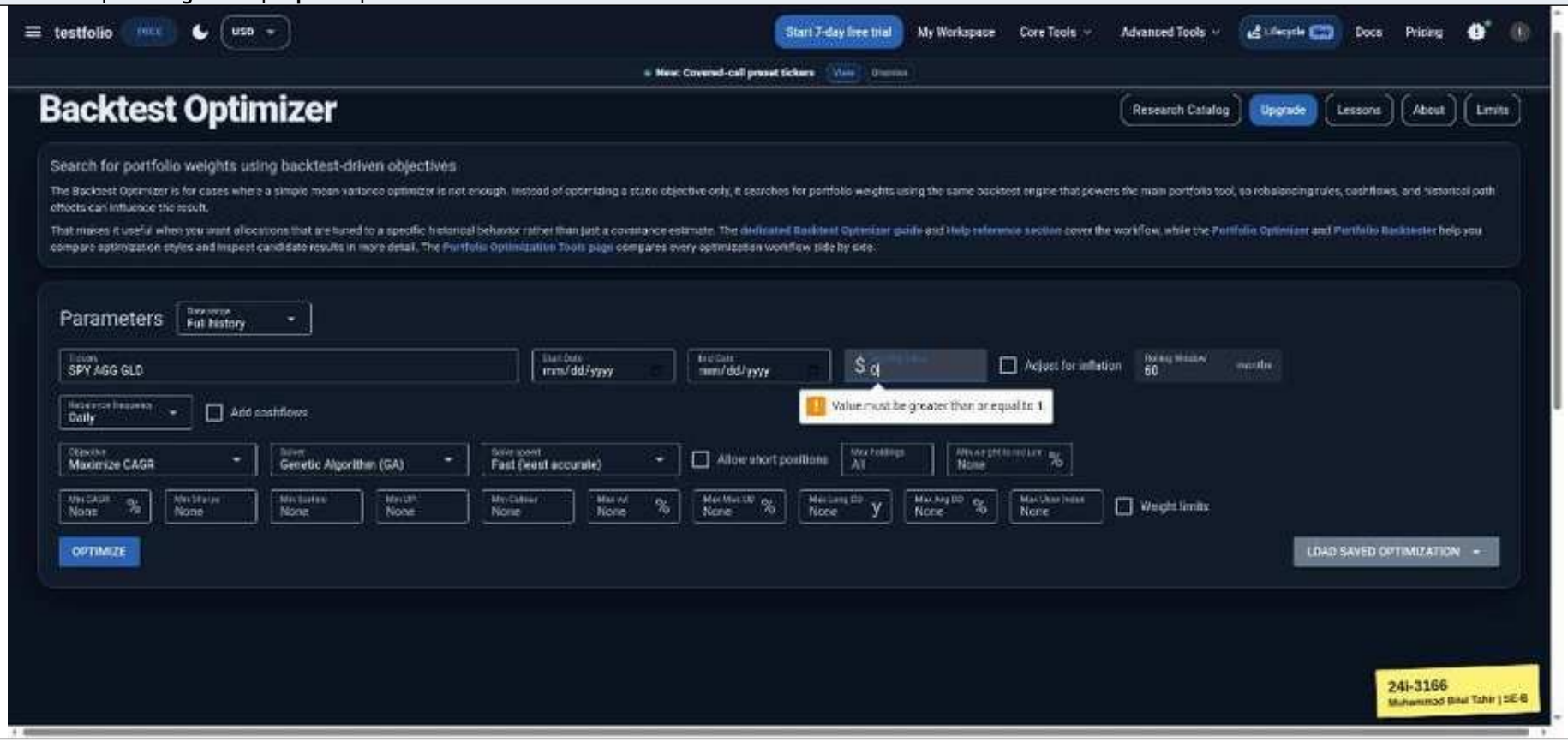


TC-BVA-062 | Phase 2 annual withdrawal % | input: 101 | result: Pass

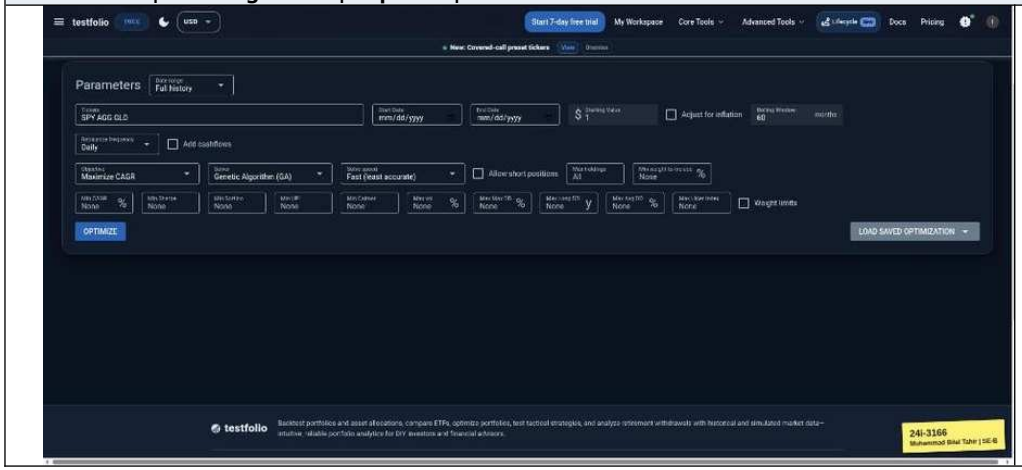


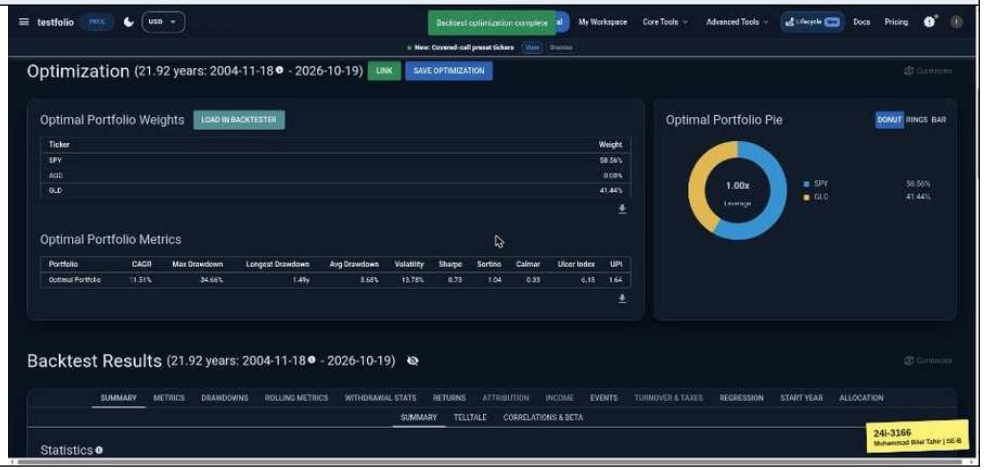
Backtest Optimizer (testfol.io/backtest-optimizer)

TC-BVA-063 | Starting Value | input: 0 | result: Pass

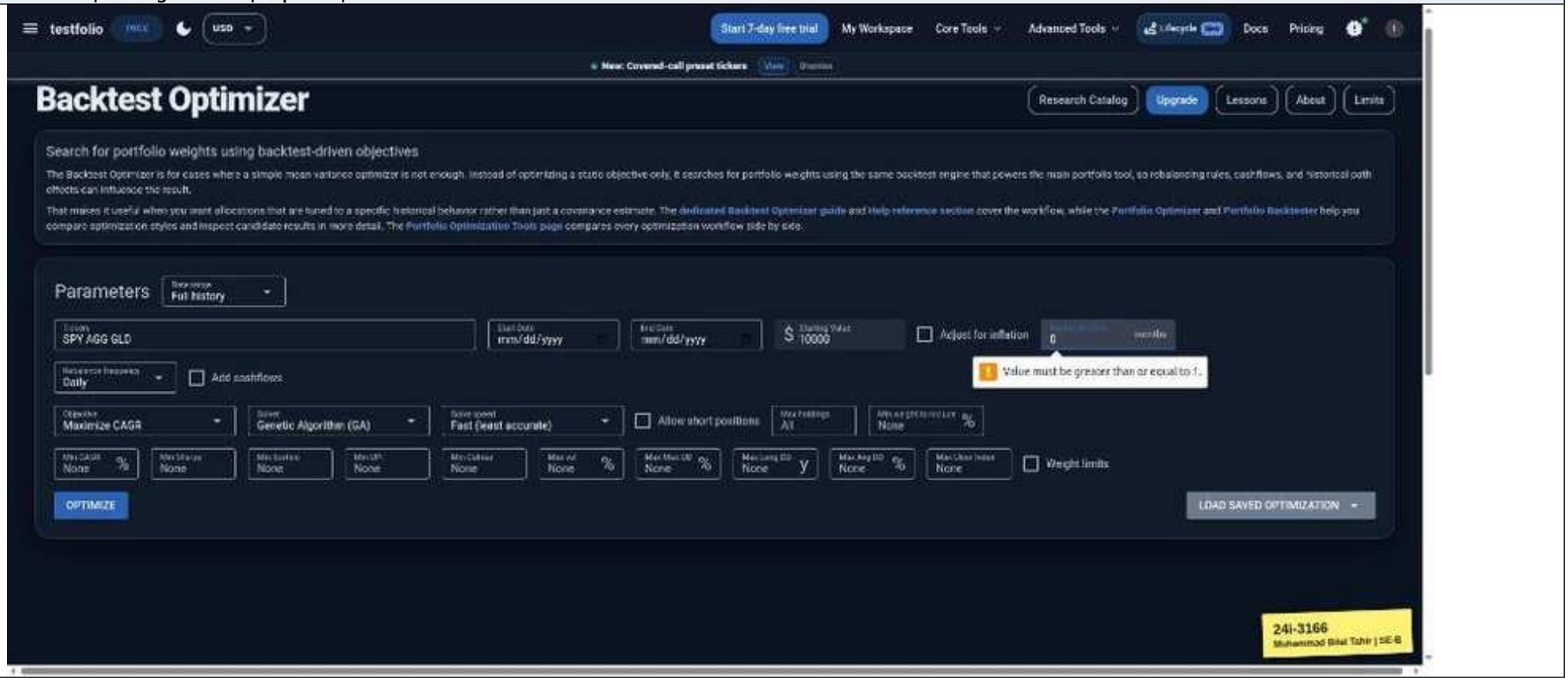


TC-BVA-064 | Starting Value | input: 1 | result: Pass

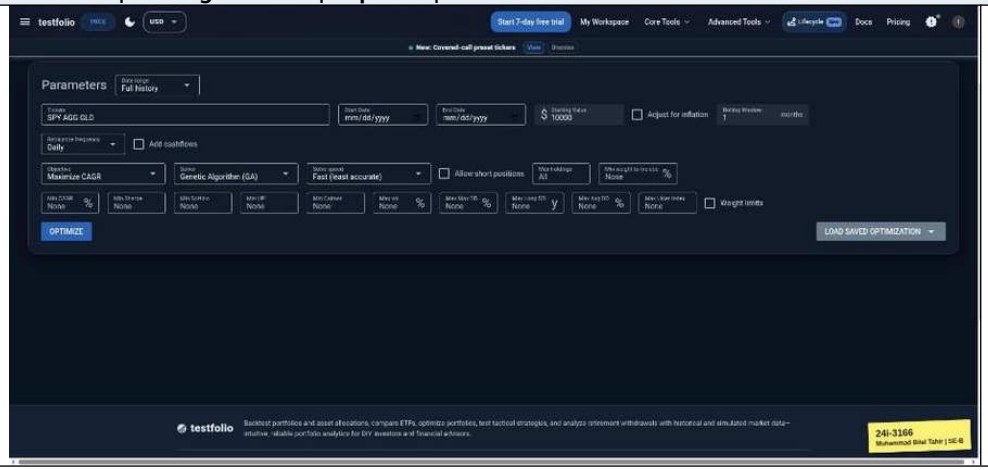


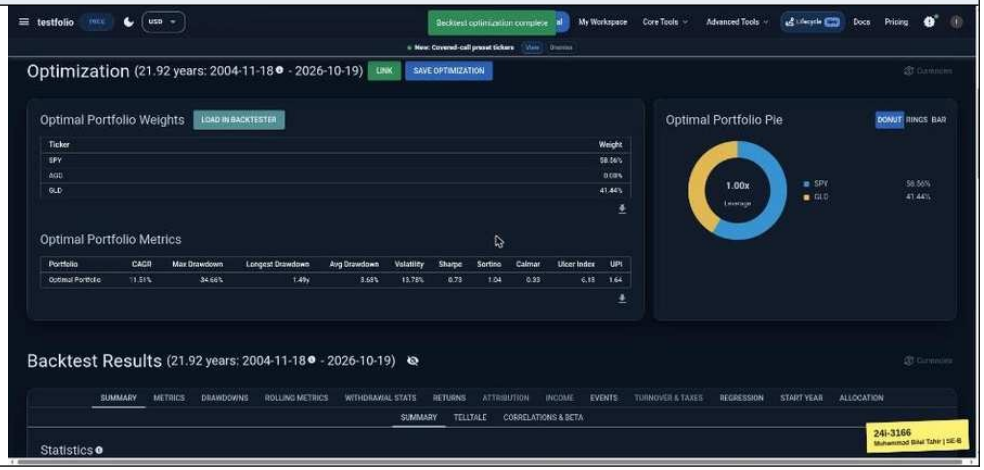


TC-BVA-065 | Rolling Window | input: 0 | result: Pass

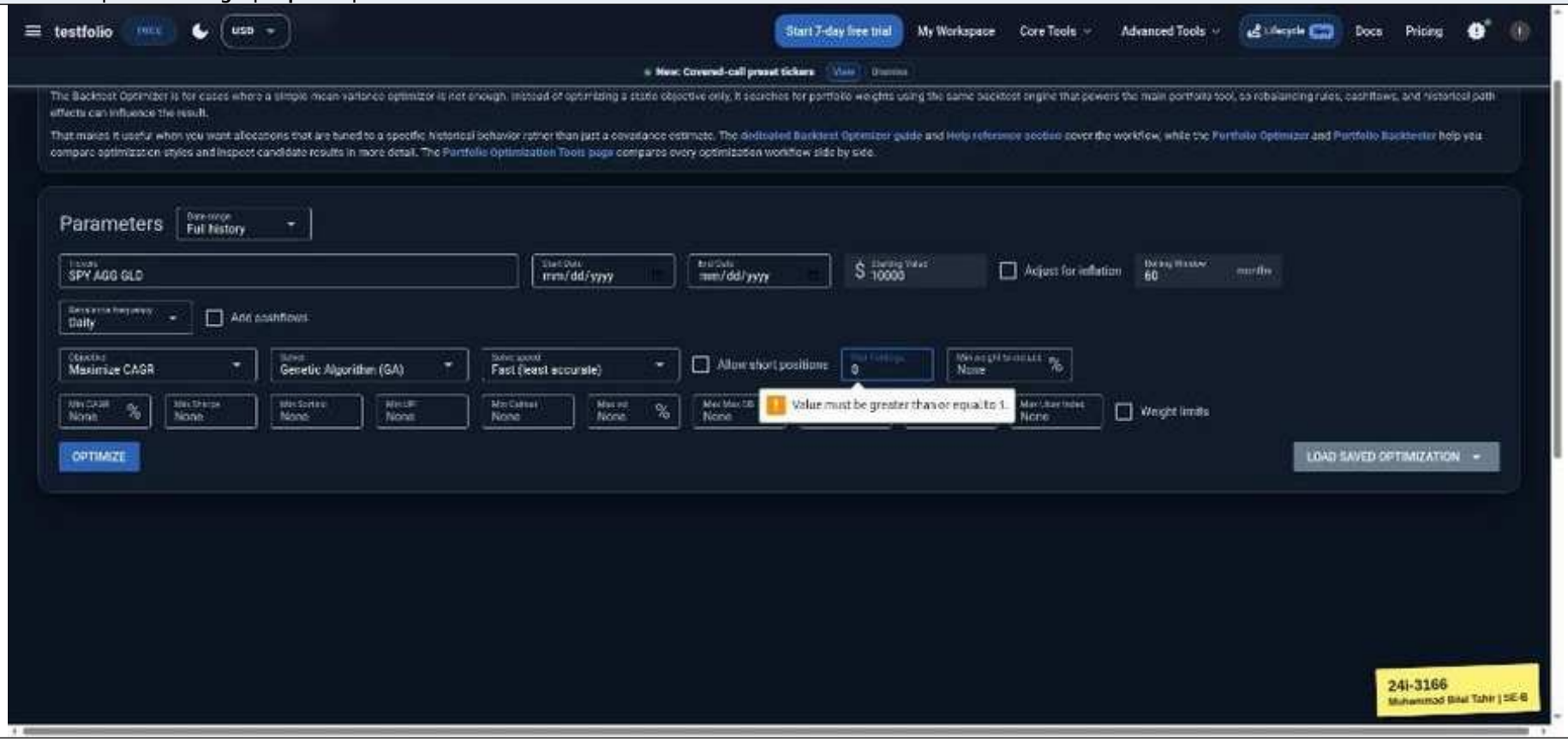


TC-BVA-066 | Rolling Window | input: 1 | result: Pass

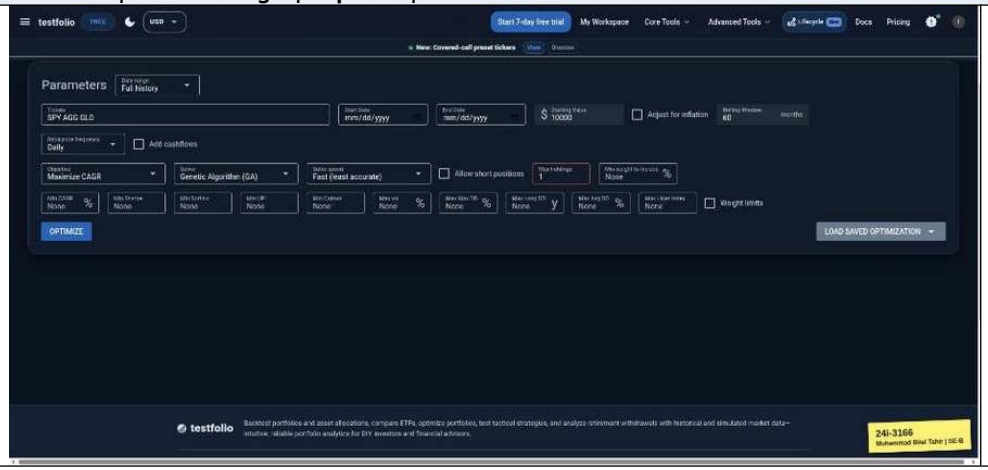


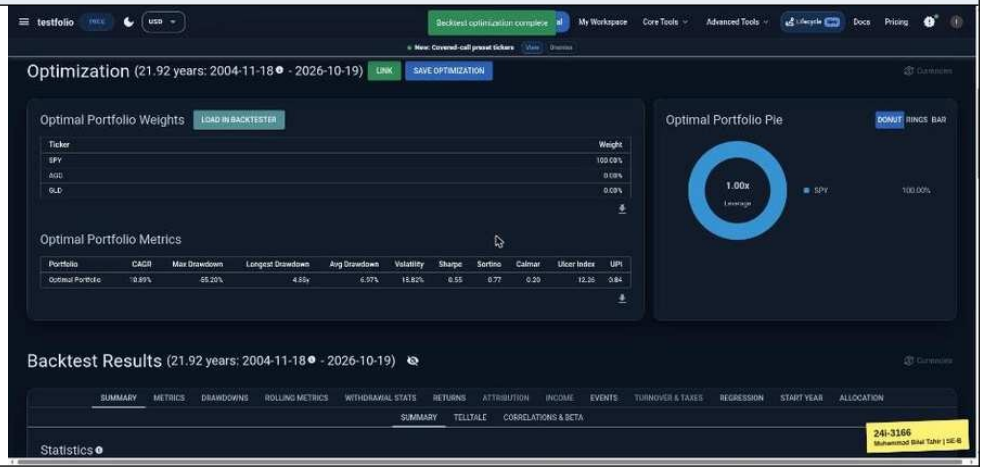


TC-BVA-067 | Max holdings | input: 0 | result: Pass

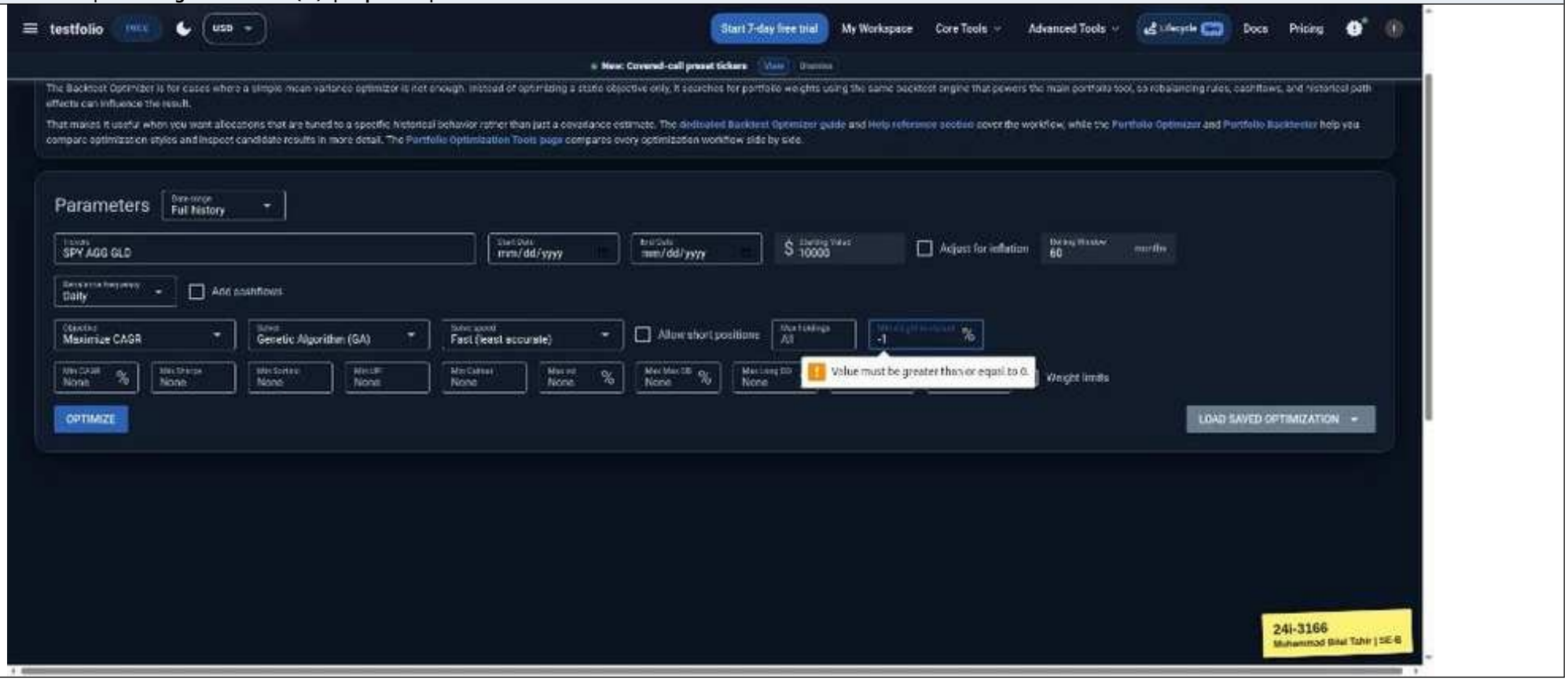


TC-BVA-068 | Max holdings | input: 1 | result: Pass

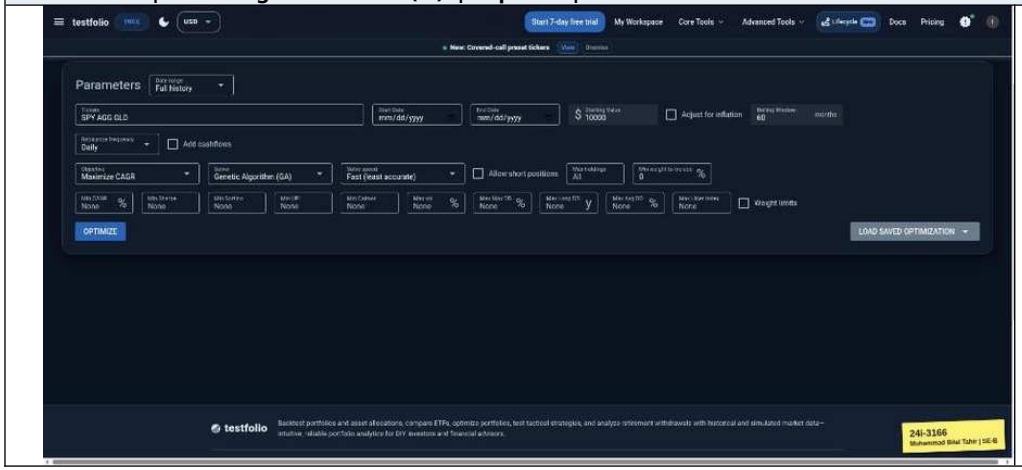


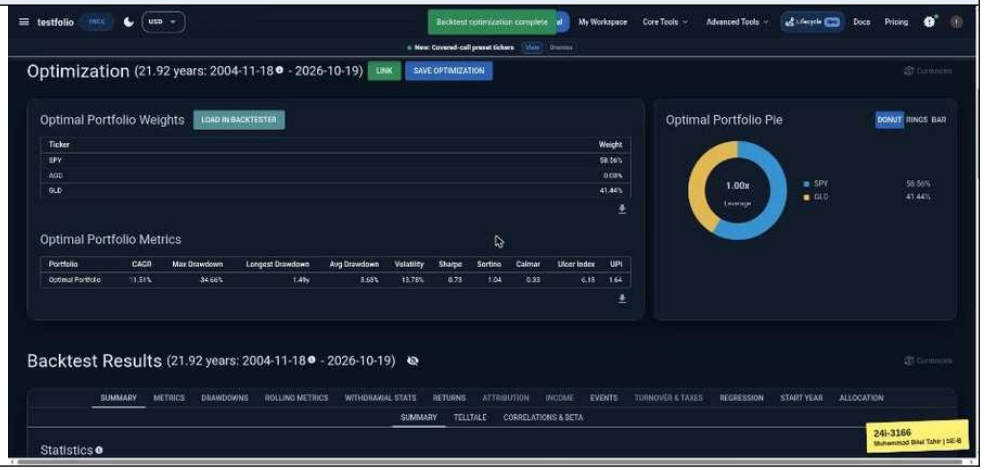


TC-BVA-069 | Min weight to include (%) | input: -1 | result: Pass

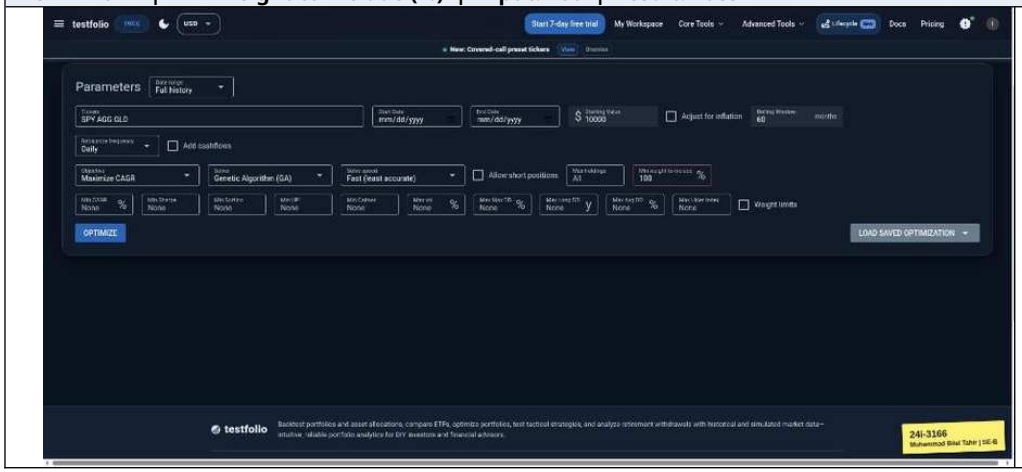


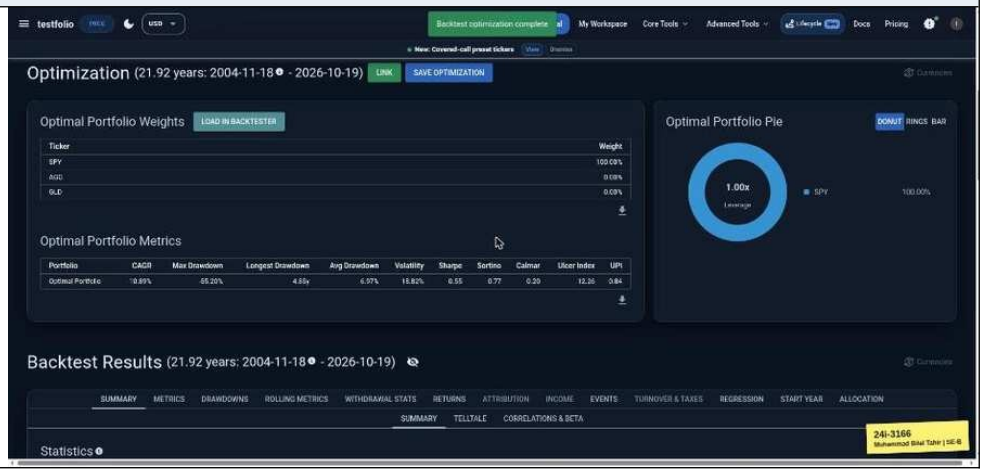
TC-BVA-070 | Min weight to include (%) | input: 0 | result: Pass



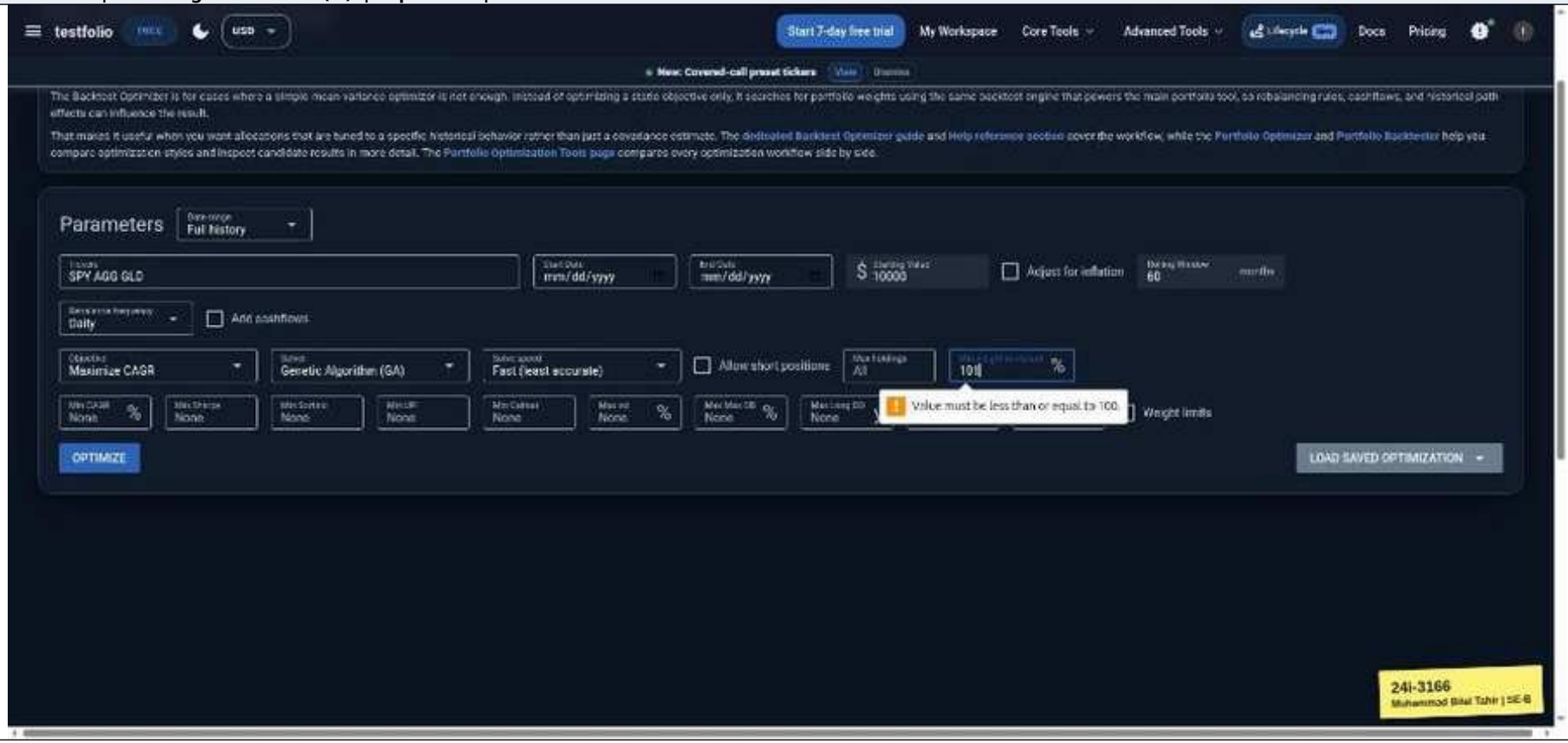


TC-BVA-071 | Min weight to include (%) | input: 100 | result: Pass

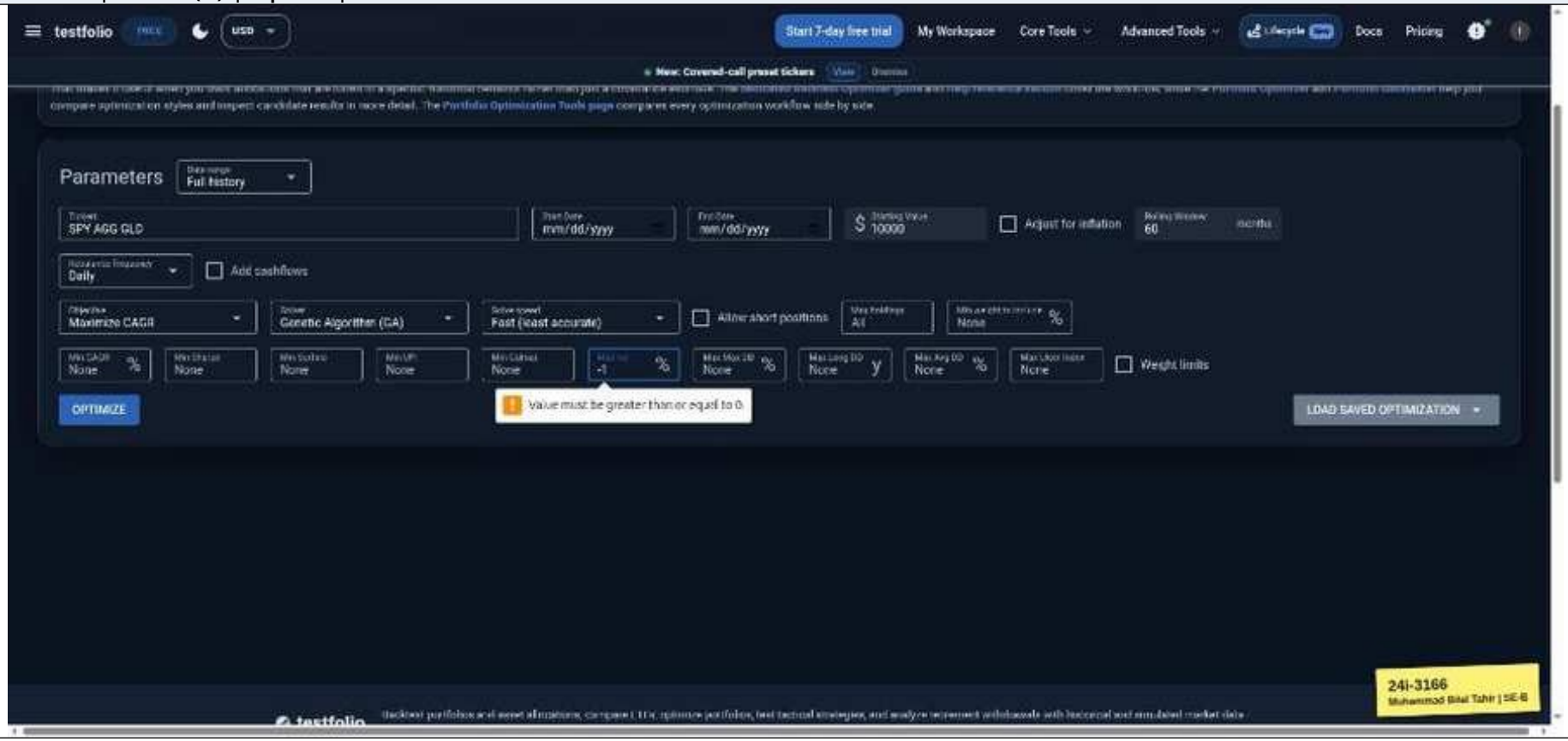




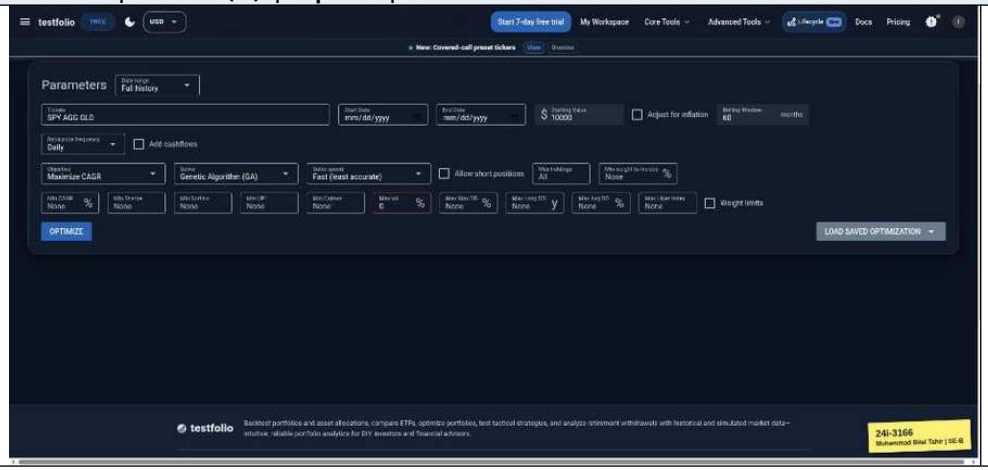
TC-BVA-072 | Min weight to include (%) | input: 101 | result: Pass

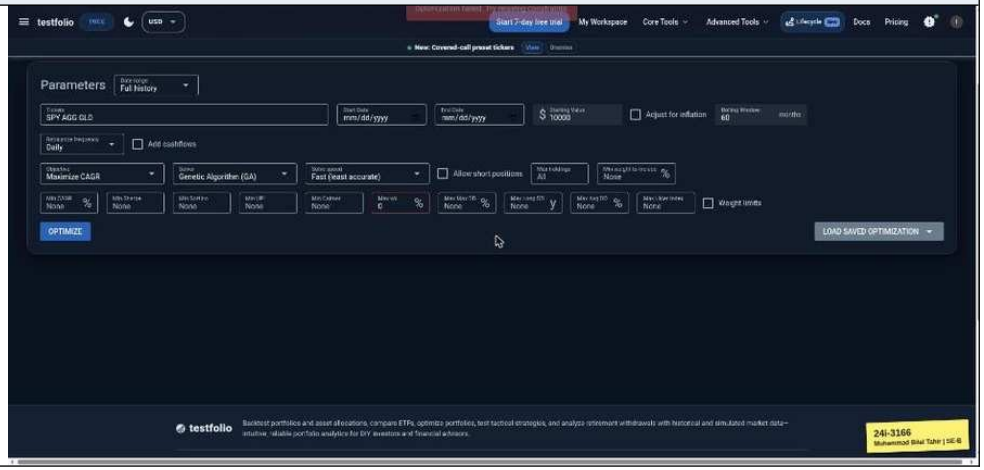


TC-BVA-073 | Max vol (%) | input: -1 | result: Pass

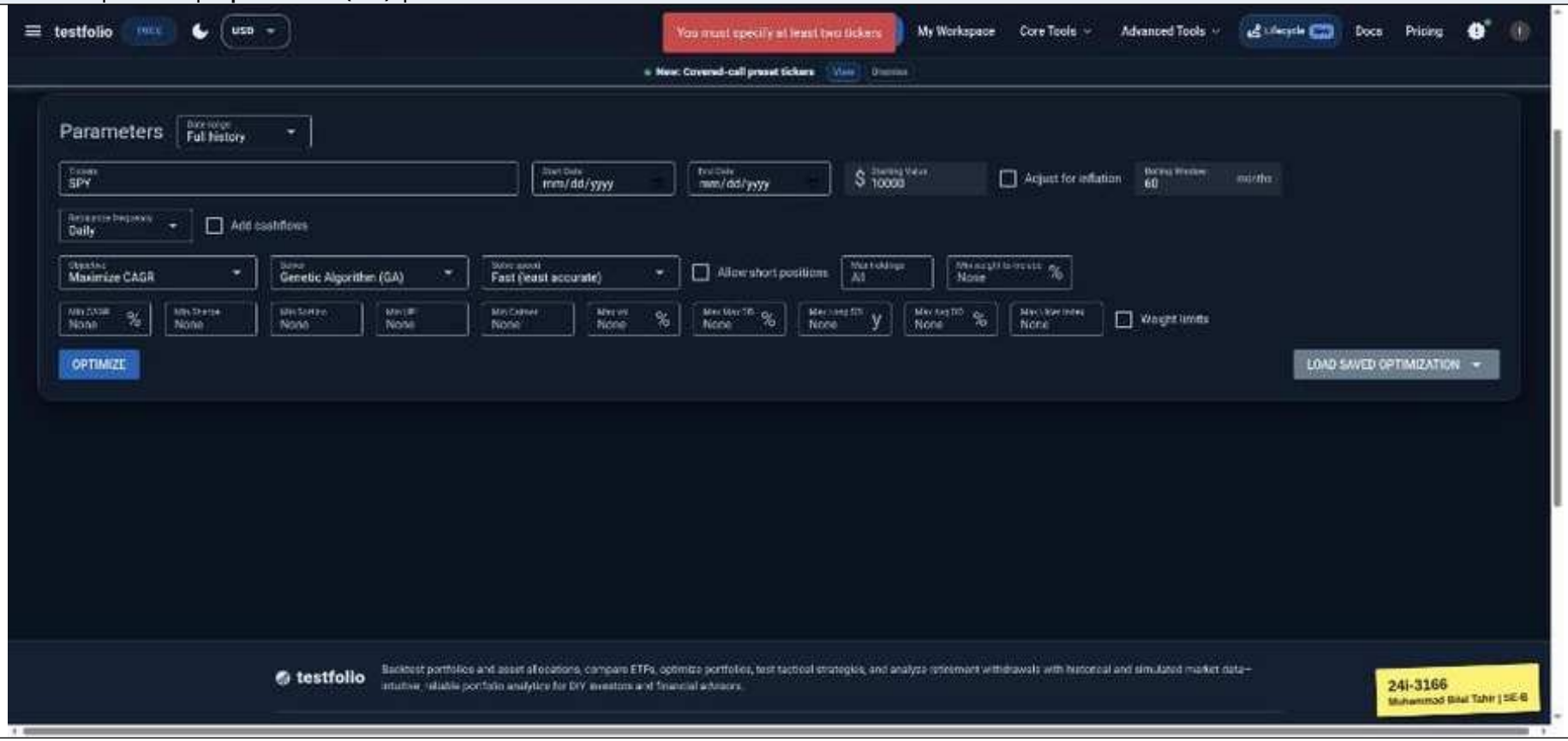


TC-BVA-074 | Max vol (%) | input: 0 | result: Pass

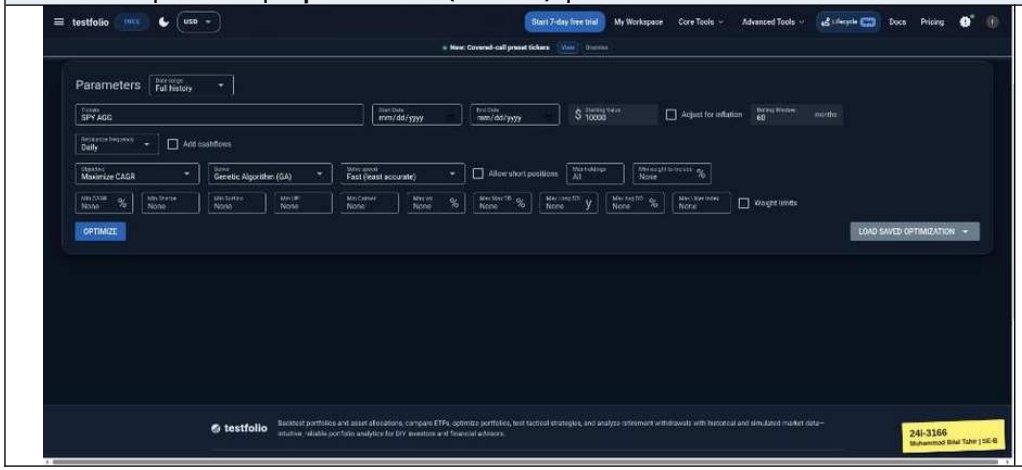


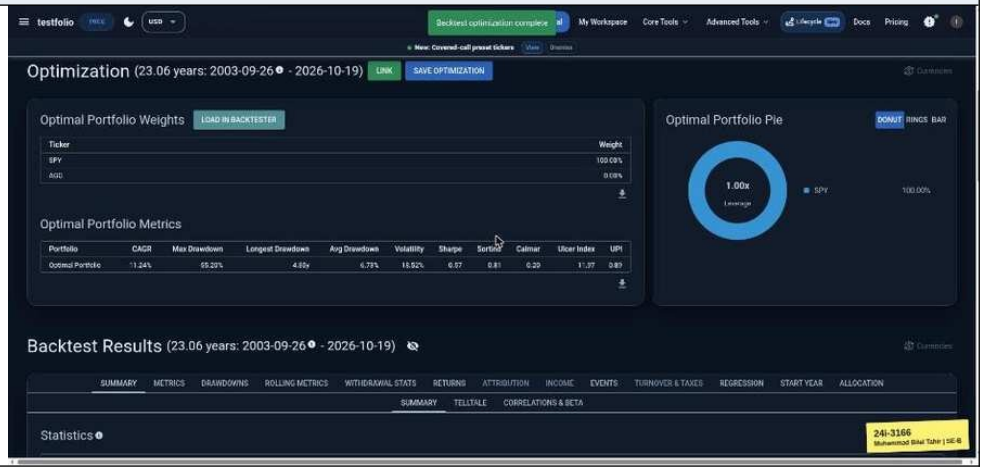


TC-BVA-075 | Tickers | input: 1 ticker (SPY) | result: Pass



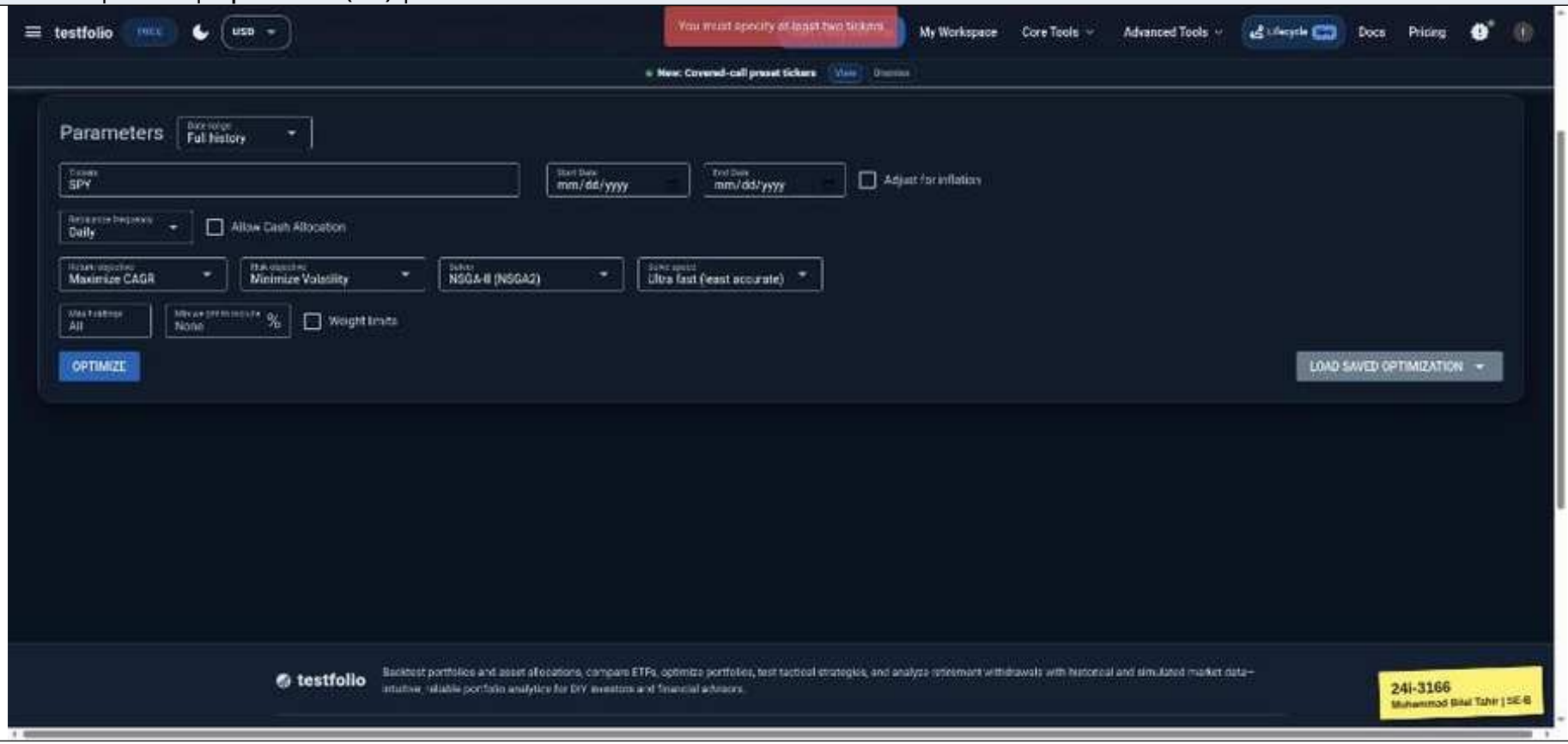
TC-BVA-076 | Tickers | input: 2 tickers (SPY AGG) | result: Pass



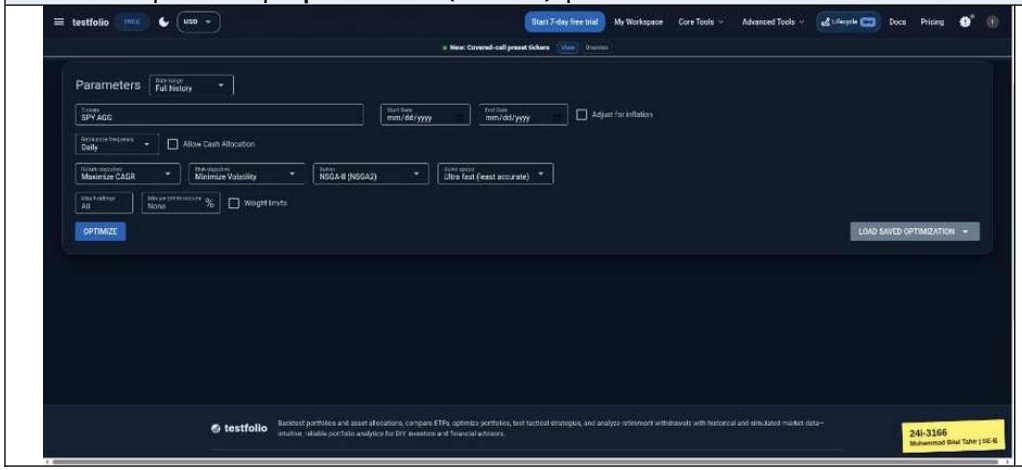


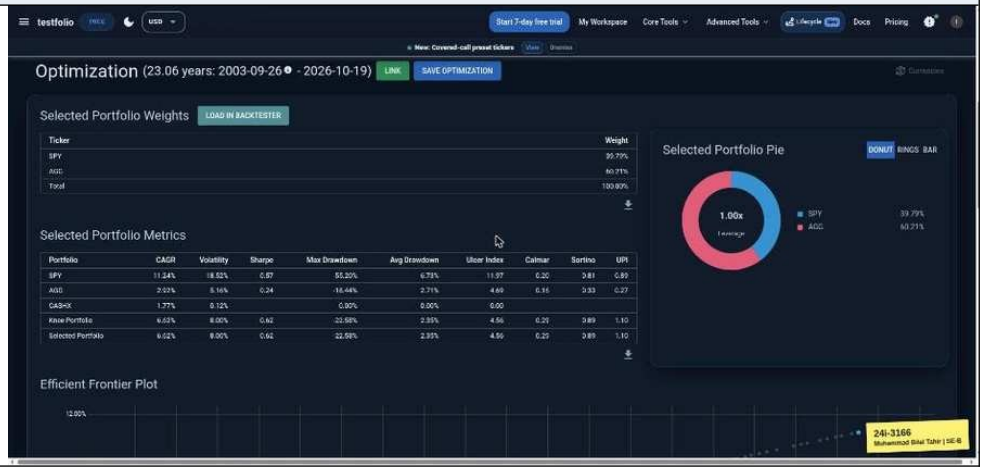
Efficient Frontier (testfol.io/efficient-frontier)

TC-BVA-077 | Tickers | input: 1 ticker (SPY) | result: Pass

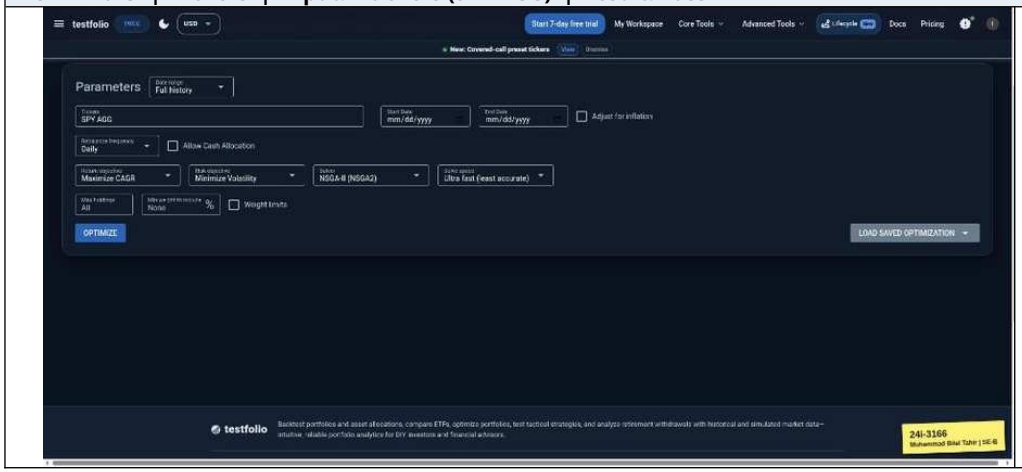


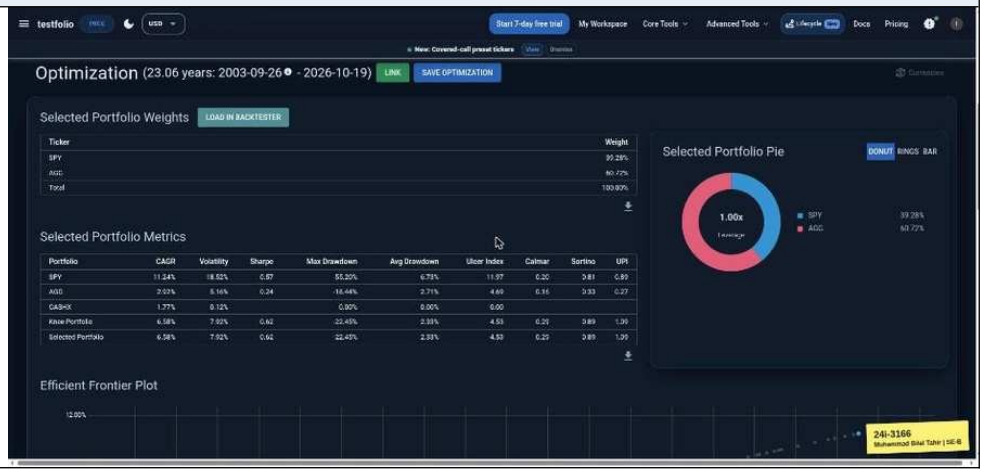
TC-BVA-078 | Tickers | input: 2 tickers (SPY AGG) | result: Pass



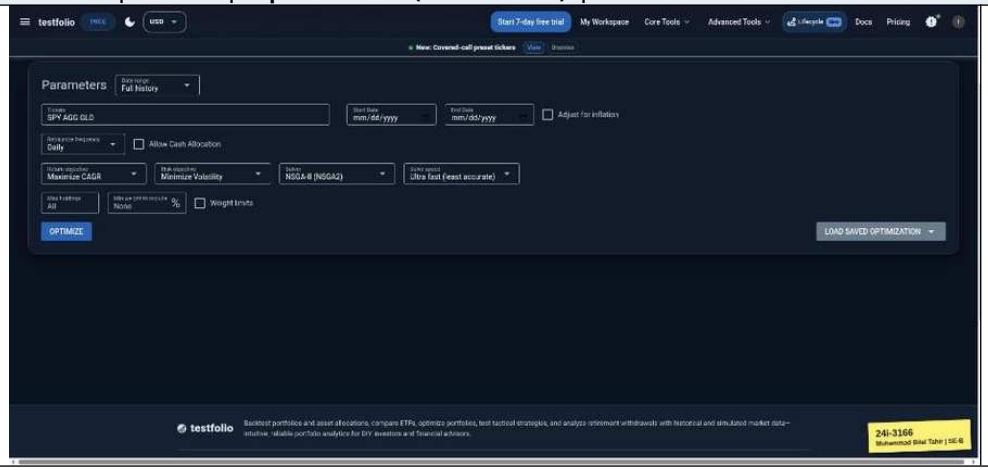


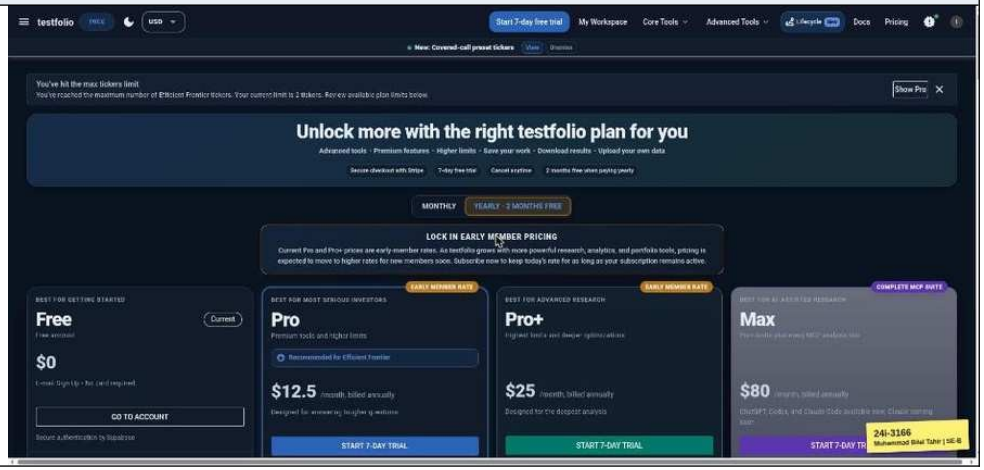
TC-BVA-079 | Tickers | input: 2 tickers (SPY AGG) | result: Pass



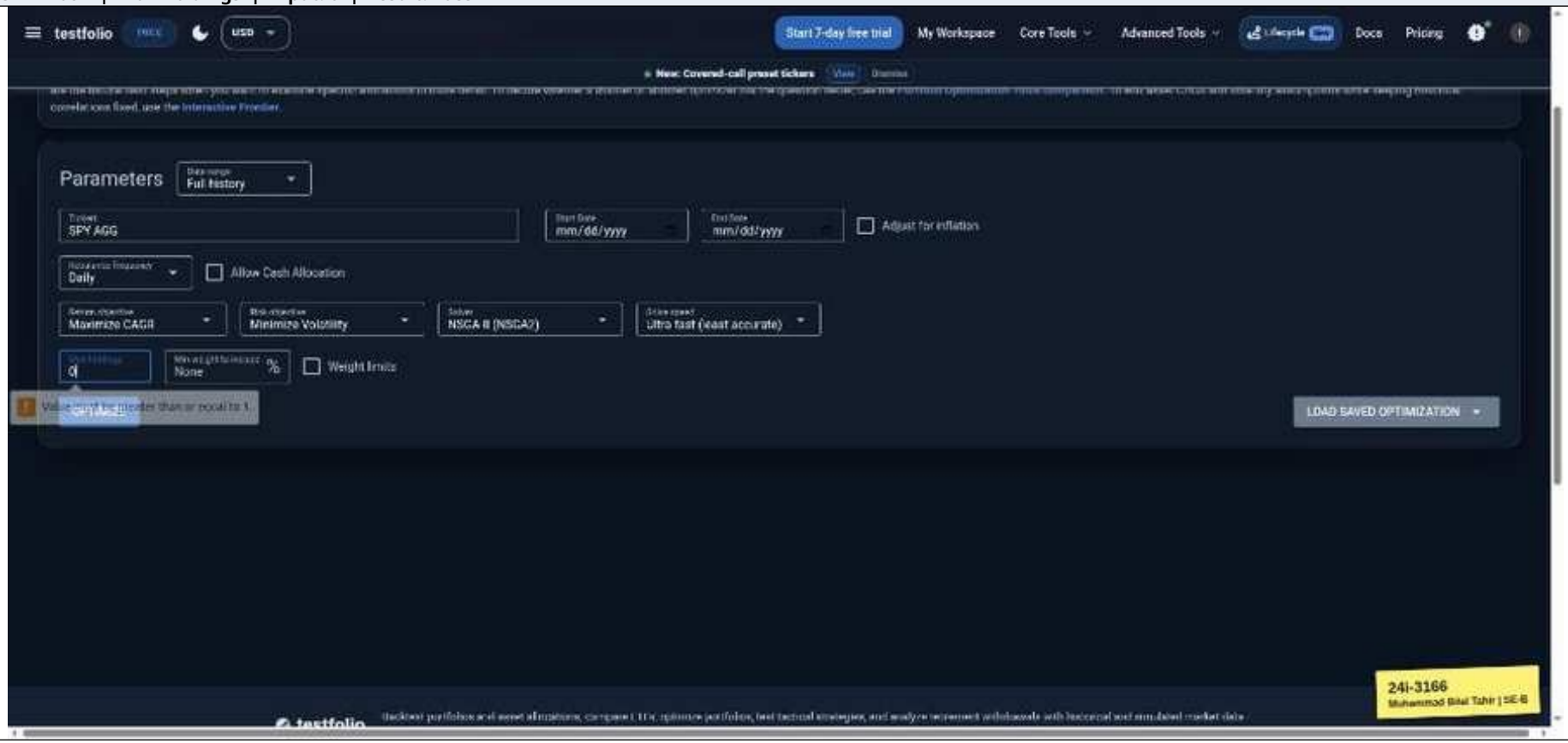


TC-BVA-080 | Tickers | input: 3 tickers (SPY AGG GLD) | result: Pass

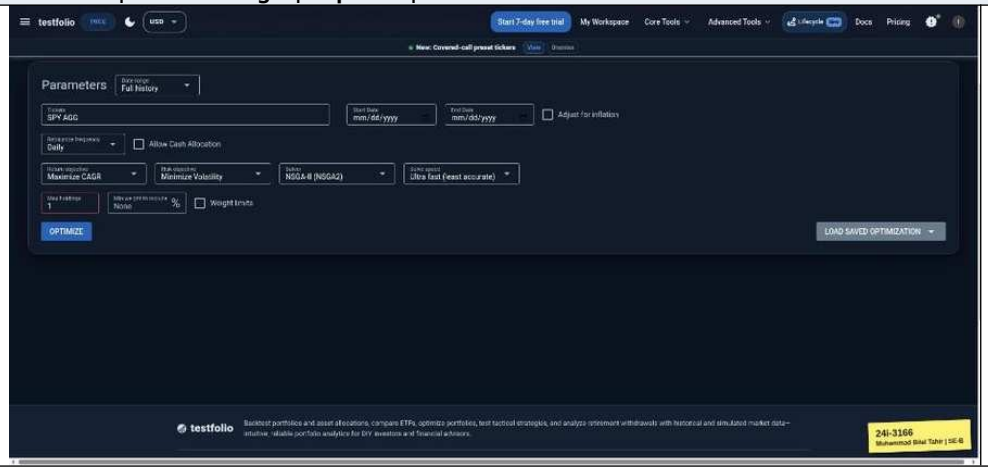


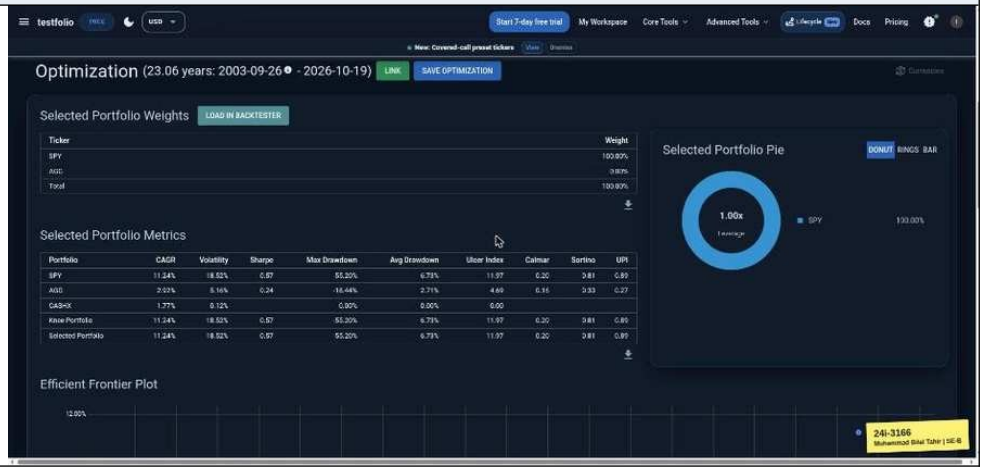


TC-BVA-081 | Max holdings | input: 0 | result: Pass

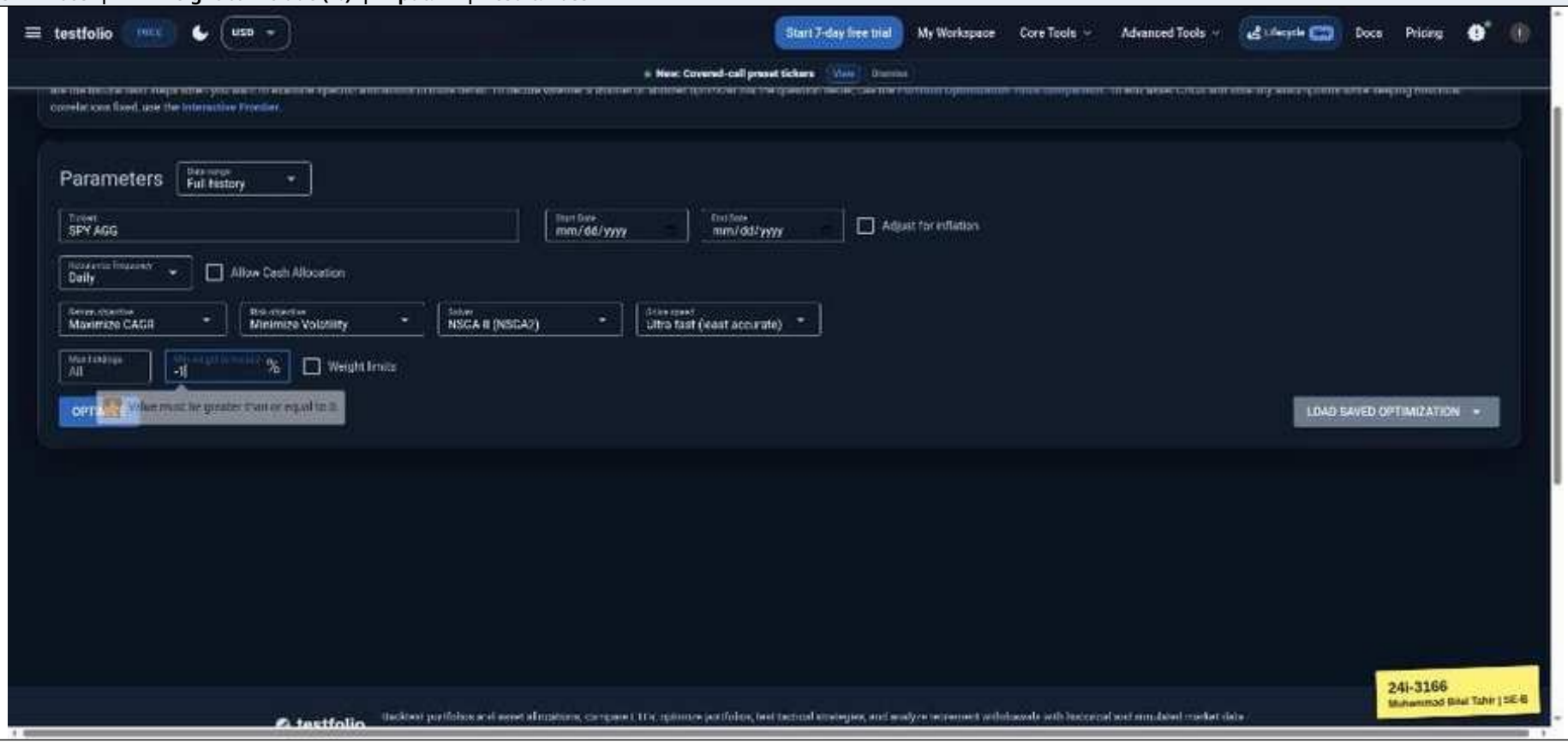


TC-BVA-082 | Max holdings | input: 1 | result: Pass

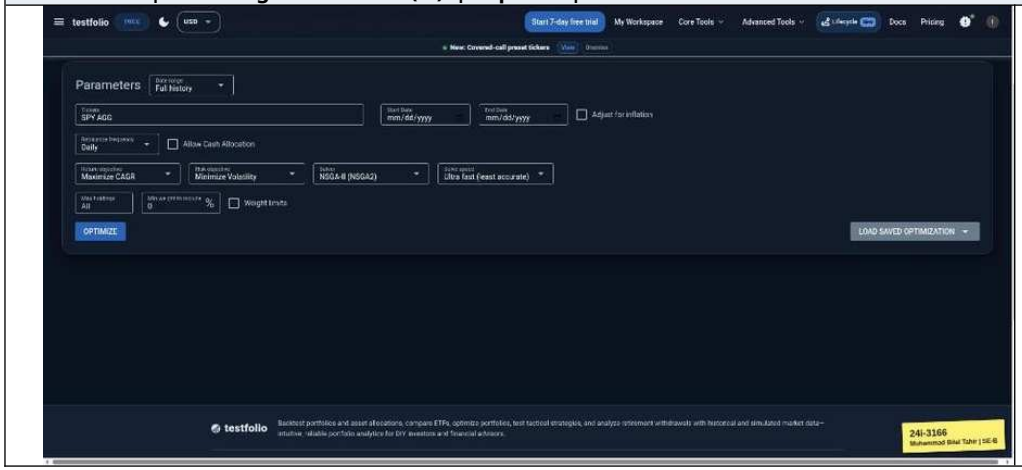


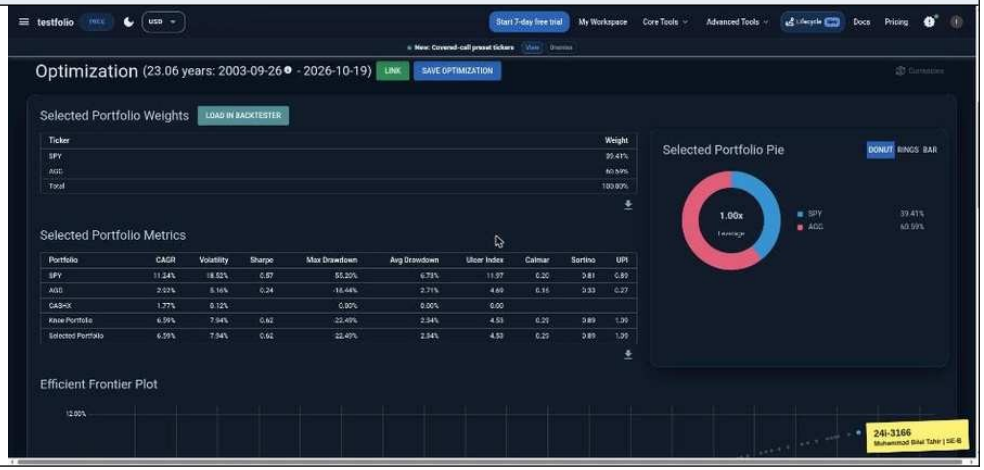


TC-BVA-083 | Min weight to include (%) | input: -1 | result: Pass

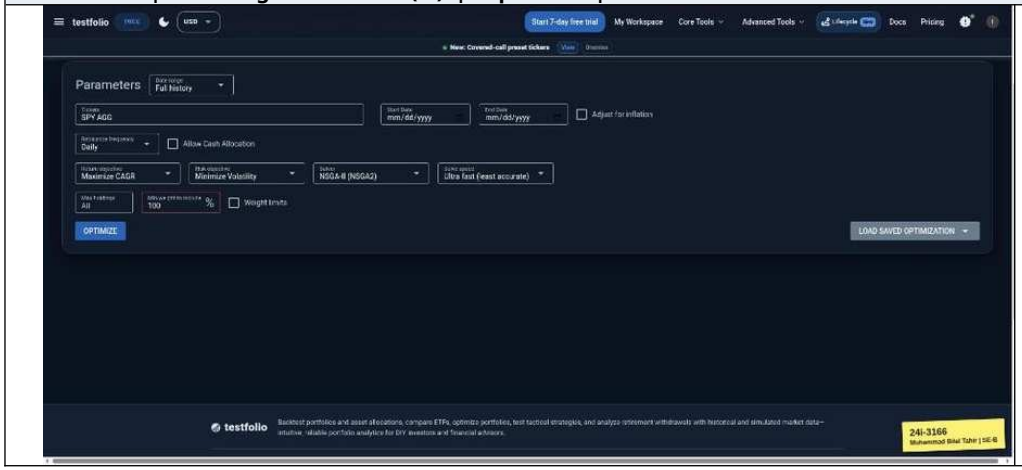


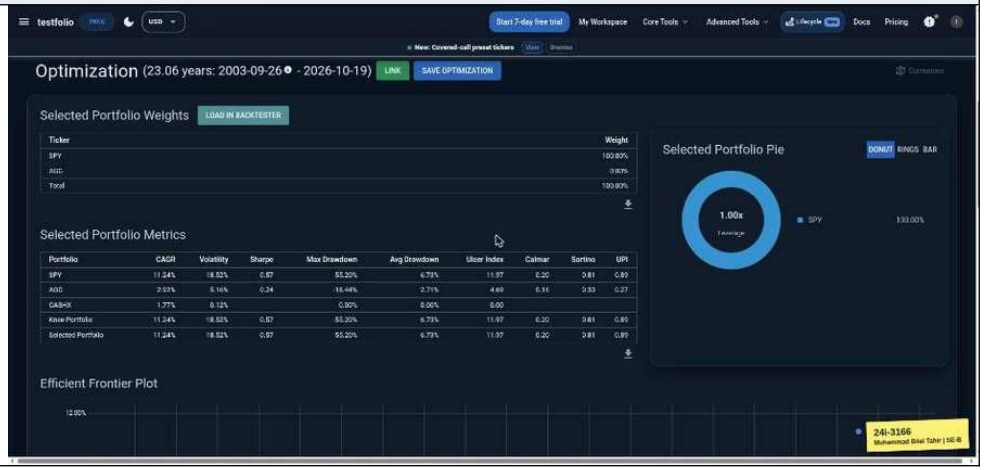
TC-BVA-084 | Min weight to include (%) | input: 0 | result: Pass



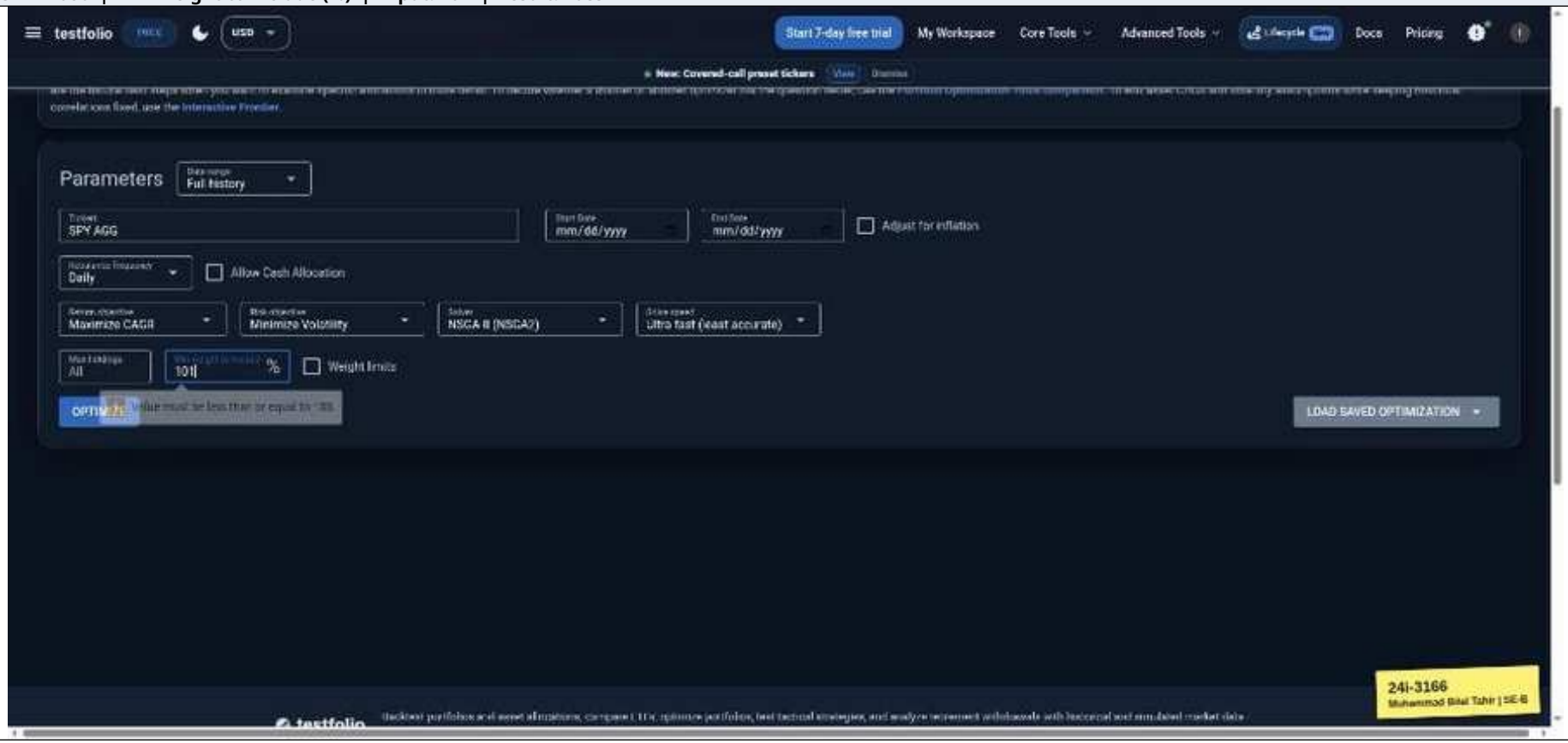


TC-BVA-085 | Min weight to include (%) | input: 100 | result: Pass



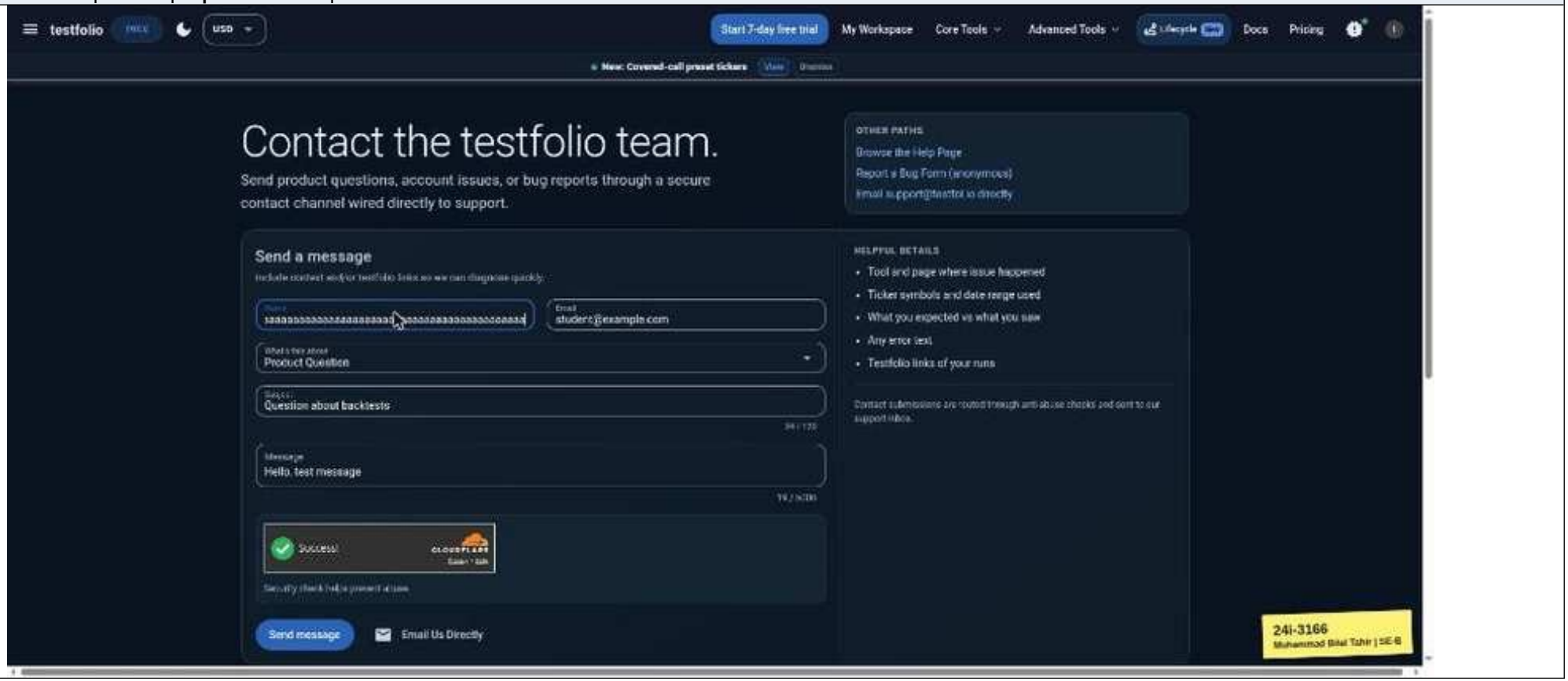


TC-BVA-086 | Min weight to include (%) | input: 101 | result: Pass

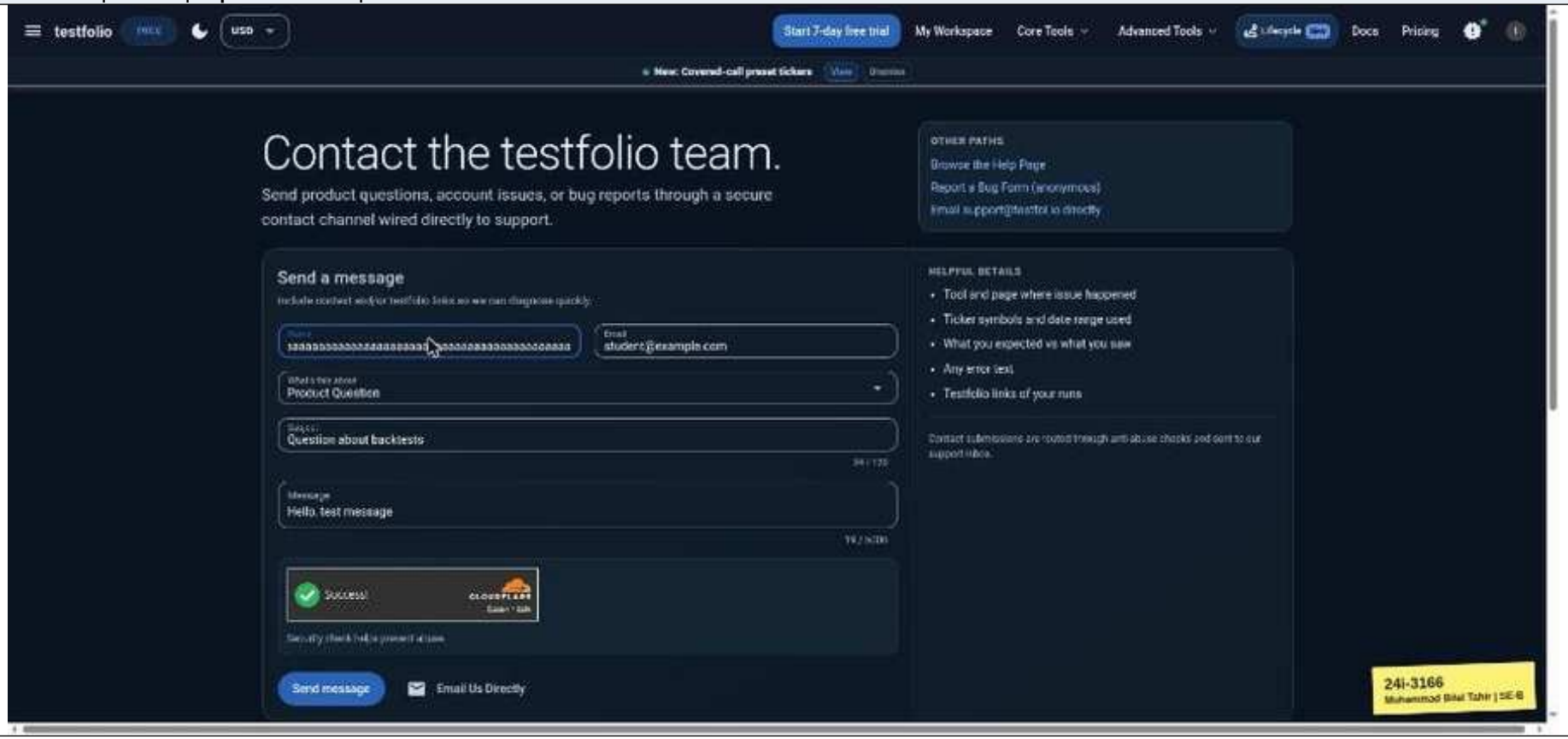


Contact Form (testfol.io/contact)

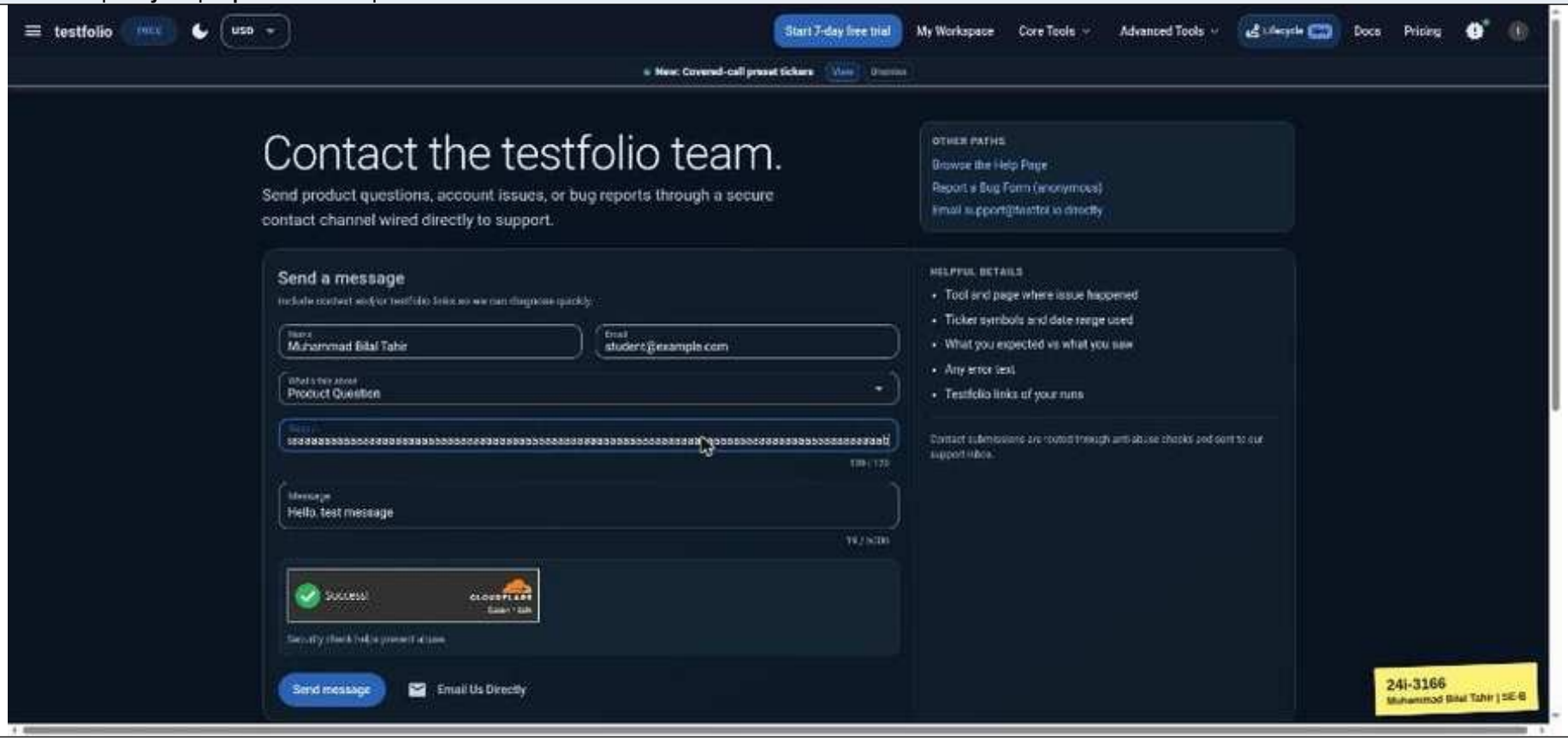
TC-BVA-087 | Name | input: 120 chars | result: Pass



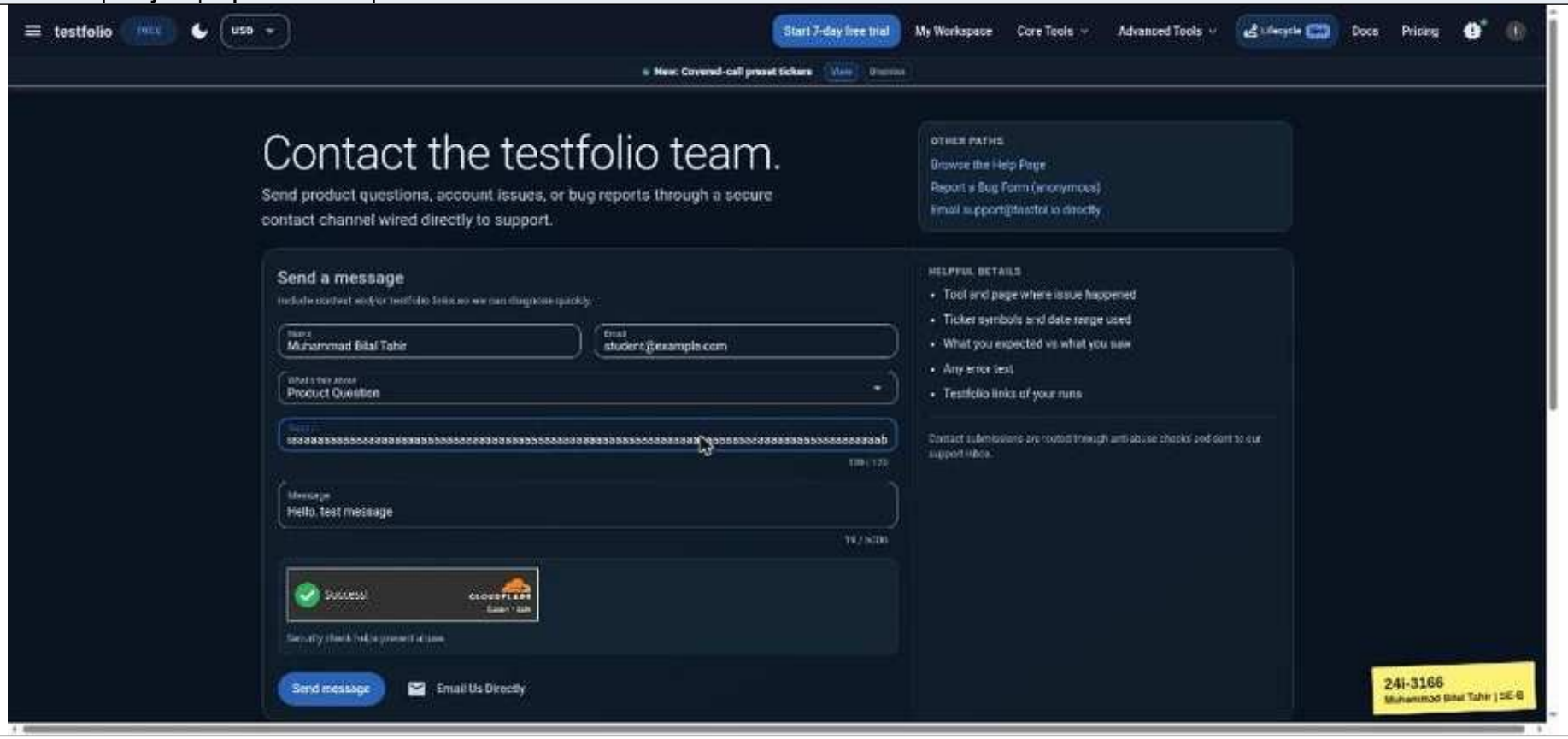
TC-BVA-088 | Name | input: 121 chars | result: Pass



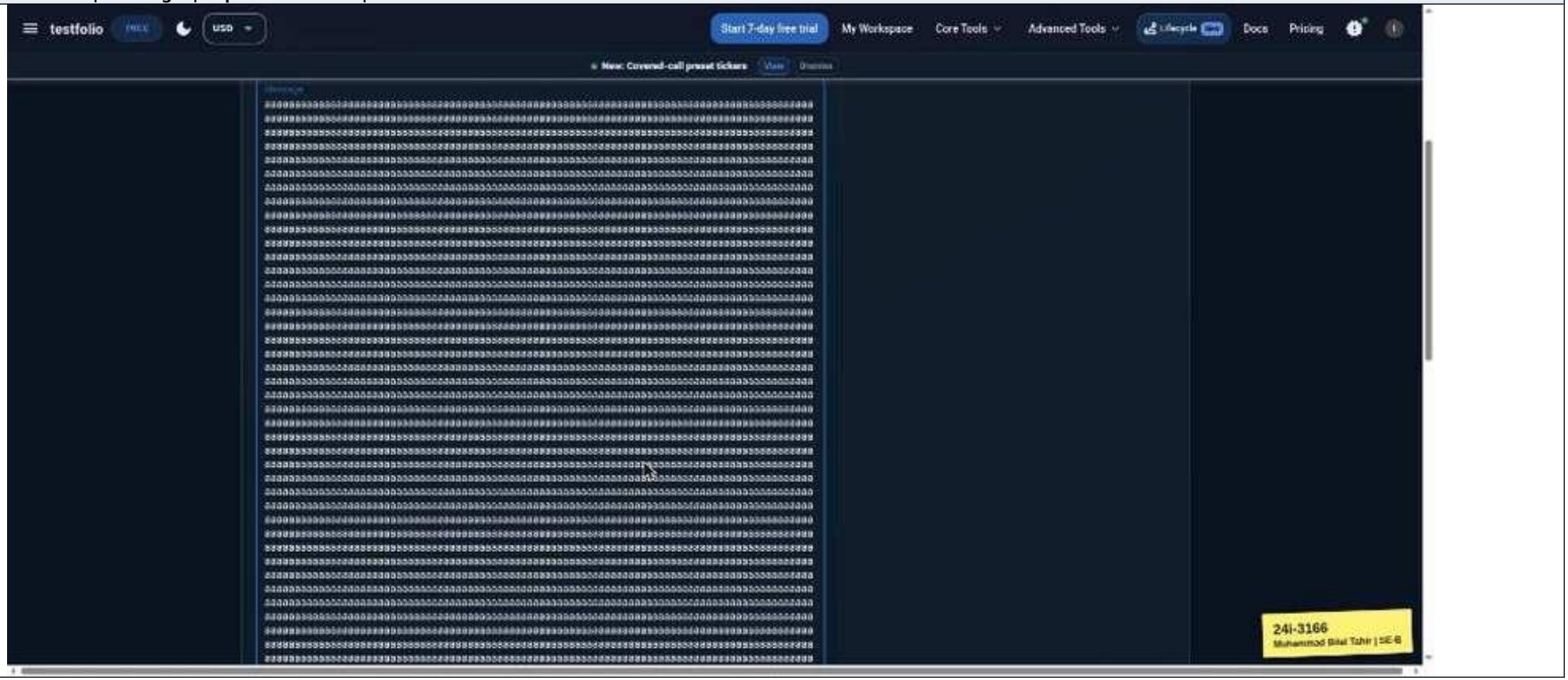
TC-BVA-089 | Subject | input: 120 chars | result: Pass



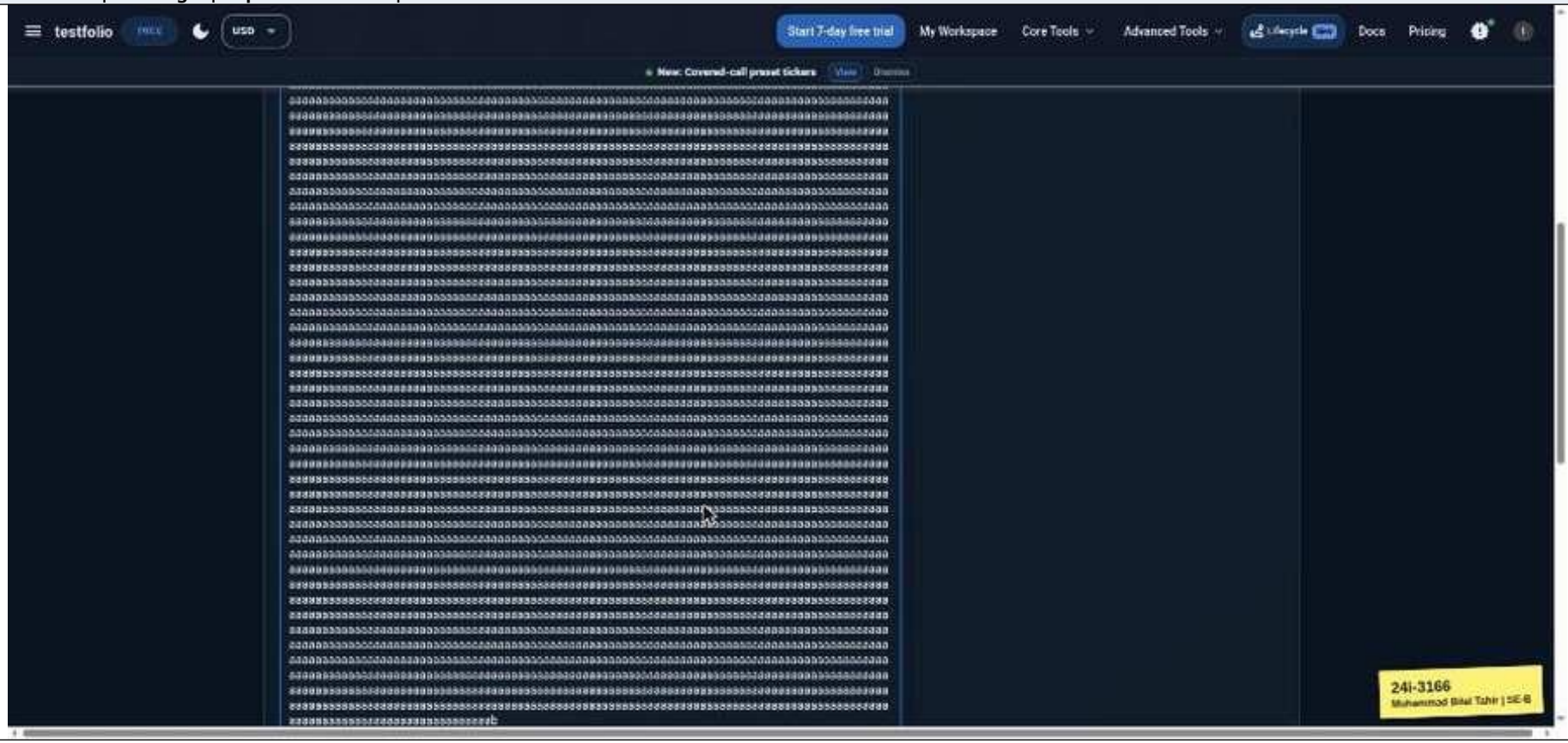
TC-BVA-090 | Subject | input: 121 chars | result: Pass



TC-BVA-091 | Message | input: 5000 chars | result: Pass

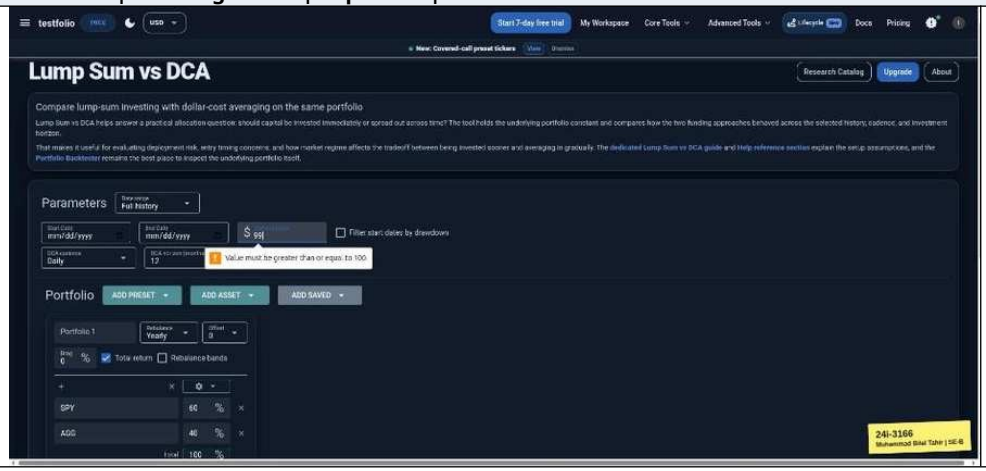


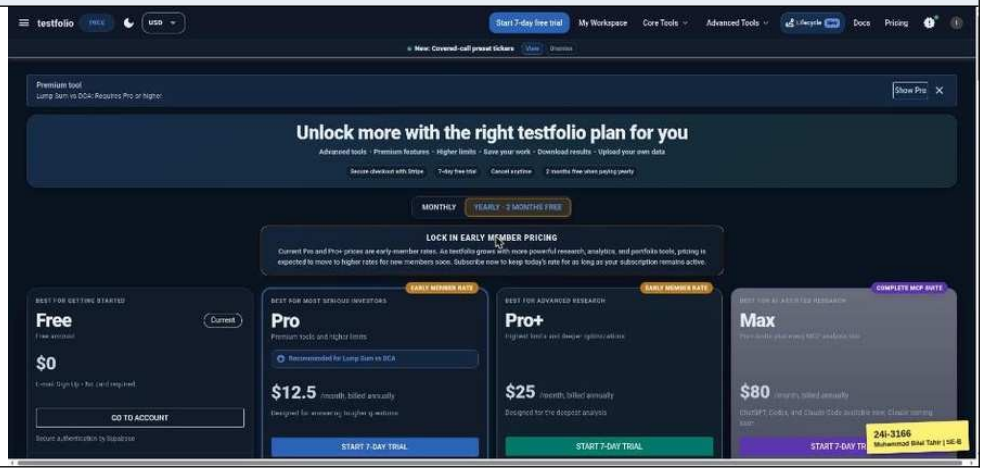
TC-BVA-092 | Message | input: 5001 chars | result: Pass



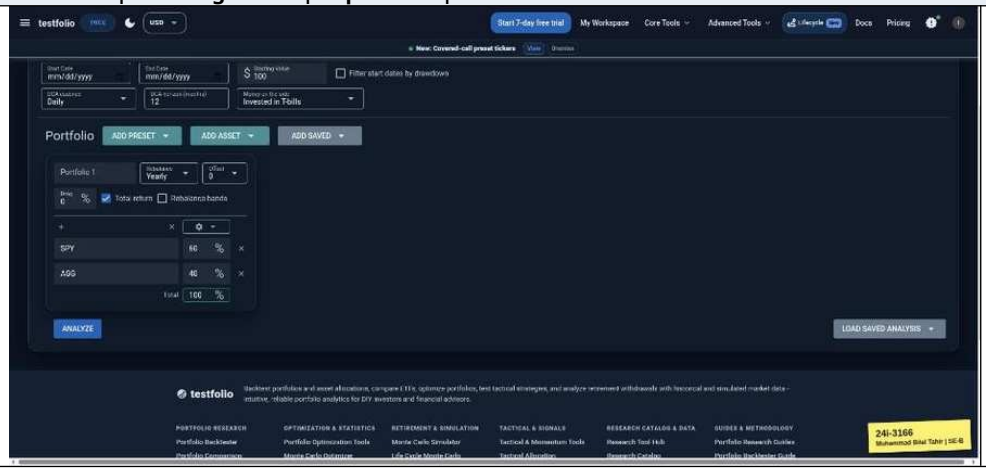
Lump Sum vs DCA (testfol.io/lumpsum-vs-dca)

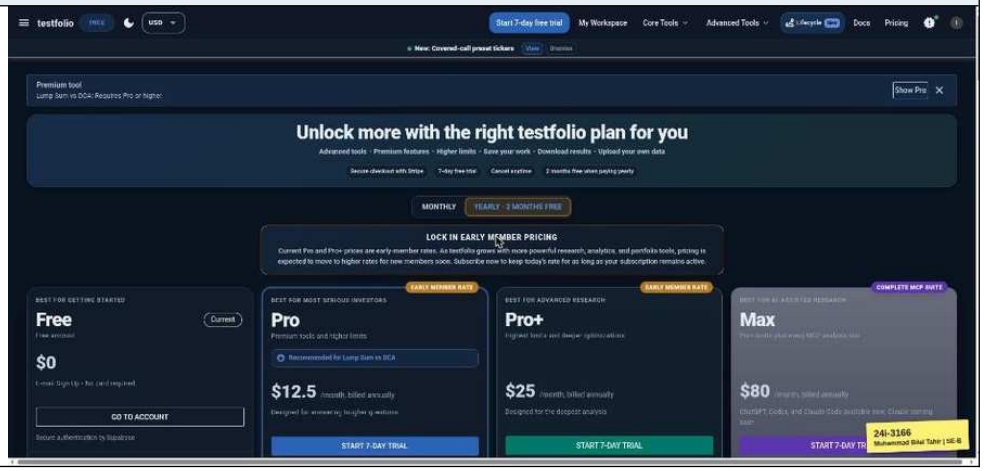
TC-BVA-093 | Starting Value | input: 99 | result: Blocked



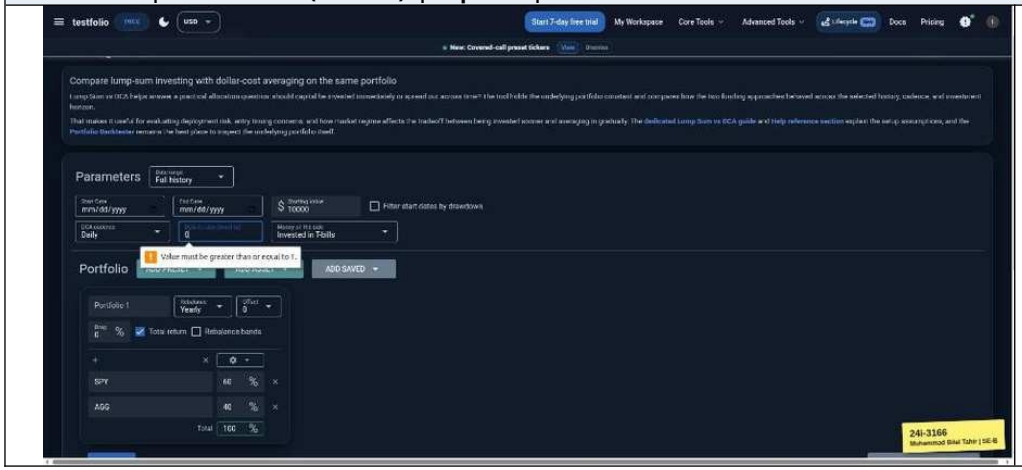


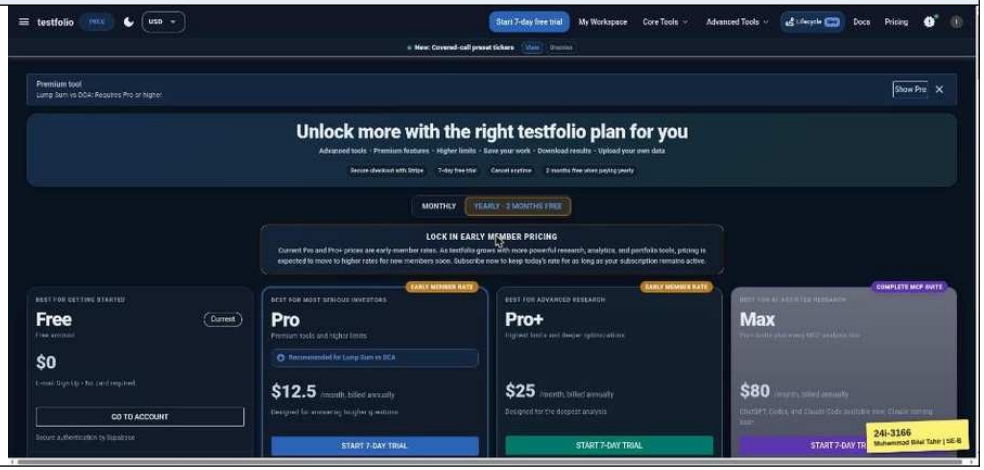
TC-BVA-094 | Starting Value | input: 100 | result: Blocked



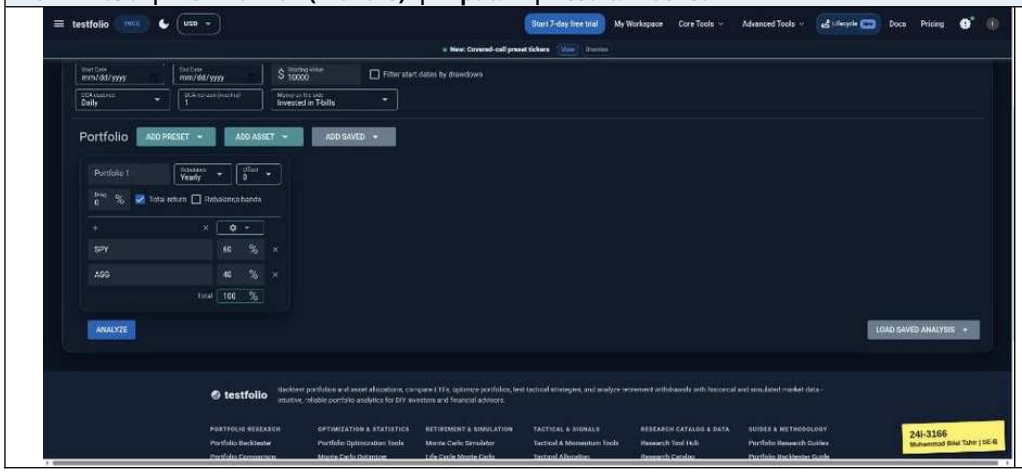


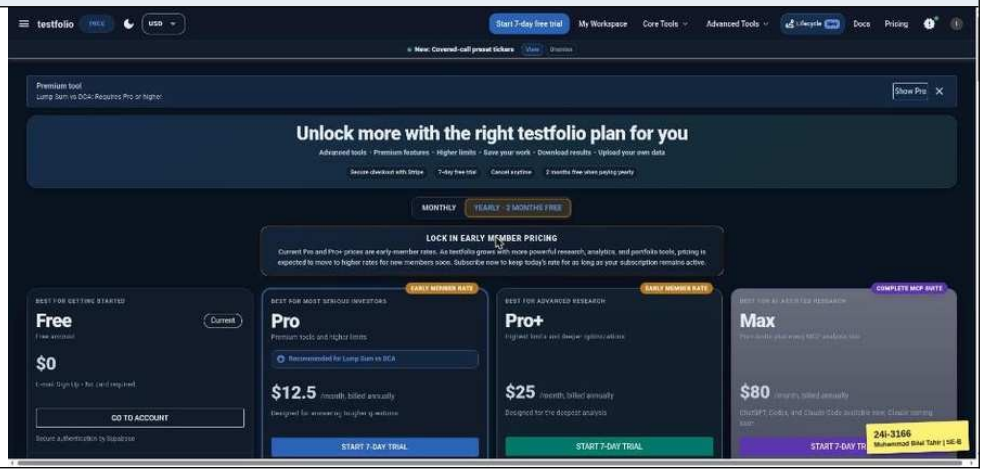
TC-BVA-095 | DCA horizon (months) | input: 0 | result: Blocked



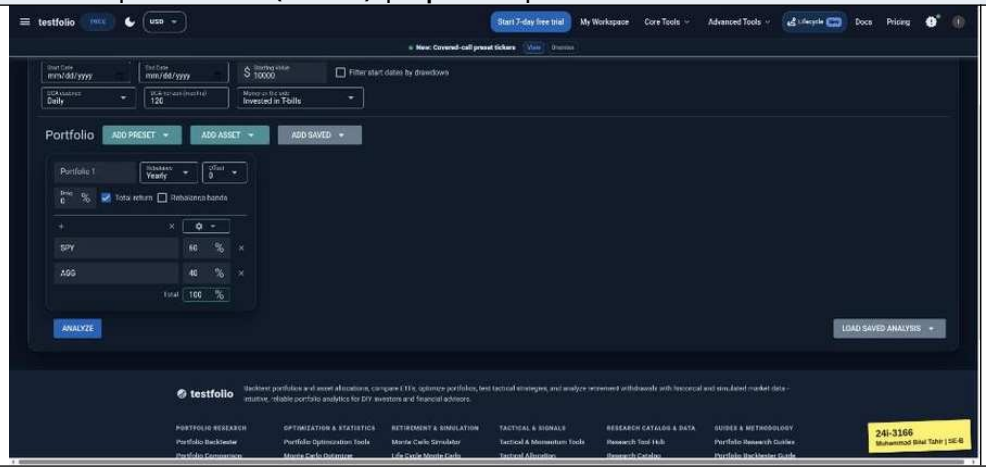


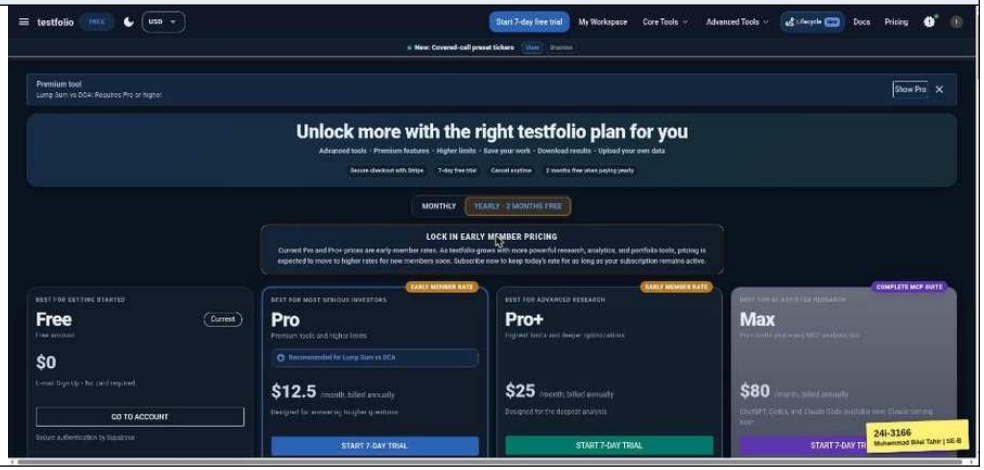
TC-BVA-096 | DCA horizon (months) | input: 1 | result: Blocked



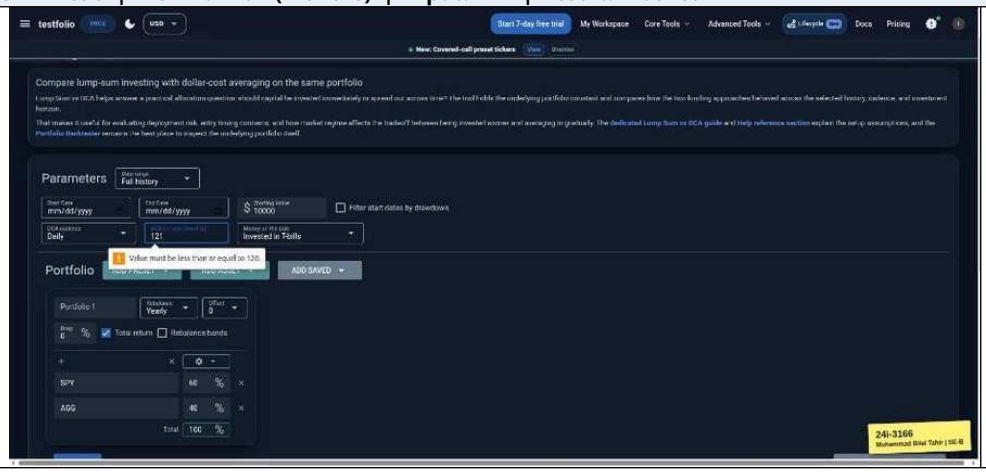


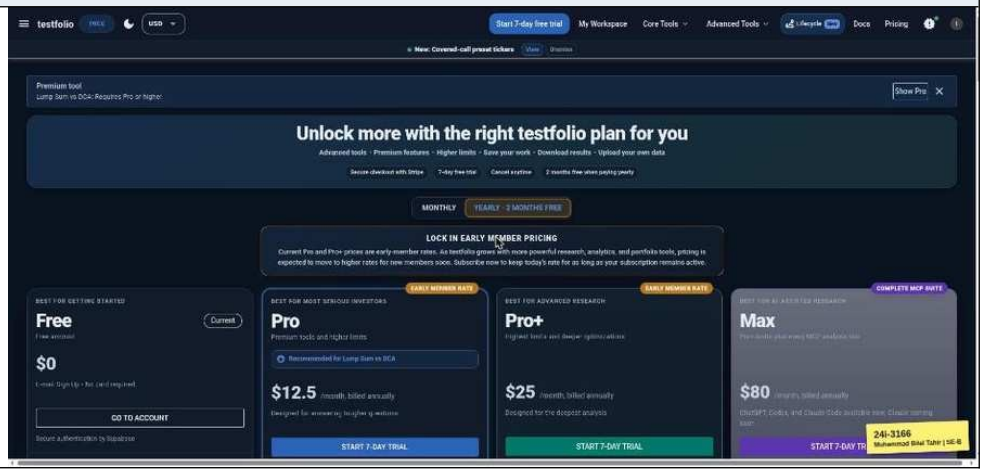
TC-BVA-097 | DCA horizon (months) | input: 120 | result: Blocked





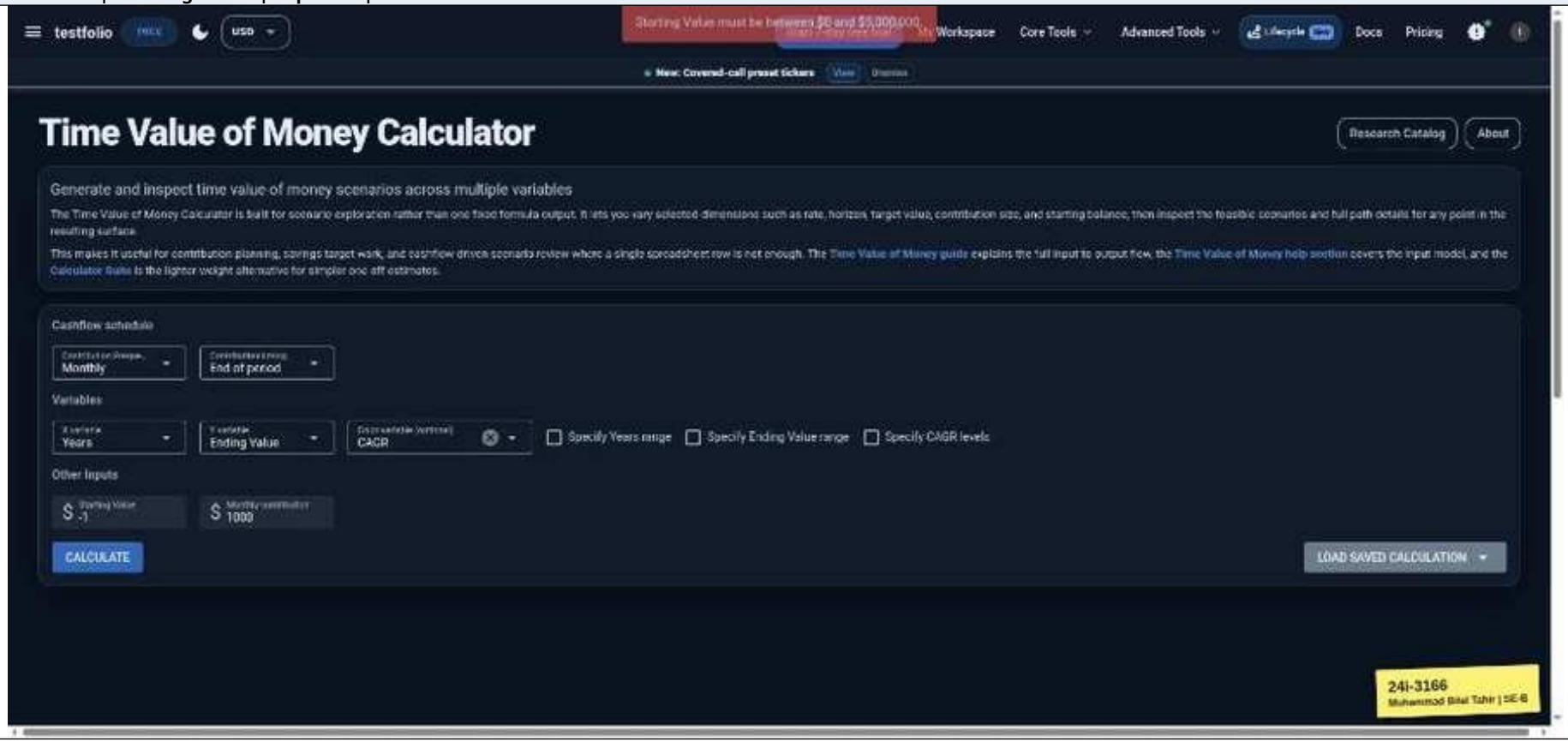
TC-BVA-098 | DCA horizon (months) | input: 121 | result: Blocked



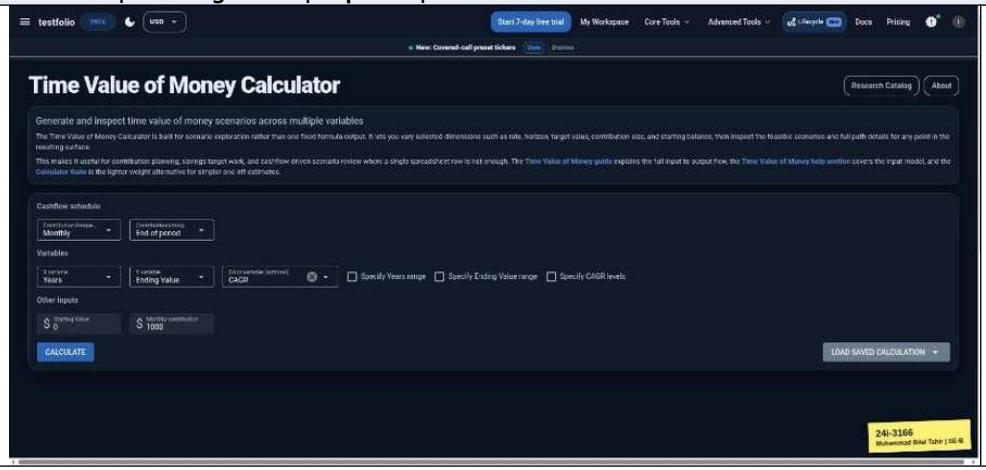


Time Value of Money Calculator (testfol.io/time-value-of-money)

TC-BVA-099 | Starting Value | input: -1 | result: Pass

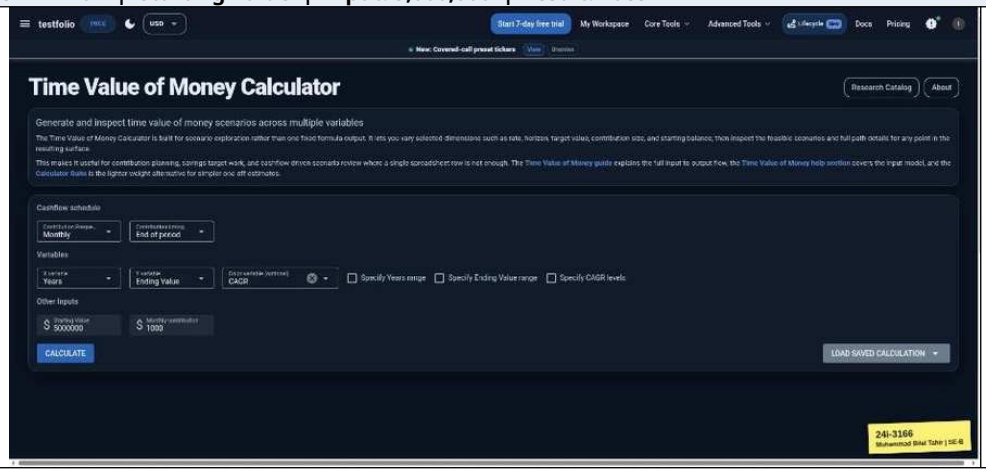


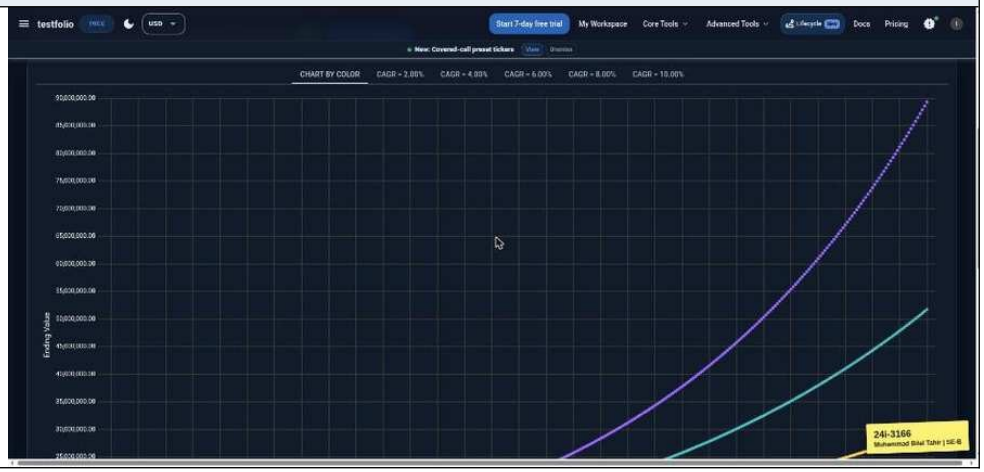
TC-BVA-100 | Starting Value | input: 0 | result: Pass



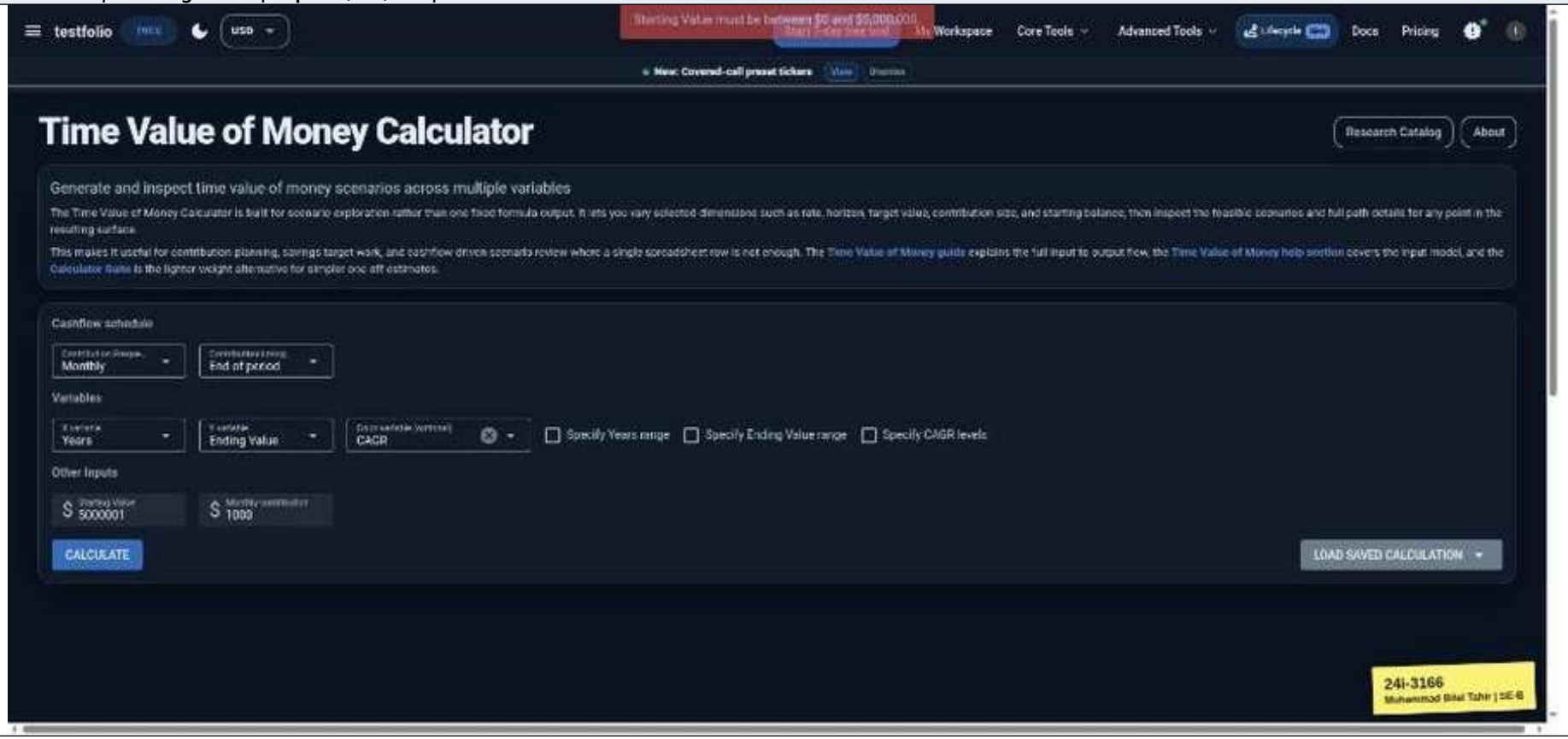


TC-BVA-101 | Starting Value | input: 5,000,000 | result: Pass

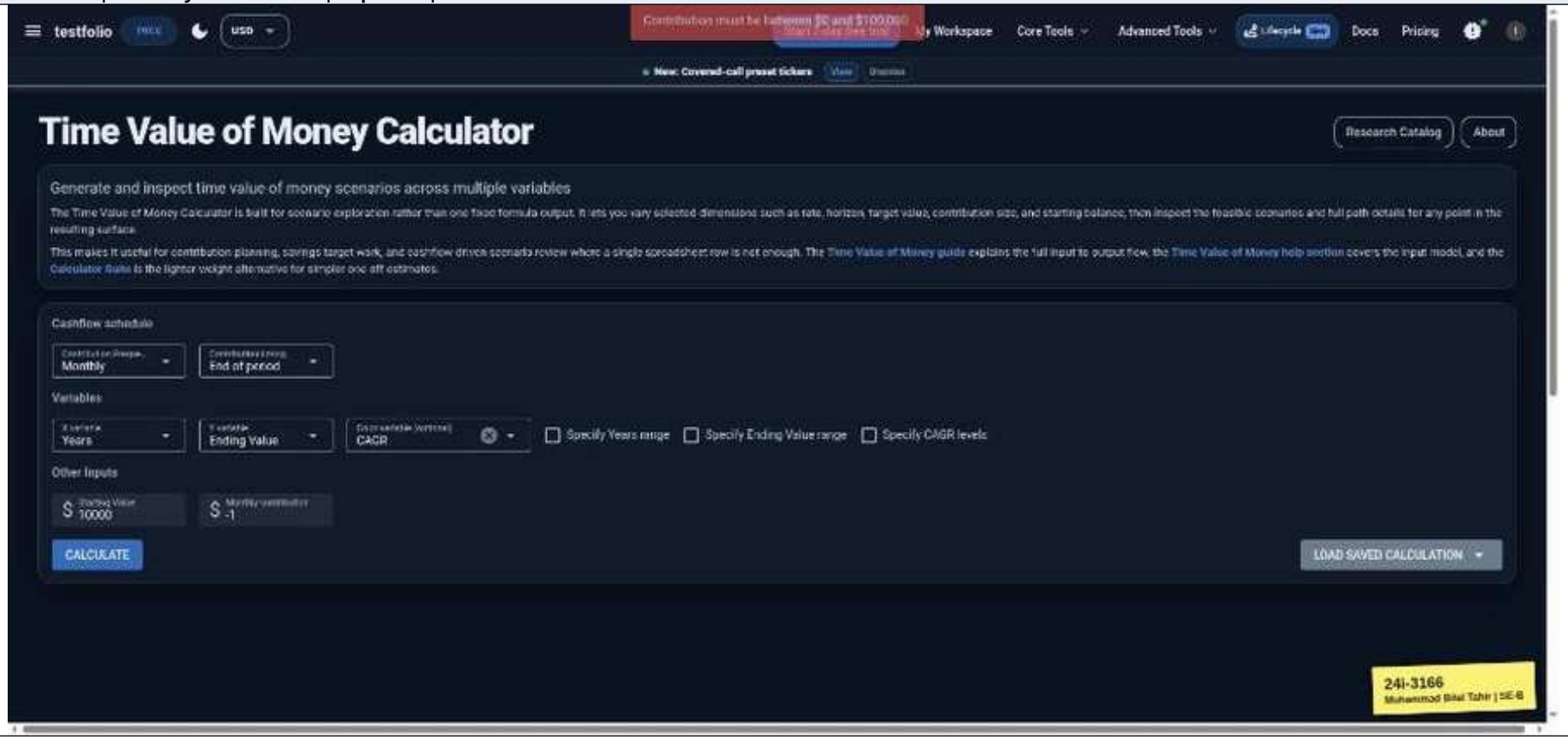




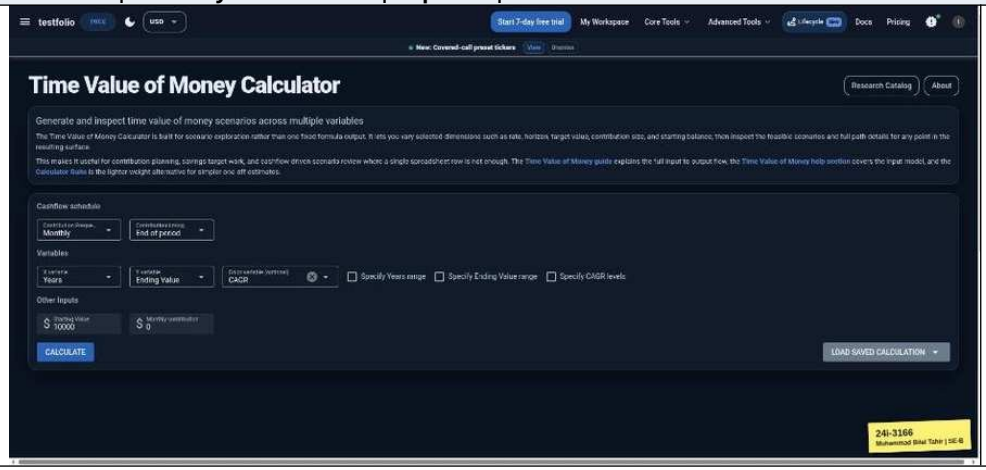
TC-BVA-102 | Starting Value | input: 5,000,001 | result: Pass



TC-BVA-103 | Monthly contribution | input: -1 | result: Pass

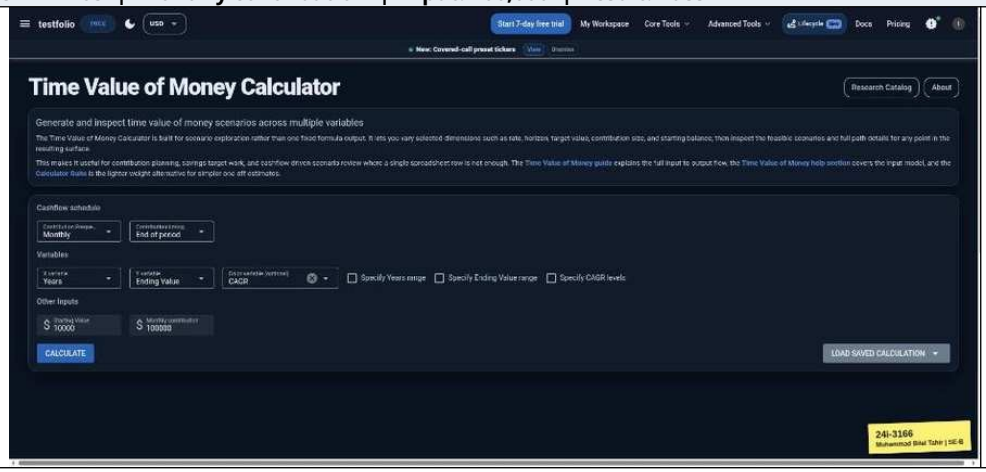


TC-BVA-104 | Monthly contribution | input: 0 | result: Pass





TC-BVA-105 | Monthly contribution | input: 100,000 | result: Pass





TC-BVA-106 | Monthly contribution | input: 100,001 | result: Pass

